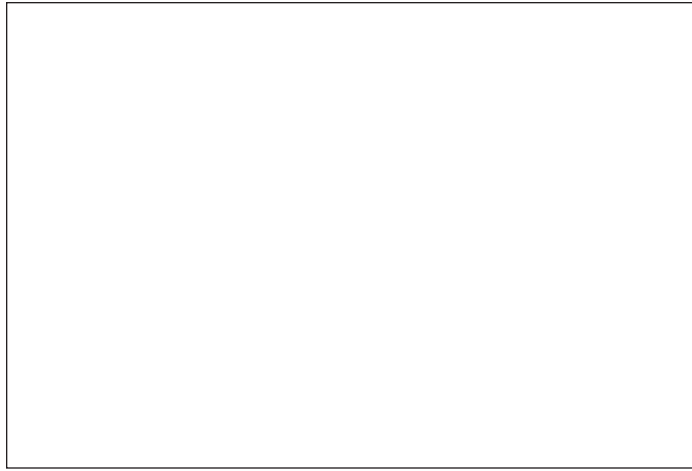




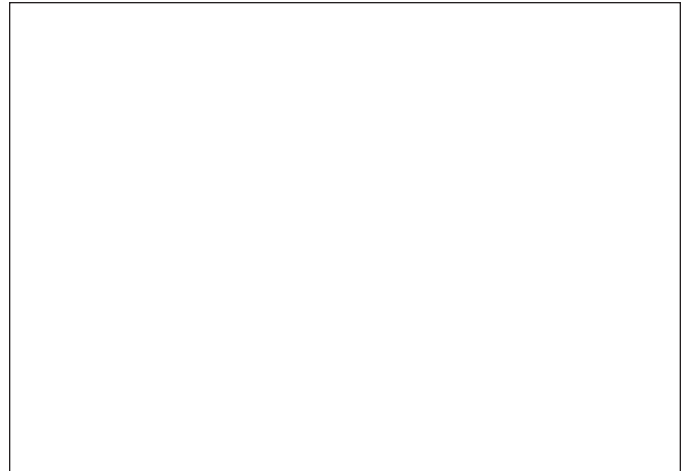
**Testing Lack's theory of optimum clutch size.** In this study from woods near Oxford, England, researchers found that the most common clutch size was 8, even though clutches of 12 produced the greatest number of surviving offspring. (After Boyce and Perrins, 1987.)

## The Experiment

Lack's theory is attractive because of its simplicity and common sense—but is it right? Many studies have been conducted to examine this hypothesis. Typically, experimenters would remove eggs from nests, and look to see if this improved the survivorship of the remaining offspring. If Lack is right, then it should, as the remaining offspring will have access to a larger share of what the parents can provide. Usually, however, removal of eggs did not seem to make any difference. Parents just adjusted down the amount of food they provided. The situation was clearly more complicated than Lack's simple theory envisioned.

One can always argue with tests such as these, however, as they involve direct interference with the nests, potentially having a major influence on how the birds behave. It is hard to believe that a bird caring for a nest of six eggs would not notice when one turned up missing. A clear test of Lack's theory would require avoiding all intervention.

Just such a test was completed in 1987 in the woods near Oxford, England. Over many years, Oxford University researchers led by Professor Mark Boyce (now at the University of Wyoming, Laramie) carefully monitored nests of a songbird, the greater tit, very common in the English countryside. They counted the number of eggs laid in each nest (the clutch size) and then watched to see how many of the offspring survived to fly away from the nest. Nothing was done to interfere with the birds. Over 22 years, they patiently examined 4489 nests.



**Two theories of optimum clutch size.** David Lack's theory predicts that optimum clutch size will be where reproductive success of the clutch is greatest. George Williams's theory predicts that optimum clutch size will be where the *net* benefit is greatest—that is, where the difference between the cost of reproduction and the reproductive success of the clutch is greatest.

## The Results

The Oxford researchers found that the average clutch size was 8 eggs, but that nests with the greatest number of surviving offspring had not 8 but 12 eggs in them! Clearly, Lack's theory is wrong. These birds are not producing as many offspring as natural selection to maximize fitness (that is, number of surviving offspring) would predict (see above *left*).

Lack's proposal had seemed eminently sensible. What was wrong? In 1966 the evolutionary theorist George Williams suggested the problem was that Lack's theory ignores the cost of reproduction (see above). If a bird spends too much energy feeding one brood, then it may not survive to raise another. Looking after a large clutch may extract too high a price in terms of future reproductive success of the parent. The clutch size actually favored by natural selection is adjusted for the wear-and-tear on the parents, so that it is almost always smaller than the number which would produce the most offspring in that nest—just what the Oxford researchers observed.

However, even William's "cost-of-reproduction" is not enough to completely explain Boyce's greater tit data. There were marked fluctuations in the weather over the years that the Oxford researchers gathered their data, and they observed that harsh years decreased survival of the young in large nests more than in small ones. This "bad-year" effect reduces the fitness of individuals laying larger clutches, and Boyce argues that it probably contributes at least as much as cost-of-reproduction in making it more advantageous, in the long term, for birds to lay clutches smaller than the Lack optimum.

# 24

## Population Ecology

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### Concept Outline

#### 24.1 Populations are individuals of the same species that live together.

**Population Ecology.** The borders of populations are determined by areas in which individuals cannot survive and reproduce. Population ranges expand and contract through time as conditions change.

**Population Dispersion.** The distribution of individuals in a population can be clumped, random, or even.

**Metapopulations.** Sometimes, populations are arranged in networks connected by the exchange of individuals.

#### 24.2 Population dynamics depend critically upon age distribution.

**Demography.** The growth rate of a population is a sensitive function of its age structure; populations with many young individuals grow rapidly as these individuals enter reproductive age.

#### 24.3 Life histories often reflect trade-offs between reproduction and survival.

**The Cost of Reproduction.** Evolutionary success is a trade-off between investment in current reproduction and in growth that promotes future reproduction.

#### 24.4 Population growth is limited by the environment.

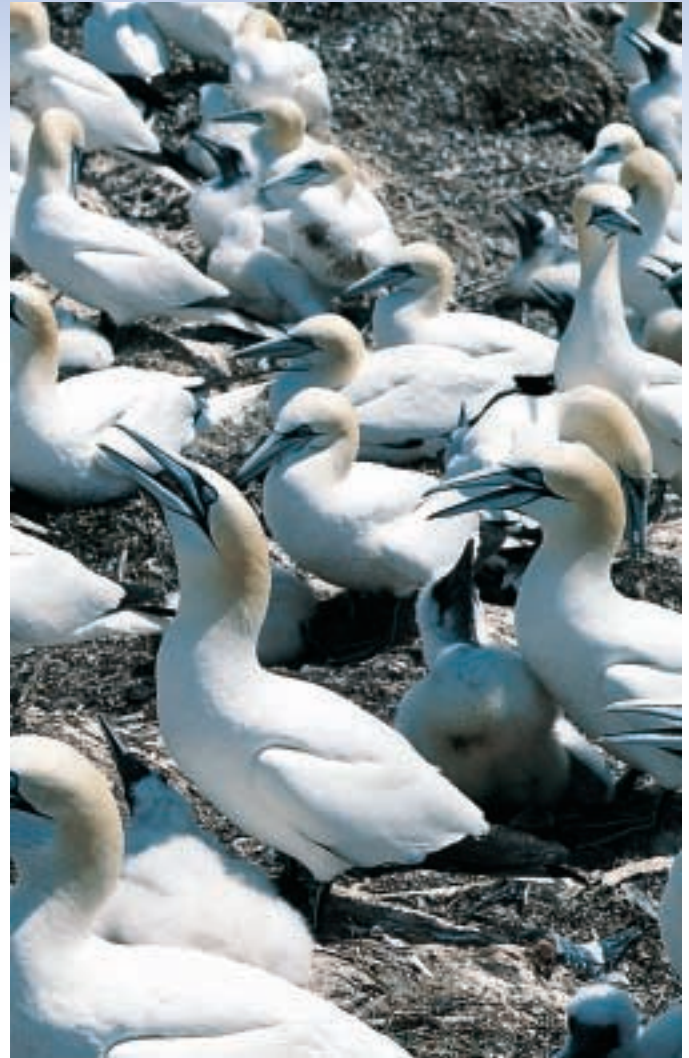
**Biotic Potential.** Populations grow if the birthrate exceeds the death rate until they reach the carrying capacity of their environment.

**The Influence of Population Density.** Some of the factors that regulate a population's growth depend upon the size of the population; others do not.

**Population Growth Rates and Life History Models.** Some species have adaptations for rapid, exponential population growth, whereas other species exhibit slower population growth and have intense competition for resources.

#### 24.5 The human population has grown explosively in the last three centuries.

**The Advent of Exponential Growth.** Human populations have been growing exponentially since the 1700s and will continue to grow in developing countries because of the number of young people entering their reproductive years.



**FIGURE 24.1**

**Life takes place in populations.** This population of gannets is subject to the rigorous effects of reproductive strategy, competition, predation, and other limiting factors.

**E**cology, the study of how organisms relate to one another and to their environments, is a complex and fascinating area of biology that has important implications for each of us. In our exploration of ecological principles, we will first consider the properties of populations, emphasizing population dynamics (figure 24.1). In chapter 25, we will discuss communities and the interactions that occur in them. Chapter 26 moves on to focus on animals and how and why they behave as they do. Chapter 27 then deals with behavior in an environmental context, the extent to which natural selection has molded behaviors adaptively.

## 24.1 Populations are individuals of the same species that live together.

### Population Ecology

Organisms live as members of **populations**, groups of individuals of a species that live together. In this chapter, we will consider the properties of populations, focusing on elements that influence whether a population will grow or shrink, and at what rate. The explosive growth of the world's human population in the last few centuries provides a focus for our inquiry.

A population consists of the individuals of a given species that occur together at one place and time. This flexible definition allows us to speak in similar terms of the world's human population, the population of protozoa in the gut of an individual termite, or the population of deer that inhabit a forest. Sometimes the boundaries defining a boundary are sharp, such as the edge of an isolated mountain lake for trout, and sometimes they are more fuzzy, such as when individuals readily move back and forth between areas, like deer in two forests separated by a cornfield.

Three aspects of populations are particularly important: the range throughout which a population occurs, the dispersion of individuals within that range, and the size a population attains.

### Population Distributions

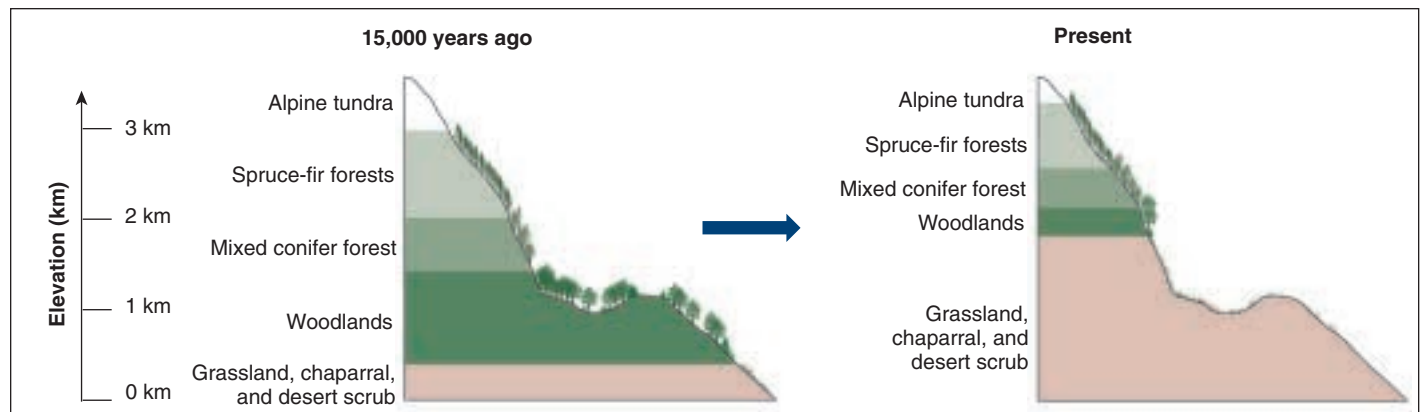
No population, not even of humans, occurs in all habitats throughout the world. Most species, in fact, have relatively limited geographic ranges. The Devil's Hole pupfish, for example, lives in a single hot water spring in southern Nevada, and the Socorro isopod is known from a single spring system in Socorro, New Mexico (figure 24.2). At the other extreme, some species are widely distributed. Populations of some whales, for example, are found throughout all of the oceans of the northern or southern hemisphere.

In chapter 29 we will discuss the variety of environmental challenges facing organisms. Suffice it to say for now that no population contains individuals adapted to live in all of the different environments on the earth. Polar bears are exquisitely adapted to survive the cold of the Arctic, but you won't find them in the tropical rain forest. Certain bacteria can live in the near boiling waters of Yellowstone's geysers, but they do not occur in cooler streams that are nearby. Each population has its own requirements—temperature, humidity, certain types of food, and a host of other factors—that determine where it can live and reproduce and where it can't. In addition, in places that are otherwise suitable, the presence of predators, competitors, or parasites may prevent a population from occupying an area.



**FIGURE 24.2**

**Species that occur in only one place.** These species, and many others, are only found in a single population. All are endangered species, and should anything happen to their single habitat, the population—and the species—would go extinct.



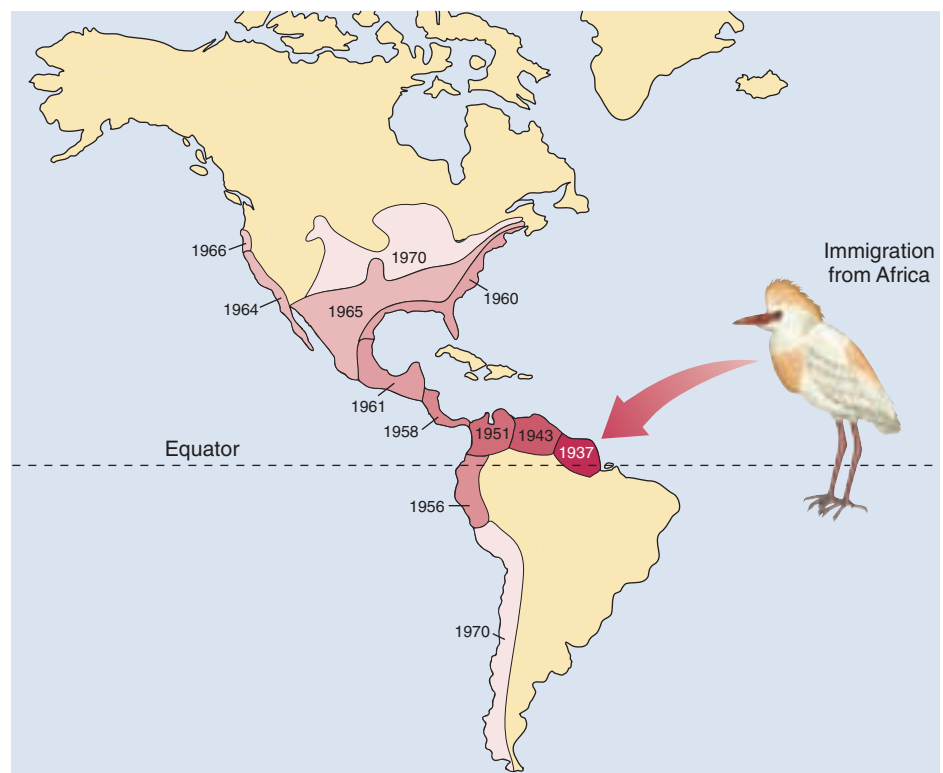
**FIGURE 24.3**

**Altitudinal shifts in population ranges.** During the glacial period 15,000 years ago, conditions were cooler than they are now. As the climate has warmed, tree species that require colder temperatures have shifted their distributional range upward in altitude so that they live in the climatic conditions to which they are adapted.

### Range Expansions and Contractions

Population ranges are not static, but, rather, change through time. These changes occur for two reasons. In some cases, the environment changes. For example, as the glaciers retreated at the end of the last ice age, approximately 10,000 years ago, many North American plant and animal populations expanded northward. At the same time, as climates have warmed, species have experienced shifts in the elevation at which they are found on mountains (figure 24.3).

In addition, populations can expand their ranges when they are able to circumvent inhospitable habitat to colonize suitable, previously unoccupied areas. For example, the cattle egret is native to Africa. Some time in the late 1800s, these birds appeared in northern South America, having made the nearly 2000-mile transatlantic crossing, perhaps aided by strong winds. Since then, they have steadily expanded their range such that they now can be found throughout most of the United States (figure 24.4).



**FIGURE 24.4**

**Range expansion of the cattle egret.** Although the cattle egret—so-named because it follows cattle and other hoofed animals, catching any insects or small vertebrates that they disturb—first arrived in South America in the late 1800s, the oldest preserved specimen dates from the 1930s. Since then, the range expansion of this species has been well documented, as it has moved westward and up into much of North America, as well as down the western side of the Andes to near the southern tip of South America.

## Population Dispersion

Another key characteristic of population structure is the way in which individuals of a population are arranged. They may be randomly spaced, uniformly spaced, or clumped (figure 24.5).

### Randomly spaced

Individuals are randomly spaced within populations when they do not interact strongly with one another or with nonuniform aspects of their microenvironment. Random distributions are not common in nature. Some species of trees, however, appear to exhibit random distributions in Amazonian rain forests.

### Uniformly spaced

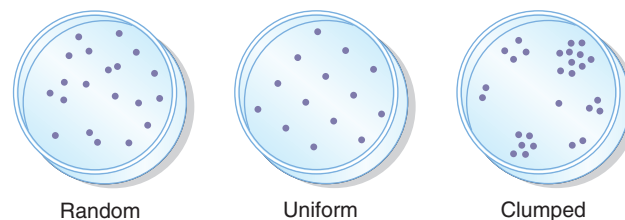
Individuals often are uniformly spaced within a population. This spacing may often, but not always, result from competition for resources. The means by which it is accomplished, however, varies.

In animals, uniform spacing often results from behavioral interactions, which we will discuss in chapter 27. In many species, individuals of one or both sexes defend a territory from which other individuals are excluded. These territories serve to provide the owner with exclusive access to resources such as food, water, hiding refuges, or mates and tend to space individuals evenly across the habitat. Even in nonterritorial species, individuals often maintain a defended space into which other animals are not allowed to intrude.

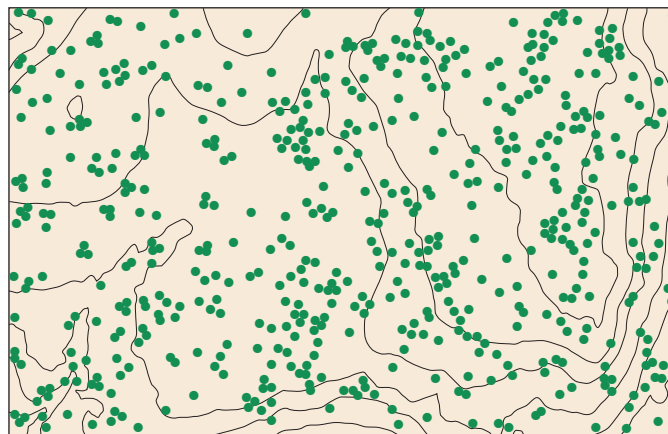
Among plants, uniform spacing also is a common result of competition for resources. In this case, however, the spacing results from direct competition for the resources. Closely spaced individual plants will contest for available sunlight, nutrients, or water. These contests can be direct, such as one plant casting a shadow over another, or indirect, such as two plants competing to see which is more efficient at extracting nutrients or water from a shared area. Only plants that are spaced an adequate distance from each other will be able to coexist, leading to uniform spacing.

**FIGURE 24.5**

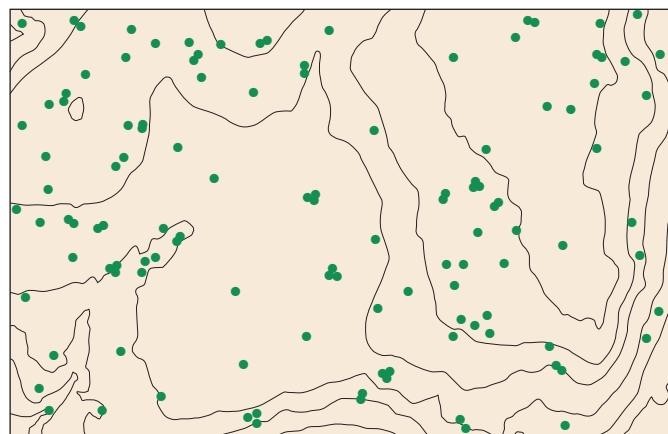
**Population dispersion.** (a) Different arrangements of bacterial colonies. The different patterns of dispersion are exhibited by three different species of trees from the same locality in the Amazonian rain forest. (b) *Brosimum alicastrum* is randomly dispersed, (c) *Coccoloba coronata* is uniformly dispersed, and (d) *Chamguava schippii* exhibits a clumped distribution.



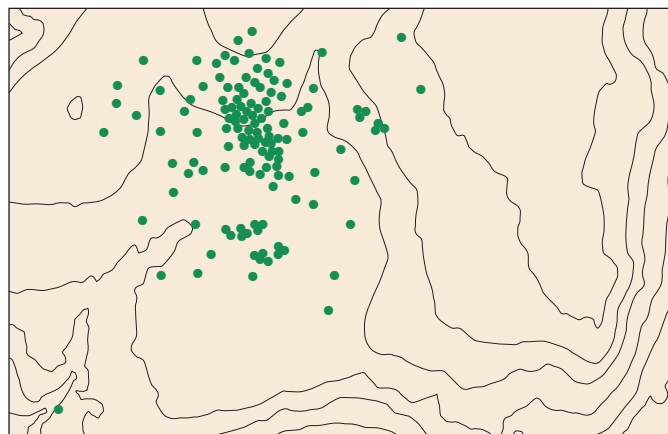
(a) Bacterial colonies



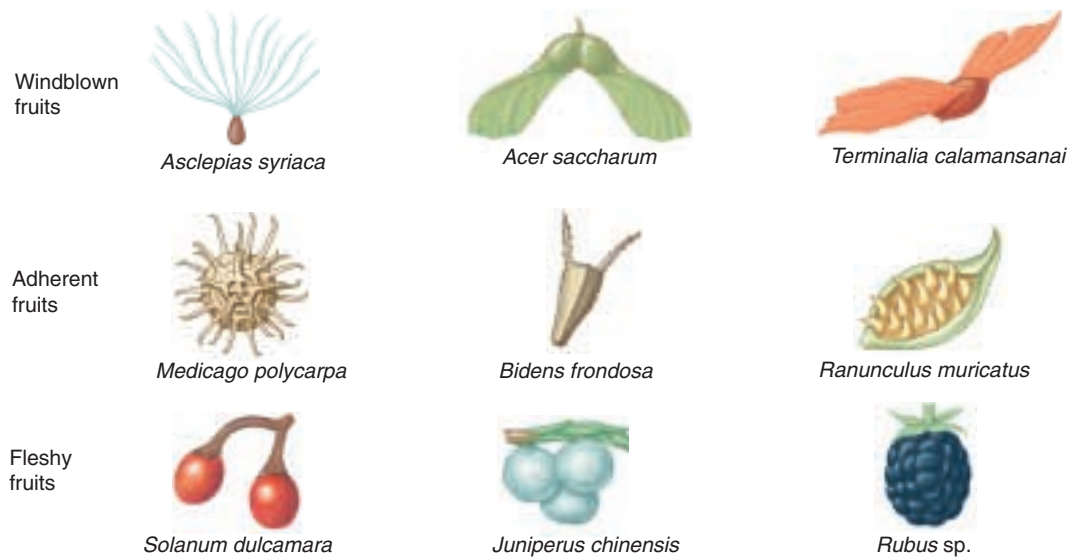
(b) Random distribution of *Brosimum alicastrum*



(c) Uniform distribution of *Coccoloba coronata*



(d) Clumped distribution of *Chamguava schippii*



**FIGURE 24.6**  
Some of the many adaptations of seeds to facilitate dispersal. Seeds have evolved a number of different means of moving long distances from their maternal plant.

## Clumped Spacing

Individuals clump into groups or clusters in response to uneven distribution of resources in their immediate environments. Clumped distributions are common in nature because individual animals, plants, and microorganisms tend to prefer microhabitats defined by soil type, moisture, or certain kinds of host trees.

Social interactions also can lead to clumped distributions. Many species live and move around in large groups, which go by a variety of names (examples include herds of antelope, flocks of birds, gaggles of geese, packs of wolves, prides of lions). Such groupings can provide many advantages, including increased awareness of and defense against predators, decreased energetic cost of moving through air and water, and access to the knowledge of all group members.

At a broader scale, populations are often most densely populated in the interior of their range and less densely distributed toward the edges. Such patterns usually result from the manner in which the environment changes in different areas. Populations are often best adapted to the conditions in the interior of their distribution. As environmental conditions change, individuals are less well adapted and thus densities decrease. Ultimately, the point is reached at which individuals cannot persist at all; this marks the edge of a population's range.

## The Human Effect

By altering the environment, we have allowed some species, such as coyotes, to expand their ranges, although, sadly, for most species the effect has been detrimental. Moreover, humans have served as an agent of dispersal for many species. Some of these transplants have been widely successful. For example, 100 starlings were introduced

into New York City in 1896 in a misguided attempt to establish every species of bird mentioned by Shakespeare. Their population steadily spread such that by 1980, they occurred throughout the United States. Similar stories could be told for countless numbers of plants and animals, and the list increases every year. Unfortunately, the success of these invaders often comes at the expense of native species.

## Dispersal Mechanisms

Dispersal to new areas can occur in many ways. Lizards, for example, have colonized many distant islands, probably by individuals or their eggs floating or drifting on vegetation. Seeds of many plants are designed to disperse in many ways (figure 24.6). Some seeds are aerodynamically designed to be blown long distances by the wind. Others have structures that stick to the fur or feathers of animals, so that they are carried long distances before falling to the ground. Still others are enclosed in fruits. These seeds can pass through the digestive systems of mammals or birds and then germinate at the spot upon which they are defecated. Finally, seeds of *Arceuthobium* are violently propelled from the base of the fruit in an explosive discharge. Although the probability of long-distance dispersal events occurring and leading to successful establishment of new populations is slim, over millions of years, many such dispersals have occurred.

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**A population is a group of individuals of the same species living together at the same place and time. The range of a population is limited by ecologically inhospitable habitats, but through time, these range boundaries can change.**

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## Metapopulations

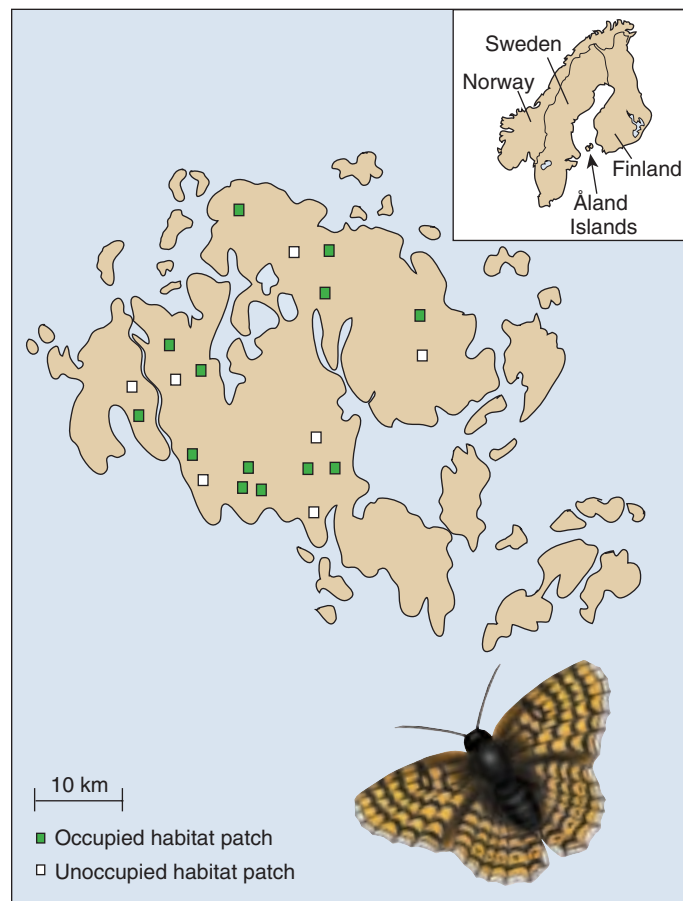
Species are often composed of a network of distinct populations that interact with each other by exchanging individuals. Such networks are termed **metapopulations** and usually occur in areas in which suitable habitat is patchily distributed and separated by intervening stretches of unsuitable habitat.

To what degree populations within a metapopulation interact depends on the amount of dispersal and is often not symmetrical: populations increasing in size may tend to send out many dispersers, whereas populations at low levels will tend to receive more immigrants than they send off. In addition, relatively isolated populations will tend to receive relatively few arrivals.

Not all suitable habitats within a metapopulation's area may be occupied at any one time. For various reasons, some individual populations may go extinct, perhaps as a result of an epidemic disease, a catastrophic fire, or inbreeding depression. However, because of dispersal from other populations, such areas may eventually be recolonized. In some cases, the number of habitats occupied in a metapopulation may represent an equilibrium in which the rate of extinction of existing populations is balanced by the rate of colonization of empty habitats.

A second type of metapopulation structure occurs in areas in which some habitats are suitable for long-term population maintenance, whereas others are not. In these situations, termed **source-sink metapopulations**, the populations in the better areas (the sources) continually send out dispersers that bolster the populations in the poorer habitats (the sinks). In the absence of such continual replenishment, sink populations would have a negative growth rate and would eventually become extinct.

Metapopulations of butterflies have been studied particularly intensively (figure 24.7). In one study, Ilkka Hanski and colleagues at the University of Helsinki sampled populations of the glanville fritillary butterfly at 1600 meadows in southwestern Finland. On average, every year, 200 populations became extinct, but 114 empty meadows were colonized. A variety of factors seemed to increase the likelihood of a population's extinction, including small population size, isolation from sources of immigrants, low resource availability (as indicated by the number of flowers on a meadow), and lack of genetic variation present within the population. The researchers attribute the greater number of extinctions than colonizations to a string of very dry summers. Because none of the populations is large enough to survive on its own, continued survival of the species in southwestern Finland would appear to require the continued existence of a metapopulation network in which new populations are continually created and existing populations are supplemented by emigrants. Continued bad weather thus may doom the species, at least in this part of its range.



**FIGURE 24.7**

**Metapopulations of butterflies.** The glanville fritillary butterfly occurs in metapopulations in southwestern Finland on the Åland Islands. None of the populations is large enough to survive for long on its own, but continual emigration of individuals from other populations allows some populations to survive. In addition, continual establishment of new populations tends to offset extinction of established populations, although in recent years, extinctions have outnumbered colonizations.

Metapopulations, where they occur, can have two important implications for the range of a species. First, by continual colonization of empty patches, they prevent long-term extinction. If no such dispersal existed, then each population might eventually perish, leading to disappearance of the species from the entire area. Moreover, in source-sink metapopulations, the species as a whole occupies a larger area than it otherwise might occupy. For these reasons, the study of metapopulations has become very important in conservation biology as natural habitats become increasingly fragmented.

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**The distribution of individuals within a population can be random, uniform, or clumped. Across broader areas, individuals may occur in populations that are loosely interconnected, termed metapopulations.**

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## 24.2 Population dynamics depend critically upon age distribution.

One of the important features of any population is its size. Population size has a direct bearing on the ability of a given population to survive: for a variety of reasons discussed in chapter 31, smaller populations are at a greater risk of disappearing than large populations. In addition, the interactions that occur between members of a population also depend critically on a population's size and density.

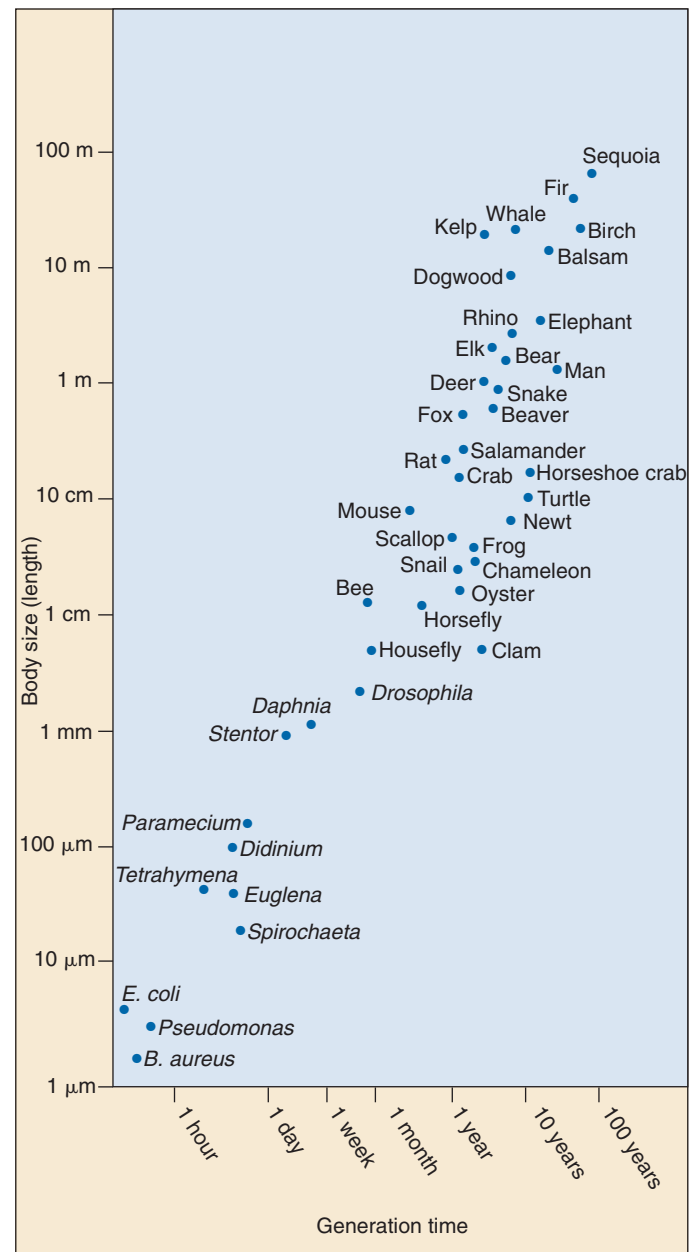
### Demography

**Demography** (from the Greek *demos*, “the people,” + *graphos*, “measurement”) is the statistical study of populations. How the size of a population changes through time can be studied at two levels. At the most inclusive level, we can study the population as a whole to determine whether it is increasing, decreasing, or remaining constant. Populations grow if births outnumber deaths and shrink if deaths outnumber births. Understanding these trends is often easier if we break a population down into its constituent parts and analyze each separately.

### Factors Affecting Population Growth Rates

The proportion of males and females in a population is its **sex ratio**. The number of births in a population is usually directly related to the number of females, but may not be as closely related to the number of males in species in which a single male can mate with several females. In many species, males compete for the opportunity to mate with females (a situation we discuss in chapter 27); consequently, a few males get many matings, whereas many males do not mate at all. In such species, a female-biased sex ratio would not affect population growth rates; reduction in the number of males simply changes the identities of the reproductive males without reducing the number of births. Among monogamous species like many birds, by contrast, in which pairs form long-lasting reproductive relationships, a reduction in the number of males can directly reduce the number of births.

**Generation time**, defined as the average interval between the birth of an individual and the birth of its offspring, can also affect population growth rates. Species differ greatly in generation time. Differences in body size can explain much of this variation—mice go through approximately 100 generations during the course of one elephant generation—but not all of it (figure 24.8). Newts, for example, are smaller than mice, but have considerably longer generation times. Everything else equal, populations with shorter generations can increase in size more quickly than populations with long generations. Con-



**FIGURE 24.8**

**The relationship between body size and generation time.** In general, larger animals have longer generation times, although there are exceptions.

versely, because generation time and life span are usually closely correlated, populations with short generation times may also diminish in size more rapidly if birthrates suddenly decrease.

Age Structure

In most species, the probability that an individual will reproduce or die varies through its life span. A group of individuals of the same age is referred to as a **cohort**. Within a population, every cohort has a characteristic birthrate, or **fecundity**, defined as the number of offspring produced in a standard time (for example, per year), and a characteristic death rate, or **mortality**, the number of individuals that die in that period.

The relative number of individuals in each cohort defines a population’s **age structure**. Because individuals of different ages have different fecundity and death rates, age structure has a critical impact on a population’s growth rate. Populations with a large proportion of young individuals, for example, tend to grow rapidly because an increasing proportion of their individuals are reproductive. Populations in many underdeveloped countries are an example, as we will discuss later in the chapter. Conversely, if a large proportion of a population is relatively old, populations may decline. This phenomenon now characterizes some wealthy countries in Europe and Japan.

Life Tables and Population Change through Time

Ecologists use **life tables** to assess how populations in nature are changing. Life tables can be constructed by following the fate of a cohort from birth until death, noting the number of offspring produced and individuals that die each year. A very nice example of a life table analysis is exhibited in a study of the meadow grass *Poa annua*. This study follows the fate of 843 individuals through time, charting how many survive in each interval and how many offspring each survivor produces (table 24.1).

In table 24.1, the first column indicates the age of the cohort (that is, the number of 3-month intervals from the start of the study). The second and third columns indicate the number of survivors and the proportion of the original cohort still alive at the beginning of that interval. The fourth column presents the **mortality rate**, the proportion of individuals that started that interval alive but died by the end of it. The fifth column indicates the average number of seeds produced by each surviving individual in that interval, and the last column presents the number of seeds produced relative to the size of the original cohort.

| Table 24.1 Life Table for a Cohort of the grass <i>Poa annua</i> |                                                  |                                                                                      |                                           |                                                                 |                                |
|------------------------------------------------------------------|--------------------------------------------------|--------------------------------------------------------------------------------------|-------------------------------------------|-----------------------------------------------------------------|--------------------------------|
| Age<br>(in 3-<br>month<br>intervals)                             | Number alive at<br>beginning of time<br>interval | Proportion of cohort<br>surviving to beginning<br>of time interval<br>(survivorship) | Mortality<br>rate during<br>time interval | Seeds<br>produced<br>per surviving<br>individual<br>(fecundity) | Fecundity<br>×<br>survivorship |
| 0                                                                | 843                                              | 1.000                                                                                | 0.143                                     | 0.00                                                            | 0.00                           |
| 1                                                                | 722                                              | 0.857                                                                                | 0.271                                     | 0.42                                                            | 0.36                           |
| 2                                                                | 527                                              | 0.625                                                                                | 0.400                                     | 1.18                                                            | 0.74                           |
| 3                                                                | 316                                              | 0.375                                                                                | 0.544                                     | 1.36                                                            | 0.51                           |
| 4                                                                | 144                                              | 0.171                                                                                | 0.626                                     | 1.46                                                            | 0.25                           |
| 5                                                                | 54                                               | 0.064                                                                                | 0.722                                     | 1.11                                                            | 0.07                           |
| 6                                                                | 15                                               | 0.018                                                                                | 0.800                                     | 2.00                                                            | 0.04                           |
| 7                                                                | 3                                                | 0.004                                                                                | 1.000                                     | 3.33                                                            | <u>0.01</u>                    |
| 8                                                                | 0                                                | 0.000                                                                                |                                           |                                                                 | Total = 1.98                   |
| Modified from Ricklefs, 1997.                                    |                                                  |                                                                                      |                                           |                                                                 |                                |

Much can be learned from examination of life tables. In this particular case, we see that the probability of dying increases steadily with age, whereas the number of offspring produced increases with age. By adding up the numbers in the last column, we get the total number of offspring produced per individual in the initial cohort. This number is almost 2, which means that for every original member of the cohort, on average two individuals have been produced. A figure of 1.0 would be the break-even number, the point at which the population was neither growing nor shrinking. In this case, the population appears to be growing rapidly.

In most cases, life table analysis is more complicated than this. First, except for organisms with short life spans, it is difficult to track the fate of a cohort from birth until death of the last individual. An alternative approach is to construct a cross-sectional study, examining the fate of all cohorts over a single year. In addition, many factors—such as offspring reproducing before all members of their parental generation's cohort have died—complicate the interpretation of whether populations are growing or shrinking.

### Survivorship Curves

One way to express some aspects of the age distribution characteristics of populations is through a **survivorship curve**. Survivorship is defined as the percentage of an original population that survives to a given age. Examples of different kinds of survivorship curves are shown in figure 24.9. In hydra, animals related to jellyfish, individuals are equally likely to die at any age, as indicated by the straight survivorship curve (type II). Oysters, like plants, produce vast numbers of offspring, only a few of which live to reproduce. However, once they become established and grow into reproductive individuals, their mortality rate is extremely low (type III survivorship curve). Finally, even though human babies are susceptible to death at relatively high rates, mortality rates in humans, as in many animals and protists, rise steeply in the postreproductive years (type I survivorship curve). Examination of the data for *Poa annua* reveals that it approximates a type II survivorship curve (figure 24.10).

The growth rate of a population is a sensitive function of its age structure. The age structure of a population and the manner in which mortality and birthrates vary among different age cohorts determine whether a population will increase or decrease in size.

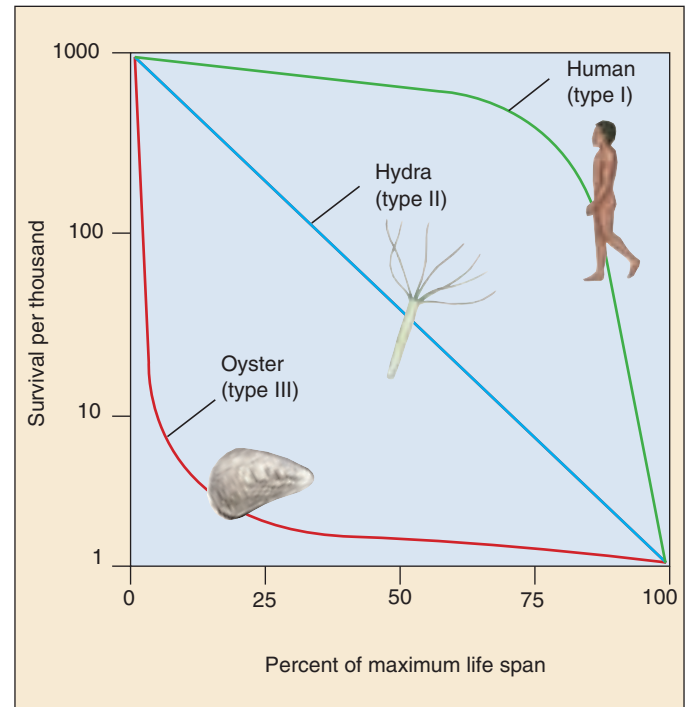


FIGURE 24.9

**Survivorship curves.** By convention, survival (the vertical axis) is plotted on a log scale. Humans have a type I life cycle, the hydra (an animal related to jellyfish) type II, and oysters type III.

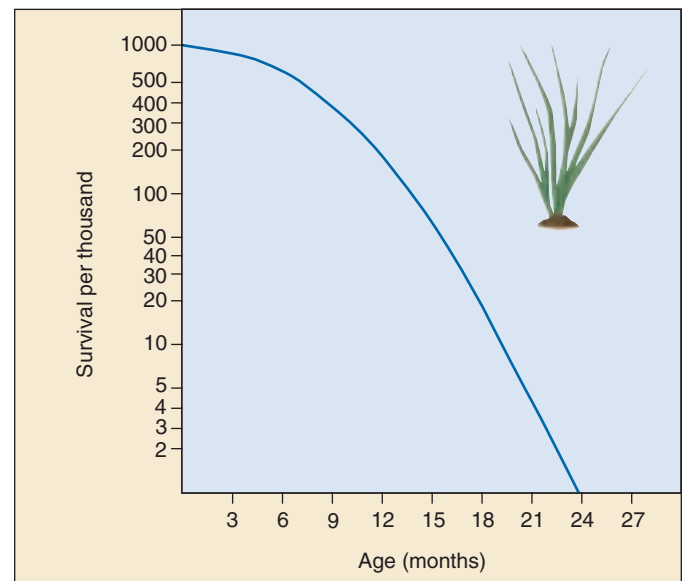


FIGURE 24.10

**Survivorship curve for a cohort of the meadow grass, *Poa annua*.** Mortality increases at a constant rate through time.

## 24.3 Life histories often reflect trade-offs between reproduction and survival.

Natural selection favors traits that maximize the number of surviving offspring left in the next generation. Two factors affect this quantity: how long an individual lives and how many young it produces each year. Why doesn't every organism reproduce immediately after its own birth, produce large families of large offspring, care for them intensively, and do this repeatedly throughout a long life, while outcompeting others, escaping predators, and capturing food with ease? The answer is that no one organism can do all of this—there are simply not enough resources available. Consequently, organisms allocate resources either to current reproduction or to increase their prospects of surviving and reproducing at later life stages.

### The Cost of Reproduction

The complete life cycle of an organism constitutes its **life history**. All life histories involve significant trade-offs. Because resources are limited, a change that increases reproduction may decrease survival and reduce future reproduction. Thus, a Douglas fir tree that produces more cones increases its current reproductive success, but it also grows more slowly; because the number of cones produced is a function of how large a tree is, this diminished growth will decrease the number of cones it can produce in the future. Similarly, birds that have more offspring each year have a higher probability of dying during that year or producing smaller clutches the following year (figure 24.11). Conversely, individuals that delay reproduction may grow faster and larger, enhancing future reproduction.

In one elegant experiment, researchers changed the number of eggs in nests of a bird, the collared flycatcher (figure 24.12). Birds whose clutch size (the number of eggs produced in one breeding event) was decreased laid more eggs the next year, whereas those given more eggs produced fewer eggs the following year. Ecologists refer to the reduction in future reproductive potential resulting from current reproductive efforts as the **cost of reproduction**.

Natural selection will favor the life history that maximizes lifetime reproductive success. When the cost of reproduction is low, individuals should invest in producing as many offspring as possible because there is little cost. Low costs of reproduction may occur when resources are abundant, such that producing offspring does not impair survival or the ability to produce many offspring in subsequent years. Costs of reproduction will also be low when overall mortality rates are high. In such cases, individuals may be unlikely to survive to the next breeding season anyway, so the incremental effect of increased reproductive efforts may not make a difference in future survival.

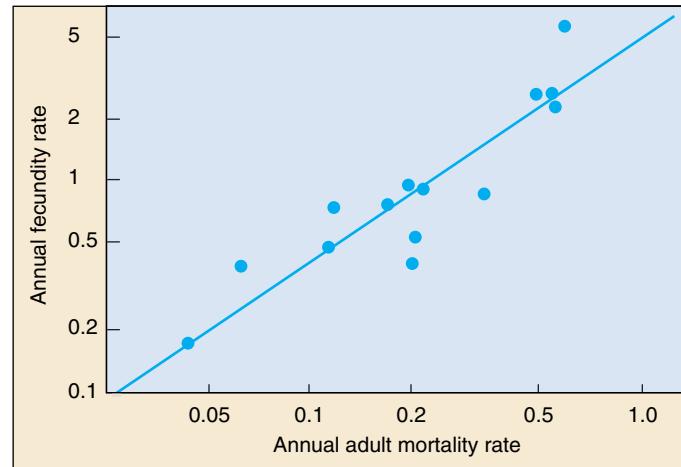


FIGURE 24.11

**Reproduction has a price.** Increased fecundity in birds correlates with higher mortality in several populations of birds ranging from albatross (low) to sparrow (high). Birds that raise more offspring per year have a higher probability of dying during that year.

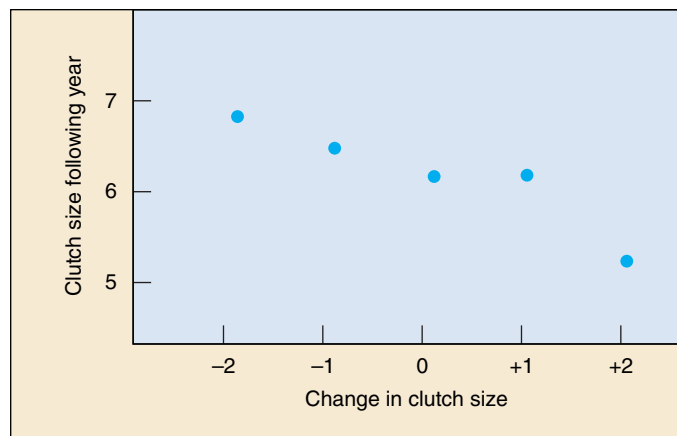


FIGURE 24.12

**Reproductive events per lifetime.** Adding eggs to nests of collared flycatchers (which increases the reproductive efforts of the female rearing the young) decreases clutch size the following year; removing eggs from the nest increases the next year's clutch size. This experiment demonstrates the tradeoff between current reproductive effort and future reproductive success.

Alternatively, when costs of reproduction are high, lifetime reproductive success may be maximized by deferring or minimizing current reproduction to enhance growth and survival rates. This may occur when costs of reproduction significantly affect the ability of an individual to survive or decrease the number of offspring that can be produced in the future.

## Investment per Offspring

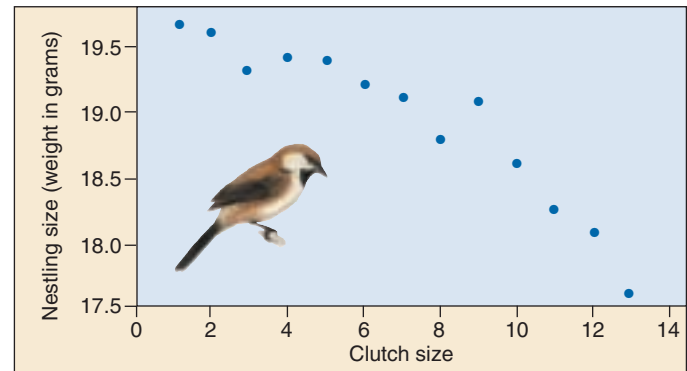
In terms of natural selection, the number of offspring produced is not as important as how many of those offspring themselves survive to reproduce.

A key reproductive trade-off concerns how many resources to invest in producing any single offspring. Assuming that the amount of energy to be invested in offspring is limited, a trade-off must exist between the number of offspring produced and the size of each offspring (figure 24.13). This trade-off has been experimentally demonstrated in the side-blotched lizard, *Uta stansburiana*, which normally lays on average four and a half eggs at a time. When some of the eggs are removed surgically early in the reproductive cycle, the female lizard produces only 1 to 3 eggs, but supplies each of these eggs with greater amounts of yolk, producing eggs that are much larger than normal.

In many species, the size of offspring critically affects their survival prospects—larger offspring have a greater chance of survival. Producing many offspring with little chance of survival might not be the best strategy, but producing only a single, extraordinarily robust offspring also would not maximize the number of surviving offspring. Rather, an intermediate situation, in which several fairly large offspring are produced, should maximize the number of surviving offspring. This example is fundamentally the same as the trade-off between clutch size and parental investment discussed above; in this case, the parental investment is simply how many resources can be invested in each offspring before they are born.

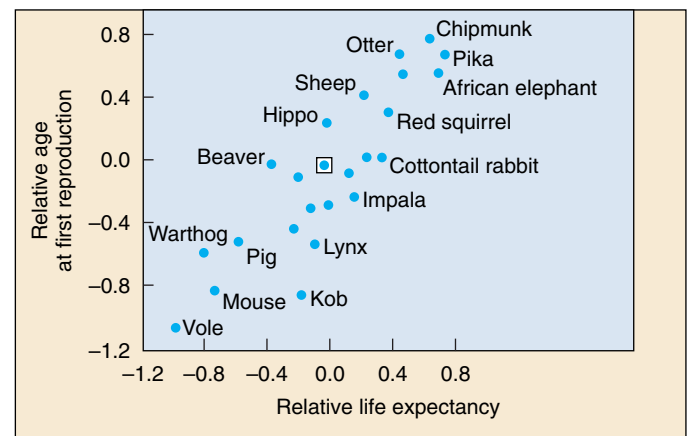
## Reproductive Events per Lifetime

The trade-off between age and fecundity plays a key role in many life histories. Annual plants and most insects focus all of their reproductive resources on a single large event and then die. This life history adaptation is called **semelparity** (from the Latin *semel*, “once,” + *parito*, “to beget”). Organisms that produce offspring several times over many seasons exhibit a life history adaptation called **iteroparity** (from the Latin *itero*, “to repeat”). Species that reproduce yearly must avoid overtaxing themselves in any one reproductive episode so that they will be able to survive and reproduce in the future. Semelparity, or “big bang” reproduction, is usually found in short-lived species in which the probability of staying alive between broods is low, such as plants growing in harsh climates. Semelparity is also favored when fecundity entails large reproductive cost, as when Pacific salmon migrate upriver to their spawning grounds. In these species, rather than investing some resources in an unlikely bid to survive until the next breeding season, individuals place all their resources into reproduction.



**FIGURE 24.13**

**The relationship between clutch size and offspring size.** In great tits, the size of nestlings is inversely related to the number of eggs laid. The more mouths they have to feed, the less the parents can provide to any one nestling.



**FIGURE 24.14**

**Age at first reproduction.** Among mammals, compensating for the effects of size, age at first reproduction increases with life expectancy at birth. Each dot represents a species. Values are relative to the species symbolized □. (After Begon et al., 1996.)

## Age at First Reproduction

Among mammals and many other animals, longer-lived species reproduce later (figure 24.14). Birds, for example, gain experience as juveniles before expending the high costs of reproduction. In long-lived animals, the relative advantage of juvenile experience outweighs the energy investment in survival and growth. In shorter-lived animals, on the other hand, quick reproduction is more critical than juvenile training, and reproduction tends to occur earlier.

**Life history adaptations involve many trade-offs between reproductive cost and investment in survival. Different kinds of animals and plants employ quite different approaches.**

## 24.4 Population growth is limited by the environment.

### Biotic Potential

Populations often remain at a relatively constant size, regardless of how many offspring they produce. As you saw in chapter 1, Darwin based his theory of natural selection partly on this seeming contradiction. Natural selection occurs because of checks on reproduction, with some individuals reproducing less often than others. To understand populations, we must consider how they grow and what factors in nature limit population growth.

### The Exponential Growth Model

The actual rate of population increase,  $r$ , is defined as the difference between the birth rate and the death rate corrected for any movement of individuals in or out of the population, whether net emigration (movement out of the area) or net immigration (movement into the area). Thus,

$$r = (b - d) + (i - e)$$

Movements of individuals can have a major impact on population growth rates. For example, the increase in human population in the United States during the closing decades of the twentieth century is mostly due to immigrants. Less than half of the increase came from the reproduction of the people already living there.

The simplest model of population growth assumes a population growing without limits at its maximal rate. This rate, called the **biotic potential**, is the rate at which a population of a given species will increase when no limits are placed on its rate of growth. In mathematical terms, this is defined by the following formula:

$$\frac{dN}{dt} = r_i N$$

where  $N$  is the number of individuals in the population,  $dN/dt$  is the rate of change in its numbers over time, and  $r_i$  is the intrinsic rate of natural increase for that population—its innate capacity for growth.

The innate capacity for growth of any population is exponential (red line in figure 24.15). Even when the *rate* of increase remains constant, the actual increase in the *number* of individuals accelerates rapidly as the size of the population grows. The result of unchecked exponential growth is a population explosion. A single pair of houseflies, laying 120 eggs per generation, could produce more than 5 trillion descendants in a year. In 10 years, their descendants would form a swarm more than 2 meters thick over the entire surface of the earth! In practice, such patterns of unrestrained growth prevail only for short periods, usually when an organism reaches a new habitat with abundant resources (figure 24.16). Natural examples include dandelions reaching the fields, lawns, and meadows

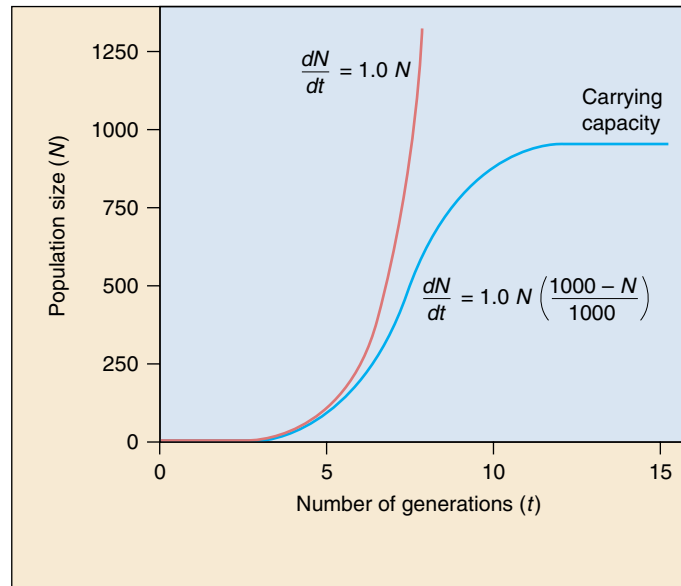


FIGURE 24.15

**Two models of population growth.** The red line illustrates the exponential growth model for a population with an  $r$  of 1.0. The blue line illustrates the logistic growth model in a population with  $r = 1.0$  and  $K = 1000$  individuals. At first, logistic growth accelerates exponentially, then, as resources become limiting, the death rate increases and growth slows. Growth ceases when the death rate equals the birthrate. The carrying capacity ( $K$ ) ultimately depends on the resources available in the environment.



FIGURE 24.16

**An example of a rapidly increasing population.** European purple loosestrife, *Lythrum salicaria*, became naturalized over thousands of square miles of marshes and other wetlands in North America. It was introduced sometime before 1860 and has had a negative impact on many native plants and animals.

of North America from Europe for the first time; algae colonizing a newly formed pond; or the first terrestrial immigrants arriving on an island recently thrust up from the sea.

## Carrying Capacity

No matter how rapidly populations grow, they eventually reach a limit imposed by shortages of important environmental factors, such as space, light, water, or nutrients. A population ultimately may stabilize at a certain size, called the **carrying capacity** of the particular place where it lives. The carrying capacity, symbolized by  $K$ , is the maximum number of individuals that a population can support.

## The Logistic Growth Model

As a population approaches its carrying capacity, its rate of growth slows greatly, because fewer resources remain for each new individual to use. The growth curve of such a population, which is always limited by one or more factors in the environment, can be approximated by the following logistic growth equation:

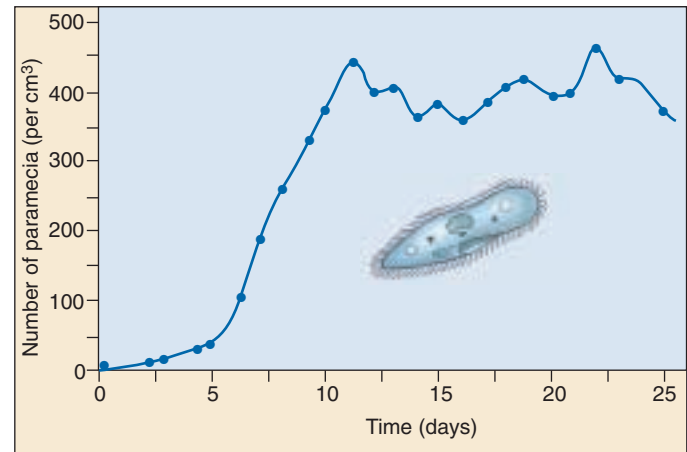
$$\frac{dN}{dt} = rN \left( \frac{K - N}{K} \right)$$

In this logistic model of population growth, the growth rate of the population ( $dN/dt$ ) equals its rate of increase ( $r$  multiplied by  $N$ , the number of individuals present at any one time), adjusted for the amount of resources available. The adjustment is made by multiplying  $rN$  by the fraction of  $K$  still unused ( $K$  minus  $N$ , divided by  $K$ ). As  $N$  increases (the population grows in size), the fraction by which  $r$  is multiplied (the remaining resources) becomes smaller and smaller, and the rate of increase of the population declines.

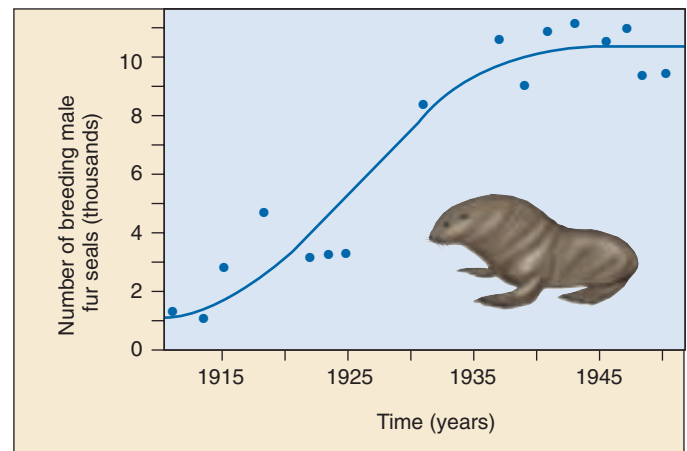
In mathematical terms, as  $N$  approaches  $K$ , the *rate* of population growth ( $dN/dt$ ) begins to slow, reaching 0 when  $N = K$  (blue line in figure 24.18). Graphically, if you plot  $N$  versus  $t$  (time) you obtain an **S-shaped sigmoid growth curve** characteristic of many biological populations. The curve is called “sigmoid” because its shape has a double curve like the letter S. As the size of a population stabilizes at the carrying capacity, its rate of growth slows down, eventually coming to a halt (figure 24.17a).

In many cases, real populations display trends corresponding to a logistic growth curve. This is true not only in the laboratory, but also in natural populations (figure 24.17b). In some cases, however, the fit is not perfect (figure 24.17c) and, as we shall see shortly, many populations exhibit other patterns.

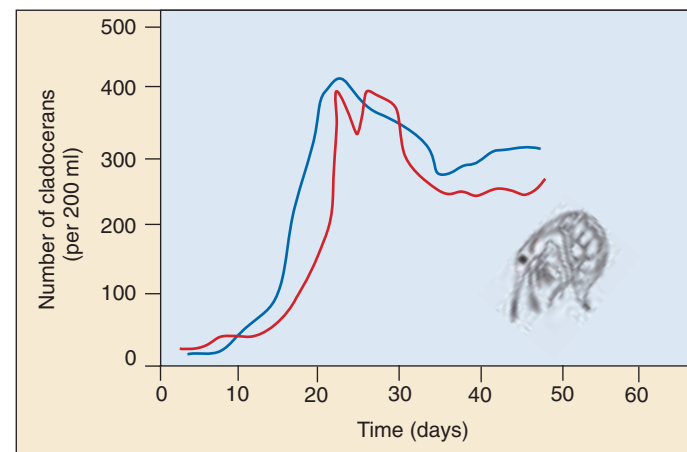
**The size at which a population stabilizes in a particular place is defined as the carrying capacity of that place for that species. Populations often grow to the carrying capacity of their environment.**



(a)



(b)



(c)

**FIGURE 24.17**

**Most natural populations exhibit logistic growth. (a)**

*Paramecium* grown in a laboratory environment. (b) A fur seal

(*Callorhinus ursinus*) population on St. Paul Island, Alaska. (c)

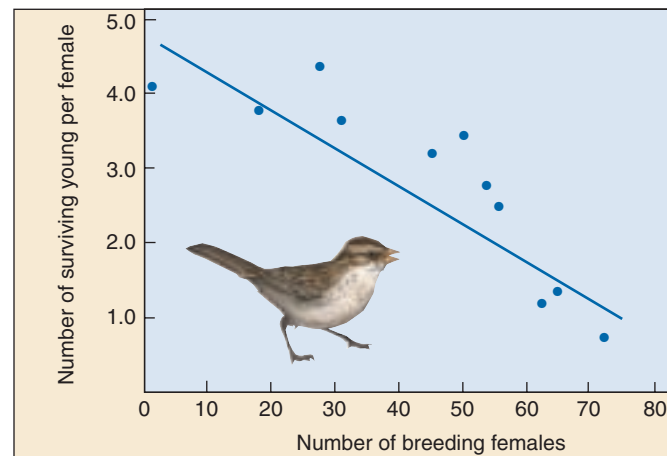
Laboratory populations of two populations of the cladoceran *Bosmina longirostris*. Note that the populations first exceeded the carrying capacity, before decreasing to a size which was then maintained.

## The Influence of Population Density

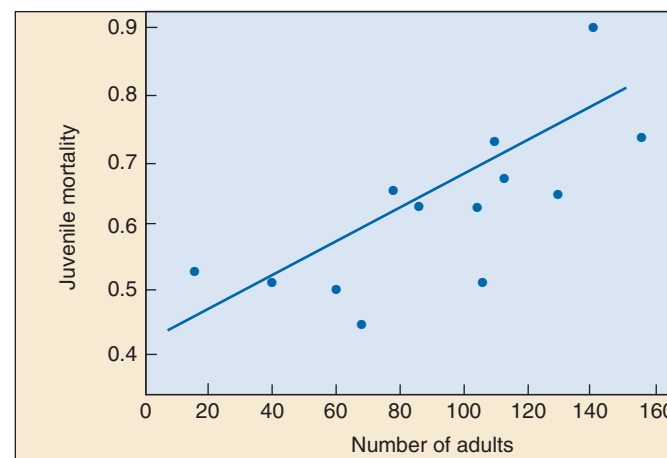
The reason that population growth rates are affected by population size is that many important processes are **density-dependent**. When populations approach their carrying capacity, competition for resources can be severe, leading both to a decreased birthrate and an increased risk of mortality (figure 24.18). In addition, predators often focus their attention on particularly common prey, which also results in increasing rates of mortality as populations increase. High population densities can also lead to an accumulation of toxic wastes, a situation to which humans are becoming increasingly accustomed.

Behavioral changes may also affect population growth rates. Some species of rodents, for example, become antisocial, fighting more, breeding less, and generally acting stressed-out. These behavioral changes result from hormonal actions, but their ultimate cause is not yet clear; most likely, they have evolved as adaptive responses to situations in which resources are scarce. In addition, in crowded populations, the population growth rate may decrease because of an increased rate of emigration of individuals attempting to find better conditions elsewhere (figure 24.19).

However, not all density-dependent factors are negatively related to population size. In some cases, growth rates increase with population size. This phenomenon is referred to as the **Allee effect** (after Warder Allee, who first described it). The Allee effect can take several forms. Most obviously, in populations that are too sparsely distributed, individuals may have difficulty finding mates. Moreover, some species may rely on large groups to deter predators or to provide the necessary stimulation for breeding activities.

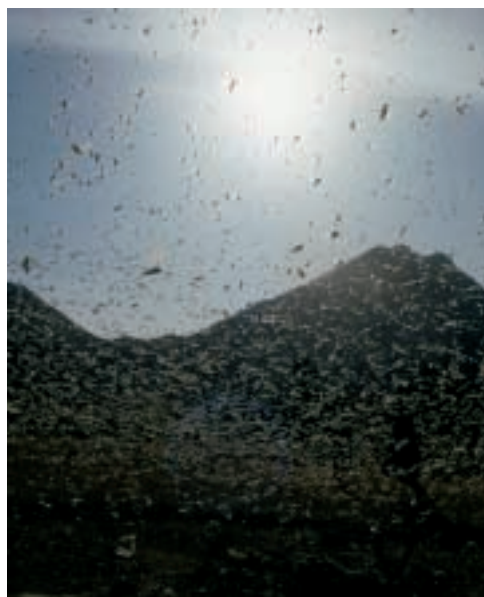
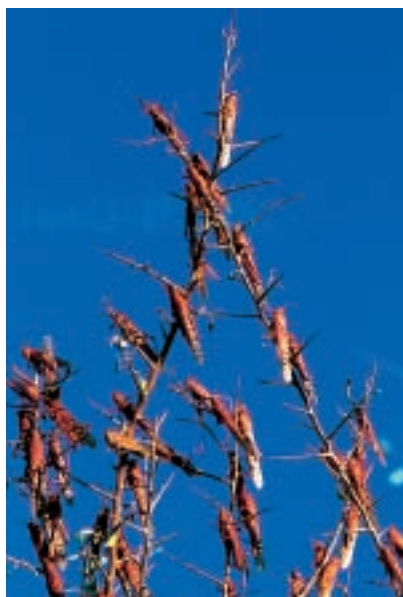


(a)



(b)

**FIGURE 24.18**  
**Density dependence in the song sparrow (*Melospiza melodia*) on Mandarte Island.** Reproductive success decreases (a) and mortality rates increase (b) as population size increases.



**FIGURE 24.19**  
**Density-dependent effects.** Migratory locusts, *Locusta migratoria*, are a legendary plague of large areas of Africa and Eurasia. At high population densities, the locusts have different hormonal and physical characteristics and take off as a swarm. The most serious infestation of locusts in 30 years occurred in North Africa in 1988.

## Density-Independent Effects

Growth rates in populations sometimes do not correspond to the logistic growth equation. In many cases, such patterns result because growth is under the control of **density-independent effects**. In other words, the rate of growth of a population at any instant is limited by something other than the size of the population.

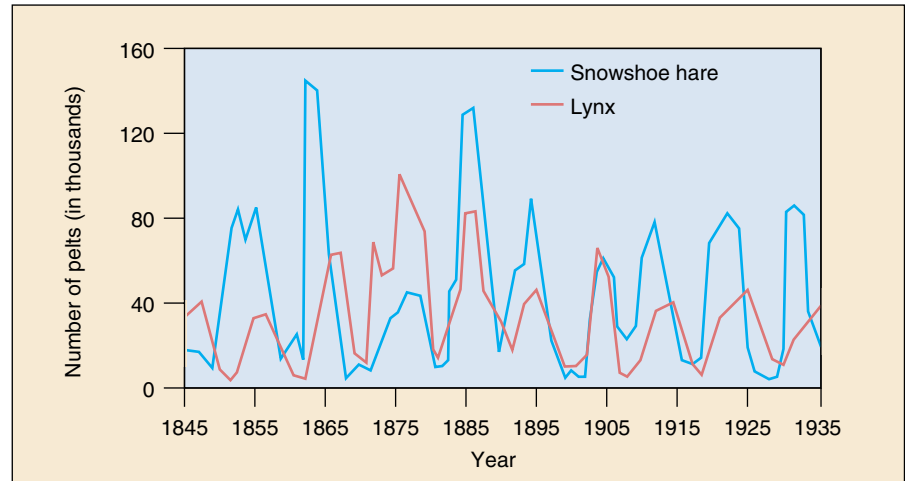
A variety of factors may affect populations in a density-independent manner. Most of these are aspects of the external environment. Extremely cold winters, droughts, storms, volcanic eruptions—individuals often will be affected by these activities regardless of the size of the population. Populations that occur in areas in which such events occur relatively frequently will display erratic population growth patterns in which the populations increase rapidly when conditions are benign, but suffer extreme reductions whenever the environment turns hostile.

## Population Cycles

Some populations exhibit another type of pattern inconsistent with simple logistic equations: they exhibit cyclic patterns of increase and decrease. Ecologists have studied cycles in hare populations since the 1920s. They have found that the North American snowshoe hare (*Lepus americanus*) follows a “10-year cycle” (in reality, it varies from 8 to 11 years). Its numbers fall tenfold to 30-fold in a typical cycle, and 100-fold changes can occur. Two factors appear to be generating the cycle: food plants and predators.

**Food plants.** The preferred foods of snowshoe hares are willow and birch twigs. As hare density increases, the quantity of these twigs decreases, forcing the hares to feed on high-fiber (low-quality) food. Lower birthrates, low juvenile survivorship, and low growth rates follow. The hares also spend more time searching for food, exposing them more to predation. The result is a precipitous decline in willow and birch twig abundance, and a corresponding fall in hare abundance. It takes two to three years for the quantity of mature twigs to recover.

**Predators.** A key predator of the snowshoe hare is the Canada lynx (*Lynx canadensis*). The Canada lynx shows a “10-year” cycle of abundance that seems remarkably entrained to the hare abundance cycle (figure 24.20). As hare numbers increase, lynx numbers do, too, rising in response to the increased availability of lynx food. When hare numbers fall, so do lynx numbers, their food supply depleted.



**FIGURE 24.20**

**Linked population cycles of the snowshoe hare and the northern lynx.** These data are based on records of fur returns from trappers in the Hudson Bay region of Canada. The lynx populations carefully track the snowshoe hares, but lag behind them slightly.

Which factor is responsible for the predator-prey oscillations? Do increasing numbers of hares lead to overharvesting of plants (a hare-plant cycle) or do increasing numbers of lynx lead to overharvesting of hares (a hare-lynx cycle)? Field experiments carried out by C. Krebs and coworkers in 1992 provide an answer. Krebs set up experimental plots in Canada’s Yukon—containing hare populations. If food is added (no food effect) and predators excluded (no predator effect) from an experimental area, hare numbers increase tenfold and stay there—the cycle is lost. However, the cycle is retained if either of the factors is allowed to operate alone: exclude predators but don’t add food (food effect alone), or add food in presence of predators (predator effect alone). Thus, both factors can affect the cycle, which, in practice, seems to be generated by the interaction between the two factors.

Population cycles traditionally have been considered to occur rarely. However, a recent review of nearly 700 long-term (25 years or more) studies of trends within populations found that cycles were not uncommon; nearly 30% of the studies—including birds, mammals, fish, and crustaceans—provided evidence of some cyclic pattern in population size through time, although most of these cycles are nowhere near as dramatic in amplitude as the snowshoe hare and lynx cycles.

**Density-dependent effects are caused by factors that come into play particularly when the population size is larger; density-independent effects are controlled by factors that operate regardless of population size.**

Population Growth Rates and Life History Models

As we have seen, some species usually have stable population sizes maintained near the carrying capacity, whereas the populations of other species fluctuate markedly and are often far below carrying capacity. As we saw in our discussion of life histories, the selective factors affecting such species will differ markedly. Populations near their carrying capacity may face stiff competition for limited resources. By contrast, resources are abundant in populations far below carrying capacity.

We have already seen the consequences of such differences. When resources are limited, the cost of reproduction often will be very high. Consequently, selection will favor individuals that can compete effectively and utilize resources efficiently. Such adaptations often come at the cost of lowered reproductive rates. Such populations are termed **K-selected** because they are adapted to thrive when the population is near its carrying capacity (*K*). Table 24.2 lists some of the typical features of *K*-selected populations. Examples of *K*-selected species include coconut palms, whooping cranes, whales, and humans.

By contrast, in populations far below the carrying capacity, resources may be abundant. Costs of reproduction will be low, and selection will favor those individuals that can produce the maximum number of offspring. Selection here favors individuals with the highest reproductive rates; such populations are termed **r-selected**. Examples of organisms displaying *r*-selected life history adaptations include dandelions, aphids, mice, and cockroaches (figure 24.21).

Most natural populations show life history adaptations that exist along a continuum ranging from completely *r*-selected traits to completely *K*-selected traits. Although these tendencies hold true as generalities, few populations are purely *r*- or *K*-selected and show all of the traits listed in table 24.2. These attributes should be treated as generalities, with the recognition that many exceptions do exist.

Some life history adaptations favor near-exponential growth, others the more competitive logistic growth. Most natural populations exhibit a combination of the two.

| Table 24.2 <i>r</i> -Selected and <i>K</i> -Selected Life History Adaptations |                                |                                |
|-------------------------------------------------------------------------------|--------------------------------|--------------------------------|
| Adaptation                                                                    | <i>r</i> -Selected Populations | <i>K</i> -Selected Populations |
| Age at first reproduction                                                     | Early                          | Late                           |
| Life span                                                                     | Short                          | Long                           |
| Maturation time                                                               | Short                          | Long                           |
| Mortality rate                                                                | Often high                     | Usually low                    |
| Number of offspring produced per reproductive episode                         | Many                           | Few                            |
| Number of reproductions per lifetime                                          | Usually one                    | Often several                  |
| Parental care                                                                 | None                           | Often extensive                |
| Size of offspring or eggs                                                     | Small                          | Large                          |

Source: Data from E. R. Pianka, *Evolutionary Ecology*, 4th edition, 1987, Harper & Row, New York.



**FIGURE 24.21**  
**The consequences of exponential growth.** All organisms have the potential to produce populations larger than those that actually occur in nature. The German cockroach (*Blatella germanica*), a major household pest, produces 80 young every six months. If every cockroach that hatched survived for three generations, kitchens might look like this theoretical culinary nightmare concocted by the Smithsonian Museum of Natural History.

## 24.5 The human population has grown explosively in the last three centuries.

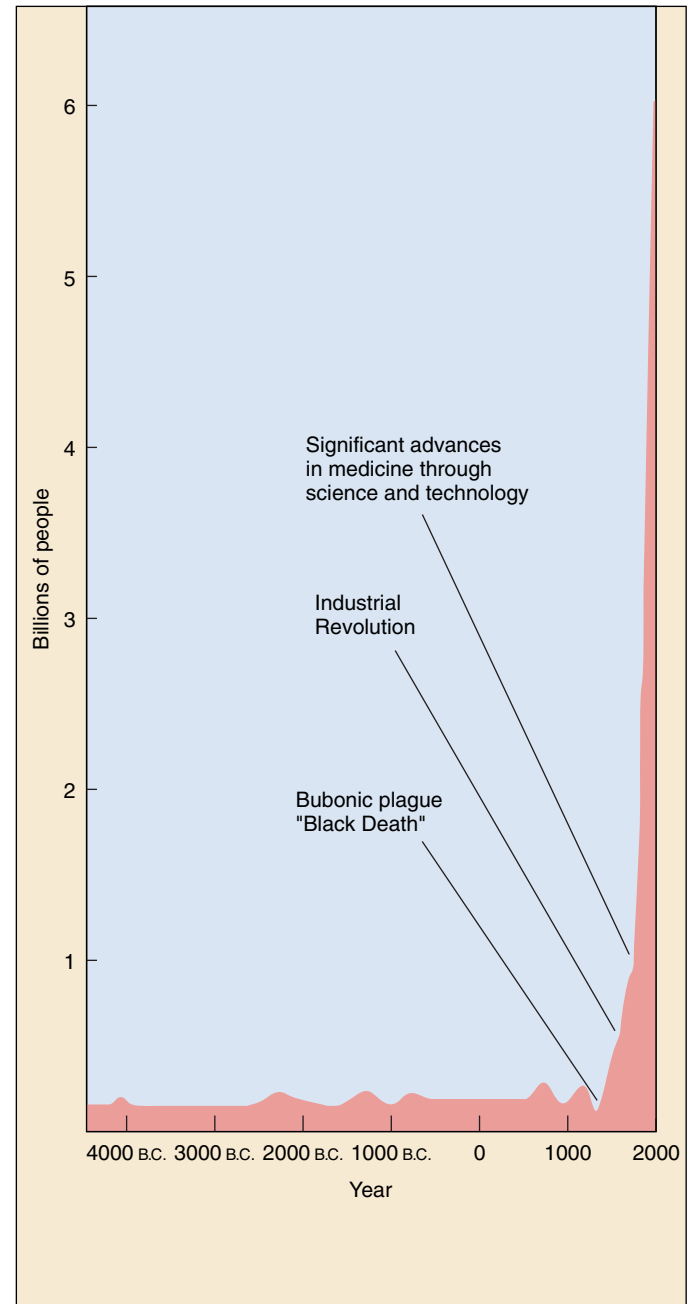
### The Advent of Exponential Growth

Humans exhibit many *K*-selected life history traits, including small brood size, late reproduction, and a high degree of parental care. These life history traits evolved during the early history of hominids, when the limited resources available from the environment controlled population size. Throughout most of human history, our populations have been regulated by food availability, disease, and predators. Although unusual disturbances, including floods, plagues, and droughts no doubt affected the pattern of human population growth, the overall size of the human population grew only slowly during our early history. Two thousand years ago, perhaps 130 million people populated the earth. It took a thousand years for that number to double, and it was 1650 before it had doubled again, to about 500 million. For over 16 centuries, the human population was characterized by very slow growth. In this respect, human populations resembled many other species with predominantly *K*-selected life history adaptations.

Starting in the early 1700s, changes in technology have given humans more control over their food supply, enabled them to develop superior weapons to ward off predators, and led to the development of cures for many diseases. At the same time, improvements in shelter and storage capabilities have made humans less vulnerable to climatic uncertainties. These changes allowed humans to expand the carrying capacity of the habitats in which they lived, and thus to escape the confines of logistic growth and reenter the exponential phase of the sigmoidal growth curve.

Responding to the lack of environmental constraints, the human population has grown explosively over the last 300 years. While the birthrate has remained unchanged at about 30 per 1000 per year over this period, the death rate has fallen dramatically, from 20 per 1000 per year to its present level of 13 per 1000 per year. The difference between birth and death rates meant that the population grew as much as 2% per year, although the rate has now declined to 1.4% per year.

A 1.4% annual growth rate may not seem large, but it has produced a current human population of 6 billion people (figure 24.22)! At this growth rate, 77 million people are added to the world population annually, and the human population will double in 39 years. As we will discuss in chapter 30, both the current human population level and the projected growth rate have potential consequences for our future that are extremely grave.



**FIGURE 24.22**

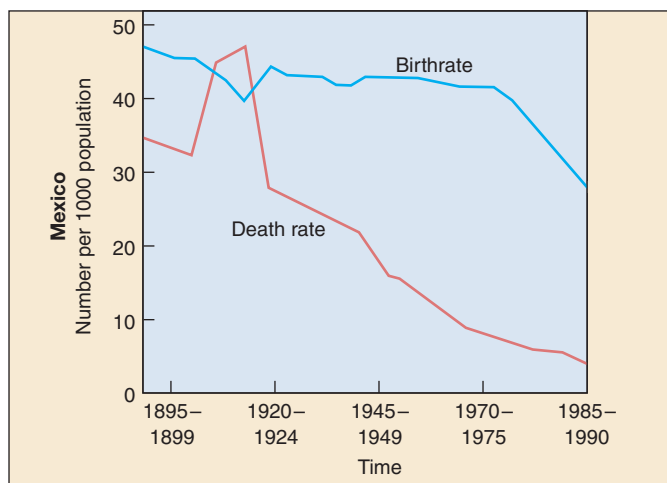
**History of human population size.** Temporary increases in death rate, even severe ones like the Black Death of the 1400s, have little lasting impact. Explosive growth began with the Industrial Revolution in the 1700s, which produced a significant long-term lowering of the death rate. The current population is 6 billion, and at the current rate will double in 39 years.

## Population Pyramids

While the human population as a whole continues to grow rapidly at the close of the twentieth century, this growth is not occurring uniformly over the planet. Some countries, like Mexico, are growing rapidly, their birthrate greatly exceeding their death rate (figure 24.23). Other countries are growing much more slowly. The rate at which a population can be expected to grow in the future can be assessed graphically by means of a **population pyramid**—a bar graph displaying the numbers of people in each age category. Males are conventionally shown to the left of the vertical age axis, females to the right. A human population pyramid thus displays the age composition of a population by sex. In most human population pyramids, the number of older females is disproportionately large compared to the number of older males, because females in most regions have a longer life expectancy than males.

Viewing such a pyramid, one can predict demographic trends in births and deaths. In general, rectangular “pyramids” are characteristic of countries whose populations are stable, their numbers neither growing nor shrinking. A triangular pyramid is characteristic of a country that will exhibit rapid future growth, as most of its population has not yet entered the child-bearing years. Inverted triangles are characteristic of populations that are shrinking.

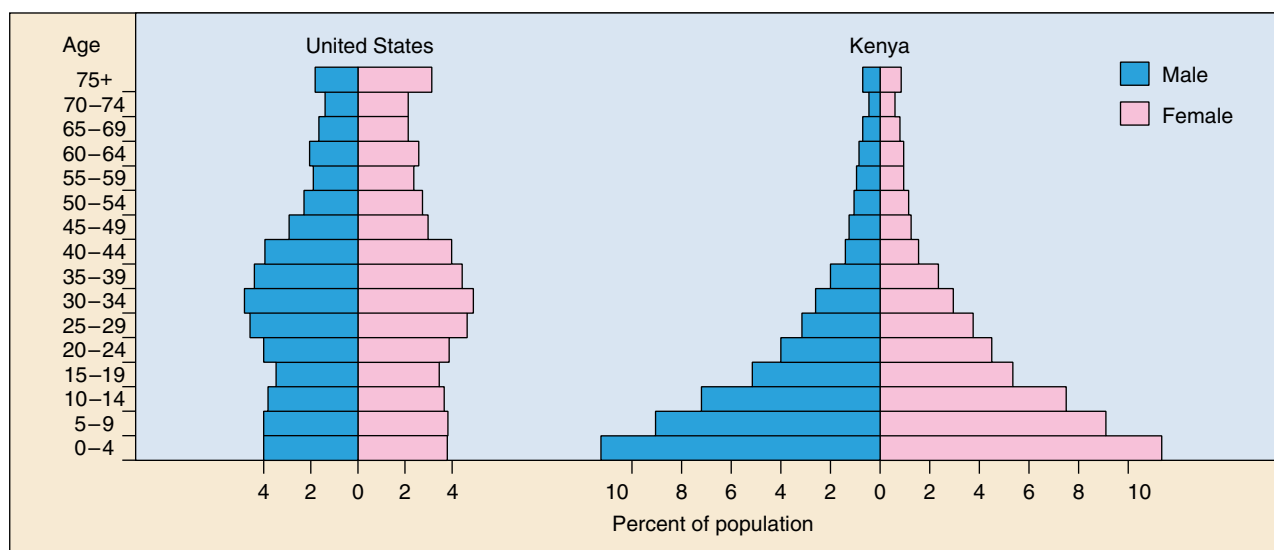
Examples of population pyramids for the United States and Kenya in 1990 are shown in figure 24.24. In the nearly rectangular population pyramid for the United States, the cohort (group of individuals) 55 to 59 years old represents people born during the Depression and is smaller in size



**FIGURE 24.23**

**Why the population of Mexico is growing.** The death rate (*red line*) in Mexico fell steadily throughout the last century, while the birthrate (*blue line*) remained fairly steady until 1970. The difference between birth and death rates has fueled a high growth rate. Efforts begun in 1970 to reduce the birthrate have been quite successful, although the growth rate remains rapid.

than the cohorts in the preceding and following years. The cohorts 25 to 44 years old represent the “baby boom.” The rectangular shape of the population pyramid indicates that the population of the United States is not expanding rapidly. The very triangular pyramid of Kenya, by contrast, predicts explosive future growth. The population of Kenya is predicted to double in less than 20 years.



**FIGURE 24.24**

**Population pyramids from 1990.** Population pyramids are graphed according to a population’s age distribution. Kenya’s pyramid has a broad base because of the great number of individuals below child-bearing age. When all of the young people begin to bear children, the population will experience rapid growth. The U.S. pyramid demonstrates a larger number of individuals in the “baby boom” cohort—the pyramid bulges because of an increase in births between 1945 and 1964.

**Table 24.3 A Comparison of 1996 Population Data in Developed and Developing Countries**

|                                         | United States<br>( <i>highly developed</i> ) | Brazil<br>( <i>moderately developed</i> ) | Ethiopia<br>( <i>poorly developed</i> ) |
|-----------------------------------------|----------------------------------------------|-------------------------------------------|-----------------------------------------|
| Fertility rate                          | 2.0                                          | 2.8                                       | 6.8                                     |
| Doubling time at current rate (yr)      | 114                                          | 41                                        | 23                                      |
| Infant mortality rate (per 1000 births) | 7.5                                          | 58                                        | 120                                     |
| Life expectancy at birth (yrs)          | 76                                           | 66                                        | 50                                      |
| Per capita GNP (U.S. \$; 1994)          | \$25,860                                     | \$3,370                                   | \$130                                   |

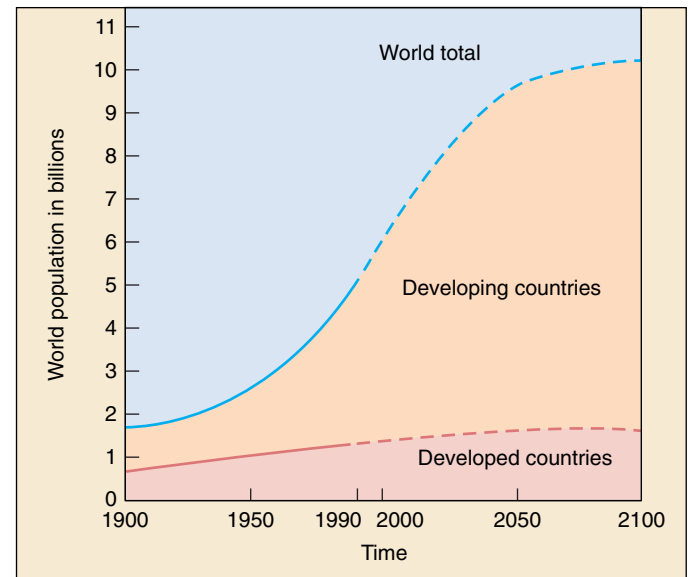
## An Uncertain Future

The earth's rapidly growing human population constitutes perhaps the greatest challenge to the future of the biosphere, the world's interacting community of living things. Humanity is adding 77 million people a year to the earth's population—a million every five days, 150 every minute! In more rapidly growing countries, the resulting population increase is staggering (table 24.3). India, for example, had a population of 853 million in 1996; by 2020 its population will exceed 1.4 billion!

A key element in the world's population growth is its uneven distribution among countries. Of the billion people added to the world's population in the 1990s, 90% live in developing countries (figure 24.25). This is leading to a major reduction in the fraction of the world's population that lives in industrialized countries. In 1950, fully one-third of the world's population lived in industrialized countries; by 1996 that proportion had fallen to one-quarter; in 2020 the proportion will have fallen to one-sixth. Thus the world's population growth will be centered in the parts of the world least equipped to deal with the pressures of rapid growth.

Rapid population growth in developing countries has the harsh consequence of increasing the gap between rich and poor. Today 23% of the world's population lives in the industrialized world with a per capita income of \$17,900, while 77% of the world's population lives in developing countries with a per capita income of only \$810. The disproportionate wealth of the industrialized quarter of the world's population is evidenced by the fact that 85% of the world's capital wealth is in the industrial world, only 15% in developing countries. Eighty percent of all the energy used today is consumed by the industrial world, only 20% by developing countries. Perhaps most worrisome for the future, fully 94% of all scientists and engineers reside in the industrialized world, only 6% in developing countries. Thus the problems created by the future's explosive population growth will be faced by countries with little of the world's scientific or technological expertise.

No one knows whether the world can sustain today's population of 6 billion people, much less the far greater populations expected in the future. As chapter 30 outlines, the world ecosystem is already under considerable



**FIGURE 24.25**

**Most of the worldwide increase in population since 1950 has occurred in developing countries.** The age structures of developing countries indicate that this trend will increase in the near future. The stabilizing of the world's population at about 10 billion (shown here) is an optimistic World Bank/United Nations prediction that assumes significant worldwide reductions in growth rate. If the world's population continues to increase at its 1996 rate, there will be over 30 billion humans by 2100!

stress. We cannot reasonably expect to continue to expand its carrying capacity indefinitely, and indeed we already seem to be stretching the limits. It seems unavoidable that to restrain the world's future population growth, birth and death rates must be equalized. If we are to avoid catastrophic increases in the death rate, the birthrates must fall dramatically. Faced with this grim dichotomy, significant efforts are underway worldwide to lower birthrates.

**The human population has been growing rapidly for 300 years, since technological innovations dramatically reduced the death rate.**



## Summary

## Questions

## Media Resources

**24.1 Populations are individuals of the same species that live together.**

- Populations are the same species in one place; communities are populations of different species that live together in a particular place. A community and the nonliving components of its environment combine to form an ecosystem.
- Populations may be dispersed in a clumped, uniform, or random manner.

1. What are the three types of dispersion in a population? Which type is most frequently seen in nature? Why?
2. What are some causes of clumped distributions?



- Introduction to Populations
- Population Characteristics



- On Science Article: Snakes in Ireland

**24.2 Population dynamics depend critically upon age distribution.**

- The growth rate of a population depends on its age structure, and to a lesser degree, sex ratio.
- Survivorship curves describe the characteristics of mortality in different kinds of populations.

3. What is survivorship? Describe the three types of survivorship curves and give examples of each.
4. What is demography? How does a life table work?



- On Science Article: Deer Hunting

**24.3 Life histories often reflect trade-offs between reproduction and survival.**

- Organisms balance investment in current reproduction with investment in growth and future reproduction.

5. Why do some birds lay fewer than the optimal number of eggs as predicted by David Lack?



- On Science Article: Science for the Future

**24.4 Population growth is limited by the environment.**

- Population size will change if birth and death rates differ, or if there is net migration into or out of the population. The intrinsic rate of increase of a population is defined as its biotic potential.
- Many populations exhibit a sigmoid growth curve, with a relatively slow start in growth, a rapid increase, and then a leveling off when the carrying capacity of the environment is reached.
- Large broods and rapid rates of population growth characterize *r*-strategists. *K*-strategists are limited in population size by the carrying capacity of their environments; they tend to have fewer offspring and slower rates of population growth.
- Density-independent factors have the same impact on a population no matter what its density.

6. Define the biotic potential of a population. What is the definition for the actual rate of population increase? What other two factors affect it?



- Stages of Population Growth

7. What is an exponential capacity for growth? When does this type of growth naturally occur? Give an example.



- Population Growth
- Size Regulation

8. What is carrying capacity? Is this a static or dynamic measure? Why?



- Scientists on Science: Coral Reefs Threatened
- Student Research: Prairie Habitat Fragmentation
- On Science Article: Tropical Songbirds Lay Fewer Eggs
- On Science Article: Was Malthus Mistaken?

9. What is the difference between *r*- and *K*-selected populations?

**24.5 The human population has grown explosively in the last three centuries.**

- Exponential growth of the world's human population is placing severe strains on the global environment.

10. How do population pyramids predict whether a population is likely to grow or shrink?



- Human Population

# 25

## Community Ecology

### Concept Outline

#### 25.1 Interactions among competing species shape ecological niches.

**The Realized Niche.** Interspecific interactions often limit the portion of their niche that they can actually use.

**Gause and the Principle of Competitive Exclusion.** No two species can occupy the same niche indefinitely without competition driving one to extinction.

**Resource Partitioning.** Species that live together partition the available resources, reducing competition.

**Detecting Interspecific Competition.** Experiments are often the best way to detect competition, but they have their limitations.

#### 25.2 Predators and their prey coevolve.

**Predation and Prey Populations.** Predators can limit the size of populations and sometimes even eliminate a species from a community.

**Plant Defenses against Herbivores.** Plants use chemicals to defend themselves against animals trying to eat them.

**Animal Defenses against Predators.** Animals defend themselves with camouflage, chemicals, and stings.

**Mimicry.** Sometimes a species copies the appearance of another protected one.

#### 25.3 Evolution sometimes fosters cooperation.

**Coevolution and Symbiosis.** Organisms have evolved many adjustments and accommodations to living together.

**Commensalism.** Some organisms use others, neither hurting or helping their benefactors.

**Mutualism.** Often species interact in ways that benefit both.

**Parasitism.** Sometimes one organism serves as the food supply of another much smaller one.

**Interactions among Ecological Processes.** Multiple processes may occur simultaneously within a community.

#### 25.4 Ecological succession may increase species richness.

**Succession.** Communities change through time.

**The Role of Disturbance.** Disturbances can disrupt successional change. In some cases, moderate amounts of disturbance increase species diversity.



FIGURE 25.1

**Communities involve interactions between disparate groups.** This clownfish is one of the few species that can nestle safely among the stinging tentacles of the sea anemone—a classic example of a symbiotic relationship.

All the organisms that live together in a place are called a community. The myriad species that inhabit a tropical rain forest are a community. Indeed, every inhabited place on earth supports its own particular array of organisms. Over time, the different species have made many complex adjustments to community living (figure 25.1), evolving together and forging relationships that give the community its character and stability. Both competition and cooperation have played key roles; in this chapter, we will look at these and other factors in community ecology.

## 25.1 Interactions among competing species shape ecological niches.

### The Realized Niche

Each organism in an ecosystem confronts the challenge of survival in a different way. The **niche** an organism occupies is the sum total of all the ways it utilizes the resources of its environment. A niche may be described in terms of space utilization, food consumption, temperature range, appropriate conditions for mating, requirements for moisture, and other factors. *Niche* is not synonymous with **habitat**, the place where an organism lives. *Habitat* is a place, *niche* a pattern of living.

Sometimes species are not able to occupy their entire niche because of the presence or absence of other species. Species can interact with each other in a number of ways, and these interactions can either have positive or negative effects. One type of interaction is **interspecific competition**, which occurs when two species attempt to utilize the same resource when there is not enough of the resource to satisfy both. Fighting over resources is referred to as **interference competition**; consuming shared resources is called **exploitative competition**.

The entire niche that a species is capable of using, based on its physiological requirements and resource needs, is called the **fundamental niche**. The actual niche the species occupies is called its **realized niche**. Because of interspecific interactions, the realized niche of a species may be considerably smaller than its fundamental niche.

In a classic study, J. H. Connell of the University of California, Santa Barbara investigated competitive interactions between two species of barnacles that grow together on rocks along the coast of Scotland. Of the two species Connell studied, *Chthamalus stellatus* lives in shallower water, where tidal action often exposed it to air, and *Semibalanus balanoides* (called *Balanus balanoides* prior to 1995) lives lower down, where it is rarely exposed to the atmosphere (figure 25.2). In the deeper zone, *Semibalanus* could always outcompete *Chthamalus* by crowding it off the rocks, undercutting it, and replacing it even where it had begun to grow, an example of interference competition. When Connell removed *Semibalanus* from the area, however, *Chthamalus* was easily able to occupy the deeper zone, indicating that no physiological or other general obstacles prevented it from becoming established there. In contrast, *Semibalanus* could not survive in the shallow-water habitats where *Chthamalus* normally occurs; it evidently does not have the special adaptations that allow *Chthamalus* to occupy this zone. Thus, the fundamental niche of the barnacle *Chthamalus* included both shallow and deeper zones, but its realized niche was much narrower because *Chthamalus* was outcompeted by *Semibalanus* in parts of its fundamental niche. By contrast, the realized and fundamental niches of *Semibalanus* appear to be identical.

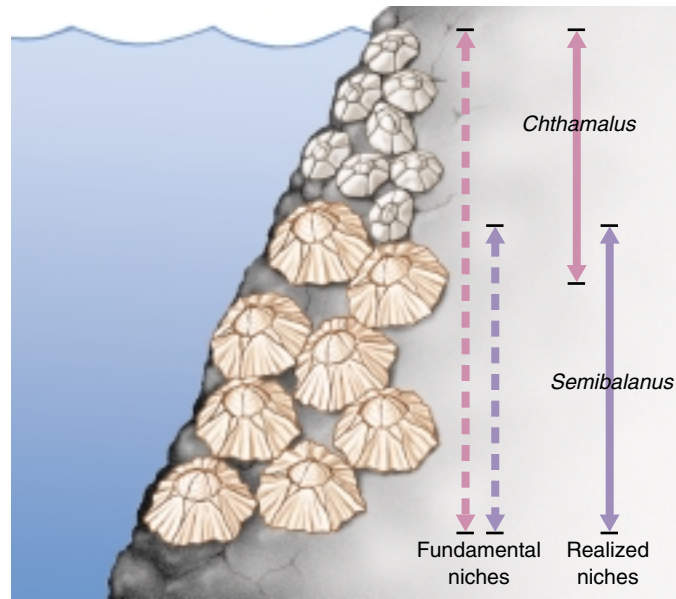


FIGURE 25.2

**Competition among two species of barnacles limits niche use.**

*Chthamalus* can live in both deep and shallow zones (its fundamental niche), but *Semibalanus* forces *Chthamalus* out of the part of its fundamental niche that overlaps the realized niche of *Semibalanus*.

Processes other than competition can also restrict the realized niche of a species. For example, a plant, the St. John's-wort, was introduced and became widespread in open rangeland habitats in California until a specialized beetle was introduced to control it. Populations of the plant quickly decreased and it is now only found in shady sites where the beetle cannot thrive. In this case, the presence of a predator limits the realized niche of a plant.

In some cases, the absence of another species leads to a smaller realized niche. For example, many North American plants depend on the American honeybee for pollination. The honeybee's population is currently declining for a variety of reasons. Conservationists are concerned that if the honeybee disappears from some habitats, the niche of these plant species will decrease or even disappear entirely. In this case, then, the absence—rather than the presence—of another species will be cause of a relatively small realized niche.

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A niche may be defined as the way in which an organism utilizes its environment. Interspecific interactions may cause a species' realized niche to be smaller than its fundamental niche. If resources are limiting, two species normally cannot occupy the same niche indefinitely.

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## Gause and the Principle of Competitive Exclusion

In classic experiments carried out in 1934 and 1935, Russian ecologist G. F. Gause studied competition among three species of *Paramecium*, a tiny protist. All three species grew well alone in culture tubes, preying on bacteria and yeasts that fed on oatmeal suspended in the culture fluid (figure 25.3a). However, when Gause grew *P. aurelia* together with *P. caudatum* in the same culture tube, the numbers of *P. caudatum* always declined to extinction, leaving *P. aurelia* the only survivor (figure 25.3b). Why? Gause found *P. aurelia* was able to grow six times faster than its competitor *P. caudatum* because it was able to better utilize the limited available resources, an example of exploitative competition.

From experiments such as this, Gause formulated what is now called the **principle of competitive exclusion**. This principle states that if two species are competing for a limited resource, the species that uses the resource more efficiently will eventually eliminate the other locally—no two species with the same niche can coexist when resources are limiting.

### Niche Overlap

In a revealing experiment, Gause challenged *Paramecium caudatum*—the defeated species in his earlier experiments—with a third species, *P. bursaria*. Because he expected these two species to also compete for the limited bacterial food supply, Gause thought one would win out, as had happened in his previous experiments. But that's not what happened. Instead, both species survived in the culture tubes; yet

*paramecia* found a way to divide the food resources. How did they do it? In the upper part of the culture tubes, where the oxygen concentration and bacterial density were high, *P. caudatum* dominated because it was better able to feed on bacteria. However, in the lower part of the tubes, the lower oxygen concentration favored the growth of a different potential food, yeast, and *P. bursaria* was better able to eat this food. The fundamental niche of each species was the whole culture tube, but the realized niche of each species was only a portion of the tube. Because the niches of the two species did not overlap too much, both species were able to survive. However, competition did have a negative effect on the participants (figure 25.3c). When grown without a competitor, both species reached densities three times greater than when they were grown with a competitor.

### Competitive Exclusion

Gause's principle of competitive exclusion can be restated to say that *no two species can occupy the same niche indefinitely when resources are limiting*. Certainly species can and do coexist while competing for some of the same resources. Nevertheless, Gause's theory predicts that when two species coexist on a long-term basis, either resources must not be limited or their niches will always differ in one or more features; otherwise, one species will outcompete the other and the extinction of the second species will inevitably result, a process referred to as **competitive exclusion**.

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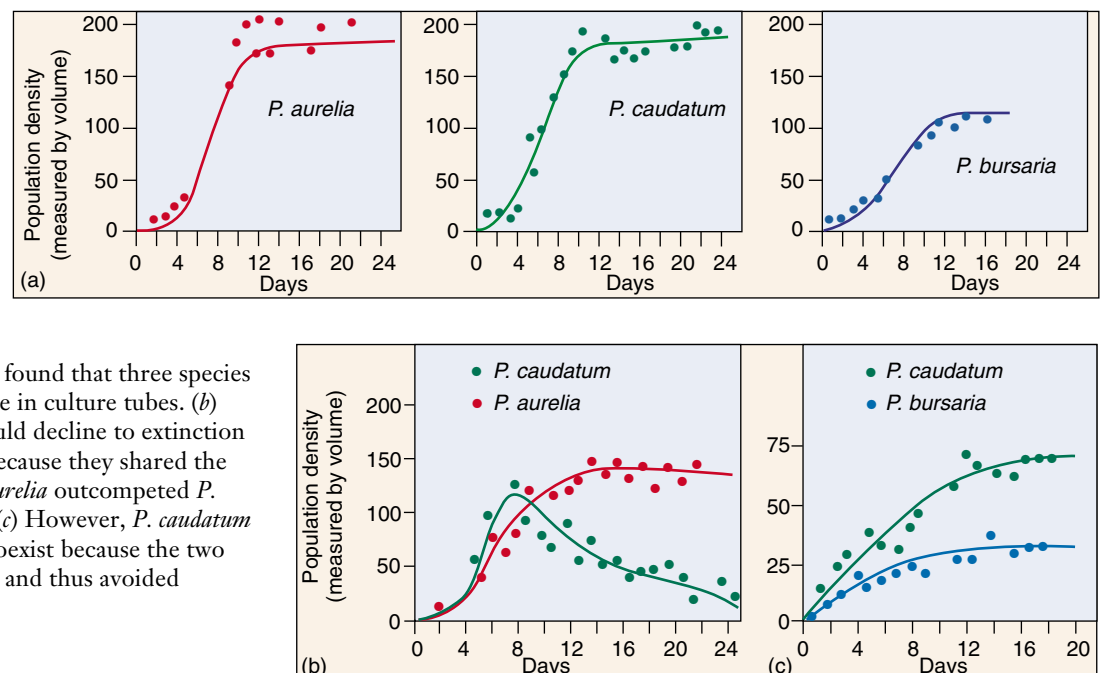
**If resources are limiting, no two species can occupy the same niche indefinitely without competition driving one to extinction.**

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**FIGURE 25.3**  
**Competitive exclusion among three species of *Paramecium*.**

In the microscopic world, *Paramecium* is a ferocious predator. *Paramecia* eat by ingesting their prey; their cell membranes surround bacterial or yeast cells, forming a food vacuole containing the prey cell.

(a) In his experiments, Gause found that three species of *Paramecium* grew well alone in culture tubes. (b) But *Paramecium caudatum* would decline to extinction when grown with *P. aurelia* because they shared the same realized niche, and *P. aurelia* outcompeted *P. caudatum* for food resources. (c) However, *P. caudatum* and *P. bursaria* were able to coexist because the two have different realized niches and thus avoided competition.

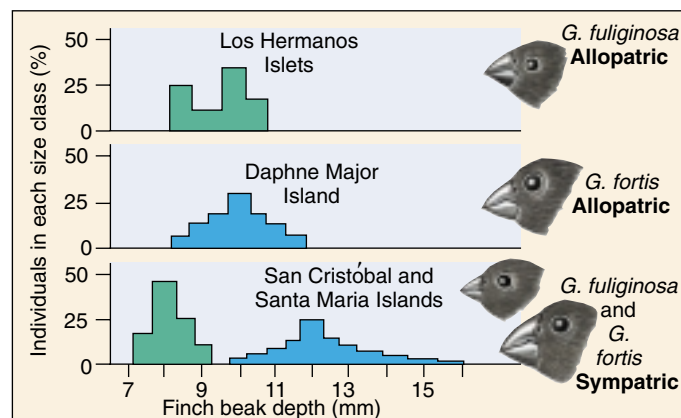


## Resource Partitioning

Gause's exclusion principle has a very important consequence: persistent competition between two species is rare in natural communities. Either one species drives the other to extinction, or natural selection reduces the competition between them. When the late Princeton ecologist Robert MacArthur studied five species of warblers, small insect-eating forest songbirds, he found that they all appeared to be competing for the same resources. However, when he studied them more carefully, he found that each species actually fed in a different part of spruce trees and so ate different subsets of insects. One species fed on insects near the tips of branches, a second within the dense foliage, a third on the lower branches, a fourth high on the trees and a fifth at the very apex of the trees. Thus, each species of warbler had evolved so as to utilize a different portion of the spruce tree resource. They *subdivided the niche*, partitioning the available resource so as to avoid direct competition with one another.

Resource partitioning is often seen in similar species that occupy the same geographical area. Such **sympatric species** often avoid competition by living in different portions of the habitat or by utilizing different food or other resources (figure 25.4). This pattern of resource partitioning is thought to result from the process of natural selection causing initially similar species to diverge in resource use in order to reduce competitive pressures.

Evidence for the role of evolution comes from comparison of species whose ranges are only partially overlapping. Where the two species co-occur, they tend to exhibit greater differences in **morphology** (the form and structure of an organism) and resource use than do their allopatric

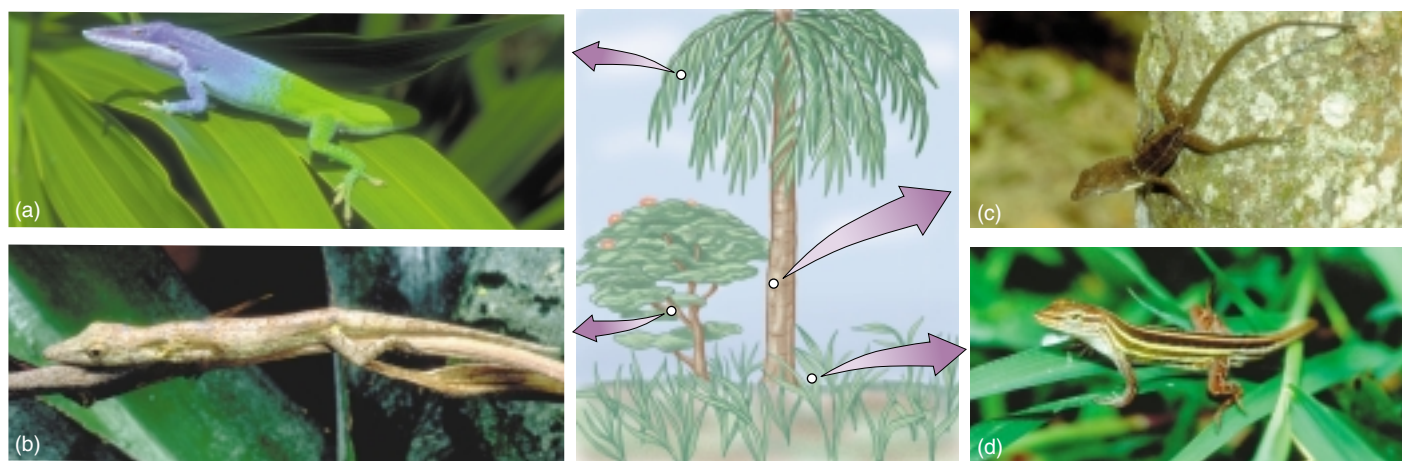


**FIGURE 25.5**

**Character displacement in Darwin's finches.** These two species of finches (genus *Geospiza*) have bills of similar size when allopatric, but different size when sympatric.

populations. Called **character displacement**, the differences evident between sympatric species are thought to have been favored by natural selection as a mechanism to facilitate habitat partitioning and thus reduce competition. Thus, the two Darwin's finches in figure 25.5 have bills of similar size where the finches are allopatric, each living on an island where the other does not occur. On islands where they are sympatric, the two species have evolved beaks of different sizes, one adapted to larger seeds, the other to smaller ones.

**Sympatric species partition available resources, reducing competition between them.**



**FIGURE 25.4**

**Resource partitioning among sympatric lizard species.** Species of *Anolis* lizards on Caribbean islands partition their tree habitats in a variety of ways. Some species of anoles occupy the canopy of trees (a), others use twigs on the periphery (b), and still others are found at the base of the trunk (c). In addition, some use grassy areas in the open (d). When two species occupy the same part of the tree, they either utilize different-sized insects as food or partition the thermal microhabitat; for example, one might only be found in the shade, whereas the other would only bask in the sun. Most interestingly, the same pattern of resource partitioning has evolved independently on different Caribbean islands.

## Detecting Interspecific Competition

It is not simple to determine when two species are competing. The fact that two species use the same resources need not imply competition if that resource is not in limited supply. If the population sizes of two species are negatively correlated, such that where one species has a large population, the other species has a small population and vice versa, the two species need not be competing for the same limiting resource. Instead, the two species might be independently responding to the same feature of the environment—perhaps one species thrives best in warm conditions and the other in cool conditions.

### Experimental Studies of Competition

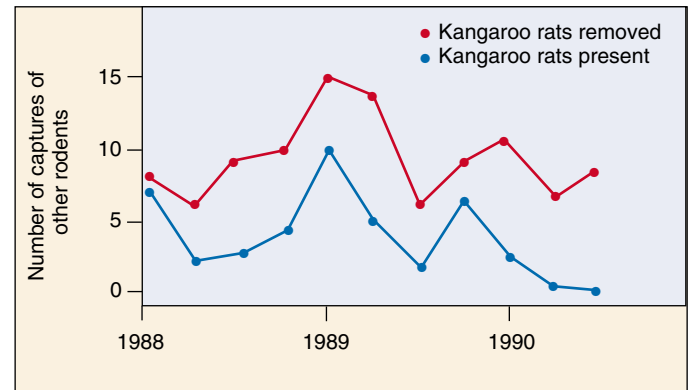
Some of the best evidence for the existence of competition comes from experimental field studies. By setting up experiments in which two species either occur alone or together, scientists can determine whether the presence of one species has a negative effect on a population of a second species. For example, a variety of seed-eating rodents occur in the Chihuahuan Desert of the southwestern part of North America. In 1988, researchers set up a series of 50 meter  $\times$  50 meter enclosures to investigate the effect of kangaroo rats on other, smaller seed-eating rodents. Kangaroo rats were removed from half of the enclosures, but not from the other enclosures. The walls of all of the enclosures had holes in them that allowed rodents to come and go, but in the kangaroo rat removal plots, the holes were too small to allow the kangaroo rats to enter. Over the course of the next three years, the researchers monitored the number of the other, smaller seed-eating rodents present in the plots. As figure 25.6 illustrates, the number of other rodents was substantially higher in the absence of kangaroo rats, indicating that kangaroo rats compete with the other rodents and limit their population sizes.

A great number of similar experiments have indicated that interspecific competition occurs between many species of plants and animals. Effects of competition can be seen in aspects of population biology other than population size, such as behavior and individual growth rates. For example, two species of *Anolis* lizards occur on the island of St. Maarten. When one of the species, *A. gingivinus*, is placed in 12 m  $\times$  12 m enclosures without the other species, individual lizards grow faster and perch lower than lizards of the same species do when placed in enclosures in which *A. pogus* is also present.

### Caution Is Necessary

Although experimental studies can be a powerful means of understanding the interactions that occur between coexisting species, they have their limitations.

First, care is necessary in interpreting the results of field experiments. Negative effects of one species on another do



**FIGURE 25.6**

**Detecting interspecific competition.** This experiment tests the effect of removal of kangaroo rats on the population size of other rodents. Immediately after kangaroo rats were removed, the number of rodents increased relative to the enclosures that still had kangaroo rats. Notice that population sizes (as estimated by number of captures) increased and decreased in synchrony in the two treatments, probably reflecting changes in the weather.

not automatically indicate the existence of competition. For example, many similar-sized fish have a negative effect on each other, but it results not from competition, but from the fact that adults of each species will prey on juveniles of the other species. In addition, the presence of one species may attract predators, which then also prey on the second species. In this case, the second species may have a lower population size in the presence of the first species due to the presence of predators, even if they are not competing at all. Thus, experimental studies are most effective when they are combined with detailed examination of the ecological mechanism causing the negative effect of one species on another species.

In addition, experimental studies are not always feasible. For example, the coyote has increased its population in the United States in recent years simultaneously with the decline of the grey wolf. Is this trend an indication that the species compete? Because of the size of the animals and the large geographic areas occupied by each individual, manipulative experiments involving fenced areas with only one or both species—with each experimental treatment replicated several times for statistical analysis—are not practical. Similarly, studies of slow-growing trees might require many centuries to detect competition between adult trees. In such cases, detailed studies of the ecological requirements of the species are our best bet to understanding interspecific interactions.

**Experimental studies can provide strong tests of the hypothesis that interspecific competition occurs, but such studies have limitations. Detailed ecological studies are important regardless of whether experiments are conducted.**

## 25.2 Predators and their prey coevolve.

**Predation** is the consuming of one organism by another. In this sense, predation includes everything from a leopard capturing and eating an antelope, to a deer grazing on spring grass. When experimental populations are set up under simple laboratory conditions, the predator often exterminates its prey and then becomes extinct itself, having nothing left to eat (figure 25.7). However, if refuges are provided for the prey, its population will drop to low levels but not to extinction. Low prey population levels will then provide inadequate food for the predators, causing the predator population to decrease. When this occurs, the prey population can recover.

### Predation and Prey Populations

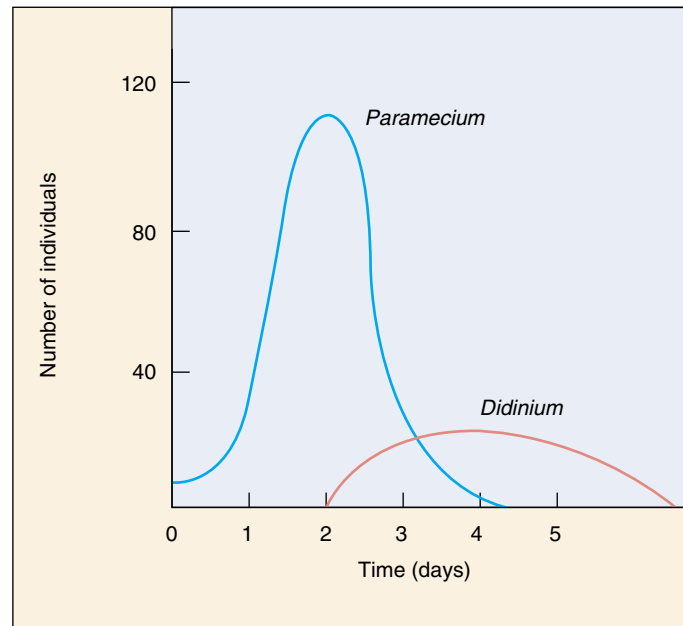
In nature, predators can often have large effects on prey populations. Some of the most dramatic examples involve situations in which humans have either added or eliminated predators from an area. For example, the elimination of large carnivores from much of the eastern United States has led to population explosions of white-tailed deer, which strip the habitat of all edible plant life. Similarly, when sea otters were hunted to near extinction on the western coast of the United States, sea urchin populations exploded.

Conversely, the introduction of rats, dogs, and cats to many islands around the world has led to the decimation of native faunas. Populations of Galápagos tortoises on several islands are endangered, for example, by introduced rats, dogs, and cats, which eat eggs and young tortoises. Similarly, several species of birds and reptiles have been eradicated by rat predation from New Zealand and now only occur on a few offshore islands that the rats have not reached. In addition, on Stephens Island, near New Zealand, every individual of the now extinct Stephen Island wren was killed by a single lighthouse keeper's cat!

A classic example of the role predation can play in a community involves the introduction of prickly pear cactus to Australia in the nineteenth century. In the absence of predators, the cactus spread rapidly, by 1925 occupying 12 million hectares of rangeland in an impenetrable morass of spines that made cattle ranching difficult. To control the cactus, a predator from its natural habitat in Argentina, the moth *Cactoblastis cactorum*, was introduced beginning in 1926. By 1940, cactus populations had been decimated, and it now generally occurs in small populations.

### Predation and Evolution

Predation provides strong selective pressures on prey populations. Any feature that would decrease the probability of



**FIGURE 25.7**

**Predator-prey in the microscopic world.** When the predatory *Didinium* is added to a *Paramecium* population, the numbers of *Didinium* initially rise, while the numbers of *Paramecium* steadily fall. When the *Paramecium* population is depleted, however, the *Didinium* individuals also die.

capture should be strongly favored. In the next three pages, we discuss a number of defense mechanisms in plants and animals. In turn, the evolution of such features will cause natural selection to favor counteradaptations in predator populations. In this way, a coevolutionary arms race may ensue in which predators and prey are constantly evolving better defenses and better means of circumventing these defenses.

One example comes from the fossil record of molluscs and gastropods and their predators. During the Mesozoic period (approximately 65 to 225 million years ago), new forms of predatory fish and crustaceans evolved that were able to crush or tear open shells. As a result, a variety of defensive measures evolved in molluscs and gastropods, including thicker shells, spines, and shells too smooth for predators to be able to grasp. In turn, these adaptations may have pressured predators to evolve ever more effective predatory adaptations and tactics.

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**Predation can have substantial effects on prey populations. As a result prey species often evolve defensive adaptations.**

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## Plant Defenses against Herbivores

Plants have evolved many mechanisms to defend themselves from herbivores. The most obvious are **morphological defenses**: thorns, spines, and prickles play an important role in discouraging browsers, and plant hairs, especially those that have a glandular, sticky tip, deter insect herbivores. Some plants, such as grasses, deposit silica in their leaves, both strengthening and protecting themselves. If enough silica is present in their cells, these plants are simply too tough to eat.



(a)



(b)

**FIGURE 25.8**

**Insect herbivores are well suited to their hosts.** (a) The green caterpillars of the cabbage butterfly, *Pieris rapae*, are camouflaged on the leaves of cabbage and other plants on which they feed. Although mustard oils protect these plants against most herbivores, the cabbage butterfly caterpillars are able to break down the mustard oil compounds. (b) An adult cabbage butterfly.

### Chemical Defenses

Significant as these morphological adaptations are, the chemical defenses that occur so widely in plants are even more crucial. Best known and perhaps most important in the defenses of plants against herbivores are **secondary chemical compounds**. These are distinguished from primary compounds, which are regular components of the major metabolic pathways, such as respiration. Many plants, and apparently many algae as well, contain very structurally diverse secondary compounds that are either toxic to most herbivores or disturb their metabolism greatly, preventing, for example, the normal development of larval insects. Consequently, most herbivores tend to avoid the plants that possess these compounds.

The mustard family (Brassicaceae) is characterized by a group of chemicals known as mustard oils. These are the substances that give the pungent aromas and tastes to such plants as mustard, cabbage, watercress, radish, and horseradish. The same tastes we enjoy signal the presence of chemicals that are toxic to many groups of insects. Similarly, plants of the milkweed family (Asclepiadaceae) and the related dogbane family (Apocynaceae) produce a milky sap that deters herbivores from eating them. In addition, these plants usually contain cardiac glycosides, molecules named for their drastic effect on heart function in vertebrates.

### The Evolutionary Response of Herbivores

Certain groups of herbivores are associated with each family or group of plants protected by a particular kind of secondary compound. These herbivores are able to feed on these plants without harm, often as their exclusive food source. For example, cabbage butterfly caterpillars (subfamily Pierinae) feed almost exclusively on plants of the mustard and caper families, as well as on a few other small families of plants that also contain mustard oils (figure

25.8). Similarly, caterpillars of monarch butterflies and their relatives (subfamily Danainae) feed on plants of the milkweed and dogbane families. How do these animals manage to avoid the chemical defenses of the plants, and what are the evolutionary precursors and ecological consequences of such patterns of specialization?

We can offer a potential explanation for the evolution of these particular patterns. Once the ability to manufacture mustard oils evolved in the ancestors of the caper and mustard families, the plants were protected for a time against most or all herbivores that were feeding on other plants in their area. At some point, certain groups of insects—for example, the cabbage butterflies—evolved the ability to break down mustard oils and thus feed on these plants without harming themselves. Having developed this ability, the butterflies were able to use a new resource without competing with other herbivores for it. Often, in groups of insects such as cabbage butterflies, sense organs have evolved that are able to detect the secondary compounds that their food plants produce. Clearly, the relationship that has formed between cabbage butterflies and the plants of the mustard and caper families is an example of **coevolution**.

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**The members of many groups of plants are protected from most herbivores by their secondary compounds. Once the members of a particular herbivore group evolve the ability to feed on them, these herbivores gain access to a new resource, which they can exploit without competition from other herbivores.**

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## Animal Defenses against Predators

Some animals that feed on plants rich in secondary compounds receive an extra benefit. When the caterpillars of monarch butterflies feed on plants of the milkweed family, they do not break down the cardiac glycosides that protect these plants from herbivores. Instead, the caterpillars concentrate and store the cardiac glycosides in fat bodies; they then pass them through the chrysalis stage to the adult and even to the eggs of the next generation. The incorporation of cardiac glycosides thus protects all stages of the monarch life cycle from predators. A bird that eats a monarch butterfly quickly regurgitates it (figure 25.9) and in the future avoids the conspicuous orange-and-black pattern that characterizes the adult monarch. Some birds, however, appear to have acquired the ability to tolerate the protective chemicals. These birds eat the monarchs.

### Defensive Coloration

Many insects that feed on milkweed plants are brightly colored; they advertise their poisonous nature using an ecological strategy known as **warning coloration**, or **aposematic coloration**. Showy coloration is characteristic of animals that use poisons and stings to repel predators, while organisms that lack specific chemical defenses are seldom brightly colored. In fact, many have **cryptic coloration**—color that blends with the surroundings and thus hides the individual from predators (figure 25.10). Camouflaged animals usually do not live together in groups because a predator that discovers one individual gains a valuable clue to the presence of others.

### Chemical Defenses

Animals also manufacture and use a startling array of substances to perform a variety of defensive functions. Bees, wasps, predatory bugs, scorpions, spiders, and many other arthropods use chemicals to defend themselves and to kill their prey. In addition, various chemical defenses have evolved among marine animals and the vertebrates, including venomous snakes, lizards, fishes, and some birds. The poison-dart frogs of the family Dendrobatidae produce toxic alkaloids in the mucus that covers their brightly colored skin (figure 25.11). Some of these toxins are so powerful that a few micrograms will kill a person if injected into the bloodstream. More than 200 different alkaloids have been isolated from these frogs, and some are playing important roles in neuromuscular research. There is an intensive investigation of marine animals, algae, and flowering plants for new drugs to fight cancer and other diseases, or as sources of antibiotics.

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**Animals defend themselves against predators with warning coloration, camouflage, and chemical defenses such as poisons and stings.**

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(a)



(b)

**FIGURE 25.9**

**A blue jay learns that monarch butterflies taste bad.** (a) This cage-reared jay had never seen a monarch butterfly before it tried eating one. (b) The same jay regurgitated the butterfly a few minutes later. This bird will probably avoid trying to capture all orange-and-black insects in the future.



**FIGURE 25.10**

**Cryptic coloration.** An inchworm caterpillar (*Necophora quernaria*) (hanging from the upper twig) closely resembles a twig.



**FIGURE 25.11**

**Vertebrate chemical defenses.** Frogs of the family Dendrobatidae, abundant in the forests of Latin America, are extremely poisonous to vertebrates. Dendrobatids advertise their toxicity with aposematic coloration, as shown here.

## Mimicry

During the course of their evolution, many species have come to resemble distasteful ones that exhibit aposematic coloration. The mimic gains an advantage by looking like the distasteful model. Two types of mimicry have been identified: Batesian and Müllerian mimicry.

### Batesian Mimicry

**Batesian mimicry** is named for Henry Bates, the British naturalist who first brought this type of mimicry to general attention in 1857. In his journeys to the Amazon region of South America, Bates discovered many instances of palatable insects that resembled brightly colored, distasteful species. He reasoned that the mimics would be avoided by predators, who would be fooled by the disguise into thinking the mimic actually is the distasteful model.

Many of the best-known examples of Batesian mimicry occur among butterflies and moths. Obviously, predators in systems of this kind must use visual cues to hunt for their prey; otherwise, similar color patterns would not matter to potential predators. There is also increasing evidence indicating that Batesian mimicry can also involve nonvisual cues, such as olfaction, although such examples are less obvious to humans.

The kinds of butterflies that provide the models in Batesian mimicry are, not surprisingly, members of groups whose caterpillars feed on only one or a few closely related plant families. The plant families on which they feed are strongly protected by toxic chemicals. The model butterflies incorporate the poisonous molecules from these plants into their bodies. The mimic butterflies, in contrast, belong to groups in which the feeding habits of the caterpillars are not so restricted. As caterpillars, these butterflies feed on a number of different plant families unprotected by toxic chemicals.

One often-studied mimic among North American butterflies is the viceroy, *Limenitis archippus* (figure 25.12a). This butterfly, which resembles the poisonous monarch, ranges from central Canada through much of the United States and into Mexico. The caterpillars feed on willows and cottonwoods, and neither caterpillars nor adults were thought to be distasteful to birds, although recent findings may dispute this. Interestingly, the Batesian mimicry seen in the adult viceroy butterfly does not extend to the caterpillars: viceroy caterpillars are camouflaged on leaves, resembling bird droppings, while the monarch's distasteful caterpillars are very conspicuous.

### Müllerian Mimicry

Another kind of mimicry, **Müllerian mimicry**, was named for German biologist Fritz Müller, who first described it in 1878. In Müllerian mimicry, several unrelated but protected animal species come to resemble one another (figure



*Danaus plexippus*

*Limenitis archippus*

(a) **Batesian mimicry:** Monarch (*Danaus*) is poisonous; viceroy (*Limenitis*) is palatable mimic



*Heliconius erato*



*Heliconius melpomene*



*Heliconius sapho*



*Heliconius cydno*

(b) **Müllerian mimicry:** two pairs of mimics; all are distasteful

#### FIGURE 25.12

**Mimicry.** (a) Batesian mimicry. Monarch butterflies (*Danaus plexippus*) are protected from birds and other predators by the cardiac glycosides they incorporate from the milkweeds and dogbanes they feed on as larvae. Adult monarch butterflies advertise their poisonous nature with warning coloration. Viceroy butterflies (*Limenitis archippus*) are Batesian mimics of the poisonous monarch. (b) Pairs of Müllerian mimics. *Heliconius erato* and *H. melpomene* are sympatric, and *H. sapho* and *H. cydno* are sympatric. All of these butterflies are distasteful. They have evolved similar coloration patterns in sympatry to minimize predation; predators need only learn one pattern to avoid.

25.12b). If animals that resemble one another are all poisonous or dangerous, they gain an advantage because a predator will learn more quickly to avoid them. In some cases, predator populations even evolve an innate avoidance of species; such evolution may occur more quickly when multiple dangerous prey look alike.

In both Batesian and Müllerian mimicry, mimic and model must not only look alike but also act alike if predators are to be deceived. For example, the members of several families of insects that closely resemble wasps behave surprisingly like the wasps they mimic, flying often and actively from place to place.

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In Batesian mimicry, unprotected species resemble others that are distasteful. Both species exhibit aposematic coloration. In Müllerian mimicry, two or more unrelated but protected species resemble one another, thus achieving a kind of group defense.

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## 25.3 Evolution sometimes fosters cooperation.

### Coevolution and Symbiosis

The plants, animals, protists, fungi, and bacteria that live together in communities have changed and adjusted to one another continually over a period of millions of years. For example, many features of flowering plants have evolved in relation to the dispersal of the plant's gametes by animals (figure 25.13). These animals, in turn, have evolved a number of special traits that enable them to obtain food or other resources efficiently from the plants they visit, often from their flowers. While doing so, the animals pick up pollen, which they may deposit on the next plant they visit, or seeds, which may be left elsewhere in the environment, sometimes a great distance from the parental plant.

Such interactions, which involve the long-term, mutual evolutionary adjustment of the characteristics of the members of biological communities, are examples of **coevolution**, a phenomenon we have already seen in predator-prey interactions.

#### Symbiosis Is Widespread

Another type of coevolution involves **symbiotic relationships** in which two or more kinds of organisms live together in often elaborate and more-or-less permanent relationships. All symbiotic relationships carry the potential for coevolution between the organisms involved, and in many instances the results of this coevolution are fascinating. Examples of symbiosis include *lichens*, which are associations of certain fungi with green algae or cyanobacteria. Lichens are discussed in more detail in chapter 36. Another important example are *mycorrhizae*, the association between fungi and the roots of most kinds of plants. The fungi expedite the plant's absorption of certain nutrients, and the plants in turn provide the fungi with carbohydrates. Similarly, root nodules that occur in legumes and certain other kinds of plants contain bacteria that fix atmospheric nitrogen and make it available to their host plants.

In the tropics, leafcutter ants are often so abundant that they can remove a quarter or more of the total leaf surface of the plants in a given area. They do not eat these leaves directly; rather, they take them to underground nests, where they chew them up and inoculate them with the spores of particular fungi. These fungi are cultivated by the ants and brought from one specially prepared bed to another, where they grow and reproduce. In turn, the fungi constitute the primary food of the ants and their larvae. The relationship between leafcutter ants and these fungi is an excellent example of symbiosis.



**FIGURE 25.13**

**Pollination by bat.** Many flowers have coevolved with other species to facilitate pollen transfer. Insects are widely known as pollinators, but they're not the only ones. Notice the cargo of pollen on the bat's snout.

#### Kinds of Symbiosis

The major kinds of symbiotic relationships include (1) **commensalism**, in which one species benefits while the other neither benefits nor is harmed; (2) **mutualism**, in which both participating species benefit; and (3) **parasitism**, in which one species benefits but the other is harmed. Parasitism can also be viewed as a form of predation, although the organism that is preyed upon does not necessarily die.

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**Coevolution is a term that describes the long-term evolutionary adjustments of species to one another. In symbiosis two or more species interact closely, with at least one species benefitting.**

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## Commensalism

Commensalism is a symbiotic relationship that benefits one species and neither hurts nor helps the other. In nature, individuals of one species are often physically attached to members of another. For example, epiphytes are plants that grow on the branches of other plants. In general, the host plant is unharmed, while the epiphyte that grows on it benefits. Similarly, various marine animals, such as barnacles, grow on other, often actively moving sea animals like whales and thus are carried passively from place to place. These “passengers” presumably gain more protection from predation than they would if they were fixed in one place, and they also reach new sources of food. The increased water circulation that such animals receive as their host moves around may be of great importance, particularly if the passengers are filter feeders. The gametes of the passenger are also more widely dispersed than would be the case otherwise.



**FIGURE 25.14**

**Commensalism in the sea.** Clownfishes, such as this *Amphiprion perideraion* in Guam, often form symbiotic associations with sea anemones, gaining protection by remaining among their tentacles and gleaning scraps from their food. Different species of anemones secrete different chemical mediators; these attract particular species of fishes and may be toxic to the fish species that occur symbiotically with other species of anemones in the same habitat. There are 26 species of clownfishes, all found only in association with sea anemones; 10 species of anemones are involved in such associations, so that some of the anemone species are host to more than one species of clownfish.

### Examples of Commensalism

The best-known examples of commensalism involve the relationships between certain small tropical fishes and sea anemones, marine animals that have stinging tentacles (see chapter 44). These fish have evolved the ability to live among the tentacles of sea anemones, even though these tentacles would quickly paralyze other fishes that touched them (figure 25.14). The anemone fishes feed on the detritus left from the meals of the host anemone, remaining uninjured under remarkable circumstances.

On land, an analogous relationship exists between birds called oxpeckers and grazing animals such as cattle or rhinoceros. The birds spend most of their time clinging to the animals, picking off parasites and other insects, carrying out their entire life cycles in close association with the host animals.

### When Is Commensalism Commensalism?

In each of these instances, it is difficult to be certain whether the second partner receives a benefit or not; there is no clear-cut boundary between commensalism

and mutualism. For instance, it may be advantageous to the sea anemone to have particles of food removed from its tentacles; it may then be better able to catch other prey. Similarly, while often thought of as commensalism, the association of grazing mammals and gleaning birds is actually an example of mutualism. The mammal benefits by having parasites and other insects removed from its body, but the birds also benefit by gaining a dependable source of food.

On the other hand, commensalism can easily transform itself into parasitism. For example, oxpeckers are also known to pick not only parasites, but also scabs off their grazing hosts. Once the scab is picked, the birds drink the blood that flows from the wound. Occasionally, the cumulative effect of persistent attacks can greatly weaken the herbivore, particularly when conditions are not favorable, such as during droughts.

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**Commensalism is the benign use of one organism by another.**

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## Mutualism

Mutualism is a symbiotic relationship among organisms in which both species benefit. Examples of mutualism are of fundamental importance in determining the structure of biological communities. Some of the most spectacular examples of mutualism occur among flowering plants and their animal visitors, including insects, birds, and bats. As we will see in chapter 37, during the course of their evolution, the characteristics of flowers have evolved in large part in relation to the characteristics of the animals that visit them for food and, in doing so, spread their pollen from individual to individual. At the same time, characteristics of the animals have changed, increasing their specialization for obtaining food or other substances from particular kinds of flowers.

Another example of mutualism involves ants and aphids. Aphids, also called greenflies, are small insects that suck fluids from the phloem of living plants with their piercing mouthparts. They extract a certain amount of the sucrose and other nutrients from this fluid, but they excrete much of it in an altered form through their anus. Certain ants have taken advantage of this—in effect, domesticating the aphids. The ants carry the aphids to new plants, where they come into contact with new sources of food, and then consume as food the “honeydew” that the aphids excrete.

### Ants and Acacias

A particularly striking example of mutualism involves ants and certain Latin American species of the plant genus *Acacia*. In these species, certain leaf parts, called stipules, are modified as paired, hollow thorns. The thorns are inhabited by stinging ants of the genus *Pseudomyrmex*, which do not nest anywhere else (figure 25.15). Like all thorns that occur on plants, the acacia horns serve to deter herbivores.

At the tip of the leaflets of these acacias are unique, protein-rich bodies called Beltian bodies, named after the nineteenth-century British naturalist Thomas Belt. Beltian bodies do not occur in species of *Acacia* that are not inhabited by ants, and their role is clear: they serve as a primary food for the ants. In addition, the plants secrete nectar from glands near the bases of their leaves. The ants consume this nectar as well, feeding it and the Beltian bodies to their larvae.

Obviously, this association is beneficial to the ants, and one can readily see why they inhabit acacias of this group. The ants and their larvae are protected within the swollen thorns, and the trees provide a balanced diet, including the sugar-rich nectar and the protein-rich Beltian bodies. What, if anything, do the ants do for the plants?

Whenever any herbivore lands on the branches or leaves of an acacia inhabited by ants, the ants, which continually patrol the acacia’s branches, immediately attack and devour the herbivore. The ants that live in the acacias also help their hosts to compete with other plants. The ants cut away



**FIGURE 25.15**

**Mutualism: ants and acacias.** Ants of the genus *Pseudomyrmex* live within the hollow thorns of certain species of acacia trees in Latin America. The nectaries at the bases of the leaves and the Beltian bodies at the ends of the leaflets provide food for the ants. The ants, in turn, supply the acacias with organic nutrients and protect the acacia from herbivores.

any branches of other plants that touch the acacia in which they are living. They create, in effect, a tunnel of light through which the acacia can grow, even in the lush deciduous forests of lowland Central America. In fact, when an ant colony is experimentally removed from a tree, the acacia is unable to compete successfully in this habitat. Finally, the ants bring organic material into their nests. The parts they do not consume, together with their excretions, provide the acacias with an abundant source of nitrogen.

As with commensalism, however, things are not always as they seem. Ant-acacia mutualisms also occur in Africa. In Kenya, several species of acacia ants occur, but only one species occurs on any tree. One species, *Crematogaster nigriceps*, is competitively inferior to two of the other species. To prevent invasion by other ant species, *C. nigriceps* prunes the branches of the acacia, preventing it from coming into contact with branches of other trees, which would serve as a bridge for invaders. Although this behavior is beneficial to the ant, it is detrimental to the tree, as it destroys the tissue from which flowers are produced, essentially sterilizing the tree. In this case, what has initially evolved as a mutualistic interaction has instead become a parasitic one.

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**Mutualism involves cooperation between species, to the mutual benefit of both.**

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## Parasitism

Parasitism may be regarded as a special form of symbiosis in which the predator, or parasite, is much smaller than the prey and remains closely associated with it. Parasitism is harmful to the prey organism and beneficial to the parasite. The concept of parasitism seems obvious, but individual instances are often surprisingly difficult to distinguish from predation and from other kinds of symbiosis.

### External Parasites

Parasites that feed on the exterior surface of an organism are external parasites, or **ectoparasites**. Many instances of external parasitism are known (figure 25.16). Lice, which live on the bodies of vertebrates—mainly birds and mammals—are normally considered parasites. Mosquitoes are not considered parasites, even though they draw food from birds and mammals in a similar manner to lice, because their interaction with their host is so brief.

**Parasitoids** are insects that lay eggs on living hosts. This behavior is common among wasps, whose larvae feed on the body of the unfortunate host, often killing it.

### Internal Parasites

Vertebrates are parasitized internally by **endoparasites**, members of many different phyla of animals and protists. Invertebrates also have many kinds of parasites that live within their bodies. Bacteria and viruses are not usually considered parasites, even though they fit our definition precisely.

Internal parasitism is generally marked by much more extreme specialization than external parasitism, as shown by the many protist and invertebrate parasites that infect humans. The more closely the life of the parasite is linked with that of its host, the more its morphology and behavior are likely to have been modified during the course of its evolution. The same is true of symbiotic relationships of all sorts. Conditions within the body of an organism are different from those encountered outside and are apt to be much more constant. Consequently, the structure of an internal parasite is often simplified, and unnecessary armaments and structures are lost as it evolves.

### Brood Parasitism

Not all parasites consume the body of their host. In brood parasitism, birds like cowbirds and European cuckoos lay their eggs in the nests of other species. The host parents raise the brood parasite as if it were one of their own clutch, in many cases investing more in feeding the imposter than in feeding their own offspring (figure 25.17). The brood parasite reduces the reproductive success of the foster parent hosts, so it is not surprising that in some cases natural selection has fostered the hosts' ability to detect



**FIGURE 25.16**

**An external parasite.** The flowering plant dodder (*Cuscuta*) is a parasite and has lost its chlorophyll and its leaves in the course of its evolution. Because it is heterotrophic, unable to manufacture its own food, dodder obtains its food from the host plants it grows on.



**FIGURE 25.17**

**Brood parasitism.** This bird is feeding a cuckoo chick in its nest. The cuckoo chick is larger than the adult bird, but the bird does not recognize that the cuckoo is not its own offspring. Cuckoo mothers sneak into the nests of other birds and lay an egg, entrusting the care of their offspring to an unwitting bird of another species.

parasite eggs and reject them. What is more surprising is that in many other species, the ability to detect parasite eggs has not evolved.

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**In parasitism, one organism serves as a host to another organism, usually to the host's disadvantage.**

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## Interactions among Ecological Processes

We have seen the different ways in which species within a community can interact with each other. In nature, however, more than one type of interaction usually occurs at the same time. In many cases, the outcome of one type of interaction is modified or even reversed when another type of interaction is also occurring.

### Predation Reduces Competition

When resources are limiting, a superior competitor can eliminate other species from a community. However, predators can prevent or greatly reduce competitive exclusion by reducing the numbers of individuals of competing species. A given predator may often feed on two, three, or more kinds of plants or animals in a given community. The predator's choice depends partly on the relative abundance of the prey options. In other words, a predator may feed on species *A* when it is abundant and then switch to species *B* when *A* is rare. Similarly, a given prey species may be a primary source of food for increasing numbers of species as it becomes more abundant. In this way, superior competitors may be prevented from outcompeting other species.

Such patterns are often characteristic of biological communities in marine intertidal habitats. For example, in preying selectively on bivalves, sea stars prevent bivalves from monopolizing such habitats, opening up space for many other organisms (figure 25.18). When sea stars are removed from a habitat, species diversity falls precipitously, the seafloor community coming to be dominated by a few species of bivalves. Because predation tends to reduce competition in natural communities, it is usually a mistake to attempt to eliminate a major predator such as wolves or mountain lions from a community. The result is to decrease rather than increase the biological diversity of the community, the opposite of what is intended.

### Parasitism May Counter Competition

Parasites may effect sympatric species differently and thus influence the outcome of interspecific interactions. In a classic experiment, Thomas Park of the University of Chicago investigated interactions between two flour beetles, *Tribolium castaneum* and *T. confusum* with a parasite, *Adelina*. In the absence of the parasite, *T. castaneum* is dominant and *T. confusum* normally goes extinct. When the parasite is present, however, the outcome is reversed and *T. castaneum* perishes. Similar effects of parasites in natural systems have been observed in many species. For example, in the *Anolis* lizards of St. Maarten mentioned previously, the competitively inferior species is resistant to malaria, whereas the other species is highly susceptible. Only in



(a)



(b)

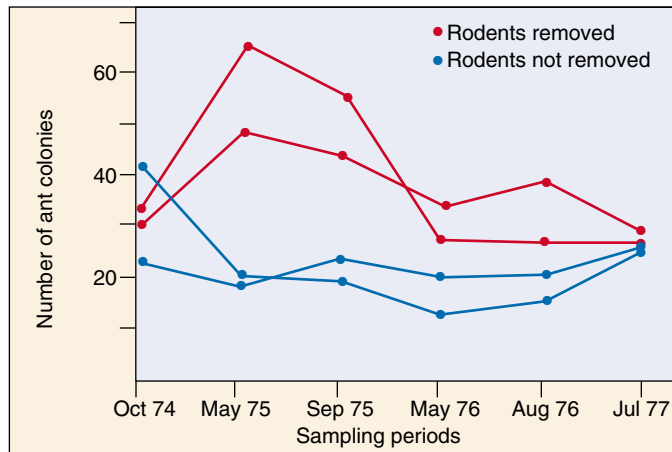
**FIGURE 25.18**

**Predation reduces competition.** (a) In a controlled experiment in a coastal ecosystem, an investigator removed a key predator (*Pisaster*). (b) In response, fiercely competitive mussels exploded in growth, effectively crowding out seven other indigenous species.

areas in which the malaria parasite occurs are the two species capable of coexisting.

### Indirect Effects

In some cases, species may not directly interact, yet the presence of one species may effect a second species by way of interactions with a third species. Such effects are termed **indirect effects**. For example, in the Chihuahuan Desert, rodents and ants both eat seeds. Thus, one might expect them to compete with each other. However, when all rodents were completely removed from large enclosures (unlike the experiment discussed above, there were no holes in



**FIGURE 25.19**

**Change in ant population size after the removal of rodents.**

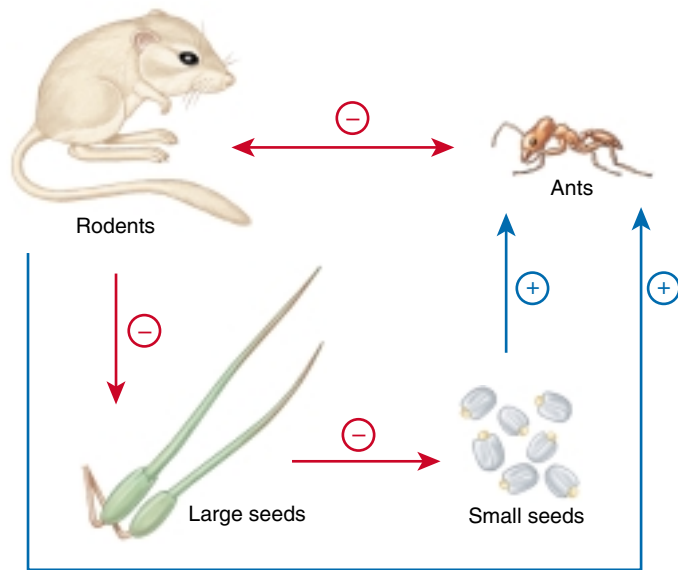
Ants initially increased in population size relative to ants in the enclosures from which rodents weren't removed, but then these ant populations declined.

the enclosure walls, so once removed, rodents couldn't get back in), ant populations first increased, but then declined (figure 25.19). The initial increase was the expected result of removing a competitor; why did it reverse? The answer reveals the intricacies of natural ecosystems (figure 25.20). Rodents prefer large seeds, whereas ants prefer smaller seeds. Further, in this system plants with large seeds are competitively superior to plants with small seeds. Thus, the removal of rodents leads to an increase in the number of plants with large seeds, which reduces the number of small seeds available to ants, which thus leads to a decline in ant populations. Thus, the effect of rodents on ants is complicated: a direct negative effect of resource competition and an indirect, positive effect mediated by plant competition.

## Keystone Species

Species that have particularly strong effects on the composition of communities are termed **keystone species**. Predators, such as the starfish, can often serve as keystone species by preventing one species from outcompeting others, thus maintaining high levels of species richness in a community.

There are, however, a wide variety of other types of keystone species. Some species manipulate the environment in ways that create new habitats for other species. Beavers, for example, change running streams into small impoundments, changing the flow of water and flooding areas (figure 25.21). Similarly, alligators excavate deep holes at the bottoms of lakes. In times of drought, these holes are the only areas in which water remains, thus allowing aquatic species that otherwise would perish to persist until the drought ends and the lake refills.



**FIGURE 25.20**

**Rodent-ant interactions.** Rodents and ants both eat seeds, so the presence of rodents has a negative effect on ants and vice versa. However, the presence of rodents has a negative effect on large seeds. In turn, the number of plants with large seeds has a negative effect on plants that produce small seeds. Hence, the presence of rodents should increase the number of small seeds. In turn, the number of small seeds has a positive effect on ant populations. Thus, indirectly, the presence of rodents has a positive effect on ant population size.



**FIGURE 25.21**

**Example of a keystone species.** Beavers, by constructing dams and transforming flowing streams into ponds, create new habitats for many plant and animal species.

Many different processes are likely to be occurring simultaneously within communities. Only by understanding how these processes interact will we be able to understand how communities function.

## 25.4 Ecological succession may increase species richness.

Even when the climate of an area remains stable year after year, ecosystems have a tendency to change from simple to complex in a process known as **succession**. This process is familiar to anyone who has seen a vacant lot or cleared woods slowly become occupied by an increasing number of plants, or a pond become dry land as it is filled with vegetation encroaching from the sides.

### Succession

If a wooded area is cleared and left alone, plants will slowly reclaim the area. Eventually, traces of the clearing will disappear and the area will again be woods. This kind of succession, which occurs in areas where an existing community has been disturbed, is called **secondary succession**.

In contrast, **primary succession** occurs on bare, lifeless substrate, such as rocks, or in open water, where organisms gradually move into an area and change its nature. Primary succession occurs in lakes left behind after the retreat of glaciers, on volcanic islands that rise above the sea, and on land exposed by retreating glaciers (figure 25.22). Primary succession on glacial moraines provides an example (figure 25.23). On bare, mineral-poor soil, lichens grow first, forming small pockets of soil. Acidic secretions from the lichens help to break down the substrate and add to the accumulation of soil. Mosses then colonize these pockets of soil, eventually building up enough nutrients in the soil for alder shrubs to take hold. Over a hundred years, the alders build up the soil nitrogen levels until spruce are able to thrive, eventually crowding out the alder and forming a dense spruce forest.

In a similar example, an **oligotrophic** lake—one poor in nutrients—may gradually, by the accumulation of organic matter, become **eutrophic**—rich in nutrients. As this occurs, the composition of communities will change, first increasing in species richness and then declining.

Primary succession in different habitats often eventually arrives at the same kinds of vegetation—vegetation characteristic of the region as a whole. This relationship led American ecologist F. E. Clements, at about the turn of the century, to propose the concept of a final **climax community**. With an increasing realization that (1) the climate keeps changing, (2) the process of succession is often very slow, and (3) the nature of a region's vegetation is being determined to an increasing extent by human activities, ecologists do not consider the concept of “climax community” to be as useful as they once did.

### Why Succession Happens

Succession happens because species alter the habitat and the resources available in it in ways that favor other species.

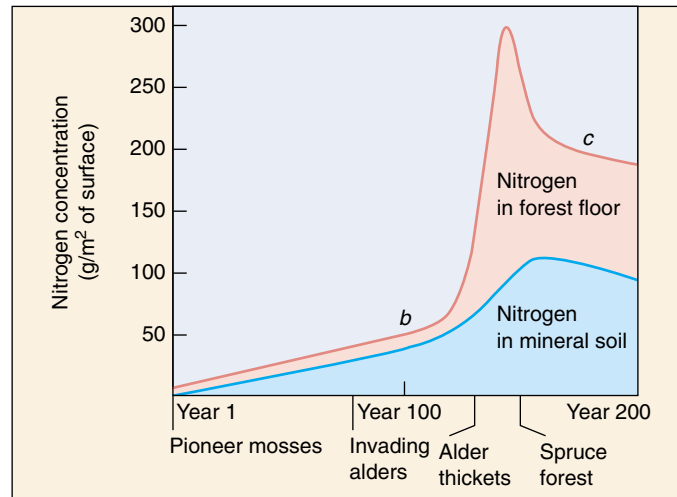


FIGURE 25.22

#### Plant succession produces progressive changes in the soil.

Initially, the glacial moraine at Glacier Bay, Alaska, portrayed in figure 25.23, had little soil nitrogen, but nitrogen-fixing alders led to a buildup of nitrogen in the soil, encouraging the subsequent growth of the conifer forest. Letters in the graph correspond to photographs in parts *b* and *c* of figure 25.23.

Three dynamic concepts are of critical importance in the process: tolerance, inhibition, and facilitation.

1. **Tolerance.** Early successional stages are characterized by weedy *r*-selected species that are tolerant of the harsh, abiotic conditions in barren areas.
2. **Facilitation.** The weedy early successional stages introduce local changes in the habitat that favor other, less weedy species. Thus, the mosses in the Glacier Bay succession convert nitrogen to a form that allows alders to invade. The alders in turn lower soil pH as their fallen leaves decompose, and spruce and hemlock, which require acidic soil, are able to invade.
3. **Inhibition.** Sometimes the changes in the habitat caused by one species, while favoring other species, inhibit the growth of the species that caused them. Alders, for example, do not grow as well in acidic soil as the spruce and hemlock that replace them.

Over the course of succession, the number of species typically increases as the environment becomes more hospitable. In some cases, however, as ecosystems mature, more *K*-selected species replace *r*-selected ones, and superior competitors force out other species, leading ultimately to a decline in species richness.

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Communities evolve to have greater total biomass and species richness in a process called succession.

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**FIGURE 25.23**

**Primary succession at Alaska's Glacier Bay.** (a) The sides of the glacier have been retreating at a rate of some 8 meters a year, leaving behind exposed soil from which nitrogen and other minerals have been leached out. The first invaders of these exposed sites are pioneer moss species with nitrogen-fixing mutualistic microbes. Within 20 years, young alder shrubs take hold. (b) Rapidly fixing nitrogen, they soon form dense thickets. As soil nitrogen levels rise, (c) spruce crowd out the mature alders, forming a forest.

## The Role of Disturbance

Disturbances often interrupt the succession of plant communities. Depending on the magnitude of the disturbance, communities may revert to earlier stages of succession or even, in extreme cases, begin at the earliest stages of primary succession. Disturbances severe enough to disrupt succession include calamities such as forest fires, drought, and floods. Animals may also cause severe disruptions. Gypsy moths can devastate a forest by consuming its trees. Unregulated deer populations may grow explosively, the deer overgrazing and so destroying the forest they live in, in the same way too many cattle overgraze a pasture by eating all available grass down to the ground.

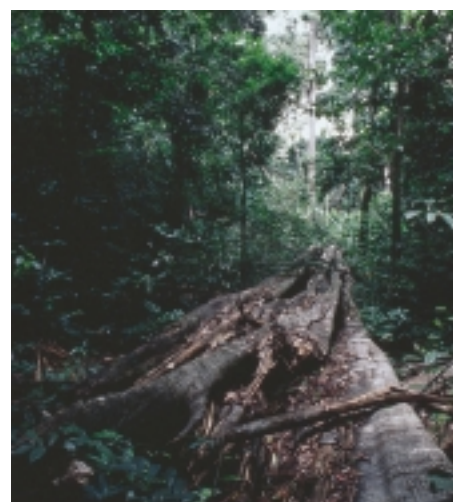
### Intermediate Disturbance Hypothesis

In some cases, disturbance may act to increase the species richness of an area. According to the intermediate disturbance hypothesis, communities experiencing moderate amounts of disturbance will have higher levels of species richness than communities experiencing either little or great amounts of disturbance. Two factors could account for this pattern. First, in communities in which moderate amounts of disturbance occur, patches of habitat will exist at different successional stages. Thus, within the area as a whole, species diversity will be greatest because the full range of species—those characteristic of all stages of succession—will be present. For example, a pattern of intermittent episodic disturbance that produces gaps in the rain forest (like when a tree falls) allows invasion of the gap by other species (figure 25.24). Eventually, the species inhabiting the gap will go through a successional sequence, one tree replacing another, until a canopy tree species comes again to occupy the gap. But if there are lots of gaps of different ages in the forest, many different species will coexist, some in young gaps, others in older ones.

Second, moderate levels of disturbance may prevent communities from reaching the final stages of succession,

**FIGURE 25.24**

**Intermediate disturbance.** A single fallen tree creates a small light gap in the tropical rain forest of Panama. Such gaps play a key role in maintaining the high species diversity of the rain forest.



in which a few dominant competitors eliminate most of the other species. On the other hand, too much disturbance might leave the community continually in the earliest stages of succession, when species richness is relatively low.

Ecologists are increasingly realizing that disturbance is the norm, rather than the exception, in many communities. As a result, the idea that communities inexorably move along a successional trajectory culminating in the development of a climax community is no longer widely accepted. Rather, predicting the state of a community in the future may be difficult because the unpredictable occurrence of disturbances will often counter successional changes. Understanding the role that disturbances play in structuring communities is currently an important area of investigation in ecology.

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**Succession is often disrupted by natural or human causes. In some cases, intermediate levels of disturbance may maximize the species richness of a community.**

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## Summary

## Questions

## Media Resources

**25.1 Interactions among competing species shape ecological niches.**

- Each species plays a specific role in its ecosystem; this role is called its niche.
- An organism's fundamental niche is the total niche that the organism would occupy in the absence of competition. Its realized niche is the actual niche it occupies in nature.
- Two species cannot occupy the same niche for long if resources are limiting; one will outcompete the other, driving it to extinction.
- Species can coexist by partitioning resources to minimize competition.

1. What is the difference between interspecific competition and intraspecific competition? What is Gause's principle of competitive exclusion?
2. Is the term niche synonymous with the term *habitat*? Why or why not? How does an organism's fundamental niche differ from its realized niche?



- Introduction to Communities
- Community Organization



- On Science Article: Killer Bees

**25.2 Predators and their prey coevolve.**

- Plants are often protected from herbivores by chemicals they manufacture.
- Warning, or aposematic, coloration is characteristic of organisms that are poisonous, sting, or are otherwise harmful. In contrast, cryptic coloration, or camouflage, is characteristic of nonpoisonous organisms.
- Predator-prey relationships are of crucial importance in limiting population sizes in nature.

3. What morphological defenses do plants use to defend themselves against herbivores?
4. Consider aposematic coloration, cryptic coloration, and Batesian mimicry. Which would be associated with an adult viceroy butterfly? Which would be associated with a larval monarch butterfly? Which would be associated with a larval viceroy butterfly?

**25.3 Evolution sometimes fosters cooperation.**

- Coevolution occurs when different kinds of organisms evolve adjustments to one another over long periods of time.
- Many organisms have coevolved to a point of dependence. In mutualism the relationship is mutually beneficial; in commensalism, only one organism benefits while the other is unharmed; and in parasitism one organism serves as a host to another, usually to the host's disadvantage.

5. Why is eliminating predators a bad idea for species richness?
6. How can predation and competition interact in regulating species diversity of a community?



- Student Research: Hermit Crab—Sea Anemone Associations

**25.4 Ecological succession may increase species richness.**

- Primary succession takes place in barren areas, like rocks or open water. Secondary succession takes place in areas where the original communities of organisms have been disturbed.
- Succession occurs because of tolerance, facilitation, and inhibition.
- Disturbance can disrupt successional changes. In some cases, disturbance can increase species richness of a community.

7. Why have scientists altered the concept of a final, climax vegetation in a given ecosystem? What types of organisms are often associated with early stages of succession? What is the role of disturbance in succession?



- Succession



- Book Review: *Guns, Germs, and Steel* by Diamond

# 26

## *Animal Behavior*

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### Concept Outline

#### 26.1 Ethology focuses on the natural history of behavior.

**Approaches to the Study of Behavior.** Field biologists focus on evolutionary aspects of behavior.

**Behavioral Genetics.** At least some behaviors are genetically determined.

#### 26.2 Comparative psychology focuses on how learning influences behavior.

**Learning.** Association plays a major role in learning.

**The Development of Behavior.** Parent-offspring interactions play a key role in the development of behavior.

**The Physiology of Behavior.** Hormones influence many behaviors, particularly reproductive ones.

**Behavioral Rhythms.** Many behaviors are governed by innate biological clocks.

#### 26.3 Communication is a key element of many animal behaviors.

**Courtship.** Animals use many kinds of signals to court one another.

**Communication in Social Groups.** Bees and other social animals communicate in complex ways.

#### 26.4 Migratory behavior presents many puzzles.

**Orientation and Migration.** Animals use many cues from the environment to navigate during migrations.

#### 26.5 To what degree animals “think” is a subject of lively dispute.

**Animal Cognition.** It is not clear to what degree animals “think.”



**FIGURE 26.1**

**Rearing offspring involves complex behaviors.** Living in groups called prides makes lions better mothers. Females share the responsibilities of nursing and protecting the pride's young, increasing the probability that the youngsters will survive into adulthood.

Organisms interact with their environment in many ways. To understand these interactions, we need to appreciate both the internal factors that shape the way an animal behaves, as well as aspects of the external environment that affect individuals and organisms. In this chapter, we explore the mechanisms that determine an animal's behavior (figure 26.1), as well as the ways in which behavior develops in an individual. In the next chapter, we will consider the field of behavioral ecology, which investigates how natural selection has molded behavior through evolutionary time.

## 26.1 Ethology focuses on the natural history of behavior.

### Approaches to the Study of Behavior

During the past two decades, the study of animal behavior has emerged as an important and diverse science that bridges several disciplines within biology. Evolution, ecology, physiology, genetics, and psychology all have natural and logical linkages with the study of behavior, each discipline adding a different perspective and addressing different questions.

Research in animal behavior has made major contributions to our understanding of nervous system organization, child development, and human communication, as well as the process of speciation, community organization, and the mechanism of natural selection itself. The study of the behavior of nonhuman animals has led to the identification of general principles of behavior, which have been applied, often controversially, to humans. This has changed the way we think about the origins of human behavior and the way we perceive ourselves.

Behavior can be defined as the way an organism responds to stimuli in its environment. These stimuli might be as simple as the odor of food. In this sense, a bacterial cell “behaves” by moving toward higher concentrations of sugar. This behavior is very simple and well-suited to the life of bacteria, allowing these organisms to live and reproduce. As animals evolved, they occupied different environments and faced diverse problems that affected their survival and reproduction. Their nervous systems and behavior concomitantly became more complex. Nervous systems perceive and process information concerning environmental stimuli and trigger adaptive motor responses, which we see as patterns of behavior.

When we observe animal behavior, we can explain it in two different ways. First, we might ask *how* it all works, that is, how the animal’s senses, nerve networks, or internal state provide a physiological basis for the behavior. In this way, we would be asking a question of **proximate causation**. To analyze the proximate cause of behavior, we might measure hormone levels or record the impulse activity of neurons in the animal. We could also ask *why* the behavior evolved, that is, what is its adaptive value? This is a question concerning **ultimate causation**. To study the ultimate cause of a behavior, we would attempt to determine how it influenced the animal’s survival or reproductive success. Thus, a male songbird may sing during the breeding season because it has a level of the steroid sex hormone, testosterone, which binds to hormone receptors in the brain and triggers the production of song; this would be the proximate cause of the male bird’s song. But the male sings to defend a territory from

other males and to attract a female to reproduce; this is the ultimate, or evolutionary, explanation for the male’s vocalization.

The study of behavior has had a long history of controversy. One source of controversy has been the question of whether behavior is determined more by an individual’s genes or its learning and experience. In other words, is behavior the result of nature (instinct) or nurture (experience)? In the past, this question has been considered an “either/or” proposition, but we now know that instinct and experience both play significant roles, often interacting in complex ways to produce the final behavior. The scientific study of instinct and learning, as well as their interrelationship, has led to the growth of several scientific disciplines, including ethology, behavioral genetics, behavioral neuroscience, and comparative psychology.

### Ethology

Ethology is the study of the natural history of behavior. Early ethologists (figure 26.2) were trained in zoology and evolutionary biology, fields that emphasize the study of animal behavior under natural conditions. As a result of this training, they believed that behavior is largely instinctive, or innate—the product of natural selection. Because behavior is often **stereotyped** (appearing in the same way in different individuals of a species), they argued that it must be based on preset paths in the nervous system. In their view, these paths are structured from genetic blueprints and cause animals to show a relatively complete behavior the first time it is produced.

The early ethologists based their opinions on behaviors such as egg retrieval by geese. Geese incubate their eggs in a nest. If a goose notices that an egg has been knocked out of the nest, it will extend its neck toward the egg, get up, and roll the egg back into the nest with a side-to-side motion of its neck while the egg is tucked beneath its bill. Even if the egg is removed during retrieval, the goose completes the behavior, as if driven by a program released by the initial sight of the egg outside the nest. According to ethologists, egg retrieval behavior is triggered by a **sign stimulus** (also called a **key stimulus**), the appearance of an egg out of the nest; a component of the goose’s nervous system, the **innate releasing mechanism**, provides the neural instructions for the motor program, or **fixed action pattern** (figure 26.3). More generally, the sign stimulus is a “signal” in the environment that triggers a behavior. The innate releasing mechanism is the sensory mechanism that detects the signal, and the fixed action pattern is the stereotyped act.



**FIGURE 26.2**

**The founding fathers of ethology: Karl von Frisch, Konrad Lorenz, and Niko Tinbergen pioneered the study of behavioral science.** In 1973, they received the Nobel Prize in Physiology or Medicine for their path-making contributions. Von Frisch led the study of honeybee communication and sensory biology. Lorenz focused on social development (imprinting) and the natural history of aggression. Tinbergen examined the functional significance of behavior and was the first behavioral ecologist.

Similarly, a frog unfolds its long, sticky tongue at the sight of a moving insect, and a male stickleback fish will attack another male showing a bright red underside. Such responses certainly appear to be programmed and instinctive, but what evidence supports the ethological view that behavior has an underlying neural basis?

### Behavior as a Response to Stimuli in the Environment

In the example of egg retrieval behavior in geese, a goose must first perceive that an egg is outside of the nest. To respond to this stimulus, it must convert one form of energy which is an input to its visual system—the energy of photons of light—into a form of energy its nervous system can understand and use to respond—the electrical energy of a nerve impulse. Animals need to respond to other stimuli in the environment as well. For an animal to orient from a food source back to its nest, it might rely on the position of the sun. To find a mate, an animal might use a particular chemical scent. The electromagnetic energy of light and the chemical energy of an odor must be converted to the electrical energy of a nerve impulse. This is done through **transduction**, the conversion of energy in the environment to an action potential, and the first step in the processing of stimuli perceived by the senses. For example, rhodopsin is responsible for the transduction of visual stimuli. Rhodopsin is made of *cis*-retinal and the protein opsin. Light is absorbed by the visual pigment *cis*-retinal causing it



**FIGURE 26.3**

**Lizard prey capture.** The complex series of movements of the tongue this chameleon uses to capture an insect represents a fixed action pattern.

to change its shape to *trans*-retinal (see chapter 55). This in turn changes the shape of the companion protein opsin, and induces the first step in a cascade of molecular events that finally triggers a nerve impulse. Sound, odor, and tastes are transduced to nerve impulses by similar processes.

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**Ethologists study behavior from an evolutionary perspective, focusing on the neural basis of behaviors.**

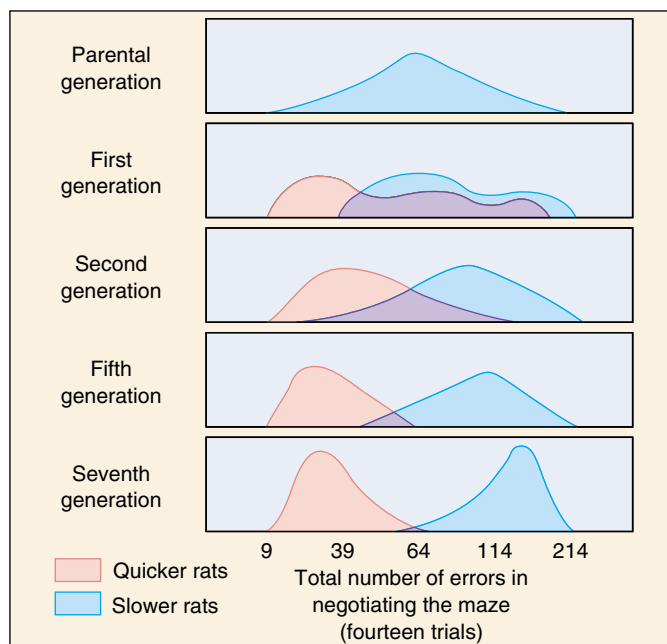
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## Behavioral Genetics

In a famous experiment carried out in the 1940s, Robert Tryon studied the ability of rats to find their way through a maze with many blind alleys and only one exit, where a reward of food awaited. Some rats quickly learned to zip right through the maze to the food, making few incorrect turns, while other rats took much longer to learn the correct path (figure 26.4). Tryon bred the fast learners with one another to establish a “maze-bright” colony, and he similarly bred the slow learners with one another to establish a “maze-dull” colony. He then tested the offspring in each colony to see how quickly they learned the maze. The offspring of maze-bright rats learned even more quickly than their parents had, while the offspring of maze-dull parents were even poorer at maze learning. After repeating this procedure over several generations, Tryon was able to produce two behaviorally distinct types of rat with very different maze-learning abilities. Clearly the ability to learn the maze was to some degree hereditary, governed by genes passed from parent to offspring. Furthermore, those genes were specific to this behavior, as the two groups of rats did not differ in their ability to perform other behavioral tasks, such as running a completely different kind of maze. Tryon’s research demonstrates how a study can reveal that behavior has a heritable component.

Further support for the genetic basis of behavior has come from studies of hybrids. William Dilger of Cornell University has examined two species of lovebird (genus *Agapornis*), which differ in the way they carry twigs, paper, and other materials used to build a nest. *A. personata* holds nest material in its beak, while *A. roseicollis* carries material tucked under its flank feathers (figure 26.5). When Dilger crossed the two species to produce hybrids, he found that the hybrids carry nest material in a way that seems intermediate between that of the parents: they repeatedly shift material between the bill and the flank feathers. Other studies conducted on courtship songs in crickets and tree frogs also demonstrate the intermediate nature of hybrid behavior.

The role of genetics can also be seen in humans by comparing the behavior of identical twins. Identical twins are, as their name implies, genetically identical. However, most sets of identical twins are raised in the same environment, so it is not possible to determine whether similarities in behavior result from their genetic similarity or from experiences shared as they grew up (the classic nature versus nurture debate). However, in some cases, twins have been separated at birth. A recent study of 50 such sets of twins revealed many similarities in personality, temperament, and even leisure-time activities, even though the twins were often raised in very different environments. These similarities indicate that genetics plays a role in determining behavior even in humans, although the relative importance of genetics versus environment is still hotly debated.



**FIGURE 26.4**

**The genetics of learning.** Tryon selected rats for their ability to learn to run a maze and demonstrated that this ability is influenced by genes. He tested a large group of rats, selected those that ran the maze in the shortest time, and let them breed with one another. He then tested their progeny and again selected those with the quickest maze-running times for breeding. After seven generations, he had succeeded in halving the average time an inexperienced rat required to negotiate the maze. Parallel “artificial selection” for slow running time more than doubled the average running time.

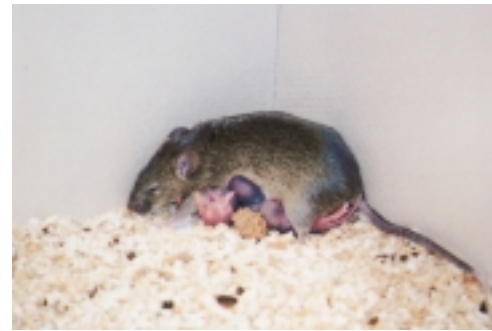


**FIGURE 26.5**

**Genetics of lovebird behavior.** Lovebirds inherit the tendency to carry nest material, such as these paper strips, under their flank feathers.

**FIGURE 26.6****Genetically caused defect in maternal care.**

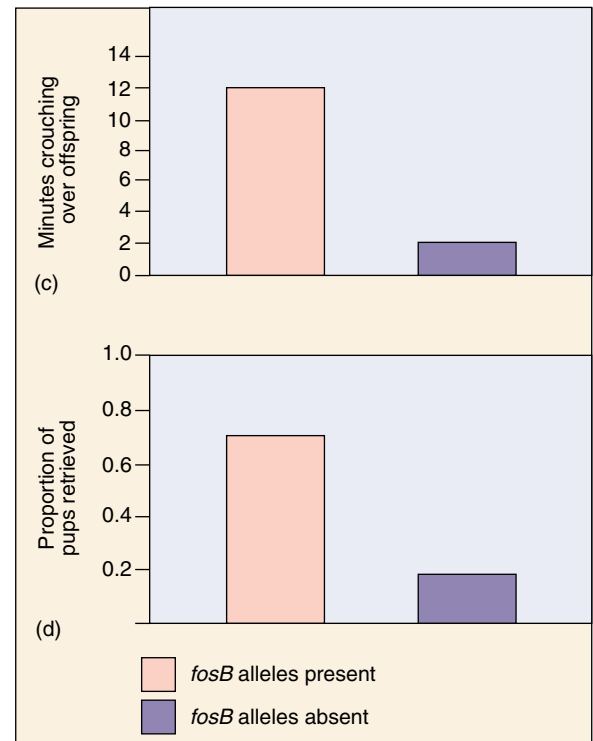
(a) In mice, normal mothers take very good care of their offspring, retrieving them if they move away and crouching over them. (b) Mothers with the mutant *fosB* allele perform neither of these behaviors, leaving their pups exposed. (c) Amount of time female mice were observed crouching in a nursing posture over offspring. (d) Proportion of pups retrieved when they were experimentally moved.



(a)



(b)

**Single Gene Effects on Behavior**

The maze-learning, hybrid, and identical twins studies just described suggest genes play a role in behavior, but recent research has provided much greater detail on the genetic basis of behavior. In the fruit fly *Drosophila*, and in mice, many mutations have been associated with particular behavioral abnormalities.

In fruit flies, for example, individuals that possess alternative alleles for a single gene differ greatly in their feeding behavior as larvae; larvae with one allele move around a great deal as they eat, whereas individuals with the alternative allele move hardly at all. A wide variety of mutations at other genes are now known in *Drosophila* which affect almost every aspect of courtship behavior.

The ways in which genetic differences affect behavior have been worked out for several mouse genes. For example, some mice with one mutation have trouble remembering information that they learned two days earlier about where objects are located. This difference appears to result because the mutant mice do not produce the enzyme  $\alpha$ -calcium-calmodulin-dependent kinase II, which plays an important role in the functioning of a part of the brain, the hippocampus, that is important for spatial learning.

Modern molecular biology techniques allow the role of genetics in behavior to be investigated with ever greater precision. For example, male mice genetically engineered (as “knock-outs”) to lack the ability to synthesize nitric oxide, a brain neurotransmitter, show increased aggressive behavior.

A particularly fascinating breakthrough occurred in 1996, when scientists using the knock-out technique discovered a new gene, *fosB*, that seems to determine whether or not female mice will nurture their young. Females with both *fosB* alleles knocked out will initially investigate their newborn babies, but then ignore them, in stark contrast to the caring and protective maternal behavior provided by normal females (figure 26.6).

The cause of this inattentiveness appears to result from a chain reaction. When mothers of new babies initially inspect them, information from their auditory, olfactory, and tactile senses are transmitted to the hypothalamus, where *fosB* alleles are activated, producing a particular protein, which in turn activates other enzymes and genes that affect the neural circuitry within the hypothalamus. These modifications within the brain cause the female to react maternally toward her offspring. In contrast, in mothers lacking the *fosB* alleles, this reaction is stopped midway. No protein is activated, the brain’s neural circuitry is not rewired, and maternal behavior does not result.

As these genetic techniques are becoming used more widely, the next few years should see similar dramatic advances in our knowledge of how genes affect behavior in many varieties of humans.

**The genetic basis of behavior is supported by artificial selection experiments, hybridization studies, and studies on the behavior of mutants. Research has also identified specific genes that control behavior.**

## 26.2 Comparative psychology focuses on how learning influences behavior.

### Learning

While ethologists were attempting to explain behavior as an instinctive process, **comparative psychologists** focused heavily on learning as the major element that shapes behavior. These behavioral scientists, working primarily on rats in laboratory settings, identified the ways in which animals learn. Learning is any modification of behavior that results from experience rather than maturation.

The simplest type of learning, **nonassociative learning**, does not require an animal to form an association between two stimuli or between a stimulus and a response. One form of nonassociative learning is **habituation**, which can be defined as a decrease in response to a repeated stimulus that has no positive or negative consequences (that is, no *reinforcement*). In many cases, the stimulus evokes a strong response when it is first encountered, but the magnitude of the response gradually declines with repeated exposure. For example, young birds see many types of objects moving overhead. At first, they may respond by crouching down and remaining still. Some of the objects, like falling leaves or members of their own species flying by, are seen very frequently and have no positive or negative consequence to the nestlings. Over time, the young birds may habituate to such stimuli and stop responding. Thus, habituation can be thought of as learning *not* to respond to a stimulus. Being able to ignore unimportant stimuli is critical to an animal confronting a barrage of stimuli in a complex environment. Another form of nonassociative learning is **sensitization**, characterized by an increased responsiveness to a stimulus. Sensitization is essentially the opposite of habituation.

A change in behavior that involves an association between two stimuli or between a stimulus and a response is termed **associative learning** (figure 26.7). The behavior is modified, or **conditioned**, through the association. This form of learning is more complex than habituation or sensitization. The two major types of associative learning are called **classical conditioning** and **operant conditioning**; they differ in the way the associations are established.

### Classical Conditioning

In classical conditioning, the paired presentation of two different kinds of stimuli causes the animal to form an association between the stimuli. Classical conditioning is also called **Pavlovian conditioning**, after Russian psychologist Ivan Pavlov, who first described it. Pavlov presented meat powder, an *unconditioned stimulus*, to a dog and noted that the dog responded by salivating, an *unconditioned response*. If an unrelated stimulus, such as the ringing of a bell, was



(a)



(b)



(c)

**FIGURE 26.7**

**Learning what is edible.** Associative learning is involved in predator-prey interactions. (a) A naive toad is offered a bumblebee as food. (b) The toad is stung, and (c) subsequently avoids feeding on bumblebees or any other insects having their black-and-yellow coloration. The toad has associated the appearance of the insect with pain, and modifies its behavior.

presented at the same time as the meat powder, over repeated trials the dog would salivate in response to the sound of the bell *alone*. The dog had learned to associate the unrelated sound stimulus with the meat powder stimulus. Its response to the sound stimulus was, therefore, conditioned, and the sound of the bell is referred to as a *conditioned stimulus*.

## Operant Conditioning

In operant conditioning, an animal learns to associate its behavioral response with a reward or punishment. American psychologist B. F. Skinner studied operant conditioning in rats by placing them in an apparatus that came to be called a “Skinner box.” As the rat explored the box, it would occasionally press a lever by accident, causing a pellet of food to appear. At first, the rat would ignore the lever, eat the food pellet, and continue to move about. Soon, however, it learned to associate pressing the lever (the behavioral response) with obtaining food (the reward). When it was hungry, it would spend all its time pressing the lever. This sort of trial-and-error learning is of major importance to most vertebrates.

Comparative psychologists used to believe that any two stimuli could be linked in classical conditioning and that animals could be conditioned to perform any learnable behavior in response to any stimulus by operant conditioning. As you will see below, this view has changed. Today, it is thought that instinct guides learning by determining what type of information can be learned through conditioning.

## Instinct

It is now clear that some animals have innate predispositions toward forming certain associations. For example, if a rat is offered a food pellet at the same time it is exposed to X rays (which *later* produces nausea), the rat will remember the taste of the food pellet but not its size. Conversely, if a rat is given a food pellet at the same time an electric shock is delivered (which *immediately* causes pain), the rat will remember the size of the pellet but not its taste. Similarly, pigeons can learn to associate *food* with colors but not with sounds; on the other hand, they can associate *danger* with sounds but not with colors.

These examples of **learning preparedness** demonstrate that what an animal can learn is biologically influenced—that is, learning is possible only within the boundaries set by instinct. Innate programs have evolved because they underscore adaptive responses. Rats, which forage at night and have a highly developed sense of smell, are better able to identify dangerous food by its odor than by its size or color. The seed a pigeon eats may have a distinctive color that the pigeon can see, but it makes no sound the pigeon can hear. The study of learn-



**FIGURE 26.8**

**The Clark's nutcracker has an extraordinary memory.** A Clark's nutcracker can remember the locations of up to 2000 seed caches months after hiding them. After conducting experiments, scientists have concluded that the birds use features of the landscape and other surrounding objects as spatial references to memorize the locations of the caches.

ing has expanded to include its ecological significance, so that we are now able to consider the “evolution of learning.” An animal's ecology, of course, is key to understanding what an animal is capable of learning. Some species of birds, like Clark's nutcracker, feed on seeds. Birds store seeds in caches they bury when seeds are abundant so they will have food during the winter. Thousands of seed caches may be buried and then later recovered. One would expect the birds to have an extraordinary spatial memory, and this is indeed what has been found (figure 26.8). Clark's nutcracker, and other seed-hoarding birds, have an unusually large hippocampus, the center for memory storage in the brain (see chapter 54).

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Habituation and sensitization are simple forms of learning in which there is no association between stimuli and responses. In contrast, associative learning (classical and operant conditioning) involves the formation of an association between two stimuli or between a stimulus and a response.

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## The Development of Behavior

Behavioral biologists now recognize that behavior has both genetic and learned components, and the schools of ethology and psychology are less polarized than they once were. Thus far in this chapter we have discussed the influence of genes and learning separately. As we will see, these factors interact during development to shape behavior.

### Parent-Offspring Interactions

As an animal matures, it may form social attachments to other individuals or form preferences that will influence behavior later in life. This process, called **imprinting**, is sometimes considered a type of learning. In **filial imprinting**, social attachments form between parents and offspring. For example, young birds of some species begin to follow their mother within a few hours after hatching, and their following response results in a bond between mother and young. However, the young birds' initial experience determines how this imprint is established. The German ethologist Konrad Lorenz showed that birds will follow the first object they see after hatching and direct their social behavior toward that object. Lorenz raised geese from eggs, and when he offered himself as a model for imprinting, the goslings treated him as if he were their parent, following him dutifully (figure 26.9). Black boxes, flashing lights, and watering cans can also be effective imprinting objects (figure 26.10). Imprinting occurs during a **sensitive phase**, or a **critical period** (roughly 13 to 16 hours after hatching in geese), when the success of imprinting is highest.

Several studies demonstrate that the social interactions that occur between parents and offspring during the critical period are key to the normal development of behavior. The psychologist Harry Harlow gave orphaned rhesus monkey infants the opportunity to form social attachments with two surrogate “mothers,” one made of soft cloth covering a wire frame and the other made only of wire. The infants chose to spend time with the cloth mother, even if only the wire mother provided food, indicating that texture and tactile contact, rather than providing food, may be among the key qualities in a mother that promote infant social attachment. If infants are deprived of normal social contact, their development is abnormal. Greater degrees of deprivation lead to greater abnormalities in social behavior during



**FIGURE 26.9**

**An unlikely parent.** The eager goslings following Konrad Lorenz think he is their mother. He is the first object they saw when they hatched, and they have used him as a model for imprinting.



(a)



(b)

**FIGURE 26.10**

**How imprinting is studied.** Ducklings will imprint on the first object they see, even (a) a black box or (b) a white sphere.

childhood and adulthood. Studies on orphaned human infants suggest that a constant “mother figure” is required for normal growth and psychological development.

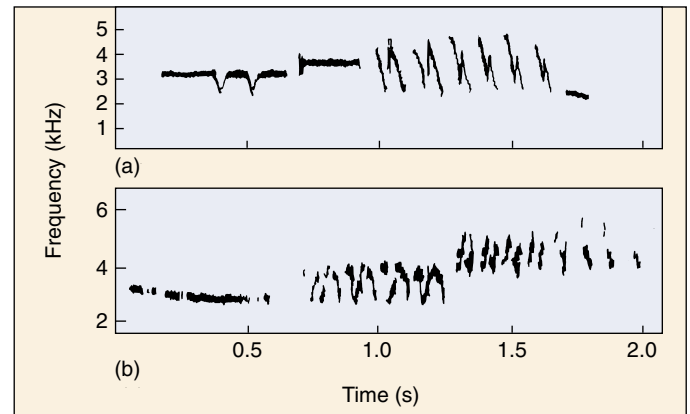
Recent research has revealed a biological need for the stimulation that occurs during parent-offspring interactions early in life. Female rats lick their pups after birth, and this stimulation inhibits the release of an endorphin (see chapter 56) that can block normal growth. Pups that receive normal tactile stimulation also have more brain receptors for glucocorticoid hormones, longer-lived brain neurons, and a greater tolerance for stress. Premature human infants who are massaged gain weight rapidly. These studies indicate that the need for normal social interaction is based in the brain and that touch and other aspects of contact between parents and offspring are important for physical as well as behavioral development.

**Sexual imprinting** is a process in which an individual learns to direct its sexual behavior at members of its own species. **Cross-fostering** studies, in which individuals of one species are raised by parents of another species, reveal that this form of imprinting also occurs early in life. In most species of birds, these studies have shown that the fostered bird will attempt to mate with members of its foster species when it is sexually mature.

## Interaction between Instinct and Learning

The work of Peter Marler and his colleagues on the acquisition of courtship song by white-crowned sparrows provides an excellent example of the interaction between instinct and learning in the development of behavior. Courtship songs are sung by mature males and are species-specific. By rearing male birds in soundproof incubators provided with speakers and microphones, Marler could control what a bird heard as it matured and record the song it produced as an adult. He found that white-crowned sparrows that heard no song at all during development, or that heard only the song of a different species, the song sparrow, sang a poorly developed song as adults (figure 26.11). But birds that heard the song of their own species, or that heard the songs of *both* the white-crowned sparrow and the song sparrow, sang a fully developed, white-crowned sparrow song as adults. These results suggest that these birds have a **genetic template**, or instinctive program, that guides them to learn the appropriate song. During a critical period in development, the template will accept the correct song as a model. Thus, song acquisition depends on learning, but only the song of the correct species can be learned. The genetic template for learning is *selective*. However, learning plays a prominent role as well. If a young white-crowned sparrow is surgically deafened *after* it hears its species' song during the critical period, it will also sing a poorly developed song as an adult. Therefore, the bird must “practice” listening to himself sing, matching what he hears to the model his template has accepted.

Although this explanation of song development stood unchallenged for many years, recent research has shown that white-crowned sparrow males *can* learn another species' song under certain conditions. If a live male strawberry finch is placed in a cage next to a young male sparrow, the young sparrow will learn to sing the strawberry finch's song! This finding indicates that social stimuli may be more effective than a tape-recorded song in overriding the innate program that guides song development. Furthermore, the males of some bird species have no opportunity to hear the song of their own species. In such cases, it appears that the males instinctively “know” their own species' song. For example, cuckoos are brood parasites; females lay their eggs in the nest of another species of bird, and the young that hatch are reared by the foster parents (figure 26.12). When the cuckoos become adults, they sing the song of their own species rather than that of their foster parents. Because male brood parasites would most likely hear the song of their host species during development, it is adaptive for them to ignore such “incorrect” stimuli. They hear no adult males of their own species singing, so no correct song models are available. In these species, natural selection has programmed the male with a genetically guided song.



**FIGURE 26.11**

**Song development in birds.** (a) The sonograms of songs produced by male white-crowned sparrows that had been exposed to their own species' song during development are different from (b) those of male sparrows that heard no song during rearing. This difference indicates that the genetic program itself is insufficient to produce a normal song.



**FIGURE 26.12**

**Brood parasite.** Cuckoos lay their eggs in the nests of other species of birds. Because the young cuckoos (large bird to the right) are raised by a different species (like this meadow pipit, smaller bird to the left), they have no opportunity to *learn* the cuckoo song; the cuckoo song they later sing is innate.

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Interactions that occur during sensitive phases of imprinting are critical to normal behavioral development. Physical contact plays an important role in the development of psychological well-being and growth.

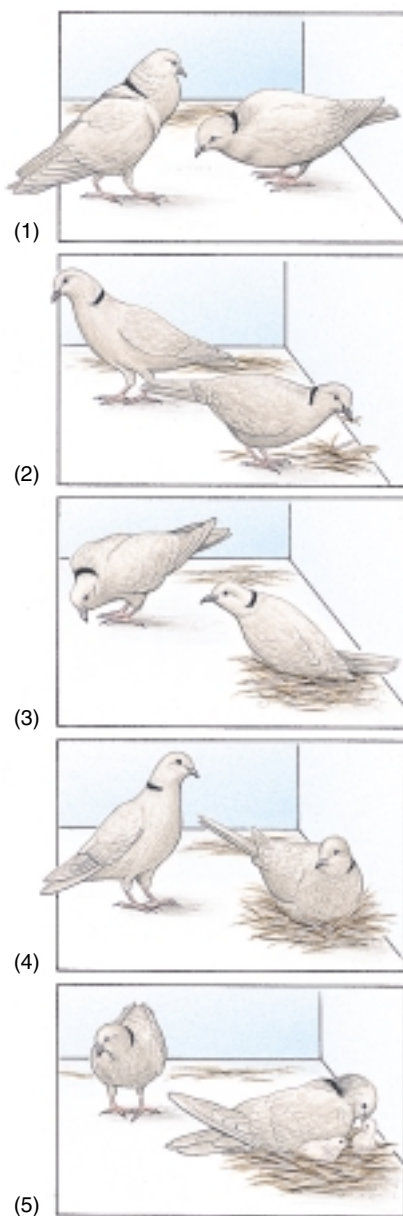
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## The Physiology of Behavior

The early ethologists' emphasis on instinct sometimes overlooked the internal factors that control behavior. If asked why a male bird defends a territory and sings only during the breeding season, they would answer that a bird sings when it is in the right *motivational state* or *mood* and has the appropriate *drive*. But what do these phrases mean in terms of physiological control mechanisms?

Part of our understanding of the physiological control of behavior has come from the study of reproductive behavior. Animals show reproductive behaviors such as courtship only during the breeding season. Research on lizards, birds, rats, and other animals has revealed that hormones play an important role in these behaviors. Changes in day length trigger the secretion of gonadotropin-releasing hormone by the hypothalamus, which stimulates the release of the gonadotropins, follicle-stimulating hormone (FSH) and luteinizing hormone, by the anterior pituitary gland. These hormones cause the development of reproductive tissues to ready the animal for breeding. The gonadotropins, in turn, stimulate the secretion of the steroid sex hormones, estrogens and progesterone in females and testosterone in males. The sex hormones act on the brain to trigger behaviors associated with reproduction. For example, birdsong and territorial behavior depend upon the level of testosterone in the male, and the receptivity of females to male courtship depends upon estrogen levels.

Hormones have both organizational and activational effects. In the example of birdsong given above, estrogen in the male causes the development of the song system, which is composed of neural tissue in the forebrain and its connections to the spinal cord and the syrinx (a structure like our larynx that allows the bird to sing). Early in a male's development, the gonads produce estrogen, which stimulates neuron growth in the brain. In the mature male, the testes produce testosterone, which activates song. Thus, the development of the neural systems that are responsible for behavior is first organized, then activated by hormones.



**FIGURE 26.13**  
**Hormonal control of reproductive behavior.** Reproduction in the ring dove involves a sequence of behaviors regulated by hormones: (1) courtship and copulation; (2) nest building; (3) egg laying; (4) incubation; and (5) feeding crop milk to the young after they hatch.

Research on the physiology of reproductive behavior shows that there are important interactions among hormones, behavior, and stimuli in both the physical and social environments of an individual. Daniel Lehrman's work on reproduction in ring doves provides an excellent example of how these factors interact (figure 26.13). Male courtship behavior is controlled by testosterone and related steroid hormones. The male's behavior causes the release of FSH in the female, and FSH promotes the growth of the ovarian follicles (see chapter 59). The developing follicles release estrogens, which affect other reproductive tissues. Nest construction follows after one or two days. The presence of the nest then triggers the secretion of progesterone in the female, initiating incubation behavior after the egg is laid. Feeding occurs once the eggs hatch, and this behavior is also hormonally controlled.

The research of Lehrman and his colleagues paved the way for many additional investigations in **behavioral endocrinology**, the study of the hormonal regulation of behavior. For example, male *Anolis* lizards begin courtship after a seasonal rise in temperature, and the male's courtship is needed to stimulate the growth of ovarian follicles in the female. These and other studies demonstrate the interactive effects of the physical environment (for example, temperature and day length) and the social environment (such as the presence of a nest and the courtship display of a mate) on the hormonal condition of an animal. Hormones are, therefore, a proximate cause of behavior. To control reproductive behavior, they must be released when the conditions are most favorable for the growth of young. Other behaviors, such as territoriality and dominance behavior, also have hormonal correlates.

Hormones may interact with neurotransmitters to alter behavior. Estrogen affects the neurotransmitter serotonin in female mice, and may be in part responsible for the "mood swings" experienced by some human females during the menstrual cycle.

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**Hormones have important influences on reproductive and social behavior.**

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## Behavioral Rhythms

Many animals exhibit behaviors that vary at regular intervals of time. Geese migrate south in the fall, birds sing in the early morning, bats fly at night rather than during the day, and most humans sleep at night and are active in the daytime. Some behaviors are timed to occur in concert with lunar or tidal cycles (figure 26.14). Why do regular repeating patterns of behavior occur, and what determines when they occur? The study of questions like these has revealed that rhythmic animal behaviors are based on both **endogenous** (internal) **rhythms** and **exogenous** (external) **timers**.

Most studies of behavioral rhythms have focused on behaviors that appear to be keyed to a daily cycle, such as sleeping. Rhythms with a period of about 24 hours are called **circadian** (“about a day”) **rhythms**. Many of these behaviors have a strong endogenous component, as if they were driven by a **biological clock**. Such behaviors are said to be *free-running*, continuing on a regular cycle even in the absence of any cues from the environment. Almost all fruit fly pupae hatch in the early morning, for example, even if they are kept in total darkness throughout their week-long development. They keep track of time with an internal clock whose pattern is determined by a single gene, called the *per* (for *period*) gene. Different mutations of the *per* gene shorten or lengthen the daily rhythm. The *per* gene produces a protein in a regular 24-hour cycle in the brain, serving as the fly’s pacemaker of activity. The protein probably affects the expression of other genes that ultimately regulate activity. As the *per* protein accumulates, it seems to turn off the gene. In mice, the *clock* gene is responsible for regulating the animal’s daily rhythm.

Most biological clocks do not exactly match the rhythms of the environment. Therefore, the behavioral rhythm of an individual deprived of external cues gradually drifts out of phase with the environment. Exposure to an environmental cue resets the biological clock and keeps the behavior properly synchronized with the environment. Light is the most common cue for resetting circadian rhythms.

The most obvious circadian rhythm in humans is the sleep-activity cycle. In controlled experiments, humans have lived for months in underground apartments, where all light is artificial and there are no external cues whatsoever indicating day length. Left to set their own schedules, most of these people adopt daily activity patterns (one phase of activity plus one phase of sleep) of about 25 hours, although there is considerable variation. Some individuals exhibit 50-hour clocks, active for as long as 36 hours during each period! Under normal circumstances, the day-night cycle resets an individual’s free-running clock every day to a cycle period of 24 hours.

What constitutes an animal’s biological clock? In some insects, the clock is thought to be located in the optic lobes of the brain, and timekeeping appears to be based on hormones. In mammals, including humans, the biological



**FIGURE 26.14**

**Tidal rhythm.** Oysters open their shells for feeding when the tide is in and close them when the tide is out.

clock lies in a specific region of the hypothalamus called the **suprachiasmatic nucleus (SCN)**. The SCN is a self-sustaining oscillator, which means it undergoes spontaneous, cyclical changes in activity. This oscillatory activity helps the SCN to act as a pacemaker for circadian rhythms, but in order for the rhythms to be entrained to external light-dark cycles, the SCN must be influenced by light. In fact, there are both direct and indirect neural projections from the retina to the SCN.

The SCN controls circadian rhythms by regulating the secretion of the hormone **melatonin** by the **pineal gland**. During the daytime, the SCN suppresses melatonin secretion. Consequently, more melatonin is secreted over a 24-hour period during short days than during long days. Variations in melatonin secretion thus serve as an indicator of seasonal changes in day length, and these variations participate in timing the seasonal reproductive behavior of many mammals. Disturbances in melatonin secretion may be partially responsible for the “jet-lag” people experience when air travel suddenly throws their internal clocks out of register with the day-night cycle.

Many important behavioral rhythms have cycle periods longer than 24 hours. For example, **circannual behaviors** such as breeding, hibernation, and migration occur on a yearly cycle. These behaviors seem to be largely timed by hormonal and other physiological changes keyed to exogenous factors such as day length. The degree to which endogenous biological clocks underlie circannual rhythms is not known, as it is very difficult to perform constant-environment experiments of several years’ duration. The mechanism of the biological clock remains one of the most tantalizing puzzles in biology today.

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**Endogenous circadian rhythms have free-running cycle periods of approximately 24 hours; they are entrained to a more exact 24-hour cycle period by environmental cues.**

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## 26.3 Communication is a key element of many animal behaviors.

Much of the research in animal behavior is devoted to analyzing the nature of communication signals, determining how they are perceived, and identifying the ecological roles they play and their evolutionary origins.

### Courtship

During courtship, animals produce signals to communicate with potential mates and with other members of their own sex. A **stimulus-response chain** sometimes occurs, in which the behavior of one individual in turn releases a behavior by another individual (figure 26.15).

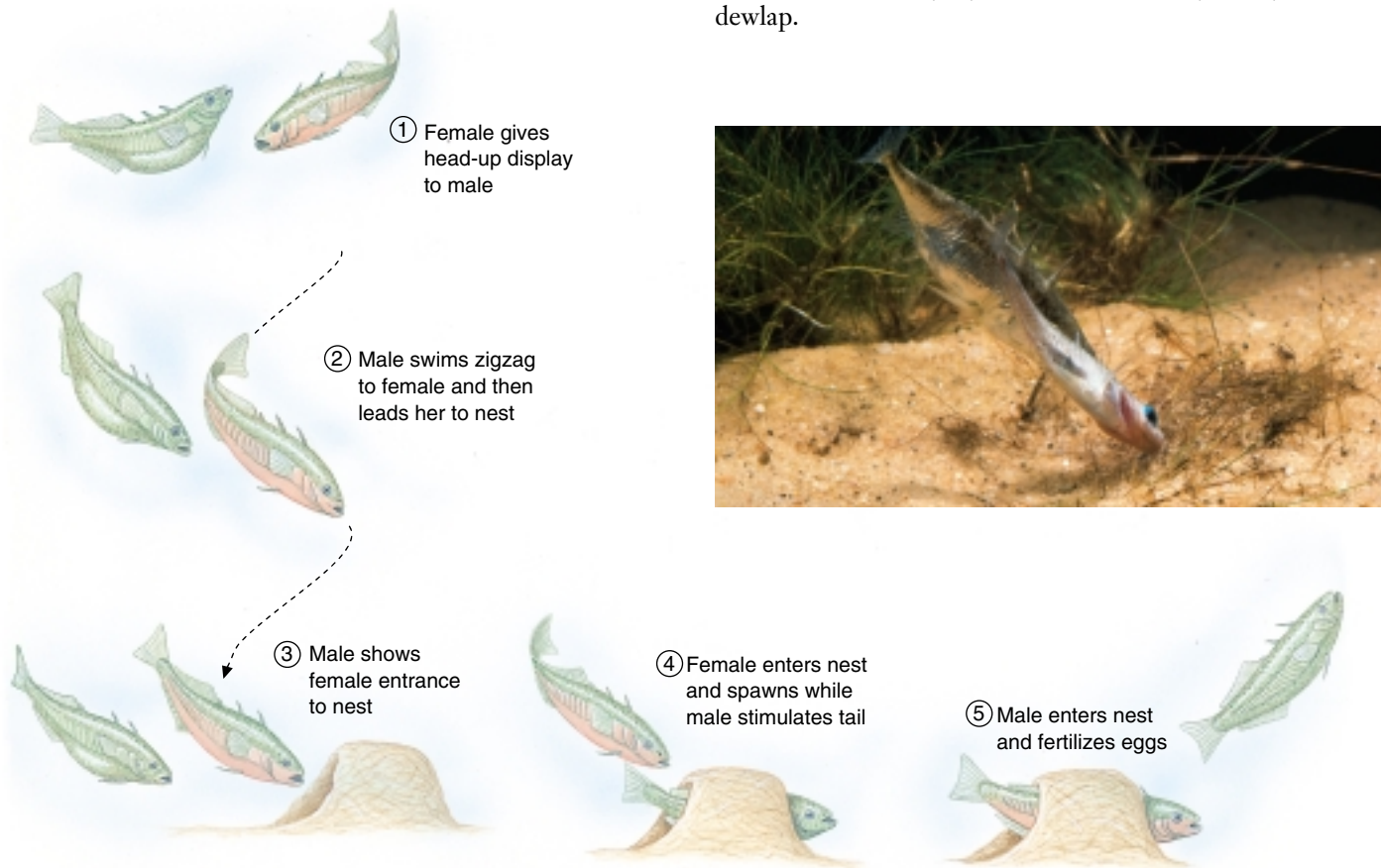
#### Courtship Signaling

A male stickleback fish will defend the nest it builds on the bottom of a pond or stream by attacking *conspecific* males (that is, males of the same species) that approach the nest. Niko Tinbergen studied the social releasers responsible for this behavior by making simple clay models. He found that a model's shape and degree of resemblance to a fish were

unimportant; any model with a red underside (like the underside of a male stickleback) could release the attack behavior. Tinbergen also used a series of clay models to demonstrate that a male stickleback recognizes a female by her abdomen, swollen with eggs.

Courtship signals are often *species-specific*, limiting communication to members of the same species and thus playing a key role in reproductive isolation. The flashes of fireflies (which are actually beetles) are such species-specific signals. Females recognize conspecific males by their flash pattern (figure 26.16), and males recognize conspecific females by their flash response. This series of reciprocal responses provides a continuous “check” on the species identity of potential mates.

Visual courtship displays sometimes have more than one component. The male *Anolis* lizard extends and retracts his fleshy and often colorful dewlap while perched on a branch in his territory (figure 26.17). The display thus involves both color and movement (the extension of the dewlap as well as a series of lizard “push-ups”). To which component of the display does the female respond? Experiments in which the dewlap color is altered with ink show that color is unimportant for some species; that is, a female can be courted successfully by a male with an atypically colored dewlap.



**FIGURE 26.15**  
A stimulus-response chain. Stickleback courtship involves a sequence of behaviors leading to the fertilization of eggs.

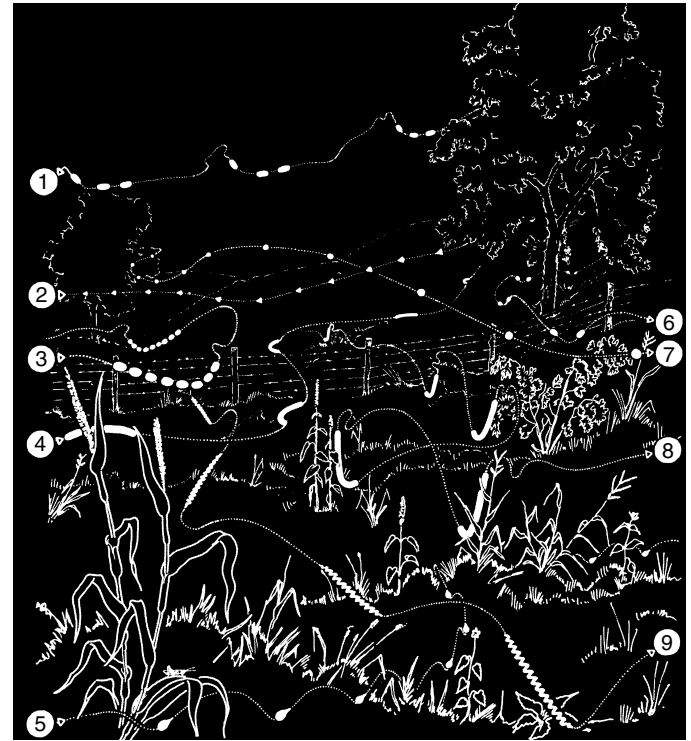
## Pheromones

Chemical signals also mediate interactions between males and females. **Pheromones**, chemical messengers used for communication between individuals of the same species, serve as sex attractants among other functions in many animals. Even the human egg produces a chemical attractant to communicate with sperm! Female silk moths (*Bombyx mori*) produce a sex pheromone called *bombykol* in a gland associated with the reproductive system. Neurophysiological studies show that the male's antennae contain numerous sensory receptors specific for bombykol. These receptors are extraordinarily sensitive, enabling the male to respond behaviorally to concentrations of bombykol as low as one molecule in  $10^{17}$  molecules of oxygen in the air!

Many insects, amphibians, and birds produce species-specific acoustic signals to attract mates. Bullfrog males call to females by inflating and discharging air from their vocal sacs, located beneath the lower jaw. The female can distinguish a conspecific male's call from the call of other frogs that may be in the same habitat and mating at the same time. Male birds produce songs, complex sounds composed of notes and phrases, to advertise their presence and to attract females. In many bird species, variations in the males' songs identify *particular* males in a population. In these species, the song is individually specific as well as species-specific.

## Level of Specificity

Why should different signals have different levels of specificity? The **level of specificity** relates to the function of the signal. Many courtship signals are species-specific to help animals avoid making errors in mating that would produce inviable hybrids or otherwise waste reproductive effort. A male bird's song is individually specific because it allows his presence (as opposed to simply the presence of an unidentifiable member of the species) to be recognized by neighboring birds. When territories are being established, males may sing and aggressively confront neighboring conspecifics to defend their space. Aggression carries the risk of injury, and it is energetically costly to sing. After territorial borders have been established, intrusions by neighbors are few because the outcome of the contests have already been determined. Each male then "knows" his neighbor by the song he sings, and also "knows" that male does not constitute a threat because they have already settled their territorial contests. So, all birds in the population can lower their energy costs by identifying their neighbors through their individualistic songs. In a similar way, mammals mark their territories with pheromones that signal individual identity, which may be encoded as a blend of a number of chemicals. Other signals, such as the mobbing and alarm calls of birds, are anonymous, conveying no information about the identity of the sender. These signals may permit communication about the presence of a predator common to several bird species.



**FIGURE 26.16**

**Firefly fireworks.** The bioluminescent displays of these lampyrid beetles are species-specific and serve as behavioral mechanisms of reproductive isolation. Each number represents the flash pattern of a male of a different species.



**FIGURE 26.17**

**Dewlap display of a male *Anolis* lizard.** Under hormonal stimulation, males extend their fleshy, colored dewlaps to court females. This behavior also stimulates hormone release and egg-laying in the female.

**Courtship behaviors are keyed to species-specific visual, chemical, and acoustic signals.**

## Communication in Social Groups

Many insects, fish, birds, and mammals live in social groups in which information is communicated between group members. For example, some individuals in mammalian societies serve as “guards.” When a predator appears, the guards give an **alarm call**, and group members respond by seeking shelter. Social insects, such as ants and honeybees, produce **alarm pheromones** that trigger attack behavior. Ants also deposit **trail pheromones** between the nest and a food source to induce cooperation during foraging (figure 26.18). Honeybees have an extremely complex **dance language** that directs nestmates to rich nectar sources.

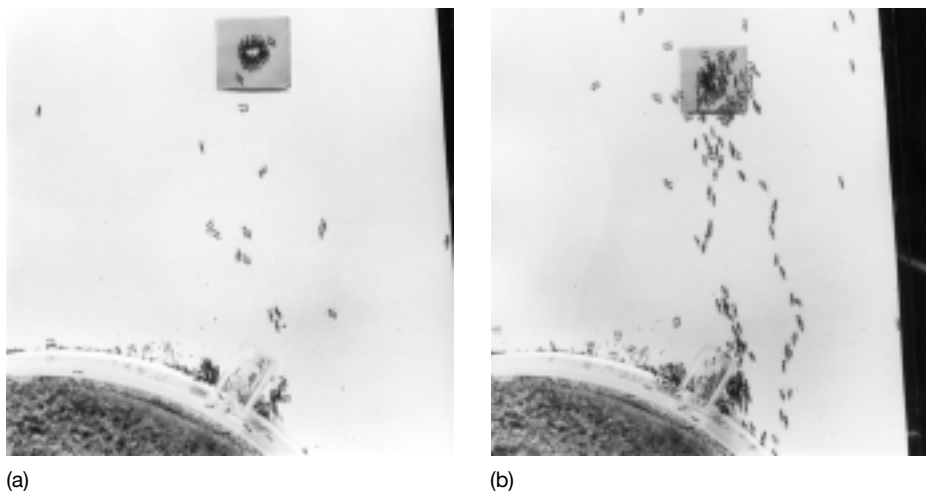


FIGURE 26.18

**The chemical control of fire ant foraging.** Trail pheromones, produced in an accessory gland near the fire ant’s sting, organize cooperative foraging. The trails taken by the first ants to travel to a food source (a) are soon followed by most of the other ants (b).

### The Dance Language of the Honeybee

The European honeybee, *Apis mellifera*, lives in hives consisting of 30,000 to 40,000 individuals whose behaviors are integrated into a complex colony. Worker bees may forage for miles from the hive, collecting nectar and pollen from a variety of plants and switching between plant species and populations on the basis of how energetically rewarding their food is. The food sources used by bees tend to occur in patches, and each patch offers much more food than a single bee can transport to the hive. A colony is able to exploit the resources of a patch because of the behavior of scout bees, which locate patches and communicate their location to hivemates through a *dance language*. Over many years, Nobel laureate Karl von Frisch was able to unravel the details of this communication system.

After a successful scout bee returns to the hive, she performs a remarkable behavior pattern called a *waggle dance* on a vertical comb (figure 26.19). The path of the bee during the dance resembles a figure-eight. On the straight part of the path, the bee vibrates or waggles her abdomen while producing bursts of sound. She may stop periodically to give her hivemates a sample of the nectar she has carried back to the hive in her crop. As she dances, she is followed closely by other bees, which soon appear as foragers at the new food source.

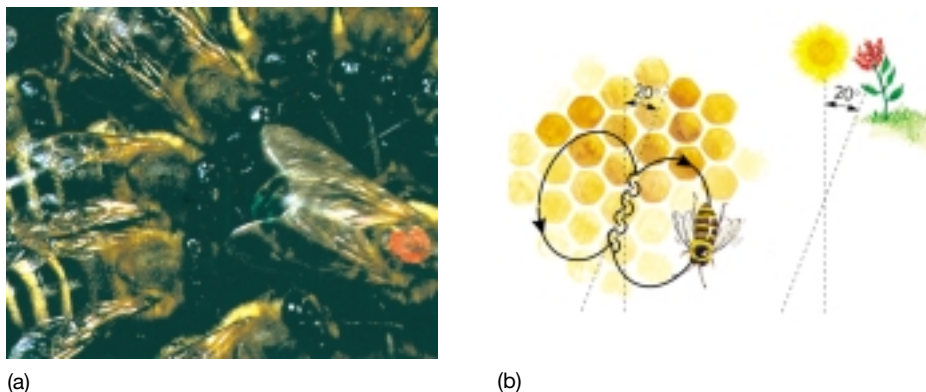


FIGURE 26.19

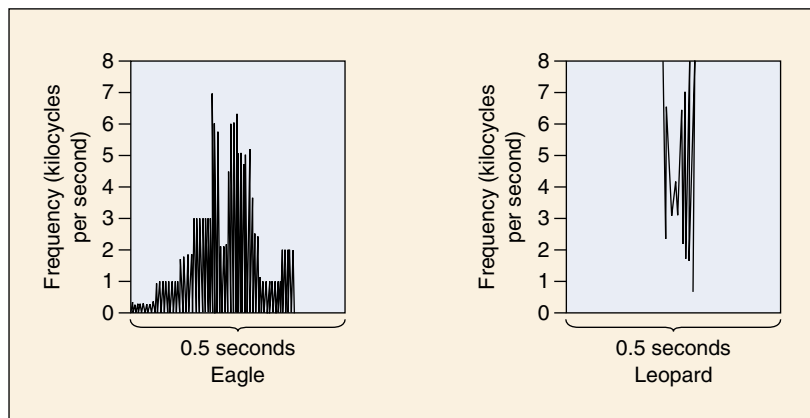
**The waggle dance of honeybees.** (a) A scout bee dances on a comb in the hive. (b) The angle between the food source and the nest is represented by a dancing bee as the angle between the straight part of the dance and vertical. The food is 20° to the right of the sun, and the straight part of the bee’s dance on the hive is 20° to the right of vertical.

Von Frisch and his colleagues claimed that the other bees use information in the waggle dance to locate the food source. According to their explanation, the scout bee indicates the *direction* of the food source by representing the angle between the food source and the hive in reference to the sun as the angle between the straight part of the dance and vertical in the hive. The *distance* to the food source is indicated by the tempo, or degree of vigor, of the dance.

Adrian Wenner, a scientist at the University of California, did not believe that the dance language communicated anything about the location of food, and he challenged von Frisch’s explanation. Wenner maintained that flower odor was the most important cue allowing recruited bees to arrive at a new food source. A heated controversy ensued as



(a)



(b)

**FIGURE 26.20**

**Primate semantics.** (a) Predators, like this leopard, attack and feed on vervet monkeys. (b) The monkeys give different alarm calls when eagles, leopards, and snakes are sighted by troupe members. Each distinctive call elicits a different and adaptive escape behavior.

the two groups of researchers published articles supporting their positions.

The “dance language controversy” was resolved (in the minds of most scientists) in the mid-1970s by the creative research of James L. Gould. Gould devised an experiment in which hive members were tricked into misinterpreting the directions given by the scout bee’s dance. As a result, Gould was able to manipulate where the hive members would go if they were using visual signals. If odor were the cue they were using, hive members would have appeared at the food source, but instead they appeared exactly where Gould predicted. This confirmed von Frisch’s ideas.

Recently, researchers have extended the study of the honeybee dance language by building robot bees whose dances can be completely controlled. Their dances are programmed by a computer and perfectly reproduce the natural honeybee dance; the robots even stop to give food samples! The use of robot bees has allowed scientists to determine precisely which cues direct hivemates to food sources.

## Primate Language

Some primates have a “vocabulary” that allows individuals to communicate the identity of specific predators. The vocalizations of African vervet monkeys, for example, distinguish eagles, leopards, and snakes (figure 26.20). Chimpanzees and gorillas can learn to recognize a large number of symbols and use them to communicate abstract concepts. The complexity of human language would at first appear to defy biological explanation, but closer examination suggests that the differences are in fact superficial—all languages share many basic structural similarities. All of the roughly 3000 languages draw from the same set of 40 con-

sonant sounds (English uses two dozen of them), and any human can learn them. Researchers believe these similarities reflect the way our brains handle abstract information, a genetically determined characteristic of all humans.

Language develops at an early age in humans. Human infants are capable of recognizing the 40 consonant sounds characteristic of speech, including those not present in the particular language they will learn, while they ignore other sounds. In contrast, individuals who have not heard certain consonant sounds as infants can only rarely distinguish or produce them as adults. That is why English speakers have difficulty mastering the throaty French “r,” French speakers typically replace the English “th” with “z,” and native Japanese often substitute “l” for the unfamiliar English “r.” Children go through a “babbling” phase, in which they learn by trial and error how to make the sounds of language. Even deaf children go through a babbling phase using sign language. Next, children quickly and easily learn a vocabulary of thousands of words. Like babbling, this phase of rapid learning seems to be genetically programmed. It is followed by a stage in which children form simple sentences which, though they may be grammatically incorrect, can convey information. Learning the rules of grammar constitutes the final step in language acquisition.

While language is the primary channel of human communication, odor and other nonverbal signals (such as “body language”) may also convey information. However, it is difficult to determine the relative importance of these other communication channels in humans.

**The study of animal communication involves analysis of the specificity of signals, their information content, and the methods used to produce and receive them.**

## 26.4 Migratory behavior presents many puzzles.

### Orientation and Migration

Animals may travel to and from a nest to feed or move regularly from one place to another. To do so, they must orient themselves by tracking stimuli in the environment.

Movement toward or away from some stimulus is called **taxis**. The attraction of flying insects to outdoor lights is an example of *positive phototaxis*. Insects that avoid light, such as the common cockroach, exhibit *negative phototaxis*. Other stimuli may be used as orienting cues. For example, trout orient themselves in a stream so as to face against the current. However, not all responses involve a specific orientation. Some animals just become more active when stimulus intensity increases, a responses called **kineses**.

Long-range, two-way movements are known as **migrations**. In many animals, migrations occur circannually. Ducks and geese migrate along flyways from Canada across the United States each fall and return each spring. Monarch butterflies migrate each fall from central and eastern North America to several small, geographically isolated areas of coniferous forest in the mountains of central Mexico (figure 26.21). Each August, the butterflies begin a flight southward to their overwintering sites. At the end of winter, the monarchs begin the return flight to their summer breeding ranges. What is amazing about the migration of the monarch, however, is that two to five generations may be produced as the butterflies fly north. The butterflies that migrate in the autumn to the precisely located overwintering grounds in Mexico have never been there before.

When colonies of bobolinks became established in the western United States, far from their normal range in the Midwest and East, they did not migrate directly to their winter range in South America. Instead, they migrated east to their ancestral range and then south along the original flyway (figure 26.22). Rather than changing the original migration pattern, they simply added a new pattern.

### How Migrating Animals Navigate

Biologists have studied migration with great interest, and we now have a good understanding of how these feats of navigation are achieved. It is important to understand the distinction between **orientation** (the ability to follow a bearing) and **navigation** (the ability to set or adjust a bearing, and then follow it). The former is analogous to using a compass, while the latter is like using a compass in conjunction with a map. Experiments on starlings indicate that inexperienced birds migrate by orientation, while older birds that have migrated previously use true navigation (figure 26.23).

Birds and other animals navigate by looking at the sun and the stars. The indigo bunting, which flies during the day and uses the sun as a guide, compensates for the movement of the sun in the sky as the day progresses by reference to the north star, which does not move in the sky. Buntings also use the positions of the constellations and the position of the pole star in the night sky, cues they learn as young birds. Starlings and certain other birds compensate for the sun's apparent movement in the

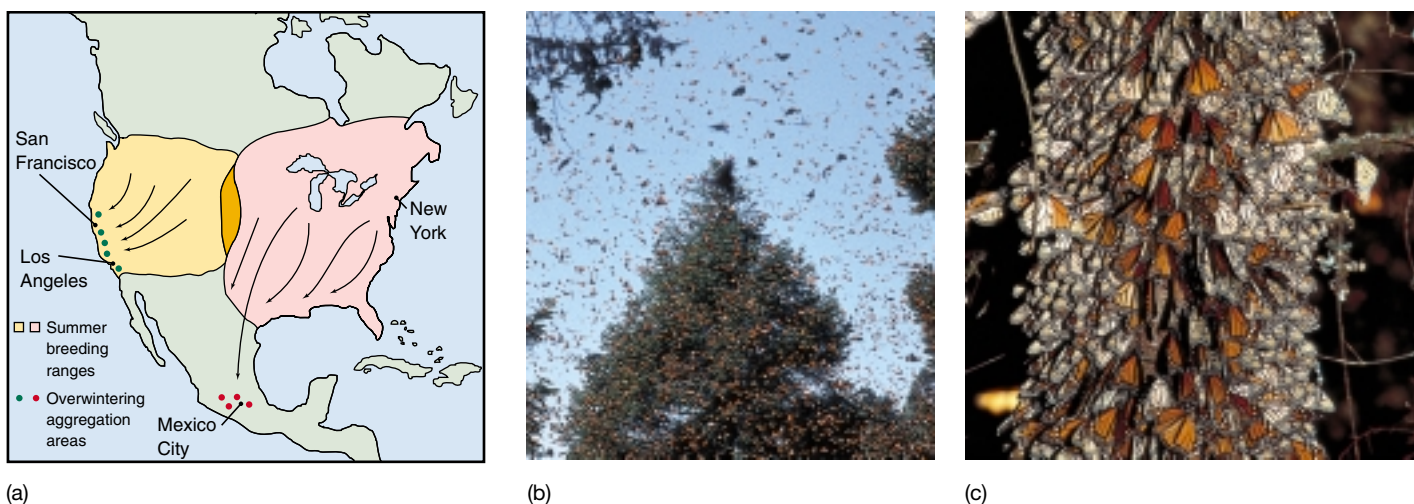
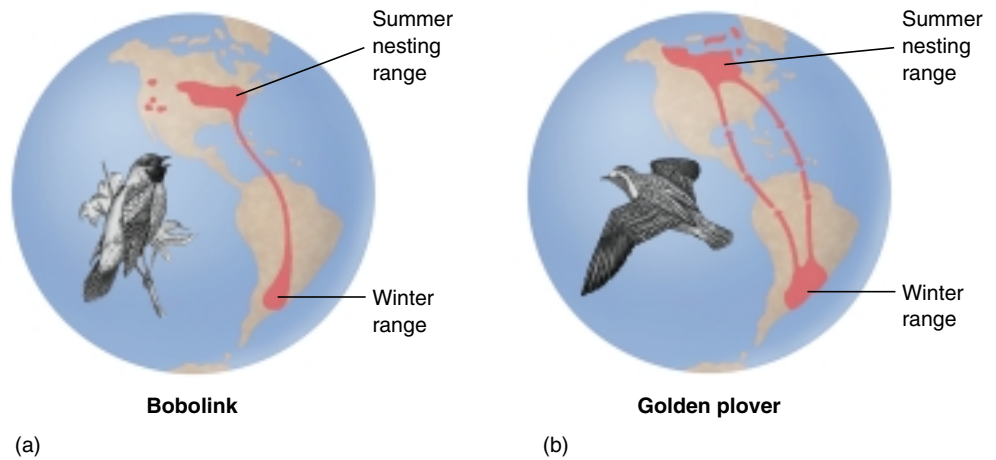


FIGURE 26.21

**Migration of monarch butterflies.** (a) Monarchs from western North America overwinter in areas of mild climate along the Pacific Coast. Those from the eastern United States and southeastern Canada migrate to Mexico, a journey of over 3000 kilometers that takes from two to five generations to complete. (b) Monarch butterflies arriving at the remote fir forests of the overwintering grounds and (c) forming aggregations on the tree trunks.

**FIGURE 26.22**

**Birds on the move.** (a) The summer range of bobolinks recently extended to the far West from their more established range in the Midwest. When they migrate to South America in the winter, bobolinks that nested in the West do not fly directly to the winter range; instead, they fly to the Midwest first and then use the ancestral flyway. (b) The golden plover has an even longer migration route that is circular. These birds fly from Arctic breeding grounds to wintering areas in southeastern South America, a distance of some 13,000 kilometers.

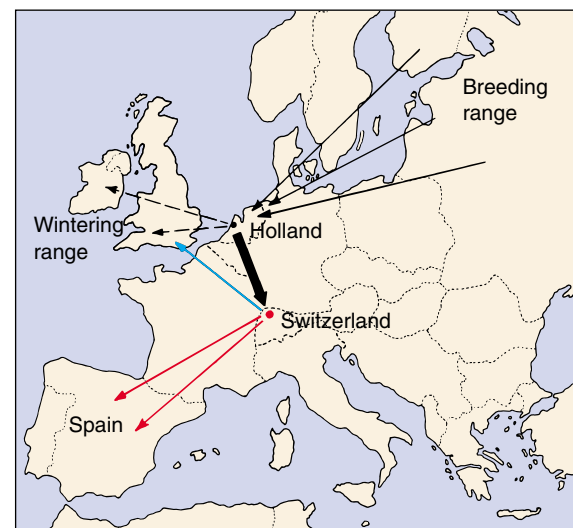


sky by using an internal clock. If such birds are shown an experimental sun in a fixed position while in captivity, they will change their orientation to it at a constant rate of about  $15^\circ$  per hour.

Many migrating birds also have the ability to detect the earth's magnetic field and to orient themselves with respect to it. In a closed indoor cage, they will attempt to move in the correct geographical direction, even though there are no visible external cues. However, the placement of a powerful magnet near the cage can alter the direction in which the birds attempt to move. Magnetite, a magnetized iron ore, has been found in the heads of some birds, but the sensory receptors birds employ to detect magnetic fields have not been identified.

It appears that the first migration of a bird is innately guided by both celestial cues (the birds fly mainly at night) and the earth's magnetic field. These cues give the same information about the general direction of the migration, but the information about direction provided by the stars seems to dominate over the magnetic information when the two cues are experimentally manipulated to give conflicting directions. Recent studies, however, indicate that celestial cues tell northern hemisphere birds to move south when they begin their migration, while magnetic cues give them the direction for the specific migratory path (perhaps a southeast turn the bird must make midroute). In short, these new data suggest that celestial and magnetic cues interact during development to fine-tune the bird's navigation.

We know relatively little about how other migrating animals navigate. For instance, green sea turtles migrate from Brazil halfway across the Atlantic Ocean to Ascension Island, where the females lay their eggs. How do they find this tiny island in the middle of the ocean, which they haven't seen for perhaps 30 years? How do the young that hatch on the island know how to find their way to Brazil? Recent studies suggest that wave action is an important cue.

**FIGURE 26.23**

**Migratory behavior of starlings.** The navigational abilities of inexperienced birds differ from those of adults who have made the migratory journey before. Starlings were captured in Holland, halfway along their full migratory route from Baltic breeding grounds to wintering grounds in the British Isles; these birds were transported to Switzerland and released. Experienced older birds compensated for the displacement and flew toward the normal wintering grounds (*blue arrow*). Inexperienced young birds kept flying in the same direction, on a course that took them toward Spain (*red arrow*). These observations imply that inexperienced birds fly by orientation, while experienced birds learn true navigation.

Many animals migrate in predictable ways, navigating by looking at the sun and stars, and in some cases by detecting magnetic fields.

## 26.5 To what degree animals “think” is a subject of lively dispute.

### Animal Cognition

It is likely each of us could tell an anecdotal story about the behavior of a pet cat or dog that would seem to suggest that the animal had a degree of reasoning ability or was capable of thinking. For many decades, however, students of animal behavior flatly rejected the notion that nonhuman animals can think. In fact, behaviorist Lloyd Morgan stated that one should never assume a behavior represents conscious thought if there is any other explanation that precludes the assumption of consciousness. The prevailing approach was to treat animals as though they responded to the environment through reflexlike behaviors.

In recent years, serious attention has been given to the topic of animal awareness. The central question is whether animals show **cognitive behavior**—that is, do they process information and respond in a manner that suggests thinking (figure 26.24)? What kinds of behavior would demonstrate cognition? Some birds in urban areas remove the foil caps from nonhomogenized milk bottles to get at the cream beneath, and this behavior is known to

have spread within a population to other birds. Japanese macaques learned to wash potatoes and float grain to separate it from sand. A chimpanzee pulls the leaves off of a tree branch and uses the stick to probe the entrance to a termite nest and gather termites. As we saw earlier, vervet monkeys have a vocabulary that identifies specific predators.

Only a few experiments have tested the thinking ability of nonhuman animals. Some of these studies suggest that animals may give false information (that is, they “lie”). Currently, researchers are trying to determine if some primates deceive others to manipulate the behavior of the other members of their troop. There are many anecdotal accounts that appear to support the idea that deception occurs in some nonhuman primate species such as baboons and chimpanzees, but it has been difficult to devise field-based experiments to test this idea. Much of this type of research on animal cognition is in its infancy, but it is sure to grow and to raise controversy. In any case, there is nothing to be gained by a dogmatic denial of the *possibility* of animal consciousness.



(a)



(b)

**FIGURE 26.24**

**Animal thinking?** (a) This chimpanzee is stripping the leaves from a twig, which it will then use to probe a termite nest. This behavior strongly suggests that the chimpanzee is consciously planning ahead, with full knowledge of what it intends to do. (b) This sea otter is using a rock as an “anvil,” against which it bashes a clam to break it open. A sea otter will often keep a favorite rock for a long time, as though it has a clear idea of what it is going to use the rock for. Behaviors such as these suggest that animals have cognitive abilities.

**FIGURE 26.25**  
**Problem solving by a chimpanzee.** Unable to get the bananas by jumping, the chimpanzee devises a solution.



In any case, some examples, particularly those involving problem-solving, are hard to explain in any way other than as a result of some sort of mental process. For example, in a series of classic experiments conducted in the 1920s, a chimpanzee was left in a room with a banana hanging from the ceiling out of reach. Also in the room were several boxes, each lying on the floor. After unsuccessful attempts to jump up and grab the bananas, the chimpanzee suddenly looks at the boxes and immediately proceeds to move them underneath the banana, stack one on top of another, and climb up to claim its prize (figure 26.25).

Perhaps it is not so surprising to find obvious intelligence in animals as closely related to us as chimpanzees. But recent studies have found that other animals also show evidence of cognition. Ravens have always been considered among the most intelligent of birds. Bernd Heinrich of the University of Vermont recently conducted an experiment using a group of hand-reared crows that lived in an outdoor aviary. Heinrich placed a piece of meat on the end of a string and hung it from a branch in the aviary. The birds liked to eat meat, but had never seen string before and were unable to get at the meat. After several hours, during which time the birds periodically looked at the meat but did nothing else, one bird flew to the branch, reached down, grabbed the string, pulled it up, and placed it under his foot. He then reached down and grabbed another piece of the string, repeating this action over and over, each time bringing the meat closer (figure 26.26). Eventually, the meat was within reach and was grasped. The raven, presented with a completely novel problem, had devised a solution. Eventually, three of the other five ravens also figured out how to get to the meat. Heinrich has conducted other similarly creative experiments that can leave little doubt that ravens have advanced cognitive abilities.



**FIGURE 26.26**  
**Problem solving by a raven.** Confronted with a problem it had never previously confronted, the raven figures out how to get the meat at the end of the string by repeatedly pulling up a bit of string and stepping on it.

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**Research on the cognitive behavior of animals is in its infancy, but some examples are compelling.**

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## Summary

## Questions

## Media Resources

## 26.1 Ethology focuses on the natural history of behavior.

- Behavior is an adaptive response to stimuli in the environment. An animal's sensory systems detect and process information about these stimuli.
- 1. How does a hybrid lovebird's method of carrying nest materials compare with that of its parents? What does this comparison suggest about whether the behavior is instinctive or learned?

## 26.2 Comparative psychology focuses on how learning influences behavior.

- Behavior is both instinctive (influenced by genes) and learned through experience. Genes are thought to limit the extent to which behavior can be modified and the types of associations that can be made.
- The simplest forms of learning involve habituation and sensitization. More complex associative learning, such as classical and operant conditioning, may be due to the strengthening or weakening of existing synapses as well as the formation of entirely new synapses.
- An animal's internal state influences when and how a behavior will occur. Hormones can change an animal's behavior and perception of stimuli in a way that facilitates reproduction.
- 2. How does associative learning differ from nonassociative learning? How does classical conditioning differ from operant conditioning?
- 3. What is filial imprinting? What is sexual imprinting? Why do some young animals imprint on objects like a moving box?
- 4. How does Marler's work on song development in white-crowned sparrows indicate that behavior is shaped by learning? How does it indicate that behavior is shaped by instinct?

## 26.3 Communication is a key element of many animal behaviors.

- Animals communicate by producing visual, acoustic, and chemical signals. These signals are involved in mating, finding food, defense against predators, and other social situations.
- 5. How do communication signals participate in reproductive isolation? Give one example of a signal that is species-specific. Why are some signals individually specific?



- On Science Article: Polyandry in Hawks

## 26.4 Migratory behavior presents many puzzles.

- Animals use cues such as the position of the sun and stars to orient during daily activities and to navigate during long-range migrations.
- 6. What is the definition of taxis? What are kinesis? What cues do migrating birds use to orient and navigate during their migrations?

## 26.5 To what degree animals "think" is a subject of lively dispute.

- Many anecdotal accounts point to animal cognition, but research is in its infancy.
- 7. What evidence would you accept that an animal is "thinking"?

# 27

## Behavioral Ecology

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### Concept Outline

#### 27.1 Evolutionary forces shape behavior.

**Behavioral Ecology.** Behavior is shaped by natural selection.

**Foraging Behavior.** Natural selection favors the most efficient foraging behavior.

**Territorial Behavior.** Animals defend territory to increase reproductive advantage and foraging efficiency.

#### 27.2 Reproductive behavior involves many choices influenced by natural selection.

**Parental Investment and Mate Choice.** The degree of parental investment strongly influences other reproductive behaviors.

**Reproductive Competition and Sexual Selection.** Mate choice affects reproductive success, and so is a target of natural selection.

**Mating Systems.** Mating systems are reproductive solutions to particular ecological challenges.

#### 27.3 There is considerable controversy about the evolution of social behavior.

**Factors Favoring Altruism and Group Living.** Many explanations have been put forward to explain the evolution of altruism.

**Examples of Kin Selection.** One explanation for altruism is that individuals can increase the extent to which their genes are passed on to the next generation by aiding their relatives.

**Group Living and the Evolution of Social Systems.** Insect societies exhibit extreme cooperation and altruism, perhaps as a result of close genetic relationship of society members.

#### 27.4 Vertebrates exhibit a broad range of social behaviors.

**Vertebrate Societies.** Many vertebrate societies exhibit altruism.

**Human Sociobiology.** Human behavior, like that of other vertebrates, is influenced by natural selection.



**FIGURE 27.1**

**A snake in the throes of death—or is it?** When threatened, many organisms feign death, as this snake is doing—foaming at the mouth and going limp or looking paralyzed.

Animal behavior can be investigated in a variety of ways. An investigator can ask, how did the behavior develop? What is the physiology behind the behavior? Or what is the function of the behavior (figure 27.1), and does it confer an advantage to the animal? The field of behavioral ecology deals with the last two questions. Specifically, *behavioral ecologists* study the ways in which behavior may be adaptive by allowing an animal to increase or even maximize its reproductive success. This chapter examines both of these aspects of behavioral ecology.

## 27.1 Evolutionary forces shape behavior.

### Behavioral Ecology

In an important essay, Nobel laureate Niko Tinbergen outlined the different types of questions biologists can ask about animal behavior. In essence, he divided the investigation of behavior into the study of its development, physiological basis, and function (evolutionary significance). One type of evolutionary analysis pioneered by Tinbergen himself was the study of the **survival value** of behavior. That is, how does an animal's behavior allow it to stay alive or keep its offspring alive? For example, Tinbergen observed that after gull nestlings hatch, the parents remove the eggshells from the nest. To understand *why* this behavior occurs, he camouflaged chicken eggs by painting them to resemble the natural background where they would lie and distributed them throughout the area in which the gulls were nesting (figure 27.2). He placed broken eggshells next to some of the eggs, and as a control, he left other camouflaged eggs alone without eggshells. He then noted which eggs were found more easily by crows. Because the crows could use the white interior of a broken eggshell as a cue, they ate more of the camouflaged eggs that were near eggshells. Thus, Tinbergen concluded that eggshell removal behavior is *adaptive*: it reduces predation and thus increases the offspring's chances of survival.

Tinbergen is credited with being one of the founders of the field of **behavioral ecology**, the study of how natural selection shapes behavior. This branch of ecology examines the **adaptive significance** of behavior, or how behavior may increase survival and reproduction. Current research in behavioral ecology focuses on the contribution behavior makes to an animal's reproductive success, or **fitness**. As we saw in chapter 26, differences in behavior among individuals often result from genetic differences. Thus, natural selection operating on behavior has the potential to produce evolutionary change. To study the relation between behavior and fitness, then, is to study the process of adaptation itself.

Consequently, the field of behavioral ecology is concerned with two questions. First, is behavior adaptive? Although it is tempting to assume that the behavior produced by individuals must in some way represent an adaptive response to the environment, this need not be the case. As we saw in chapter 20, traits can evolve for many reasons other than natural selection, such as genetic drift or gene flow. Moreover, traits may be present in a population because they evolved as adaptations in the past, but no longer are useful. These possibilities hold true for behavioral traits as much as they do for any other kind of trait.

If a trait is adaptive, the question then becomes: how is it adaptive? Although the ultimate criterion is reproductive



**FIGURE 27.2**

**The adaptive value of egg coloration.** Niko Tinbergen painted chicken eggs to resemble the mottled brown camouflage of gull eggs. The eggs were used to test the hypothesis that camouflaged eggs are more difficult for predators to find and thus increase the young's chances of survival.

success, behavioral ecologists are interested in how a trait can lead to greater reproductive success. By enhancing energy intake, thus increasing the number of offspring produced? By improving success in getting more matings? By decreasing the chance of predation? The job of a behavioral ecologist is to determine the effect of a behavioral trait on each of these activities and then to discover whether increases in, for example, foraging efficiency, translate into increased fitness.

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**Behavioral ecology is the study of how natural selection shapes behavior.**

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## Foraging Behavior

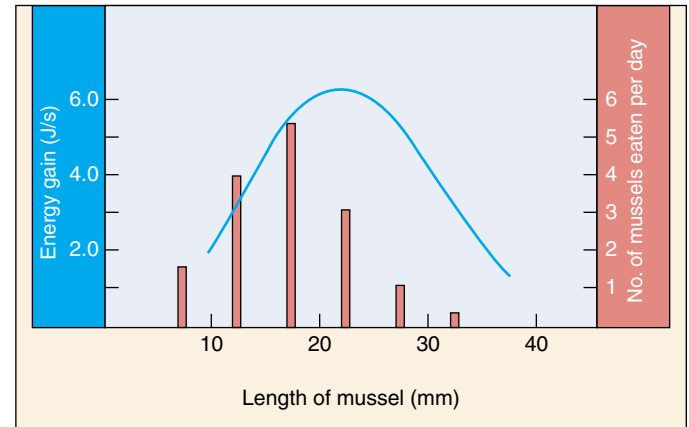
The best introduction to behavioral ecology is the examination of one well-defined behavior in detail. While many behaviors might be chosen, we will focus on foraging behavior. For many animals, food comes in a variety of sizes. Larger foods may contain more energy but may be harder to capture and less abundant. In addition, some types of food may be farther away than other types. Hence, foraging for these animals involves a trade-off between a food's energy content and the cost of obtaining it. The *net energy* (in calories or Joules) gained by feeding on each size prey is simply the energy content of the prey minus the energy costs of pursuing and handling it. According to **optimal foraging theory**, natural selection favors individuals whose foraging behavior is as energetically efficient as possible. In other words, animals tend to feed on prey that maximize their net energy intake per unit of foraging time.

A number of studies have demonstrated that foragers do preferentially utilize prey that maximize the energy return. Shore crabs, for example, tend to feed primarily on intermediate-sized mussels which provide the greatest energetic return; larger mussels provide more energy, but also take considerably more energy to crack open (figure 27.3).

This optimal foraging approach makes two assumptions. First, natural selection will only favor behavior that maximizes energy acquisition if increased energy reserves lead to increases in reproductive success. In some cases, this is true. For example, in both Columbian ground squirrels and captive zebra finches, a direct relationship exists between net energy intake and the number of offspring raised; similarly, the reproductive success of orb-weaving spiders is related to how much food they can capture.

However, animals have other needs beside energy acquisition, and sometimes these needs come in conflict. One obvious alternative is avoiding predators: oftentimes the behavior that maximizes energy intake is not the one that minimizes predation risk. Thus, the behavior that maximizes fitness often may reflect a trade-off between obtaining the most energy at the least risk of being eaten. Not surprisingly, many studies have shown that a wide variety of animal species alter their foraging behavior when predators are present. Still another alternative is finding mates: males of many species, for example, will greatly reduce their feeding rate in order to enhance their ability to attract and defend females.

The second assumption of optimal foraging theory is that it has resulted from natural selection. As we have seen, natural selection can lead to evolutionary change only when differences among individuals have a genetic basis. Few studies have investigated whether differences among individuals in their ability to maximize energy in-



**FIGURE 27.3**

**Optimal diet.** The shore crab selects a diet of energetically profitable prey. The curve describes the net energy gain (equal to energy gained minus energy expended) derived from feeding on different sizes of mussels. The bar graph shows the numbers of mussels of each size in the diet. Shore crabs most often feed on those mussels that provide the most energy.

take is the result of genetic differences, but there are some exceptions. For example, one study found that female zebra finches that were particularly successful in maximizing net energy intake tended to have offspring that were similarly successful. Because birds were removed from their mothers before they left the nest, this similarity likely reflected a genetic basis for foraging behavior, rather than being a result of young birds learning to forage from their mothers.

Differences among individuals in foraging behavior may also be a function of age. Inexperienced yellow-eyed juncos (a small North American bird), for example, have not learned how to handle large prey items efficiently. As a result, the energetic costs of eating such prey are higher than the benefits, and as a result they tend to focus on smaller prey. Only when the birds are older and more experienced do they learn to easily dispatch these prey, which are then included in the diet.

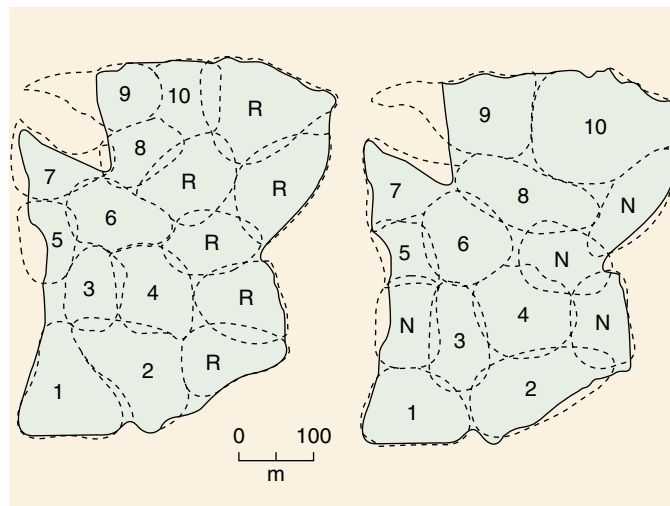
**Natural selection may favor the evolution of foraging behaviors that maximize the amount of energy gained per unit time spent foraging. Animals that acquire energy efficiently during foraging may increase their fitness by having more energy available for reproduction, but other considerations, such as avoiding predators, also are important in determining reproductive success.**

## Territorial Behavior

Animals often move over a large area, or **home range**, during their daily course of activity. In many animal species, the home ranges of several individuals may overlap in time or in space, but each individual defends a *portion* of its home range and uses it *exclusively*. This behavior, in which individual members of a species maintain exclusive use of an area that contains some limiting resource, such as foraging ground, food, or potential mates, is called **territoriality** (figure 27.4). The critical aspect of territorial behavior is *defense* against intrusion by other individuals. Territories are defended by displays that advertise that the territories are occupied and by overt aggression. A bird sings from its perch within a territory to prevent a takeover by a neighboring bird. If an intruder is not deterred by the song, it may be attacked. However, territorial defense has its costs. Singing is energetically expensive, and attacks can lead to injury. In addition, advertisement through song or visual display can reveal one's position to a predator.

Why does an animal bear the costs of territorial defense? Over the past two decades, it has become increasingly clear that an *economic* approach can be useful in answering this question. Although there are costs to defending a territory, there are also benefits; these benefits may take the form of increased food intake, exclusive access to mates, or access to refuges from predators. Studies of nectar-feeding birds like hummingbirds and sunbirds provide an example (figure 27.5). A bird benefits from having the exclusive use of a patch of flowers because it can efficiently harvest the nectar they produce. In order to maintain exclusive use, however, the bird must actively defend the flowers. The benefits of exclusive use outweigh the costs of defense only under certain conditions. Sunbirds, for example, expend 3000 calories per hour chasing intruders from a territory. Whether or not the benefit of defending a territory will exceed this cost depends upon the amount of nectar in the flowers and how efficiently the bird can collect it. If flowers are very scarce or nectar levels are very low, for example, a nectar-feeding bird may not gain enough energy to balance the energy used in defense. Under this circumstance, it is not advantageous to be territorial. Similarly, if flowers are very abundant, a bird can efficiently meet its daily energy requirements without behaving territorially and adding the costs of defense. From an energetic standpoint, defending abundant resources isn't worth the cost. Territoriality thus only occurs at intermediate levels of flower availability and higher levels of nectar production, where the benefits of defense outweigh the costs.

In many species, exclusive access to females is a more important determinant of territory size of males than is food availability. In some lizards, for example, males maintain enormous territories during the breeding season. These territories, which encompass the territories of



**FIGURE 27.4**

**Competition for space.** Territory size in birds is adjusted according to the number of competitors. When six pairs of great tits (*Parus major*) were removed from their territories (indicated by R in the left figure), their territories were taken over by other birds in the area and by four new pairs (indicated by N in the right figure). Numbers correspond to the birds present before and after.



**FIGURE 27.5**

**The benefit of territoriality.** Sunbirds increase nectar availability by defending flowers.

several females, are much larger than what is required to supply enough food and are defended vigorously. In the nonbreeding season, by contrast, male territory size decreases dramatically, as does aggressive territorial behavior.

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An economic approach can be used to explain the evolution and ecology of reproductive behaviors such as territoriality. This approach assumes that animals that gain more energy from a behavior than they expend will have an advantage in survival and reproduction over animals that behave in less efficient ways.

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## 27.2 Reproductive behavior involves many choices influenced by natural selection.

Searching for a place to nest, finding a mate, and rearing young involve a collection of behaviors loosely referred to as reproductive behavior. These behaviors often involve seeking and defending a particular territory, making choices about mates and about the amount of energy to devote to the rearing of young. Mate selection, in particular, often involves intense natural selection. We will look briefly at each of these components of reproductive behavior.

During the breeding season, animals make several important “decisions” concerning their choice of mates, how many mates to have, and how much time and energy to devote to rearing offspring. These decisions are all aspects of an animal’s **reproductive strategy**, a set of behaviors that presumably have evolved to maximize reproductive success. Reproductive strategies have evolved partly in response to the energetic costs of reproduction and the way food resources, nest sites, and members of the opposite sex are spatially distributed in the environment.

### Parental Investment and Mate Choice

Males and females usually differ in their reproductive strategies. Darwin was the first to observe that females often do not simply mate with the first male they encounter, but instead seem to evaluate a male’s quality and then decide whether to mate. This behavior, called **mate choice**, has since been described in many invertebrate and vertebrate species.

By contrast, mate choice by males is much less common. Why should this be? Many of the differences in reproductive strategies between the sexes can be understood by comparing the parental investment made by males and females. **Parental investment** refers to the contributions each sex makes in producing and rearing offspring; it is, in effect, an estimate of the energy expended by males and females in each reproductive event.

Many studies have shown that parental investment is high in females. One reason is that eggs are much larger than sperm—195,000 times larger in humans! Eggs contain proteins and lipids in the yolk and other nutrients for the developing embryo, but sperm are little more than mobile DNA. Furthermore, in some groups of animals, females are responsible for gestation and lactation, costly reproductive functions only they can carry out.

The consequence of such great disparities in reproductive investment is that the sexes should face very different selective pressures. Because any single reproductive event is relatively cheap for males, they can best increase their fitness by mating with as many females as possible—male fit-



FIGURE 27.6

**The advantage of male mate choice.** Male mormon crickets choose heavier females as mates, and larger females have more eggs. Thus, male mate selection increases fitness.

ness is rarely limited by the amount of sperm they can produce. By contrast, each reproductive event for females is much more costly and the number of eggs that can be produced often does limit reproductive success. For this reason, females have an incentive to be choosy, trying to pick the male they can provide the greatest benefit to her offspring. As we shall see, this benefit can take a number of different forms.

These conclusions only hold when female reproductive investment is much greater than that of males. In species with parental care, males may contribute equally to the cost of raising young; in this case, the degree of mate choice should be equal between the sexes.

Furthermore, in some cases, male investment exceeds that of females. For example, male mormon crickets transfer a protein-containing spermatophore to females during mating. Almost 30% of a male’s body weight is made up by the spermatophore, which provides nutrition for the female, and helps her develop her eggs. As one might expect, in this case it is the females that compete with each other for access to males, and the males that are the choosy sex. Indeed, males are quite selective, favoring heavier females. The selective advantage of this strategy results because heavier females have more eggs; thus, males that choose larger females leave more offspring (figure 27.6).

**Reproductive investment by the sexes is strongly influenced by differences in the degree of parental investment.**

## Reproductive Competition and Sexual Selection

In chapter 20, we learned that the reproductive success of an individual is determined by a number of factors: how long the individual lives, how successful it is in obtaining matings, and how many offspring it produces per mating. The second of these factors, competition for mating opportunities, has been termed **sexual selection**. Some people consider sexual selection to be distinctive from natural selection, but others see it as a subset of natural selection, just one of a number of ways in which organisms can increase their fitness.

Sexual selection involves both **intrasexual selection**, or interactions between members of one sex (“the power to conquer other males in battle,” as Darwin put it), and **intersexual selection**, essentially mate choice (“the power to charm”). Sexual selection thus leads to the evolution of structures used in combat with other males, such as a deer’s antlers and a ram’s horns, as well as ornamentation used to “persuade” members of the opposite sex to mate, such as long tail feathers and bright plumage (figure 27.7*a*). These traits are called **secondary sexual characteristics**.

### Intrasexual Selection

In many species, individuals of one sex—usually males—compete with each other for the opportunity to mate with individuals of the other sex. These competitions may take place over ownership of a territory in which females reside or direct control of the females themselves. The latter case is exemplified by many species, such as impala, in which fe-

males travel in large groups with a single male that gets exclusive rights to mate with the females and thus strives vigorously to defend these rights against other males which would like to supplant him.

In mating systems such as these, a few males may get an inordinate number of matings and most males do not mate at all. In elephant seals, in which males control territories on the breeding beaches, a few dominant males do most of the breeding. On one beach, for example, eight males impregnated 348 females, while the remaining males got very little action (or, we could say, while the remaining males mated rarely, if at all).

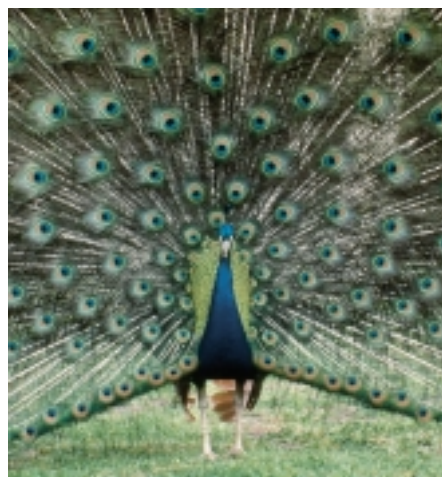
For this reason, selection will strongly favor any trait that confers greater ability to outcompete other males. In many cases, size determines mating success: the larger male is able to dominate the smaller one. As a result, in many territorial species, males have evolved to be considerably larger than females, for the simple reason that the largest males are the ones that get to mate. Such differences between the sexes are referred to as **sexual dimorphism**. In other species, males have evolved structures used for fighting, such as horns, antlers, and large canine teeth. These traits are also often sexually dimorphic and may have evolved because of the advantage they give in intrasexual conflicts.

### Intersexual Selection

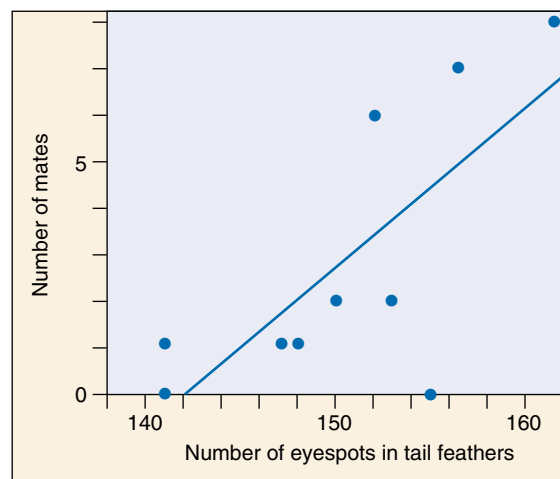
Peahens prefer to mate with peacocks that have more spots in their long tail feathers (figure 27.7*b,c*). Similarly, female frogs prefer to mate with males with more complex calls. Why did such mating preferences evolve?



(a)



(b)



(c)

**FIGURE 27.7**

**Products of sexual selection.** Attracting mates with long feathers is common in bird species such as the African paradise whydah (*a*) and the peacock (*b*), which show pronounced sexual dimorphism. (*c*) Female peahens prefer to mate with males with greater numbers of eyespots in their tail feathers.

## The Benefits of Mate Choice

In some cases, the benefits are obvious. In many species of birds and mammals, and some species of other types of animals, males help raise the offspring. In these cases, females would benefit by choosing the male that can provide the best care—the better the parent, the more offspring she is likely to rear.

In other species, males provide no care, but maintain territories that provide food, nesting sites, and predator refuges. In such species, females that choose males with the best territories will maximize their reproductive success.

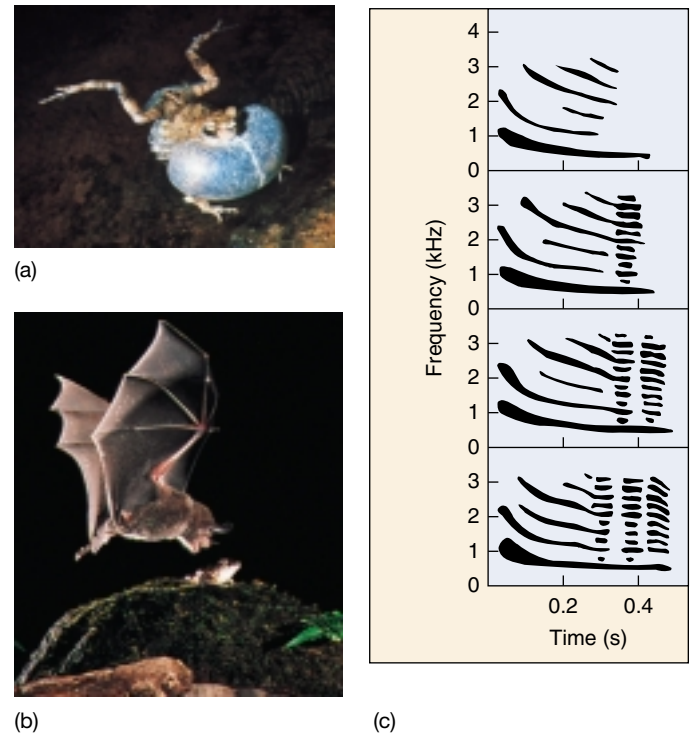
## Indirect Benefits

In other species, however, males provide no direct benefits of any kind to females. In such cases, it is not intuitively obvious what females have to gain by being choosy. Moreover, what could be the possible benefit of choosing a male with an extremely long tail or a complex song?

A number of theories have been proposed to explain the evolution of such preferences. One idea is that females choose the male that is the healthiest or oldest. Large males, for example, have probably been successful at living long, acquiring a lot of food and resisting parasites and disease. Similarly, in guppies and some birds, the brightness of a male's color is a reflection of the quality of his diet and overall health. Females may gain two benefits from mating with large or colorful males. First, to the extent that the males' success in living long and prospering is the result of a good genetic makeup, the female will be ensuring that her offspring receive good genes from their father. Indeed, several studies have demonstrated that males that are preferred by females produce offspring that are more vigorous and survive better than offspring of males that are not preferred. Second, healthy males are less likely to be carrying diseases, which might be transmitted to the female during mating.

A variant of this theory goes one step further. In some cases, females prefer mates with traits that are detrimental to survival (figure 27.8). The long tail of the peacock is a hindrance in flying and makes males more vulnerable to predators. Why should females prefer males with such traits? The **handicap hypothesis** states that only genetically superior mates can survive with such a handicap. By choosing a male with the largest handicap, the female is ensuring that her offspring will receive these quality genes. Of course, the male offspring will also inherit the genes for the handicap. For this reason, evolutionary biologists are still debating the merits of this hypothesis.

Other courtship displays appear to have evolved from a predisposition in the female's sensory system to be stimulated by a certain type of stimulus. For example, females may be better able to detect particular colors or sounds at a certain frequency. Sensory exploitation involves the evolution in males of an attractive signal that “exploits” these



**FIGURE 27.8**

**The benefits and costs of vocalizing.** (a) The male Túngara frog, *Physalaemus pustulosus*. (b) The males' calls attract females as well as predatory bats. Calls of greater complexity are represented from top to bottom in (c). Females prefer more complex calls, but these calls are detected particularly well by bats. Consequently, males that females prefer are at the greatest risk of being captured.

preexisting biases—if females are particularly adept at detecting red objects, for example, then males will evolve red coloration. Consider the vocalizations of the Túngara frog (*Physalaemus pustulosus*) (see figure 27.8). Unlike related species, males include a “chuck” in their calls. Recent research suggests that even females of related species are particularly attracted to calls of this sort, even though males of these species do not produce “chucks.” Why this preference evolved is unknown, but males of the Túngara frog have evolved to take advantage of it.

A great variety of other theories have been proposed to explain the evolution of mating preferences. Many of these theories may be correct in some circumstances and none seems capable of explaining all of the variation in mating behavior in the animal world. This is an area of vibrant research with new discoveries appearing regularly.

**Natural selection has favored the evolution of behaviors that maximize the reproductive success of males and females. By evaluating and selecting mates with superior qualities, an animal can increase its reproductive success.**

## Mating Systems

The number of individuals with which an animal mates during the breeding season varies throughout the animal kingdom. Mating systems such as monogamy (one male mates with one female); polygyny (one male mates with more than one female; figure 27.9), and polyandry (one female mates with more than one male) are aspects of male and female reproductive strategy that concern how many mates an individual has during the breeding season. Like mate choice, mating systems have evolved to maximize reproductive fitness. Much research has shown that mating systems are strongly influenced by ecology. For instance, a male may defend a territory that holds nest sites or food sources necessary for a female to reproduce, and the territory might have resources sufficient for more than one female. If males differ in the quality of the territories they hold, a female's fitness will be maximized if she mates with a male holding a high-quality territory. Such a male may already have a mate, but it is still more advantageous for the female to breed with that male than with an unmated male that defends a low-quality territory. In this way, natural selection would favor the evolution of polygyny.

Mating systems are also constrained by the needs of offspring. If the presence of both parents is necessary for young to be reared successfully, then monogamy may be favored. This is generally the case in birds, in which over 90% of all species are monogamous. A male may either remain with his mate and provide care for the offspring or desert that mate to search for others; both strategies may increase his fitness. The strategy that natural selection will favor depends upon the requirement for male assistance in feeding or defending the offspring. In some species, offspring are **altricial**—they require prolonged and extensive care. In these species, the need for care by two parents will reduce the tendency for the male to desert his mate and seek other matings. In species where the young are **precocial** (requiring little parental care), males may be more likely to be polygynous.

Although polygyny is much more common, polyandrous systems—in which one female mates with several males—are known in a variety of animals. For example, in spotted sandpipers, males take care of all incubation and parenting, and females mate and leave eggs with two or more males.

In recent years, researchers have uncovered many unexpected aspects of animal reproductive systems. Some of these discoveries have resulted from the application of new technologies, whereas others have come from detailed and intensive field studies.

### Extra-Pair Copulations

In chapter 19, we saw how DNA fingerprinting can be used to identify blood samples. Another common use of this technology is to establish paternity. DNA fingerprints are



**FIGURE 27.9**

**Female defense polygyny in bats.** The male at the lower right is guarding a group of females.

so variable that each individual's is unique. Thus, by comparing the DNA of a man and a child, experts can establish with a relatively high degree of confidence whether the man is the child's father.

This approach is now commonly used in paternity lawsuits, but it has also become a standard weapon in the arsenal of behavioral ecologists. By establishing paternity, researchers can precisely quantify the reproductive success of individual males and thus assess how successful their particular reproductive strategies have been (figure 27.10a). In one classic study of red-winged blackbirds (figure 27.10b), researchers established that half of all nests contained at least one bird fertilized by a male other than the territory owner; overall, 20% of the offspring were the result of such **extra-pair copulations** (EPCs).

Studies such as this have established that EPCs—"cheating"—are much more pervasive in the bird world than originally suspected. Even in some species that were believed to be monogamous on the basis of behavioral observations, the incidence of offspring being fathered by a male other than the female's mate is sometimes surprisingly high.

Why do individuals have extra-pair copulations? For males, the answer is obvious: increased reproductive success. For females, it is less clear, as in most cases, it does not result in an increased number of offspring. One possi-

bility is that females mate with genetically superior individuals, thus enhancing the genes passed on to their offspring. Another possibility is that females can increase the amount of help they get in raising their offspring. If a female mates with more than one male, each male may help raise the offspring. This is exactly what happens in a common English bird, the dunnock. Females mate not only with the territory owner, but also with subordinate males that hang around the edge of the territory. If these subordinates mate enough with a female, they will help raise her young, presumably because some of these young may have been fathered by this male.

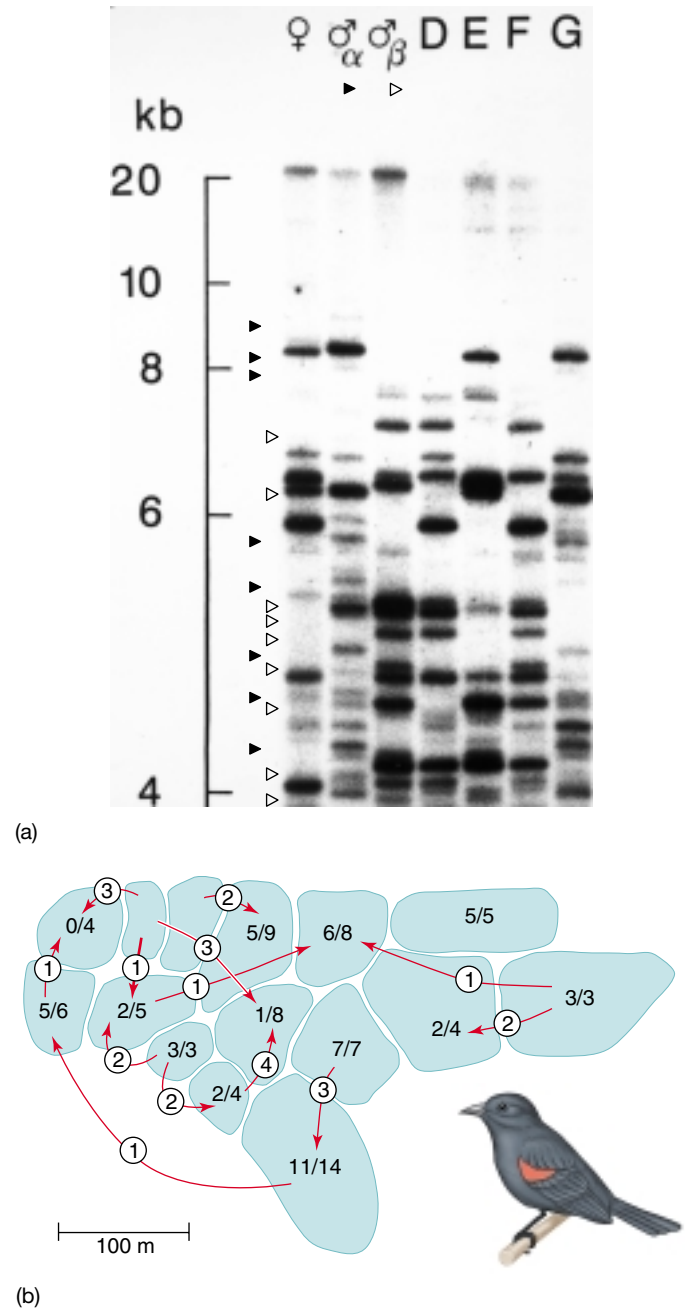
### Alternative Mating Tactics

Natural selection has led to the evolution of a variety of other means of increasing reproductive success. For example, in many species of fish, there are two genetic classes of males. One group is large and defends territories to obtain matings. The other type of male is small and adopts a completely different strategy. They do not maintain territories, but hang around the edge of the territories of large males. Just at the end of a male's courtship, when the female is laying her eggs and the territorial male is depositing sperm, the smaller male will dart in and release its own sperm into the water, thus fertilizing some of the eggs. If this strategy is successful, natural selection will favor the evolution of these two different male reproductive strategies.

Similar patterns are seen in other organisms. In some dung beetles, territorial males have large horns that they use to guard the chambers in which females reside, whereas genetically small males don't have horns; instead, they dig side tunnels and attempt to intercept the female inside her chamber. In isopods, there are three genetic size classes. The medium-sized males pass for females and enter a large male's territory in this way; the smallest class are so tiny, they are able to sneak in completely undetected.

This is just a glimpse of the rich diversity in mating systems that have evolved. The bottom line is: if there is a way of increasing reproductive success, natural selection will favor its evolution.

**Mating systems represent reproductive adaptations to ecological conditions. The need for parental care, the ability of both sexes to provide it, and the timing of female reproduction are important influences on the evolution of monogamy, polygyny, and polyandry. Detailed study of animal mating systems, along with the use of modern molecular techniques, are revealing many surprises in animal mating systems. This diversity is a testament to the power of natural selection to favor any trait that increases an animal's fitness.**



**FIGURE 27.10**  
**The study of paternity.** (a) A DNA fingerprinting gel from the dunnock. The bands represent fragments of DNA of different lengths. The four nestlings (D-G) were in the nest of the female. By comparing the bands present in the two males, we can determine which male fathered which offspring. The triangles point to the bands which are diagnostic for one male and not the other. In this case, the beta male fathered three of the four offspring. (b) Results of a DNA fingerprinting study in red-winged blackbirds. Fractions indicate the proportion of offspring fathered by the male in whose territory the nest occurred. Arrows indicate how many offspring were fathered by particular males outside of each territory. Nests on some territories were not sampled.

## 27.3 There is considerable controversy about the evolution of social behavior.

### Factors Favoring Altruism and Group Living

Altruism—the performance of an action that benefits another individual at a cost to the actor—occurs in many guises in the animal world. In many bird species, for example, parents are assisted in raising their young by other birds, which are called **helpers at the nest**. In species of both mammals and birds, individuals that spy a predator will give an alarm call, alerting other members of their group, even though such an act would seem to call the predator's attention to the caller. Finally, lionesses with cubs will allow all cubs in the pride to nurse, including cubs of other females.

The existence of altruism has long perplexed evolutionary biologists. If altruism imposes a cost to an individual, how could an allele for altruism be favored by natural selection? One would expect such alleles to be at a disadvantage and thus their frequency in the gene pool should decrease through time.

A number of explanations have been put forward to explain the evolution of altruism. One suggestion often heard on television documentaries is that such traits evolve for the good of the species. The problem with such explanations is that natural selection operates on individuals within species, not on species themselves. Thus, it is even possible for traits to evolve that are detrimental to the species as a whole, as long as they benefit the individual. In some cases, selection can operate on groups of individuals, but this is rare. For example, if an allele for supercannibalism evolved within a population, individuals with that allele would be favored, as they would have more to eat; however, the group might eventually eat itself to extinction, and the allele would be removed from the species. In certain circumstances, such **group selection** can occur, but the conditions for it to occur are rarely met in nature. In most cases, consequently, the “good of the species” cannot explain the evolution of altruistic traits.

Another possibility is that seemingly altruistic acts aren't altruistic after all. For example, helpers at the nest are often young and gain valuable parenting experience by assisting established breeders. Moreover, by hanging around an area, such individuals may inherit the territory when the established breeders die. Similarly, alarm callers may actually be beneficial by causing other animals to panic. In the ensuing confusion, the caller may be able to slip off undetected. Detailed field studies in recent years have demonstrated that some acts truly are altruistic, but others are not as they seemed.

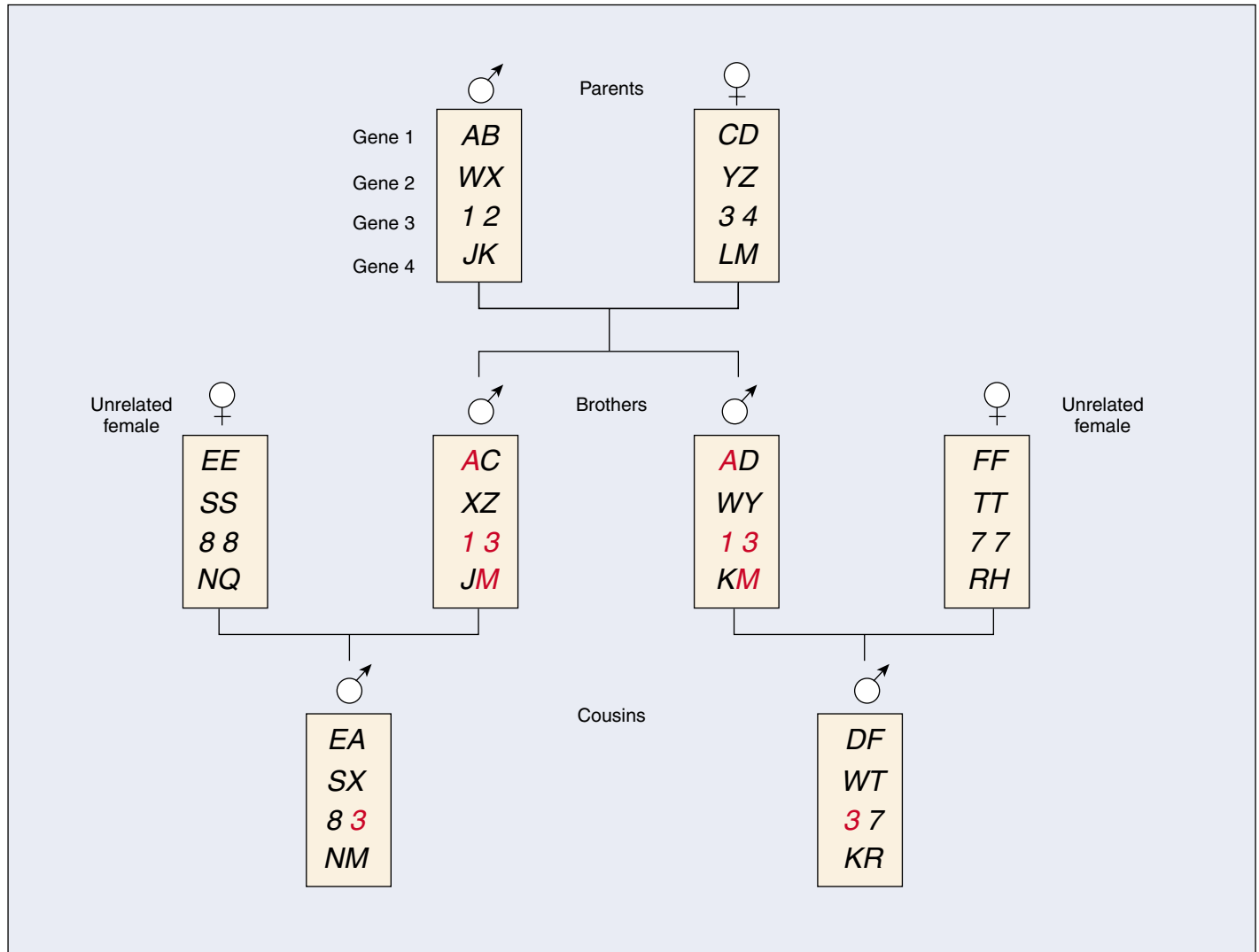
### Reciprocity

Robert Trivers, now of Rutgers University, proposed that individuals may form “partnerships” in which mutual exchanges of altruistic acts occur, because it benefits both participants to do so. In the evolution of such **reciprocal altruism**, “cheaters” (nonreciprocators) are discriminated against and are cut off from receiving future aid. According to Trivers, if the altruistic act is relatively inexpensive, the small benefit a cheater receives by not reciprocating is far outweighed by the potential cost of not receiving future aid. Under these conditions, cheating should not occur.

Vampire bats roost in hollow trees in groups of 8 to 12 individuals. Because these bats have a high metabolic rate, individuals that have not fed recently may die. Bats that have found a host imbibe a great deal of blood; giving up a small amount presents no great energy cost to the donor, and it can keep a roostmate from starvation. Vampire bats tend to share blood with past reciprocators. If an individual fails to give blood to a bat from which it had received blood in the past, it will be excluded from future bloodsharing.

### Kin Selection

The most influential explanation for the origin of altruism was presented by William D. Hamilton in 1964. It is perhaps best introduced by quoting a passing remark made in a pub in 1932 by the great population geneticist J. B. S. Haldane. Haldane said that he would willingly lay down his life for *two brothers* or *eight first cousins*. Evolutionarily speaking, Haldane's statement makes sense, because for each allele Haldane received from his parents, his brothers each had a 50% chance of receiving the same allele (figure 27.11). Consequently, it is statistically expected that two of his brothers would pass on as many of Haldane's particular combination of alleles to the next generation as Haldane himself would. Similarly, Haldane and a first cousin would share an eighth of their alleles (see figure 27.11). Their parents, which are siblings, would each share half their alleles, and each of their children would receive half of these, of which half on the average would be in common:  $\text{one-half} \times \text{one-half} \times \text{one-half} = \text{one-eighth}$ . Eight first cousins would therefore pass on as many of those alleles to the next generation as Haldane himself would. Hamilton saw Haldane's point clearly: natural selection will favor any strategy that increases the net flow of an individual's alleles to the next generation.



**FIGURE 27.11**

**Hypothetical example of genetic relationship.** On average, full siblings will share half of their alleles. By contrast, cousins will, on average, only share one-eighth of their alleles.

Hamilton showed that by directing aid toward kin, or close genetic relatives, an altruist may increase the reproductive success of its relatives enough to compensate for the reduction in its own fitness. Because the altruist's behavior increases the propagation of alleles in relatives, it will be favored by natural selection. Selection that favors altruism directed toward relatives is called **kin selection**. Although the behaviors being favored are cooperative, the genes are actually "behaving selfishly," because they encourage the organism to support copies of themselves in other individuals.

Hamilton's kin selection model predicts that altruism is likely to be directed toward close relatives. The more closely related two individuals are, the greater the poten-

tial genetic payoff. This relationship is described by **Hamilton's rule**, which states that altruistic acts are favored when  $b/c > 1/r$ . In this expression,  $b$  and  $c$  are the benefits and costs of the altruistic act, respectively, and  $r$  is the coefficient of relatedness, the proportion of alleles shared by two individuals through common descent. For example, an individual should be willing to have one less child if such actions allow a half-sibling, which shares one-quarter of its genes, to have more than four additional offspring.

**Many factors could be responsible for the evolution of altruistic behaviors.**

## Examples of Kin Selection

Many examples of kin selection are known from the animal world. For example, Belding's ground squirrel give alarm calls when they spot a predator such as a coyote or a badger. Such predators may attack a calling squirrel, so giving a signal places the caller at risk. The social unit of a ground squirrel colony consists of a female and her daughters, sisters, aunts, and nieces. Males in the colony are not genetically related to these females. By marking all squirrels in a colony with an individual dye pattern on their fur and by recording which individuals gave calls and the social circumstances of their calling, researchers found that females who have relatives living nearby are more likely to give alarm calls than females with no kin nearby. Males tend to call much less frequently as would be expected as they are not related to most colony members.

Another example of kin selection comes from a bird called the white-fronted bee-eater which lives along rivers in Africa in colonies of 100 to 200 birds. In contrast to the ground squirrels, it is the males that usually remain in the colony in which they were born, and the females that disperse to join new colonies. Many bee-eaters do not raise their own offspring, but rather help others. Many of these birds are relatively young, but helpers also include older birds whose nesting attempts have failed. The presence of a single helper, on average, doubles the number of offspring that survive. Two lines of evidence support the idea that kin selection is important in determining helping behavior in this species. First, helpers are usually males, which are usually related to other birds in the colony, and not females, which are not related. Second, when birds have the choice of helping different parents, they almost invariably choose the parents to which they are most closely related.

### Haplodiploidy and Hymenopteran Social Evolution

Probably the most famous application of kin selection theory has been to social insects. A hive of honeybees consists of a single queen, who is the sole egg-layer, and up to 50,000 of her offspring, nearly all of whom are female workers with nonfunctional ovaries (figure 27.12), a situation termed **eusociality**. The sterility of the workers is altruistic: these offspring gave up their personal reproduction to help their mother rear more of their sisters.

Hamilton explained the origin of altruism in hymenopterans (that is, bees, wasps, and ants) with his kin selection model. In these insects, males are haploid and females are diploid. This unusual system of sex determination, called **haplodiploidy**, leads to an unusual situation. If the queen is fertilized by a single male, then all



**FIGURE 27.12**  
**Reproductive division of labor in honeybees.** The queen (shown here with a red spot painted on her thorax) is the sole egg-layer. Her daughters are sterile workers.

female offspring will inherit exactly the same alleles from their father (because he is haploid and has only one copy of each allele). These female offspring will also share among themselves, on average, half of the alleles they get from the queen. Consequently, each female offspring will share on average 75% of her alleles with each sister (to verify this, rework figure 27.11, but allow the father to only have one allele for each gene). By contrast, should she have offspring of her own, she would only share half of her alleles with these offspring (the other half would come from their father). Thus, because of this close genetic relatedness, *workers propagate more alleles by giving up their own reproduction to assist their mother in rearing their sisters, some of whom will be new queens and start new colonies and reproduce*. Thus, this unusual haplodiploid system may have set the stage for the evolution of eusociality in hymenopterans and, indeed, such systems have evolved as many as 12 or many times in the Hymenoptera.

One wrinkle in this theory, however, is that eusocial systems have evolved in several other groups, including thrips, termites, and naked mole rats. Although thrips are also haplodiploid, both termites and naked mole rats are not. Thus, although haplodiploidy may have facilitated the evolution of eusociality, it is not a necessary prerequisite.

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**Kin selection is a potent force favoring, in some situations, the evolution of altruism and even complex social systems.**

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## Group Living and the Evolution of Social Systems

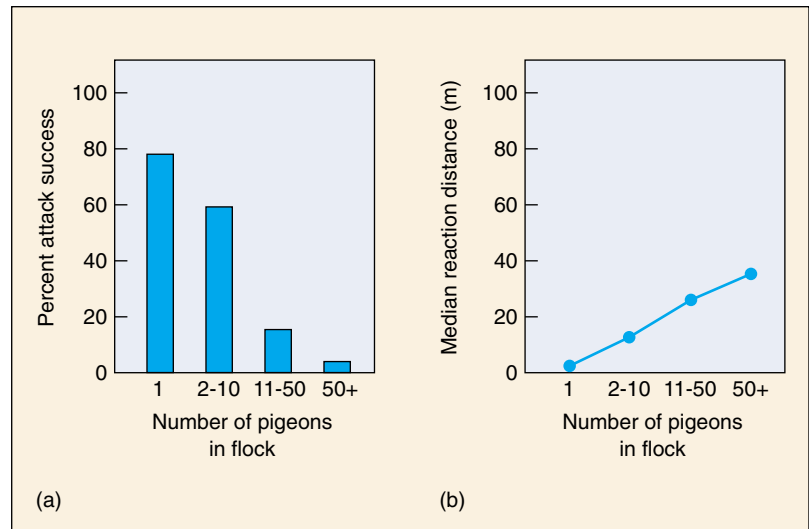
Organisms as diverse as bacteria, cnidarians, insects, fish, birds, prairie dogs, lions, whales, and chimpanzees exist in social groups. To encompass the wide variety of social phenomena, we can broadly define a society as *a group of organisms of the same species that are organized in a cooperative manner*.

Why have individuals in some species given up a solitary existence to become members of a group? We have just seen that one explanation is kin selection: groups may be composed of close relatives. In other cases, individuals may benefit directly from social living. For example, a bird that joins a flock may receive greater protection from predators. As flock size increases, the risk of predation decreases because there are more individuals to scan the environment for predators (figure 27.13). A member of a flock may also increase its feeding success if it can acquire information from other flock members about the location of new, rich food sources. In some predators, hunting in groups can increase success and allow the group to tackle prey too large for any one individual.

### Insect Societies

In insects, sociality has chiefly evolved in two orders, the Hymenoptera (ants, bees, and wasps) and the Isoptera (termites), although a few other insect groups include social species. All ants, some bees, some wasps, and all termites are **eusocial** (truly social): they have a division of labor in reproduction (a fertile queen and sterile workers), cooperative care of brood and an overlap of generations so that the queen lives alongside her offspring. Social insect colonies are composed of different **castes** of workers that differ in size and morphology and have different tasks they perform, such as workers and soldiers.

In honeybees, the queen maintains her dominance in the hive by secreting a pheromone, called “queen substance,” that suppresses development of the ovaries in other females, turning them into sterile workers. Drones (male bees) are produced only for purposes of mating. When the colony grows larger in the spring, some members do not receive a sufficient quantity of queen substance, and the colony begins preparations for swarming. Workers make several new queen chambers, in which new queens begin to develop. Scout workers look for a new nest site and communicate its location to the colony. The old queen and a swarm of female workers then move to the new site. Left behind, a new queen emerges, kills the



**FIGURE 27.13**

**Flocking behavior decreases predation.** (a) As the size of a pigeon flock increases, hawks are less successful at capturing pigeons. (b) When more pigeons are present in the flock, they can detect hawks at greater distances, thus allowing more time for the pigeons to escape.

other potential queens, flies out to mate, and returns to assume “rule” of the hive.

The leafcutter ants provide another fascinating example of the remarkable lifestyles of social insects. Leafcutters live in colonies of up to several million individuals, growing crops of fungi beneath the ground. Their mound-like nests are underground “cities” covering more than 100 square meters, with hundreds of entrances and chambers as deep as 5 meters beneath the ground. The division of labor among the worker ants is related to their size. Every day, workers travel along trails from the nest to a tree or a bush, cut its leaves into small pieces, and carry the pieces back to the nest. Smaller workers chew the leaf fragments into a mulch, which they spread like a carpet in the underground fungus chambers. Even smaller workers implant fungal hyphae in the mulch. Soon a luxuriant garden of fungi is growing. While other workers weed out undesirable kinds of fungi, nurse ants carry the larvae of the nest to choice spots in the garden, where the larvae graze. This elaborate social system has evolved to produce reproductive queens that will disperse from the parent nest and start new colonies, repeating the cycle.

**Eusocial insect workers exhibit an advanced social structure that includes division of labor in reproduction and workers with different tasks.**

## 27.4 Vertebrates exhibit a broad range of social behaviors.

### Vertebrate Societies

In contrast to the highly structured and integrated insect societies and their remarkable forms of altruism, vertebrate social groups are usually less rigidly organized and cohesive. It seems paradoxical that vertebrates, which have larger brains and are capable of more complex behaviors, are generally less altruistic than insects. Nevertheless, in some complex vertebrate social systems individuals may be exhibiting both reciprocity and kin-selected altruism. But vertebrate societies also display more conflict and aggression among group members than do insect societies. Conflict in vertebrate societies generally centers on access to food and mates.

Vertebrate societies, like insect societies, have particular types of organization. Each social group of vertebrates has a certain size, stability of members, number of breeding males and females, and type of mating system. Behavioral ecologists have learned that the way a group is organized is influenced most often by ecological factors such as food type and predation (figure 27.14).

African weaver birds, which construct nests from vegetation, provide an excellent example to illustrate the relationship between ecology and social organization. Their roughly 90 species can be divided according to the type of social group they form. One set of species lives in the forest and builds camouflaged, solitary nests. Males and females are monogamous; they forage for insects to feed their young. The second group of species nests in colonies in trees on the savanna. They are polygynous and feed in flocks on seeds. The feeding and nesting habits of these two sets of species are correlated with their mating systems. In the forest, insects are hard to find, and both parents must cooperate in feeding the young. The camouflaged nests do not call the attention of predators to their brood. On the open savanna, building a hidden nest is not an option. Rather, savanna-dwelling weaver birds protect their young from predators by nesting in trees which are not very abundant. This shortage of safe nest sites means that birds must nest together in colonies. Because seeds occur abundantly, a female can acquire all the food needed to rear young without a male's help. The male, free from the duties of parenting, spends his time courting many females—a polygynous mating system.

One exception to the general rule that vertebrate societies are not organized like those of insects is the naked mole rat, a small, hairless rodent that lives in and near East Africa. Unlike other kinds of mole rats, which live alone or in small family groups, naked mole rats form large underground colonies with a far-ranging system of tunnels and a central nesting area. It is not unusual for a colony to contain 80 individuals.



**FIGURE 27.14**

**Foraging and predator avoidance.** A meerkat sentinel on duty. Meerkats, *Suricata suricata*, are a species of highly social mongoose living in the semiarid sands of the Kalahari Desert. This meerkat is taking its turn to act as a lookout for predators. Under the security of its vigilance, the other group members can focus their attention on foraging.

Naked mole rats feed on bulbs, roots and tubers, which they locate by constant tunneling. As in insect societies, there is a division of labor among the colony members, with some mole rats working as tunnelers while others perform different tasks, depending upon the size of their body. Large mole rats defend the colony and dig tunnels.

Naked mole rat colonies have a *reproductive* division of labor similar to the one normally associated with the eusocial insects. All of the breeding is done by a single female or “queen,” who has one or two male consorts. The workers, consisting of both sexes, keep the tunnels clear and forage for food.

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**Social behavior in vertebrates is often characterized by kin-selected altruism. Altruistic behavior is involved in cooperative breeding in birds and alarm-calling in mammals.**

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## Human Sociobiology

As a social species, humans have an unparalleled complexity. Indeed, we are the only species with the intelligence to contemplate the social behavior of other animals. Intelligence is just one human trait. If an ethologist were to take an inventory of human behavior, he or she would list kin-selected altruism; reciprocity and other elaborate social contracts; extensive parental care; conflicts between parents and offspring; violence and warfare; infanticide; a variety of mating systems, including monogamy, polygyny, and polyandry; along with sexual behaviors such as extra-pair copulation (“adultery”) and homosexuality; and behaviors like adoption that appear to defy evolutionary explanation. This incredible variety of behaviors occurs *in one species*, and any trait can change within *any individual*. Are these behaviors rooted in human biology?

### Biological and Cultural Evolution

During the course of human evolution and the emergence of civilization, two processes have led to adaptive change. One is **biological evolution**. We have a primate heritage, reflected in the extensive amount of genetic material we share with our closest relatives, the chimpanzees. Our upright posture, bipedal locomotion, and powerful, precise hand grips are adaptations whose origins are traceable through our primate ancestors. Kin-selected and reciprocal altruism, as well as other shared traits like aggression and different types of mating systems, can also be seen in nonhuman primates, in whom we can demonstrate that these social traits are adaptive. We may speculate, based on various lines of evidence, that similar traits evolved in early humans. If individuals with certain social traits had an advantage in reproduction over other individuals that lacked the traits, and if these traits had a genetic basis, then the alleles for their expression would now be expected to be part of the human genome and to influence our behavior.

The second process that has underscored the emergence of civilization and led to adaptive change is **cultural evolution**, the transfer across generations of information necessary for survival. This is a nongenetic mode of adaptation. Many adaptations—the use of tools, the formation of cooperative hunting groups, the construction of shelters, and marriage practices—do not follow Mendelian rules of inheritance and are passed from generation to generation by *tradition*. Nonetheless, cultural inheritance is as valid a way to convey adaptations across generations as genetic inheritance. Human cultures are also extraordinarily diverse. The ways in which children are socialized among Trobriand Islanders, Pygmies, and Yanomamo In-

dians are very different. Again, we must remember that this fantastic variation occurs within one species, and that individual behavior is very flexible.

### Identifying the Biological Components of Human Behavior

Given this great flexibility, how can the biological components of human behavior be identified? One way is to look for common patterns that appear in a wide variety of cultures, that is, to study behaviors that are cross-cultural. In spite of cultural variation, there are some traits that characterize all human societies. For example, all cultures have an incest taboo, forbidding marriages between close relatives. Incestuous matings lead to a greater chance of exposing disorders such as mental retardation and hemophilia. Natural selection may have acted to create a behavioral disposition against incest, and that disposition is now a cultural norm. Genes responsible for guiding this behavior may have become fixed in human populations because of their adaptive effects. Genes thus guide the direction of culture.

Although human mating systems vary, polygyny is found to be the most common among all cultures. Because most mammalian species are polygynous, the human pattern seems to reflect our mammalian evolutionary heritage and thus is a part of our biology. This conclusion is drawn from using the comparative approach, common in evolutionary science. Nonverbal communication patterns, like smiling and raising the hand in a greeting, also occur in many cultures. Perhaps these behaviors represent a common human heritage.

The explanations sociobiology offers to understand human behavior have been and continue to be controversial. For example, the new discipline of **evolutionary psychology** seeks to understand the origins of the human mind. Human behaviors are viewed as being extensions of our genes. The diversity of human cultures are thought to have a common core of characteristics that are generated by our psychology, which evolved as an adaptation to the lifestyle of our hunter-gatherer ancestors during the Pleistocene. Much of human behavior is seen as reflecting ancient, adaptive traits, now expressed in the context of modern civilization. In this controversial view, human behaviors such as jealousy and infidelity are viewed as adaptations; these behaviors increased the fitness of our ancestors, and thus are now part of the human psyche.

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**Sociobiology offers general explanations of human behavior that are controversial, but are becoming more generally accepted than in the past.**

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## Summary

## Questions

## Media Resources

## 27.1 Evolutionary forces shape behavior.

- Many behaviors are ecologically important and serve as adaptations.
- Foraging and territorial behaviors have evolved because they allow animals to use resources efficiently.

1. What does optimal foraging theory predict about an animal's foraging behavior? What factors unrelated to this theory may also influence an animal's foraging choices?

2. What are the benefits of territorial behavior, and what are its costs? Under what circumstances is territorial behavior disadvantageous?

## 27.2 Reproductive behavior involves many choices influenced by natural selection.

- Male and female animals maximize their fitness with different reproductive behaviors. The differences relate to the extent to which each sex provides care for offspring.
- Usually, males are competitive and females show mate choice because females have higher reproductive costs.
- A species' mating system is related to its ecology.

3. Why does natural selection favor mate choice? What factor is most important in determining which sex exhibits mate choice?

4. In birds, how does the amount of parental care required by the offspring affect the evolution of a species' mating system?



- Bioethics Case Study: Behavior Disordered Students
- On Science Article: Repetition and Learning
- On Science Article: Flipper, A Senseless Killer?

## 27.3 There is considerable controversy about the evolution of social behavior.

- Many animals show altruistic, or self-sacrificing, behavior. Altruism may evolve through reciprocity or be directed toward relatives. Cooperative behavior often increases an individual's inclusive fitness.
- Individuals form social groups because it is advantageous for them to do so.
- The benefits of living in a group, such as enhanced feeding success, are often balanced by the cost of increased incidence of disease and parasitism.
- Animal societies are characterized by cooperation and conflict. The organization of a society is related to the ecology of a species.

5. What is reciprocal altruism? What is kin selection? How does kin selection increase an individual's success in passing its genes on to the next generation?

## 27.4 Vertebrates exhibit a broad range of social behaviors.

- Human behavior is extremely rich and varied and may result from both biology and culture.
- Evolutionary theory can give us important insight into human nature, but such an approach to the study of human behavior may have political consequences.

6. In vertebrate societies, what are the costs to an individual who makes an alarm call? Based on research in ground squirrels, which individuals are most likely to make alarm calls, and what benefits do they receive by doing so?

# Part VIII

## The Global Environment

### Identifying the Environmental Culprit Harming Amphibians

What started out as a relatively standard field trip in 1995 turned into a bizarre experience for a group of middle-school science students in Minnesota. Their assignment: to collect frogs for their biology class. What they found in local ponds were not frogs like you are accustomed to seeing, frogs like the one shown here. What they found looked more like the result of some bizarre genetic experiment! Approximately half of the animals collected were deformed, with extra legs or missing legs or no eyes. Turning to the Internet, they soon discovered that the problem was not isolated to Minnesota. Neighboring states were reporting the same phenomenon—an alarming number of deformed frogs, all across the United States and Canada.

Although deformed frogs such as those collected by the Minnesota students received national attention, a different problem affecting amphibians has received even more. During the past 30 years, there has been a worldwide catastrophic decline in amphibian populations, with many local populations becoming extinct. The problem is a focus of intensive research, which indicates that four factors are contributing in a major way to the worldwide amphibian decline: (1) habitat destruction, particularly loss of wetlands, (2) the introduction of exotic species that outcompete local amphibian populations, (3) alteration of habitats by toxic chemicals or other human activities (clear-cutting of trees, for example, drastically reduces humidity), and (4) infection of amphibians by chytrid fungi or ranavirus, both of which are fatal to them.

The developmental deformities reported in frogs are also a worldwide problem, but seem to arise from a different set of factors than those producing global declines in amphibian populations. The increase in deformities seems to reflect the fact that amphibians are particularly sensitive to their environment. Their semi-aquatic mode of living, depending on a watery environment to reproduce and keep their skin moist, means that they are exposed to all types of environmental changes.

Amphibians are particularly vulnerable during early development, when their fertilized eggs lay in water, exposed to potential infection by trematodes that can disrupt development, to acid introduced to ponds by acid rain, to toxic



**Disappearing amphibians.** Populations of amphibians, like this Cascades frog (*Rana cascadae*), are declining in numbers in many regions

chemical pollutants, and to increased levels of UV-B radiation produced by ozone depletion.

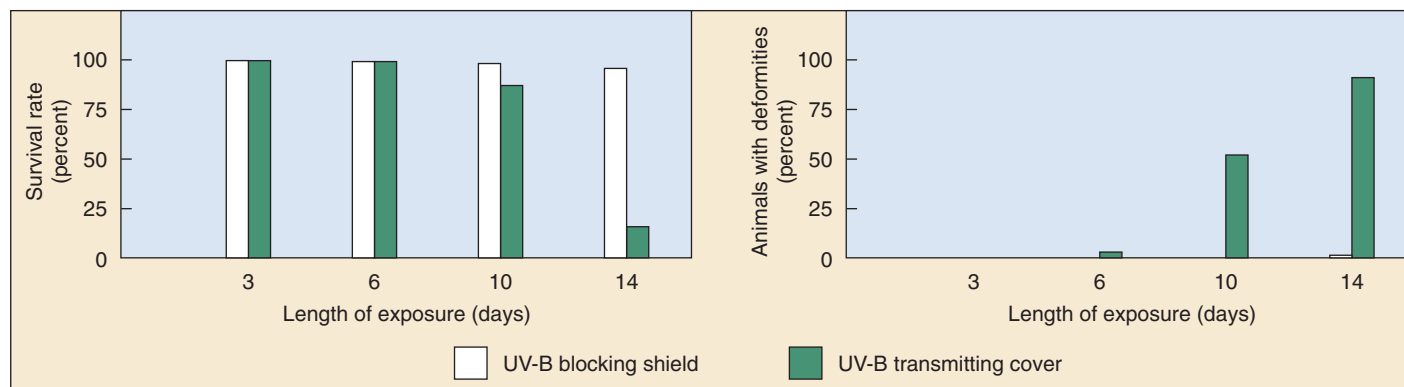
While numerous experiments performed under laboratory conditions have demonstrated the power of these factors to produce developmental deformities, and in so doing to reduce population survival rates, it is important to understand that “can” does not equal “does.” To learn what is in fact going on, scientists have examined the effects of these factors on amphibian development in the natural environment.

Some environmental scientists suspected that toxic chemical pollutants in the water might be causing the deformities and that the widespread occurrence of deformed frogs might well be an early warning of potential future problems in other species, including humans.

Other scientists cautioned that a different factor might be responsible. Although chemicals such as pesticides certainly *could* produce deformities in localized situations, say near a chemical spill, so too could other environmental factors affecting local habitats, particularly parasitic infections by trematodes. Demonstrating this point, researchers in 1999 showed that the multilimb and missing limb phenomenon in frogs can be caused by trematodes that infect the developing tadpoles, disrupting development of their limbs.

Responding to this alternative suggestion, those scientists nominating toxins as the principal culprit have cautioned that showing trematode parasites can have a significant effect on local populations is not the same thing as demonstrating that they have in fact done so. And, they add, it certainly doesn’t rule out a major contribution to the problem by toxic environmental pollutants, or by any of the other potential disruptors of amphibian development.

In a particularly clear example of the sort of investigation that will be needed to sort out this complex issue, Andrew Blaustein of Oregon State University headed a team of scientists that set out to examine the effects of UV-B radiation on amphibians in natural populations. In a series of experiments carried out in the field, they attempted to assess the degree to which UV-B radiation



**Blaustein's UV-B experiment.** In the group of salamanders whose eggs were protected from UV-B radiation, hatching rates were higher and deformity rates were lower.

promoted amphibian developmental deformities under natural conditions.

Laboratory experiments examining the affects of UV-B on amphibian development had already shown a significant increase in embryonic mortality in some amphibian species, and not in others. Why only in some?

Perhaps behaviors shared by many amphibian species might lead to an increased susceptibility to damage from UV-B radiation, behaviors such as laying eggs in open, shallow waters that offer significant exposure to UV-B radiation. Perhaps physiological traits of certain species make them particularly susceptible to damage from UV-B radiation, traits such as low levels of photolyase, an enzyme that removes harmful photoproducts induced by UV light.

Blaustein's group selected a specimen that exhibits these two factors, the long-toed salamander, *Ambystoma macrodactylum*.

## The Experiment

The goal of Blaustein's field experiment was to allow fertilized eggs to develop in their natural environment with and without a UV-B protective shield. Eggs in both groups were monitored for the appearance of deformities and for survival rates. Eggs were collected from natural shallow water sites (<20 cm deep) and randomly placed within enclosures containing either a UV-B blocking Mylar shield or a UV-B transmitting acetate cover (50 eggs per each enclosure replicated four times). The enclosures were placed in small, unperforated plastic pools containing pond water and the pools were placed back in the pond, thereby exposing the eggs and developing embryos to ambient conditions. The UV-B blocking Mylar shield filtered out more than 94% of ambient UV-B radiation, while the UV-B transmitting acetate cover allowed about 90% of ambient UV-B radiation to pass through.

## The Results

Embryos under the UV-B shields had significantly higher hatching rates and fewer deformities compared with those

under the UV-B transmitting acetate covers. Of the 29 UV-B exposed individuals that hatched, 25 had deformities. This is significant compared to the 190 UV-B shielded individuals that hatched and only 1 showed deformities. These results support the hypothesis that naturally occurring UV-B radiation can adversely affect development in some amphibians, inducing deformities.

Blaustein's group speculates that the higher mortality rates and deformities in frogs and other amphibian species might in fact be due to lower than normal levels of photolyase activity in their developing eggs and embryos, low levels such as found naturally in salamanders.

Laboratory and field experiments seem to support this idea. For one thing, frog species that are not sensitive to UV-B have very high photolyase activity levels. Evaluating 10 different species, Blaustein's team found a strong correlation between species exhibiting little UV-B radiation effects and higher levels of photolyase activity in developing eggs and embryos.

In these experiments, the Pacific tree frog (*Hyla regilla*)—whose populations have not shown deformities or decline—exhibited the highest photolyase activity and was not affected by UV-B radiation, showing no significant increases in mortality rates in UV-B exposed individuals.

In parallel experiments, the Cascades frog (*Rana cascadae*) and the Western toad (*Bufo boreas*)—both of whose populations have been experiencing deformities and markedly declining populations—had less than one-third the photolyase activity seen in *Hyla*, and were strongly affected by UV-B radiation, showing significant increases in mortality rates when exposed to UV-B radiation.

These results suggest that increased level of UV-B radiation resulting from ozone depletion may indeed be a major contributor to amphibian deformities and decline—in populations with low photolyase activity. Could chemical pollutants be acting to lower activity levels of this key enzyme? The investigation continues. Undoubtedly, many factors are contributing to deformities in amphibian population, and there are not going to be many simple answers.

# 28

## *Dynamics of Ecosystems*

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### Concept Outline

#### 28.1 Chemicals cycle within ecosystems.

**The Water Cycle.** Water cycles between the atmosphere and the oceans, although deforestation has broken the cycle in some ecosystems.

**The Carbon Cycle.** Photosynthesis captures carbon from the atmosphere; respiration returns it.

**The Nitrogen Cycle.** Nitrogen is captured from the atmosphere by the metabolic activities of bacteria; other bacteria degrade organic nitrogen, returning it to the atmosphere.

**The Phosphorus Cycle.** Of all nutrients that plants require, phosphorus tends to be the most limiting.

**Biogeochemical Cycles Illustrated: Recycling in a Forest Ecosystem.** In a classic experiment, the role of forests in retaining nutrients is assessed.

#### 28.2 Ecosystems are structured by who eats whom.

**Trophic Levels.** Energy passes through ecosystems in a limited number of steps, typically three or four.

#### 28.3 Energy flows through ecosystems.

**Primary Productivity.** Plants produce biomass by photosynthesis, while animals produce biomass by consuming plants or other animals.

**The Energy in Food Chains.** As energy passes through an ecosystem, a good deal is lost at each step.

**Ecological Pyramids.** The biomass of a trophic level is less, the farther it is from the primary production of photosynthesizers.

**Interactions among Different Trophic Levels.**

Processes on one trophic level can have effects on higher or lower levels of the food chain.

#### 28.4 Biodiversity promotes ecosystem stability.

**Effects of Species Richness.** Species-rich communities are more productive and resistant to disturbance.

**Causes of Species Richness.** Ecosystem productivity, spatial heterogeneity, and climate all affect the number of species in an ecosystem.

**Biogeographic Patterns of Species Diversity.** Many more species occur in the tropics than in temperate regions.

**Island Biogeography.** Species richness on islands may be a dynamic equilibrium between extinction and colonization.



FIGURE 28.1

**Mushrooms serve a greater function than haute cuisine.**

Mushrooms and other organisms are crucial recyclers in ecosystems, breaking down dead and decaying material and releasing critical elements such as carbon and nitrogen back into nutrient cycles.

The earth is a closed system with respect to chemicals, but an open system in terms of energy. Collectively, the organisms in ecosystems regulate the capture and expenditure of energy and the cycling of chemicals (figure 28.1). As we will see in this chapter, all organisms, including humans, depend on the ability of other organisms—plants, algae, and some bacteria—to recycle the basic components of life. In chapter 29, we consider the many different types of ecosystems that constitute the biosphere. Chapters 30 and 31 then discuss the many threats to the biosphere and the species it contains.

## 28.1 Chemicals cycle within ecosystems.

All of the chemical elements that occur in organisms cycle through ecosystems in **biogeochemical cycles**, cyclical paths involving both biological and chemical processes. On a global scale, only a very small portion of these substances is contained within the bodies of organisms; almost all exists in nonliving reservoirs: the atmosphere, water, or rocks. Carbon (in the form of carbon dioxide), nitrogen, and oxygen enter the bodies of organisms primarily from the atmosphere, while phosphorus, potassium, sulfur, magnesium, calcium, sodium, iron, and cobalt come from rocks. All organisms require carbon, hydrogen, oxygen, nitrogen, phosphorus, and sulfur in relatively large quantities; they require other elements in smaller amounts.

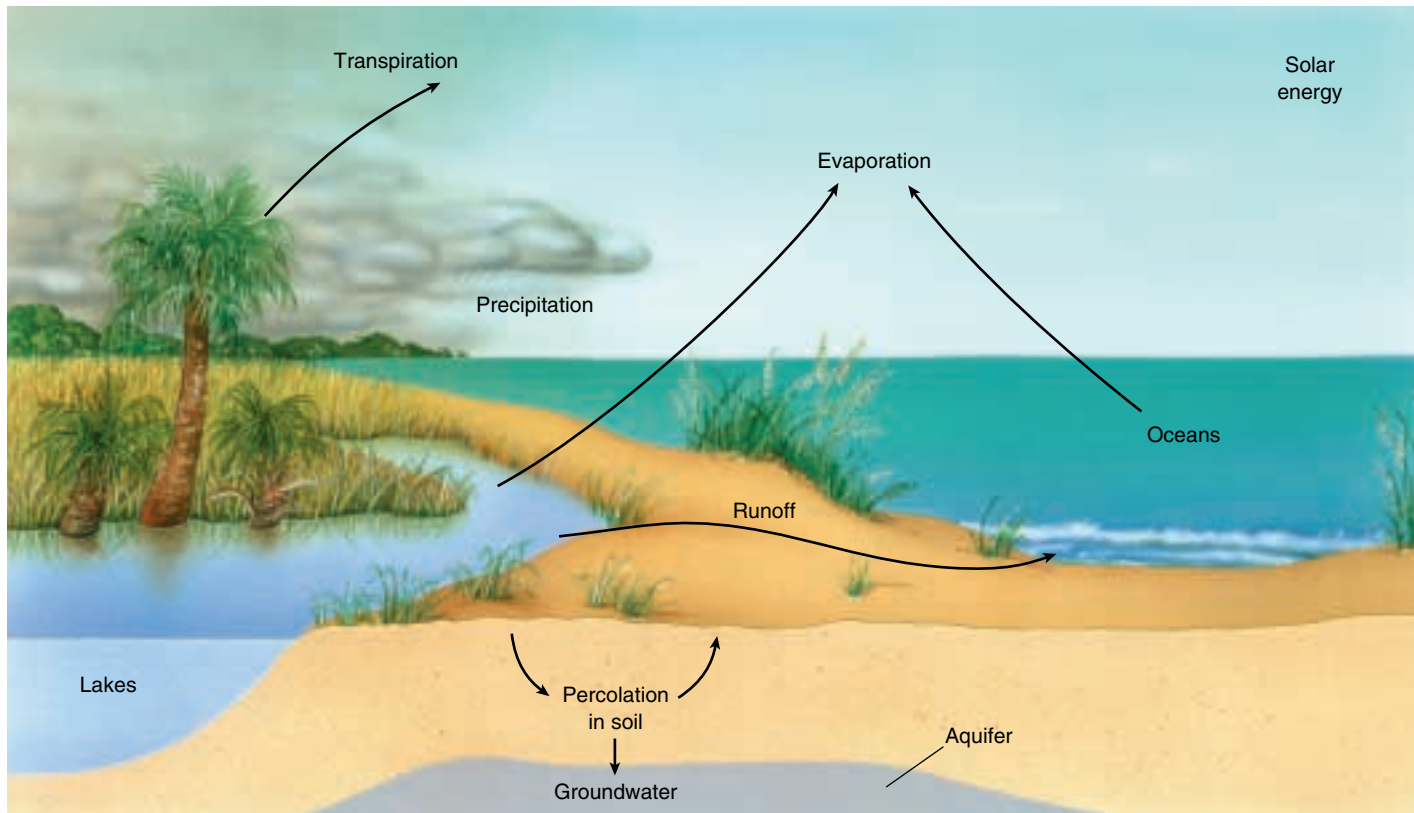
The cycling of materials in ecosystems begins when these chemicals are incorporated into the bodies of organisms from nonliving reservoirs such as the atmosphere or the waters of oceans or rivers. Many minerals, for example, first enter water from weathered rock, then pass into organisms when they drink the water. Materials pass from the organisms that first acquire them into the bodies of other organisms that eat them, until ultimately, through decomposition, they complete the cycle and return to the nonliving world.

### The Water Cycle

The water cycle (figure 28.2) is the most familiar of all biogeochemical cycles. All life depends directly on the presence of water; the bodies of most organisms consist mainly of this substance. Water is the source of hydrogen ions, whose movements generate ATP in organisms. For that reason alone, it is indispensable to their functioning.

#### The Path of Free Water

The oceans cover three-fourths of the earth's surface. From the oceans, water evaporates into the atmosphere, a process powered by energy from the sun. Over land approximately 90% of the water that reaches the atmosphere is moisture that evaporates from the surface of plants through a process called transpiration (see chapter 40). Most precipitation falls directly into the oceans, but some falls on land, where it passes into surface and subsurface bodies of fresh water. Only about 2% of all the water on earth is captured in any form—frozen, held in the soil, or incorporated into the bodies of organisms. All of the rest is free water, circulating between the atmosphere and the oceans.



**FIGURE 28.2**  
The water cycle. Water circulates from atmosphere to earth and back again.

## The Importance of Water to Organisms

Organisms live or die on the basis of their ability to capture water and incorporate it into their bodies. Plants take up water from the earth in a continuous stream. Crop plants require about 1000 kilograms of water to produce one kilogram of food, and the ratio in natural communities is similar. Animals obtain water directly or from the plants or other animals they eat. The amount of free water available at a particular place often determines the nature and abundance of the living organisms present there.

### Groundwater

Much less obvious than surface water, which we see in streams, lakes, and ponds, is groundwater, which occurs in aquifers—permeable, saturated, underground layers of rock, sand, and gravel. In many areas, groundwater is the most important reservoir of water. It amounts to more than 96% of all fresh water in the United States. The upper, unconfined portion of the groundwater constitutes the water table, which flows into streams and is partly accessible to plants; the lower confined layers are generally out of reach, although they can be “mined” by humans. The water table is recharged by water that percolates through the soil from precipitation as well as by water that seeps downward from ponds, lakes, and streams. The deep aquifers are recharged very slowly from the water table.

Groundwater flows much more slowly than surface water, anywhere from a few millimeters to a meter or so per day. In the United States, groundwater provides about 25% of the water used for all purposes and provides about 50% of the population with drinking water. Rural areas tend to depend almost exclusively on wells to access groundwater, and its use is growing at about twice the rate of surface water use. In the Great Plains of the central United States, the extensive use of the Ogallala Aquifer as a source of water for agricultural needs as well as for drinking water is depleting it faster than it can be naturally recharged. This seriously threatens the agricultural production of the area and similar problems are appearing throughout the drier portions of the globe.

Because of the greater rate of groundwater use, and because it flows so slowly, the increasing chemical pollution of groundwater is also a very serious problem. It is estimated that about 2% of the groundwater in the United States is already polluted, and the situation is worsening. Pesticides, herbicides, and fertilizers have become a serious threat to water purity. Another key source of groundwater pollution consists of the roughly 200,000 surface pits, ponds, and lagoons that are actively used for the disposal of chemical wastes in the United States alone. Because of the large volume of water, its slow rate of turnover, and its inaccessibility, removing pollutants from aquifers is virtually impossible.



**FIGURE 28.3**

**Deforestation breaks the water cycle.** As time goes by, the consequences of tropical deforestation may become even more severe, as the extensive erosion in this deforested area of Madagascar shows.

### Breaking the Water Cycle

In dense forest ecosystems such as tropical rainforests, more than 90% of the moisture in the ecosystem is taken up by plants and then transpired back into the air. Because so many plants in a rainforest are doing this, the vegetation is the primary source of local rainfall. In a very real sense, these plants create their own rain: the moisture that travels up from the plants into the atmosphere falls back to earth as rain.

Where forests are cut down, the organismic water cycle is broken, and moisture is not returned to the atmosphere. Water drains away from the area to the sea instead of rising to the clouds and falling again on the forest. As early as the late 1700s, the great German explorer Alexander von Humboldt reported that stripping the trees from a tropical rainforest in Colombia prevented water from returning to the atmosphere and created a semiarid desert. It is a tragedy of our time that just such a transformation is occurring in many tropical areas, as tropical and temperate rainforests are being clear-cut or burned in the name of “development” (figure 28.3). Much of Madagascar, a California-sized island off the east coast of Africa, has been transformed in this century from lush tropical forest into semiarid desert by deforestation. Because the rain no longer falls, there is no practical way to reforest this land. The water cycle, once broken, cannot be easily reestablished.

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**Water cycles between oceans and atmosphere. Some 96% of the fresh water in the United States consists of groundwater, which provides 25% of all the water used in this country.**

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## The Carbon Cycle

The **carbon cycle** is based on carbon dioxide, which makes up only about 0.03% of the atmosphere (figure 28.4). Worldwide, the synthesis of organic compounds from carbon dioxide and water through photosynthesis (see chapter 10) utilizes about 10% of the roughly 700 billion metric tons of carbon dioxide in the atmosphere each year. This enormous amount of biological activity takes place as a result of the combined activities of photosynthetic bacteria, protists, and plants. All terrestrial heterotrophic organisms obtain their carbon indirectly from photosynthetic organisms. When the bodies of dead organisms decompose, microorganisms release carbon dioxide back to the atmosphere. From there, it can be reincorporated into the bodies of other organisms.

Most of the organic compounds formed as a result of carbon dioxide fixation in the bodies of photosynthetic organisms are ultimately broken down and released back into the atmosphere or water. Certain carbon-containing compounds, such as cellulose, are more resistant to breakdown than others, but certain bacteria and fungi, as well as a few kinds of insects, are able to accomplish this feat. Some cellulose, however, accumulates as undecomposed

organic matter such as peat. The carbon in this cellulose may eventually be incorporated into fossil fuels such as oil or coal.

In addition to the roughly 700 billion metric tons of carbon dioxide in the atmosphere, approximately 1 trillion metric tons are dissolved in the ocean. More than half of this quantity is in the upper layers, where photosynthesis takes place. The fossil fuels, primarily oil and coal, contain more than 5 trillion additional metric tons of carbon, and between 600 million and 1 trillion metric tons are locked up in living organisms at any one time. In global terms, photosynthesis and respiration (see chapters 9 and 10) are approximately balanced, but the balance has been shifted recently because of the consumption of fossil fuels. The combustion of coal, oil, and natural gas has released large stores of carbon into the atmosphere as carbon dioxide. The increase of carbon dioxide in the atmosphere appears to be changing global climates, and may do so even more rapidly in the future, as we will discuss in chapter 30.

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**About 10% of the estimated 700 billion metric tons of carbon dioxide in the atmosphere is fixed annually by the process of photosynthesis.**

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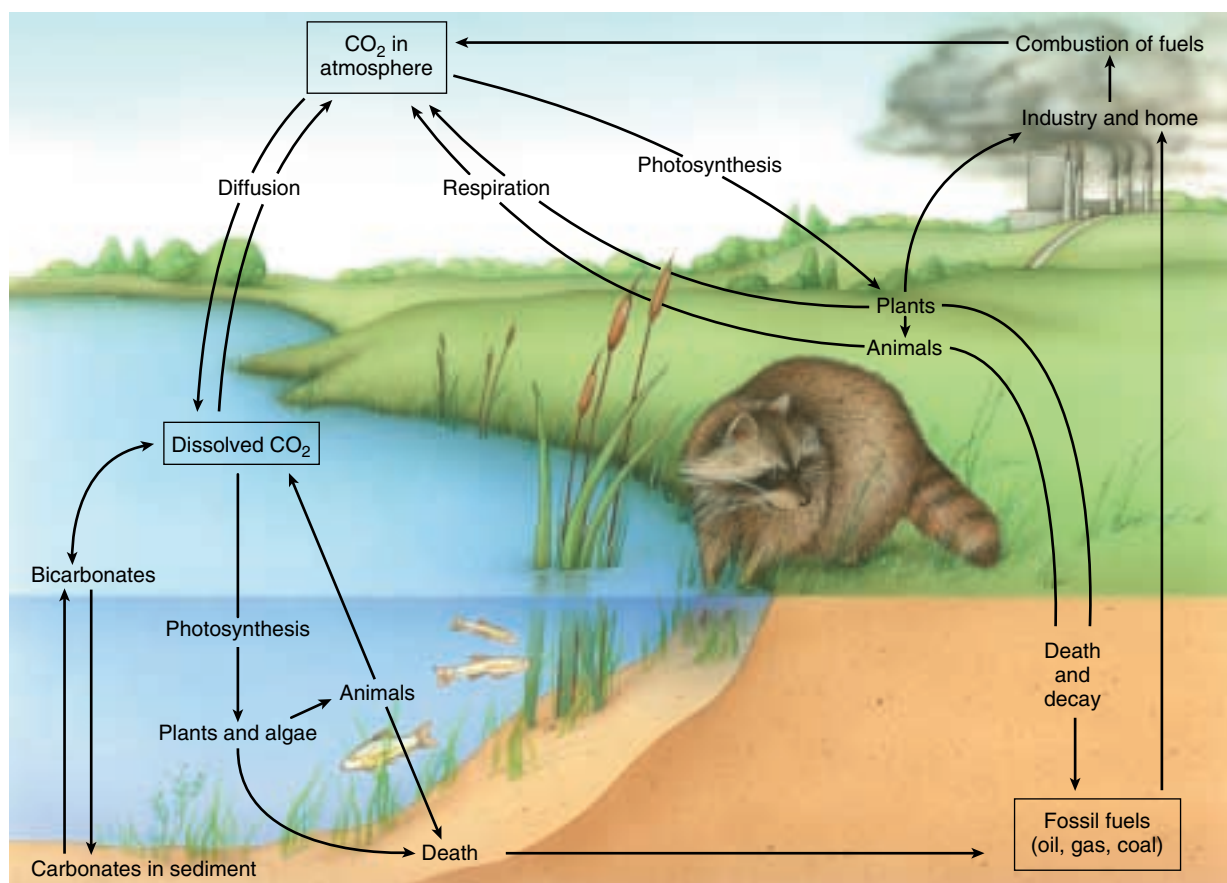
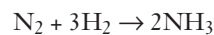


FIGURE 28.4

**The carbon cycle.** Photosynthesis captures carbon; respiration returns it to the atmosphere.

## The Nitrogen Cycle

Relatively few kinds of organisms—all of them bacteria—can convert, or fix, atmospheric nitrogen (78% of the earth's atmosphere) into forms that can be used for biological processes via the **nitrogen cycle** (figure 28.5). The triple bond that links together the two atoms that make up diatomic atmospheric nitrogen ( $\text{N}_2$ ) makes it a very stable molecule. In living systems the cleavage of atmospheric nitrogen is catalyzed by a complex of three proteins—ferredoxin, nitrogen reductase, and nitrogenase. This process uses ATP as a source of energy, electrons derived from photosynthesis or respiration, and a powerful reducing agent. The overall reaction of nitrogen fixation is written:



Some genera of bacteria have the ability to fix atmospheric nitrogen. Most are free-living, but some form symbiotic relationships with the roots of legumes (plants of the pea family, Fabaceae) and other plants. Only the symbiotic bacteria fix enough nitrogen to be of major significance in nitrogen production. Because of the activities of such organisms in the past, a large reservoir of ammonia and nitrates

now exists in most ecosystems. This reservoir is the immediate source of much of the nitrogen used by organisms.

Nitrogen-containing compounds, such as proteins in plant and animal bodies, are decomposed rapidly by certain bacteria and fungi. These bacteria and fungi use the amino acids they obtain through decomposition to synthesize their own proteins and to release excess nitrogen in the form of ammonium ions ( $\text{NH}_4^+$ ), a process known as **ammonification**. The ammonium ions can be converted to soil nitrites and nitrates by certain kinds of organisms and which then can be absorbed by plants.

A certain proportion of the fixed nitrogen in the soil is steadily lost. Under anaerobic conditions, nitrate is often converted to nitrogen gas ( $\text{N}_2$ ) and nitrous oxide ( $\text{N}_2\text{O}$ ), both of which return to the atmosphere. This process, which several genera of bacteria carry out, is called **denitrification**.

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**Nitrogen becomes available to organisms almost entirely through the metabolic activities of bacteria, some free-living and others which live symbiotically in the roots of legumes and other plants.**

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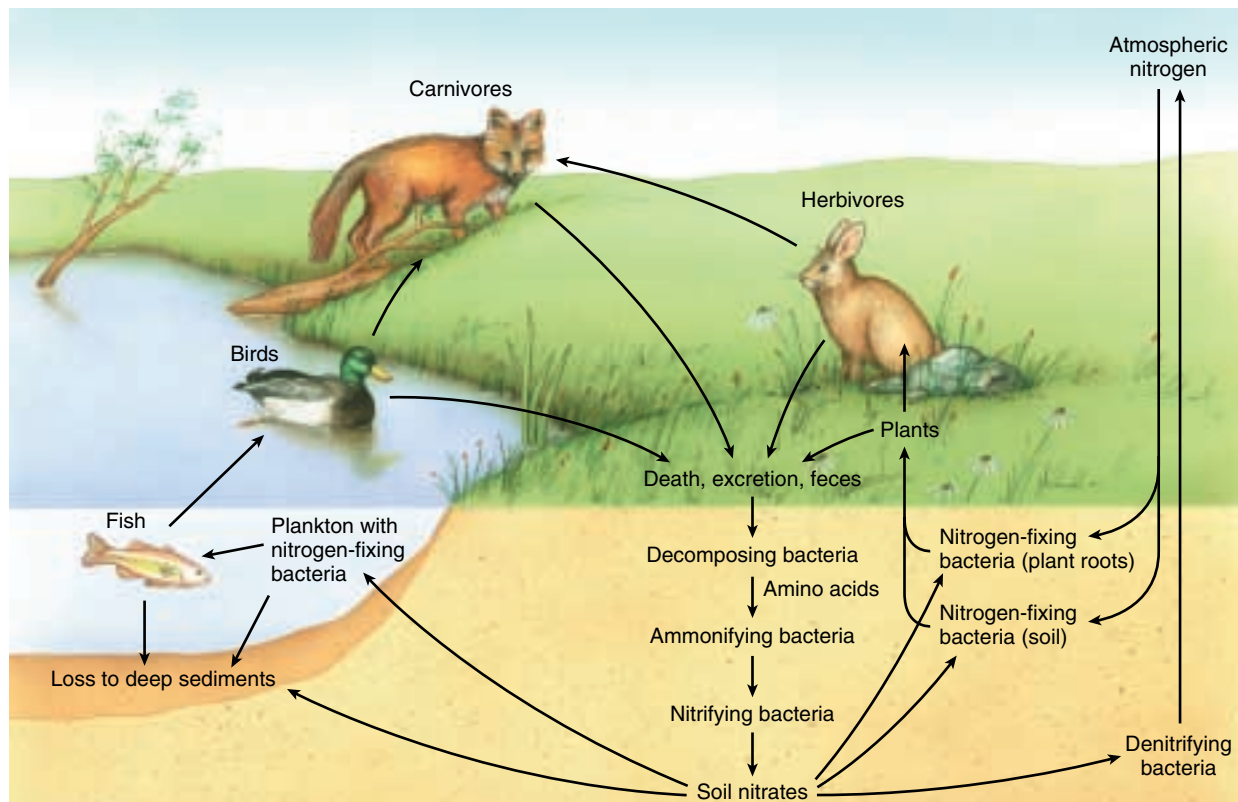


FIGURE 28.5

**The nitrogen cycle.** Certain bacteria fix atmospheric nitrogen, converting it to a form living organisms can use. Other bacteria decompose nitrogen-containing compounds from plant and animal materials, returning it to the atmosphere.

## The Phosphorus Cycle

In all biogeochemical cycles other than those involving water, carbon, oxygen, and nitrogen, the reservoir of the nutrient exists in mineral form, rather than in the atmosphere. The **phosphorus cycle** (figure 28.6) is presented as a representative example of all other mineral cycles. Phosphorus, a component of ATP, phospholipids, and nucleic acid, plays a critical role in plant nutrition.

Of all the required nutrients other than nitrogen, phosphorus is the most likely to be scarce enough to limit plant growth. Phosphates, in the form of phosphorus anions, exist in soil only in small amounts. This is because they are relatively insoluble and are present only in certain kinds of rocks. As phosphates weather out of soils, they are transported by rivers and streams to the oceans, where they accumulate in sediments. They are naturally brought back up again only by the uplift of lands, such as occurs along the Pacific coast of North and South America, creating upwelling currents. Phosphates brought to the surface are assimilated by algae, and then by fish, which are in turn eaten by birds. Seabirds deposit enormous amounts of guano

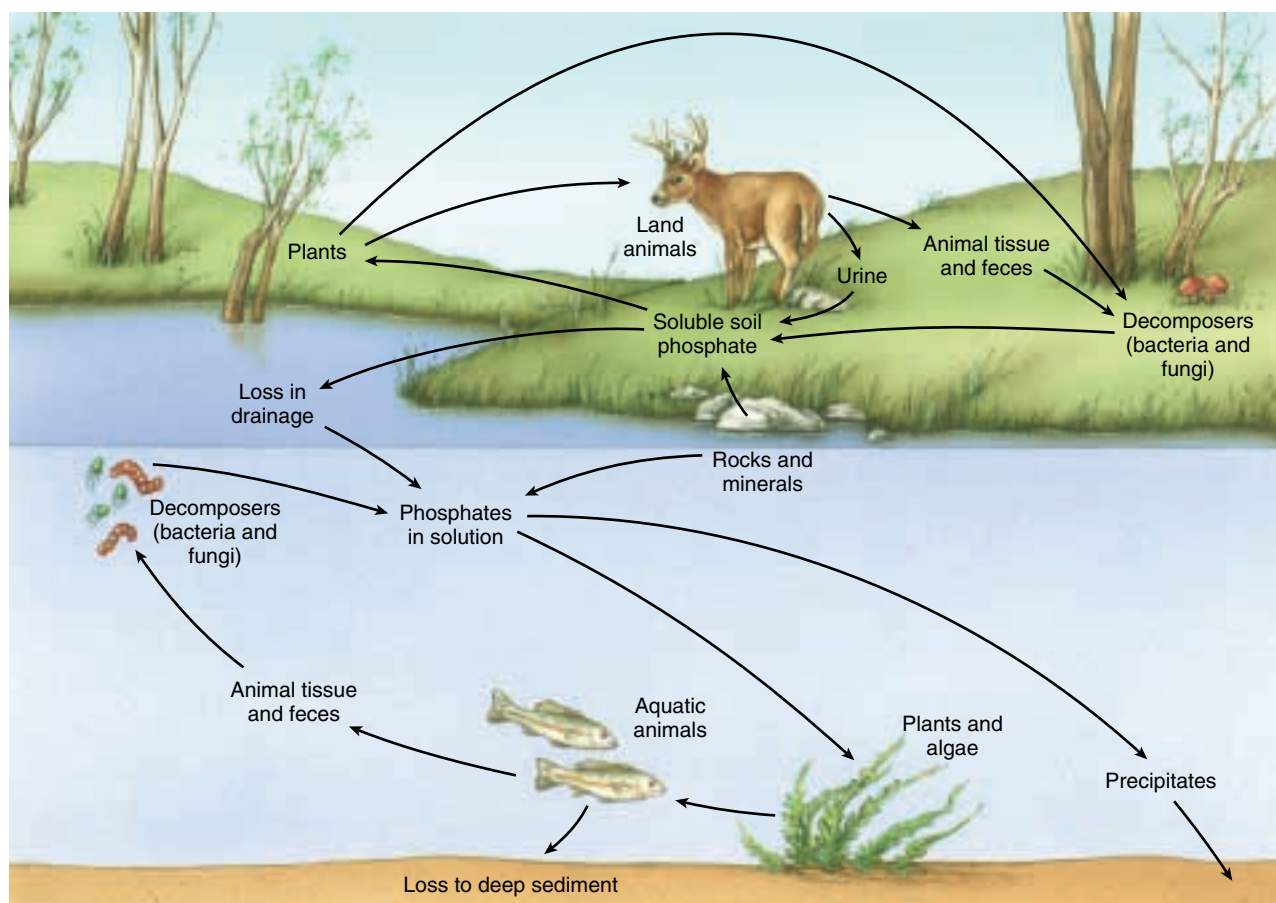
(feces) rich in phosphorus along certain coasts. Guano deposits have traditionally been used for fertilizer. Crushed phosphate-rich rocks, found in certain regions, are also used for fertilizer. The seas are the only inexhaustible source of phosphorus, making deep-seabed mining look increasingly commercially attractive.

Every year, millions of tons of phosphate are added to agricultural lands in the belief that it becomes fixed to and enriches the soil. In general, three times more phosphate than a crop requires is added each year. This is usually in the form of **superphosphate**, which is soluble calcium dihydrogen phosphate,  $\text{Ca}(\text{H}_2\text{PO}_4)_2$ , derived by treating bones or apatite, the mineral form of calcium phosphate, with sulfuric acid. But the enormous quantities of phosphates that are being added annually to the world's agricultural lands are not leading to proportionate gains in crops. Plants can apparently use only so much of the phosphorus that is added to the soil.

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**Phosphates are relatively insoluble and are present in most soils only in small amounts. They often are so scarce that their absence limits plant growth.**

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**FIGURE 28.6**

**The phosphorus cycle.** Phosphates weather from soils into water, enter plants and animals, and are redeposited in the soil when plants and animals decompose.

## Biogeochemical Cycles Illustrated: Recycling in a Forest Ecosystem

An ongoing series of studies conducted at the Hubbard Brook Experimental Forest in New Hampshire has revealed in impressive detail the overall recycling pattern of nutrients in an ecosystem. The way this particular ecosystem functions, and especially the way nutrients cycle within it, has been studied since 1963 by Herbert Bormann of the Yale School of Forestry and Environmental Studies, Gene Likens of the Institute of Ecosystem Studies, and their colleagues. These studies have yielded much of the available information about the cycling of nutrients in forest ecosystems. They have also provided the basis for the development of much of the experimental methodology that is being applied successfully to the study of other ecosystems.

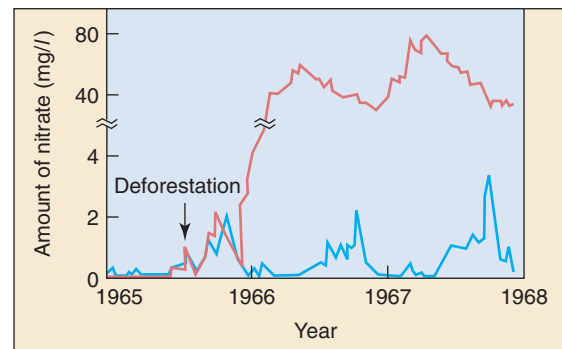
Hubbard Brook is the central stream of a large watershed that drains a region of temperate deciduous forest. To measure the flow of water and nutrients within the Hubbard Brook ecosystem, concrete weirs with V-shaped notches were built across six tributary streams. All of the water that flowed out of the valleys had to pass through the notches, as the weirs were anchored in bedrock. The researchers measured the precipitation that fell in the six valleys, and determined the amounts of nutrients that were present in the water flowing in the six streams. By these methods, they demonstrated that the undisturbed forests in this area were very efficient at retaining nutrients; the small amounts of nutrients that precipitated from the atmosphere with rain and snow were approximately equal to the amounts of nutrients that ran out of the valleys. These quantities were very low in relation to the total amount of nutrients in the system. There was a small net loss of calcium—about 0.3% of the total calcium in the system per year—and small net gains of nitrogen and potassium.

In 1965 and 1966, the investigators felled all the trees and shrubs in one of the six watersheds and then prevented regrowth by spraying the area with herbicides. The effects were dramatic. The amount of water running out of that valley increased by 40%. This indicated that water that previously would have been taken up by vegetation and ultimately evaporated into the atmosphere was now running off. For the four-month period from June to September 1966, the runoff was four times higher than it had been during comparable periods in the preceding years. The amounts of nutrients running out of the system also greatly increased; for example, the loss of calcium was 10 times higher than it had been previously. Phosphorus, on the other hand, did not increase in the stream water; it apparently was locked up in the soil.

The change in the status of nitrogen in the disturbed valley was especially striking (figure 28.7). The undisturbed ecosystem in this valley had been accumulating nitrogen at a rate of about 2 kilograms per hectare per year,



(a)



(b)

**FIGURE 28.7**

**The Hubbard Brook experiment.** (a) A 38-acre watershed was completely deforested, and the runoff monitored for several years. (b) Deforestation greatly increased the loss of minerals in runoff water from the ecosystem. The red curve represents nitrate in the runoff water from the deforested watershed; the blue curve, nitrate in runoff water from an undisturbed neighboring watershed.

but the deforested ecosystem *lost* nitrogen at a rate of about 120 kilograms per hectare per year. The nitrate level of the water rapidly increased to a level exceeding that judged safe for human consumption, and the stream that drained the area generated massive blooms of cyanobacteria and algae. In other words, the fertility of this logged-over valley decreased rapidly, while at the same time the danger of flooding greatly increased. This experiment is particularly instructive at the start of the twenty-first century, as large areas of tropical rain forest are being destroyed to make way for cropland, a topic that will be discussed further in chapter 30.

When the trees and shrubs in one of the valleys in the Hubbard Brook watershed were cut down and the area was sprayed with herbicide, water runoff and the loss of nutrients from that valley increased. Nitrogen, which had been accumulating at a rate of about 2 kilograms per hectare per year, was lost at a rate of 120 kilograms per hectare per year.

## 28.2 Ecosystems are structured by who eats whom.

### Trophic Levels

An ecosystem includes autotrophs and heterotrophs. **Autotrophs** are plants, algae, and some bacteria that are able to capture light energy and manufacture their own food. To support themselves, **heterotrophs**, which include animals, fungi, most protists and bacteria, and nongreen plants, must obtain organic molecules that have been synthesized by autotrophs. Autotrophs are also called **primary producers**, and heterotrophs are also called **consumers**.

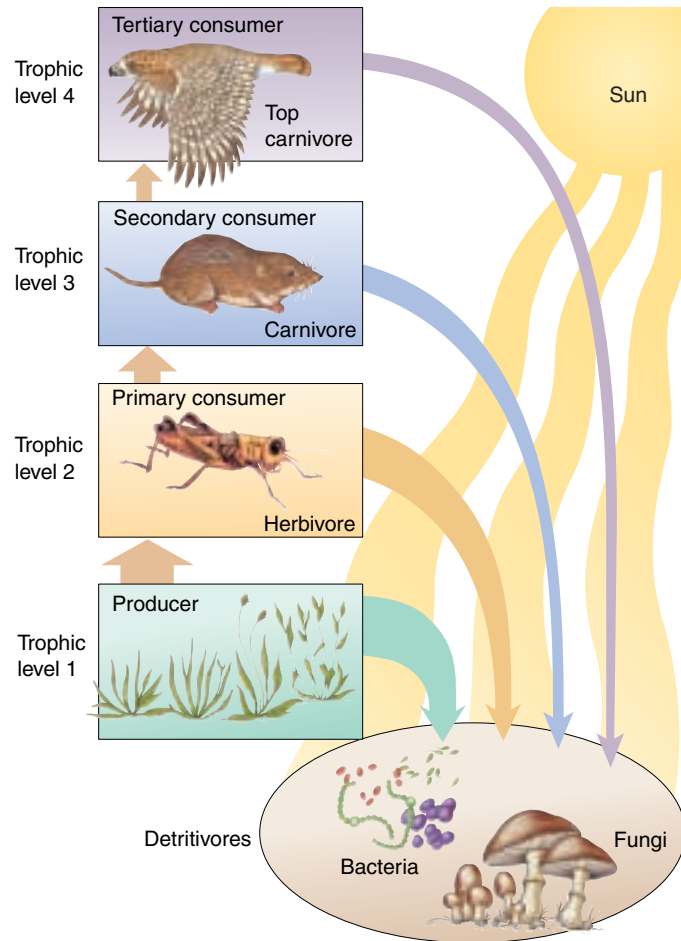
Once energy enters an ecosystem, usually as the result of photosynthesis, it is slowly released as metabolic processes proceed. The autotrophs that first acquire this energy provide all of the energy heterotrophs use. The organisms that make up an ecosystem delay the release of the energy obtained from the sun back into space.

Green plants, the primary producers of a terrestrial ecosystem, generally capture about 1% of the energy that falls on their leaves, converting it to food energy. In especially productive systems, this percentage may be a little higher. When these plants are consumed by other organisms, only a portion of the plant's accumulated energy is actually converted into the bodies of the organisms that consume them.

Several different levels of consumers exist. The **primary consumers**, or herbivores, feed directly on the green plants. **Secondary consumers**, carnivores and the parasites of animals, feed in turn on the herbivores. **Decomposers** break down the organic matter accumulated in the bodies of other organisms. Another more general term that includes decomposers is **detritivores**. Detritivores live on the refuse of an ecosystem. They include large scavengers, such as crabs, vultures, and jackals, as well as decomposers.

All of these categories occur in any ecosystem. They represent different **trophic levels**, from the Greek word *trophos*, which means "feeder." Organisms from each trophic level, feeding on one another, make up a series called a **food chain** (figure 28.8). The length and complexity of food chains vary greatly. In real life, it is rather rare for a given kind of organism to feed only on one other type of organism. Usually, each organism feeds on two or more kinds and in turn is eaten by several other kinds of organisms. When diagrammed, the relationship appears as a series of branching lines, rather than a straight line; it is called a **food web** (figure 28.9).

A certain amount of the chemical-bond energy ingested by the organisms at a given trophic level goes toward staying alive (for example, carrying out mechanical motion). Using the chemical-bond energy converts it to heat, which organisms cannot use to do work. Another portion of the chemical-bond energy taken in is retained as chemical-bond energy within the organic molecules produced by growth. Usually 40% or less of the energy ingested is

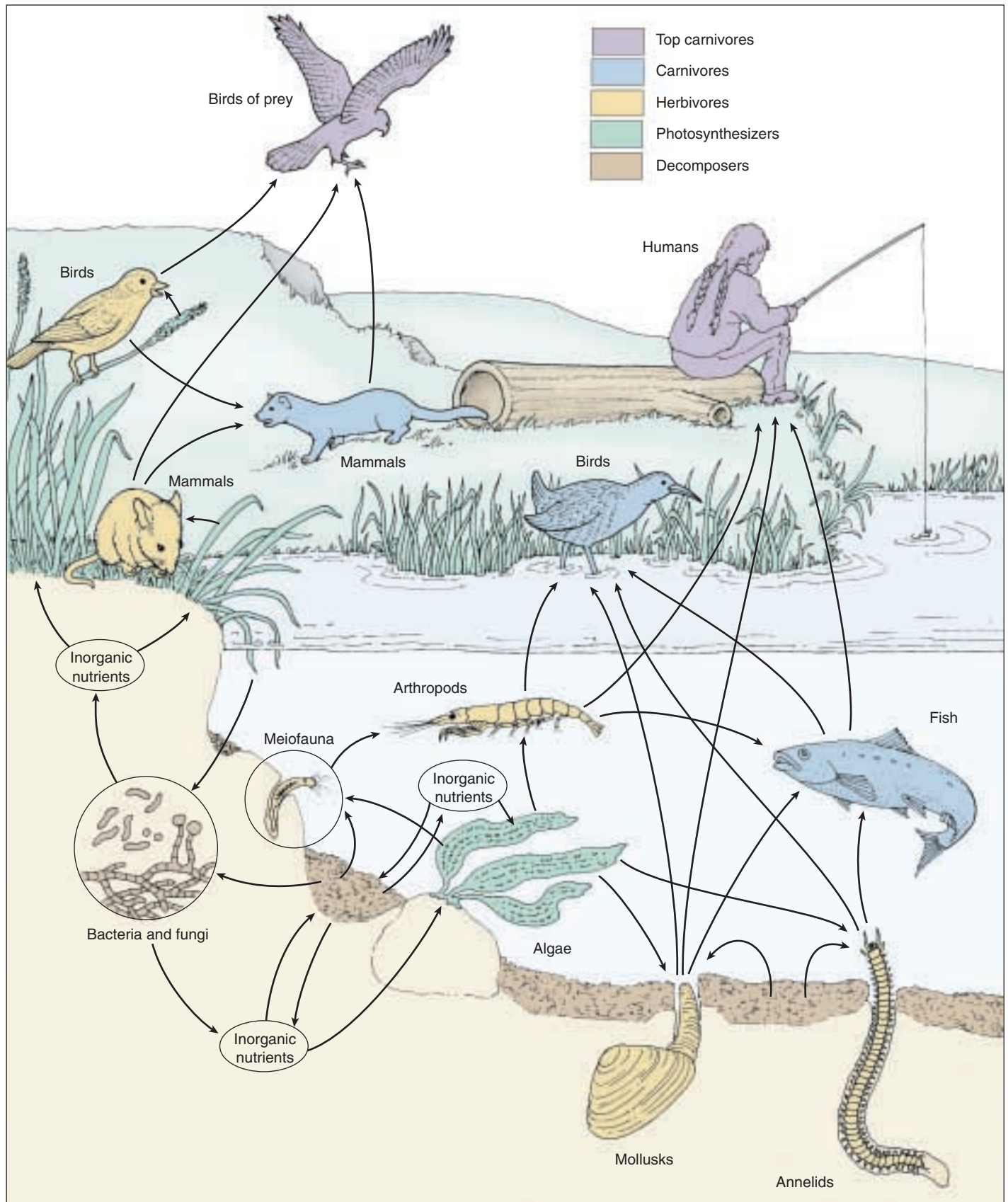


**FIGURE 28.8**

**Trophic levels within a food chain.** Plants obtain their energy directly from the sun, placing them at trophic level 1. Animals that eat plants, such as grasshoppers, are primary consumers or herbivores and are at trophic level 2. Animals that eat plant-eating animals, such as shrews, are carnivores and are at trophic level 3 (secondary consumers); animals that eat carnivorous animals, such as hawks, are tertiary consumers at trophic level 4. Detritivores use all trophic levels for food.

stored by growth. An invertebrate typically uses about a quarter of this 40% for growth; in other words, about 10% of the food an invertebrate eats is turned into its own body and thus into potential food for its predators. Although the comparable figure varies from approximately 5% in carnivores to nearly 20% for herbivores, 10% is a good average value for the amount of organic matter that reaches the next trophic level.

**Energy passes through ecosystems, a good deal being lost at each step.**



**FIGURE 28.9**

The food web in a salt marsh shows the complex interrelationships among organisms. The meiofauna are very small animals that live between the grains of sand.

## 28.3 Energy flows through ecosystems.

### Primary Productivity

Approximately 1 to 5% of the solar energy that falls on a plant is converted to the chemical bonds of organic material. **Primary production** or **primary productivity** are terms used to describe the amount of organic matter produced from solar energy in a given area during a given period of time. **Gross primary productivity** is the total organic matter produced, including that used by the photosynthetic organism for respiration. **Net primary productivity (NPP)**, therefore, is a measure of the amount of organic matter produced in a community in a given time that is available for heterotrophs. It equals the gross primary productivity minus the amount of energy expended by the metabolic activities of the photosynthetic organisms. The net weight of all of the organisms living in an ecosystem, its **biomass**, increases as a result of its net production.

### Productive Biological Communities

Some ecosystems have a high net primary productivity. For example, tropical forests and wetlands normally produce between 1500 and 3000 grams of organic material per square meter per year. By contrast, corresponding figures for other communities include 1200 to 1300 grams for temperate forests, 900 grams for savanna, and 90 grams for deserts (table 28.1).

### Secondary Productivity

The rate of production by heterotrophs is called **secondary productivity**. Because herbivores and carnivores cannot carry out photosynthesis, they do not manufacture biomolecules directly from CO<sub>2</sub>. Instead, they obtain them by eating plants or other heterotrophs. Secondary productivity by herbivores is approximately an order of magnitude less than the primary productivity upon which it is based. Where does all the energy in plants that is not captured by herbivores go (figure 28.10)? First, much of the biomass is not consumed by herbivores and instead supports the decomposer community (bacteria, fungi and detritivorous animals). Second, some energy is not assimilated by the herbivore's body but is passed on as feces to the decomposers. Third, not all the chemical-bond energy which herbivores assimilate is retained as chemical-bond energy in the organic molecules of their tissues. Some of it is lost as heat produced by work.

**Primary productivity occurs as a result of photosynthesis, which is carried out by green plants, algae, and some bacteria. Secondary productivity is the production of new biomass by heterotrophs.**

Table 28.1 Terrestrial Ecosystem Productivity Per Year

| Ecosystem Type                      | Net Primary Productivity (NPP)        |                                  |
|-------------------------------------|---------------------------------------|----------------------------------|
|                                     | NPP per Unit Area (g/m <sup>2</sup> ) | World NPP (10 <sup>9</sup> tons) |
| Extreme desert, rock, sand, and ice | 3                                     | 0.07                             |
| Desert and semidesert shrub         | 90                                    | 1.6                              |
| Tropical rain forest                | 2200                                  | 37.4                             |
| Savanna                             | 900                                   | 13.5                             |
| Cultivated land                     | 650                                   | 9.1                              |
| Boreal forest                       | 800                                   | 9.6                              |
| Temperate grassland                 | 600                                   | 5.4                              |
| Woodland and shrubland              | 700                                   | 6.0                              |
| Tundra and alpine                   | 140                                   | 1.1                              |
| Tropical seasonal forest            | 1600                                  | 12.0                             |
| Temperate deciduous forest          | 1200                                  | 8.4                              |
| Temperate evergreen forest          | 1300                                  | 6.5                              |
| Wetlands                            | 2000                                  | 4.0                              |

Source: After Whittaker, 1975.

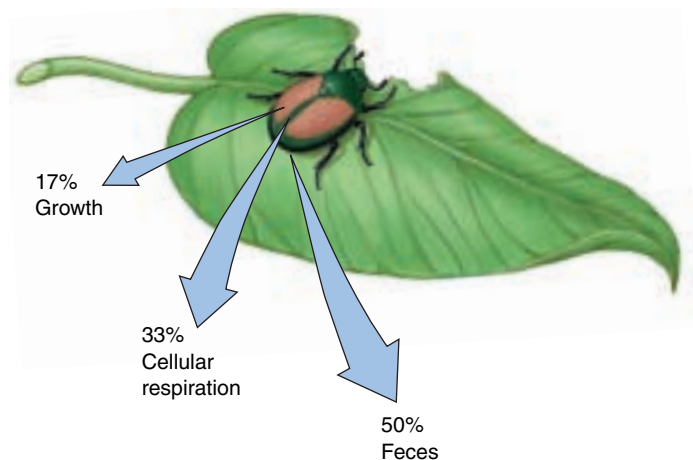
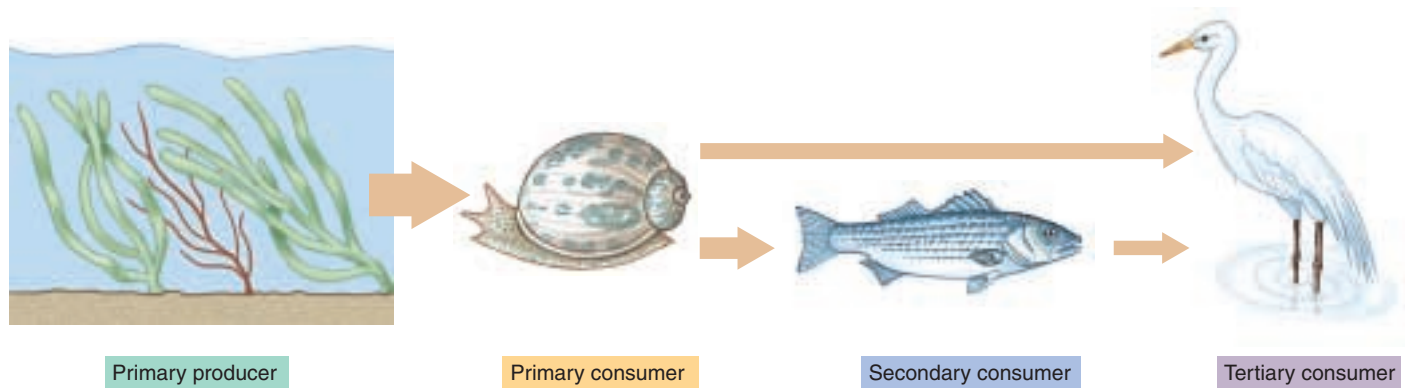


FIGURE 28.10

**How heterotrophs utilize food energy.** A heterotroph assimilates only a fraction of the energy it consumes. For example, if the “bite” of a herbivorous insect comprises 500 Joules of energy (1 Joule = 0.239 calories), about 50%, 250 J, is lost in feces, about 33%, 165 J, is used to fuel cellular respiration, and about 17%, 85 J, is converted into insect biomass. Only this 85 J is available to the next trophic level.



**FIGURE 28.11**

**A food chain.** Because so much energy is lost at each step, food chains usually consist of just three or four steps.

## The Energy in Food Chains

Food chains generally consist of only three or four steps (figure 28.11). So much energy is lost at each step that very little usable energy remains in the system after it has been incorporated into the bodies of organisms at four successive trophic levels.

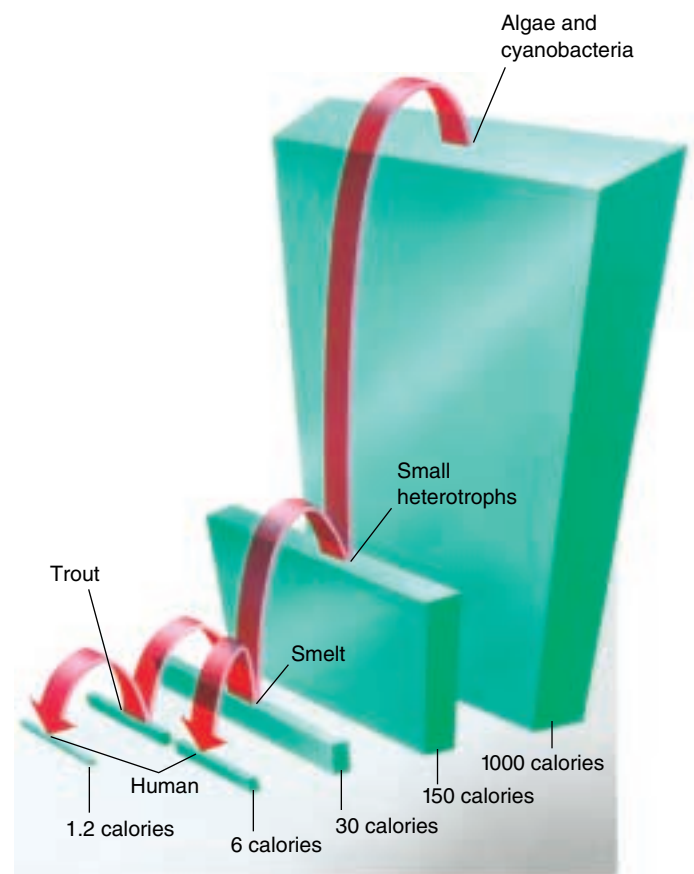
### Community Energy Budgets

Lamont Cole of Cornell University studied the flow of energy in a freshwater ecosystem in Cayuga Lake in upstate New York. He calculated that about 150 of each 1000 calories of potential energy fixed by algae and cyanobacteria are transferred into the bodies of small heterotrophs (figure 28.12). Of these, about 30 calories are incorporated into the bodies of smelt, small fish that are the principal secondary consumers of the system. If humans eat the smelt, they gain about 6 of the 1000 calories that originally entered the system. If trout eat the smelt and humans eat the trout, humans gain only about 1.2 calories.

### Factors Limiting Community Productivity

Communities with higher productivity can in theory support longer food chains. The limit on a community's productivity is determined ultimately by the amount of sunlight it receives, for this determines how much photosynthesis can occur. This is why in the deciduous forests of North America the net primary productivity increases as the growing season lengthens. NPP is higher in warm climates than cold ones not only because of the longer growing seasons, but also because more nitrogen tends to be available in warm climates, where nitrogen-fixing bacteria are more active.

**Considerable energy is lost at each stage in food chains, which limits their length. In general, more productive food chains can support longer food chains.**



**FIGURE 28.12**

**The food web in Cayuga Lake.** Autotrophic plankton (algae and cyanobacteria) fix the energy of the sun, heterotrophic plankton feed on them, and are both consumed by smelt. The smelt are eaten by trout, with about a fivefold loss in fixed energy; for humans, the amount of smelt biomass is at least five times greater than that available in trout, although humans prefer to eat trout.

## Ecological Pyramids

A plant fixes about 1% of the sun's energy that falls on its green parts. The successive members of a food chain, in turn, process into their own bodies about 10% of the energy available in the organisms on which they feed. For this reason, there are generally far more individuals at the lower trophic levels of any ecosystem than at the higher levels. Similarly, the biomass of the primary producers present in a given ecosystem is greater than the biomass of the primary consumers, with successive trophic levels having a lower and lower biomass and correspondingly less potential energy.

These relationships, if shown diagrammatically, appear as pyramids (figure 28.13). We can speak of "pyramids of biomass," "pyramids of energy," "pyramids of number," and so forth, as characteristic of ecosystems.

### Inverted Pyramids

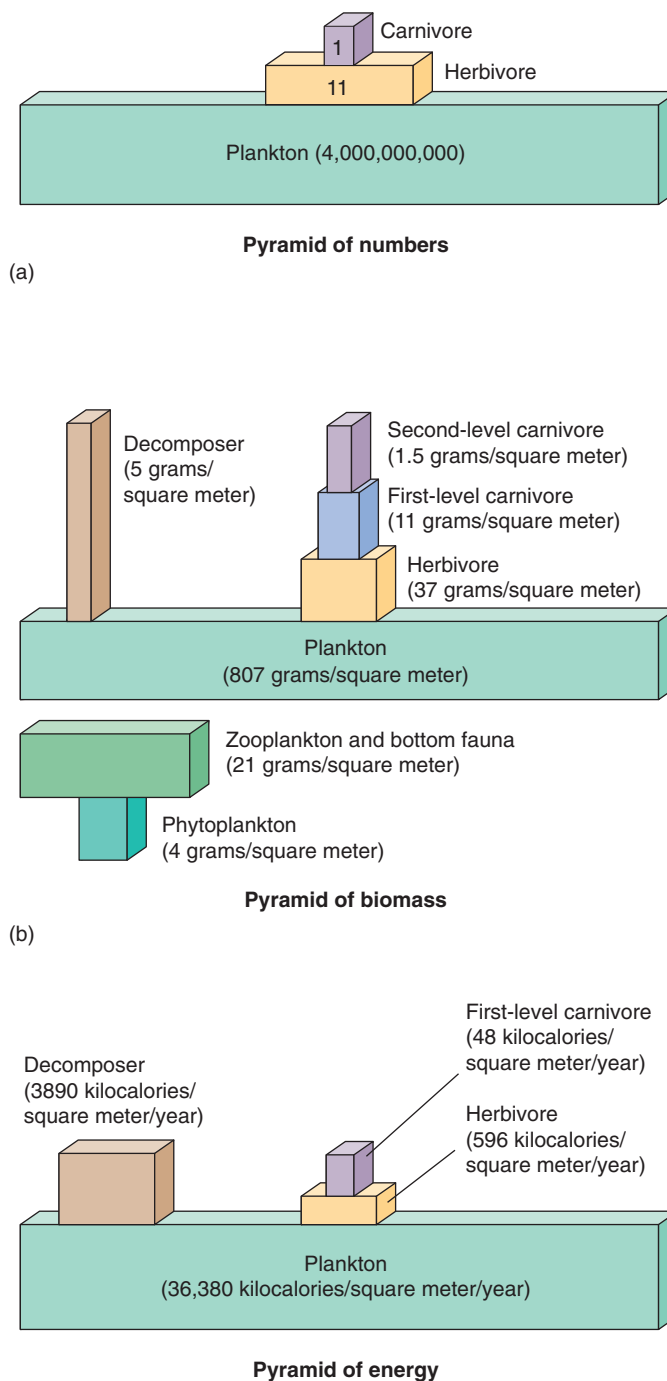
Some aquatic ecosystems have inverted biomass pyramids. For example, in a planktonic ecosystem—dominated by small organisms floating in water—the turnover of photosynthetic phytoplankton at the lowest level is very rapid, with zooplankton consuming phytoplankton so quickly that the phytoplankton (the producers at the base of the food chain) can never develop a large population size. Because the phytoplankton reproduce very rapidly, the community can support a population of heterotrophs that is larger in biomass and more numerous than the phytoplankton (see figure 28.13*b*).

### Top Carnivores

The loss of energy that occurs at each trophic level places a limit on how many top-level carnivores a community can support. As we have seen, only about one-thousandth of the energy captured by photosynthesis passes all the way through a three-stage food chain to a tertiary consumer such as a snake or hawk. This explains why there are no predators that subsist on lions or eagles—the biomass of these animals is simply insufficient to support another trophic level.

In the pyramid of numbers, top-level predators tend to be fairly large animals. Thus, the small residual biomass available at the top of the pyramid is concentrated in a relatively small number of individuals.

**Because energy is lost at every step of a food chain, the biomass of primary producers (photosynthesizers) tends to be greater than that of the herbivores that consume them, and herbivore biomass greater than the biomass of the predators that consume them.**



**FIGURE 28.13**

**Ecological pyramids.** Ecological pyramids measure different characteristics of each trophic level. (a) Pyramid of numbers. (b) Pyramids of biomass, both normal (*top*) and inverted (*bottom*). (c) Pyramid of energy.

## Interactions among Different Trophic Levels

The existence of food webs creates the possibility of interactions among species at different trophic levels. Predators will not only have effects on the species upon which they prey, but also, indirectly, upon the plants eaten by these prey. Conversely, increases in primary productivity will not only provide more food for herbivores but, indirectly, lead also to more food for carnivores.

### Trophic Cascades

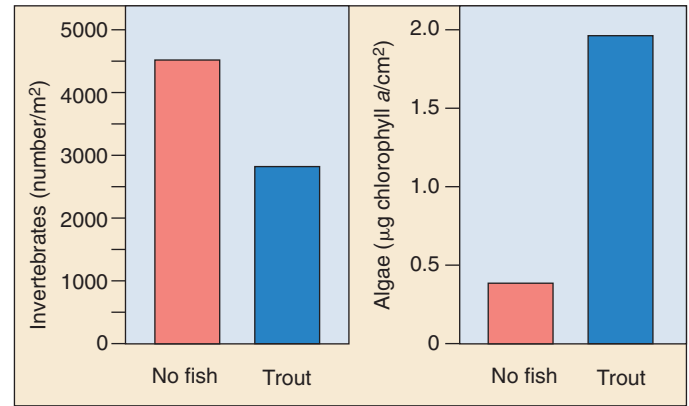
When we look at the world around us, we see a profusion of plant life. Why is this? Why don't herbivore populations increase to the extent that all available vegetation is consumed? The answer, of course, is that predators keep the herbivore populations in check, thus allowing plant populations to thrive. This phenomenon, in which the effect of one trophic level flows down to lower levels, is called a **trophic cascade**.

Experimental studies have confirmed the existence of trophic cascades. For example, in one study in New Zealand, sections of a stream were isolated with a mesh that prevented fish from entering. In some of the enclosures, brown trout were added, whereas other enclosures were left without large fish. After 10 days, the number of invertebrates in the trout enclosures was one-half of that in the controls (figure 28.14). In turn, the biomass of algae, which invertebrates feed upon, was five times greater in the trout enclosures than in the controls.

The logic of trophic cascades leads to the prediction that a fourth trophic level, carnivores that preyed on other carnivores, would also lead to cascading effects. In this case, the top predators would keep lower-level predator populations in check, which should lead to a profusion of herbivores and a paucity of vegetation. In an experiment similar to the one just described, enclosures were created in free-flowing streams in northern California. In this case, large predatory fish were added to some enclosures and not others. In the large fish enclosures, the number of smaller predators, such as damselfly nymphs was greatly reduced, leading to an increase in their prey, including algae-eating insects, which lead, in turn, to decreases in the biomass of algae (figure 28.15).

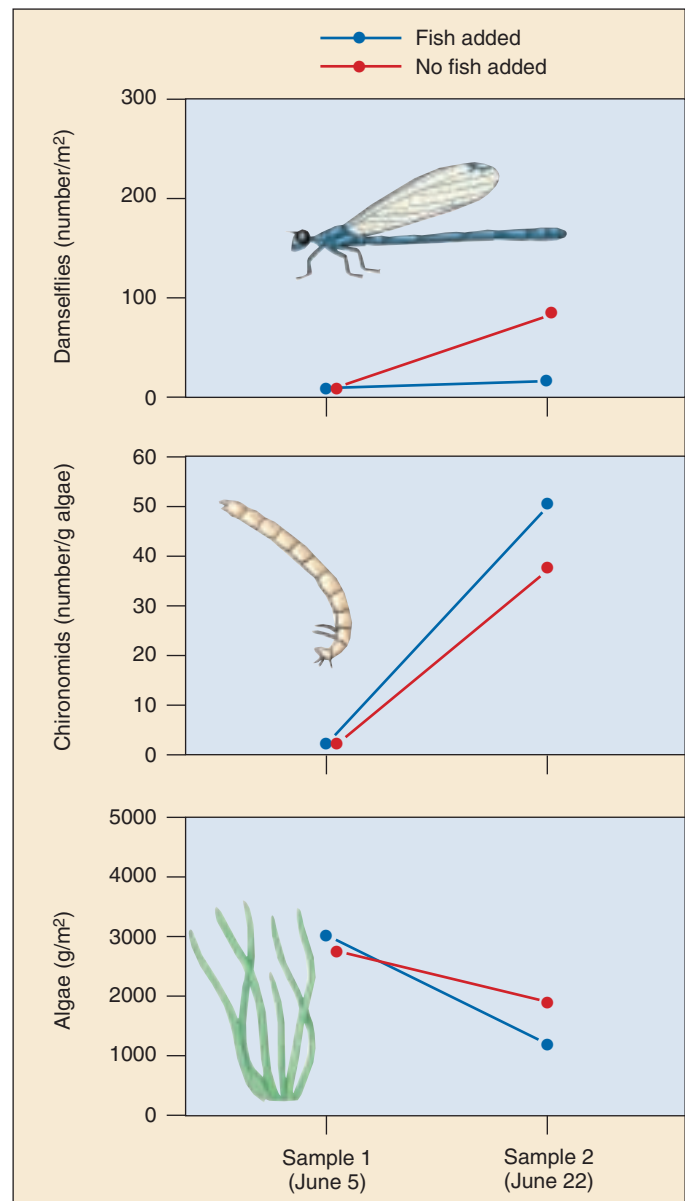
**FIGURE 28.15**

**Four-level trophic cascades.** Streams with fish have fewer lower-level predators, such as damselflies, more herbivorous insects (exemplified by the number of chironomids, a type of aquatic insect), and lower levels of algae.



**FIGURE 28.14**

**Trophic cascades.** Streams with trout have fewer herbivorous invertebrates and more algae than streams without trout.



## Human Effects on Trophic Cascades

Humans have inadvertently created a test of the trophic cascade hypothesis by removing top predators from ecosystems. The great naturalist Aldo Leopold captured the results long before the trophic cascade hypothesis had ever been scientifically articulated when he wrote in the *Sand County Almanac*:

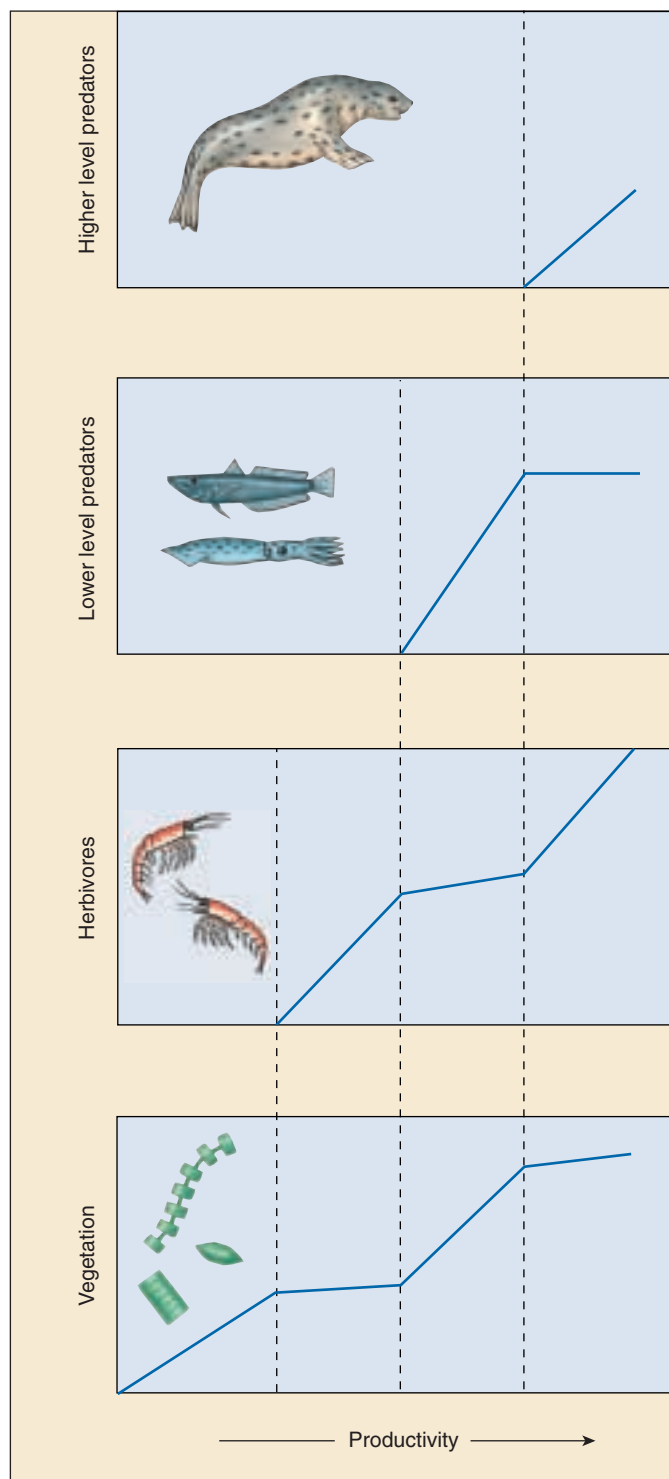
“I have lived to see state after state extirpate its wolves. I have watched the face of many a new wolfless mountain, and seen the south-facing slopes wrinkle with a maze of new deer trails. I have seen every edible bush and seedling browsed, first to anemic desuetude, and then to death. I have seen every edible tree defoliated to the height of a saddle horn.”

Many similar examples exist in nature in which the removal of predators has led to cascading effects on lower trophic levels. On Barro Colorado Island, a hilltop turned into an island by the construction of the Panama Canal at the beginning of the last century, large predators such as jaguars and mountain lions are absent. As a result, smaller predators whose populations are normally held in check—including monkeys, peccaries (a relative of the pig), coatimundis and armadillos—have become extraordinarily abundant. These animals will eat almost anything they find. Ground-nesting birds are particularly vulnerable, and many species have declined; at least 15 bird species have vanished from the island entirely. Similarly, in woodlots in the mid-western United States, raccoons, opossums, and foxes have become abundant due to the elimination of larger predators, and populations of ground-nesting birds have declined greatly.

## Bottom-Up Effects

Conversely, factors acting at the bottom of food webs may have consequences that ramify to higher trophic levels, leading to what are termed **bottom-up effects**. The basic idea is when the productivity of an ecosystem is low, herbivore populations will be too small to support any predators. Increases in productivity will be entirely devoured by the herbivores, whose populations will increase in size. At some point, herbivore populations will become large enough that predators can be supported. Thus, further increases in productivity will not lead to increases in herbivore populations, but, rather to increases in predator populations. Again, at some level, top predators will become established that can prey on lower-level predators. With the lower-level predator populations in check, herbivore populations will again increase with increasing productivity (figure 28.16).

Experimental evidence for the role of bottom-up effects was provided in an elegant study conducted on the Eel River in northern California. Enclosures were constructed that excluded large fish. A roof was placed above each enclosure. Some roofs were clear and let light pass



**FIGURE 28.16**

**Bottom-up effects.** At low levels of productivity, herbivore populations cannot be maintained. Above some threshold, increases in productivity lead to increases in herbivore biomass; vegetation biomass no longer increases with productivity because it is converted into herbivore biomass. Similarly, above another threshold, herbivore biomass gets converted to carnivore biomass. At this point, vegetation biomass is no longer constrained by herbivores, and so again increases with increasing productivity.

through, whereas others produced light or deep shade. The result was that the enclosures differed in the amount of sunlight reaching them. As one might expect, the primary productivity differed and was greatest in the unshaded enclosures. This increased productivity led to both more vegetation and more predators, but the trophic level sandwiched in between, the herbivores, did not increase, precisely as the bottom-up hypothesis predicted (figure 28.17).

### Relative Importance of Trophic Cascades and Bottom-Up Effects

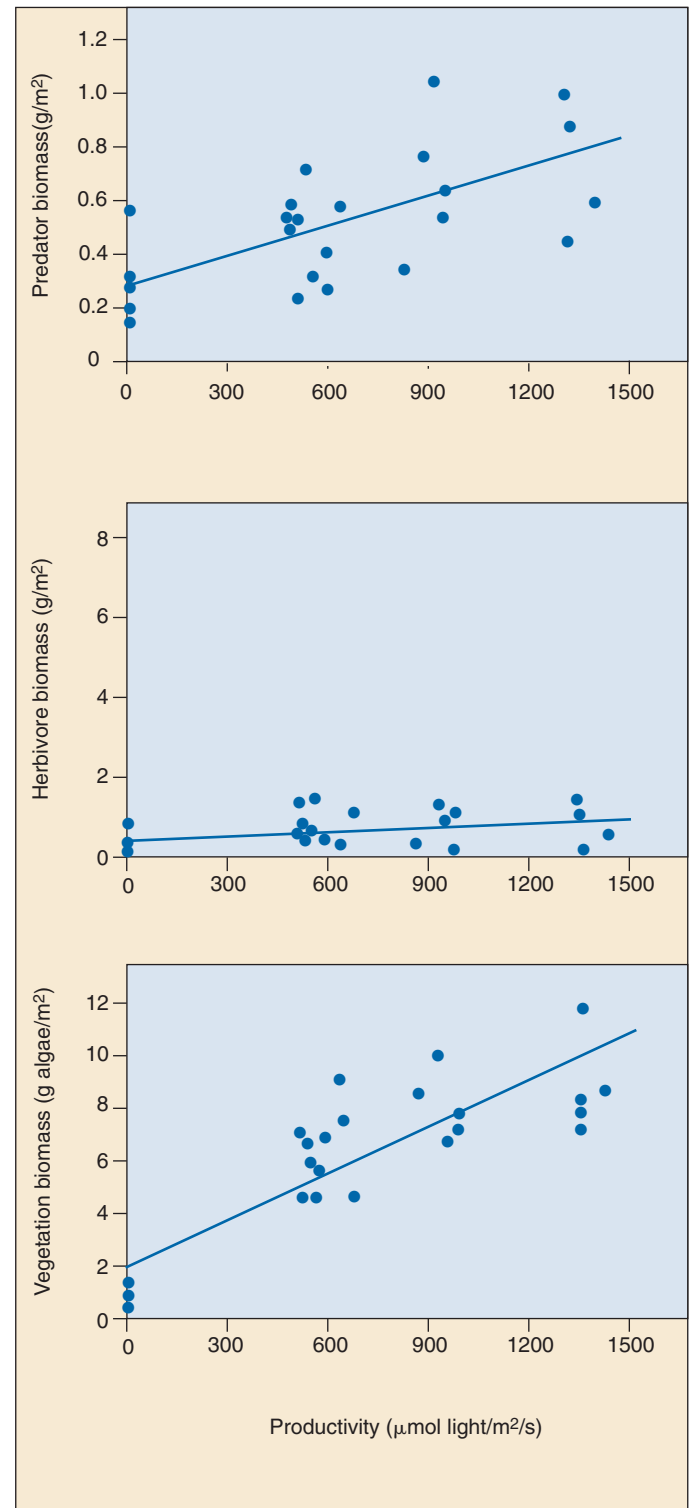
Neither trophic cascades nor bottom-up effects are inevitable. For example, if two species of herbivores exist in an ecosystem and compete strongly, and if one species is much more vulnerable to predation than the other, then top-down effects will not propagate to the next lower trophic level. Rather, increased predation will simply decrease the population of the vulnerable species while increasing the population of its competitor, with potentially no net change on the vegetation in the next lower trophic level.

Similarly, productivity increases might not move up through all trophic levels. In some cases, for example, prey populations increase so quickly that their predators cannot control them. In such cases, increases in productivity would not move up the food chain.

In other cases, trophic cascades and bottom-up effects may reinforce each other. In one experiment, large fish were removed from one lake, leaving only minnows, which ate most of the algae-eating zooplankton. By contrast, in the other lake, there were few minnows and much zooplankton. The researchers then added nutrients to both lakes. In the minnow lake, there were few zooplankton, so the resulting increase in algal productivity did not propagate up the food chain and large mats of algae formed. By comparison, in the large fish lake, increased productivity moved up the food chain and algae populations were controlled. In this case, both top-down and bottom-up processes were operating.

Nature, of course, is not always so simple. In some cases, species may simultaneously operate on multiple trophic levels, such as the jaguar which eats both smaller carnivores and herbivores, or the bear which eats both fish and berries. Nature is often much more complicated than a simple, linear food chain, as figure 28.9 indicates. Ecologists are currently working to apply theories of food chain interactions to these more complicated situations.

**Because of the linked nature of food webs, species on different trophic levels will effect each other, and these effects can promulgate both up and down the food web.**



**FIGURE 28.17**

**Bottom-up effects on a stream ecosystem.** As predicted, increases in productivity—which are a function of the amount of light hitting the stream and leading to photosynthesis—lead to increases in the amount of vegetation. However, herbivore biomass does not increase with increased productivity because it is converted into predator biomass.

## 28.4 Biodiversity promotes ecosystem stability.

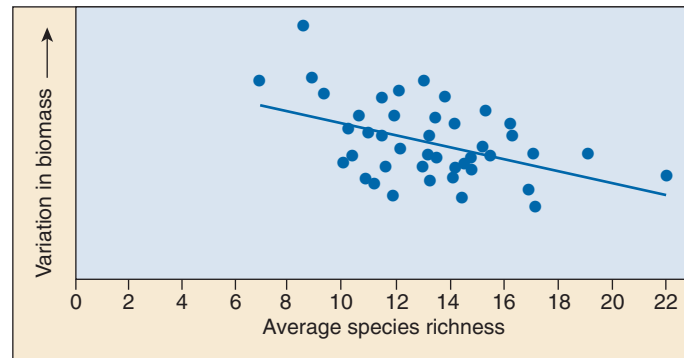
### Effects of Species Richness

Ecologists have long debated what are the consequences of differences in species richness among communities. One theory is that more species-rich communities are more stable; that is, more constant in composition and better able to resist disturbance. This hypothesis has been elegantly studied by David Tilman and colleagues at the University of Minnesota's Cedar Creek Natural History Area. These workers monitored 207 small rectangular plots of land (8 to 16 m<sup>2</sup>) for 11 years. In each plot, they counted the number of prairie plant species and measured the total amount of plant biomass (that is, the mass of all plants on the plot). Over the course of the study, plant species richness was related to community stability—plots with more species showed less year-to-year variation in biomass (figure 28.18). Moreover, in two drought years, the decline in biomass was negatively related to species richness; in other words, plots with more species were less affected. In a related experiment, when seeds of other plant species were added to different plots, the ability of these species to become established was negatively related to species richness. More diverse communities, in other words, are more resistant to invasion by new species, another measure of community stability.

Species richness may also have effects on other ecosystem processes. In a follow-up study, Tilman established another 147 plots in which they experimentally varied the number of plant species. Each of the plots was monitored to estimate how much growth was occurring and how much nitrogen the growing plants were taking up from the soil. Tilman found that the more species a plot had, the greater nitrogen uptake and total amount of biomass produced. In his study, increased biodiversity clearly leads to greater productivity (figure 28.19).

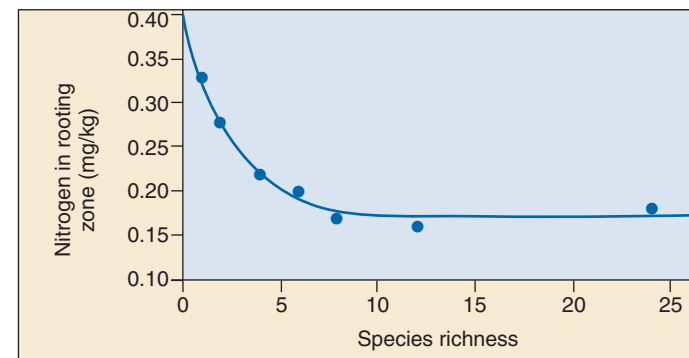
Laboratory studies on artificial ecosystems have provided similar results. In one elaborate study, ecosystems covering 1 m<sup>2</sup> were constructed in growth chambers that controlled temperature, light levels, air currents, and atmospheric gas concentrations. A variety of plants, insects, and other animals were introduced to construct ecosystems composed of 9, 15, or 31 species with the lower diversity treatments containing a subset of the species in the higher diversity enclosures. As with Tilman's experiments, the amount of biomass produced was related to species richness, as was the amount of carbon dioxide consumed, a measure of respiration occurring in the ecosystem.

Tilman's conclusion that healthy ecosystems depend on diversity is not accepted by all ecologists. Critics question the validity and relevance of these biodiversity studies, claiming their experimental design is critically flawed. Tilman's Cedar Creek result was a statistical arti-



**FIGURE 28.18**

**Effect of species richness on ecosystem stability.** In the Cedar Creek experimental fields, each square is a 100-square-foot experimental plot. Experimental plots with more plant species seem to show less variation in the total amount of biomass produced, and thus more community stability.



**FIGURE 28.19**

**Effect of species richness on productivity.** In Tilman's experimental studies, plots with more species took up more nitrogen from the soil, leaving less in the rooting zone. The increased amount of nitrogen absorption is an indicator of increased growth, increased biomass, and thus increased productivity.

fact, they argue—the more species you add to a mix, the greater the probability that you will add a highly productive one. Adding taller or highly productive plants of course increases productivity, they explain. To show a real benefit from diversity, experimental plots would have to exhibit “overyielding”—plot productivity would have to be greater than that of the single most productive species grown in isolation. The long-simmering debate continues.

**Controversial experimental field studies support the hypothesis that species-rich communities are more stable.**

## Causes of Species Richness

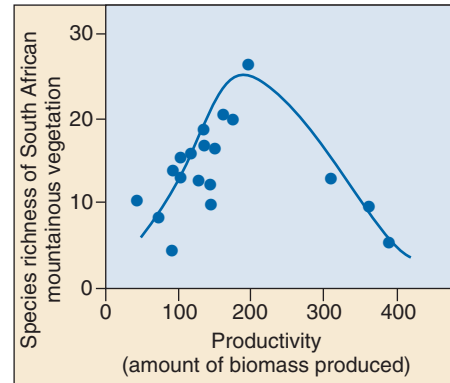
While ecologists still argue about why some ecosystems are more stable than others—better able to avoid permanent change and return to normal after disturbances like land clearing, fire, invasion by plagues of insects, or severe storm damage—most ecologists now accept as a working hypothesis that biologically diverse ecosystems are generally more stable than simple ones. Ecosystems with many different kinds of organisms support a more complex web of interactions, and an alternative niche is thus more likely to exist to compensate for the effect of a disruption.

### Factors Promoting Species Richness

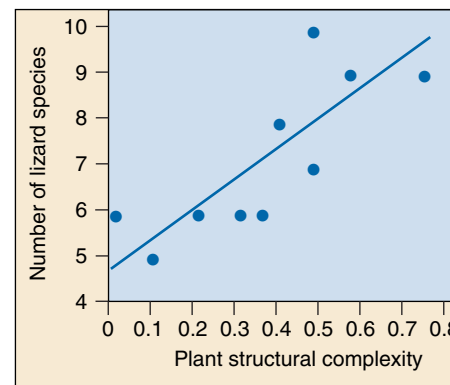
How does the number of species in a community affect the functioning of the ecosystem? How does ecosystem functioning affect the number of species in a community? It is often extremely difficult to sort out the relative contributions of different factors. With regard to determinants of species richness in a community, of the many variables that may play a role, we will discuss three: ecosystem productivity, spatial heterogeneity, and climate. Two additional factors that may play an important role, the evolutionary age of the community and the degree to which the community has been disturbed, will be examined later in this chapter.

**Ecosystem Productivity.** Ecosystems differ in productivity, which is a measure of how much new growth they can produce. Surprisingly, the relationship between productivity and species richness is not linear. Rather, ecosystems with intermediate levels of productivity tend to have the most species (figure 28.20). Why this is so is a topic of considerable current debate. One possibility is that levels of productivity are linked to numbers of predators. At low productivity, there are few predators and superior competitors eliminate most species, whereas at high productivity, there are so many predators that only the most predation-resistant species survive. At intermediate levels, however, predators may act as keystone species, maintaining species richness.

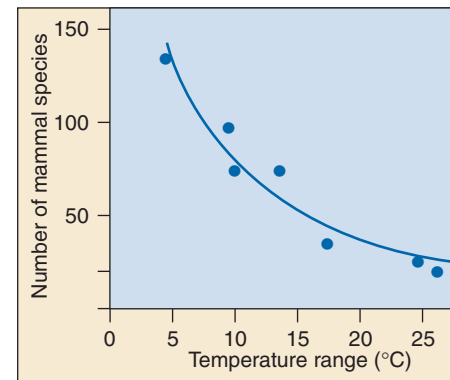
**Spatial Heterogeneity.** Environments that are more spatially heterogeneous—that contain more soil types, topographies, and other habitat variations—can be expected to accommodate more species because they provide a greater variety of microhabitats, microclimates, places to hide from predators, and so on. In general, the species richness of animals tends to reflect the species richness of the plants in their community, while plant species richness reflects the spatial heterogeneity of the ecosystem. The plants provide a biologically derived spatial heterogeneity of microhabitats to the animals. Thus, the number of lizard species in the American Southwest mirrors the structural diversity of the plants (figure 28.21).



**FIGURE 28.20 Productivity.** In fynbos plant communities of mountainous areas of South Africa, species richness of plants peaks at intermediate levels of productivity (biomass).



**FIGURE 28.21 Spatial heterogeneity.** The species richness of desert lizards is positively correlated with the structural complexity of the plant cover in desert sites in the American Southwest.



**FIGURE 28.22 Climate.** The species richness of mammals is inversely correlated with monthly mean temperature range along the west coast of North America.

**Climate.** The role of climate is more difficult to assess. On the one hand, more species might be expected to coexist in a seasonal environment than in a constant one, because a changing climate may favor different species at different times of the year. On the other hand, stable environments are able to support specialized species that would be unable to survive where conditions fluctuate. Thus, the number of mammal species along the west coast of North America increases as the temperature range decreases (figure 28.22).

**Species richness promotes ecosystem productivity and is fostered by spatial heterogeneity and stable climate.**

## Biogeographic Patterns of Species Diversity

Since before Darwin, biologists have recognized that there are more different kinds of animals and plants in the tropics than in temperate regions. For many species, there is a steady increase in species richness from the arctic to the tropics. Called a **species diversity cline**, such a biogeographic gradient in numbers of species correlated with latitude has been reported for plants and animals, including birds (figure 28.23), mammals, reptiles.

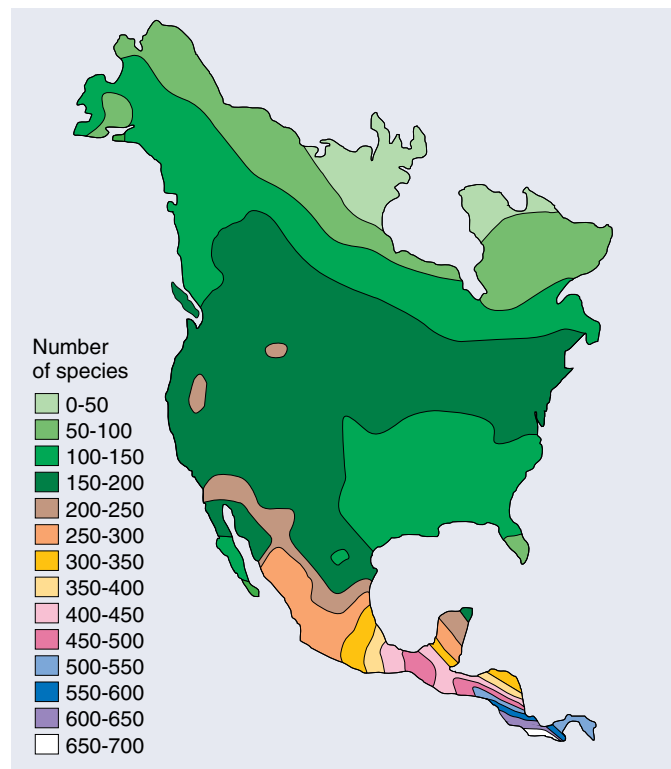
### Why Are There More Species in the Tropics?

For the better part of a century, ecologists have puzzled over the cline in species diversity from the arctic to the tropics. The difficulty has not been in forming a reasonable hypothesis of why there are more species in the tropics, but rather in sorting through the many reasonable hypotheses that suggest themselves. Here we will consider five of the most commonly discussed suggestions:

**Evolutionary age.** It has often been proposed that the tropics have more species than temperate regions because the tropics have existed over long and uninterrupted periods of evolutionary time, while temperate regions have been subject to repeated glaciations. The greater age of tropical communities would have allowed complex population interactions to coevolve within them, fostering a greater variety of plants and animals in the tropics.

However, recent work suggests that the long-term stability of tropical communities has been greatly exaggerated. An examination of pollen within undisturbed soil cores reveals that during glaciations the tropical forests contracted to a few small refuges surrounded by grassland. This suggests that the tropics have not had a continuous record of species richness over long periods of evolutionary time.

**Higher productivity.** A second often-advanced hypothesis is that the tropics contain more species because this part of the earth receives more solar radiation than temperate regions do. The argument is that more solar energy, coupled to a year-round growing season, greatly increases the overall photosynthetic activity of plants in the tropics. If we visualize the tropical forest as a pie (total resources) being cut into slices (species niches), we can see that a larger pie accommodates more slices. However, many field studies have indicated that species richness is highest at intermediate levels of productivity. Accordingly, increasing productivity would be expected to lead to lower, not higher, species richness. Perhaps the long column of vegetation down through which light passes in a tropical forest produces a wide range of frequencies and intensities, creating a greater variety of light environments and so promoting species diversity.



**FIGURE 28.23**

**A latitudinal cline in species richness.** Among North and Central American birds, a marked increase in the number of species occurs as one moves toward the tropics. Fewer than 100 species are found at arctic latitudes, while more than 600 species live in southern Central America.

**Predictability.** There are no winters in the tropics. Tropical temperatures are stable and predictable, one day much like the next. These unchanging environments might encourage specialization, with niches subdivided to partition resources and so avoid competition. The expected result would be a larger number of more specialized species in the tropics, which is what we see. Many field tests of this hypothesis have been carried out, and almost all support it, reporting larger numbers of narrower niches in tropical communities than in temperate areas.

**Predation.** Many reports indicate that predation may be more intense in the tropics. In theory, more intense predation could reduce the importance of competition, permitting greater niche overlap and thus promoting greater species richness.

**Spatial heterogeneity.** As noted earlier, spatial heterogeneity promotes species richness. Tropical forests, by virtue of their complexity, create a variety of microhabitats and so may foster larger numbers of species.

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**No one really knows why there are more species in the tropics, but there are plenty of suggestions.**

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## Island Biogeography

One of the most reliable patterns in ecology is the observation that larger islands contain more species than smaller islands. In 1967, Robert MacArthur of Princeton University and Edward O. Wilson of Harvard University proposed that this **species-area relationship** was a result of the effect of area on the likelihood of species extinction and colonization.

### The Equilibrium Model

MacArthur and Wilson reasoned that species are constantly being dispersed to islands, so islands have a tendency to accumulate more and more species. At the same time that new species are added, however, other species are lost by extinction. As the number of species on an initially empty island increases, the rate of colonization must decrease as the pool of potential colonizing species not already present on the island becomes depleted. At the same time, the rate of extinction should increase—the more species on an island, the greater the likelihood that any given species will perish. As a result, at some point, the number of extinctions and colonizations should be equal and the number of species should then remain constant. Every island of a given size, then, has a characteristic equilibrium number of species that tends to persist through time (the intersection point in figure 28.24*a*), although the individual species will change as species become extinct and new species colonize.

MacArthur and Wilson's equilibrium theory proposes that island species richness is a dynamic equilibrium between colonization and extinction. Both island size and distance from the mainland would play important roles. We would expect smaller islands to have higher rates of extinction because their population sizes would, on average, be

smaller. Also, we would expect fewer colonizers to reach islands that lie farther from the mainland. Thus, small islands far from the mainland have the fewest species; large islands near the mainland have the most (figure 28.24*b*).

The predictions of this simple model bear out well in field data. Asian Pacific bird species (figure 28.24*c*) exhibit a positive correlation of species richness with island size, but a negative correlation of species richness with distance from the mainland.

### Testing the Equilibrium Model

Field studies in which small islands have been censused, cleared, and allowed to recolonize tend to support the equilibrium model. However, long-term experimental field studies are suggesting that the situation is more complicated than MacArthur and Wilson envisioned. Their theory predicts a high level of **species turnover** as some species perish and others arrive. However, studies on island birds and spiders indicate that very little turnover occurs from year to year. Moreover, those species that do come and go are a subset of species that never attain high populations. A substantial proportion of the species appears to maintain high populations and rarely go extinct. These studies, of course, have only been going on for 20 years or less. It is possible that over periods of centuries, rare species may become common and vice versa so that, over such spans of time, the equilibrium theory is a good description of what determines island species richness. Future research is necessary to fully understand the dynamics of species richness.

Species richness on islands is a dynamic equilibrium between colonization and extinction.

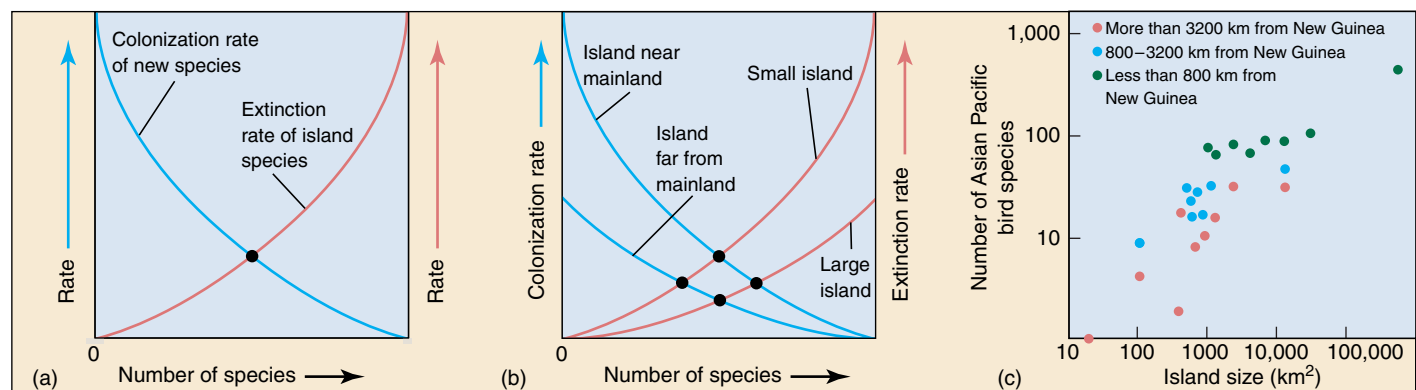


FIGURE 28.24

**The equilibrium model of island biogeography.** (a) Island species richness reaches an equilibrium (black dot) when the colonization rate of new species equals the extinction rate of species on the island. (b) The equilibrium shifts when the colonization rate is affected by distance from the mainland and when the extinction rate is affected by size of the island. Species richness is positively correlated with island size and inversely correlated with distance from the mainland. (c) The effect of distance from a larger island—which can be the source of colonizing species—is readily apparent. More distant islands have fewer species compared to nearer islands of the same size.



## Summary

## Questions

## Media Resources

## 28.1 Chemicals cycle within ecosystems.

- Fully 98% of the water on earth cycles through the atmosphere. In the United States, 96% of the fresh water is groundwater.
- About 10% of the roughly 700 billion metric tons of free carbon dioxide in the atmosphere is fixed each year through photosynthesis. About as much carbon exists in living organisms at any one time as is present in the atmosphere.
- Carbon, nitrogen, and oxygen have gaseous or liquid reservoirs, as does water. All of the other nutrients, such as phosphorus, are contained in solid mineral reservoirs.
- Phosphorus is a key component of many biological molecules; it weathers out of soils and is transported to the world's oceans.

1. What are the primary reservoirs for the chemicals in biogeochemical cycles? Are more of the life-sustaining chemicals found in these reservoirs or in the earth's living organisms?

2. What is denitrification? Which organisms carry it out?

3. How is the phosphorus cycle different from the water, carbon, nitrogen, and oxygen cycles? What are the natural sources for phosphorus?

4. What effect does deforestation have on the water cycle and overall fertility of the land?



- Activity: Nutrient Cycle
- Activity: Carbon Cycle
- Ecosystem Introduction
- Ecosystem Concept Quiz
- Water Cycle
- Ground Water
- Water Quality
- Nutrient Cycles



- Carbon Cycle
- Nitrogen Cycle

## 28.2 Ecosystems are structured by who eats whom.

- Plants convert about 1 to 5% of the light energy that falls on their leaves to food energy. Producers, the herbivores that eat them, and the carnivores that eat the herbivores constitute three trophic levels.
- At each level, only about 10% of the energy available in the food is fixed in the body of the consumer. For this reason, food chains are always relatively short.

5. How might an increase in the number of predators affect lower levels of a food chain. How might an increase in nutrients affect upper levels?



- Activity: Energy flow
- Energy Flow

## 28.3 Energy flows through ecosystems.

- The primary productivity of a community is a measure of the biomass photosynthesis produces within it.
- As energy passes through the trophic levels of an ecosystem, much is lost at each step. Ecological pyramids reflect this energy loss.

6. What is the difference between primary productivity, gross primary productivity, and net primary productivity?

7. Which type of diet, carnivorous or herbivorous, provides more food value to any given living organism?



- Exponential Population Growth



- Student Research: Assessing Paleoenvironments

## 28.4 Biodiversity promotes ecosystem stability.

- Increasing the number of species in a community seems to promote ecosystem productivity. Controversial experiments suggest that communities with increased species richness are more stable and less vulnerable to disturbance.

8. Why might rain forests have high levels of species diversity?

9. Why do distant islands tend to have fewer species than nearer islands of the same size? Why do different-sized islands tend to differ in species number?



- On Science Article: Is Biodiversity Good?
- Bioethics Case Study: Wolves in Yellowstone
- Book review: *Island of the Colorblind* by Sacks
- *The Song of the Dodo* by Quammen

# 29

## *The Biosphere*

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### Concept Outline

#### 29.1 Organisms must cope with a varied environment.

**The Environmental Challenge.** Habitats vary in ways important to survival. Organisms cope with environmental variation with physiological, morphological, and behavioral adaptations.

#### 29.2 Climate shapes the character of ecosystems.

**The Sun and Atmospheric Circulation.** The sun powers major movements in atmospheric circulation.

**Atmospheric Circulation, Precipitation, and Climate.** Latitude and elevation have important effects on climate, although other factors affect regional climate.

#### 29.3 Biomes are widespread terrestrial ecosystems.

**The Major Biomes.** Characteristic communities called biomes occur in different climatic regions. Variations in temperature and precipitation are good predictors of what biomes will occur where. Major biomes include tropical rain forest, savanna, desert, grassland, temperate deciduous forest, temperate evergreen forest, taiga, and tundra.

#### 29.4 Aquatic ecosystems cover much of the earth.

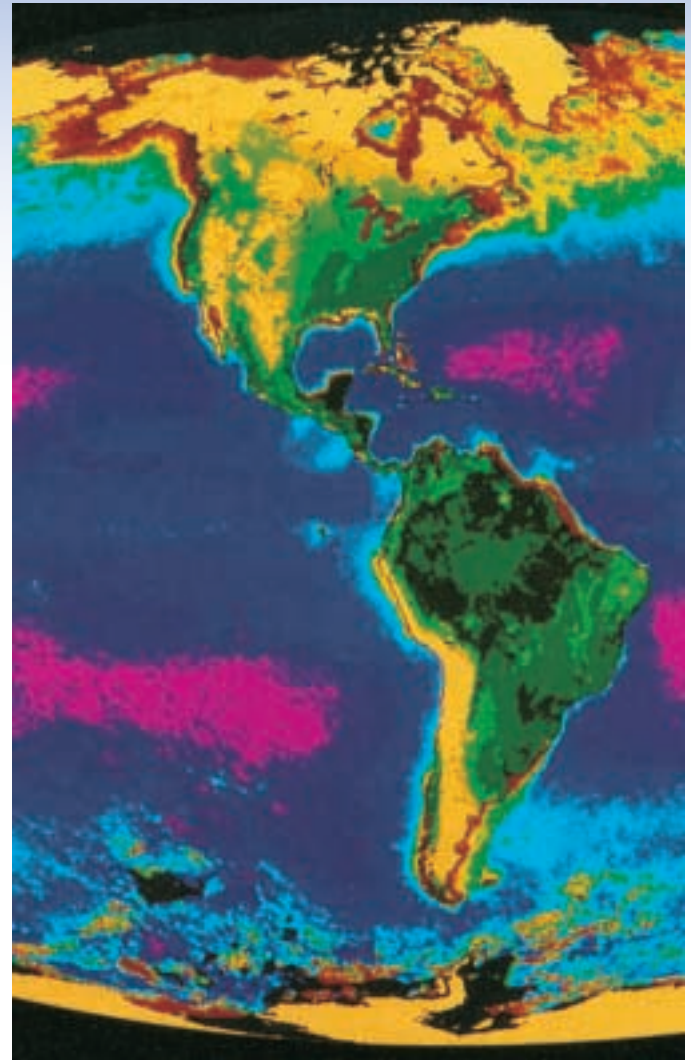
**Patterns of Circulation in the Oceans.** The world's oceans circulate in huge circles deflected by the continents.

**Life in the Oceans.** Most of the major groups of organisms originated and are still represented in the sea.

**Marine Ecosystems.** The communities of the ocean are delineated primarily by depth.

**Freshwater Habitats.** Like miniature oceans, ponds and lakes support different communities at different depths.

**Productivity of Freshwater Ecosystems.** Freshwater ecosystems are often highly productive.



**FIGURE 29.1**

**Life in the biosphere.** In this satellite image, orange zones are largely arid. Almost every environment on earth can be described in terms of temperature and moisture. These physical parameters have great bearing on the forms of life that are able to inhabit a particular region.

The biosphere includes all living communities on earth, from the profusion of life in the tropical rain forests to the photosynthetic phytoplankton in the world's oceans. In a very general sense, the distribution of life on earth reflects variations in the world's environments, principally in temperature and the availability of water. Figure 29.1 is a satellite image of North and South America, collected over eight years, the colors keyed to the relative abundance of chlorophyll, a good indicator of rich biological communities. Phytoplankton and algae produce the dark red zones in the oceans and along the seacoasts. Green and dark green areas on land are dense forests, while orange areas like the deserts of western South America are largely barren of life.

## 29.1 Organisms must cope with a varied environment.

### The Environmental Challenge

#### How Environments Vary

The nature of the physical environment in large measure determines what organisms live in a place. Key elements include:

**Temperature.** Most organisms are adapted to live within a relatively narrow range of temperatures and will not thrive if temperatures are colder or warmer. The growing season of plants, for example, is importantly influenced by temperature.

**Water.** Plants and all other organisms require water. On land, water is often scarce, so patterns of rainfall have a major influence on life.

**Sunlight.** Almost all ecosystems rely on energy captured by photosynthesis; the availability of sunlight influences the amount of life an ecosystem can support, particularly below the surface in marine communities.

**Soil.** The physical consistency, pH, and mineral composition of soil often severely limit plant growth, particularly the availability of nitrogen and phosphorus.

#### Active and Passive Approaches to Coping with Environmental Variation

An individual encountering environmental variation may choose to maintain a “steady-state” internal environment, an approach known as maintaining **homeostasis**. Many animals and plants actively employ physiological, morphological, or behavioral mechanisms to maintain homeostasis. The beetle in figure 29.2 is using a behavioral mechanism to cope with drastic changes in water availability. Other animals and plants simply conform to the environment in which they find themselves, their bodies adopting the temperature, salinity, and other aspects of their surroundings.

Responses to environmental variation can be seen over both the short and the long term. In the short term, spanning periods of a few minutes to an individual’s lifetime, organisms have a variety of ways of coping with environmental change. Over longer periods, natural selection can operate to make a population better adapted to the environment.

#### Individual Response to Environmental Change

**Physiology.** Many organisms are able to adapt to environmental change by making physiological adjustments. Thus, your body constricts the blood vessels on the surface of your face on a cold day, reducing heat loss (and also giving your face a “flush”). Similarly, humans who visit high



**FIGURE 29.2**  
**Meeting the challenge of obtaining moisture in a desert.** On the dry sand dunes of the Namib Desert in southwestern Africa, the beetle *Onymacris unguicularis* collects moisture from the fog by holding its abdomen up at the crest of a dune to gather condensed water.

**Table 29.1 Physiological changes at high altitude that increase the amount of oxygen delivered to body tissues**

|                                                                                                                |
|----------------------------------------------------------------------------------------------------------------|
| Increased rate of breathing                                                                                    |
| Increased erythrocyte production, increasing the amount of hemoglobin in the blood                             |
| Decreased binding capacity of hemoglobin, thus increasing the rate at which oxygen is unloaded in body tissues |
| Increased density of mitochondria, capillaries, and muscle myoglobin                                           |

Based on Table 14-11 in A. J. Vander, J. H. Sherman, and D. S. Luciano, *Human Physiology*, 5th Ed., McGraw-Hill, 1990.

altitudes, such as in the Andes, initially experience altitude sickness—the symptoms of which include heart palpitations, nausea, fatigue, headache, mental impairment and, in serious cases, pulmonary edema—because of the lower atmospheric pressure and consequent lower oxygen availability in the air. After several days, however, the same people will feel fine, because of a number of physiological changes that increase the delivery of oxygen to body (table 29.1).

Some insects avoid freezing in the winter by adding glycerol “antifreeze” to their blood; others tolerate freezing by converting much of their glycogen reserves into alcohols that protect their cell membranes from freeze damage.

**Morphology.** Animals that maintain a constant internal temperature (endotherms) in a cold environment have adaptations that tend to minimize energy expenditure. Many other mammals grow thicker coats during the winter, utilizing their fur as insulation to retain body heat during the winter. In general, the thicker the fur, the greater the insulation (figure 29.3). Thus, a wolf’s fur is some three times as thick in winter as summer and insulates more than twice as well. Other mammals escape some of the costs of maintaining a constant body temperature during winter by hibernating during the coldest season, behaving, in effect, like conformers.

**Behavior.** Many animals deal with variation in the environment by moving from one patch of habitat to another, avoiding areas that are unsuitable. The tropical lizards in figure 29.4 manage to maintain a fairly uniform body temperature in an open habitat by basking in patches of sun, retreating to the shade when they become too hot. By contrast, in shaded forests, the same lizards do not have the opportunity to regulate their body temperature through behavioral means. Thus, they become conformers and adopt the temperature of their surroundings.

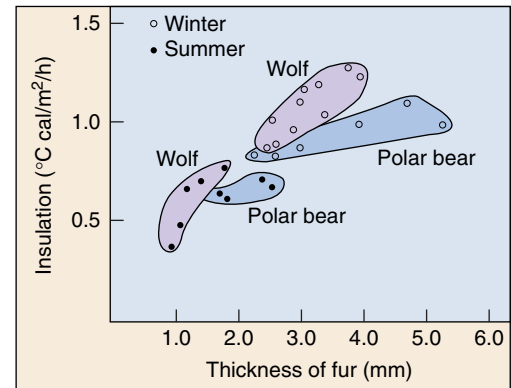
Behavioral adaptations can be extreme. The spadefoot toad *Scaphiophus*, which lives in the deserts of North America, can burrow nearly a meter below the surface and remain there for as long as nine months of each year, its metabolic rate greatly reduced, living on fat reserves. When moist cool conditions return, the toads emerge and breed. The young toads mature rapidly and burrow back underground.

## Evolutionary Responses to Environmental Variation

These examples represent different ways in which organisms may adjust to changing environmental conditions. The ability of an individual to alter its physiology, morphology, or behavior is itself an evolutionary adaptation, the result of natural selection. The results of natural selection can also be detected by comparing closely related species that live in different environments. In such cases, species often exhibit striking adaptations to the particular environment in which they live.

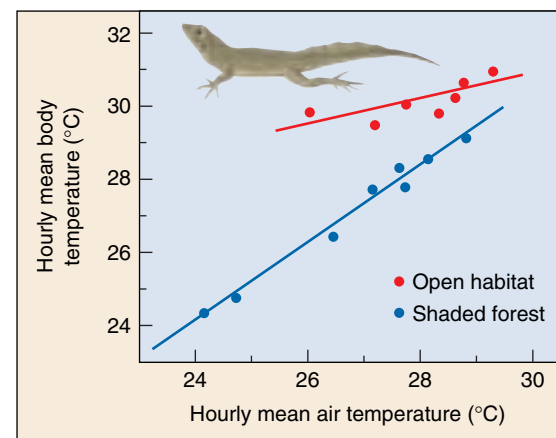
For example, animals that live in different climates show many differences. Mammals from colder climates tend to have shorter ears and limbs (Allen's Rule) and larger bodies (Bergmann's Rule) to limit heat loss. Both mechanisms reduce the surface area across which animals lose heat. Lizards that live in different climates exhibit physiological adaptations for coping with life at different temperatures. Desert lizards are unaffected by high temperatures that would kill a lizard from northern Europe, but the northern lizards are capable of running, capturing prey, and digesting food at cooler temperatures at which desert lizards would be completely immobilized.

Many species also exhibit adaptations to living in areas where water is scarce. Everyone knows of the camel, and other desert animals, which can go extended periods without drinking water. Another example of desert adaptation is seen in frogs. Most frogs have moist skins through which water permeates readily. Such animals could not survive in arid climates because they would rapidly dehydrate and dry. However, some frogs have solved this problem by greatly reducing the rate of water loss through the skin. One species, for example, secretes a waxy substance from specialized glands that waterproofs its skin and reduces rates of water loss by 95%.



**FIGURE 29.3**

**Morphological adaptation.** Fur thickness in North American mammals has a major impact on the degree of insulation the fur provides.



**FIGURE 29.4**

**Behavioral adaptation.** The Puerto Rican lizard *Anolis cristatellus* maintains a relatively constant temperature by seeking out and basking in patches of sunlight; in shaded forests, this behavior is not possible and body temperature conforms to the surroundings.

Adaptation to different environments can also be studied experimentally. For example, when strains of *E. coli* are grown at high temperatures (42°C), the speed at which resources are utilized improves through time. After 2000 generations, this ability increased 30% over what it had been when the experiment started. The mechanism by which efficiency of resource use was increased is still unknown and is the focus of current research.

**Organisms use a variety of physiological, morphological, and behavioral mechanisms to adjust to environmental variation. Over time, species evolve adaptations to living in different environments.**

## 29.2 Climate shapes the character of ecosystems.

The distribution of biomes (see discussion later in this chapter) results from the interaction of the features of the earth itself, such as different soil types or the occurrence of mountains and valleys, with two key physical factors: (1) the amount of solar heat that reaches different parts of the earth and seasonal variations in that heat; and (2) global atmospheric circulation and the resulting patterns of oceanic circulation. Together these factors dictate local climate, and so determine the amounts and distribution of precipitation.

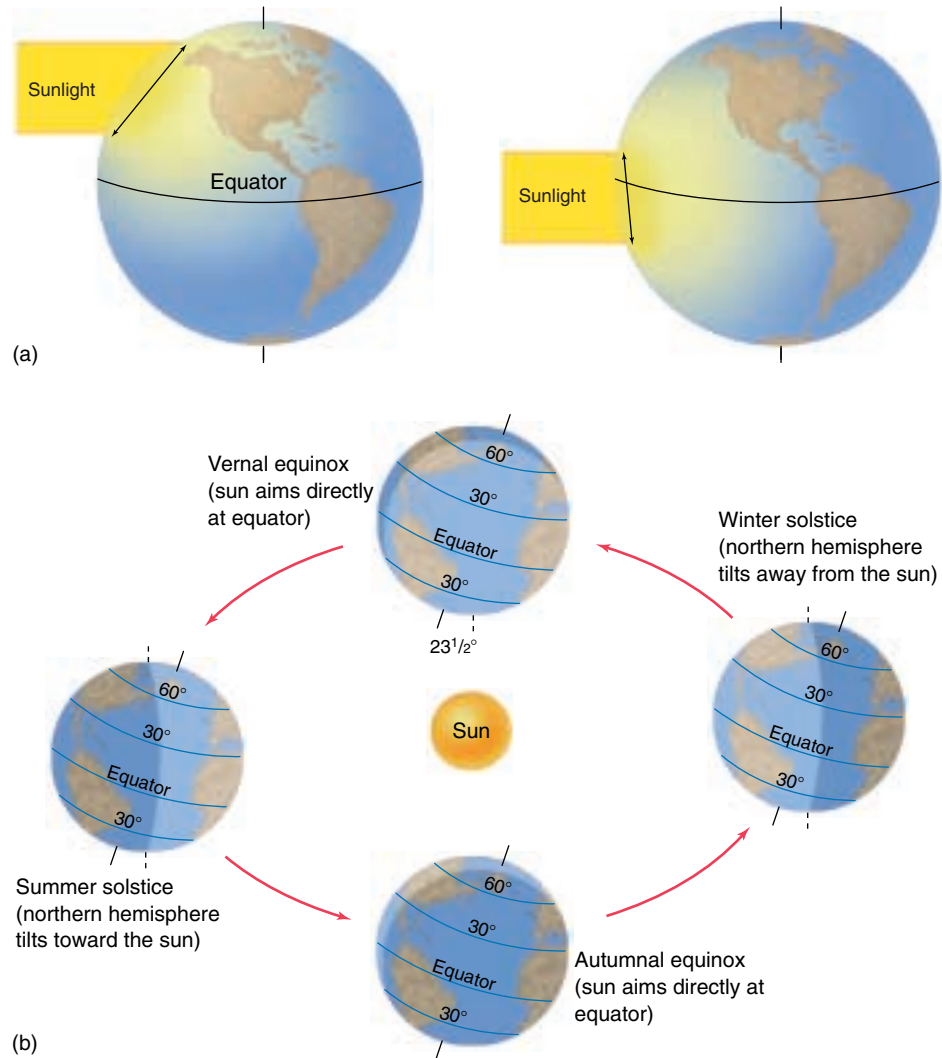
### The Sun and Atmospheric Circulation

The earth receives an enormous quantity of heat from the sun in the form of shortwave radiation, and it radiates an equal amount of heat back to space in the form of long-wave radiation. About  $10^{24}$  calories arrive at the upper surface of the earth's atmosphere each year, or about 1.94 calories per square centimeter per minute. About half of this energy reaches the earth's surface. The wavelengths that reach the earth's surface are not identical to those that reach the outer atmosphere. Most of the ultraviolet radiation is absorbed by the oxygen ( $O_2$ ) and ozone ( $O_3$ ) in the atmosphere. As we will see in chapter 30, the depletion of the ozone layer, apparently as a result of human activities, poses serious ecological problems.

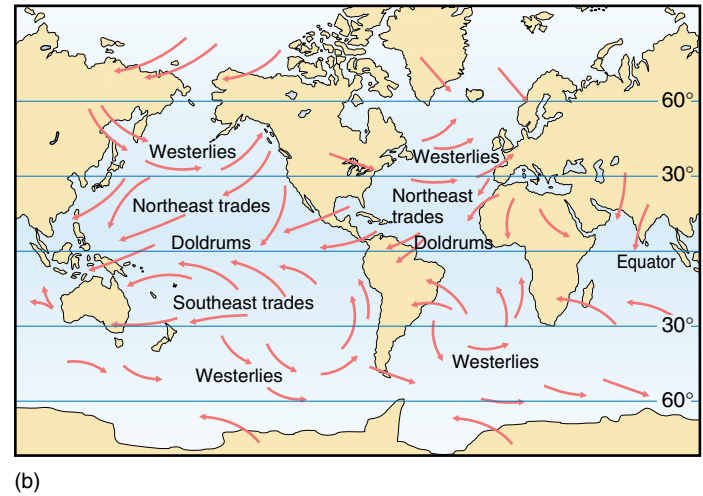
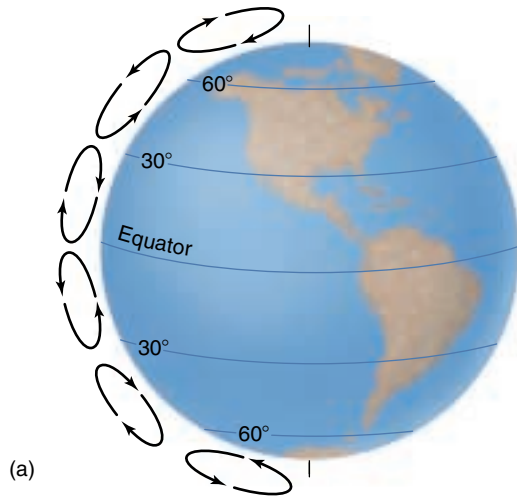
### Why the Tropics Are Warmer

The world contains a great diversity of biomes because its climate varies so much from place to place. On a given day, Miami, Florida, and Bangor, Maine, often have very different weather. There is no mystery about this. Because the earth is a sphere, some parts of it receive more energy from the sun than others. This variation is responsible for many of the major climatic differences that

occur over the earth's surface, and, indirectly, for much of the diversity of biomes. The tropics are warmer than temperate regions because the sun's rays arrive almost perpendicular to regions near the equator. Near the poles the angle of incidence of the sun's rays spreads them out over a much greater area, providing less energy per unit area (figure 29.5a).



**FIGURE 29.5**  
**Relationships between the earth and the sun are critical in determining the nature and distribution of life on earth.** (a) A beam of solar energy striking the earth in the middle latitudes spreads over a wider area of the earth's surface than a similar beam striking the earth near the equator. (b) The rotation of the earth around the sun has a profound effect on climate. In the northern and southern hemispheres, temperatures change in an annual cycle because the earth tilts slightly on its axis in relation to the path around the sun.



**FIGURE 29.6**

**General patterns of atmospheric circulation.** (a) The pattern of air movement toward and away from the earth's surface. (b) The major wind currents across the face of the earth.

The earth's annual orbit around the sun and its daily rotation on its own axis are both important in determining world climate (figure 29.5b). Because of the annual cycle, and the inclination of the earth's axis at approximately  $23.5^\circ$  from its plane of revolution around the sun, there is a progression of seasons in all parts of the earth away from the equator. One pole or the other is tilted closer to the sun at all times except during the spring and autumn equinoxes.

### Major Atmospheric Circulation Patterns

The moisture-holding capacity of air increases when it warms and decreases when it cools. High temperatures near the equator encourage evaporation and create warm, moist air. As this air rises and flows toward the poles, it cools and loses most of its moisture (figure 29.6). Consequently, the greatest amounts of precipitation on earth fall near the equator. This equatorial region of rising air is one of low pressure, called the **doldrums**, which draws air from both north and south of the equator. When the air masses that have risen reach about  $30^\circ$  north and south latitude, the dry air, now cooler, sinks and becomes reheated. As the air reheats, its evaporative capacity increases, creating a zone of decreased precipitation. The air, still warmer than in the polar regions, continues to flow toward the poles. It rises again at about  $60^\circ$  north and south latitude, producing another zone of high precipitation. At this latitude there is another low-pressure area, the polar front. Some of this rising air flows back to

the equator and some continues north and south, descending near the poles and producing another zone of low precipitation before it returns to the equator.

### Air Currents Generated by the Earth's Rotation

Related to these bands of north-south circulation are three major air currents generated mainly by the interaction of the earth's rotation with patterns of worldwide heat gain. Between about  $30^\circ$  north latitude and  $30^\circ$  south latitude, the trade winds blow, from the east-southeast in the southern hemisphere and from the east-northeast in the northern hemisphere. The trade winds blow all year long and are the steadiest winds found anywhere on earth. They are stronger in winter and weaker in summer. Between  $30^\circ$  and  $60^\circ$  north and south latitude, strong prevailing westerlies blow from west to east and dominate climatic patterns in these latitudes, particularly along the western edges of the continents. Weaker winds, blowing from east to west, occur farther north and south in their respective hemispheres.

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Warm air rises near the equator, descends and produces arid zones at about  $30^\circ$  north and south latitude, flows toward the poles, then rises again at about  $60^\circ$  north and south latitude, and moves back toward the equator. Part of this air, however, moves toward the poles, where it produces zones of low precipitation.

---

## Atmospheric Circulation, Precipitation, and Climate

As we have discussed, precipitation is generally low near 30° north and south latitude, where air is falling and warming, and relatively high near 60° north and south latitude, where it is rising and cooling. Partly as a result of these factors, all the great deserts of the world lie near 30° north or south latitude. Other major deserts are formed in the interiors of large continents. These areas have limited precipitation because of their distance from the sea, the ultimate source of most moisture.

### Rain Shadows

Other deserts occur because mountain ranges intercept moisture-laden winds from the sea. When this occurs, the air rises and the moisture-holding capacity of the air decreases, resulting in increased precipitation on the windward side of the mountains—the side from which the wind is blowing. As the air descends the other side of the mountains, the leeward side, it is warmed, and its moisture-holding capacity increases, tending to block precipitation. In California, for example, the eastern sides of the Sierra Nevada Mountains are much drier than the western sides, and the vegetation is often very different. This phenomenon is called the **rain shadow effect** (figure 29.7).

### Regional Climates

Four relatively small areas, each located on a different continent, share a climate that resembles that of the Mediterranean region. So-called Mediterranean climates are found in portions of Baja, California, and Oregon; in central Chile; in southwestern Australia; and in the Cape region of South Africa. In all of these areas, the prevailing westerlies blow during the summer from a cool ocean onto warm land. As a result, the air's moisture-holding capacity increases, the air absorbing moisture and creating hot rainless summers. Such climates are unusual on a world scale. In the five regions where they occur, many unique kinds of plants and animals, often local in distribution, have evolved. Because of the prevailing westerlies, the great deserts of the world (other than those in the interiors of continents) and the areas of Mediterranean climate lie on the western sides of the continents.

Another kind of regional climate occurs in southern Asia. The monsoon climatic conditions characteristic of India and southern Asia occur during the summer months. During the winter, the trade winds blow from the east-northeast off the cool land onto the warm sea. From June to October, though, when the land is heated, the direction of the air flow reverses, and the winds veer around to blow

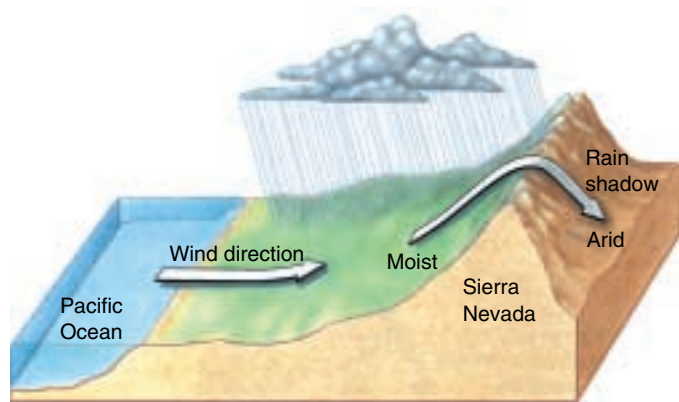


FIGURE 29.7

**The rain shadow effect.** Moisture-laden winds from the Pacific Ocean rise and are cooled when they encounter the Sierra Nevada Mountains. As their moisture-holding capacity decreases, precipitation occurs, making the middle elevation of the range one of the snowiest regions on earth; it supports tall forests, including those that include the famous giant sequoias (*Sequoiadendron giganteum*). As the air descends on the east side of the range, its moisture-holding capacity increases again, and the air picks up rather than releases moisture from its surroundings. As a result, desert conditions prevail on the east side of the mountains.

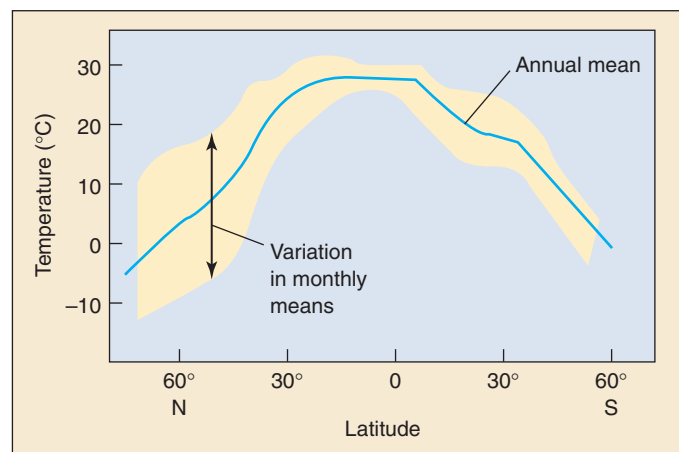
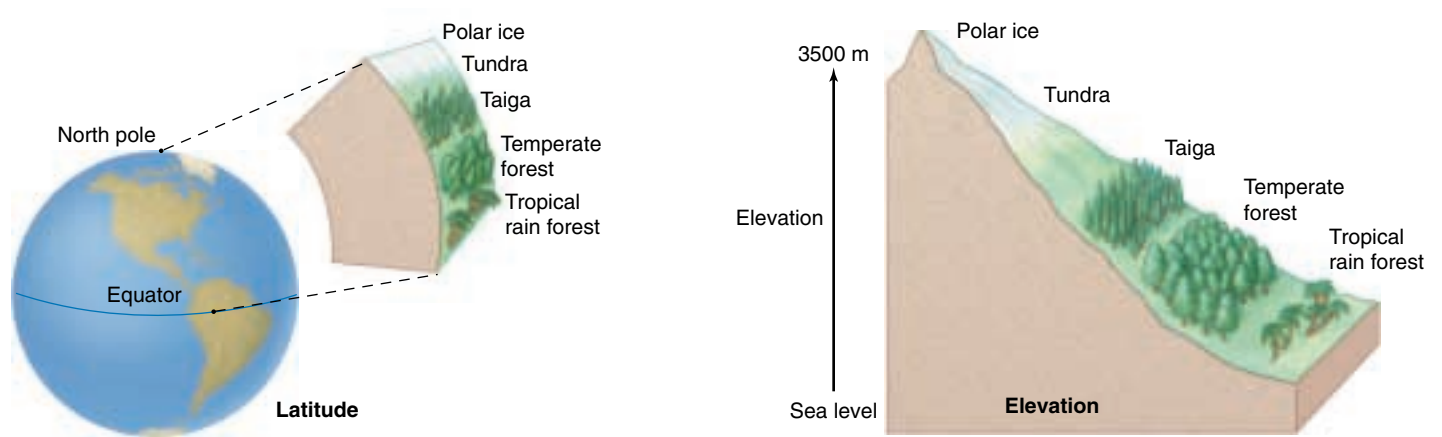


FIGURE 29.8

**Temperature varies with latitude.** The blue line represents the annual mean temperature at latitudes from the North Pole to Antarctica.

onto the Indian subcontinent and adjacent areas from the southwest bringing rain. The duration and strength of the monsoon winds spell the difference between food sufficiency and starvation for hundreds of millions of people in this region each year.



**FIGURE 29.9**

**Elevation affects the distribution of biomes much as latitude does.** Biomes that normally occur far north and far south of the equator at sea level also occur in the tropics at high mountain elevations. Thus, on a tall mountain in southern Mexico or Guatemala, one might see a sequence of biomes like the one illustrated here.

## Latitude

Temperatures are higher in tropical ecosystems for a simple reason: more sunlight per unit area falls on tropical latitudes. Solar radiation is most intense when the sun is directly overhead, and this only occurs in the tropics, where sunlight strikes the equator perpendicularly. As figure 29.8 shows, the highest mean global temperatures occur near the equator (that is, 0 latitude). Because there are no seasons in the tropics, there is little variation in mean monthly temperature in tropical ecosystems. As you move from the equator into temperate latitudes, sunlight strikes the earth at a more oblique angle, so that less falls on a given area. As a result, mean temperatures are lower. At temperate latitudes, temperature variation increases because of the increasingly marked seasons.

Seasonal changes in wind circulation produce corresponding changes in ocean currents, sometimes causing nutrient-rich cold water to well up from ocean depths. This produces “blooms” among the plankton and other organisms living near the surface. Similar turnover occurs seasonally in freshwater lakes and ponds, bringing nutrients from the bottom to the surface in the fall and again in the spring.

## Elevation

Temperature also varies with elevation, with higher altitudes becoming progressively colder. At any given lati-

tude, air temperature falls about  $6^{\circ}\text{C}$  for every 1000-meter increase in elevation. The ecological consequences of temperature varying with elevation are the same as temperature varying with latitude (figure 29.9). Thus, in North America a 1000-meter increase in elevation results in a temperature drop equal to that of an 880-kilometer increase in latitude. This is one reason “timberline” (the elevation above which trees do not grow) occurs at progressively lower elevations as one moves farther from the equator.

## Microclimate

Climate also varies on a very fine scale within ecosystems. Within the litter on a forest floor, there is considerable variation in shading, local temperatures, and rates of evaporation from the soil. Called **microclimate**, these very localized climatic conditions can be very different from those of the overhead atmosphere. Gardeners spread straw over newly seeded lawns to create such a moisture-retaining microclimate.

**The great deserts and associated arid areas of the world mostly lie along the western sides of continents at about  $30^{\circ}$  north and south latitude. Mountain ranges tend to intercept rain, creating deserts in their shadow. In general, temperatures are warmer in the tropics and at lower elevations.**

## 29.3 Biomes are widespread terrestrial ecosystems.

### The Major Biomes

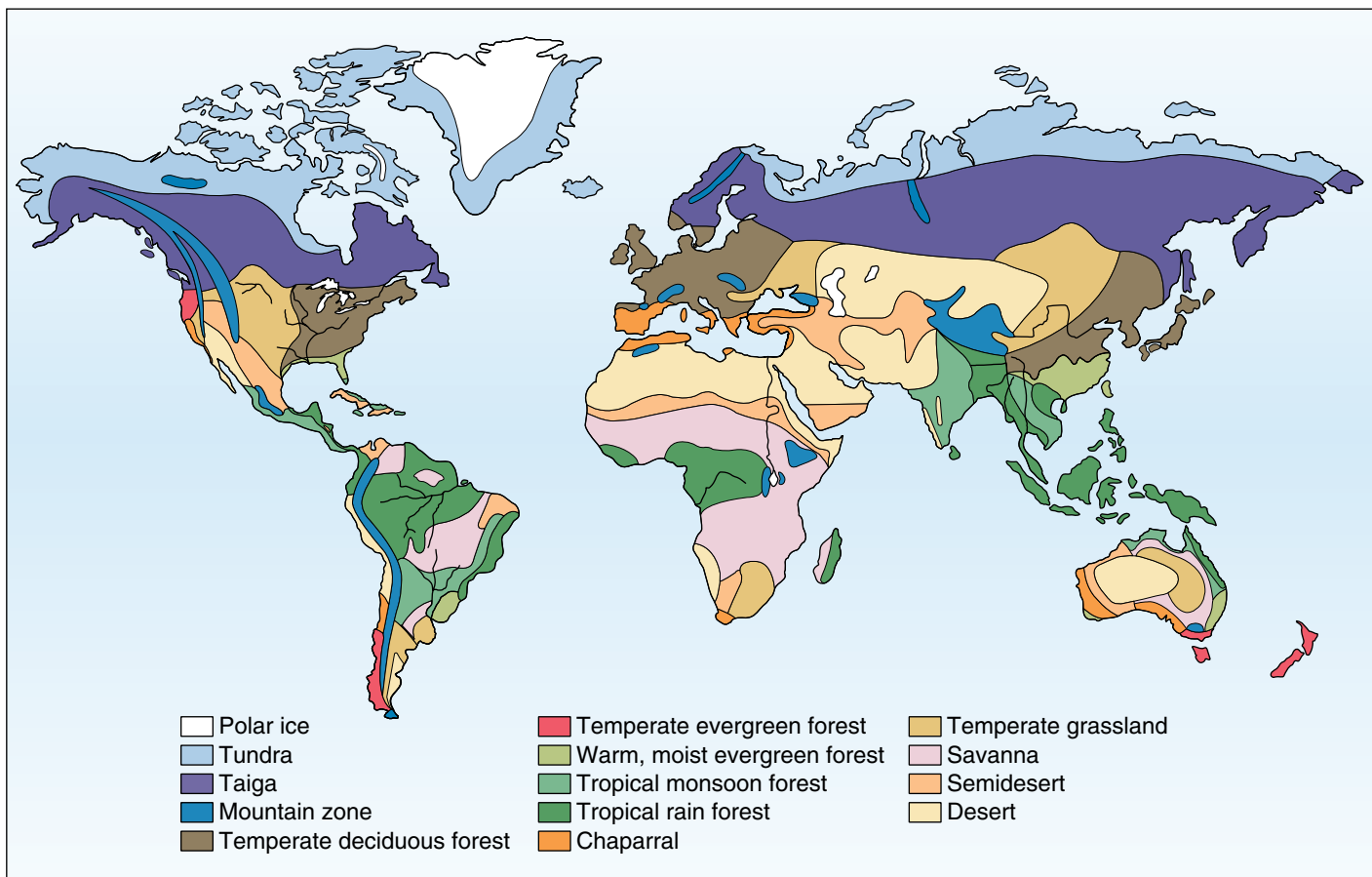
**Biomes** are major communities of organisms that have a characteristic appearance and that are distributed over a wide land area defined largely by regional variations in climate. As you might imagine from such a broad definition, there are many ways to classify biomes, and different ecologists may assign the same community to different biomes. There is little disagreement, however, about the reality of biomes as major biological communities—only about how to best describe them.

### Distribution of the Major Biomes

Eight major biome categories are presented in this text: tropical rain forest, savanna, desert, temperate grassland, temperate deciduous forest, temperate evergreen forest, taiga, and tundra. These biomes occur worldwide, occupying large regions that can be defined by rainfall and temperature.

Six additional biomes are considered by some ecologists to be subsets of the eight major ones: polar ice, mountain zone, chaparral, warm moist evergreen forest, tropical monsoon forest, and semidesert. They vary remarkably from one another because they have evolved in regions with very different climates.

Distributions of the 14 biomes are mapped in figure 29.10. Although each is by convention named for the dominant vegetation (deciduous forest, evergreen forest, grassland, and so on) each biome is also characterized by particular animals, fungi, and microorganisms adapted to live as members of that community. Wolves, caribou or reindeer, polar bears, hares, lynx, snowy owls, deer flies, and mosquitoes inhabit the tundra all over the world and are as much a defining characteristic of the tundra biomes as the low, shrubby, matlike vegetation.

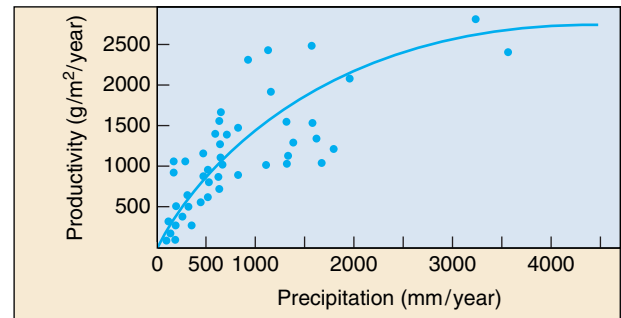


**FIGURE 29.10**  
The distribution of biomes. Each biome is similar in structure and appearance wherever it occurs on earth.

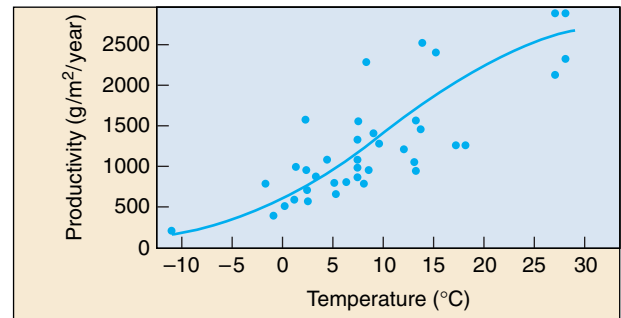
## Biomes and Climate

Many different environmental factors play a role in determining which biomes are found where. Two key parameters are available moisture and temperature. Figure 29.11 presents data on ecosystem productivity as a function of annual precipitation and of annual mean temperature: ecosystem productivity is strongly influenced by both. This is not to say that other factors such as soil structure and its mineral composition (discussed in detail in chapter 39), or seasonal versus constant climate, are not also important. Different places with the same annual precipitation and temperature sometimes support different biomes, so other factors must also be important. Nevertheless, these two variables do a fine job of predicting what biomes will occur in most places, as figure 29.12 illustrates.

If there were no mountains and no climatic effects caused by the irregular outlines of continents and by different sea temperatures, each biome would form an even belt around the globe, defined largely by latitude. In truth, these other factors also greatly affect the distribution of biomes. Distance from the ocean has a major impact on rainfall, and elevation affects temperature—the summits of the Rocky Mountains are covered with a vegetation type that resembles the tundra which normally occurs at a much higher latitude.



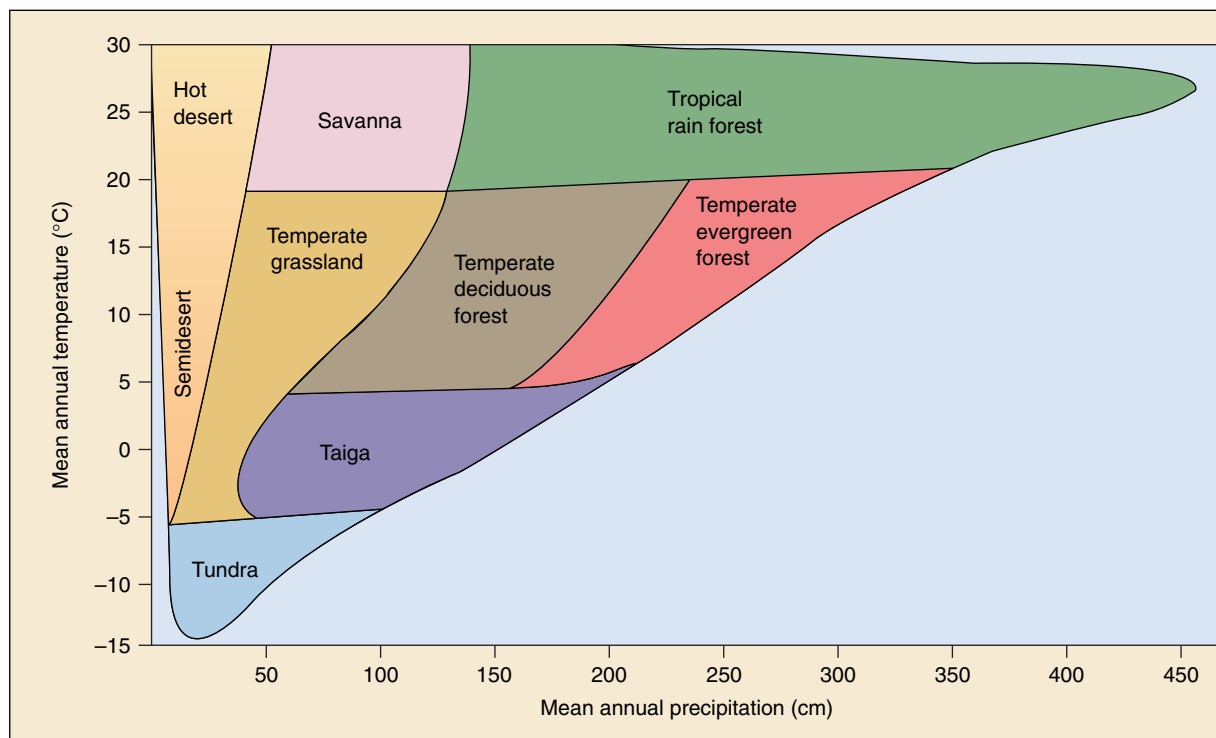
(a)



(b)

**FIGURE 29.11**

**The effects of precipitation and temperature on primary productivity.** The net primary productivity of ecosystems at 52 locations around the globe depends significantly upon (a) mean annual precipitation and (b) mean annual temperature.



**FIGURE 29.12**

**Temperature and precipitation are excellent predictors of biome distribution.** At mean annual precipitations between 50 and 150 cm, other factors such as seasonal drought, fire, and grazing also have a major influence on biome distribution.

## Tropical Rain Forests

**Rain forests**, which receive 140 to 450 centimeters of rain a year, are the richest ecosystems on earth (figure 29.13). They contain at least half of the earth's species of terrestrial plants and animals—more than 2 million species! In a single square mile of tropical forest in Rondonia, Brazil, there are 1200 species of butterflies—twice the total number found in the United States and Canada combined. The communities that make up tropical rain forests are diverse in that each kind of animal, plant, or microorganism is often represented in a given area by very few individuals. There are extensive tropical rain forests in South America, Africa, and Southeast Asia. But the world's rain forests are being destroyed, and countless species, many of them never seen by humans, are disappearing with them. A quarter of the world's species will disappear with the rain forests during the lifetime of many of us.



**FIGURE 29.13**  
Tropical rain forest.

## Savannas

In the dry climates that border the tropics are the world's great grasslands, called **savannas**. Savanna landscapes are open, often with widely spaced trees, and rainfall (75 to 125 centimeters annually) is seasonal. Many of the animals and plants are active only during the rainy season. The huge herds of grazing animals that inhabit the African savanna are familiar to all of us. Such animal communities lived in North America during the Pleistocene epoch but have persisted mainly in Africa. On a global scale, the savanna biome is transitional between tropical rain forest and desert. As these savannas are increasingly converted to agricultural use to feed rapidly expanding human populations in subtropical areas, their inhabitants are struggling to survive. The elephant and rhino are now endangered species; lion, giraffe, and cheetah will soon follow.

## Deserts

In the interior of continents are the world's great deserts, especially in Africa (the Sahara), Asia (the Gobi) and Australia (the Great Sandy Desert). **Deserts** are dry places where less than 25 centimeters of rain falls in a year—an amount so low that vegetation is sparse and survival depends on water conservation. Plants and animals may restrict their activity to favorable times of the year, when water is present. To avoid high temperatures, most desert

vertebrates live in deep, cool, and sometimes even somewhat moist burrows. Those that are active over a greater portion of the year emerge only at night, when temperatures are relatively cool. Some, such as camels, can drink large quantities of water when it is available and then survive long, dry periods. Many animals simply migrate to or through the desert, where they exploit food that may be abundant seasonally.

## Temperate Grasslands

Halfway between the equator and the poles are temperate regions where rich **grasslands** grow. These grasslands once covered much of the interior of North America, and they were widespread in Eurasia and South America as well. Such grasslands are often highly productive when converted to agricultural use. Many of the rich agricultural lands in the United States and southern Canada were originally occupied by **prairies**, another name for temperate grasslands. The roots of perennial grasses characteristically penetrate far into the soil, and grassland soils tend to be deep and fertile. Temperate grasslands are often populated by herds of grazing mammals. In North America, huge herds of bison and pronghorns once inhabited the prairies. The herds are almost all gone now, with most of the prairies having been converted to the richest agricultural region on earth.

## Temperate Deciduous Forests

Mild climates (warm summers and cool winters) and plentiful rains promote the growth of **deciduous** (hardwood) **forests** in Eurasia, the northeastern United States, and eastern Canada (figure 29.14). A deciduous tree is one that drops its leaves in the winter. Deer, bears, beavers, and raccoons are familiar animals of the temperate regions. Because the temperate deciduous forests represent the remnants of more extensive forests that stretched across North America and Eurasia several million years ago, the remaining areas in eastern Asia and eastern North America share animals and plants that were once more widespread. Alligators, for example, are found today only in China and in the southeastern United States. The deciduous forest in eastern Asia is rich in species because climatic conditions have historically remained constant. Many perennial herbs live in areas of temperate deciduous forest.



**FIGURE 29.14**  
Temperate deciduous forest.

## Temperate Evergreen Forests

Temperate evergreen forests occur in regions where winters are cold and there is a strong, seasonal dry period. The pine forests of the western United States and California oak woodlands are typical temperate evergreen forests. Temperate evergreen forests are characteristic of regions with nutrient-poor soils. Temperate-mixed evergreen forests represent a broad transitional zone between temperate deciduous forests to the south and taiga to the north. Many of these forests are endangered by overlogging, particularly in the western United States.

## Taiga

A great ring of northern forests of coniferous trees (spruce, hemlock, and fir) extends across vast areas of Asia and North America. Coniferous trees are ones with leaves like needles that are kept all year long. This ecosystem, called **taiga**, is one of the largest on earth. Here, the winters are long and cold, and most of the limited precipitation falls in the summer. Because the taiga has too short a growing season for farming, few people live there. Many large mammals, including elk, moose, deer, and such carnivores as wolves, bears, lynx, and wolverines, live in the taiga. Traditionally, fur trapping has been extensive in this region, as has lumber production. Marshes, lakes, and ponds are common and are often fringed by willows or birches. Most of the trees occur in dense stands of one or a few species.

## Tundra

In the far north, above the great coniferous forests and south of the polar ice, few trees grow. There the grassland, called **tundra**, is open, windswept, and often boggy. Enormous in extent, this ecosystem covers one-fifth of the earth's land surface. Very little rain or snow falls. When rain does fall during the brief arctic summer, it sits on frozen ground, creating a sea of boggy ground. **Permafrost**, or permanent ice, usually exists within a meter of the surface. Trees are small and are mostly confined to the margins of streams and lakes. As in taiga, herbs of the tundra are perennials that grow rapidly during the brief summers. Large grazing mammals, including musk-oxen, caribou, reindeer, and carnivores, such as wolves, foxes, and lynx, live in the tundra. Lemming populations rise and fall on a long-term cycle, with important effects on the animals that prey on them.

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Major biological communities called biomes can be distinguished in different climatic regions. These communities, which occur in regions of similar climate, are much the same wherever they are found. Variation in annual mean temperature and precipitation are good predictors of what biome will occur where.

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## 29.4 Aquatic ecosystems cover much of the earth.

### Patterns of Circulation in the Oceans

Patterns of ocean circulation are determined by the patterns of atmospheric circulation, but they are modified by the locations of landmasses. Oceanic circulation is dominated by huge surface gyres (figure 29.15), which move around the subtropical zones of high pressure between approximately 30° north and 30° south latitude. These gyres move clockwise in the northern hemisphere and counterclockwise in the southern hemisphere. The ways they redistribute heat profoundly affects life not only in the oceans but also on coastal lands. For example, the Gulf Stream, in the North Atlantic, swings away from North America near Cape Hatteras, North Carolina, and reaches Europe near the southern British Isles. Because of the Gulf Stream, western Europe is much warmer and more temperate than eastern North America at similar latitudes.

As a general principle, western sides of continents in temperate zones of the northern hemisphere are warmer than their eastern sides; the opposite is true of the southern hemisphere. In addition, winds passing over cold water onto warm land increase their moisture-holding capacity, limiting precipitation.

In South America, the Humboldt Current carries phosphorus-rich cold water northward up the west coast. Phosphorus is brought up from the ocean depths by the upwelling of cool water that occurs as offshore winds blow from the mountainous slopes that border the Pacific Ocean. This nutrient-rich current helps make possible the abundance of marine life that supports the fisheries of Peru and northern Chile. Marine birds, which feed on these organisms, are responsible for the commercially important, phosphorus-rich, guano deposits on the sea-coasts of these countries.

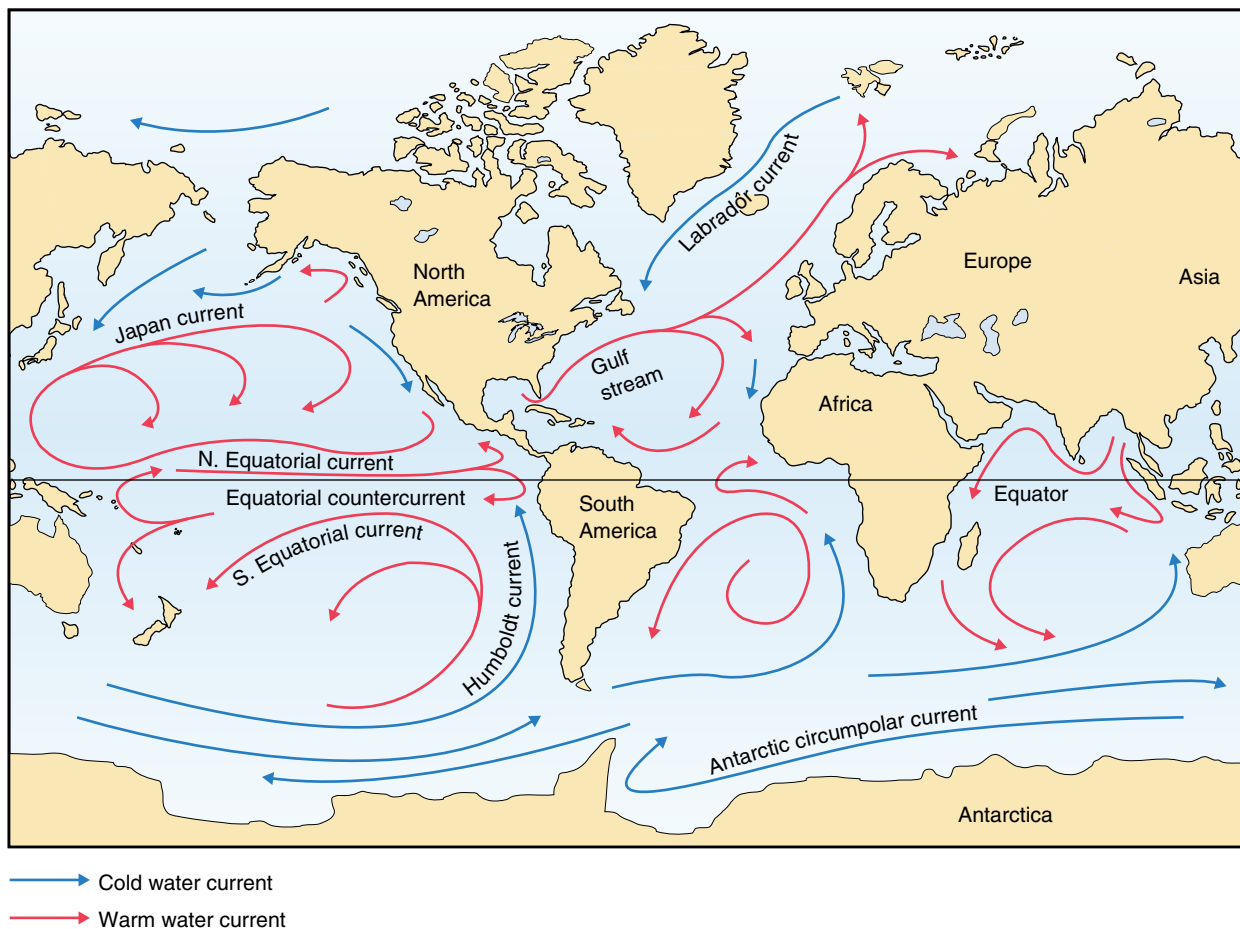
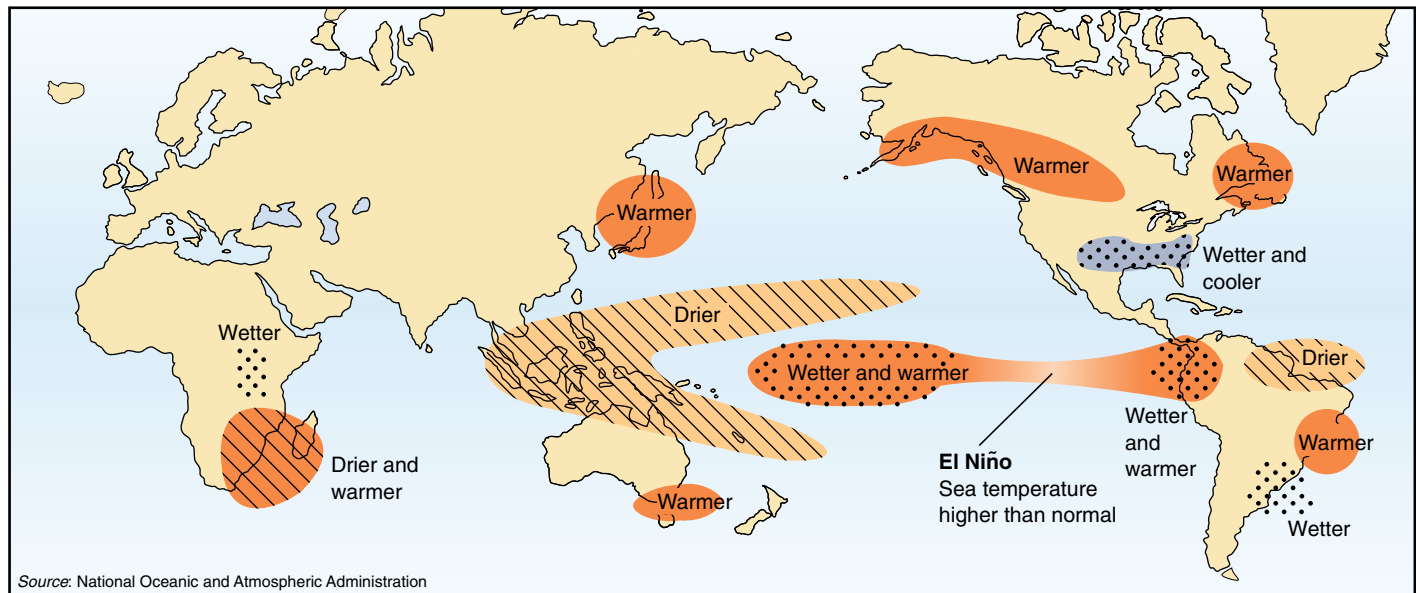


FIGURE 29.15

**Ocean circulation.** Water moves in the oceans in great surface spiral patterns called gyres; they profoundly affect the climate on adjacent lands.



**FIGURE 29.16**

**An El Niño winter.** El Niño currents produce unusual weather patterns all over the world as warm waters from the western Pacific move eastward.

### El Niño and Ocean Ecology

Every Christmas a tepid current sweeps down the coast of Peru and Ecuador from the tropics, reducing the fish population slightly and giving local fishermen some time off. The local fishermen named this Christmas current *El Niño* (literally, “the child,” after “the Christ Child”). Now, though, the term is reserved for a catastrophic version of the same phenomenon, one that occurs every two to seven years and is felt not only locally but on a global scale.

Scientists now have a pretty good idea of what goes on in an El Niño. Normally the Pacific Ocean is fanned by constantly blowing east-to-west trade winds that push warm surface water away from the ocean’s eastern side (Peru, Ecuador, and Chile) and allow cold water to well up from the depths in its place, carrying nutrients that feed plankton and hence fish. This surface water piles up in the west, around Australia and the Philippines, making it several degrees warmer and a meter or so higher than the eastern side of the ocean. But if the winds slacken briefly, warm water begins to slosh back across the ocean.

Once this happens, ocean and atmosphere conspire to ensure it keeps happening. The warmer the eastern ocean gets, the warmer and lighter the air above it becomes, and hence more similar to the air on the western side. This reduces the difference in pressure across the ocean. Because a pressure difference is what makes winds blow, the easterly trades weaken further, letting the warm water continue its eastward advance.

The end result is to shift the weather systems of the western Pacific Ocean 6000 km eastward. The tropical rainstorms that usually drench Indonesia and the Philip-

pinas are caused when warm seawater abutting these islands causes the air above it to rise, cool, and condense its moisture into clouds. When the warm water moves east, so do the clouds, leaving the previously rainy areas in drought. Conversely, the western edge of South America, its coastal waters usually too cold to trigger much rain, gets a soaking, while the upwelling slows down. During an El Niño, commercial fish stocks virtually disappear from the waters of Peru and northern Chile, and plankton drop to a twentieth of their normal abundance.

That is just the beginning. El Niño’s effects are propagated across the world’s weather systems (figure 29.16). Violent winter storms lash the coast of California, accompanied by flooding, and El Niño produces colder and wetter winters than normal in Florida and along the Gulf Coast. The American midwest experiences heavier-than-normal rains, as do Israel and its neighbors.

Though the effects of El Niños are now fairly clear, what triggers them still remains a mystery. Models of these weather disturbances suggest that the climatic change that triggers El Niño is “chaotic.” Wind and ocean currents return again and again to the same condition, but never in a regular pattern, and small nudges can send them off in many different directions—including an El Niño.

The world’s oceans circulate in huge gyres deflected by continental landmasses. Circulation of ocean water redistributes heat, warming the western side of continents. Disturbances in ocean currents like El Niño can have profound influences on world climate.

## Life in the Oceans

Nearly three-quarters of the earth's surface is covered by ocean. Oceans have an *average* depth of more than 3 kilometers, and they are, for the most part, cold and dark. Heterotrophic organisms inhabit even the greatest ocean depths, which reach nearly 11 kilometers in the Marianas Trench of the western Pacific Ocean. Photosynthetic organisms are confined to the upper few hundred meters of water. Organisms that live below this level obtain almost all of their food indirectly, as a result of photosynthetic activities that occur above.

The supply of oxygen can often be critical in the ocean, and as water temperatures become warmer, the water holds less oxygen. For this reason, the amount of available oxygen becomes an important limiting factor for organisms in warmer marine regions of the globe. Carbon dioxide, in contrast, is almost never limited in the oceans. The distribution of minerals is much more uniform in the ocean than it is on land, where individual soils reflect the composition of the parent rocks from which they have weathered.

Frigid and bare, the floors of the deep sea have long been considered a biological desert. Recent close-up looks taken by marine biologists, however, paint a different picture (figure 29.17). The ocean floor is teeming with life. Often miles deep, thriving in pitch darkness under enormous pressure, crowds of marine invertebrates have been found in hundreds of deep samples from the Atlantic and Pacific. Rough estimates of deep-sea diversity have soared to millions of species. Many appear endemic (local). The diversity of species is so high it may rival that of tropical rain forests! This profusion is unexpected. New species usually require some kind of barrier in order to diverge (see chapter 22), and the ocean floor seems boringly uniform. However, little migration occurs among deep populations, thus allowing populations to diverge and encouraging local specialization and species formation. A patchy environment may also contribute to species formation there; deep-sea ecologists find evidence that fine but nonetheless formidable resource barriers arise in the deep sea.

Another conjecture is that the extra billion years or so that life has been evolving in the sea compared with land may be a factor in the unexpected biological richness of its deep recesses.

Despite the many new forms of small invertebrates now being discovered on the seafloor, and the huge biomass that occurs in the sea, more than 90% of all *described* species of organisms occur on land. Each of the largest groups of organisms, including insects, mites, nematodes, fungi, and plants has marine representatives, but they constitute only a very small fraction of the total number of described species. There are two reasons for this. First, barriers between habitats are sharper on land, and variations in elevation, parent rock, degree of exposure, and



**FIGURE 29.17**

**Food comes to the ocean floor from above.** Looking for all the world like some undersea sunflower, the two sea anemones (actually animals) use a glass-sponge stalk to catch “marine snow,” food particles raining down on the ocean floor from the ocean surface miles above.

other factors have been crucial to the evolution of the millions of species of terrestrial organisms. Second, there are simply few taxonomists actively classifying the profusion of ocean floor life being brought to the surface.

In terms of higher level diversity, the pattern is quite different. Of the major groups of organisms—phyla—most originated in the sea, and almost every one has representatives in the sea. Only a few phyla have been successful on land or in freshwater habitats, but these have given rise to an extraordinarily large number of described species.

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**Although representatives of almost every phylum occur in the sea, an estimated 90% of living species of organisms are terrestrial. This is because of the enormous evolutionary success of a few phyla on land, where the boundaries between different habitats are sharper than they are in the sea.**

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## Marine Ecosystems

The marine environment consists of three major habitats: (1) the **neritic zone**, the zone of shallow waters along the coasts of continents; (2) the **pelagic zone**, the area of water above the ocean floor; and (3) the **benthic zone**, the actual ocean floor (figure 29.18). The part of the ocean floor that drops to depths where light does not penetrate is called the **abyssal zone**.

### The Neritic Zone

The **neritic zone** of the ocean is the area less than 300 meters below the surface along the coasts of continents and islands. The zone is small in area, but it is inhabited by large numbers of species (figure 29.19). The intense and sometimes violent interaction between sea and land in this zone gives a selective advantage to well-secured organisms that can withstand being washed away by the continual beating of the waves. Part of this zone, the **intertidal region**, sometimes called the **littoral region**, is exposed to the air whenever the tides recede.

The world's great fisheries are in shallow waters over continental shelves, either near the continents themselves or in the open ocean, where huge banks come near the surface. Nutrients, derived from land, are much more abundant in coastal and other shallow regions, where upwelling from the depths occurs, than in the open ocean. This accounts for the great productivity of the continental shelf fisheries. The preservation of these fisheries, a source of high-quality protein exploited throughout the world, has become a growing concern. In Chesapeake Bay, where complex systems of rivers enter the ocean from heavily populated areas, environmental stresses have become so severe that they not only threaten the continued existence of formerly highly productive fisheries, but also diminish the quality of human life in these regions. Increased runoff from farms and sewage effluent in areas like Chesapeake Bay add large amounts of nutrients to the water. This increased nutrient supply allows an increase in the numbers of some marine organisms. The increased populations then use up more and more of the oxygen in the water and thus may disturb established populations of organisms such as oysters. Climatic shifts may magnify these effects, and large numbers of marine animals die suddenly as a result.

About three-fourths of the surface area of the world's oceans are located in the tropics. In these waters, where the water temperature remains about 21°C, coral reefs can grow. These highly productive ecosystems can successfully concentrate nutrients, even from the relatively nutrient-poor waters characteristic of the tropics.

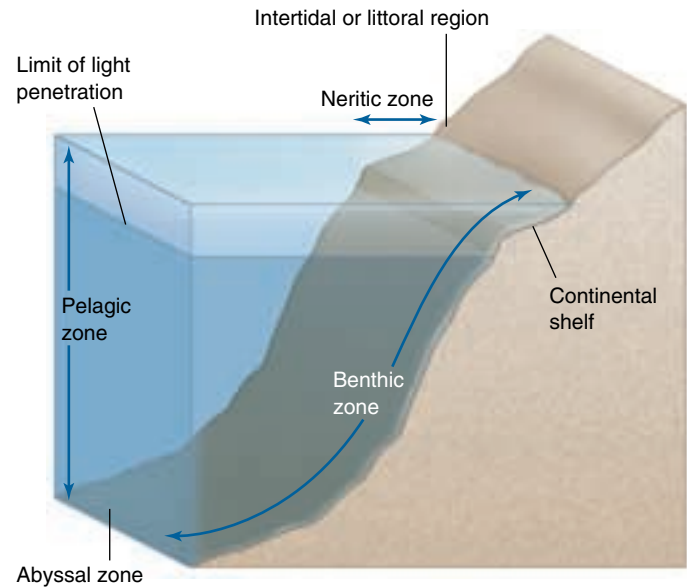


FIGURE 29.18

**Marine ecosystems.** Ecologists classify marine communities into neritic, pelagic, benthic, and abyssal zones, according to depth (which affects how much light penetrates) and distance from shore.



FIGURE 29.19

**Diversity is great in coastal regions.** Fishes and many other kinds of animals find food and shelter among the kelp beds in the coastal waters of temperate regions.

## The Pelagic Zone

Drifting freely in the upper waters of the pelagic zone, a diverse biological community exists, primarily consisting of microscopic organisms called **plankton**. Fish and other larger organisms that swim in these waters constitute the *nekton*, whose members feed on plankton and one another. Some members of the plankton, including protists and some bacteria, are photosynthetic. Collectively, these organisms account for about 40% of all photosynthesis that takes place on earth. Most plankton live in the top 100 meters of the sea, the zone into which light from the surface penetrates freely. Perhaps half of the total photosynthesis in this zone is carried out by organisms less than 10 micrometers in diameter—at the lower limits of size

for organisms—including cyanobacteria and algae, organisms so small that their abundance and ecological importance have been unappreciated until relatively recently.

Many heterotrophic protists and animals live in the plankton and feed directly on photosynthetic organisms and on one another. Gelatinous animals, especially jellyfish and ctenophores, are abundant in the plankton. The largest animals that have ever existed on earth, baleen whales, graze on plankton and nekton as do a number of other organisms, such as fishes and crustaceans.

Populations of organisms that make up plankton can increase rapidly, and the turnover of nutrients in the sea is great, although the productivity in these systems is quite low. Because nitrogen and phosphorus are often present in only small amounts and organisms may be relatively scarce, this productivity reflects rapid use and recycling rather than an abundance of these nutrients.

## The Benthic Zone

The seafloor at depths below 1000 meters, the **abyssal zone**, has about twice the area of all the land on earth. The seafloor itself, sometimes called the **benthic zone**, is a thick blanket of mud, consisting of fine particles that have settled from the overlying water and accumulated over millions of years. Because of high pressures (an additional atmosphere of pressure for every 10 meters of depth), cold temperatures (2° to 3°C), darkness, and lack of food, biologists thought that nothing could live on the seafloor. In fact, recent work has shown that the number of species that live at great depth is quite high. Most of these animals are only a few millimeters in size, although larger ones also occur in these regions. Some of the larger ones are bioluminescent (figure 29.20a) and thus are able to communicate with one another or attract their prey.



(a)



(b)

**FIGURE 29.20**

**Life in the abyssal and benthic zones.** (a) The luminous spot below the eye of this deep-sea fish results from the presence of a symbiotic colony of luminous bacteria. Similar luminous signals are a common feature of deep-sea animals that move about. (b) These giant bearded worms live along vents where water jets from fissures at 350°C and then cools to the 2°C of the surrounding water.

Animals on the sea bottom depend on the meager leftovers from organisms living kilometers overhead. The low densities and small size of most deep-sea bottom animals is in part a consequence of this limited food supply. In 1977, oceanographers diving in a research submarine were surprised to find dense clusters of large animals living on geothermal energy at a depth of 2500 meters. These deep-sea oases occur where seawater circulates through porous rock at sites where molten material from beneath the earth's crust comes close to the rocky surface. A series of these areas occur on the Mid-Ocean Ridge, where basalt erupts through the ocean floor.

This water is heated to temperatures in excess of 350°C and, in the process, becomes rich in reduced compounds. These compounds, such as hydrogen sulfide, provide energy for bacterial primary production through chemosynthesis instead of photosynthesis. Mussels, clams, and large red-plumed worms in a phylum unrelated to any shallow-water invertebrates cluster around the vents (figure 29.20b). Bacteria live symbiotically within the tissues of these animals. The animal supplies a place for the bacteria to live and transfers CO<sub>2</sub>, H<sub>2</sub>S, and O<sub>2</sub> to them for their growth; the bacteria supply the animal with organic compounds to use as food. Polychaete worms (see chapter 46), anemones, and limpets live on free-living chemosynthetic bacteria. Crabs act as scavengers and predators, and some of the fish are also predators. This is one of the few ecosystems on earth that does not depend on the sun's energy.

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About 40% of the world's photosynthetic productivity is estimated to occur in the oceans. The turnover of nutrients in the plankton is much more rapid than in most other ecosystems, and the total amounts of nutrients are very low.

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## Freshwater Habitats

Freshwater habitats are distinct from both marine and terrestrial ones, but they are limited in area. Inland lakes cover about 1.8% of the earth's surface, and running water (streams and rivers) covers about 0.3%. All freshwater habitats are strongly connected with terrestrial ones, with marshes and swamps constituting intermediate habitats. In addition, a large amount of organic and inorganic material continuously enters bodies of fresh water from communities growing on the land nearby (figure 29.21a). Many kinds of organisms are restricted to freshwater habitats (figure 29.21b,c). When organisms live in rivers and streams, they must be able to swim against the current or attach themselves in such a way as to resist the effects of current, or risk being swept away.



(a)



(b)



(c)

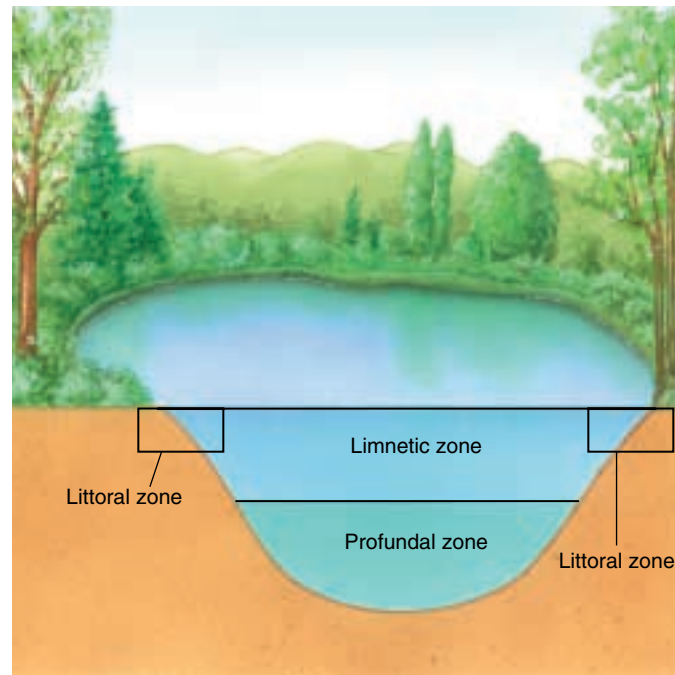
**FIGURE 29.21**

**A nutrient-rich stream.** (a) Much organic material falls or seeps into streams from communities along the edges. This input increases the stream's biological productivity. (b) This speckled darter and (c) this giant waterbug with eggs on its back can only live in freshwater habitats.

## Ponds and Lakes

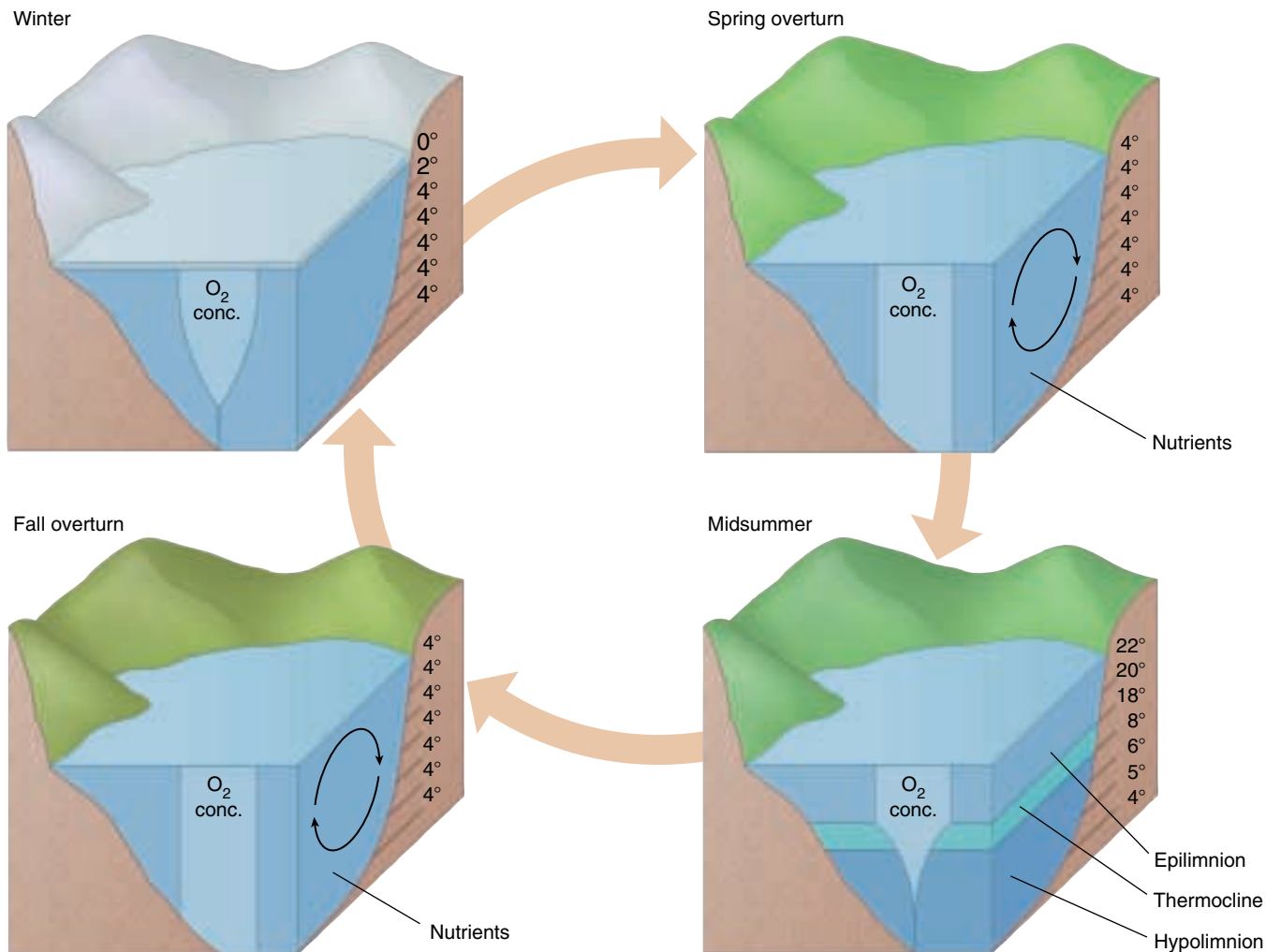
Small bodies of fresh water are called ponds, and larger ones lakes. Because water absorbs light passing through it at wavelengths critical to photosynthesis (every meter absorbs 40% of the red and about 2% of the blue), the distribution of photosynthetic organisms is limited to the upper **photic zone**; heterotrophic organisms occur in the lower **aphotic zone** where very little light penetrates.

Ponds and lakes, like the ocean, have three zones where organisms occur, distributed according to the depth of the water and its distance from shore (figure 29.22). The *littoral zone* is the shallow area along the shore. The *limnetic zone* is the well-illuminated surface water away from the shore, inhabited by plankton and other organisms that live in open water. The *profundal zone* is the area below the limits where light can effectively penetrate.



**FIGURE 29.22**

**The three zones in ponds and lakes.** A shallow “edge” (littoral) zone lines the periphery of the lake, where attached algae and their insect herbivores live. An open-water surface (limnetic) zone lies across the entire lake and is inhabited by floating algae, zooplankton, and fish. A dark, deep-water (profundal) zone overlies the sediments at the bottom of the lake and contains numerous bacteria and wormlike organisms that consume dead debris settling at the bottom of the lake.



**FIGURE 29.23**

**Stratification in fresh water.** The pattern stratification in a large pond or lake in temperate regions is upset in the spring and fall overturns. Of the three layers of water shown, the hypolimnion consists of the densest water, at 4°C; the epilimnion consists of warmer water that is less dense; and the thermocline is the zone of abrupt change in temperature that lies between them. If you have dived into a pond in temperate regions in the summer, you have experienced the existence of these layers.

## Thermal Stratification

Thermal stratification is characteristic of larger lakes in temperate regions (figure 29.23). In summer, warmer water forms a layer at the surface known as the *epilimnion*. Cooler water, called the *hypolimnion* (about 4°C), lies below. An abrupt change in temperature, the thermocline, separates these two layers. Depending on the climate of the particular area, the epilimnion may become as much as 20 meters thick during the summer.

In autumn the temperature of the epilimnion drops until it is the same as that of the hypolimnion, 4°C. When this occurs, epilimnion and hypolimnion mix—a process called fall overturn. Because water is densest at about 4°C, further cooling of the water as winter progresses creates a layer of cooler, lighter water, which freezes to form a layer of ice at the surface. Below the ice, the water tem-

perature remains between 0° and 4°C, and plants and animals can survive. In spring, the ice melts, and the surface water warms up. When it warms back to 4°C, it again mixes with the water below. This process is known as spring overturn. When lake waters mix in the spring and fall, nutrients formerly held in the depths of the lake are returned to the surface, and oxygen from surface waters is carried to the depths.

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**Freshwater habitats include several distinct life zones. These zones shift seasonally in temperate lakes and ponds. In the spring and fall, when their temperatures are equal, shallower and deeper waters of the lake mix, with oxygen being carried to the depths and nutrients being brought to the surface.**

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## Productivity of Freshwater Ecosystems

Some aquatic communities, such as fast-moving streams, are not highly productive. Because the moving water washes away plankton, the photosynthesis that supports the community is limited to algae attached to the surface and to rooted plants.

### The Productivity of Lakes

Lakes can be divided into two categories based on their production of organic matter. **Eutrophic lakes** contain an abundant supply of minerals and organic matter. As the plentiful organic material drifts below the thermocline from the well-illuminated surface waters of the lake, it provides a source of energy for other organisms. Most of these are oxygen-requiring organisms that can easily deplete the oxygen supply below the thermocline during the summer months. The oxygen supply of the deeper waters cannot be replenished until the layers mix in the fall. This lack of oxygen in the deeper waters of some lakes may have profound effects, such as allowing relatively harmless materials such as sulfates and nitrates to convert into toxic materials such as hydrogen sulfide and ammonia.

In **oligotrophic lakes**, organic matter and nutrients are relatively scarce. Such lakes are often deeper than eutrophic lakes and have very clear blue water. Their hypolimnetic water is always rich in oxygen.

Human activities can transform oligotrophic lakes into eutrophic ones. In many lakes, phosphorus is in short supply and is the nutrient that limits growth. When excess phosphorus from sources such as fertilizer runoff, sewage, and detergents enters lakes, it can quickly lead to harmful effects. In many cases, this leads to perfect conditions for the growth of blue-green algae, which proliferate immensely. Soon, larger plants are outcompeted and disappear, along with the animals that live on them. In addition, as these phytoplankton die and decompose, oxygen in the water is used up, killing the natural fish and invertebrate populations. This situation can be remedied if the continual input of phosphorus can be diminished. Given time, lakes can recover and return to pre-pollution states, as happened with Lake Washington pictured in figure 29.24.

### The Productivity of Wetlands

Swamps, marshes, bogs, and other **wetlands** covered with water support a wide variety of water-tolerant plants, called hydrophytes (“water plants”), and a rich diversity of invertebrates, birds, and other animals. Wetlands are among the most productive ecosystems on earth (table 29.2). They also play a key ecological role by providing water storage basins that moderate flooding. Many wetlands are being disrupted by human “development” of

Table 29.2 The Most Productive Ecosystems

| Ecosystem                           | Net Primary Productivity per Unit Area (g/m <sup>2</sup> ) |
|-------------------------------------|------------------------------------------------------------|
| Coral reefs                         | 2500                                                       |
| Tropical rain forest                | 2200                                                       |
| Wetlands                            | 2000                                                       |
| Tropical seasonal forest            | 1600                                                       |
| Estuaries                           | 1500                                                       |
| Temperate evergreen forest          | 1300                                                       |
| Temperate deciduous forest          | 1200                                                       |
| Savanna                             | 900                                                        |
| Boreal forest                       | 800                                                        |
| Cultivated land                     | 650                                                        |
| Continental shelf                   | 360                                                        |
| Lake and stream                     | 250                                                        |
| Open ocean                          | 125                                                        |
| Extreme desert, rock, sand, and ice | 3                                                          |

Source: Whitaker, 1975.



FIGURE 29.24

**Oligotrophic lakes are highly susceptible to pollution.** Lake Washington is an oligotrophic lake near Seattle, Washington. The drainage from fertilizers applied to the plantings around residences, business concerns, and recreational facilities bordering the lake poses an ever-present threat to its deep blue water. By supplying phosphorus, the drainage promotes algal growth. Aerobic bacteria decomposing dead algae deplete the lake's oxygen, killing much of the lake's life.

what is sometimes perceived as otherwise useless land, but government efforts are now underway to protect the remaining wetlands.

**The most productive freshwater ecosystems are wetlands. Most lakes are far less productive, limited by lack of nutrients.**



## Summary

## Questions

## Media Resources

## 29.1 Organisms must cope with a varied environment.

- Organisms employ physiological, morphological, and behavioral mechanisms to cope with variations in the environment.

1. What are several ways that individual organisms adjust to changes in temperature during the course of a year?



- Biosphere Introduction
- Biosphere Quiz
- Climate

## 29.2 Climate shapes the character of ecosystems.

- Warm air rises near the equator and flows toward the poles, descending at about 30° north and south latitude. Because the air falls in these regions, it is warmed, and its moisture-holding capacity increases. The great deserts of the world are formed in these drier latitudes.

2. Why are the majority of great deserts located near 30° north and south latitude? Is it more likely that a desert will form in the interior or at the edge of a continent? Explain why.



- Four Seasons
- Global Air Circulation
- Rainshadow Effect



- Weather

## 29.3 Biomes are widespread terrestrial ecosystems.

- The world's major biomes, or terrestrial communities, can be grouped into eight major categories. These are (1) tropical rain forest; (2) savanna; (3) desert; (4) temperate grassland; (5) temperate deciduous forest; (6) temperate evergreen forest; (7) taiga; and (8) tundra.

3. What is a biome? What are the two key physical factors that affect the distribution of biomes across the earth?



- Soils
- Land Biomes
- Tropical Forests
- Temperate Forests



- Book Review: *Savages* by Kane

## 29.4 Aquatic ecosystems cover much of the earth.

- The ocean contains three major environments: the neritic zone, the pelagic zone, and the benthic zone.
- The neritic zone, which lies along the coasts, is small in area but very productive and rich in species.
- The surface layers of the pelagic zone are home to plankton (drifting organisms) and nekton (actively swimming ones). The productivity of this zone has been underestimated because of the very small size (less than 10 mm) of many of its key organisms and because of its rapid turnover of nutrients.
- The benthic zone is home to a surprising number of species.
- Freshwater habitats constitute only about 2.1% of the earth's surface; most are ponds and lakes. These possess a littoral zone, a limnetic zone, and a profundal zone. The waters in these zones mix seasonally, delivering oxygen to the bottom and nutrients to the surface.

4. What is the difference between plankton and nekton in the ocean's pelagic zone? How important are the photosynthetic plankton to the survival of the earth? Is the turnover of nutrients in the surface zone slow or fast?



- El Nino Southern Oscillation



- Aquatic Systems
- Weather



5. What conditions of the abyssal zone led early deep-sea biologists to believe nothing lived there? What provides the energy for the deep-sea communities found around thermal vents? What kind of organisms live there?

6. Does much diversity occur in the abyssal zone? How are such ecosystems supported in the absence of light?

7. What is the difference between a eutrophic and an oligotrophic lake? Why have humans increased the frequency of lakes becoming eutrophied?

- Student Research: Exotic Species and Freshwater Ecology
- On Science Article: Cold Winter in St. Louis

# 30

## *The Future of the Biosphere*

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### Concept Outline

#### 30.1 The world's human population is growing explosively.

**A Growing Population.** The world's population of 6 billion people is growing rapidly and at current rates will double in 39 years.

#### 30.2 Improvements in agriculture are needed to feed a hungry world.

**The Future of Agriculture.** Much of the effort in searching for new sources of food has focused on improving the productivity of existing crops.

#### 30.3 Human activity is placing the environment under increasing stress.

**Nuclear Power.** Nuclear power, a plentiful source of energy, is neither cheap nor safe.

**Carbon Dioxide and Global Warming.** The world's industrialization has led to a marked increase in the atmosphere's level of CO<sub>2</sub>, with resulting warming of climates.

**Pollution.** Human industrial and agricultural activity introduces significant levels of many harmful chemicals into ecosystems.

**Acid Precipitation.** Burning of cheap high-sulfur coal has introduced sulfur to the upper atmosphere, where it combines with water to form sulfuric acid that falls back to earth, harming ecosystems.

**The Ozone Hole.** Industrial chemicals called CFCs are destroying the atmosphere's ozone layer, removing an essential shield from the sun's UV radiation.

**Destruction of the Tropical Forests.** Much of the world's tropical forest is being destroyed by human activity.

#### 30.4 Solving environmental problems requires individual involvement.

**Environmental Science.** The commitment of one person often makes a key difference in solving environmental problems.

**Preserving Nonreplaceable Resources.** Three key nonreplaceable resources are topsoil, groundwater, and biodiversity.



**FIGURE 30.1**  
New York City by satellite.

The view of New York City in figure 30.1 was photographed from a satellite in the spring of 1985. At the moment this picture was taken, millions of people within its view were talking, hundreds of thousands of cars were struggling through traffic, hearts were being broken, babies born, and dead people buried. Our futures and those of everyone on the planet are linked to the unseen millions in this photograph, for we share the earth with them. A lot of people consume a lot of food and water, use a great deal of energy and raw materials, and produce a great deal of waste. They also have the potential to solve the problems that arise in an increasingly crowded world. In this chapter, we will study how human life affects the environment and how the efforts being mounted can lessen the adverse impact and increase the potential benefits of our burgeoning population.

## 30.1 The world's human population is growing explosively.

### A Growing Population

The current world population of 6 billion people is placing severe strains on the biosphere. How did it grow so large? For the past 300 years, the human birthrate (as a global average) has remained nearly constant, at about 30 births per year per 1000 people. Today it is about 25 births per year per 1000 people. However, at the same time, better sanitation and improved medical techniques have caused the death rate to fall steadily, from about 29 deaths per 1000 people per year to 13 per 1000 per year. Thus, while the birthrate has remained fairly constant and may have even decreased slightly, the tremendous fall in the death rate has produced today's enormous population. The difference between the birth and death rates amounts to an annual worldwide increase of approximately 1.4%. This rate of increase may seem relatively small, but it would double the world's population in only 39 years!

The *annual* increase in world population today is nearly 77 million people, about equal to the current population of Germany. 210,000 people are added to the world each day, or more than 140 every minute! The world population is expected to continue beyond its current level of 6 billion people, perhaps stabilizing at a figure anywhere between 8.5 billion and 20 billion during the next century.

### The Future Situation

About 60% of the people in the world live in tropical or subtropical regions (figure 30.2). An additional 20% will be living in China, and the remaining 20% in the developed or industrialized countries: Europe, the successor states of the Soviet Union, Japan, United States, Canada, Australia, and New Zealand. Although populations of industrialized countries are growing at an annual rate of about 0.3%, those of the developing, mostly tropical countries (excluding China) are growing at an annual rate estimated in 1995 to be about 2.2%. For every person living in an industrialized country like the United States in 1950, there were two people living elsewhere; in 2020, just 70 years later, there will be five.

As you learned in chapter 24, the age structure of a population determines how fast the population will grow. To predict the future growth patterns of a population, it is essential to know what proportion of its individuals have not yet reached childbearing age. In industrialized countries such as the United States, about a fifth of the population is under 15 years of age; in developing countries such as Mexico, the proportion is typically about twice as high. Even if most tropical and subtropical countries consistently carry out the policies they have established to limit population growth, their populations will continue to grow well into the twenty-first century (figure 30.3), and indus-

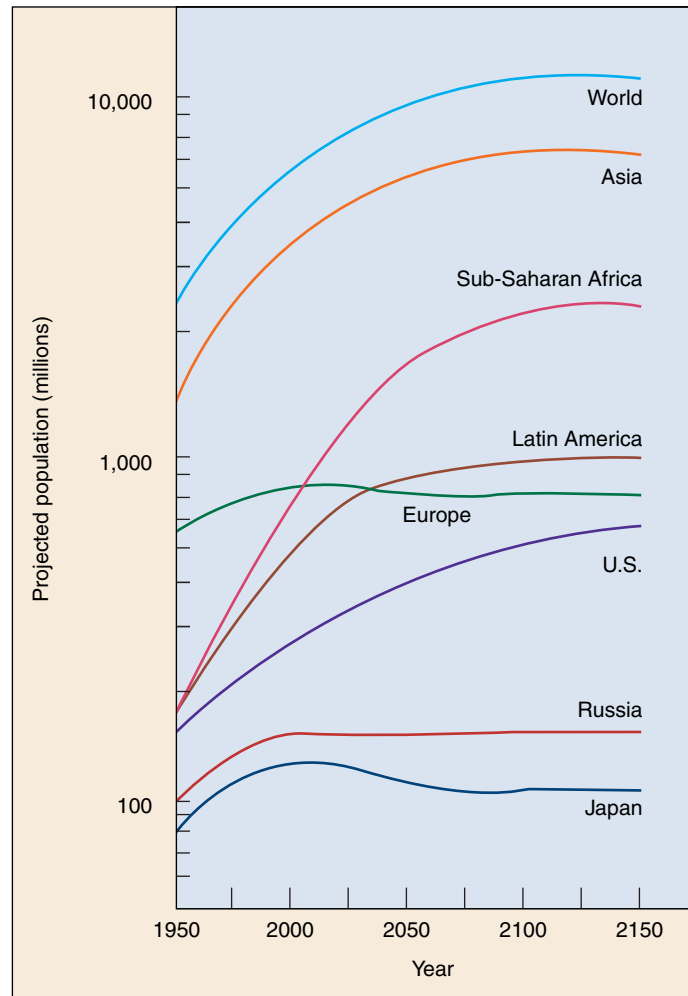


FIGURE 30.2

**Anticipated growth of the global human population.** Despite considerable progress in lowering birthrates, the human population will continue to grow for another century (data are presented above on a log scale). Much of the growth will center in sub-Saharan Africa, the poorest region on the globe, where the population could reach over 2 billion. Fertility rates there currently range from 3 to more than 5 children per woman, compared to fewer than 2.1 in Europe and the United States.

trialized countries will constitute a smaller and smaller proportion of the world's population. If India, with a 1995 population level of about 930 million people (36% under 15 years old), managed to reach a simple replacement reproductive rate by the year 2000, its population would still not stop growing until the middle of the twenty-first century. At present rates of growth, India will have a population of nearly 1.4 billion people by 2025 and will still be growing rapidly.

## Population Growth Rate Starting to Decline

The United Nations has announced that the world population growth rate continues to decline, down from a high of 2.0% in the period 1965–1970 to 1.4% in 1998. Nonetheless, because of the larger population, this amounts to an increase of 77 million people per year to the world population, compared to 53 million per year in the 1960s.

The U.N. attributes the decline to increased family planning efforts and the increased economic power and social status of women. While the U.N. applauds the United States for leading the world in funding family planning programs abroad, some oppose spending money on international family planning. The opposition states that money is better spent on improving education and the economy in other countries, leading to an increased awareness and lowered fertility rates. The U.N. certainly supports the improvement of education programs in developing countries, but, interestingly, it has reported increased education levels *following* a decrease in family size as a result of family planning.

Most countries are devoting considerable attention to slowing the growth rate of their populations, and there are genuine signs of progress. If these efforts are maintained, the world population may stabilize sometime in the next century. No one knows how many people the planet can support, but we clearly already have more people than can be sustainably supported with current technologies.

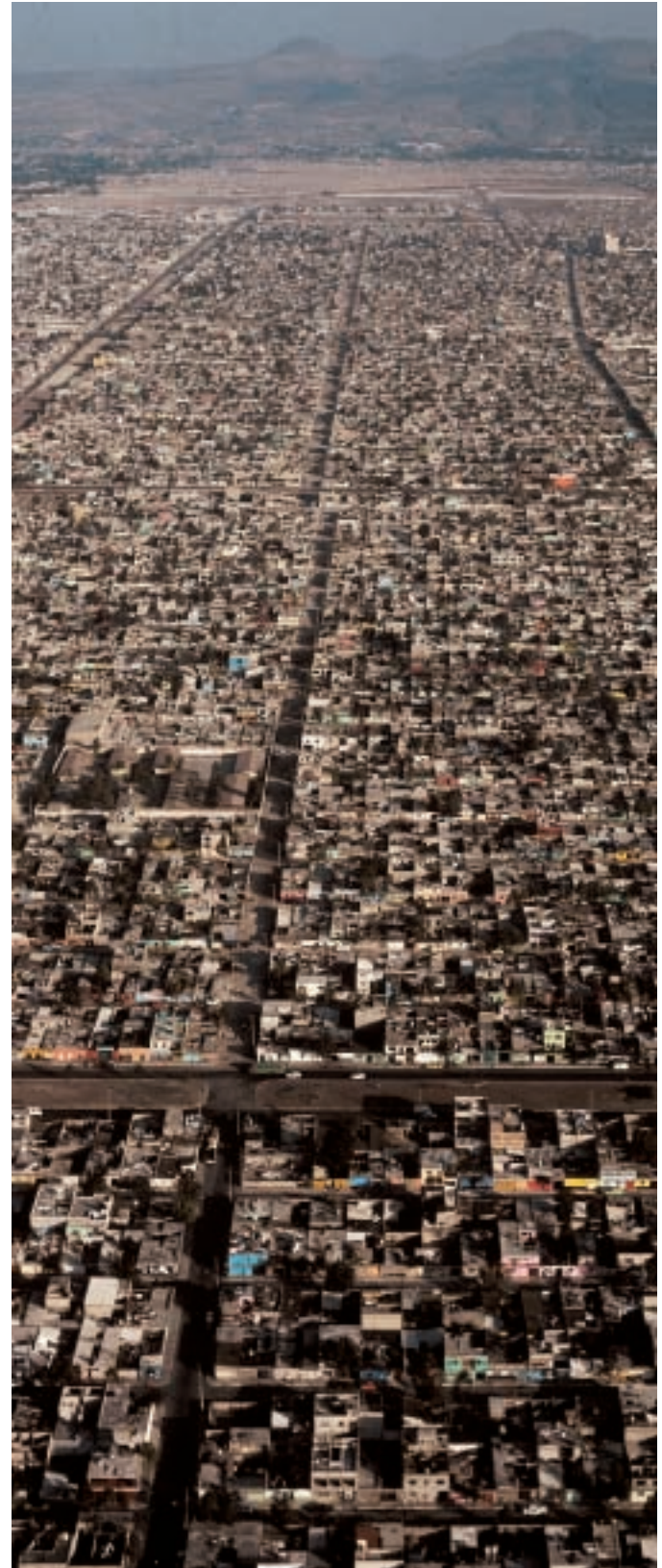
However, population size is not the only factor that determines resource use; per capita consumption is also important. In this respect, we in the developing world need to pay more attention to lessening the impact each of us makes, because, even though the vast majority of the world's population is in developing countries, the vast majority of resource consumption occurs in the developed world. Indeed, the wealthiest 20% of the world's population accounts for 86% of the world's consumption of resources and produces 53% of the world's carbon dioxide emissions, whereas the poorest 20% of the world is responsible for only 1.3% of consumption and 3% of CO<sub>2</sub> emissions. Looked at another way, in terms of resource use, a child born today in the developed world will consume as many resources over the course of his or her life as 30 to 50 children born in the developing world.

Building a sustainable world is the most important task facing humanity's future. The quality of life available to our children in the next century will depend to a large extent on our success both in limiting population growth and the amount of per capita resource consumption.

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**In 1998, the global human population of 6 billion people was growing at a rate of approximately 1.4% annually. At that rate, the population would double in 39 years. Growth rates, however, are declining, but consumption per capita in the developing world is also a significant drain on resources.**

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**FIGURE 30.3**  
Population growth is highest in tropical and subtropical countries. Mexico City, the world's largest city, has well over 20 million inhabitants.

## 30.2 Improvements in agriculture are needed to feed a hungry world.

### The Future of Agriculture

One of the greatest and most immediate challenges facing today's world is producing enough food to feed our expanding population. This problem is often not appreciated by economists, who estimate that world food production has expanded 2.6 times since 1950, more rapidly than the human population. However, virtually all land that can be cultivated is already in use, and much of the world is populated by large numbers of hungry people who are rapidly destroying the sustainable productivity of the lands they inhabit. Well over 20% of the world's topsoil has been lost from agricultural lands since 1950. In the face of these massive problems, we need to consider what the prospects are for increased agricultural productivity in the future.

#### Finding New Food Plants

How many food plants do we use at present? Just three species—rice, wheat, and corn—supply more than half of all human energy requirements. Just over 100 kinds of plants supply over 90% of the calories we consume. Only about 5000 have ever been used for food. There may be tens of thousands of additional kinds of plants, among the 250,000 known species, that could be used for human food

if their properties were fully explored and they were brought into cultivation (figure 30.4).

Agricultural scientists are attempting to identify such new crops, especially ones that will grow well in the tropics and subtropics, where the world's population is expanding most rapidly. Nearly all major crops now grown in the world have been cultivated for hundreds or even thousands of years. Only a few, including rubber and oil palms, have entered widespread cultivation since 1800.

One key feature for which nearly all of our important crops were first selected was ease of growth by relatively simple methods. Today, however, techniques of cultivation are far more sophisticated and are able to improve soil fertility and combat pests. This enables us to consider many more plants as potential crops. Agricultural scientists are searching systematically for new crops that fit the multiple needs of modern society, in ways that would not have been considered earlier.

#### Improving the Productivity of Today's Crops

Searching for new crops is not a quick process. While the search proceeds, the most promising strategy to quickly expand the world food supply is to improve the productivity of crops that are already being grown. Much of the im-



(a)



(b)

**FIGURE 30.4**

**New food plants.** (a) Grain amaranths (*Amaranthus* spp.) were important crops in the Latin American highlands during the days of the Incas and Aztecs. Grain amaranths are fast-growing plants that produce abundant grain rich in lysine, an amino acid rare in plant proteins but essential for animal nutrition. (b) The winged bean (*Psophocarpus tetragonolobus*) is a nitrogen-fixing tropical vine that produces highly nutritious leaves and tubers whose seeds produce large quantities of edible oil. First cultivated in New Guinea and Southeast Asia, the winged bean has spread since the 1970s throughout the tropics.

provement in food production must take place in the tropics and subtropics, where the rapidly growing majority of the world's population lives, including most of those enduring a life of extreme poverty. These people cannot be fed by exports from industrial nations, which contribute only about 8% of their total food at present and whose agricultural lands are already heavily exploited. During the 1950s and 1960s, the so-called Green Revolution introduced new, improved strains of wheat and rice. The production of wheat in Mexico increased nearly tenfold between 1950 and 1970, and Mexico temporarily became an exporter of wheat rather than an importer. During the same decades, food production in India was largely able to outstrip even a population growth of approximately 2.3% annually, and China became self-sufficient in food.

Despite the apparent success of the Green Revolution, improvements were limited. Raising the new agricultural strains of plants requires the expenditure of large amounts of energy and abundant supplies of fertilizers, pesticides, and herbicides, as well as adequate machinery. For example, in the United States it requires about 1000 times as much energy to produce the same amount of wheat produced from traditional farming methods in India.

Biologists are playing a crucial role in improving existing crops and in developing new ones by applying traditional methods of plant breeding and selection to many new, nontraditional crops in the tropics and subtropics (see figure 30.4).

### Genetic Engineering to Improve Crops

Genetic engineering techniques (discussed in chapter 19) make it possible to produce plants resistant to specific herbicides. These herbicides can then control weeds much more effectively, without damaging crop plants. Genetic engineers are also developing new strains of plants that will grow successfully in areas where they previously could not grow. Desirable characteristics are being introduced into important crop plants. Genetically modified rice, for example, is no longer deficient in ascorbic acid and iron, providing a major improvement in human nutrition. Other modifications allow crops to tolerate irrigation with salt water, fix nitrogen, carry out C<sub>4</sub> photosynthesis, and produce substances that deter pests and diseases.

Genetically modified crops (so-called GM foods) have proven to be a highly controversial issue, one currently being debated in legislative bodies all over the globe. Critics fear loss of genetic diversity, escape of engineered varieties into the environment, harm to insects feeding near GM crops, undo influence of seed companies, and many other real or imagined potential problems. The issue of risk is assessed in chapter 19. While risks must be carefully considered, the ability to transfer genes between organisms, first accomplished in a laboratory in 1973, has tremendous potential for the improvement of crop plants as the twenty-first century opens.

### New Approaches to Cultivation

Several new approaches may improve crop production. “No-till” agriculture, spreading widely in the United States and elsewhere in the 1990s, conserves topsoil and so is a desirable agricultural practice for many areas. On the other hand, **hydroponics**, the cultivation of plants in water containing an appropriate mixture of nutrients, holds less promise. It does not differ remarkably in its requirements and challenges from growing plants on land. It requires as much fertilizer and other chemicals, as well as the water itself.

The oceans were once regarded as an inexhaustible source of food, but overexploitation of their resources is limiting the world catch more each year, and these catches are costing more in terms of energy. Mismanagement of fisheries, mainly through overfishing, local pollution, and the destruction of fish breeding and feeding grounds, has lowered the catch of fish in the sea by about 20% from maximum levels. Many fishing areas that were until recently important sources of food have been depleted or closed. For example, the Grand Banks in the North Atlantic Ocean off Newfoundland, a major source of cod and other fish, are now nearly depleted. In 1994, the Canadian government prohibited all cod fishing there indefinitely, throwing 27,000 fishermen out of work, and the United States government banned all fishing on Georges Bank and other defined New England waters. Populations of Atlantic bluefin tuna have dropped 90% since 1975. The United Nations Food and Agriculture organization estimated in 1993 that 13 of 17 major ocean fisheries are in trouble, with the annual marine fish catch dropping from 86 million metric tons in 1986 to 82.5 million tons by 1992 and continuing to fall each year as the intensity of the fishing increases.

The development of new kinds of food, such as microorganisms cultured in nutrient solutions, should definitely be pursued. For example, the photosynthetic, nitrogen-fixing cyanobacterium *Spirulina* is being investigated in several countries as a possible commercial food source. It is a traditional food in Africa, Mexico, and other regions. *Spirulina* thrives in very alkaline water, and it has a higher protein content than soybeans. Ponds in which it grows are 10 times more productive than wheat fields. Such protein-rich concentrates of microorganisms could provide important nutritional supplements. However, psychological barriers must be overcome to persuade people to eat such foods, and the processing required tends to be energy-expensive.

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**Just over 100 kinds of plants, out of the roughly 250,000 known, supply more than 90% of all the calories we consume. Many more could be developed by a careful search for new crops.**

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### 30.3 Human activity is placing the environment under increasing stress.

The simplest way to gain a feeling for the dimensions of the global environmental problem we face is simply to scan the front pages of any newspaper or news magazine or to watch television. Although they are only a sampling, features selected by these media teach us a great deal about the scale and complexity of the challenge we face. We will discuss a few of the most important issues here.

#### Nuclear Power

At 1:24 A.M. on April 26, 1986, one of the four reactors of the Chernobyl nuclear power plant blew up. Located in Ukraine 100 kilometers north of Kiev, Chernobyl was one of the largest nuclear power plants in Europe, producing 1000 megawatts of electricity, enough to light a medium-sized city. Before dawn on April 26, workers at the plant hurried to complete a series of tests of how Reactor Number 4 performed during a power reduction and took a foolish short-cut: they shut off all the safety systems. Reactors at Chernobyl were graphite reactors designed with a series of emergency systems that shut the reactors down at low power, because the core is unstable then—and these are the emergency systems the workers turned off. A power surge occurred during the test, and there was nothing to dampen it. Power zoomed to hundreds of times the maximum, and a white-hot blast with the force of a ton of dynamite partially melted the fuel rods and heated a vast head of steam that blew the reactor apart.

The explosion and heat sent up a plume 5 kilometers high, carrying several tons of uranium dioxide fuel and fission products. The blast released over 100 megacuries of radioactivity, making it the largest nuclear accident ever reported; by comparison, the Three Mile Island accident in Pennsylvania in 1979 released 17 curies, millions of times less. This cloud traveled first northwest, then southeast, spreading the radioactivity in a band across central Europe from Scandinavia to Greece. Within a 30-kilometer radius of the reactor, at least one-fifth of the population, some 24,000 people, received serious radiation doses (greater than 45 rem). Thirty-one individuals died as a direct result of radiation poisoning, most of them firefighters who succeeded in preventing the fire from spreading to nearby reactors.

The rest of Europe received a much lower but still significant radiation dose. Data indicate that, because of the large numbers of people exposed, radiation outside of the immediate Chernobyl area can be expected to cause from 5000 to 75,000 cancer deaths.

#### The Promise of Nuclear Power

Our industrial society has grown for over 200 years on a diet of cheap energy. Until recently, much of this energy

has been derived from burning wood and fossil fuels: coal, gas, and oil. However, as these sources of fuel become increasingly scarce and the cost of locating and extracting new deposits becomes more expensive, modern society is being forced to look elsewhere for energy. The great promise of nuclear power is that it provides an alternative source of plentiful energy. Although nuclear power is not cheap—power plants are expensive to build and operate—its raw material, uranium ore, is so common in the earth's crust that it is unlikely we will ever run out of it.

Burning coal and oil to obtain energy produces two undesirable chemical by-products: sulfur and carbon dioxide. The sulfur emitted from burning coal is a principal cause of acid rain, while the CO<sub>2</sub> produced from burning all fossil fuels is a major greenhouse gas (see the discussion of global warming in the next section). For these reasons, we need to find replacements for fossil fuels.

For all of its promise of plentiful energy, nuclear power presents several new problems that must be addressed before its full potential can be realized. First, safe operation of the world's approximately 390 nuclear reactors must be ensured. A second challenge is the need to safely dispose of the radioactive wastes produced by the plants and to safely decommission plants that have reached the end of their useful lives (about 25 years). In 1997, over 35 plants were more than 25 years old, and not one has been safely decommissioned, its nuclear wastes disposed of. A third challenge is the need to guard against terrorism and sabotage, because the technology of nuclear power generation is closely linked to that of nuclear weapons.

For these reasons, it is important to continue to investigate and develop other promising alternatives to fossil fuels, such as solar energy and wind energy. The generation of electricity by burning fossil fuels accounts for up to 15% of global warming gas emissions in the United States. As much as 75% of the electricity produced in the United States and Canada currently is wasted through the use of inefficient appliances, according to scientists at Lawrence Berkeley Laboratory. Using highly efficient motors, lights, heaters, air conditioners, refrigerators, and other technologies already available could save huge amounts of energy and greatly reduce global warming gas emission. For example, a new, compact fluorescent light bulb uses only 20% of the amount of electricity a conventional light bulb uses, provides equal or better lighting, lasts up to 13 times longer, and provides substantial cost savings.

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**Nuclear power offers plentiful energy for the world's future, but its use involves significant problems and dangers.**

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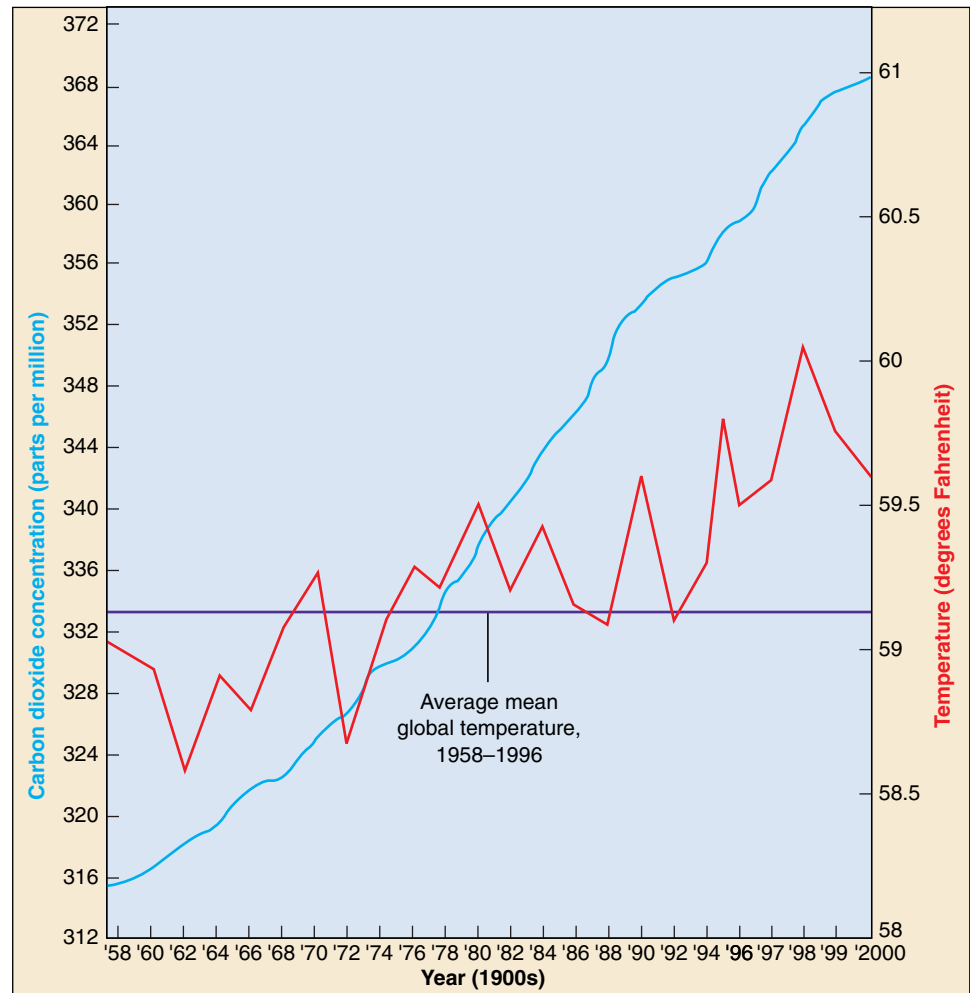
## Carbon Dioxide and Global Warming

By studying earth's history and making comparisons with other planets, scientists have determined that concentrations of gases in the atmosphere, particularly carbon dioxide, maintain the average temperature on earth about 25°C higher than it would be if these gases were absent. Carbon dioxide and other gases trap the longer wavelengths of infrared light, or heat, radiating from the surface of the earth, creating what is known as a **greenhouse effect** (figure 30.5). The atmosphere acts like the glass of a gigantic greenhouse surrounding the earth.

Roughly seven times as much carbon dioxide is locked up in fossil fuels as exists in the atmosphere today. Before industrialization, the concentration of carbon dioxide in the atmosphere was approximately 260 to 280 parts per million (ppm). Since the extensive use of fossil fuels, the amount of carbon dioxide in the atmosphere has been increasing rapidly. During the 25-year period starting in 1958, the concentration of carbon dioxide increased from 315 ppm to more than 340 ppm and continues to rise. Climatologists have calculated that the actual mean global temperature has increased about 1°C since 1900, a change known as **global warming**.

In a recent study, the U.S. National Research Council estimated that the concentration of carbon dioxide in the atmosphere would pass 600 ppm (roughly double the current level) by the third quarter of the next century, and might exceed that level as soon as 2035. These concentrations of carbon dioxide, if actually reached, would warm global surface air by between 1.5° and 4.5°C. The actual increase might be considerably greater, however, because a number of trace gases, such as nitrous oxide, methane, ozone, and chlorofluorocarbons, are also increasing rapidly in the atmosphere as a result of human activities. These gases have warming, or “greenhouse,” effects similar to those of carbon dioxide. One, methane, increased from 1.14 ppm in the atmosphere in 1951 to 1.68 ppm in 1986—nearly a 50% increase.

Major problems associated with climatic warming include rising sea levels. Sea levels may have already risen



**FIGURE 30.5**

**The greenhouse effect.** The concentration of carbon dioxide in the atmosphere has steadily increased since the 1950s (*blue line*). The red line shows the general increase in average global temperature for the same period of time.

Source: Data from Geophysical Monograph, American Geophysical Union, National Academy of Sciences, and National Center for Atmospheric Research.

2 to 5 centimeters from global warming. If the climate becomes so warm that the polar ice caps melt, sea levels would rise by more than 150 meters, flooding the entire Atlantic coast of North America for an average distance of several hundred kilometers inland.

Changes in the distribution of precipitation are difficult to model. Certainly, changing climatic patterns are likely to make some of the best farmlands much drier than they are at present. If the climate warms as rapidly as many scientists project, the next 50 years may see greatly altered weather patterns, a rising sea level, and major shifts of deserts and fertile regions.

**As the global concentration of carbon dioxide increases, the world's temperature is rising, with great potential impact on the world's climate.**

## Pollution

The River Rhine is a broad ribbon of water that runs through the heart of Europe. From high in the Alps that separate Italy and Switzerland, the Rhine flows north across the industrial regions of Germany before reaching Holland and the sea. Where it crosses the mountains between Mainz and Coblenz, Germany, the Rhine is one of the most beautiful rivers on earth. On the first day of November 1986, the Rhine almost died.

The blow that struck at the life of the Rhine did not at first seem deadly. Firefighters were battling a blaze that morning in Basel, Switzerland. The fire was gutting a huge warehouse, into which firefighters shot streams of water to dampen the flames. The warehouse belonged to Sandoz, a giant chemical company. In the rush to contain the fire, no one thought to ask what chemicals were stored in the warehouse. By the time the fire was out, streams of water had washed 30 tons of mercury and pesticides into the Rhine.

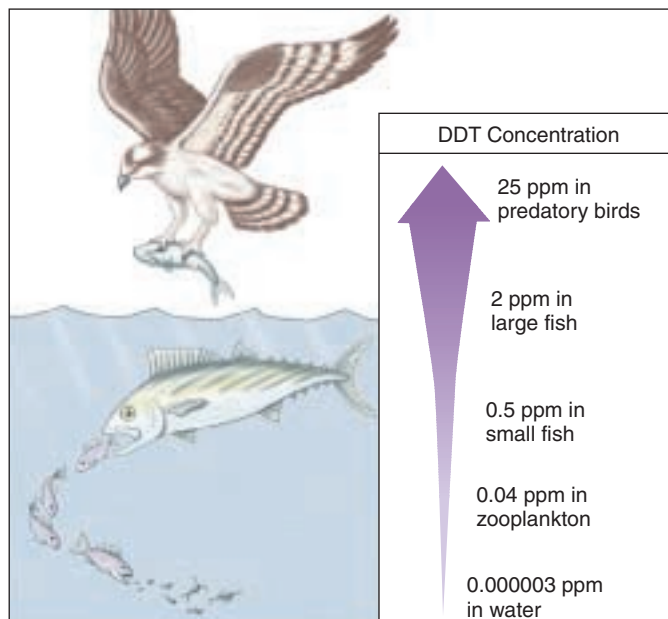
Flowing downriver, the deadly wall of poison killed everything it passed. For hundreds of kilometers, dead fish blanketed the surface of the river. Many cities that use the water of the Rhine for drinking had little time to make other arrangements. Even the plants in the river began to die. All across Germany, from Switzerland to the sea, the river reeked of rotting fish, and not one drop of water was safe to drink.

Six months later, Swiss and German environmental scientists monitoring the effects of the accident were able to report that the blow to the Rhine was not mortal. Enough small aquatic invertebrates and plants had survived to provide a basis for the eventual return of fish and other water life, and the river was rapidly washing out the remaining residues from the spill. A lesson difficult to ignore, the spill on the Rhine has caused the governments of Germany and Switzerland to intensify efforts to protect the river from future industrial accidents and to regulate the growth of chemical and industrial plants on its shores.

### The Threat of Pollution

The pollution of the Rhine is a story that can be told countless times in different places in the industrial world, from Love Canal in New York to the James River in Virginia to the town of Times Beach in Missouri. Nor are all pollutants that threaten the sustainability of life immediately toxic. Many forms of pollution arise as by-products of industry. For example, the polymers known as plastics, which we produce in abundance, break down slowly in nature. Much is burned or otherwise degraded to smaller vinyl chloride units. Virtually all of the plastic that has ever been produced is still with us, in one form or another. Collectively, it constitutes a new form of pollution.

Widespread agriculture, carried out increasingly by modern methods, introduces large amounts of many new kinds of chemicals into the global ecosystem, including



**FIGURE 30.6**

**Biological magnification of DDT.** Because DDT accumulates in animal fat, the compound becomes increasingly concentrated in higher levels of the food chain. Before DDT was banned in the United States, predatory bird populations drastically declined because DDT made their eggshells thin and fragile enough to break during incubation.

pesticides, herbicides, and fertilizers. Industrialized countries like the United States now attempt to carefully monitor side effects of these chemicals. Unfortunately, large quantities of many toxic chemicals no longer manufactured still circulate in the ecosystem.

For example, the chlorinated hydrocarbons, a class of compounds that includes DDT, chlordane, lindane, and dieldrin, have all been banned for normal use in the United States, where they were once widely used. They are still manufactured in the United States, however, and exported to other countries, where their use continues. Chlorinated hydrocarbon molecules break down slowly and accumulate in animal fat. Furthermore, as they pass through a food chain, they become increasingly concentrated in a process called **biological magnification** (figure 30.6). DDT caused serious problems by leading to the production of thin, fragile eggshells in many predatory bird species in the United States and elsewhere until the late 1960s, when it was banned in time to save the birds from extinction. Chlorinated compounds have other undesirable side effects and exhibit hormonelike activities in the bodies of animals, disrupting normal hormonal cycles with sometimes potentially serious consequences.

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**Chemical pollution is causing ecosystems to accumulate many harmful substances, as the result of spills and runoff from agricultural or urban use.**

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## Acid Precipitation

The Four Corners power plant in New Mexico burns coal, sending smoke up high into the atmosphere through its smokestacks, each over 65 meters tall. The smoke the stacks belch out contains high concentrations of sulfur dioxide and other sulfates, which produce acid when they combine with water vapor in the air. The intent of those who designed the plant was to release the sulfur-rich smoke high in the atmosphere, where winds would disperse and dilute it, carrying the acids far away.

Environmental effects of this acidity are serious. Sulfur introduced into the upper atmosphere combines with water vapor to produce sulfuric acid, and when the water later falls as rain or snow, the precipitation is acid. Natural rainwater rarely has a pH lower than 5.6; in the northeastern United States, however, rain and snow now have a pH of about 3.8, roughly 100 times as acid (figure 30.7).

Acid precipitation destroys life. Thousands of lakes in southern Sweden and Norway no longer support fish; these lakes are now eerily clear. In the northeastern United States and eastern Canada, tens of thousands of lakes are dying biologically as a result of acid precipitation. At pH levels below 5.0, many fish species and other aquatic animals die, unable to reproduce. In southern Sweden and elsewhere, groundwater now has a pH between 4.0 and 6.0, as acid precipitation slowly filters down into the underground reservoirs.

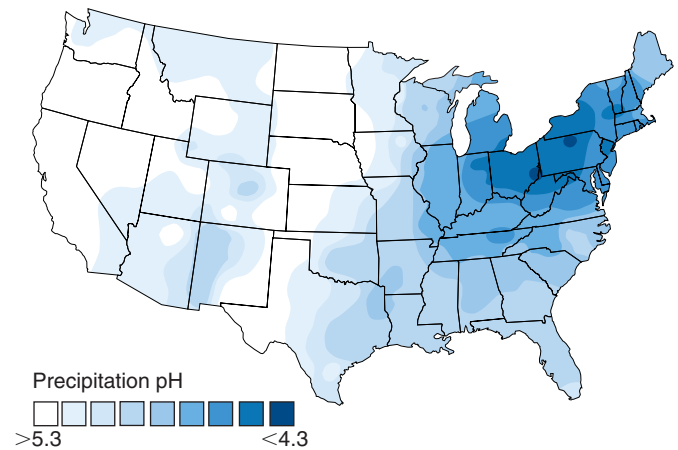
There has been enormous forest damage in the Black Forest in Germany and in the forests of the eastern United States and Canada. It has been estimated that at least 3.5 million hectares of forest in the northern hemisphere are being affected by acid precipitation (figure 30.8), and the problem is clearly growing.

Its solution at first seems obvious: capture and remove the emissions instead of releasing them into the atmosphere. However, there are serious difficulties in executing this solution. First, it is expensive. The costs of installing and maintaining the necessary “scrubbers” in the United States are estimated to be 4 to 5 billion dollars per year. An additional difficulty is that the polluter and the recipient of the pollution are far from each other, and neither wants to pay for what they view as someone else’s problem. The Clean Air Act revisions of 1990 addressed this problem in the United States significantly for the first time, and substantial worldwide progress has been made in implementing a solution.

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Industrial pollutants such as nitric and sulfuric acids, introduced into the upper atmosphere by factory smokestacks, are spread over wide areas by the prevailing winds and fall to earth with precipitation called “acid rain,” lowering the pH of water on the ground and killing life.

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**FIGURE 30.7**  
**pH values of rainwater in the United States.** Precipitation in the United States, especially in the Northeast, is more acidic than that of natural rainwater, which has a pH of about 5.6.



**FIGURE 30.8**  
**Damage to trees at Clingman's Dome, Tennessee.** Acid precipitation weakens trees and makes them more susceptible to pests and predators.

## The Ozone Hole

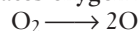
The swirling colors of the satellite photos in figure 30.9 represent different concentrations of **ozone** ( $O_3$ ), a different form of oxygen gas than  $O_2$ . As you can see, over Antarctica there is an “ozone hole” three times the size of the United States, an area within which the ozone concentration is much less than elsewhere. The ozone thinning appeared for the first time in 1975. The hole is not a permanent feature, but rather becomes evident each year for a few months during Antarctic winter. Every September from 1975 onward, the ozone “hole” has reappeared. Each year the layer of ozone is thinner and the hole is larger.

The major cause of the ozone depletion had already been suggested in 1974 by Sherwood Roland and Mario Molina, who were awarded the Nobel Prize for their work in 1995. They proposed that chlorofluorocarbons (CFCs), relatively inert chemicals used in cooling systems, fire extinguishers, and Styrofoam containers, were percolating up through the atmosphere and reducing  $O_3$  molecules to  $O_2$ . One chlorine atom from a CFC molecule could destroy 100,000 ozone molecules in the following mechanism:

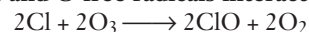
**UV radiation causes CFCs to release Cl atoms:**



**UV creates oxygen free radicals:**



**Cl atoms and O free radicals interact with ozone:**



**Net reaction:**  $2O_3 \longrightarrow 3O_2$

Although other factors have also been implicated in ozone depletion, the role of CFCs is so predominant that worldwide agreements have been signed to phase out their production. The United States banned the production of CFCs and other ozone-destroying chemicals after 1995. Nonetheless, the CFCs that were manufactured earlier are moving slowly upward through the atmosphere. The ozone layer will be further depleted before it begins to form again.

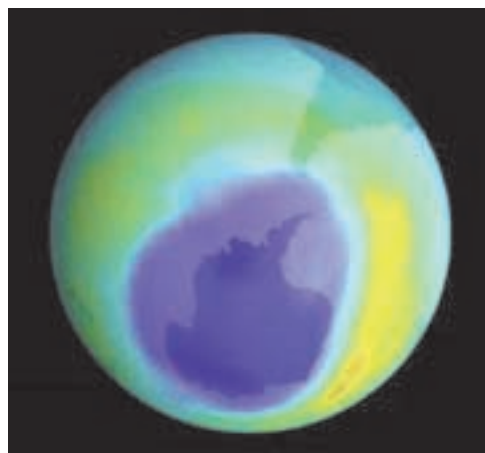
Thinning of the ozone layer in the stratosphere, 25 to 40 kilometers above the surface of the earth, is a matter of serious concern. This layer protects life from the harmful ultraviolet rays that bombard the earth continuously from the sun. Life appeared on land only after the oxygen layer was sufficiently thick to generate enough ozone to shield the surface of the earth from these destructive rays.

Ultraviolet radiation is a serious human health concern. Every 1% drop in atmospheric ozone is estimated to lead to a 6% increase in the incidence of skin cancers. At middle latitudes, the approximately 3% drop that has already occurred worldwide is estimated to have increased skin cancers by as much as 20%. A type of skin cancer (melanoma) is one of the more lethal human diseases.

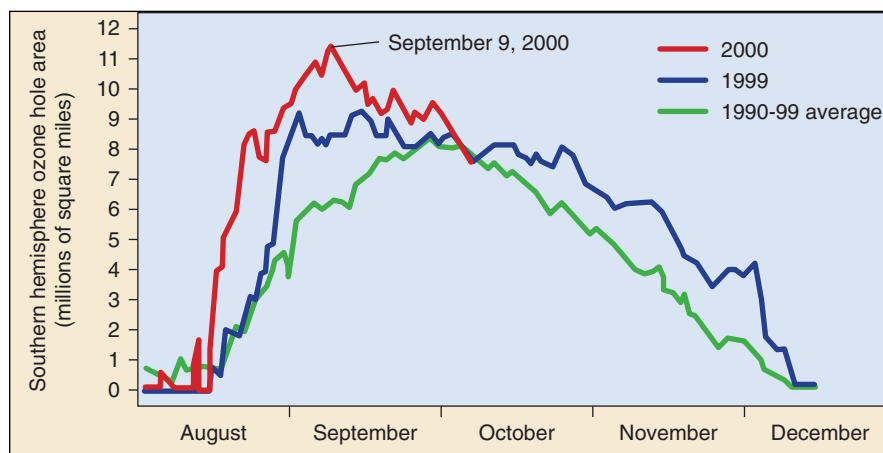
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**Industrial CFCs released into the atmosphere react at very cold temperatures with ozone, converting it to oxygen gas. This has the effect of destroying the earth's ozone shield and exposing the earth's surface to increased levels of harmful UV radiation.**

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(a)



(b)

**FIGURE 30.9**

**The ozone hole over Antarctica is still growing.** For decades NASA satellites have tracked the extent of ozone depletion over Antarctica. Every year since 1979 an ozone “hole” has appeared in August when sunlight triggers chemical reactions in cold air trapped over the South Pole during Antarctic winter. The hole intensifies during September before tailing off as temperatures rise in November–December. In 2000, the 11.4 million square-mile hole (dark blue in the satellite image) covered an area larger than the United States, Canada, and Mexico combined, the largest hole ever recorded. In September 2000, the hole extended over Punta Arenas, a city of about 120,000 people southern Chile, exposing residents to very high levels of UV radiation.

## Destruction of the Tropical Forests

More than half of the world's human population lives in the tropics, and this percentage is increasing rapidly. For global stability, and for the sustainable management of the world ecosystem, it will be necessary to solve the problems of food production and regional stability in these areas. World trade, political and economic stability, and the future of most species of plants, animals, fungi, and microorganisms depend on our addressing these problems.

### Rain Forests Are Rapidly Disappearing

Tropical rain forests are biologically the richest of the world's biomes. Most other kinds of tropical forest, such as seasonally dry forests and savanna forests, have already been largely destroyed—because they tend to grow on more fertile soils, they were exploited by humans a long time ago. Now the rain forests, which grow on poor soils, are being destroyed. In the mid-1990s, it is estimated that only about 5.5 million square kilometers of tropical rain forest still exist in a relatively undisturbed form. This area, about two-thirds of the size of the United States (excluding Alaska), represents about half of the original extent of the rain forest. From it, about 160,000 square kilometers are being clear-cut every year, with perhaps an equivalent amount severely disturbed by shifting cultivation, firewood gathering, and the clearing of land for cattle ranching. The total area of tropical rain forest destroyed—and therefore permanently removed from the world total—amounts to an area greater than the size of Indiana each year. At this rate, all of the tropical rain forest in the world will be gone in about 30 years; but in many regions, the rate of destruction is much more rapid. As a result of this overexploitation, experts predict there will be little undisturbed tropical forest left anywhere in the world by early in the next century. Many areas now occupied by dense, species-rich forests may still be tree-covered, but the stands will be sparse and species-poor.

### A Serious Matter

Not only does the disappearance of tropical forests represent a tragic loss of largely unknown biodiversity, but the loss of the forests themselves is ecologically a serious mat-



(a)



(b)

**FIGURE 30.10**

**Destroying the tropical forests.** (a) When tropical forests are cleared, the ecological consequences can be disastrous. These fires are destroying rain forest in Brazil and clearing it for cattle pasture. (b) The consequences of deforestation can be seen on these middle-elevation slopes in Ecuador, which now support only low-grade pastures and permit topsoil to erode into the rivers (note the color of the water, stained brown by high levels of soil erosion). These areas used to support highly productive forest, which protected the watersheds of the area, in the 1970s.

ter. Tropical forests are complex, productive ecosystems that function well in the areas where they have evolved. When people cut a forest or open a prairie in the north temperate zone, they provide farmland that we know can be worked for generations. In most areas of the tropics, people are unable to engage in continuous agriculture. When they clear a tropical forest, they engage in a one-time consumption of natural resources that will never be available again (figure 30.10). The complex ecosystems built up over millions of years are now being dismantled, in almost complete ignorance, by humans.

What biologists must do is to learn more about the construction of sustainable agricultural ecosystems that will meet human needs in tropical and subtropical regions. The ecological concepts we have been reviewing in the last three chapters are universal principles. The undisturbed tropical rain forest has one of the highest rates of net primary productivity of any plant community on earth, and it is therefore imperative to develop ways that it can be harvested for human purposes in a sustainable, intelligent way.

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**More than half of the tropical rain forests have been destroyed by human activity, and the rate of loss is accelerating.**

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## 30.4 Solving environmental problems requires individual involvement.

### Environmental Science

**Environmental scientists** attempt to find solutions to environmental problems, considering them in a broad context. Unlike biology or ecology, sciences that seek to learn general principles about how life functions, environmental science is an applied science dedicated to solving practical problems. Its basic tools are derived from ecology, geology, meteorology, social sciences, and many other areas of knowledge that bear on the functioning of the environment and our management of it. Environmental science addresses the problems created by rapid human population growth: an increasing need for energy, a depletion of resources, and a growing level of pollution.

### Solving Environmental Problems

The problems our severely stressed planet faces are not insurmountable. A combination of scientific investigation and public action, when brought to bear effectively, can solve environmental problems that seem intractable. Viewed simply, there are five components to solving any environmental problem:

1. **Assessment.** The first stage in addressing any environmental problem is scientific analysis, the gathering of information. Data must be collected and experiments performed to construct a model that describes the situation. This model can be used to make predictions about the future course of events.
2. **Risk analysis.** Using the results of scientific analysis as a tool, it is possible to analyze what could be expected to happen if a particular course of action were followed. It is necessary to evaluate not only the potential for solving the environmental problem, but also any adverse effects a plan of action might create.
3. **Public education.** When a clear choice can be made among alternative courses of action, the public must be informed. This involves explaining the problem in terms the public can understand, presenting the alternatives available, and explaining the probable costs and results of the different choices.
4. **Political action.** The public, through its elected officials selects a course of action and implements it. Choices are particularly difficult to implement when environmental problems transcend national boundaries.
5. **Follow-through.** The results of any action should be carefully monitored to see whether the environmental problem is being solved as well as to evaluate and improve the initial modeling of the problem. Every environmental intervention is an experiment, and we need the knowledge gained from each one to better address future problems.

### Individuals Can Make the Difference

The development of appropriate solutions to the world's environmental problems must rest partly on the shoulders of politicians, economists, bankers, engineers—many kinds of public and commercial activity will be required. However, it is important not to lose sight of the key role often played by informed individuals in solving environmental problems. Often one person has made the difference; two examples serve to illustrate the point.

**The Nashua River.** Running through the heart of New England, the Nashua River was severely polluted by mills established in Massachusetts in the early 1900s. By the 1960s, the river was clogged with pollution and declared ecologically dead. When Marion Stoddart moved to a town along the river in 1962, she was appalled. She approached the state about setting aside a “greenway” (trees running the length of the river on both sides), but the state wasn’t interested in buying land along a filthy river. So Stoddart organized the Nashua River Cleanup Committee and began a campaign to ban the dumping of chemicals and wastes into the river. The committee presented bottles of dirty river water to politicians, spoke at town meetings, recruited businesspeople to help finance a waste treatment plant, and began to clean garbage from the Nashua’s banks. This citizen’s campaign, coordinated by Stoddart, greatly aided passage of the Massachusetts Clean Water Act of 1966. Industrial dumping into the river is now banned, and the river has largely recovered.

**Lake Washington.** A large, 86 km<sup>2</sup> freshwater lake east of Seattle, Lake Washington became surrounded by Seattle suburbs in the building boom following the Second World War. Between 1940 and 1953, a ring of 10 municipal sewage plants discharged their treated effluent into the lake. Safe enough to drink, the effluent was believed “harmless.” By the mid-1950s a great deal of effluent had been dumped into the lake (try multiplying 80 million liters/day  $\times$  365 days/year  $\times$  10 years). In 1954, an ecology professor at the University of Washington in Seattle, W. T. Edmondson, noted that his research students were reporting filamentous blue-green algae growing in the lake. Such algae require plentiful nutrients, which deep freshwater lakes usually lack—the sewage had been fertilizing the lake! Edmondson, alarmed, began a campaign in 1956 to educate public officials to the danger: bacteria decomposing dead algae would soon so deplete the lake’s oxygen that the lake would die. After five years, joint municipal taxes financed the building of a sewer to carry the effluent out to sea. The lake is now clean.

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**In solving environmental problems, the commitment of one person can make a critical difference.**

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## Preserving Nonreplaceable Resources

Among the many ways ecosystems are suffering damage, one class of problem stands out as more serious than the rest: consuming or destroying resources that we cannot replace in the future. While a polluted stream can be cleaned up, no one can restore an extinct species. In the United States, we are consuming three nonreplaceable resources at alarming rates: topsoil, groundwater, and biodiversity. We will briefly discuss the first two of these in this chapter, with a more detailed discussion of biodiversity in the following chapter.

### Topsoil

The United States is one of the most productive agricultural countries on earth, largely because much of it is covered with particularly fertile soils. Our midwestern farm belt sits astride what was once a great prairie. The **topsoil** of that ecosystem accumulated bit by bit from countless generations of animals and plants until, by the time humans began to plow it, the rich soil extended down several feet.

We can never replace this rich topsoil, the capital upon which our country's greatness is built, yet we are allowing it to be lost at a rate of centimeters every decade. By repeatedly tilling (turning the soil over) to eliminate weeds, we permit rain to wash more and more of the topsoil away, into rivers, and eventually out to sea. Our country has lost one-quarter of its topsoil since 1950! New approaches are desperately needed to lessen our reliance on intensive cultivation. Some possible solutions include using genetic engineering to make crops resistant to weed-killing herbicides and terracing to recapture lost topsoil.

### Groundwater

A second resource we cannot replace is **groundwater**, water trapped beneath the soil within porous rock reservoirs called aquifers (figure 30.11). This water seeped into its underground reservoir very slowly during the last ice age over 12,000 years ago. We should not waste this treasure, for we cannot replace it.

In most areas of the United States, local governments exert relatively little control over the use of groundwater. As a result, a large portion is wasted watering lawns, washing cars, and running fountains. A great deal more is inadvertently polluted by poor disposal of chemical wastes—and once pollution enters the groundwater, there is no effective means of removing it.

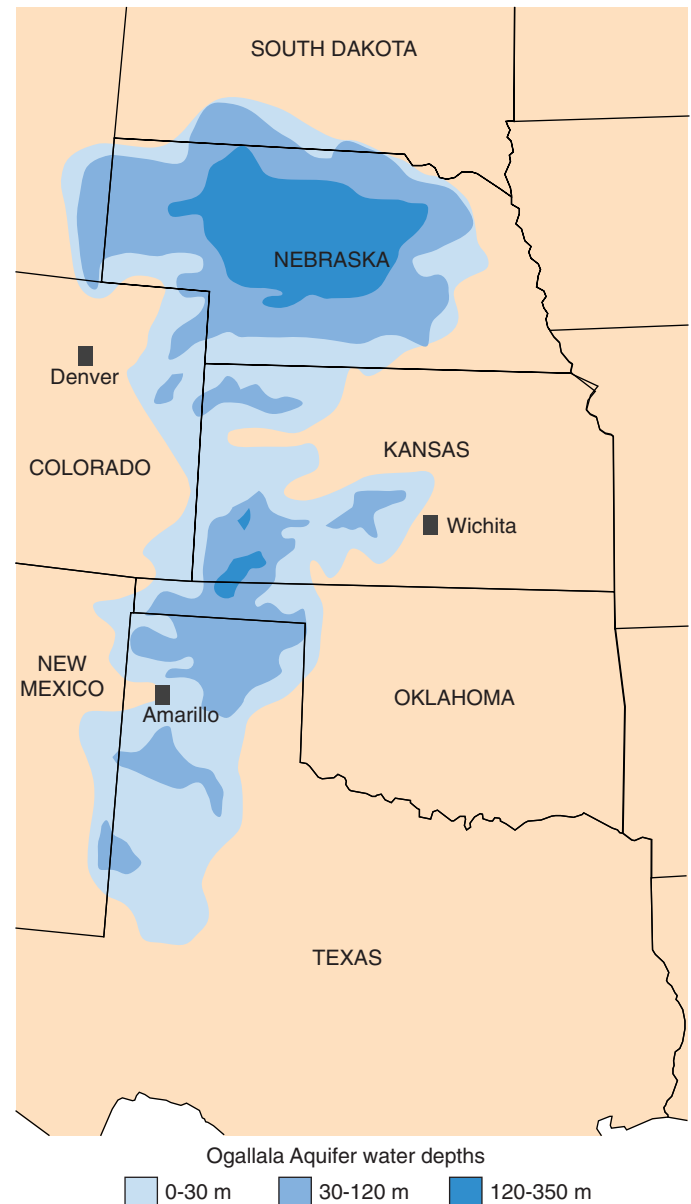


FIGURE 30.11

**The Ogallala Aquifer.** This massive deposit of groundwater lies under eight states, mainly Texas, Kansas, and Nebraska. Excessive pumping of this water for irrigation and other uses has caused the water level to drop 30 meters in some places. Continued excessive use of this kind endangers the survival of the Ogallala Aquifer, as it takes hundreds or even thousands of years for aquifers to recharge.

Topsoil and groundwater are essential for agriculture and other human activities. Replenishment of these resources occurs at a very slow rate. Current levels of consumption are not sustainable and will cause serious problems in the near future.

**Summary****Questions****Media Resources****30.1 The world's human population is growing explosively.**

- Population growth rates are declining throughout much of the world, but still the human population increases by 77 million people per year. At this rate, the global population will double in 39 years.
- An explosively growing human population is placing considerable stress on the environment. People in the developed world consume resources at a vastly higher rate than those in the nondeveloped world. Such high levels of consumption are not sustainable and are as important a problem as global overpopulation.

1. What biological event fostered the rapid growth of human populations? How did this event affect the location in which humans lived? What major cultural event eventually took place?
2. Why, in some respects, is the population size of the developed world more of a consideration in discussing resource use than the population of the nondeveloped world?



- History of Human Population

**30.2 Improvements in agriculture are needed to feed a hungry world.**

- Much current effort is focused on improving the productivity of existing crops, although the search for new crops continues.

3. What three species supply more than half of the human energy requirements on earth? How many plants supply over 90%?



- Food Needs
- Food Production
- Future Agricultural Prospects
- Land Degradation
- Agriculture-Related Problems

**30.3 Human activity is placing the environment under increasing stress.**

- Human activities present many challenges to the environment, including the release of harmful materials into the environment.
- Burning fossil fuels releases carbon dioxide, which may increase the world's temperature and alter weather and ocean levels.
- Release of pollutants into rivers may make the water unfit for aquatic life and human consumption.
- Release of industrial smoke into the upper atmosphere leads to acid precipitation that kills forests and lakes.
- Release of chemicals such as chlorofluorocarbons may destroy the atmosphere's ozone and expose the world to dangerous levels of ultraviolet radiation.
- Cutting and burning the tropical rain forests of the world to make pasture and cropland is producing a massive wave of extinction.

4. What problems must we master before nuclear power's full potential can be realized?
5. Why were chlorinated hydrocarbons banned in the United States? Why can you still find them as contaminants on fruits and vegetables?
6. How does acid precipitation form? Why has it been difficult to implement solutions to this problem?
7. What is the ozone layer? How is it formed? What are the harmful effects of decreasing the earth's ozone layer? What may be the primary cause of this damage?



- Global Warming
- Bioaccumulation
- Acid Rain
- Ozone Layer Depletion



- Student Research: Plants and Global Warming
- On Science Article: Shrinking Sea Ice

**30.4 Solving environmental problems requires individual involvement.**

- All of these challenges to our future can and must be addressed. Today, environmental scientists and concerned citizens are actively searching for constructive solutions to these problems.

8. What sort of action might you take that would make a significant contribution to solving the world's environmental problems?



- Scientists on Science: History of Life
- On Science Article: Do-It-Yourself Environmentalism

# 31

## Conservation Biology

### Concept Outline

#### 31.1 The new science of conservation biology is focused on conserving biodiversity.

**Overview of the Biodiversity Crisis.** In prehistoric times, humans decimated the faunas of many areas. Worldwide extinction rates are accelerating.

**Species Endemism and Hot Spots.** Some geographic areas are particularly rich in species that occur nowhere else.

**What's So Bad about Losing Biodiversity?** Biodiversity is of considerable direct economic value, and provides key support to the biosphere.

#### 31.2 Vulnerable species are more likely to become extinct.

**Predicting Which Species Are Vulnerable to Extinction.** Biologists carry out population viability analyses to assess danger of extinction.

**Dependence upon Other Species.** Extinction of one species can have a cascading effect throughout the food web, making other species vulnerable as well.

**Categories of Vulnerable Species.** Declining population size, loss of genetic variation, and commercial value all tend to increase a species' vulnerability.

#### 31.3 Causes of endangerment usually reflect human activities.

**Factors Responsible for Extinction.** Most recorded extinctions can be attributed to a few causes.

**Habitat Loss.** Without a place to live, species cannot survive.

**Case Study: Overexploitation**

**Case Study: Introduced Species**

**Case Study: Disruption of Ecological Relationships**

**Case Study: Loss of Genetic Variation**

**Case Study: Habitat Loss and Fragmentation**

#### 31.4 Successful recovery plans will need to be multidimensional.

**Many Approaches Exist for Preserving Endangered Species.** Species recovery requires restoring degraded habitats, breeding in captivity, maintaining population diversity, and maintaining keystone species.

**Conservation of Ecosystems.** Maintaining large preserves and focusing on the health of the entire ecosystem may be the best means of preserving biodiversity.



FIGURE 31.1

**Endangered.** The Siberian tiger is in grave danger of extinction, hunted for its pelt and having its natural habitat greatly reduced. A concerted effort is being made to save it, using many of the approaches discussed in this chapter.

Among the greatest challenges facing the biosphere is the accelerating pace of species extinctions—not since the Cretaceous have so many species become extinct in so short a period of time (figure 31.1). This challenge has led to the emergence in the last decade of the new discipline of conservation biology. Conservation biology is an applied discipline that seeks to learn how to preserve species, communities, and ecosystems. It both studies the causes of declines in species richness and attempts to develop methods to prevent such declines. In this chapter we will first examine the biodiversity crisis and its importance. Then, we will assess the sorts of species which seem vulnerable to extinction. Using case histories, we go on to identify and study five factors that have played key roles in many extinctions. We finish with a review of recovery efforts at the species and community level.

## 31.1 The new science of conservation biology is focused on conserving biodiversity.

### Overview of the Biodiversity Crisis

Extinction is a fact of life, as normal and necessary as species formation is to a stable world ecosystem. Most species, probably all, go extinct eventually. More than 99% of species known to science (most from the fossil record) are now extinct. However, current rates are alarmingly high. Taking into account the rapid and accelerating loss of habitat that is occurring at present, especially in the tropics, it has been calculated that as much as 20% of the world's biodiversity may be lost during the next 30 years. In addition, many of these species may be lost before we are even aware of their extinction. Scientists estimate that no more than 15% of the world's eukaryotic organisms have been discovered and given scientific names, and this proportion probably is much lower for tropical species.

These losses will not just affect poorly known groups. As many as 50,000 species of the world's total of 250,000 species of plants, 4000 of the world's 20,000 species of butterflies, and nearly 2000 of the world's 9000 species of birds could be lost during this short period of time. Considering that our species has been in existence for only 500,000 years of the world's 4.5-billion-year history, and that our ancestors developed agriculture only about 10,000 years ago, this is an astonishing—and dubious—accomplishment.

#### Extinctions Due to Prehistoric Humans

A great deal can be learned about current rates of extinction by studying the past, and in particular the impact of human-caused extinctions. In prehistoric times, *Homo sapiens* wreaked havoc whenever they entered a new area. For example, at the end of the last ice age, approximately 12,000 years ago, the fauna of North America was composed of a diversity of large mammals similar to Africa today: mammoths and mastodons, horses, camels, giant ground-sloths, saber-toothed cats, and lions, among others (figure 31.2). Shortly after humans arrived, 74 to 86% of the megafauna (that is, animals weighing more than 100 pounds) became extinct. These extinctions are thought to have been caused by hunting, and indirectly by burning and clearing forests (some scientists attribute these extinctions to climate change, but that hypothesis doesn't explain why the end of earlier ice ages was not associated with mass extinctions, nor does it explain why extinctions occurred primarily among larger animals, with smaller species relatively unaffected).



**FIGURE 31.2**

**North America before human inhabitants.** Animals found in North America prior to the migration of humans included large mammals and birds such as the ancient North American camel, saber-toothed cat, giant ground-sloth, and the teratorn vulture.

Similar results have followed the arrival of humans around the globe. Forty thousand years ago, Australia was occupied by a wide variety of large animals, including marsupials similar in size and ecology to hippos and leopards, a kangaroo nine feet tall, and a 20-foot-long monitor lizard. These all disappeared, at approximately the same time as humans arrived. Smaller islands have also been devastated. On Madagascar, at least 15 species of lemurs, including one the size of a gorilla, a pygmy hippopotamus, and the flightless elephant bird, *Aepyornis*, the largest bird to ever live (more than 3 meters tall and weighing 450 kilograms) all perished. On New Zealand, 30 species of birds went extinct, including all 13 species of moas, another group of large, flightless birds. Interestingly, one continent that seems to have been spared these megafaunal extinctions is Africa. Scientists speculate that this lack of extinction in prehistoric Africa may have resulted because much of human evolution occurred in Africa. Consequently, other African species had been coevolving with humans for several million years and thus had evolved counteradaptations to human predation.

#### Extinctions in Historical Time

Historical extinction rates are best known for birds and mammals because these species are conspicuous—relatively large and well studied. Estimates of extinction rates for other species are much rougher. The data presented

| Table 31.1 Recorded Extinctions since 1600 a.d. |                      |        |       |                             |                          |         |
|-------------------------------------------------|----------------------|--------|-------|-----------------------------|--------------------------|---------|
| Taxon                                           | Recorded Extinctions |        |       | Approximate Number of Total | Percent of Taxon Species | Extinct |
|                                                 | Mainland             | Island | Ocean |                             |                          |         |
| Mammals                                         | 30                   | 51     | 4     | 85                          | 4,000                    | 2.1     |
| Birds                                           | 21                   | 92     | 0     | 113                         | 9,000                    | 1.3     |
| Reptiles                                        | 1                    | 20     | 0     | 21                          | 6,300                    | 0.3     |
| Amphibians*                                     | 2                    | 0      | 0     | 2                           | 4,200                    | 0.05    |
| Fish                                            | 22                   | 1      | 0     | 23                          | 19,100                   | 0.1     |
| Invertebrates                                   | 49                   | 48     | 1     | 98                          | 1,000,000+               | 0.01    |
| Flowering plants                                | 245                  | 139    | 0     | 384                         | 250,000                  | 0.2     |

Source: Reid and Miller, 1989; data from various sources.

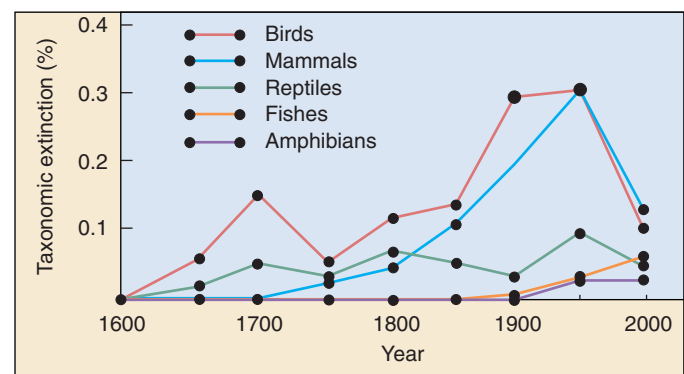
\*There has been an alarming decline in amphibian populations recently, and many species may be on the verge of extinction.

in table 31.1, based on the best available evidence, shows recorded extinctions from 1600 to the present. These estimates indicate that about 85 species of mammals and 113 species of birds have become extinct since the year 1600. That is about 2.1% of known mammal species and 1.3% of known birds. The majority of extinctions have come in the last 150 years. The extinction rate for birds and mammals was about one species every decade from 1600 to 1700, but it rose to one species every year during the period from 1850 to 1950, and four species per year between 1986 and 1990 (figure 31.3). It is this increase in the rate of extinction that is the heart of the biodiversity crisis.

The majority of historic extinctions—though by no means all of them—have occurred on islands. For example, of the 90 species of mammals that have gone extinct in the last 500 years, 73% lived on islands (and another 19% on Australia). The particular vulnerability of island species probably results from a number of factors: such species have often evolved in the absence of predators and so have lost their ability to escape both humans and introduced predators such as rats and cats. In addition, humans have introduced competitors and diseases (avian malaria, for example has devastated the bird fauna of the Hawaiian Islands). Finally, island populations are often relatively small, and thus particularly vulnerable to extinction, as we shall see later in the chapter.

In recent years, however, the extinction crisis has moved from islands to continents. Most species now threatened with extinction occur on continents, and it is these areas which will bear the brunt of the extinction crisis in this century.

Some people have argued that we should not be concerned because extinctions are a natural event and mass extinctions have occurred in the past. Indeed, as we saw in chapter 21, mass extinctions have occurred several times over the past half billion years. However, the current mass extinction event is notable in several respects. First, it is the only such event triggered by a single



**FIGURE 31.3**

**Trends in species loss.** The graphs above present data on recorded animal extinctions since 1600. The majority of extinctions have occurred on islands, with birds and mammals particularly affected (although this may reflect to some degree our more limited knowledge of other groups).

species. Moreover, although species diversity usually recovers after a few million years, this is a long time to deny our descendants the benefits and joys of biodiversity. In addition, it is not clear that biodiversity will rebound this time. After previous mass extinction events, new species have evolved to utilize resources available due to species extinctions. Today, however, such resources are unlikely to be available, because humans are destroying the habitats and taking the resources for their own use.

**Biologists estimate rates of extinction both by studying recorded extinction events and by analyzing trends in habitat loss and disruption. Since prehistoric times, humans have had a devastating effect on biodiversity almost everywhere in the world.**

# Species Endemism and Hot Spots

A species found naturally in only one geographic area and no place else is said to be **endemic** to that area. The area over which an endemic species is found may be very large. The black cherry tree (*Prunus serotina*), for example, is endemic to all of temperate North America. More typically, however, endemic species occupy restricted ranges. The Komodo dragon (*Varanus komodoensis*) lives only on a few small islands in the Indonesian archipelago, while the Mauna Kea silversword (*Argyroxiphium sandwicense*) lives in a single volcano crater on the island of Hawaii.

Isolated geographical areas, such as oceanic islands, lakes, and mountain peaks, often have high percentages of endemic species, often in significant danger of extinction. The number of endemic plant species varies greatly in the United States from one state to another. Thus, 379 plant species are found in Texas and nowhere else, whereas New York has only one endemic plant species. California, with its varied array of habitats, including deserts, mountains, seacoast, old growth forests, grasslands, and many others, is home to more endemic species than any other state.

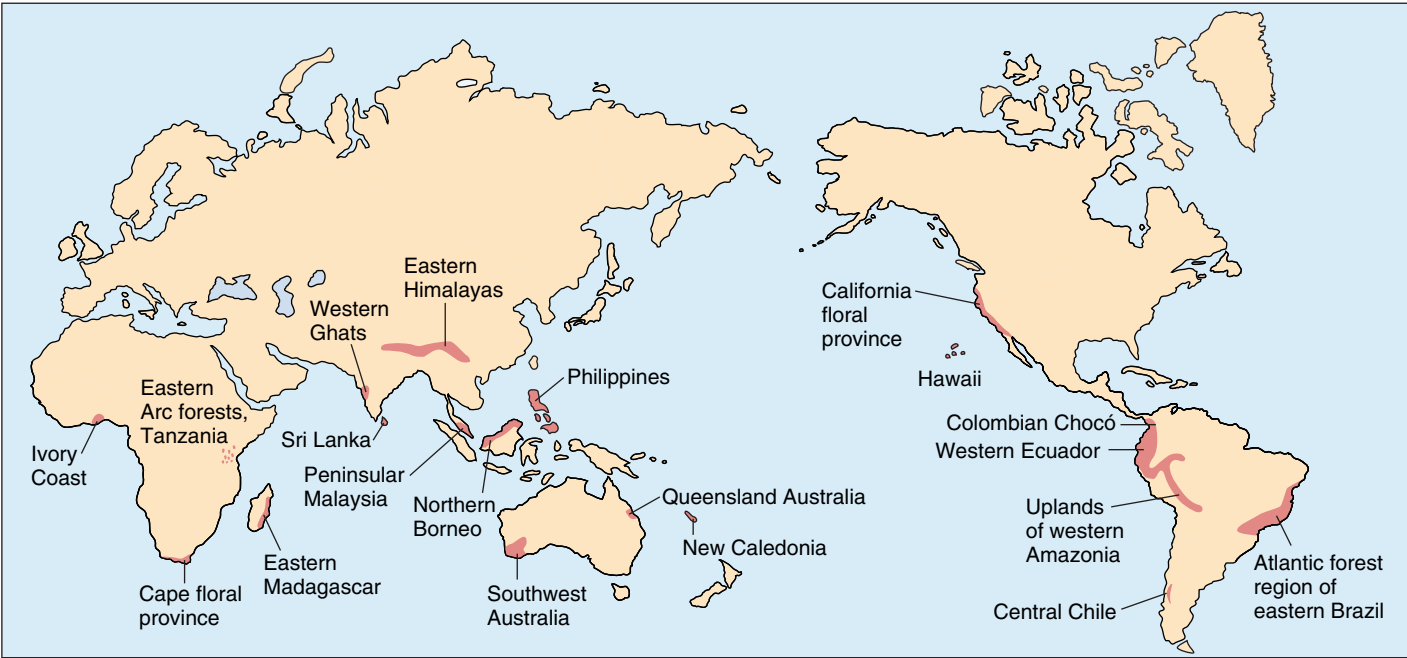
Worldwide, notable concentrations of endemic species occur in particular “hot spots” of high endemism. Such hot spots are found in Madagascar, in a variety of tropical rain forests, in the eastern Himalayas, in areas with Mediterranean climates like California, South Africa, and Australia, and in several other climatic areas (figure 31.4 and table 31.2). Unfortunately, many of these areas are experiencing high rates of habitat destruction with consequent species extinctions. In Madagascar, it is estimated that 90% of the

| Table 31.2 Numbers of Endemic Vertebrate Species in Some “Hot Spot” Areas |         |          |            |
|---------------------------------------------------------------------------|---------|----------|------------|
| Region                                                                    | Mammals | Reptiles | Amphibians |
| Atlantic coastal Brazil                                                   | 40      | 92       | 168        |
| Colombian Chocó                                                           | 8       | 137      | 111        |
| Philippines                                                               | 98      | 120      | 41         |
| Northern Borneo                                                           | 42      | 69       | 47         |
| Southwestern Australia                                                    | 10      | 25       | 22         |
| Madagascar                                                                | 86      | 234      | 142        |
| Cae region (South Africa)                                                 | 16      | 43       | 23         |
| California Floristic Province                                             | 15      | 25       | 7          |
| New Cledonia                                                              | 2       | 21       | 0          |
| Eastern Himalayas                                                         | —       | 20       | 25         |

Source: Data from Myers 1988; World Conservation and Monitoring Center 1992.

original forest has already been lost, this in an island in which 85% of the species are found nowhere else in the world. In the forests of the Atlantic coast of Brazil, the extent of deforestation is even higher: 95% of the original forest is gone.

Some areas of the earth have particularly high levels of species endemism. Unfortunately, many of these areas are currently in great jeopardy due to habitat destruction with correspondingly high rates of species extinction.



**FIGURE 31.4**  
“Hot spots” of high endemism. These areas are rich in endemic species under significant threat of imminent extinction.

## What's So Bad about Losing Biodiversity?

What's so bad about losing species? What is the value of biodiversity? Its value can be divided into three principal components: (1) *direct economic value* of products we obtain from species of plants, animals, and other groups; (2) *indirect economic value* of benefits produced by species without our consuming them; and (3) *ethical* and *aesthetic* value.

### Direct Economic Value

Many species have direct value, as sources of food, medicine, clothing, biomass (for energy and other purposes), and shelter. Most of the world's food, for example, is derived from a small number of plants that were originally domesticated from wild plants in tropical and semi-arid regions. In the future, wild strains of these species may be needed for their genetic diversity if we are to improve yields, or find a way to breed resistance to new pests.

About 40% of the prescription and nonprescription drugs used today have active ingredients extracted from plants or animals. Aspirin, the world's most widely used drug, was first extracted from the leaves of the tropical willow, *Salix alba*. The rosy periwinkle, *Catharanthus roseus*, from Madagascar has yielded potent drugs for combating leukemia (figure 31.5).

Only in the last few decades have biologists perfected the techniques that make possible the transfer of genes from one kind of organism to another. We are just beginning to be able to use genes obtained from other species to our advantage, as explored at length in chapter 19. So-called **gene prospecting** of the genomes of plants and animals for useful genes has only begun. We have been able to examine only a minute proportion of the world's organisms so far, to see whether any of their genes have useful properties. By conserving biodiversity we maintain the option of finding useful benefit in the future.

### Indirect Economic Value

Diverse biological communities are of vital importance to healthy ecosystems, in maintaining the chemical quality of natural water, in buffering ecosystems against floods and drought, in preserving soils and preventing loss of minerals and nutrients, in moderating local and regional climate, in absorbing pollution, and in promoting the breakdown of organic wastes and the cycling of minerals. By destroying biodiversity, we are creating conditions of instability and lessened productivity and promoting desertification, waterlogging, mineralization, and many other undesirable outcomes throughout the world.

Given the major role played by many species in maintaining healthy ecosystems, it is alarming how little we know about the details of how ecosystems and communities function. It is impossible to predict all the conse-



**FIGURE 31.5**

**The rosy periwinkle.** Two drugs extracted from the Madagascar periwinkle *Catharanthus roseus*, vinblastine and vincristine, effectively treat common forms of childhood leukemia, increasing chances of survival from 20% to over 95%.

quences of removing a species, or to be sure that some of them will not be catastrophic. Imagine taking a part list for an airliner, and randomly changing a digit in one of the part numbers: you might change a cushion to a roll of toilet paper—but you might as easily change a key bolt holding up the wing to a pencil. The point is, you shouldn't gamble if you cannot afford to lose, and in removing biodiversity we are gambling with the future of ecosystems upon which we depend, and upon whose functioning we only little understand.

### Ethical and Aesthetic Value

Many people believe that preserving biodiversity is an ethical issue, feeling that every species is of value in its own right, even if humans are not able to exploit or benefit from it. It is clear that humans have the power to exploit and destroy other species, but it is not as ethically clear that they have the *right* to do so. Many people believe that along with power comes responsibility: as the only organisms capable of eliminating species and entire ecosystems, and as the only organisms capable of reflecting upon what we are doing, we should act as guardians or stewards for the diversity of life around us.

Almost no one would deny the aesthetic value of biodiversity, of a beautiful flower or noble elephant, but how do we place a value on beauty? Perhaps the best we can do is to appreciate the deep sense of lack we feel at its permanent loss.

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**Biodiversity is of great value, for the products with which it provides us, for its contributions to the health of the ecosystems upon which we all depend, and for the beauty it provides us, as well as being valuable in its own right.**

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## 31.2 Vulnerable species are more likely to become extinct.

### Predicting Which Species Are Vulnerable to Extinction

How can a biologist assess whether a particular species is vulnerable to extinction? To get some handle on this, conservation biologists look for changes in population size and habitat availability. Species whose populations are shrinking rapidly, whose habitats are being destroyed (figure 31.6), or which are endemic to small areas can be considered to be endangered.

#### Population Viability Analysis

Quantifying the risk faced by a particular species is not a simple or precise enterprise. Increasingly, conservation biologists make a rough estimate of a population's risk of local extinction in terms of a **minimum viable population** (MVP), the estimated number or density of individuals necessary for the population to maintain or increase its numbers.

Some small populations are at high risk of extinction, while other populations equally small are at little or no risk. Conservation biologists carry out a **population viability analysis** (PVA) to assess how the size of a population influences its risk of becoming extinct over a specific time period, often 100 years. Many factors must be taken into account in a PVA. Two components of particular importance are *demographic stochasticity* (the amount of random variation in birth and death rates) and *genetic stochasticity* (fluctuations in a population's level of genetic variation). Demographic stochasticity refers to random events that affect a population. The smaller the population, the more likely it is that a random event, such as a disease epidemic or an environmental disturbance (such as a flood or a fire) could decimate a population and lead to extinction. Similarly, small populations are most likely to lose genetic variation due to genetic drift (see chapter 20) and thus be vulnerable to both the short- and long-term consequences of genetic uniformity. For these reasons, small populations are at particularly great risk of extinction.

Many species are distributed as metapopulations, collections of small populations each occupying a suitable patch of habitat in an otherwise unsuitable landscape (see chapter 24). Each individual subpopulation may be quite small and in real threat of extinction, but the metapopulation may be quite safe from extinction so long as individuals from other populations repopulate the habitat patches vacated by extinct populations. The extent of this rescue effect is an important component of the PVA of



**FIGURE 31.6**

**Habitat removal.** In this clear-cut lumbering of National Forest land in Washington State, few if any trees have been left standing, removing as well the home of the deer, birds, and other animal inhabitants of temperate forest. Until a replacement habitat is provided by replanting, this is a truly “lost” habitat.

such species; if rates of population extinction increase, there may not be enough surviving populations to found new populations, and the species as a whole may slide toward extinction.

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**Small populations are particularly in danger of extinction. To assess the risk of local extinction of a particular species, conservation biologists carry out a population viability analysis that takes into account demographic and genetic variation.**

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## Dependence upon Other Species

Species often become vulnerable to extinction when their web of ecological interactions becomes seriously disrupted. A recent case in point are the sea otters that live in the cold waters off Alaska and the Aleutian Islands. A keystone species in the kelp forest ecosystem, the otter populations have declined sharply in recent years. In a 500-mile stretch of coastline, otter numbers had dropped to an estimated 6000 from 53,000 in the 1970s, a plunge of nearly 90%. Investigating this catastrophic decline, marine ecologists uncovered a chain of interactions among the species of the ocean and kelp forest ecosystems, a falling domino chain of lethal effects.

The first in a series of events leading to the sea otter's decline seems to have been the heavy commercial harvesting of whales (see the case history later in this chapter). Without whales to keep their numbers in check, ocean zooplankton thrived, leading in turn to proliferation of a species of fish called pollock that feed on the now-abundant zooplankton. Given this ample food supply, the pollock proved to be very successful competitors of other northern Pacific fish like herring and ocean perch, so that levels of these other fish fell steeply in the 1970s.

Now the falling chain of dominos begins to accelerate. The decline in the nutritious forage fish led to an ensuing crash in Alaskan populations of Steller's sea lions and harbor seals, for which pollock did not provide sufficient nourishment. Numbers of these pinniped species have fallen precipitously since the 1970s.

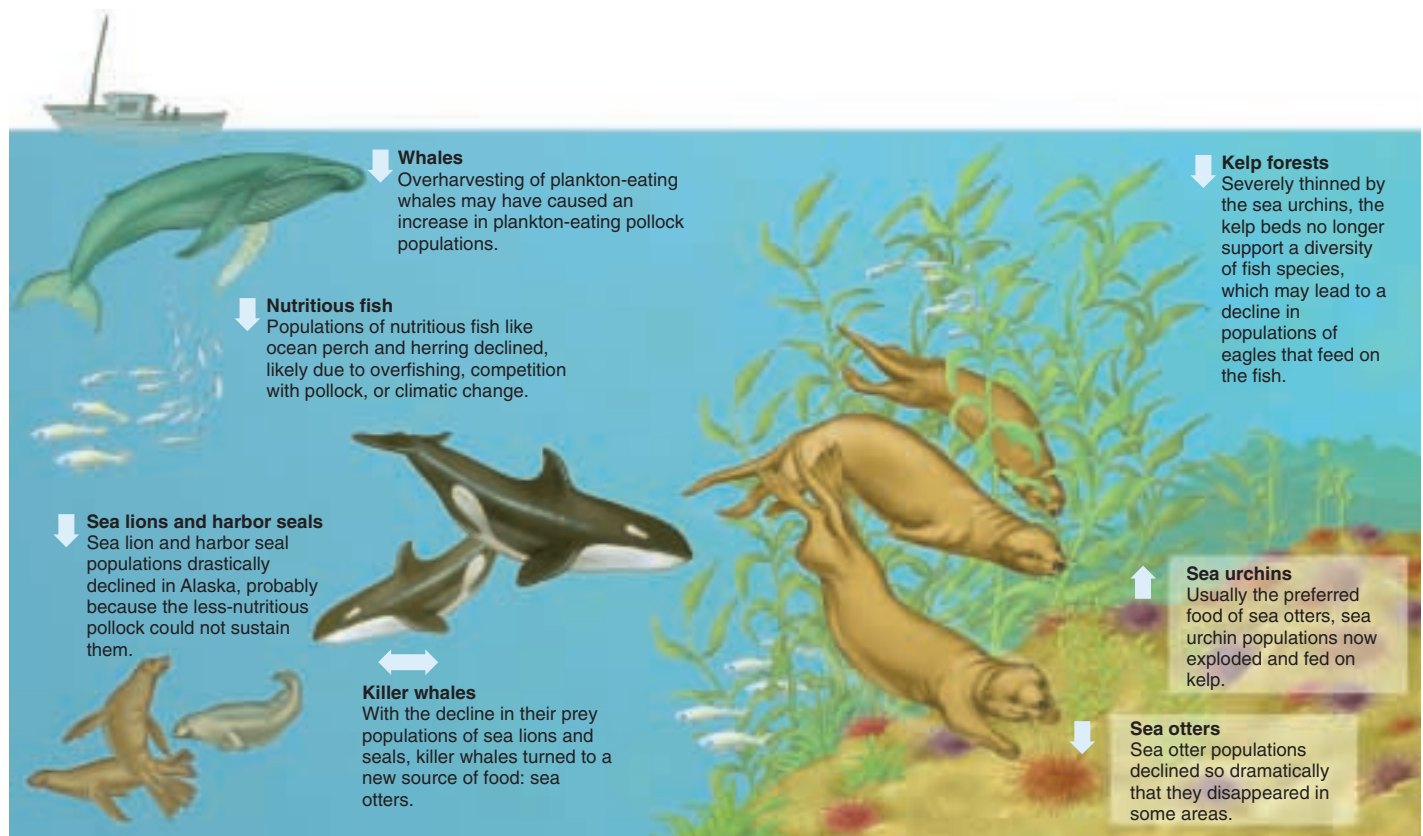
Pinnipeds are the major food of orcas, also called killer whales. Faced with a food shortage, some orcas seem to have turned to the next best thing: sea otters. In one bay where the entrance from the sea was too narrow and shallow for orcas to enter, only 12% of the sea otters have disappeared, while in a similar bay which orcas could enter easily, two-thirds of the otters disappeared in a year's time.

Without otters to eat them, the population of sea urchins in the ecosystem exploded, eating the kelp and so "deforesting" the kelp forests and denuding the ecosystem (figure 31.7). As a result, fish species that live in the kelp forest, like sculpins and greenlings (a cod relative), are declining. This chain reaction demonstrates why sea otters are considered to be a keystone species.

---

**Commercial whaling appears to have initiated a series of changes that have led to orcas feeding on sea otters, with disastrous effects on their kelp forest ecosystem.**

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**FIGURE 31.7**

**Disruption of the kelp forest ecosystem.** Overharvesting by commercial whalers altered the balance of fish in the ocean ecosystem, inducing killer whales to feed on sea otters, a keystone species of the kelp forest ecosystem.

## Categories of Vulnerable Species

Studying past extinctions of species and using population viability analyses of threatened ones, conservation biologists have observed that some categories of species are particularly vulnerable to extinction.

### Local Endemic Distribution

Local endemic species typically occur at only one or a few sites in a restricted geographical range, which makes them particularly vulnerable to anything that harms the site, such as destruction of habitat by human activity. Bird species on oceanic islands have often become extinct as humans affect the island habitats. Many endemic fish species confined to a single lake undergo similar fates.

Local endemic species often have small population sizes, placing them at particular risk of extinction because of their greater vulnerability to demographic and genetic fluctuations. Indeed, population size by itself seems to be one of the best predictors of the extinction risk of populations.

Local endemic species often have quite specialized niche requirements. Once a habitat is altered, it may no longer be able to support a particular local endemic, while remaining satisfactory for species with less particular requirements. For example, wetlands plants that require very specific and regular changes in water level may be rapidly eliminated when human activity affects the hydrology of an area.

### Declining Population Size

Species in which population size is declining are often at grave risk of extinction, particularly if the decline in numbers of individuals is severe. Although there is no hard rule, population trends in nature tend to continue, so a population showing significant signs of decline should be considered at risk of extinction unless the cause of the decline is identified and corrected. Darwin makes this point very clearly in *On the Origin of Species*:

“To admit that species generally become rare before they become extinct, to feel no surprise at the rarity of the species, and yet to marvel greatly when the species ceases to exist, is much the same as to admit that sickness in the individual is the forerunner of death—to feel no surprise at sickness, but when the sick man dies, to wonder and to suspect that he dies of some deed of violence.”

Although long-term trends toward smaller population numbers suggest that a species may be at risk in future years, abrupt recent declines in population numbers, particularly when the population is small or locally endemic, fairly scream of risk of extinction. It is for this reason that PVA is best carried out with data on population sizes gathered over a period of time.

### Lack of Genetic Variability

Species with little genetic variability are generally at significantly greater risk of extinction than more variable species, simply because they have a more limited arsenal with which to respond to the vagaries of environmental change. Species with extremely low genetic variability are particularly vulnerable when faced with a new disease, predator, or other environmental challenge. For example, the African cheetah (*Acinonyx jubatus*) has almost no genetic variability. This lack of genetic variability is considered to be a significant contributing factor to a lack of disease resistance in the cheetah—diseases that are of little consequence to other cat species can wipe out a colony of cheetahs (although environmental factors also seem to have played a key role in the cheetah's decline).

### Hunted or Harvested by People

Species that are hunted or harvested by people have historically been at grave risk of extinction. Overharvesting of natural populations can rapidly reduce the population size of a species, even when that species is initially very abundant. A century ago the skies of North America were darkened by huge flocks of passenger pigeons; hunted as free and tasty food, they were driven to extinction. The buffalo that used to migrate in enormous herds across the central plains of North America only narrowly escaped the same fate, a few individuals preserved from this catastrophic exercise in overhunting founding today's modest herds.

The existence of a commercial market often leads to overexploitation of a species. The international trade in furs, for example, has severely reduced the numbers of chinchilla, vicuna, otter, and many wild cat species. The harvesting of commercially valuable trees provides another telling example: almost all West Indies mahogany (*Swietenia mahoganii*) have been logged from the Caribbean islands, and the extensive cedar forests of Lebanon, once widespread at high elevations in the Middle East, now survive in only a few isolated groves.

A particularly telling example of overharvesting of a so-called commercial species is the commercial harvesting of fish in the North Atlantic. Fishing fleets continued to harvest large amounts of cod off Newfoundland during the 1980s, even as the population numbers declined precipitously. By 1992 the cod population had dropped to less than 1% of their original numbers. The American and Canadian governments have closed the fishery, but no one can predict if the fish populations will recover. The Atlantic bluefin tuna has experienced a 90% population decline in the last 10 years. The swordfish has declined even further. In both cases, the drop has led to even more intense fishing of the remaining populations.

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**A variety of factors can make a species particularly vulnerable to extinction.**

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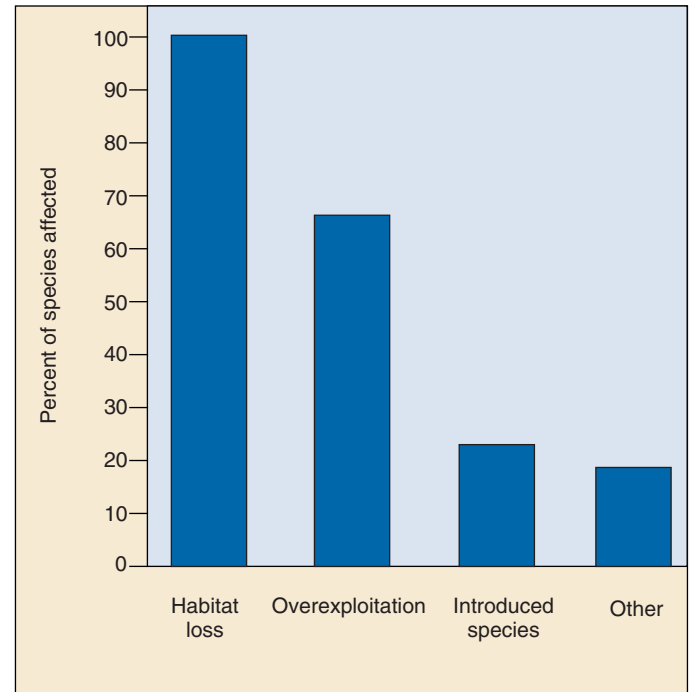
### 31.3 Causes of endangerment usually reflect human activities.

#### Factors Responsible For Extinction

Because a species is rare does not necessarily mean that it is in danger of extinction. The habitat it utilizes may simply be in short supply, preventing population numbers from growing. In a similar way, shortage of some other resource may be limiting the size of populations. Secondary carnivores, for example, are usually rare because so little energy is available to support their populations. Nor are vulnerable species such as those categories discussed in the previous section always threatened with extinction. Many local endemics are quite stable and not at all threatened.

If it's not just size or vulnerability, what factors are responsible for extinction? Studying a wide array of recorded extinctions and many species currently threatened with extinction, conservation biologists have identified a few factors that seem to play a key role in many extinctions: overexploitation, introduced species, disruption of ecological relationships, loss of genetic variability, and habitat loss and fragmentation (figure 31.8 and table 31.3).

**Most recorded extinctions can be attributed to one of five causes: overexploitation, introduced species, ecodisruption, loss of genetic variability, and habitat loss and fragmentation.**



**FIGURE 31.8**  
**Factors responsible for animal extinction.** These data represent known extinctions of mammals in Australasia and the Americas.

| Table 31.3 Causes of Extinctions |                                                       |                  |                      |           |       |         |
|----------------------------------|-------------------------------------------------------|------------------|----------------------|-----------|-------|---------|
| Group                            | Percentage of Species Influenced by the Given Factor* |                  |                      |           |       |         |
|                                  | Habitat Loss                                          | Overexploitation | Species Introduction | Predators | Other | Unknown |
| EXTINCTIONS                      |                                                       |                  |                      |           |       |         |
| Mammals                          | 19                                                    | 23               | 20                   | 1         | 1     | 36      |
| Birds                            | 20                                                    | 11               | 22                   | 0         | 2     | 37      |
| Reptiles                         | 5                                                     | 32               | 42                   | 0         | 0     | 21      |
| Fish                             | 35                                                    | 4                | 30                   | 0         | 4     | 48      |
| THREATENED EXTINCTIONS           |                                                       |                  |                      |           |       |         |
| Mammals                          | 68                                                    | 54               | 6                    | 8         | 12    | —       |
| Birds                            | 58                                                    | 30               | 28                   | 1         | 1     | —       |
| Reptiles                         | 53                                                    | 63               | 17                   | 3         | 6     | —       |
| Amphibians                       | 77                                                    | 29               | 14                   | —         | 3     | —       |
| Fish                             | 78                                                    | 12               | 28                   | —         | 2     | —       |

\*Some species may be influenced by more than one factor; thus, some rows may exceed 100%.

Source: Reid and Miller, 1989.

## Habitat Loss

As figure 31.8 and table 31.3 indicate, habitat loss is the single most important cause of extinction. Given the tremendous amounts of ongoing destruction of all types of habitat, from rain forest to ocean floor, this should come as no surprise. Natural habitats may be adversely affected by human influences in four ways: (1) destruction, (2) pollution, (3) human disruption, and (4) habitat fragmentation.

### Destruction

A proportion of the habitat available to a particular species may simply be destroyed. This is a common occurrence in the “clear-cut” harvesting of timber, in the burning of tropical forest to produce grazing land, and in urban and industrial development. Forest clearance has been, and is, by far the most pervasive form of habitat disruption (figure 31.9). Many tropical forests are being cut or burned at a rate of 1% or more per year.

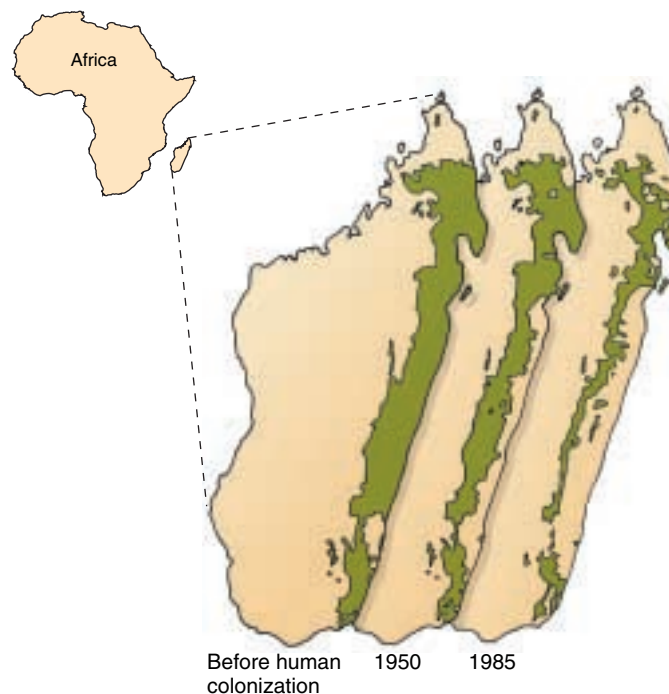
Biologists often use the well-established observation that larger areas support more species (see figure 29.24) to estimate the effect of reductions in habitat available to a species. As we saw in chapter 30, a relationship usually exists between the size of an area and the number of species it contains. Although this relationship varies according to geographic area, type of organism, and type of area (for example, oceanic islands, patches of habitat on the mainland), a general result is that a tenfold increase in area usually leads to approximately a doubling in number of species. This relationship suggests, conversely, that if the area of a habitat is reduced by 90%, so that only 10% remains, then half of all species will be lost. Evidence for this theory comes from a study of extinction rates of birds on habitat islands (that is, islands of a particular type of habitat surrounded by unsuitable habitat) in Finland where the extinction rate was found to be inversely proportional to island size (figure 31.10).

### Pollution

Habitat may be degraded by pollution to the extent that some species can no longer survive there. Degradation occurs as a result of many forms of pollution, from acid rain to pesticides. Aquatic environments are particularly vulnerable; many northern lakes in both Europe and North America, for example, have been essentially sterilized by acid rain.

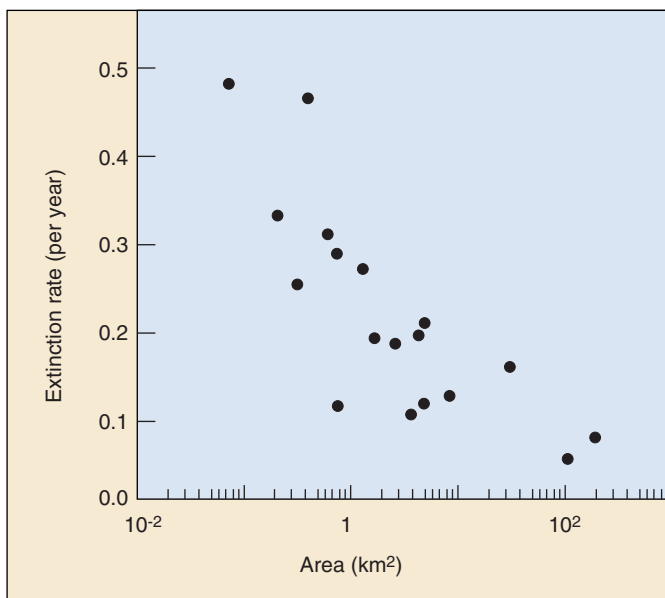
### Human Disruption

Habitat may be so disturbed by human activities as to make it untenable for some species. For example, visitors to caves in Alabama and Tennessee produced significant population declines in bats over an eight-year period, some as great as 100%. When visits were fewer than one



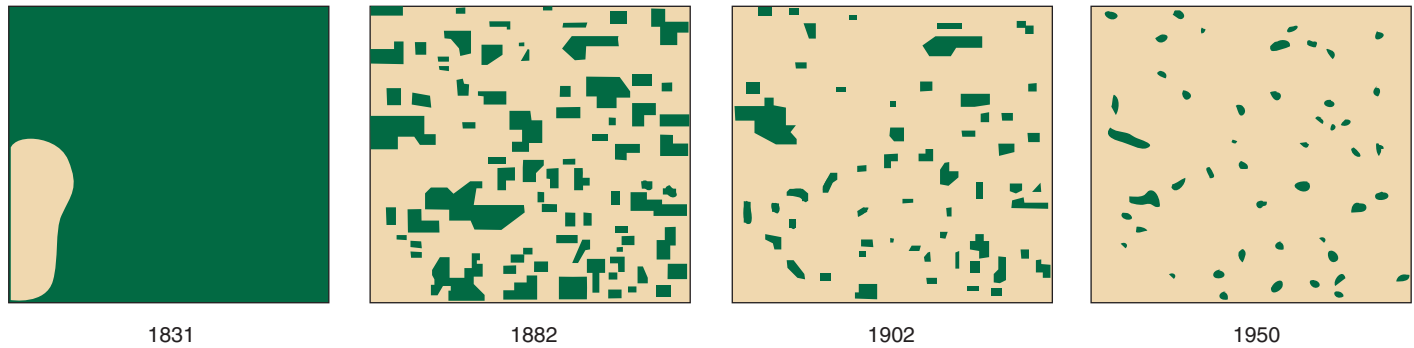
**FIGURE 31.9**

**Extinction and habitat destruction.** The rain forest covering the eastern coast of Madagascar, an island off the coast of East Africa, has been progressively destroyed as the island’s human population has grown. Ninety percent of the original forest cover is now gone. Many species have become extinct, and many others are threatened, including 16 of Madagascar’s 31 primate species.



**FIGURE 31.10**

**Extinction and the species-area relationship.** The data present percent extinction rates as a function of habitat area for birds on a series of Finnish islands. Smaller islands experience far greater local extinction rates.



**FIGURE 31.11**

**Fragmentation of woodland habitat.** From the time of settlement of Cadiz Township, Wisconsin, the forest has been progressively reduced from a nearly continuous cover to isolated woodlots covering less than 1% of the original area.

per month, less than 20% of bats were lost, but caves with more than four visits per month suffered population declines of between 86 and 95%.

### Habitat Fragmentation

Loss of habitat by a species frequently results not only in a lowering of population numbers, but also in fragmentation of the population into unconnected patches (figure 31.11).

A habitat may become fragmented in unobvious ways, such as when roads and habitation intrude into forest. The effect is to carve up the populations living in the habitat into a series of smaller populations, often with disastrous consequences. Although detailed data are not available, fragmentation of wildlife habitat in developed temperate areas is thought to be very substantial.

As habitats become fragmented and shrink in size, the relative proportion of the habitat that occurs on the boundary, or edge, increases. **Edge effects** can significantly degrade a population's chances of survival. Changes in microclimate (temperature, wind, humidity, etc.) near the edge may reduce appropriate habitat for many species more than the physical fragmentation suggests. In isolated fragments of rain forest, for example, trees on the edge are exposed to direct sunlight and, consequently, hotter and drier conditions than they are accustomed to in the cool, moist forest interior. As a result, within 100 meters of the forest edge, tree biomass decreased by 36% in the first 17 years after fragment isolation in one study.

Also, increasing habitat edges opens up opportunities for parasites and predators, both more effective at edges. As fragments decrease in size, the proportion of habitat that is distant from any edge decreases and, consequently, more and more of the habitat is within the range of these predators. Habitat fragmentation is thought to have been responsible for local extinctions in a wide range of species.

The impact of habitat fragmentation can be seen clearly in a major study done in Manaus, Brazil, as the rain forest was commercially logged. Landowners agreed to preserve



**FIGURE 31.12**

**A study of habitat fragmentation.** Biodiversity was monitored in the isolated patches of rain forest in Manaus, Brazil, before and after logging. Fragmentation led to significant species loss within patches.

patches of rain forest of various sizes, and censuses of these patches were taken before the logging started, while they were still part of a continuous forest. After logging, species began to disappear from the now-isolated patches (figure 31.12). First to go were the monkeys, which have large home ranges. Birds that prey on ant colonies followed, disappearing from patches too small to maintain enough ant colonies to support them.

Because some species like monkeys require large patches, this means that large fragments are indispensable if we wish to preserve high levels of biodiversity. The take-home lesson is that preservation programs will need to provide suitably large habitat fragments to avoid this impact.

**Habitat loss is probably the greatest cause of extinction. As habitats are destroyed, remaining habitat becomes fragmented, increasing the threat to many species.**

## Case Study: Overexploitation—Whales

Whales, the largest living animals, are rare in the world's oceans today, their numbers driven down by commercial whaling. Commercial whaling began in the sixteenth century, and reached its apex in the nineteenth and early twentieth centuries. Before the advent of cheap high-grade oils manufactured from petroleum in the early twentieth century, oil made from whale blubber was an important commercial product in the worldwide marketplace. In addition, the fine lattice-like structure used by baleen whales to filter-feed plankton from seawater (termed "baleen," but sometimes called "whalebone" even though it is actually made of keratin, like fingernails) was used in undergarments. Because a whale is such a large animal, each individual captured is of significant commercial value.

Right whales were the first to bear the brunt of commercial whaling. They were called right whales because they were slow, easy to capture, and provided up to 150 barrels of blubber oil and abundant whalebone, making them the "right" whale for a commercial whaler to hunt.

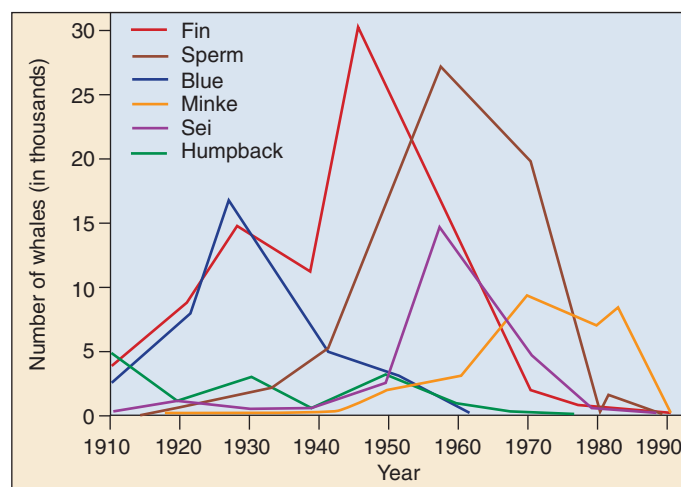
As the right whale declined in the eighteenth century, whalers turned to other species, the gray, humpback (figure 31.13), and bowhead. As their numbers declined, whalers turned to the blue, largest of all whales, and when they were decimated, to smaller whales: the fin, then the Sei, then the sperm whales. As each species of whale became the focus of commercial whaling, its numbers inevitably began a steep decline (figure 31.14).

Hunting of right whales was made illegal in 1935. By then, all three species had been driven to the brink of extinction, their numbers less than 5% of what they used to be. Protected since, their numbers have not recovered in either the North Atlantic or North Pacific. By 1946 several other species faced imminent extinction, and whaling nations formed the International Whaling Commission (IWC) to regulate commercial whale hunting. Like having the fox guard the hen house, the IWC for decades did little to limit whale harvests, and whale numbers continued a steep decline. Finally, in 1974, when numbers of all but the small minke whales had been driven down, the IWC banned hunting of blue, gray, and humpback whales, and instituted partial bans on other species. The rule was violated so often, however, that the IWC in 1986 instituted a worldwide moratorium on all commercial killing of whales. While some commercial whaling continues, often under the guise of harvesting for scientific studies, annual whale harvests have dropped dramatically in the last 15 years.

Some species appear to be recovering, while others do not. Humpback numbers have more than doubled since the early 1960s, increasing nearly 10% annually, and Pacific gray whales have fully recovered to their previous numbers of about 20,000 animals after being hunted to less than



**FIGURE 31.13**  
**A humpback whale.** Only 5000 to 10,000 humpback whales remain, out of a world population estimated to have been 100,000.



**FIGURE 31.14**  
**A history of commercial whaling.** These data show the world catch of whales in the twentieth century. Each species in turn is hunted until its numbers fall so low that hunting it becomes commercially unprofitable.

1000. Right, sperm, fin, and blue whales have not recovered, and no one knows whether they will.

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**Commercial whaling, by overharvesting, has driven most large whale species to the brink of extinction. Stopping the harvest has allowed recovery of some but not all species.**

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## Case Study: Introduced Species— Lake Victoria Cichlids

Lake Victoria, an immense shallow freshwater sea about the size of Switzerland in the heart of equatorial East Africa, had until 1954 been home to an incredibly diverse collection of over 300 species of cichlid fishes (figure 31.15). These small, perchlike fishes range from 2 to 10 inches in length, with males coming in endless varieties of colors. Today, all of these cichlid species are threatened, endangered, or extinct.

What happened to bring about the abrupt loss of so many endemic cichlid species? In 1954, the Nile perch, *Lates niloticus*, a commercial fish with a voracious appetite, was introduced on the Ugandan shore of Lake Victoria. Nile perch, which grow to over 4 feet in length, were to form the basis of a new fishing industry (figure 31.16). For decades, these perch did not seem to have a significant impact—over 30 years later, in 1978, Nile perch still made up less than 2% of the fish harvested from the lake.

Then something happened to cause the Nile perch to explode and to spread rapidly through the lake, eating their way through the cichlids. By 1986, Nile perch constituted nearly 80% of the total catch of fish from the lake, and the endemic cichlid species were virtually gone. Over 70% of cichlid species disappeared, including all open-water species.

So what happened to kick-start the mass extinction of the cichlids? The trigger seems to have been eutrophication. Before 1978, Lake Victoria had high oxygen levels at all depths, down to the bottom layers exceeding 60 meters depth. However, by 1989 high inputs of nutrients from agricultural runoff and sewage from towns and villages had led to algal blooms that severely depleted oxygen levels in deeper parts of the lake. Cichlids feed on algae, and initially their population numbers are thought to have risen in response to this increase in their food supply, but unlike similar algal blooms of the past, the Nile perch was now present to take advantage of the situation. With a sudden increase in its food supply (cichlids), the numbers of Nile perch exploded, and the greater numbers of them simply ate all available cichlids.

Since 1990 the situation has been compounded by a second factor, the introduction into Lake Victoria of a floating water weed from South America, the water hyacinth *Eichhornia crassipes*. Extremely fecund under eutrophic conditions, thick mats of water hyacinth soon covered entire bays and inlets, choking off the coastal habitats of non-open-water cichlids.

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**Lake Victoria's diverse collection of cichlid species is being driven to extinction by an introduced species, the Nile perch. A normal increase in cichlid numbers due to algal blooms led to an explosive increase in perch, which then ate their way through the cichlids.**

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**FIGURE 31.15**

**Lake Victoria cichlids.** Cichlid fishes are extremely diverse and occupy different niches. Some species feed on arthropods, others on dense stands of plants; there are fish-eaters, and still other species feed on fish eggs and larvae.



**FIGURE 31.16**

**Victor and vanquished.** The Nile perch (larger fishes in foreground), a commercial fish introduced into Lake Victoria as a potential food source, is responsible for the virtual extinction of hundreds of species of cichlid fishes (smaller fishes in tub).

Case Study: Disruption of Ecological Relationships—Black-Footed Ferrets

The black-footed ferret (*Mustela nigripes*) is one of the most attractive weasels of North America. A highly specialized predator, black-footed ferrets prey on prairie dogs, which live in large underground colonies connected by a maze of tunnels. These ferrets have experienced a dramatic decline in their North American range during this century, as agricultural development has destroyed their prairie habitat, and particularly the prairie dogs on which they feed (figure 31.17). Prairie dogs once roamed freely over 100 million acres of the Great Plains states, but are now confined to under 700,000 acres (table 31.4). Their ecological niche devastated, populations of the black-footed ferret collapsed. Increasingly rare in the second half of the century, the black-footed ferret was thought to have gone extinct in the late 1970s, when the only known wild population—a small colony in South Dakota—died out.

In 1981, a colony of 128 animals was located in Meeteetse, Wyoming. Left undisturbed for four years, the number of ferrets dropped by 50%, and the entire population seemed in immediate danger of extinction. A decision was made to capture some animals for a captive breeding program. The first six black-footed ferrets captured died of canine distemper, a disease present in the colony and probably responsible for its rapid decline.

At this point, drastic measures seemed called for. In the next year, a concerted effort was made to capture all the remaining ferrets in the Meeteetse colony. A captive population of 18 individuals was established before the Meeteetse colony died out. The breeding program proved a great success, the population jumping to 311 individuals by 1991.

In 1991, biologists began to attempt to reintroduce black-footed ferrets to the wild, releasing 49 animals in Wyoming. An additional 159 were released over the next two years. Six litters were born that year in the wild, and the reintroduction seemed a success. However, the released animals then underwent a drastic decline, and only ten individuals were still alive in the wild five years later in 1998. The reason for the decline is not completely understood, but predators such as coyotes appear to have played a large role. Current attempts at reintroduction involve killing the local coyotes. It is important that these attempts succeed, as numbers of offspring in the captive breeding colony are declining, probably as a result of the intensive inbreeding. The black-footed ferret still teeters at the brink of extinction.

Loss of its natural prey has eliminated black-footed ferrets from the wild; attempts to reintroduce them have not yet proven successful.



FIGURE 31.17 Teetering on the brink. The black-footed ferret is a predator of prairie dogs, and loss of prairie dog habitat as agriculture came to dominate the plains states in this century has led to a drastic decline in prairie dogs, and an even more drastic decline in the black-footed ferrets that feed on them. Attempts are now being made to reestablish natural populations of these ferrets, which have been extinct in the wild since 1986.

| Table 31.4 Acres of Prairie Dog Habitat |             |                |
|-----------------------------------------|-------------|----------------|
| State                                   | 1899-1990   | 1998           |
| Arizona                                 | unknown     | extinct        |
| Colorado                                | 7,000,000   | 44,000         |
| Kansas                                  | 2,500,000   | 36,000         |
| Montana                                 | 6,000,000   | 65,000         |
| Nebraska                                | 6,000,000   | 60,000         |
| New Mexico                              | 12,000,000  | 15,000         |
| North Dakota                            | 2,000,000   | 20,400         |
| Oklahoma                                | 950,000     | 9,500*         |
| South Dakota                            | 1,757,000   | 244,500        |
| Texas                                   | 56,833,000  | 22,650         |
| Wyoming                                 | 16,000,000  | 70,000-180,000 |
| U.S. Total                              | 111,000,000 | 700,000        |

Source: National Wildlife Federation and U.S. Fish and Wildlife Report, 1998.  
\*1990.

## Case Study: Loss of Genetic Variation—Prairie Chickens

The greater prairie chicken (*Tympanuchus cupido pinnatus*) is a showy 2-pound wild bird renowned for its flamboyant mating rituals (figure 31.18). Abundant in many midwestern states, the prairie chickens in Illinois have in the last six decades undergone a population collapse. Once, enormous numbers of birds covered the state, but with the introduction of the steel plow in 1837, the first that could slice through the deep dense root systems of prairie grasses, the Illinois prairie began to be replaced by farmland, and by the turn of the century the prairie had vanished. By 1931, the subspecies known as the heath hen (*Tympanuchus cupido cupido*) became extinct in Illinois. The greater prairie chicken fared little better in Illinois, its numbers falling to 25,000 statewide in 1933, then to 2000 in 1962. In surrounding states with less intensive agriculture, it continued to prosper.

In 1962, a sanctuary was established in an attempt to preserve the prairie chicken, and another in 1967. But privately owned grasslands kept disappearing, with their prairie chickens, and by the 1980s the birds were extinct in Illinois except for the two preserves. And there they were not doing well. Their numbers kept falling. By 1990, the egg hatching rate, which had averaged between 91 and 100%, had dropped to an extremely low 38%. By the mid-1990s, the count of males dropped to as low as six in each sanctuary.

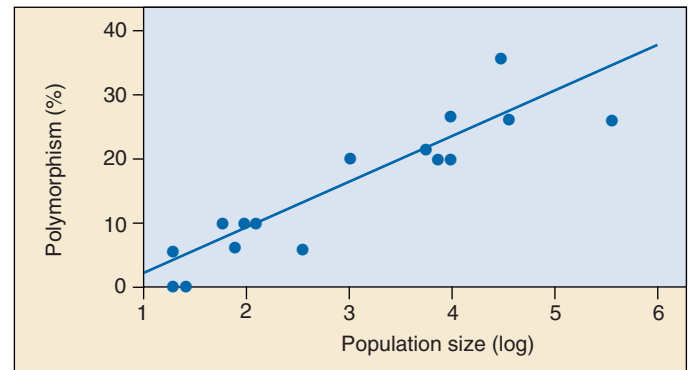
What was wrong with the sanctuary populations? One reasonable suggestion was that because of very small population sizes and a mating ritual where one male may dominate a flock, the Illinois prairie chickens had lost so much genetic variability as to create serious inbreeding problems. To test this idea, biologists at the University of Illinois compared DNA from frozen tissue samples of birds that died in Illinois between 1974 and 1993 and found that in recent years, Illinois birds had indeed become genetically less diverse. Extracting DNA from tissue in the roots of feathers from stuffed birds collected in the 1930s from the same population, the researchers found little genetic difference between the Illinois birds of the 1930s and present-day prairie chickens of other states. However, present-day Illinois birds had lost fully one-third of the genetic diversity of birds living in the same place before the population collapse of the 1970s.

Now the stage was set to halt the Illinois prairie chicken's race toward extinction. Wildlife managers began to transplant birds from genetically diverse populations of Minnesota, Kansas, and Nebraska to Illinois. Between 1992 and 1996, a total of 518 out-of-state prairie chickens were brought in to interbreed with the Illinois birds, and hatching rates were back up to 94% by 1998. It looks like the Illinois prairie chickens have been saved from extinction.



**FIGURE 31.18**

**A male prairie chicken performing a mating ritual.** He inflates bright orange air sacs, part of his esophagus, into balloons on each side of his head. As air is drawn into the sacs, it creates a three-syllable “boom-boom-boom” that can be heard for miles.



**FIGURE 31.19**

**Small populations lose much of their genetic variability.** The percentage of polymorphic genes in isolated populations of the tree *Halocarpus bidwilli* in the mountains of New Zealand is a sensitive function of population size.

The key lesson to be learned is the importance of not allowing things to go too far, not to drop down to a single isolated population (figure 31.19). Without the outlying genetically different populations, the prairie chickens in Illinois could not have been saved. The black-footed ferrets discussed on the previous page are particularly endangered because they exist as a single isolated population.

**When their numbers fell, Illinois prairie chickens lost much of their genetic variability, resulting in reproductive failure and the threat of immediate extinction. Breeding with genetically more variable birds appears to have reversed the decline.**

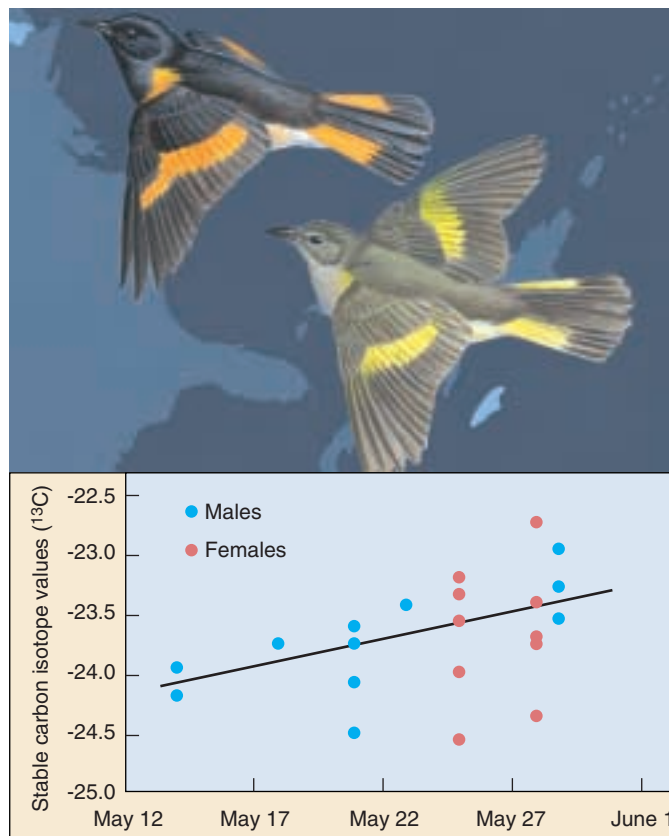
## Case Study: Habitat Loss and Fragmentation—Songbirds

Every year since 1966, the U.S. Fish and Wildlife Service has organized thousands of amateur ornithologists and bird watchers in an annual bird count called the Breeding Bird Survey. In recent years, a shocking trend has emerged. While year-round residents that prosper around humans, like robins, starlings, and blackbirds, have increased their numbers and distribution over the last 30 years, forest songbirds have declined severely. The decline has been greatest among long-distance migrants such as thrushes, orioles, tanagers, catbirds, vireos, buntings, and warblers. These birds nest in northern forests in the summer, but spend their winters in South or Central America or the Caribbean Islands.

In many areas of the eastern United States, more than three-quarters of the neotropical migrant bird species have declined significantly. Rock Creek Park in Washington, D.C., for example, has lost 90 percent of its long distance migrants in the last 20 years. Nationwide, American redstarts declined about 50% in the single decade of the 1970s. Studies of radar images from National Weather Service stations in Texas and Louisiana indicate that only about half as many birds fly over the Gulf of Mexico each spring compared to numbers in the 1960s. This suggests a loss of about half a billion birds in total, a devastating loss.

The culprit responsible for this widespread decline appears to be habitat fragmentation and loss. Fragmentation of breeding habitat and nesting failures in the summer nesting grounds of the United States and Canada have had a major negative impact on the breeding of woodland songbirds. Many of the most threatened species are adapted to deep woods and need an area of 25 acres or more per pair to breed and raise their young. As woodlands are broken up by roads and developments, it is becoming increasingly difficult to find enough contiguous woods to nest successfully.

A second and perhaps even more important factor seems to be the availability of critical winter habitat in Central and South America. Living in densely crowded limited areas, the availability of high-quality food is critical. Studies of the American redstart clearly indicate that birds with better winter habitat have a superior chance of successfully migrating back to their breeding grounds in the spring. Peter Marra and Richard Holmes of Dartmouth College and Keith Hobson of the Canadian Wildlife Service captured birds, took blood samples, and measured the levels of the stable carbon isotope  $^{13}\text{C}$ . Plants growing in the best overwintering habitats in Jamaica and Honduras (mangroves and wetland forests) have low levels of  $^{13}\text{C}$ , and so do the redstarts that feed on them. Sixty-five percent of the wet forest birds maintained or gained weight over the winter. Plants growing in substandard dry scrub, by contrast, have high levels of  $^{13}\text{C}$ , and so do the redstarts that feed on them. Scrub-dwelling birds lost up to 11% of their body mass over the winter. Now here's the key: birds that winter



**FIGURE 31.20**  
**The American redstart, a migratory songbird whose numbers are in serious decline.** The graph presents data on the level of  $^{13}\text{C}$  in redstarts arriving at summer breeding grounds. Early arrivals, with the best shot at reproductive success, have lower levels of  $^{13}\text{C}$ , indicating they wintered in more favorable mangrove-wetland forest habitats.

in the substandard scrub leave later in the spring on the long flight to northern breeding grounds, arrive at their summer homes, and have fewer young. You can see this clearly in the redstart study (figure 31.20): the proportion of  $^{13}\text{C}$  carbon in birds arriving in New Hampshire breeding grounds increases as spring wears on and scrub-overwintering stragglers belatedly arrive. Thus, loss of mangrove habitat in the neotropics is having a real negative impact. As the best habitat disappears, overwintering birds fare poorly, and this leads to population declines. Unfortunately, the Caribbean lost about 10% of its mangroves in the 1980s, and continues to lose about 1% a year. This loss of key habitat appears to be a driving force in the looming extinction of songbirds.

**Fragmentation of summer breeding grounds and loss of high-quality overwintering habitat seem both to be contributing to a marked decline in migratory songbird species.**

## 31.4 Successful recovery plans will need to be multidimensional.

### Many Approaches Exist for Preserving Endangered Species

Once you understand the reasons why a particular species is endangered, it becomes possible to think of designing a recovery plan. If the cause is commercial overharvesting, regulations can be designed to lessen the impact and protect the threatened species. If the cause is habitat loss, plans can be instituted to restore lost habitat. Loss of genetic variability in isolated subpopulations can be countered by transplanting individuals from genetically different populations. Populations in immediate danger of extinction can be captured, introduced into a captive breeding program, and later reintroduced to other suitable habitat.

Of course, all of these solutions are extremely expensive. As Bruce Babbitt, Interior Secretary in the Clinton administration, noted, it is much more economical to prevent such “environmental trainwrecks” from occurring than it is to clean them up afterwards. Preserving ecosystems and monitoring species before they are threatened is the most effective means of protecting the environment and preventing extinctions.

#### Habitat Restoration

Conservation biology typically concerns itself with preserving populations and species in danger of decline or extinction. Conservation, however, requires that there be something left to preserve, while in many situations, conservation is no longer an option. Species, and in some cases whole communities, have disappeared or have been irretrievably modified. The clear-cutting of the temperate forests of Washington State leaves little behind to conserve; nor does converting a piece of land into a wheat field or an asphalt parking lot. Redeeming these situations requires restoration rather than conservation.

Three quite different sorts of habitat restoration programs might be undertaken, depending very much on the cause of the habitat loss.

**Pristine Restoration.** In situations where all species have been effectively removed, one might attempt to restore the plants and animals that are believed to be the natural inhabitants of the area, when such information is available. When abandoned farmland is to be restored to prairie (figure 31.21), how do you know what to plant? Although it is in principle possible to reestablish each of the original species in their original proportions, rebuilding a community requires that you know the identity of all of the original inhabitants, and the ecologies of each of the species. We rarely ever have this much information, so no restoration is truly pristine.



(a)



(b)

**FIGURE 31.21**

**The University of Wisconsin–Madison Arboretum has pioneered restoration ecology.** (a) The restoration of the prairie was at an early stage in November, 1935. (b) The prairie as it looks today. This picture was taken at approximately the same location as the 1935 photograph.

**Removing Introduced Species.** Sometimes the habitat of a species has been destroyed by a single introduced species. In such a case, habitat restoration involves removal of the introduced species. Restoration of the once-diverse cichlid fishes to Lake Victoria will require more than breeding and restocking the endangered species. Eutrophication will have to be reversed, and the introduced water hyacinth and Nile perch populations brought under control or removed.

It is important to act quickly if an introduced species is to be removed. When aggressive African bees (the so-called “killer bees”) were inadvertently released in Brazil, they remained in the local area only one season. Now they occupy much of the Western hemisphere.

**Cleanup and Rehabilitation.** Habitats seriously degraded by chemical pollution cannot be restored until the pollution is cleaned up. The successful restoration of the Nashua River in New England, discussed in chapter 30, is one example of how a concerted effort can succeed in restoring a heavily polluted habitat to a relatively pristine condition.

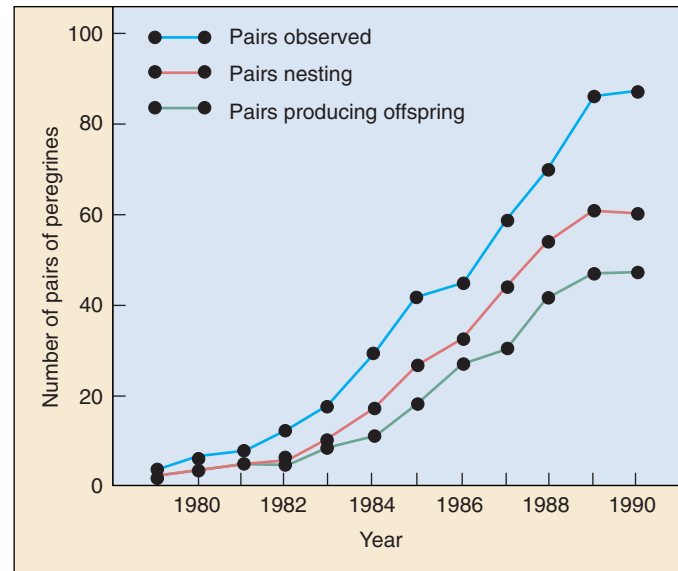
## Captive Propagation

Recovery programs, particularly those focused on one or a few species, often must involve direct intervention in natural populations to avoid an immediate threat of extinction. Earlier we learned how introducing wild-caught individuals into captive breeding programs is being used in an attempt to save ferret and prairie chicken populations in immediate danger of disappearing. Several other such captive propagation programs have had significant success.

**Case History: The Peregrine Falcon.** American populations of birds of prey such as the peregrine falcon (*Falco peregrinus*) began an abrupt decline shortly after World War II. Of the approximately 350 breeding pairs east of the Mississippi River in 1942, all had disappeared by 1960. The culprit proved to be the chemical pesticide DDT (dichlorodiphenyltrichloroethane) and related organochlorine pesticides. Birds of prey are particularly vulnerable to DDT because they feed at the top of the food chain, where DDT becomes concentrated. DDT interferes with the deposition of calcium in the bird's eggshells, causing most of the eggs to break before they hatch.

The use of DDT was banned by federal law in 1972, causing levels in the eastern United States to fall quickly. There were no peregrine falcons left in the eastern United States to reestablish a natural population, however. Falcons from other parts of the country were used to establish a captive breeding program at Cornell University in 1970, with the intent of reestablishing the peregrine falcon in the eastern United States by releasing offspring of these birds. By the end of 1986, over 850 birds had been released in 13 eastern states, producing an astonishingly strong recovery (figure 31.22).

**Case History: The California Condor.** Numbers of the California condor (*Gymnogyps californianus*), a large vulturelike bird with a wingspan of nearly 3 meters, have been declining gradually for the last 200 years. By 1985 condor numbers had dropped so low the bird was on the verge of extinction. Six of the remaining 15 wild birds disappeared that year alone. The entire breeding population of the species consisted of the 6 birds remaining in the wild, and an additional 21 birds in captivity. In a last-ditch attempt to save the condor from extinction, the remaining birds were captured and placed in a captive breeding population. The breeding program was set up in zoos, with the goal of releasing offspring on a large 5300-ha ranch in prime condor habitat. Birds were isolated from human contact as much as possible, and closely related individuals were prevented from breeding. By the end of 1999 the captive population of California condors had reached over 110 individuals. Twenty-nine captive-reared condors have been released successfully in California at two sites in the mountains north of Los Angeles, after extensive prerelease training to avoid power poles and people, all of the released birds seem to be



**FIGURE 31.22**

**Captive propagation.** The peregrine falcon has been reestablished in the eastern United States by releasing captive-bred birds over a period of 10 years.

doing well. Twenty additional birds released into the Grand Canyon have adapted well. Biologists are waiting to see if the released condors will breed in the wild and successfully raise a new generation of wild condors.

**Case History: Yellowstone Wolves.** The ultimate goal of captive breeding programs is not simply to preserve interesting species, but rather to restore ecosystems to a balanced functional state. Yellowstone Park has been an ecosystem out of balance, due in large part to the systematic extermination of the gray wolf (*Canis lupus*) in the park early in this century. Without these predators to keep their numbers in check, herds of elk and deer expanded rapidly, damaging vegetation so that the elk themselves starve in times of scarcity. In an attempt to restore the park's natural balance, two complete wolf packs from Canada were released into the park in 1995 and 1996. The wolves adapted well, breeding so successfully that by 1998 the park contained nine free-ranging packs, a total of 90 wolves.

While ranchers near the park have been unhappy about the return of the wolves, little damage to livestock has been noted, and the ecological equilibrium of Yellowstone Park seems well on the way to recovery. Elk are congregating in larger herds, and their populations are not growing as rapidly as in years past. Importantly, wolves are killing coyotes and their pups, driving them out of some areas. Coyotes, the top predators in the absence of wolves, are known to attack cattle on surrounding ranches, so reintroduction of wolves to the park may actually benefit the cattle ranchers that are opposed to it.

## Sustaining Genetic Diversity

One of the chief obstacles to a successful species recovery program is that a species is generally in serious trouble by the time a recovery program is instituted. When populations become very small, much of their genetic diversity is lost (see figure 31.19), as we have seen clearly in our examination of the case histories of prairie chickens and black-footed ferrets. If a program is to have any chance of success, every effort must be made to sustain as much genetic diversity as possible.

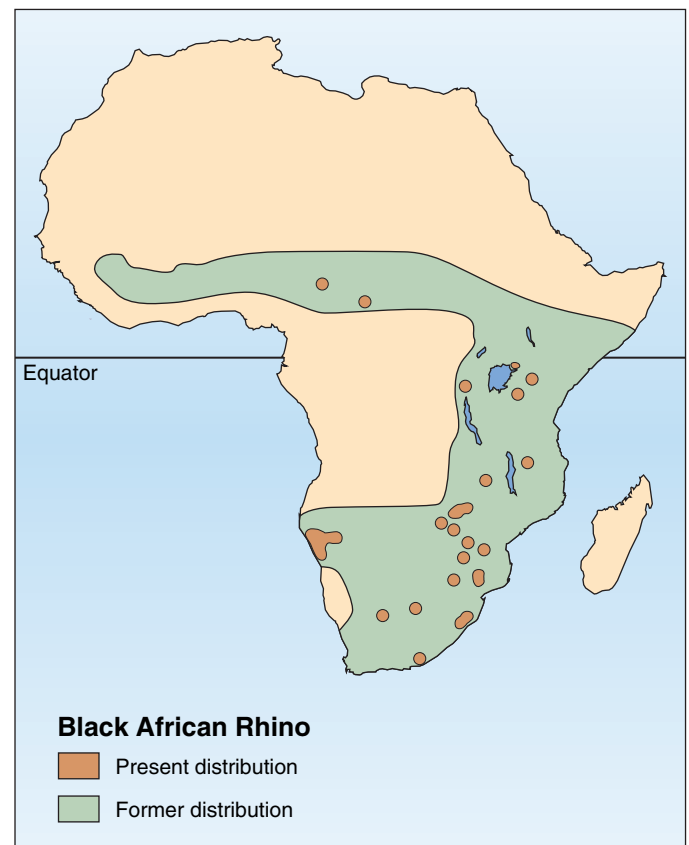
**Case History: The Black Rhino.** All five species of rhinoceros are critically endangered. The three Asian species live in forest habitat that is rapidly being destroyed, while the two African species are illegally killed for their horns. Fewer than 11,000 individuals of all five species survive today. The problem is intensified by the fact that many of the remaining animals live in very small, isolated populations. The 2400 wild-living individuals of the black rhino, *Diceros bicornis*, live in approximately 75 small widely separated groups (figure 31.23) consisting of six subspecies adapted to local conditions throughout the species' range. All of these subspecies appear to have low genetic variability; in three of the subspecies, only a few dozen animals remain. Analysis of mitochondrial DNA suggests that in these populations most individuals are genetically very similar.

This lack of genetic variability represents the greatest challenge to the future of the species. Much of the range of the black rhino is still open and not yet subject to human encroachment. To have any significant chance of success, a species recovery program will have to find a way to sustain the genetic diversity that remains in this species. Heterozygosity could be best maintained by bringing all black rhinos together in a single breeding population, but this is not a practical possibility. A more feasible solution would be to move individuals between populations. Managing the black rhino populations for genetic diversity could fully restore the species to its original numbers and much of its range.

Placing black rhinos from a number of different locations together in a sanctuary to increase genetic diversity raises a potential problem: local subspecies may be adapted in different ways to their immediate habitats—what if these local adaptations are crucial to their survival? Homogenizing the black rhino populations by pooling their genes risks destroying such local adaptations, if they exist, perhaps at great cost to survival.



(a)



(b)

**FIGURE 31.23**

**Sustaining genetic diversity.** The black rhino (a) is highly endangered, living in 75 small, widely separated populations (b). Only about 2400 individuals survive in the wild.

## Preserving Keystone Species

Keystone species are species that exert a particularly strong influence on the structure and functioning of a particular ecosystem. The sea otters of figure 31.7 are a keystone species of the kelp forest ecosystem, and their removal can have disastrous consequences. There is no hard-and-fast line that allows us to clearly identify keystone species. It is rather a qualitative concept, a statement that a species plays a particularly important role in its community. Keystone species are usually characterized by measuring the strength of their impact on their community. **Community importance** measures the change in some quantitative aspect of the ecosystem (species richness, productivity, nutrient cycling) per unit of change in the abundance of a species.

**Case History: Flying Foxes.** The severe decline of many species of pteropodid bats, or “flying foxes,” in the Old World tropics is an example of how the loss of a keystone species can have dramatic effects on the other species living within an ecosystem, sometimes even leading to a cascade of further extinctions (figure 31.24).

These bats have very close relationships with important plant species on the islands of the Pacific and Indian Oceans. The family Pteropodidae contains nearly 200 species, approximately a quarter of them in the genus *Pteropus*, and is widespread on the islands of the South Pacific, where they are the most important—and often the only—pollinators and seed dispersers. A study in Samoa found that 80 to 100% of the seeds landing on the ground during the dry season were deposited by flying foxes. Many species are entirely dependent on these bats for pollination. Some have evolved features like night-blooming flowers that prevent any other potential pollinators from taking over the role of the fruit bats.

In Guam, where the two local species of flying fox have recently been driven extinct or nearly so, the impact on the ecosystem appears to be substantial. Botanists have found some plant species are not fruiting, or are doing so only marginally, with fewer fruits than normal. Fruits are not being dispersed away from parent plants, so offspring shoots are being crowded out by the adults.

Flying foxes are being driven to extinction by human hunting. They are hunted for food, for sport, and by orchard farmers, who consider them pests. Flying foxes are particularly vulnerable because they live in large, easily seen groups of up to a million individuals. Because they



**FIGURE 31.24**

**Preserving keystone species.** The flying fox is a keystone species in many Old World tropical islands. It pollinates many of the plants, and is a key disperser of seeds. Its elimination by hunting and habitat loss is having a devastating effect on the ecosystems of many South Pacific islands.

move in regular and predictable patterns and can be easily tracked to their home roost, hunters can easily bag thousands at a time.

Species preservation programs aimed at preserving particular species of flying foxes are only just beginning. One particularly successful example is the program to save the Rodrigues fruit bat, *Pteropus rodricensis*, which occurs only on Rodrigues Island in the Indian Ocean near Madagascar. The population dropped from about 1000 individuals in 1955 to fewer than 100 by 1974, the drop reflecting largely the loss of the fruit bat’s forest habitat to farming. Since 1974 the species has been legally protected, and the forest area of the island is being increased through a tree-planting program. Eleven captive breeding colonies have been established, and the bat population is now increasing rapidly. The combination of legal protection, habitat restoration, and captive breeding has in this instance produced a very effective preservation program.

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**Recovery programs at the species level must deal with habitat loss and fragmentation, and often with a marked reduction in genetic diversity. Captive breeding programs that stabilize genetic diversity and pay careful attention to habitat preservation and restoration are typically involved in successful recoveries.**

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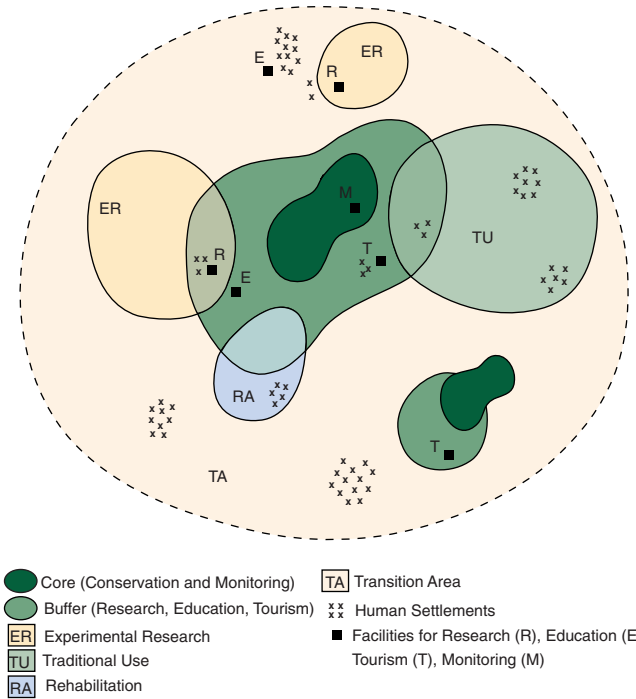
## Conservation of Ecosystems

Habitat fragmentation is one of the most pervasive enemies of biodiversity conservation efforts. As we have seen, some species simply require large patches of habitat to thrive, and conservation efforts that cannot provide suitable habitat of such a size are doomed to failure. As it has become clear that isolated patches of habitat lose species far more rapidly than large preserves do, conservation biologists have promoted the creation, particularly in the tropics, of so-called megareserves, large areas of land containing a core of one or more undisturbed habitats (figure 31.25). The key to devoting such large tracts of land to reserves successfully over a long period of time is to operate the reserve in a way compatible with local land use. Thus, while no economic activity is allowed in the core regions of the megareserve, the remainder of the reserve may be used for nondestructive harvesting of resources. Linking preserved areas to carefully managed land zones creates a much larger total “patch” of habitat than would otherwise be economically practical, and thus addresses the key problem created by habitat fragmentation. Pioneering these efforts, a series of eight such megareserves have been created in Costa Rica (figure 31.26) to jointly manage biodiversity and economic activity.

In addition to this focus on maintaining large enough reserves, in recent years, conservation biologists also have recognized that the best way to preserve biodiversity is to focus on preserving intact ecosystems, rather than focusing on particular species. For this reason, attention in many cases is turning to identifying those ecosystems most in need of preservation and devising the means to protect not only the species within the ecosystem, but the functioning of the ecosystem itself.

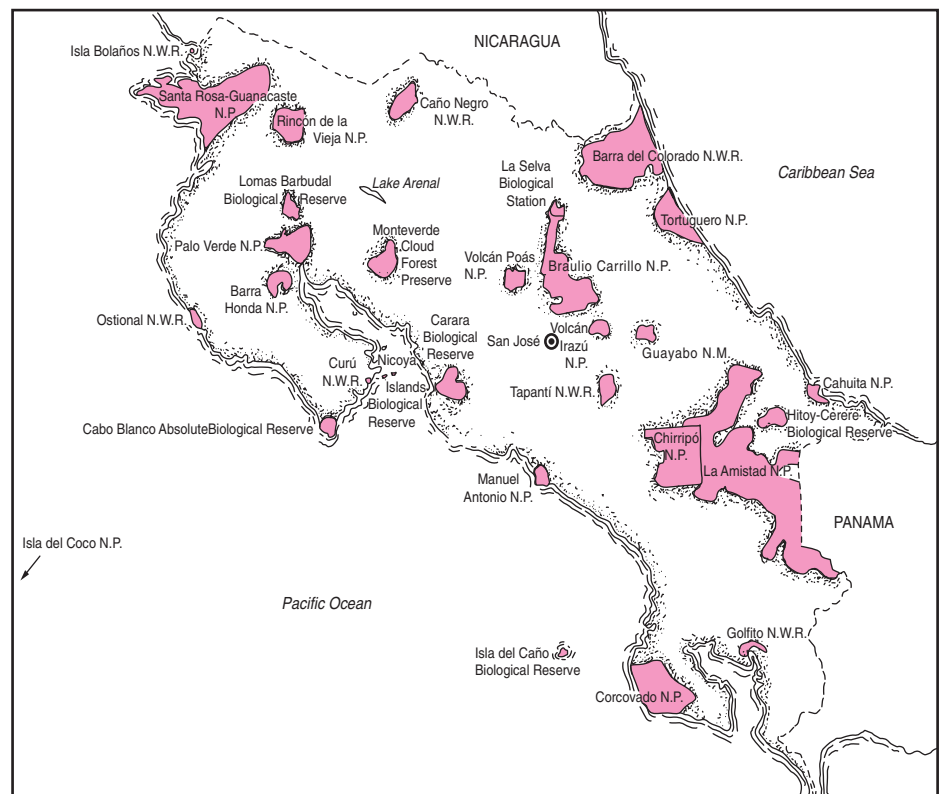
Efforts are being undertaken worldwide to preserve biodiversity in megareserves designed to counter the influences of habitat fragmentation. Focusing on the health of entire ecosystems, rather than of particular species, can often be a more effective means of preserving biodiversity.

**FIGURE 31.26**  
**Biopreserves in Costa Rica.** Costa Rica has placed about 12% of its land into national parks and eight megareserves.



**FIGURE 31.25**

**Design of a megareserve.** A megareserve, or biosphere reserve, recognizes the need for people to have access to resources. Critical ecosystems are preserved in the core zone. Research and tourism is allowed in the buffer zone. Sustainable resource harvesting and permanent habitation is allowed in the multiple-use areas surrounding the buffer.





## Summary

## Questions

## Media Resources

**31.1 The new science of conservation biology is focusing on conserving biodiversity.**

- Early humans caused many extinctions when they appeared in new areas, but rates of extinction have increased in modern times.
- Some areas are particularly rich in species diversity and particularly merit conservation attention.

1. Are some areas particularly important for conserving biodiversity?
2. Describe some of the indirect economic values of biodiversity.



- Biodiversity
- Species
- Extinction
- Wetlands

**31.2 Vulnerable species are more likely to become extinct.**

- Interdependence among species in an ecosystem leads to the possibility of cascading extinctions if removal of one species has major effects throughout the food web.
- Species are particularly vulnerable when they have localized distributions, are declining in population size, lack genetic variability, or are harvested or hunted by humans.

3. What factors contribute to the extinction rate on a particular piece of land?
4. How does a low genetic variability contribute to a species' greater risk of extinction?



- Extinction

**31.3 Causes of endangerment usually reflect human activities.**

- Habitat loss is the single most important cause of species extinction.
- As suggested by the species-area relationship, a reduced habitat will support fewer numbers of species.
- This reduction in habitat can occur in four different ways: a habitat can be completely removed or destroyed, a habitat can become fragmented and disjunct, a habitat can be degraded or altered, or a habitat can become too frequently used by humans so as to disturb the species there.

5. How can problems resulting from lack of genetic diversity within a population be solved?
6. How can extinction of a keystone species be particularly disruptive to an ecosystem?



- Deoxygenation of Lakes



- On Science Article: What's Killing the Frogs?
- Book Review: *The End of the Game* by Beard
- Book Review: *West with the Night* by Markham

**31.4 Successful recovery plans will need to be multidimensional.**

- Pristine restoration of a habitat may be attempted, but removing introduced species, rehabilitating the habitat, and cleaning up the habitat may be more feasible.
- Captive propagation, sustaining genetic variability, and preserving keystone species have been effective in preserving biodiversity.
- Megareserves have been successfully designed in many parts of the world to contain core areas of undisturbed habitat surrounded by managed land.

7. Why is maintaining large preserves particularly important?
8. Is captive propagation always an answer to species vulnerability?
9. Why is it important to attempt to eradicate introduced species soon after they appear?



- Activity: Biomagnification



- On Science Article: Biodiversity Behind Bars

## Viruses and Simple Organisms

### Discovering the Virus Responsible for Hepatitis C

You may not be aware that our country is in the midst of an epidemic of this potentially fatal liver disease. Almost 4 million Americans are infected with the hepatitis C virus, most of them without knowing it. Some 9000 people will die this year in the United States from liver cancer and chronic liver failure brought on by the virus, and the number is expected to triple in the next decade. In the first years of the new century, the number of annual U.S. deaths caused by hepatitis C is predicted to overtake deaths caused by AIDS.

Hepatitis is inflammation of the liver. Researchers in the 1940s identified two distinct forms. One, called infectious hepatitis or hepatitis A, is transmitted by contact with feces from infected individuals. A second form of hepatitis, called serum hepatitis or hepatitis B, is passed only through the blood. Hepatitis B virus was isolated in the mid-1960s, hepatitis A virus a decade later. This led in the 1970s to the development of tests for the two viruses. Disturbingly, a substantial proportion of hepatitis cases did not appear to be caused by either of these two viruses.

Clearly another virus was at work. At first, investigators thought it wouldn't be long before it was isolated. However, it was not until 1990 that researchers succeeded in isolating the virus responsible for these "non-A, non-B" cases, a virus that we now call hepatitis C virus (HCV).

HCV was difficult to isolate because it cannot be grown reliably in a laboratory culture of cells. Making the problem even more difficult, HCV is a strictly primate virus. It infects only humans and our close relatives—chimpanzees and tamarins. Because it is very expensive to maintain these animals in research laboratories, only small numbers of animals can be employed in any one study. Thus, the virus could not be isolated by the traditional means of purification from extracts of infected cells. What finally succeeded, after 15 years of failed attempts at isolation, was molecular technology. HCV was the first virus isolated entirely by cloning the infectious nucleic acid.

The successful experiment was carried out by Michael Houghton and fellow researchers at Chiron, a California

biotechnology company. What they did was shotgun clone the DNA of infected cells, and then screen for HCV.

The genetic material of HCV, like that of many other viruses, is RNA. So the first step was to convert HCV RNA to DNA, so that it could be cloned. There was no need to attempt to achieve entire faithful copies, a touchy and difficult task, because they did not wish to replicate HCV, only identify it. So the researchers took the far easier route of copying the virus RNA as a series of segments, each carrying some part of the virus genome.

Next, they inserted these DNA copies of HCV genes into a bacteriophage, and allowed the bacteriophage to infect *Escherichia coli* bacteria. In such a "shotgun" experiment, millions of bacterial cells are infected with bacteriophages. The researchers grew individual infected cells to form discrete colonies on plates of solid culture media. The colonies together constituted a "clone library." The problem then is to screen the library for colonies that had successfully received HCV.

To understand how they did this, focus on the quarry, a cell infected with an HCV gene. Once inside a bacterial cell, an HCV gene fragment becomes just so much more DNA, not particularly different from all the rest. The cellular machinery of the bacteria reads it just like bacterial genes, manufacturing the virus protein that the inserted HCV gene encodes. The secret is to look for cells with HCV proteins.

How to identify an HCV protein from among a background of thousands of bacterial proteins? Houghton and his colleagues tested each colony for its ability to cause a visible immune reaction with serum isolated from HCV-infected chimpanzees.

The test is a very simple and powerful one, because its success does not depend on knowing the identity of the genes you seek. The serum of HCV-infected animals contains antibodies directed against a broad range of HCV proteins encountered while combating the animal's HCV infection. The serum can thus be used as a probe for the presence of HCV proteins in other cells.

Out of a million bacterial clones tested, just one was found that reacted with the chimp HCV serum, but not with serum from the same chimp before infection.

Electron micrograph of hepatitis C virus.

**How the hepatitis C virus was discovered.** Michael Houghton and fellow researchers identified the virus responsible for hepatitis C by making DNA copies of RNA from the cells of infected chimpanzees. They then cloned this DNA, using bacteriophages to carry it into bacterial cells. Colonies of the bacteria were then tested with serum from infected chimps. Any colony that produced an immune reaction would have to contain the virus.

Using this clone as a toehold, the researchers were able to go back and fish out the rest of the virus genome from infected cells. From the virus genome, it was a straightforward matter to develop a diagnostic antibody test for the presence of the HCV virus.

Using the diagnostic test, researchers found hepatitis C to be far more common than had been supposed. This is a problem of major proportions, because hepatitis C virus is unlike hepatitis A or B in a very important respect: it causes chronic disease. Most viruses cause a brief, intense infection and then are done. Hepatitis A, for example, typically lasts a few weeks. Ninety percent of people with hepatitis C have it for years, many of them for decades.

All during these long years of infection, damage is being done to the liver. Cells of the immune system called cytotoxic T cells recognize hepatitis C virus proteins on the surface of liver cells, and kill the infected cells. Over the years, many dead liver cells accumulate, and in response the cells around them begin to secrete collagen and other proteins to cover the mess. This eventually produces protein fibers interlacing the liver, fibers which disrupt the flow of materials through the liver's many internal passages. Imagine dropping bricks and rubble on a highway—it gets more and more difficult for traffic to move as the rubble accumulates.

If this fibrosis progresses far enough, it results in complete blockage, cirrhosis, a serious condition which may induce fatal liver failure, and which often induces primary liver cancer. About 20% of patients develop cirrhosis within 20 years of infection.

Luckily, hepatitis C is a very difficult virus to transmit. Direct blood contact is the only known path of direct transmission. Sexual transmission does not seem likely, although the possibility is still being investigated. Married partners of infected individuals rarely get the virus, and its incidence among promiscuous gay men is no higher than among the population at large.

Why not move vigorously to produce a vaccine directed against hepatitis C? This turns out to be particularly difficult for this virus, because antibodies directed against it appear to be largely ineffective. Those few individuals who do succeed in clearing the virus from their bodies gain no immunity to subsequent infection. They produce antibodies directed against the virus, but the antibodies don't protect them. It appears that hepatitis C virus evades our antibody defenses by high mutation rates, just as the AIDS virus does. By the time antibodies are being produced against one version of the virus, some of the viruses have already mutated to a different form that the antibody does not recognize. Like chasing a burglar who is constantly changing his disguise, the antibodies never learn to recognize the newest version of the virus.

To date, attempts to develop a drug to combat hepatitis C virus focus on the virus itself. This virus carries just one gene, a very big one. When it infects liver cells, this gene is translated into a single immense "polyprotein." Enzymes then cut the polyprotein into 10 functional pieces. Each piece plays a key role in building new viruses in infected liver cells. Some of these proteins form parts of the virus body, others are enzymes needed to replicate the virus gene. As you might expect, each of these 10 proteins is being investigated as a potential target for a drug to fight the virus, although no success is reported as yet.

Other attempts to fight hepatitis C focus on the part of our immune system that attacks infected liver cells. Unlike the ineffective antibody defense, our bodies' cytotoxic T cells clearly are able to detect and attack cells carrying hepatitis C proteins. A vaccine that stimulates these cytotoxic T cells might eliminate all infected cells at the start of an infection, stopping the disease in its tracks before it got started. A serious effort is being made to develop such a vaccine.

It doesn't look like an effective remedy is going to be available anytime soon. In the meantime, as the death rates from hepatitis C exceed those for AIDS in the next few years, we can hope research will further intensify.

# 32

## *How We Classify Organisms*

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### Concept Outline

#### 32.1 Biologists name organisms in a systematic way.

**The Classification of Organisms.** Biologists name organisms using a binomial system.

**Species Names.** Every kind of organism is assigned a unique name.

**The Taxonomic Hierarchy.** The higher groups into which an organism is placed reveal a great deal about the organism.

**What Is a Species?** Species are groups of similar organisms that tend not to interbreed with individuals of other groups.

#### 32.2 Scientists construct phylogenies to understand the evolutionary relationships among organisms.

**Evolutionary Classifications.** Traditional and cladistic interpretations of evolution differ in the emphasis they place on particular traits.

#### 32.3 All living organisms are grouped into one of a few major categories.

**The Kingdoms of Life.** Living organisms are grouped into three great groups called domains, and within domains into kingdoms.

**Domain Archaea (Archaeobacteria).** The oldest domain consists of primitive bacteria that often live in extreme environments.

**Domain Bacteria (Eubacteria).** Too small to see with the unaided eye, eubacteria are more numerous than any other organism.

**Domain Eukarya (Eukaryotes).** There are four kingdoms of eukaryotes, three of them entirely or predominantly multicellular. Two of the most important characteristics to have evolved among the eukaryotes are multicellularity and sexuality.

**Viruses: A Special Case.** Viruses are not organisms, and thus do not belong to any kingdom.



FIGURE 32.1

**Biological diversity.** All living things are assigned to particular classifications based on characteristics such as their anatomy, development, mode of nutrition, level of organization, and biochemical composition. Coral reefs, like the one seen here, are home to a variety of living things.

All organisms share many biological characteristics. They are composed of one or more cells, carry out metabolism and transfer energy with ATP, and encode hereditary information in DNA. All species have evolved from simpler forms and continue to evolve. Individuals live in populations. These populations make up communities and ecosystems, which provide the overall structure of life on earth. So far, we have stressed these common themes, considering the general principles that apply to all organisms. Now we will consider the diversity of the biological world and focus on the differences among groups of organisms (figure 32.1). For the rest of the text, we will examine the different kinds of life on earth, from bacteria and amoebas to blue whales and sequoia trees.

## 32.1 Biologists name organisms in a systematic way.

### The Classification of Organisms

Organisms were first classified more than 2000 years ago by the Greek philosopher Aristotle, who categorized living things as either plants or animals. He classified animals as either land, water, or air dwellers, and he divided plants into three kinds based on stem differences. This simple classification system was expanded by the Greeks and Romans, who grouped animals and plants into basic units such as cats, horses, and oaks. Eventually, these units began to be called **genera** (singular, **genus**), the Latin word for “groups.” Starting in the Middle Ages, these names began to be systematically written down, using Latin, the language used by scholars at that time. Thus, cats were assigned to the genus *Felis*, horses to *Equus*, and oaks to *Quercus*—names that the Romans had applied to these groups. For genera that were not known to the Romans, new names were invented.

The classification system of the Middle Ages, called the *polynomial system*, was used virtually unchanged for hundreds of years.

#### The Polynomial System

Until the mid-1700s, biologists usually added a series of descriptive terms to the name of the genus when they wanted to refer to a particular kind of organism, which they called a **species**. These phrases, starting with the name of the genus, came to be known as polynomials (*poly*, “many”; *nomial*, “name”), strings of Latin words and phrases consisting of up to 12 or more words. One name for the European honeybee, for example, was *Apis pubescens, thorace subgriseo, abdomine fusco, pedibus posticis glabris utrinque margine ciliatis*. As you can imagine, these polynomial names were cumbersome. Even more worrisome, the names were altered at will by later authors, so that a given organism really did not have a single name that was its alone.

#### The Binomial System

A much simpler system of naming animals, plants, and other organisms stems from the work of the Swedish biologist Carolus Linnaeus (1707–1778). Linnaeus devoted his life to a challenge that had defeated many biologists before him—cataloging all the different kinds of organisms. In the 1750s he produced several major works that, like his earlier

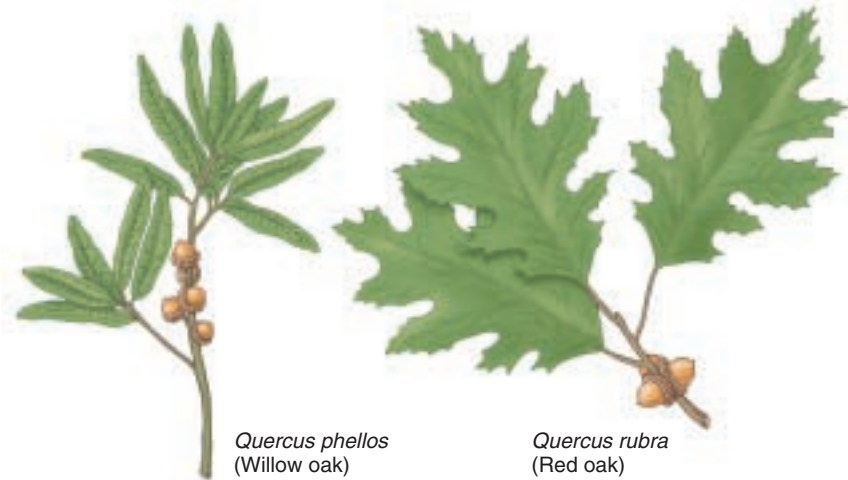


FIGURE 32.2

**Two species of oaks.** (a) Willow oak, *Quercus phellos*. (b) Red oak, *Quercus rubra*. Although they are both oaks (*Quercus*), these two species differ sharply in leaf shape and size and in many other features, including geographical range.

books, employed the polynomial system. But as a kind of shorthand, Linnaeus also included in these books a two-part name for each species. For example, the honeybee became *Apis mellifera*. These two-part names, or **binomials** (*bi*, “two”) have become our standard way of designating species.

#### A Closer Look at Linnaeus

To illustrate Linnaeus’s work further, let’s consider how he treated two species of oaks from North America, which by 1753 had been described by scientists. He grouped all oaks in the genus *Quercus*, as had been the practice since Roman times. Linnaeus named the willow oak of the southeastern United States (figure 32.2a) *Quercus foliis lanceolatis integerrimis glabris* (“oak with spear-shaped, smooth leaves with absolutely no teeth along the margins”). For the common red oak of eastern temperate North America (figure 32.2b), Linnaeus devised a new name, *Quercus foliis obtuse-sinuatis setaceo-mucronatis* (“oak with leaves with deep blunt lobes bearing hairlike bristles”). For each of these species, he also presented a shorthand designation, the binomial names *Quercus phellos* and *Quercus rubra*. These have remained the official names for these species since 1753, even though Linnaeus did not intend this when he first used them in his book. He considered the polynomials the true names of the species.

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**Two-part (“binomial”) Latin names, first utilized by Linnaeus, are now universally employed by biologists to name particular organisms.**

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## Species Names

**Taxonomy** is the science of classifying living things, and a group of organisms at a particular level in a classification system is called a **taxon** (plural, **taxa**). By agreement among taxonomists throughout the world, no two organisms can have the same name. So that no one country is favored, a language spoken by no country—Latin—is used for the names. Because the scientific name of an organism is the same anywhere in the world, this system provides a standard and precise way of communicating, whether the language of a particular biologist is Chinese, Arabic, Spanish, or English. This is a great improvement over the use of common names, which often vary from one place to the next. As you can see in figure 32.3, corn in Europe refers to the plant Americans call wheat; a bear is a large placental omnivore in the United States but a koala (a vegetarian marsupial) in Australia; and a robin is a very different bird in Europe and North America.

Also by agreement, the first word of the binomial name is the genus to which the organism belongs. This word is always capitalized. The second word refers to the particular species and is not capitalized. The two words together are called the **scientific name** and are written in italics or distinctive print: for example, *Homo sapiens*. Once a genus has been used in the body of a text, it is often abbreviated in later uses. For example, the dinosaur *Tyrannosaurus rex* becomes *T. rex*, and the potentially dangerous bacterium *Escherichia coli* is known as *E. coli*. The system of naming animals, plants, and other organisms established by Linnaeus has served the science of biology well for nearly 230 years.

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By convention, the first part of a binomial species name identifies the genus to which the species belongs, and the second part distinguishes that particular species from other species in the genus.

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(a)



(b)



(c)



**FIGURE 32.3**

**Common names make poor labels.** The common names corn (a), bear (b), and robin (c) bring clear images to our minds (photos on left), but the images are very different to someone living in Europe or Australia (photos on right). There, the same common names are used to label very different species.

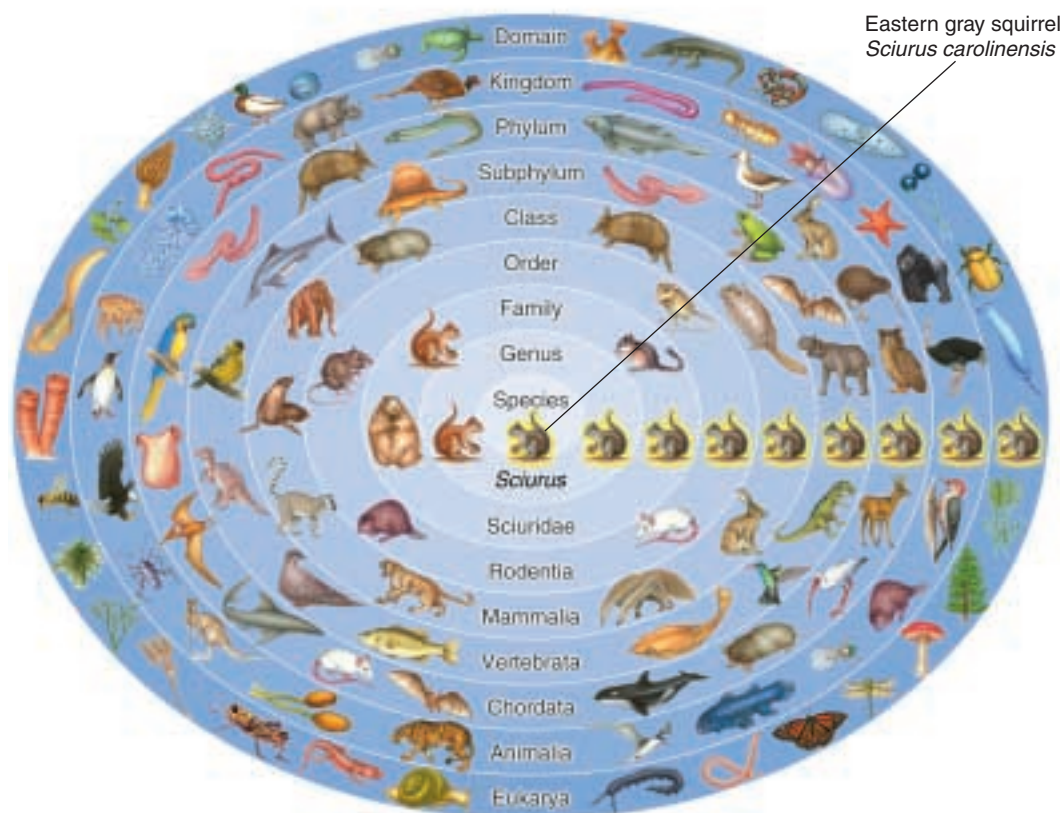
## The Taxonomic Hierarchy

In the decades following Linnaeus, taxonomists began to group organisms into larger, more inclusive categories. Genera with similar properties were grouped into a cluster called a **family**, and similar families were placed into the same **order** (figure 32.4). Orders with common properties were placed into the same **class**, and classes with similar characteristics into the same **phylum** (plural, **phyla**). For historical reasons, phyla may also be called *divisions* among plants, fungi, and algae. Finally, the phyla were assigned to one of several great groups, the **kingdoms**. Biologists currently recognize six kingdoms: two kinds of bacteria (Archaea and Eubacteria), a largely unicellular group of eukaryotes (Protista), and three multicellular groups (Fungi, Plantae, and Animalia). In order to remember the seven categories of the taxonomic hierarchy in their proper order, it may prove useful to memorize a phrase such as “**kindly pay cash or furnish good security**” (**k**ingdom–**p**hylum–**c**lass–**o**rders–**f**amilies–**g**enus–**s**pecies).

In addition, an eighth level of classification, called *domains*, is sometimes used. Biologists recognize three domains, which will be discussed later in this chapter. The scientific names of the taxonomic units higher than the genus level are capitalized but not printed distinctively, italicized, or underlined.

The categories at the different levels may include many, a few, or only one taxon. For example, there is only one living genus of the family Hominidae, but several living genera of Fagaceae. To someone familiar with classification or with access to the appropriate reference books, each taxon implies both a set of characteristics and a group of organisms belonging to the taxon. For example, a honeybee has the species (level 1) name *Apis mellifera*. Its genus name (level 2) *Apis* is a member of the family Apidae (level 3). All members of this family are bees, some solitary, others living in hives as *A. mellifera* does. Knowledge of its order (level 4), Hymenoptera, tells you that *A. mellifera* is likely able to sting and may live in colonies. Its class (level 5) Insecta indicates that *A. mellifera* has three major body segments, with wings and three pairs of legs attached to the middle segment. Its phylum (level 6), Arthropoda, tells us that the honeybee has a hard cuticle of chitin and jointed appendages. Its kingdom (level 7), Animalia, tells us that *A. mellifera* is a multicellular heterotroph whose cells lack cell walls.

**Species are grouped into genera, genera into families, families into orders, orders into classes, and classes into phyla. Phyla are the basic units within kingdoms; such a system is hierarchical.**



**FIGURE 32.4**  
**The hierarchical system used in classifying an organism.** The organism is first recognized as a eukaryote (domain: Eukarya). Second, within this domain, it is an animal (kingdom: Animalia). Among the different phyla of animals, it is a vertebrate (phylum: Chordata, subphylum: Vertebrata). The organism’s fur characterizes it as a mammal (class: Mammalia). Within this class, it is distinguished by its gnawing teeth (order: Rodentia). Next, because it has four front toes and five back toes, it is a squirrel (family: Sciuridae). Within this family, it is a tree squirrel (genus: *Sciurus*), with gray fur and white-tipped hairs on the tail (species: *Sciurus carolinensis*, the eastern gray squirrel).

## What Is a Species?

In the previous section we discussed how species are named and grouped, but how do biologists decide when one organism is distinct enough from another to be called its own species? In chapter 22, we reviewed the nature of species and saw there are no absolute criteria for the definition of this category. Looking different, for example, is not a useful criterion: different individuals that belong to the same species (for example, dogs) may look very unlike one another, as different as a Chihuahua and a St. Bernard. These very different-appearing individuals are fully capable of hybridizing with one another.

The **biological species concept** (figure 32.5) essentially says that two organisms that cannot interbreed and produce fertile offspring are different species. This definition of a species can be useful in describing sexually reproducing species that regularly **outcross**—interbreed with individuals other than themselves. However, in many groups of organisms, including bacteria, fungi, and many plants and animals, **asexual reproduction**—reproduction without sex—predominates. Among them, hybridization cannot be used as a criterion for species recognition.

### Defining Species

Despite such difficulties, biologists generally agree on the organisms they classify as species based on the similarity of morphological features and ecology. As a practical definition, we can say that species are groups of organisms that remain relatively constant in their characteristics, can be distinguished from other species, and do not normally interbreed with other species in nature.

### Evolutionary Species Concept

This simple definition of species leaves many problems unsolved. How, for instance, are we to compare living species with seemingly similar ones now extinct? Much of the disagreement among alternative species concepts relates to solving this problem. When do we assign fossil specimens a unique species name, and when do we assign them to species living today? If we trace the lineage of two sister species backwards through time, how far must we go before the two species converge on their common ancestor? It is often very hard to know where to draw a sharp line between two closely related species.

To address this problem, biologists have added an evolutionary time dimension to the biological species concept. A current definition of an evolutionary species is *a single lineage of populations that maintains its distinctive identity from other such lineages*. Unlike the biological species concept, the evolutionary species concept applies to both asexual and sexually reproducing forms. Abrupt changes in diagnostic features mark the boundaries of different species in evolutionary time.



(a)



(b)



(c)

**FIGURE 32.5**

**The biological species concept.** Horses (a) and donkeys (b) are not the same species, because the offspring they produce when they interbreed, mules (c), are sterile.

### How Many Species Are There?

Scientists have described and named a total of 1.5 million species, but doubtless many more actually exist. Some groups of organisms, such as flowering plants, vertebrate animals, and butterflies, are relatively well known with an estimated 90% of the total number of species that actually exist in these groups having already been described. Many other groups, however, are very poorly known. It is generally accepted that only about 5% of all species have been recognized for bacteria, nematodes (roundworms), fungi, and mites (a group of organisms related to spiders).

By taking representative samples of organisms from different environments, such as the upper branches of tropical trees or the deep ocean, scientists have estimated the total numbers of species that may actually exist to be about 10 million, about 15% of them marine organisms.

### Most Species Live in the Tropics

Most species, perhaps 6 or 7 million, are tropical. Presently only 400,000 species have been named in tropical Asia, Africa, and Latin America combined, well under 10% of all species that occur in the tropics. This is an incredible gap in our knowledge concerning biological diversity in a world that depends on biodiversity for its sustainability.

These estimates apply to the number of eukaryotic organisms only. There is no functional way of estimating the numbers of species of prokaryotic organisms, although it is clear that only a very small fraction of all species have been discovered and characterized so far.

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**Species are groups of organisms that differ from one another in recognizable ways and generally do not interbreed with one another in nature.**

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## 32.2 Scientists construct phylogenies to understand the evolutionary relationships among organisms.

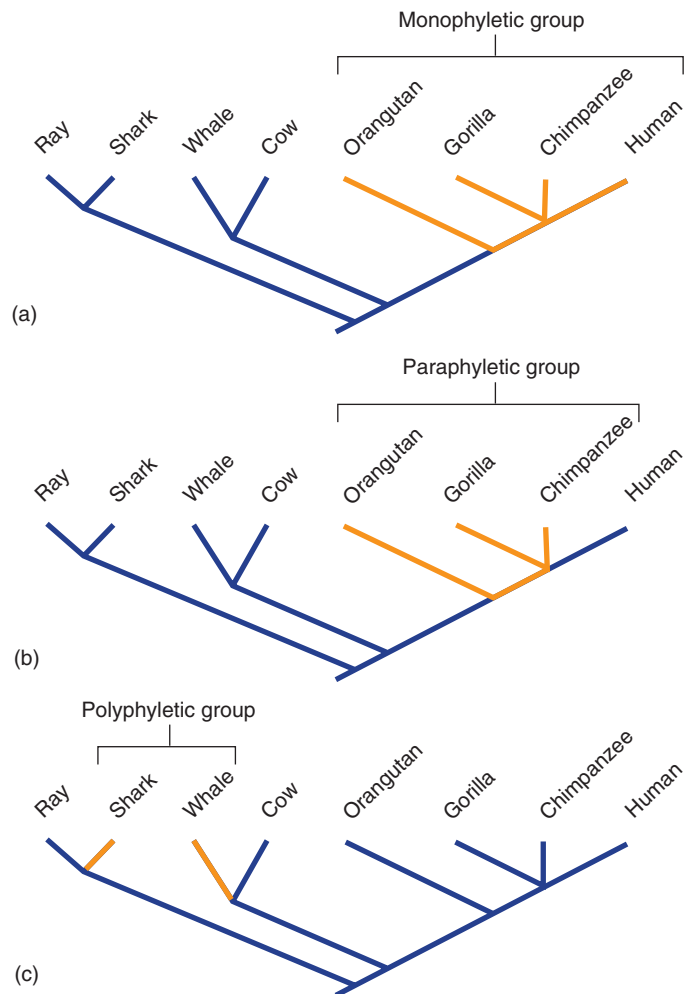
### Evolutionary Classifications

After naming and classifying some 1.5 million organisms, what have biologists learned? One very important advantage of being able to classify particular species of plants, animals, and other organisms is that individuals of species that are useful to humans as sources of food and medicine can be identified. For example, if you cannot tell the fungus *Penicillium* from *Aspergillus*, you have little chance of producing the antibiotic penicillin. In a thousand ways, just having names for organisms is of immense importance in our modern world.

Taxonomy also enables us to glimpse the evolutionary history of life on earth. The more similar two taxa are, the more closely related they are likely to be. By looking at the differences and similarities between organisms, biologists can construct an evolutionary tree, or **phylogeny**, inferring which organisms evolved from which other ones, in what order, and when. The reconstruction and study of phylogenies is called **systematics**. Within a phylogeny, a grouping can be either monophyletic, paraphyletic, or polyphyletic. A **monophyletic** group includes the most recent common ancestor of the group and all of its descendants. A **paraphyletic** group includes the most recent common ancestor of the group but not all of its descendants. And, a **polyphyletic** group does not include the most recent common ancestor of all the members of the group. Monophyletic groups are commonly assigned names, but systematists will not assign a taxonomic classification to a polyphyletic group. Paraphyletic groups may be considered taxa by some scientists, although they do not accurately represent the evolutionary relationships among the members of the group (figure 32.6).

### Cladistics

A simple and objective way to construct a phylogenetic tree is to focus on key characters that a group of organisms share because they have inherited them from a common ancestor. A **clade** is a group of organisms related by descent, and this approach to constructing a phylogeny is called **cladistics**. Cladistics infers phylogeny (that is, builds family trees) according to similarities derived from a common ancestor, so-called derived characters. A derived character that is unique to a particular clade is sometimes called a **synapomorphy**. The key to the approach is being able to identify morphological, physiological, or behavioral traits that differ among the organisms being studied and can be attributed to a common ancestor. By examining the distribution of these traits among the organisms, it is possible to construct a **clado-**



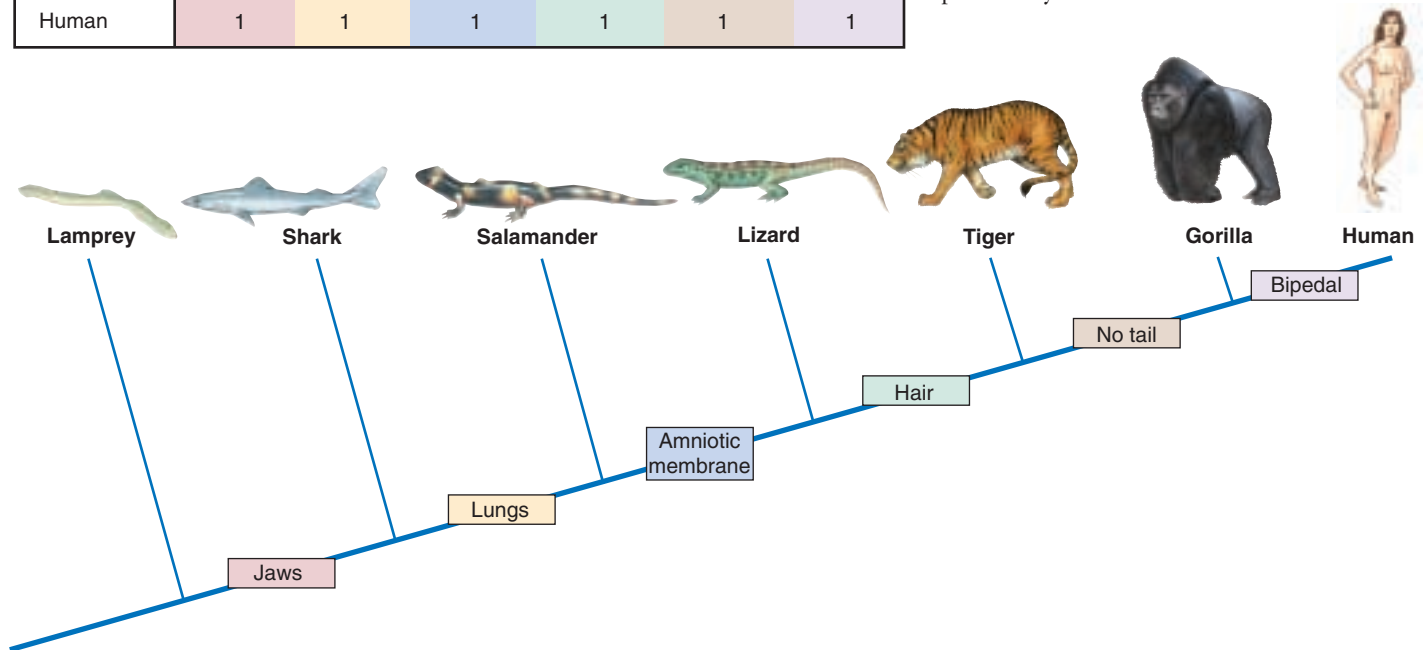
**FIGURE 32.6**

(a) A monophyletic group consists of the most recent common ancestor and all of its descendants. All taxonomists accept monophyletic groups in their classifications and in the above example would give the name “Apes” to the orangutans, gorillas, chimpanzees, and humans. (b) A paraphyletic group consists of the most recent common ancestor and some of its descendants. Taxonomists differ in their acceptance of paraphyletic groups. For example, some taxonomists arbitrarily group orangutans, gorillas, and chimpanzees into the paraphyletic family Pongidae, separate from humans. Other taxonomists do not use the family Pongidae in their classifications because gorillas and chimpanzees are more closely related to humans than to orangutans. (c) A polyphyletic group does not contain the most recent common ancestor of the group, and taxonomists do not assign taxa to polyphyletic groups. For example, sharks and whales could be classified in the same group because they have similar shapes, anatomical features, and habitats. However, their similarities reflect convergent evolution, not common ancestry.

| Traits:<br>Organism | Jaws | Lungs | Amniotic<br>membrane | Hair | No tail | Bipedal |
|---------------------|------|-------|----------------------|------|---------|---------|
| Lamprey             | 0    | 0     | 0                    | 0    | 0       | 0       |
| Shark               | 1    | 0     | 0                    | 0    | 0       | 0       |
| Salamander          | 1    | 1     | 0                    | 0    | 0       | 0       |
| Lizard              | 1    | 1     | 1                    | 0    | 0       | 0       |
| Tiger               | 1    | 1     | 1                    | 1    | 0       | 0       |
| Gorilla             | 1    | 1     | 1                    | 1    | 1       | 0       |
| Human               | 1    | 1     | 1                    | 1    | 1       | 1       |

**FIGURE 32.7**

**A cladogram.** Morphological data for a group of seven vertebrates is tabulated. A “1” indicates the presence of a trait, or derived character, and a “0” indicates the absence of the trait. A tree, or cladogram, diagrams the proposed evolutionary relationships among the organisms based on the presence of derived characters. The derived characters between the cladogram branch points are shared by all organisms above the branch point and are not present in any below it. The outgroup, in this case the lamprey, does not possess any of the derived characters.

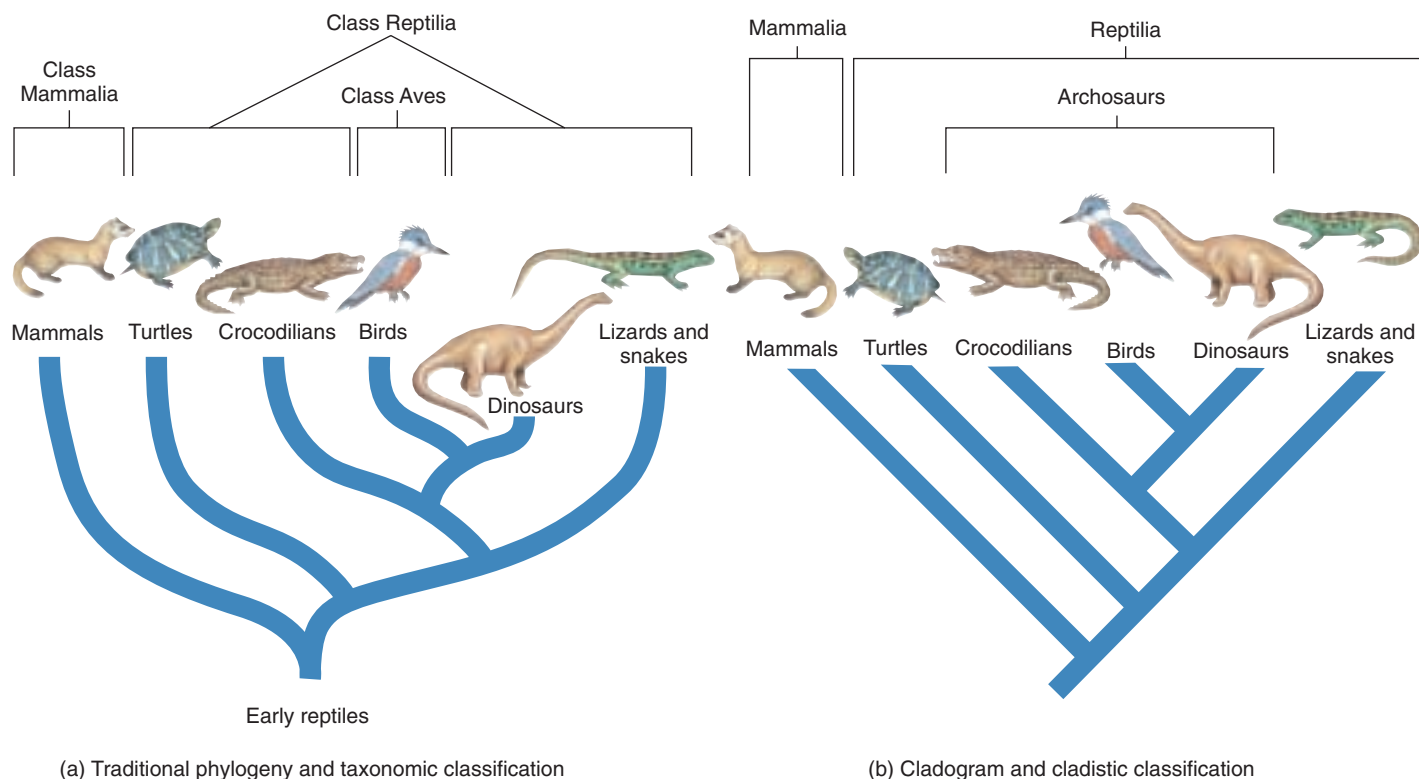


**gram** (figure 32.7), a branching diagram that represents the phylogeny.

In traditional phylogenies, proposed ancestors will often be indicated at the nodes between branches, and the lengths of branches correspond to evolutionary time, with extinct groups having shorter branches. In contrast, cladograms are not true family trees in that they do not identify ancestors, and the branch lengths do not reflect evolutionary time (see figure 32.6). Instead, they convey comparative information about relative relationships. Organisms that are closer together on a cladogram simply share a more recent common ancestor than those that are farther apart. Because the analysis is comparative, it is necessary to have something to anchor the comparison to, some solid ground against which the comparisons can be made. To achieve this, each cladogram must contain an **outgroup**, a rather different organism (but not too different) to serve as a baseline for comparisons among the other organisms being evaluated, the **ingroup**. For example, in figure 32.7, the lamprey is the outgroup to the clade of animals that have jaws.

Cladistics is a relatively new approach in biology and has become popular among students of evolution. This is because it does a very good job of portraying the order in which a series of evolutionary events have occurred. The great strength of a cladogram is that it can be completely objective. In fact, most cladistic analyses involve many characters, and computers are required to make the comparisons.

Sometime it is necessary to “weight” characters, or take into account the variation in the “strength” of a character, such as the size or location of a fin or the effectiveness of a lung. To reduce a systematist’s bias even more, many analyses will be run through the computer with the traits weighted differently each time. Under this procedure, several different cladograms will be constructed, the goal being to choose the one that is the most **parsimonious**, or simplest and thus most likely. Reflecting the importance of evolutionary processes to all fields of biology, most taxonomy today includes at least some element of cladistic analysis.



**FIGURE 32.8**

**Traditional and cladistic interpretations of vertebrate classification.** Traditional and cladistic taxonomic analyses of the same set of data often produce different results: in these two classifications of vertebrates, notice particularly the placement of the birds. (a) In the traditional analysis, key characteristics such as feathers and hollow bones are weighted more heavily than others, placing the birds in their own group and the reptiles in a paraphyletic group. (b) Cladistic analysis gives equal weight to these and many other characters and places birds in the same grouping with crocodiles, reflecting the close evolutionary relationship between the two. Also, in the traditional phylogeny, the branch leading to the dinosaurs is shorter because the length corresponds to evolutionary time. In cladograms, branch lengths do not correspond to evolutionary time.

## Traditional Taxonomy

Weighting characters lies at the core of **traditional taxonomy**. In this approach, taxa are assigned based on a vast amount of information about the morphology and biology of the organism gathered over a long period of time. Traditional taxonomists consider both the common descent and amount of adaptive evolutionary change when grouping organisms. The large amount of information used by traditional taxonomists permits a knowledgeable weighting of characters according to their biological significance. In traditional taxonomy, the full observational power and judgment of the biologist is brought to bear—and also any biases he or she may have. For example, in classifying the terrestrial vertebrates, traditional taxonomists place birds in their own class (Aves), giving great weight to the characters that made powered flight possible, such as feathers. However, cladists (figure 32.8) lumps birds in among the reptiles with crocodiles. This accurately reflects their true an-

cestry but ignores the immense evolutionary impact of a derived character such as feathers.

Overall, classifications based on traditional taxonomy are information-rich, while classifications based on cladograms need not be. Traditional taxonomy is often used when a great deal of information is available to guide character weighting, while cladistics is a good approach when little information is available about how the character affects the life of the organism. DNA sequence comparisons, for example, lend themselves well to cladistics—you have a great many derived characters (DNA sequence differences) but little or no idea of what impact the sequence differences have on the organism.

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**A phylogeny may be represented as a cladogram based on the order in which groups evolved. Traditional taxonomists weight characters according to assumed importance.**

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## 32.3 All living organisms are grouped into one of a few major categories.

### The Kingdoms of Life

The earliest classification systems recognized only two kingdoms of living things: animals and plants (figure 32.9*a*). But as biologists discovered microorganisms and learned more about other organisms, they added kingdoms in recognition of fundamental differences discovered among organisms (figure 32.9*b*). Most biologists now use a six-kingdom system first proposed by Carl Woese of the University of Illinois (figure 32.9*c*).

In this system, four kingdoms consist of eukaryotic organisms. The two most familiar kingdoms, **Animalia** and **Plantae**, contain only organisms that are multicellular during most of their life cycle. The kingdom **Fungi** contains multicellular forms and single-celled yeasts, which are thought to have multicellular ancestors. Fundamental differences divide these three kingdoms. Plants are mainly stationary, but some have motile sperm; fungi have no motile cells; animals are mainly motile. Animals ingest their food, plants manufacture it, and fungi digest it by means of secreted extracellular enzymes. Each of these kingdoms probably evolved from a different single-celled ancestor.

The large number of unicellular eukaryotes are arbitrarily grouped into a single kingdom called **Protista** (see chapter 35). This kingdom includes the algae, all of which are unicellular during parts of their life cycle.

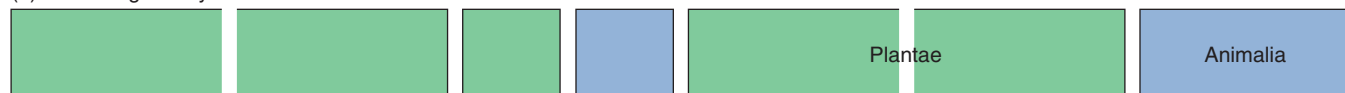
The remaining two kingdoms, **Archaeobacteria** and **Eubacteria**, consist of prokaryotic organisms, which are vastly different from all other living things (see chapter 34). Archaeobacteria are a diverse group including the methanogens and extreme thermophiles, and differ from the other bacteria, members of the kingdom Eubacteria.

### Domains

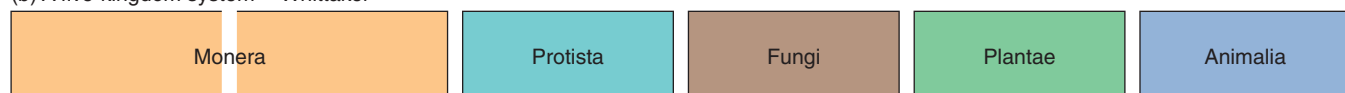
As biologists have learned more about the archaeobacteria, it has become increasingly clear that this ancient group is very different from all other organisms. When the full genomic DNA sequences of an archaeobacterium and a eubacterium were first compared in 1996, the differences proved striking. Archaeobacteria are as different from eubacteria as eubacteria are from eukaryotes. Recognizing this, biologists are increasingly adopting a classification of living organisms that recognizes three **domains**, a taxonomic level higher than kingdom (figure 32.9*d*). Archaeobacteria are in one domain, eubacteria in a second, and eukaryotes in the third.

Living organisms are grouped into three general categories called domains. One of the domains, the eukaryotes, is subdivided into four kingdoms: protists, fungi, plants, and animals.

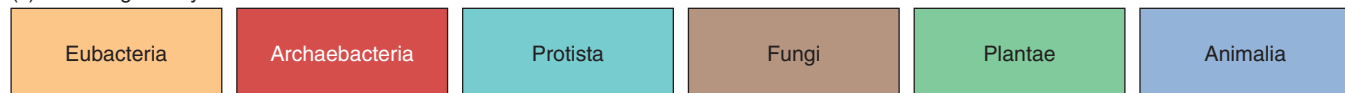
(a) A two-kingdom system—Linnaeus



(b) A five-kingdom system—Whittaker



(c) A six-kingdom system—Woese



(d) A three-domain system—Woese



FIGURE 32.9

**Different approaches to classifying living organisms.** (a) Linnaeus popularized a two-kingdom approach, in which the fungi and the photosynthetic protists were classified as plants, and the nonphotosynthetic protists as animals; when bacteria were described, they too were considered plants. (b) Whittaker in 1969 proposed a five-kingdom system that soon became widely accepted. (c) Woese has championed splitting the bacteria into two kingdoms for a total of six kingdoms, or even assigning them separate domains (d).

## Domain Archaea (Archaeobacteria)

The term *archaeobacteria* (Greek, *archaio*, ancient) refers to the ancient origin of this group of bacteria, which seem to have diverged very early from the eubacteria (figure 32.10). This conclusion comes largely from comparisons of genes that encode ribosomal RNAs. The last several years have seen an explosion of DNA sequence information from microorganisms, information which paints a more complex picture. It had been thought that by sequencing numerous microbes we could eventually come up with an accurate picture of the phylogeny of the earliest organisms on earth. The new whole-genome DNA sequence data described in chapter 19 tells us that it will not be that simple. Comparing whole-genome sequences leads evolutionary biologists to a variety of trees, some of which contradict each other. It appears that during their early evolution microorganisms have swapped genetic information, making constructing phylogenetic trees very difficult.

As an example of the problem, we can look at *Thermotoga*, a thermophile found on Volcano Island off Italy. The sequence of one of its RNAs places it squarely within the eubacteria near an ancient microbe called *Aquifex*. Recent DNA sequencing, however, fails to support any consistent relationship between the two microbes. There is disagreement as to the serious effect of gene swapping on the ability of evolutionary biologists to provide accurate phylogenies from molecular data. For now, we will provisionally accept the tree presented in figure 32.10. Over the next few years we can expect to see considerable change in accepted viewpoints as more and more data is brought to bear.

Today, archaeobacteria inhabit some of the most extreme environments on earth. Though a diverse group, all archaeobacteria share certain key characteristics (table 32.1). Their cell walls lack peptidoglycan (an important component of the cell walls of eubacteria), the lipids in the cell membranes of archaeobacteria have a different structure than those in all other organisms, and archaeobacteria have distinctive ribosomal RNA sequences. Some of their genes possess introns, unlike those of other bacteria.

The archaeobacteria are grouped into three general categories, methanogens, extremophiles, and nonextreme archaeobacteria, based primarily on the environments in which they live or their specialized metabolic pathways.

**Methanogens** obtain their energy by using hydrogen gas ( $H_2$ ) to reduce carbon dioxide ( $CO_2$ ) to methane gas ( $CH_4$ ). They are strict anaerobes, poisoned by even traces of oxygen. They live in swamps, marshes, and the intestines of mammals. Methanogens release about 2 billion tons of methane gas into the atmosphere each year.

**Extremophiles** are able to grow under conditions that seem extreme to us.

*Thermophiles* (“heat lovers”) live in very hot places, typically from 60° to 80°C. Many thermophiles are autotrophs and have metabolisms based on sulfur. Some

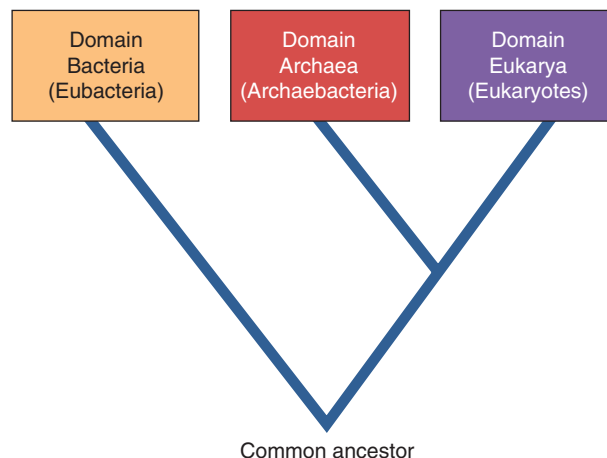


FIGURE 32.10

**An evolutionary relationship among the three domains.**

Eubacteria are thought to have diverged early from the evolutionary line that gave rise to the archaeobacteria and eukaryotes.

thermophilic archaeobacteria form the basis of food webs around deep-sea thermal vents where they must withstand extreme temperatures and pressures. Other types, like *Sulfolobus*, inhabit the hot sulfur springs of Yellowstone National Park at 70° to 75°C. The recently described *Pyrolobus fumarii* holds the current record for heat stability, with a 106°C temperature optimum and 113°C maximum—it is so heat tolerant that it is not killed by a one-hour treatment in an autoclave (121°C)!

*Halophiles* (“salt lovers”) live in very salty places like the Great Salt Lake in Utah, Mono Lake in California, and the Dead Sea in Israel. Whereas the salinity of seawater is around 3%, these bacteria thrive in, and indeed require, water with a salinity of 15 to 20%.

*pH-tolerant* archaeobacteria grow in highly acidic (pH = 0.7) and very basic (pH = 11) environments.

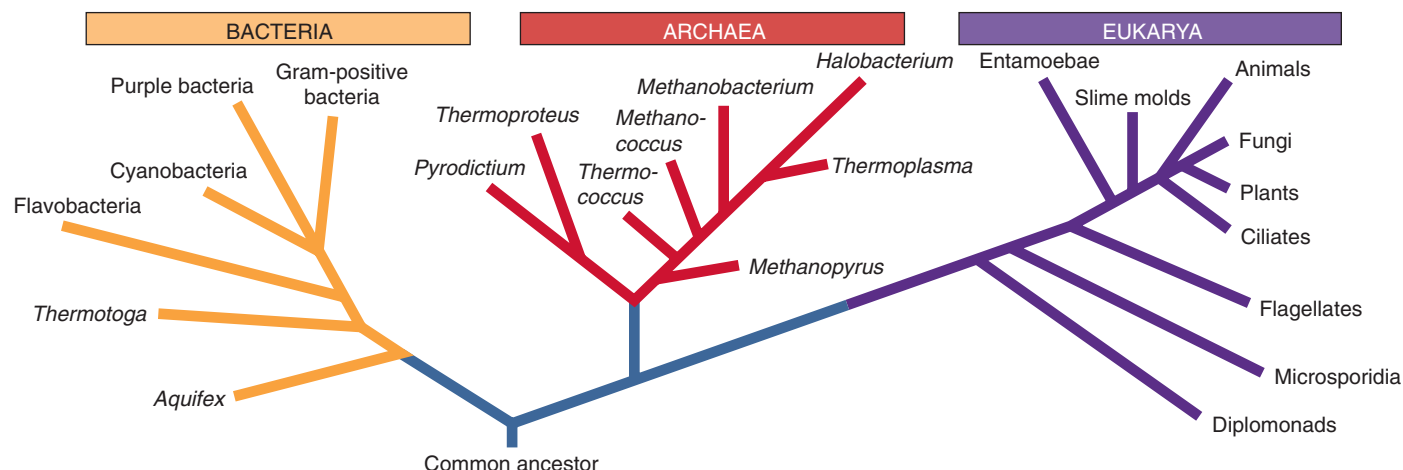
*Pressure-tolerant* archaeobacteria have been isolated from ocean depths that require at least 300 atmospheres of pressure to survive, and tolerate up to 800 atmospheres!

**Nonextreme archaeobacteria** grow in the same environments eubacteria do. As the genomes of archaeobacteria have become better known, microbiologists have been able to identify **signature sequences** of DNA present in all archaeobacteria and in no other organisms. When samples from soil or seawater are tested for genes matching these signal sequences, many of the bacteria living there prove to be archaeobacteria. Clearly, archaeobacteria are not restricted to extreme habitats, as microbiologists used to think.

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**Archaeobacteria are poorly understood bacteria that inhabit diverse environments, some of them extreme.**

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**FIGURE 32.11**

**A tree of life.** This phylogeny, prepared from rRNA analyses, shows the evolutionary relationships among the three domains. The base of the tree was determined by examining genes that are duplicated in all three domains, the duplication presumably having occurred in the common ancestor. When one of the duplicates is used to construct the tree, the other can be used to root it. This approach clearly indicates that the root of the tree is within the eubacterial domain. Archaeobacteria and eukaryotes diverged later and are more closely related to each other than either is to eubacteria.

## Domain Bacteria (Eubacteria)

The eubacteria are the most abundant organisms on earth. There are more living eubacteria in your mouth than there are mammals living on earth. Although too tiny to see with the unaided eye, eubacteria play critical roles throughout the biosphere. They extract from the air all the nitrogen used by organisms, and play key roles in cycling carbon and sulfur. Much of the world's photosynthesis is carried out by eubacteria. However, certain groups of eubacteria are also responsible for many forms of disease. Understanding their metabolism and genetics is a critical part of modern medicine.

There are many different kinds of eubacteria, and the evolutionary links between them are not well understood. While there is considerable disagreement among taxonomists about the details of bacterial classification, most recognize 12 to 15 major groups of eubacteria. Comparisons of the nucleotide sequences of ribosomal RNA (rRNA) molecules are beginning to reveal how these groups are related to one another and to the other two domains. One view of our current understanding of the "Tree of Life" is presented in figure 32.11. The oldest divergences represent the deepest rooted branches in the tree. The root of the tree is within the eubacterial domain. The archaeobacteria and eukaryotes are more closely related to each other than to eubacteria and are on a separate evolutionary branch of the tree, even though archaeobacteria and eubacteria are both prokaryotes.

**Eubacteria are as different from archaeobacteria as from eukaryotes.**

**Table 32.1 Features of the Domains of Life**

| Feature                                                      | Domain                |                   |                      |
|--------------------------------------------------------------|-----------------------|-------------------|----------------------|
|                                                              | Archaea               | Bacteria          | Eukarya              |
| Amino acid that initiates protein synthesis                  | Methionine            | Formyl-methionine | Methionine           |
| Introns                                                      | Present in some genes | Absent            | Present              |
| Membrane-bounded organelles                                  | Absent                | Absent            | Present              |
| Membrane lipid structure                                     | Branched              | Unbranched        | Unbranched           |
| Nuclear envelope                                             | Absent                | Absent            | Present              |
| Number of different RNA polymerases                          | Several               | One               | Several              |
| Peptidoglycan in cell wall                                   | Absent                | Present           | Absent               |
| Response to the antibiotics streptomycin and chloramphenicol | Growth not inhibited  | Growth inhibited  | Growth not inhibited |

## Domain Eukarya (Eukaryotes)

For at least 2 billion years, bacteria ruled the earth. No other organisms existed to eat them or compete with them, and their tiny cells formed the world's oldest fossils. The third great domain of life, the eukaryotes, appear in the fossil record much later, only about 1.5 billion years ago. Metabolically, eukaryotes are more uniform than bacteria. Each of the two domains of prokaryotic organisms has far more metabolic diversity than all eukaryotic organisms taken together. However, despite the metabolic similarity of eukaryotic cells, their structure and function allowed larger cell sizes and, eventually, multicellular life to evolve.

### Four Kingdoms of Eukaryotes

The first eukaryotes were unicellular organisms. A wide variety of unicellular eukaryotes exist today, grouped together in the kingdom Protista on the basis that they do not fit into any of the other three kingdoms of eukaryotes. Protists are a fascinating group containing many organisms of intense interest and great biological significance. They vary from the relatively simple, single-celled amoeba to multicellular organisms like kelp that can be 20 meters long.

Fungi, plants, and animals are largely multicellular kingdoms, each a distinct evolutionary line from a single-celled ancestor that would be classified in the kingdom Protista.

Because of the size and ecological dominance of plants, animals, and fungi, and because they are predominantly multicellular, we recognize them as kingdoms distinct from Protista, even though the amount of diversity among the protists is much greater than that within or between the fungi, plants, and animals.

### Symbiosis and the Origin of Eukaryotes

The hallmark of eukaryotes is complex cellular organization, highlighted by an extensive endomembrane system that subdivides the eukaryotic cell into functional compartments. Not all of these compartments, however, are derived from the endomembrane system. With few exceptions, all modern eukaryotic cells possess energy-producing organelles, the mitochondria, and some eukaryotic cells possess chloroplasts, which are energy-harvesting organelles. Mitochondria and chloroplasts are both believed to have entered early eukaryotic cells by a process called **endosymbiosis** (*endo*, inside). We discussed the theory of the endosymbiotic origin of mitochondria and chloroplasts in chapter 5; also see figure 32.12. Both organelles contain their own ribosomes, which are more similar to bacterial ribosomes than to eukaryotic cytoplasmic ribosomes. They divide independently of the cell and contain chromosomes similar to those in bacteria. Mitochondria are about the size of bacteria and contain DNA. Comparison of the nucleotide sequence of this DNA with that of a variety of organisms

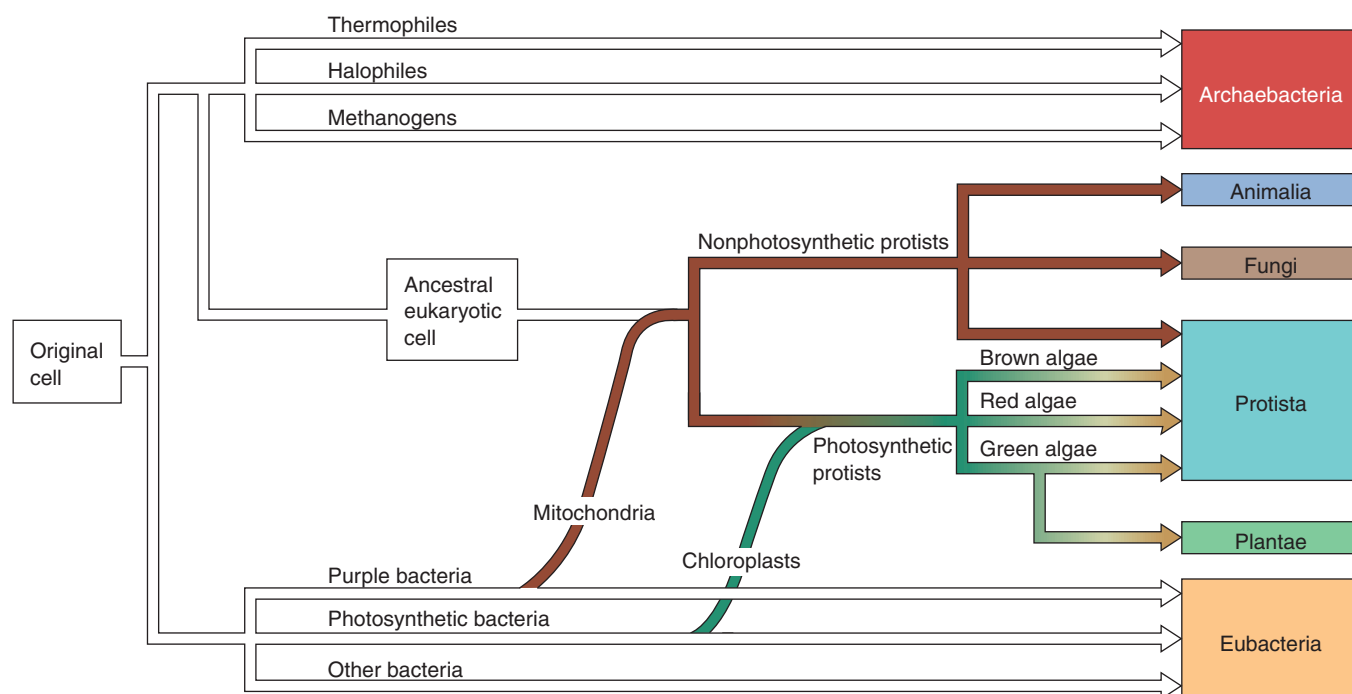


FIGURE 32.12

Diagram of the evolutionary relationships among the six kingdoms of organisms. The colored lines indicate symbiotic events.

indicates clearly that mitochondria are the descendants of purple bacteria that were incorporated into eukaryotic cells early in the history of the group. Chloroplasts are derived from cyanobacteria that became symbiotic in several groups of protists early in their history.

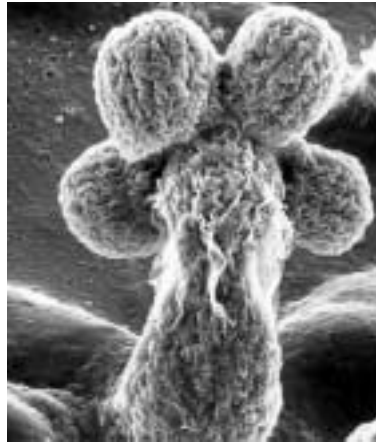
Some biologists suggest that basal bodies, centrioles, flagella, and cilia may have arisen from endosymbiotic spirochaete-like bacteria. Even today, so many bacteria and unicellular protists form symbiotic alliances that the incorporation of smaller organisms with desirable features into eukaryotic cells appears to be a relatively common process.

## Key Characteristics of Eukaryotes

**Multicellularity.** The unicellular body plan has been tremendously successful, with unicellular prokaryotes and eukaryotes constituting about half of the biomass on earth. Yet a single cell has limits. The evolution of multicellularity allowed organisms to deal with their environments in novel ways. Distinct types of cells, tissues, and organs can be differentiated within the complex bodies of multicellular organisms. With such a functional division within its body, a multicellular organism can do many things, like protect itself, resist drought efficiently, regulate its internal conditions, move about, seek mates and prey, and carry out other activities on a scale and with a complexity that would be impossible for its unicellular ancestors. With all these advantages, it is not surprising that multicellularity has arisen independently so many times.

True multicellularity, in which the activities of individual cells are coordinated and the cells themselves are in contact, occurs only in eukaryotes and is one of their major characteristics. The cell walls of bacteria occasionally adhere to one another, and bacterial cells may also be held together within a common sheath. Some bacteria form filaments, sheets, or three-dimensional aggregates (figure 32.13), but the individual cells remain independent of each other, reproducing and carrying on their metabolic functions and without coordinating with the other cells. Such bacteria are considered colonial, but none are truly multicellular. Many protists also form similar colonial aggregates of many cells with little differentiation or integration.

Other protists—the red, brown, and green algae, for example—have independently attained multicellularity. Cer-



**FIGURE 32.13**  
**Colonial bacteria.** No bacteria are truly multicellular. These gliding bacteria, *Stigmatella aurantiaca*, have aggregated into a structure called a fruiting body; within, some cells transform into spores.

tain forms of multicellular green algae were ancestors of the plants (see chapters 35 and 37), and, like the other photosynthetic protists, are considered plants in some classification schemes. In the system adopted here, the plant kingdom includes only multicellular land plants, a group that arose from a single ancestor in terrestrial habitats and that has a unique set of characteristics. Aquatic plants are recent derivatives.

Fungi and animals arose from unicellular protist ancestors with different characteristics. As we will see in subsequent chapters, the groups that seem to have given rise to each of these kingdoms are still in existence.

**Sexuality.** Another major characteristic of eukaryotic organisms as a group is sexuality. Although some interchange of genetic material occurs in bacteria (see chapter 34), it is certainly not a regular, predictable mechanism in the same sense that sex is in eukaryotes. The sexual cycle characteristic of eukaryotes alternates between

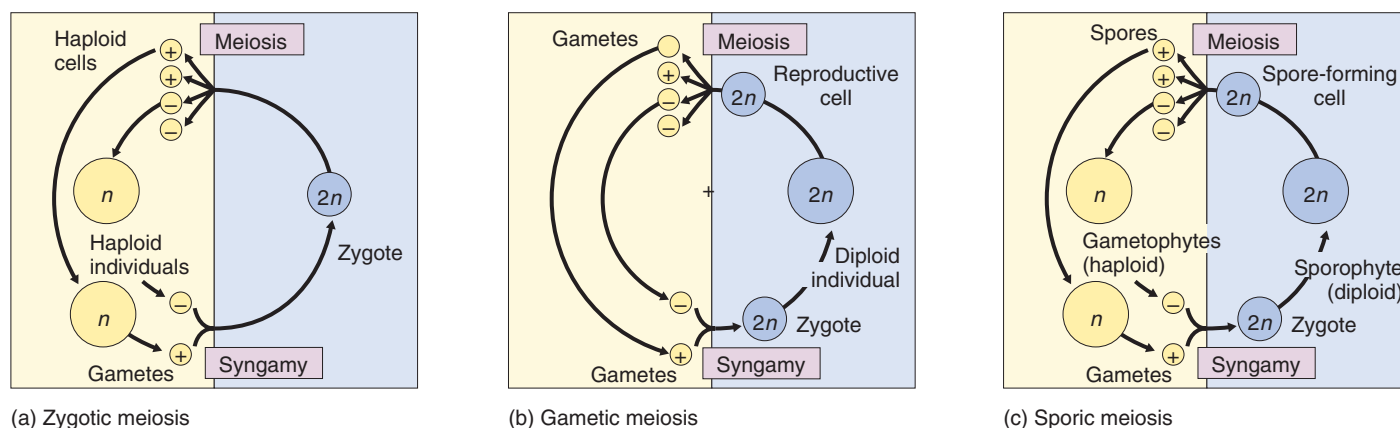
**syngamy**, the union of male and female gametes producing a cell with two sets of chromosomes, and meiosis, cell division producing daughter cells with one set of chromosomes. This cycle differs sharply from any exchange of genetic material found in bacteria.

Except for gametes, the cells of most animals and plants are diploid, containing two sets of chromosomes, during some part of their life cycle. A few eukaryotes complete their life cycle in the haploid condition, with only one set of chromosomes in each cell. As we have seen, in diploid cells, one set of chromosomes comes from the male parent and one from the female parent. These chromosomes segregate during meiosis. Because crossing over frequently occurs during meiosis (see chapter 12), no two products of a single meiotic event are ever identical. As a result, the offspring of sexual, eukaryotic organisms vary widely, thus providing the raw material for evolution.

Sexual reproduction, with its regular alternation between syngamy and meiosis, produces genetic variation. Sexual organisms can adapt to the demands of their environments because they produce a variety of progeny.

In many of the unicellular phyla of protists, sexual reproduction occurs only occasionally. Meiosis may have originally evolved as a means of repairing damage to DNA, producing an organism better adapted to survive changing environmental conditions. The first eukaryotes were probably haploid. Diploids seem to have arisen on a number of separate occasions by the fusion of haploid cells, which then eventually divided by meiosis.

Key:  Haploid  Diploid



**FIGURE 32.14**  
Diagrams of the three major kinds of life cycles in eukaryotes. (a) Zygotic meiosis, (b) gametic meiosis, and (c) sporic meiosis.

## Eukaryotic Life Cycles

Eukaryotes are characterized by three major types of life cycles (figure 32.14):

1. In the simplest cycle, found in algae, the zygote is the only diploid cell. Such a life cycle is said to be characterized by **zygotic meiosis**, because the zygote immediately undergoes meiosis.
2. In most animals, the gametes are the only haploid cells. Animals exhibit **gametic meiosis**, meiosis producing gametes which fuse, giving rise to a zygote.
3. Plants show a regular *alternation of generations* between a multicellular haploid phase and a multicellular diploid phase. The diploid phase undergoes meiosis producing haploid spores that give rise to the haploid phase, and the haploid phase produces

gametes that fuse to form the zygote. The zygote is the first cell of the multicellular diploid phase. This kind of life cycle is characterized by **alternation of generations** and has **sporic meiosis**.

The characteristics of the six kingdoms are outlined in table 32.2.

Eukaryotic cells acquired mitochondria and chloroplasts by endosymbiosis, mitochondria being derived from purple bacteria and chloroplasts from cyanobacteria. The complex differentiation that we associate with advanced life-forms depends on multicellularity and sexuality, which must have been highly advantageous to have evolved independently so often.

**Table 32.2** Characteristics of the Six Kingdoms

| Kingdom                        | Cell Type   | Nuclear Envelope | Mitochondria      | Chloroplasts                                  | Cell Wall                                      |
|--------------------------------|-------------|------------------|-------------------|-----------------------------------------------|------------------------------------------------|
| Archaeobacteria and Eubacteria | Prokaryotic | Absent           | Absent            | None (photosynthetic membranes in some types) | Noncellulose (polysaccharide plus amino acids) |
| Protista                       | Eukaryotic  | Present          | Present or absent | Present (some forms)                          | Present in some forms, various types           |
| Fungi                          | Eukaryotic  | Present          | Present or absent | Absent                                        | Chitin and other noncellulose polysaccharides  |
| Plantae                        | Eukaryotic  | Present          | Present           | Present                                       | Cellulose and other polysaccharides            |
| Animalia                       | Eukaryotic  | Present          | Present           | Absent                                        | Absent                                         |

## Viruses: A Special Case

Viruses pose a challenge to biologists as they do not possess the fundamental characteristics of living organisms. Viruses appear to be fragments of nucleic acids originally derived from the genome of a living cell. Unlike all living organisms, viruses are acellular—that is, they are not cells and do not consist of cells. They do not have a metabolism; in other words, viruses do not carry out photosynthesis, cellular respiration, or fermentation. The one characteristic of life that they do display is reproduction, which they do by hijacking the metabolism of living cells.

Viruses thus present a special classification problem. Because they are not organisms, we cannot logically place them in any of the kingdoms. Viruses are really just complicated associations of molecules, bits of nucleic acids usually surrounded by a protein coat. But, despite their simplicity, viruses are able to invade cells and direct the genetic machinery of these cells to manufacture more of the molecules that make up the virus (figure 32.15). Viruses can infect organisms at all taxonomic levels.

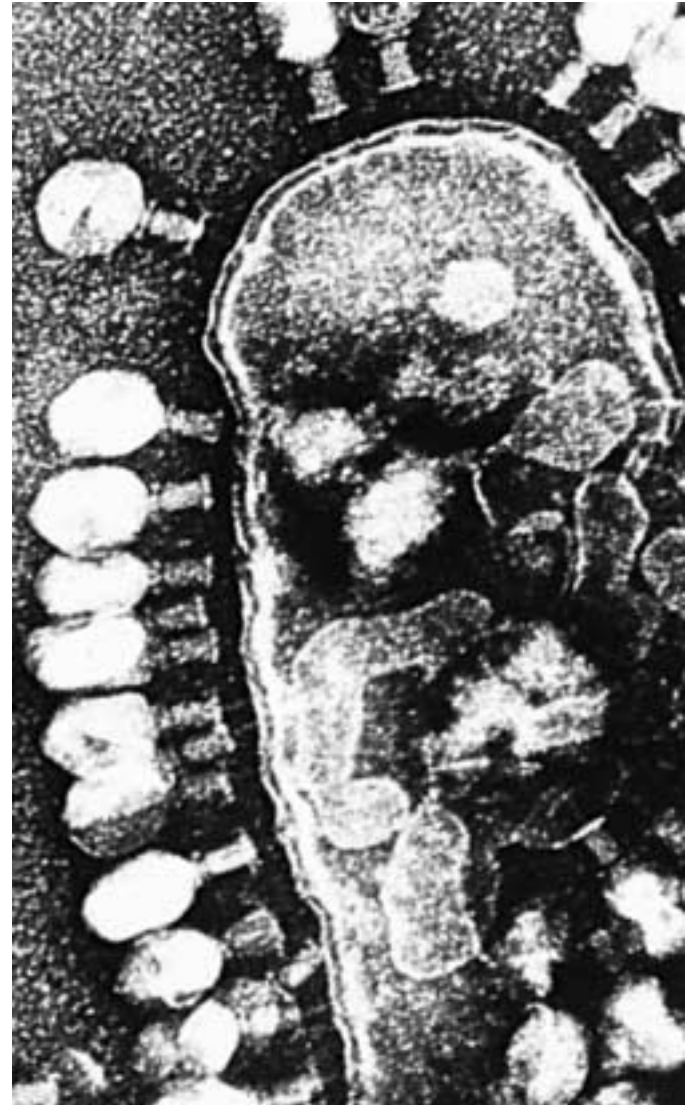
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**Viruses are not organisms and are not classified in the kingdoms of life.**

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**FIGURE 32.15**

**Viruses are cell parasites.** In this micrograph, several T4 bacteriophages (viruses) are attacking an *Escherichia coli* bacterium. Some of the viruses have already entered the cell and are reproducing within it.



**Table 32.2** Characteristics of the Six Kingdoms

| Means of Genetic Recombination, if Present | Mode of Nutrition                                             | Motility                                                              | Multicellularity      | Nervous System                                            |
|--------------------------------------------|---------------------------------------------------------------|-----------------------------------------------------------------------|-----------------------|-----------------------------------------------------------|
| Conjugation, transduction, transformation  | Autotrophic (chemosynthetic, photosynthetic) or heterotrophic | Bacterial flagella, gliding or nonmotile                              | Absent                | None                                                      |
| Fertilization and meiosis                  | Photosynthetic or heterotrophic, or combination of both       | 9 + 2 cilia and flagella; amoeboid, contractile fibrils               | Absent in most forms  | Primitive mechanisms for conducting stimuli in some forms |
| Fertilization and meiosis                  | Absorption                                                    | Nonmotile                                                             | Present in most forms | None                                                      |
| Fertilization and meiosis                  | Photosynthetic chlorophylls <i>a</i> and <i>b</i>             | None in most forms, 9 + 2 cilia and flagella in gametes of some forms | Present in all forms  | None                                                      |
| Fertilization and meiosis                  | Digestion                                                     | 9 + 2 cilia and flagella, contractile fibrils                         | Present in all forms  | Present, often complex                                    |



## Summary

## Questions

## Media Resources

**32.1 Biologists name organisms in a systematic way.**

- Biologists give every species a two-part (binomial) name that consists of the name of its genus plus a distinctive specific epithet.
- In the hierarchical system of classification used in biology, genera are grouped into families, families into orders, orders into classes, classes into phyla, and phyla into kingdoms.
- There are perhaps 10 million species of plants, animals, fungi, and eukaryotic microorganisms, but only about 1.5 million of them have been assigned names. About 15% of the total number of species are marine; the remainder are mostly terrestrial.

1. What was the polynomial system? Why didn't this system become the standard for naming particular species?
2. From the most specific to the most general, what are the names of the groups in the hierarchical taxonomic system? Which two are given special consideration in the way in which they are printed? What are these distinctions?



- Hierarchies



- Book Reviews:  
*Ship Fever* by Barrett

**32.2 Scientists construct phylogenies to understand the evolutionary relationships among organisms.**

- Taxonomists may use different approaches to classify organisms.
- Cladistic systems of classification arrange organisms according to evolutionary relatedness based on the presence of shared, derived traits.
- Traditional taxonomy classifies organisms based on large amounts of information, giving due weight to the evolutionary significance of certain characters.

3. What types of features are emphasized in a cladistic classification system? What is the resulting relationship of organisms that are classified in this manner?
4. What does it mean when characters are weighted?

**32.3 All living organisms are grouped into one of a few major categories.**

- A fundamental division among organisms is between prokaryotes, which lack a true nucleus, and eukaryotes, which have a true nucleus and several membrane-bound organelles.
- Prokaryotes, or bacteria, are assigned to two quite different kingdoms, Archaeobacteria and Eubacteria.
- The eukaryotic kingdoms are more closely related than are the two kingdoms of prokaryotes. Many distinctive evolutionary lines of unicellular eukaryotes exist, most are in the Protista kingdom.
- Three of the major evolutionary lines of eukaryotic organisms that consist principally or entirely of multicellular organisms are recognized as separate kingdoms: Plantae, Animalia, and Fungi.
- True multicellularity and sexuality are found only among eukaryotes. Multicellularity confers the advantages of functional specialization. Sexuality permits genetic variation among descendants.
- Viruses are not organisms and are not included in the classification of organisms. They are self-replicating portions of the genomes of organisms.

5. Is there a greater fundamental difference between plants and animals or between prokaryotes and eukaryotes? Explain.



- Art Activity:  
Organism  
Classification

6. From which of the four eukaryotic kingdoms have the other three evolved?



- Kingdoms
- Three Domains
- Phylogeny

7. What is the apparent origin of the organelles found in almost all eukaryotes?



- Book Review:  
*Throwim Way Leg*  
by Flannery

8. What defines if a collection of cells is truly multicellular? Did multicellularity arise once or many times in the evolutionary process? What advantages do multicellular organisms have over unicellular ones?

9. What are the three major types of life cycles in eukaryotes? Describe the major events of each.

# 33

## Viruses

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### Concept Outline

#### 33.1 Viruses are strands of nucleic acid encased within a protein coat.

**The Discovery of Viruses.** The first virus to be isolated proved to consist of two chemicals, one a protein and the other a nucleic acid.

**The Nature of Viruses.** Viruses occur in all organisms. Able to reproduce only within living cells, viruses are not themselves alive.

#### 33.2 Bacterial viruses exhibit two sorts of reproductive cycles.

**Bacteriophages.** Some bacterial viruses, called bacteriophages, rupture the cells they infect, while others integrate themselves into the bacterial chromosome to become a stable part of the bacterial genome.

**Cell Transformation and Phage Conversion.** Integrated bacteriophages sometimes modify the host bacterium they infect.

#### 33.3 HIV is a complex animal virus.

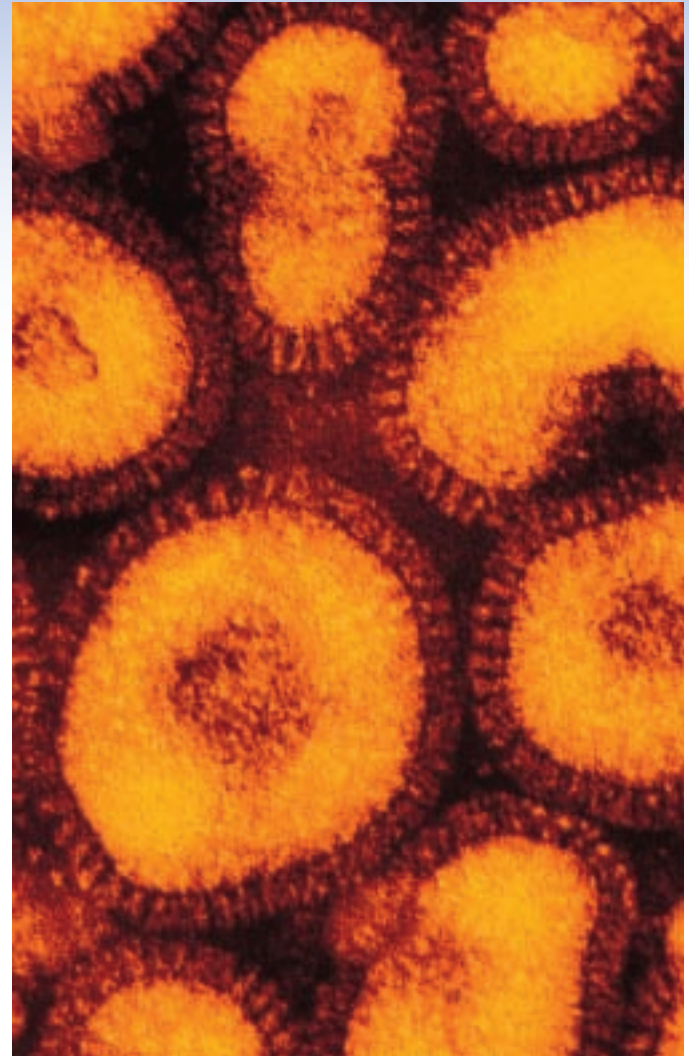
**AIDS.** The animal virus HIV infects certain key cells of the immune system, destroying the ability of the body to defend itself from cancer and disease. The HIV infection cycle is typically a lytic cycle, in which the HIV RNA first directs the production of a corresponding DNA, and this DNA then directs the production of progeny virus particles.

**The Future of HIV Treatment.** Combination therapies and chemokines offer promising avenues of AIDS therapy.

#### 33.4 Nonliving infectious agents are responsible for many human diseases.

**Disease Viruses.** Some of the most serious viral diseases have only recently infected human populations, the result of transfer from other hosts.

**Prions and Viroids.** In some instances, proteins and “naked” RNA molecules can also transmit diseases.



**FIGURE 33.1**

**Influenza viruses.** A virus has been referred to as “a piece of bad news wrapped up in a protein.” How can something as “simple” as a virus have such a profound effect on living organisms? (30,000 $\times$ )

We start our exploration of the diversity of life with viruses. Viruses are genetic elements enclosed in protein and are not considered to be organisms, as they cannot reproduce independently. Because of their disease-producing potential, viruses are important biological entities. The virus particles you see in figure 33.1 produce the important disease influenza. Other viruses cause AIDS, polio, flu, and some can lead to cancer. Many scientists have attempted to unravel the nature of viral genes and how they work. For more than four decades, viral studies have been thoroughly intertwined with those of genetics and molecular biology. In the future, it is expected that viruses will be one of the principal tools used to experimentally carry genes from one organism to another. Already, viruses are being employed in the treatment of human genetic diseases.

## 33.1 Viruses are strands of nucleic acid encased within a protein coat.

### The Discovery of Viruses

The border between the living and the nonliving is very clear to a biologist. Living organisms are cellular and able to grow and reproduce independently, guided by information encoded within DNA. The simplest creatures living on earth today that satisfy these criteria are bacteria. Even simpler than bacteria are viruses. As you will learn in this section, viruses are so simple that they do not satisfy the criteria for “living.”

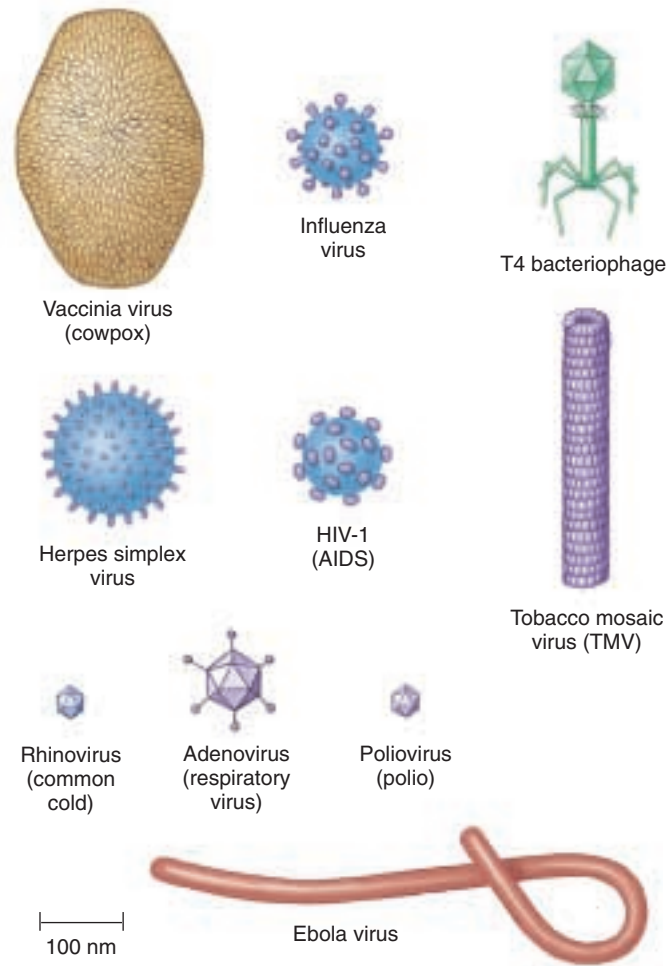
**Viruses** possess only a portion of the properties of organisms. Viruses are literally “parasitic” chemicals, segments of DNA or RNA wrapped in a protein coat. They cannot reproduce on their own, and for this reason they are not considered alive by biologists. They can, however, reproduce within cells, often with disastrous results to the host organism. Earlier theories that viruses represent a kind of halfway point between life and nonlife have largely been abandoned. Instead, viruses are now viewed as detached fragments of the genomes of organisms due to the high degree of similarity found among some viral and eukaryotic genes.

Viruses vary greatly in appearance and size. The smallest are only about 17 nanometers in diameter, and the largest are up to 1000 nanometers (1 micrometer) in their greatest dimension (figure 33.2). The largest viruses are barely visible with a light microscope, but viral morphology is best revealed using the electron microscope. Viruses are so small that they are comparable to molecules in size; a hydrogen atom is about 0.1 nanometer in diameter, and a large protein molecule is several hundred nanometers in its greatest dimension.

Biologists first began to suspect the existence of viruses near the end of the nineteenth century. European scientists attempting to isolate the infectious agent responsible for hoof-and-mouth disease in cattle concluded that it was smaller than a bacterium. Investigating the agent further, the scientists found that it could not multiply in solution—it could only reproduce itself within living host cells that it infected. The infecting agents were called viruses.

The true nature of viruses was discovered in 1933, when the biologist Wendell Stanley prepared an extract of a plant virus called *tobacco mosaic virus* (TMV) and attempted to purify it. To his great surprise, the purified TMV preparation precipitated (that is, separated from solution) in the form of crystals. This was surprising because precipitation is something that only chemicals do—the TMV virus was acting like a chemical off the shelf rather than an organism. Stanley concluded that TMV is best regarded as just that—chemical matter rather than a living organism.

Within a few years, scientists disassembled the TMV virus and found that Stanley was right. TMV was not cellu-



**FIGURE 33.2**

**Viral diversity.** A sample of the extensive diversity and small size of viruses is depicted. At the scale these viruses are shown, a human hair would be nearly 8 meters thick.

lar but rather chemical. Each particle of TMV virus is in fact a mixture of two chemicals: RNA and protein. The TMV virus has the structure of a Twinkie, a tube made of an RNA core surrounded by a coat of protein. Later workers were able to separate the RNA from the protein and purify and store each chemical. Then, when they reassembled the two components, the reconstructed TMV particles were fully able to infect healthy tobacco plants and so clearly *were* the virus itself, not merely chemicals derived from it. Further experiments carried out on other viruses yielded similar results.

**Viruses are chemical assemblies that can infect cells and replicate within them. They are not alive.**

# The Nature of Viruses

## Viral Structure

All viruses have the same basic structure: a core of nucleic acid surrounded by protein. Individual viruses contain only a single type of nucleic acid, either DNA or RNA. The DNA or RNA genome may be linear or circular, and single-stranded or double-stranded. Viruses are frequently classified by the nature of their genomes. RNA-based viruses are known as **retroviruses**.

Nearly all viruses form a protein sheath, or **capsid**, around their nucleic acid core. The capsid is composed of one to a few different protein molecules repeated many times (figure 33.3) In some viruses, specialized enzymes are stored within the capsid. Many animal viruses form an **envelope** around the capsid rich in proteins, lipids, and glycoprotein molecules. While some of the material of the envelope is derived from the host cell's membrane, the envelope does contain proteins derived from viral genes as well.

Viruses occur in virtually every kind of organism that has been investigated for their presence. However, each type of virus can replicate in only a very limited number of cell types. The suitable cells for a particular virus are collectively referred to as its **host range**. The size of the host range reflects the coevolved histories of the virus and its potential hosts. A recently discovered herpesvirus turned lethal when it expanded its host range from the African elephant to the Indian elephant, a situation made possible through cross-species contacts between elephants in zoos. Some viruses wreak havoc on the cells they infect; many others produce no disease or other outward sign of their infection. Still other viruses remain dormant for years until a specific signal triggers their expression. A given organism often has more than one kind of virus. This suggests that there may be many more kinds of viruses than there are kinds of organisms—perhaps millions of them. Only a few thousand viruses have been described at this point.

## Viral Replication

An infecting virus can be thought of as a set of instructions, not unlike a computer program. A computer's operation is directed by the instructions in its operating program, just as a cell is directed by DNA-encoded instructions. A new program can be introduced into the computer that will cause the computer to cease what it is doing and devote all of its energies to another activity, such as making copies of the introduced program. The

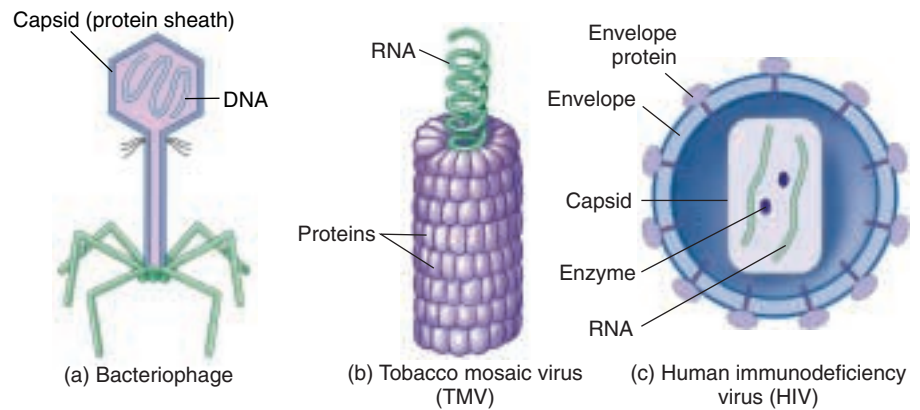


FIGURE 33.3

**The structure of a bacterial, plant, and animal virus.** (a) Bacterial viruses, called bacteriophages, often have a complex structure. (b) TMV infects plants and consists of 2130 identical protein molecules (*purple*) that form a cylindrical coat around the single strand of RNA (*green*). The RNA backbone determines the shape of the virus and is protected by the identical protein molecules packed tightly around it. (c) In the human immunodeficiency virus (HIV), the RNA core is held within a capsid that is encased by a protein envelope.

new program is not itself a computer and cannot make copies of itself when it is outside the computer, lying on the desk. The introduced program, like a virus, is simply a set of instructions.

Viruses can reproduce only when they enter cells and utilize the cellular machinery of their hosts. Viruses code their genes on a single type of nucleic acid, either DNA or RNA, but viruses lack ribosomes and the enzymes necessary for protein synthesis. Viruses are able to reproduce because their genes are translated into proteins by the cell's genetic machinery. These proteins lead to the production of more viruses.

## Viral Shape

Most viruses have an overall structure that is either **helical** or **isometric**. Helical viruses, such as the tobacco mosaic virus, have a rodlike or threadlike appearance. Isometric viruses have a roughly spherical shape whose geometry is revealed only under the highest magnification.

The only structural pattern found so far among isometric viruses is the **icosahedron**, a structure with 20 equilateral triangular facets, like the adenovirus shown in figure 33.2. Most viruses are icosahedral in basic structure. The icosahedron is the basic design of the geodesic dome. It is the most efficient symmetrical arrangement that linear subunits can take to form a shell with maximum internal capacity.

**Viruses occur in all organisms and can only reproduce within living cells. Most are icosahedral in structure.**

## 33.2 Bacterial viruses exhibit two sorts of reproductive cycles.

### Bacteriophages

**Bacteriophages** are viruses that infect bacteria. They are diverse both structurally and functionally, and are united solely by their occurrence in bacterial hosts. Many of these bacteriophages, called *phages* for short, are large and complex, with relatively large amounts of DNA and proteins. Some of them have been named as members of a “T” series (T1, T2, and so forth); others have been given different kinds of names. To illustrate the diversity of these viruses, T3 and T7 phages are icosahedral and have short tails. In contrast, the so-called T-even phages (T2, T4, and T6) have an icosahedral head, a capsid that consists primarily of three proteins, a connecting neck with a collar and long “whiskers,” a long tail, and a complex base plate (figure 33.4).

#### The Lytic Cycle

During the process of bacterial infection by T4 phage, at least one of the tail fibers of the phage—they are normally held near the phage head by the “whiskers”—contacts the lipoproteins of the host bacterial cell wall. The other tail fibers set the phage perpendicular to the surface of the bacterium and bring the base plate into contact with the cell surface. The tail contracts, and the tail tube passes through an opening that appears in the base plate, piercing the bacterial cell wall. The contents of the head, mostly DNA, are then injected into the host cytoplasm.

When a virus kills the infected host cell in which it is replicating, the reproductive cycle is referred to as a **lytic**

**cycle** (figure 33.5). The T-series bacteriophages are all **virulent viruses**, multiplying within infected cells and eventually lysing (rupturing) them. However, they vary considerably as to when they become virulent within their host cells.

#### The Lysogenic Cycle

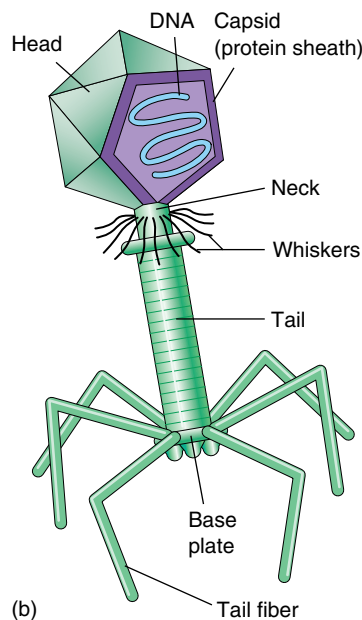
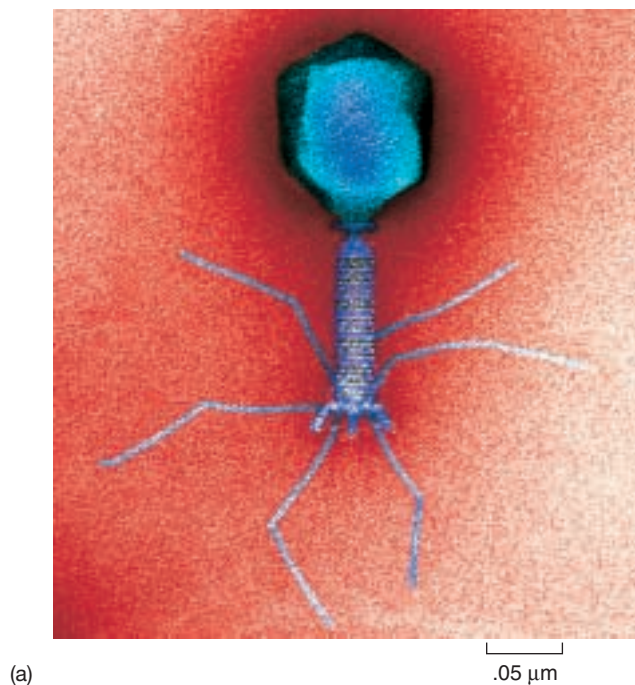
Many bacteriophages do not immediately kill the cells they infect, instead integrating their nucleic acid into the genome of the infected host cell. While residing there, it is called a **prophage**. Among the bacteriophages that do this is the lambda ( $\lambda$ ) phage of *Escherichia coli*. We know as much about this bacteriophage as we do about virtually any other biological particle; the complete sequence of its 48,502 bases has been determined. At least 23 proteins are associated with the development and maturation of lambda phage, and many other enzymes are involved in the integration of these viruses into the host genome.

The integration of a virus into a cellular genome is called **lysogeny**. At a later time, the prophage may exit the genome and initiate virus replication. This sort of reproductive cycle, involving a period of genome integration, is called a **lysogenic cycle**. Viruses that become stably integrated within the genome of their host cells are called **lysogenic viruses** or **temperate viruses**.

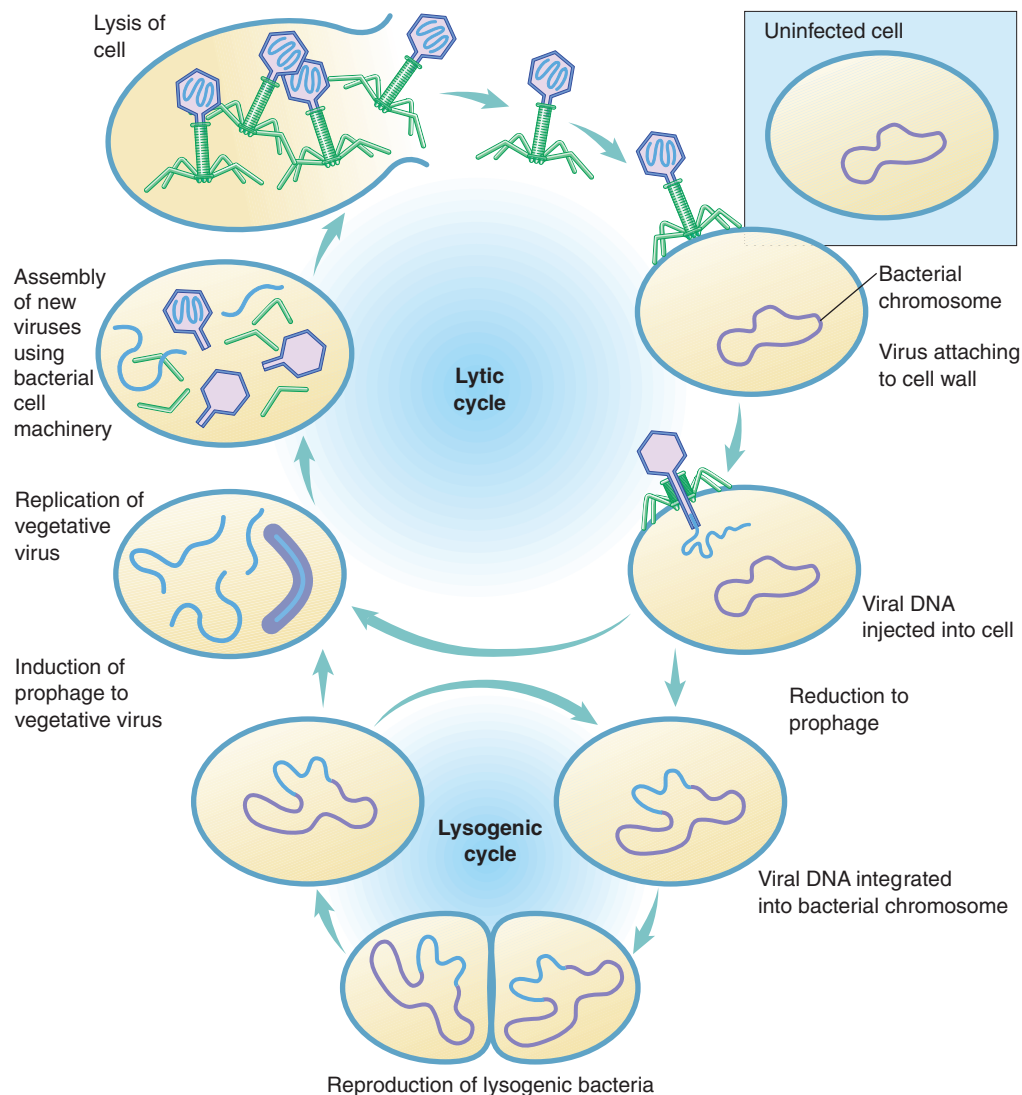
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**Bacteriophages are a diverse group of viruses that attack bacteria. Some kill their host in a lytic cycle; others integrate into the host's genome, initiating a lysogenic cycle.**

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**FIGURE 33.4**  
**A bacterial virus.**  
Bacteriophages exhibit a complex structure. (a) Electron micrograph and (b) diagram of the structure of a T4 bacteriophage.



**FIGURE 33.5**

**Lytic and lysogenic cycles of a bacteriophage.** In the lytic cycle, the bacteriophage exists as viral DNA free in the bacterial host cell's cytoplasm; the viral DNA directs the production of new viral particles by the host cell until the virus kills the cell by lysis. In the lysogenic cycle, the bacteriophage DNA is integrated into the large, circular DNA molecule of the host bacterium and is reproduced along with the host DNA as the bacterium replicates. It may continue to replicate and produce lysogenic bacteria or enter the lytic cycle and kill the cell. Bacteriophages are much smaller relative to their hosts than illustrated in this diagram.

## Cell Transformation and Phage Conversion

During the integrated portion of a lysogenic reproductive cycle, virus genes are often expressed. The RNA polymerase of the host cell reads the viral genes just as if they were host genes. Sometimes, expression of these genes has an important effect on the host cell, altering it in novel ways. The genetic alteration of a cell's genome by the introduction of foreign DNA is called **transformation**. When the foreign DNA is contributed by a bacterial virus, the alteration is called phage conversion.

### Phage Conversion of the Cholera-Causing Bacterium

An important example of this sort of phage conversion directed by viral genes is provided by the bacterium responsible for an often-fatal human disease. The disease-causing

bacteria *Vibrio cholerae* usually exists in a harmless form, but a second disease-causing, virulent form also occurs. In this latter form, the bacterium causes the deadly disease cholera, but how the bacteria changed from harmless to deadly was not known until recently. Research now shows that a bacteriophage that infects *V. cholerae* introduces into the host bacterial cell a gene that codes for the cholera toxin. This gene becomes incorporated into the bacterial chromosome, where it is translated along with the other host genes, thereby converting the benign bacterium to a disease-causing agent. The transfer occurs through bacterial pili (see chapter 34); in further experiments, mutant bacteria that did not have pili were resistant to infection by the bacteriophage. This discovery has important implications in efforts to develop vaccines against cholera, which have been unsuccessful up to this point.

**Bacteriophages convert *Vibrio cholerae* bacteria from harmless gut residents into disease-causing agents.**

### 33.3 HIV is a complex animal virus.

#### AIDS

A diverse array of viruses occur among animals. A good way to gain a general idea of what they are like is to look at one animal virus in detail. Here we will look at the virus responsible for a comparatively new and fatal viral disease, acquired immunodeficiency syndrome (AIDS). AIDS was first reported in the United States in 1981. It was not long before the infectious agent, a retrovirus called human immunodeficiency virus (HIV), was identified by laboratories in France and the United States. Study of HIV revealed it to be closely related to a chimpanzee virus, suggesting a recent host expansion to humans in central Africa from chimpanzees.

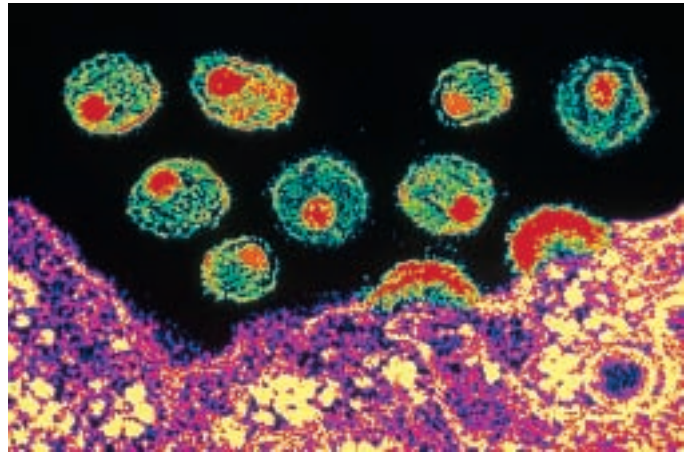
Infected humans have little resistance to infection, and nearly all of them eventually die of diseases that noninfected individuals easily ward off. Few who contract AIDS survive more than a few years untreated. The risk of HIV transmission from an infected individual to a healthy one in the course of day-to-day contact is essentially nonexistent. However, the transfer of body fluids, such as blood, semen, or vaginal fluid, or the use of nonsterile needles, between infected and healthy individuals poses a severe risk. In addition, HIV-infected mothers can pass the virus on to their unborn children during fetal development.

The incidence of AIDS is growing very rapidly in the United States. It is estimated that over 33 million people worldwide are infected with HIV. Many—perhaps all of them—will eventually come down with AIDS. Over 16 million people have died already since the outbreak of the epidemic. AIDS incidence is already very high in many African countries and is growing at 20% worldwide. The AIDS epidemic is discussed further in chapter 57.

#### How HIV Compromises the Immune System

In normal individuals, an army of specialized cells patrols the bloodstream, attacking and destroying any invading bacteria or viruses. In AIDS patients, this army of defenders is vanquished. One special kind of white blood cell, called a CD4<sup>+</sup> T cell (discussed further in chapter 57) is required to rouse the defending cells to action. In AIDS patients, the virus homes in on CD4<sup>+</sup> T cells, infecting and killing them until none are left (figure 33.6). Without these crucial immune system cells, the body cannot mount a defense against invading bacteria or viruses. AIDS patients die of infections that a healthy person could fight off.

Clinical symptoms typically do not begin to develop until after a long latency period, generally 8 to 10 years after the initial infection with HIV. During this long interval, carriers of HIV have no clinical symptoms but are apparently fully in-



**FIGURE 33.6**

**The AIDS virus.** HIV particles exit an infected CD4<sup>+</sup> T cell (both shown in false color). The free virus particles are able to infect neighboring CD4<sup>+</sup> T cells.

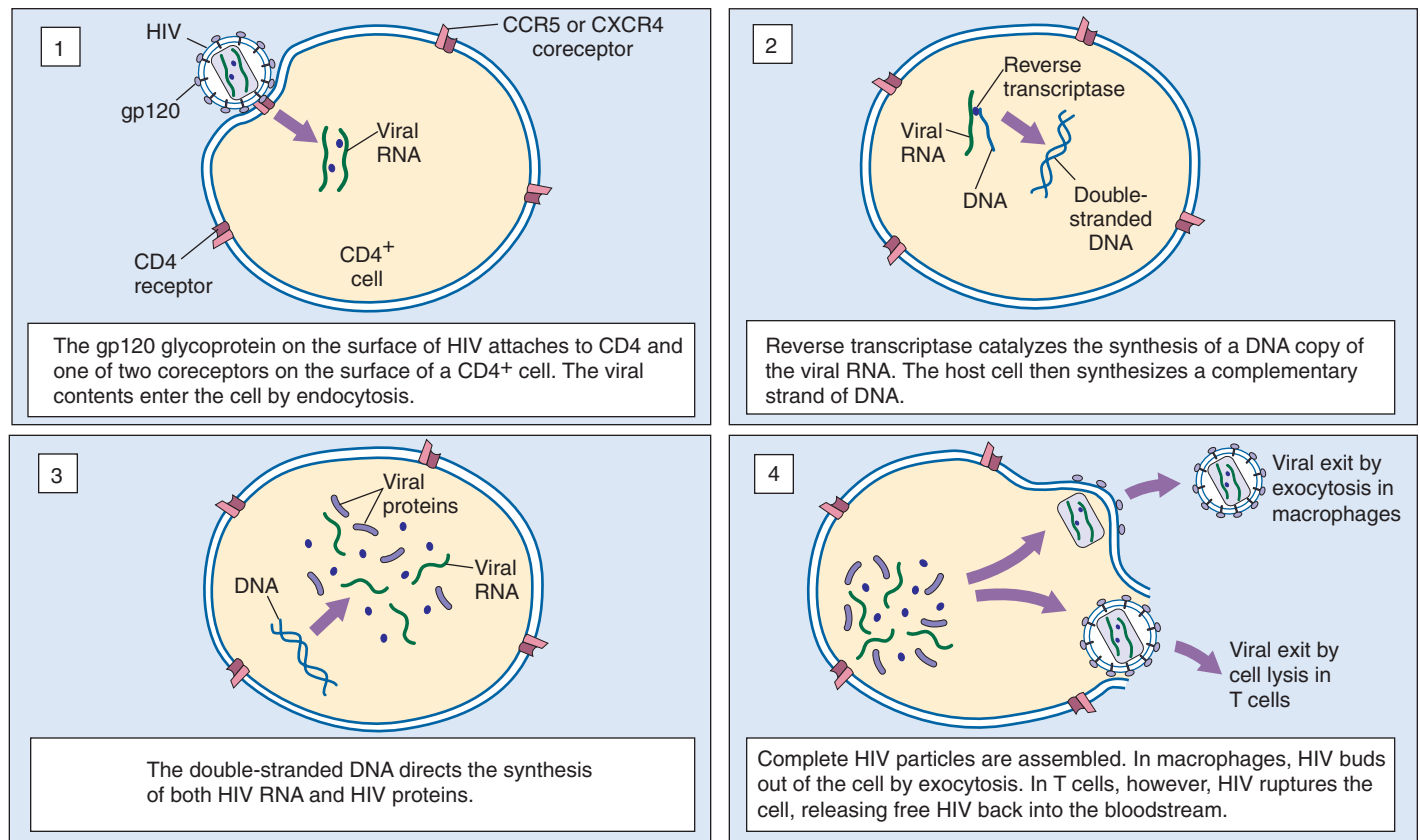
fectious, which makes the spread of HIV very difficult to control. The reason why HIV remains hidden for so long seems to be that its infection cycle continues throughout the 8- to 10-year latent period without doing serious harm to the infected person. Eventually, however, a random mutational event in the virus allows it to quickly overcome the immune defense, starting AIDS.

#### The HIV Infection Cycle

The HIV virus infects and eliminates key cells of the immune system, destroying the body's ability to defend itself from cancer and infection. The way HIV infects humans (figure 33.7) provides a good example of how animal viruses replicate. Most other viral infections follow a similar course, although the details of entry and replication differ in individual cases.

**Attachment.** When HIV is introduced into the human bloodstream, the virus particle circulates throughout the entire body but will only infect CD4<sup>+</sup> cells. Most other animal viruses are similarly narrow in their requirements; hepatitis goes only to the liver, and rabies to the brain.

How does a virus such as HIV recognize a specific kind of target cell? Recall from chapter 7 that every kind of cell in the human body has a specific array of cell-surface glycoprotein markers that serve to identify them to other, similar cells. Each HIV particle possesses a glycoprotein (called **gp120**) on its surface that precisely fits a cell-surface marker protein called **CD4** on the surfaces of immune system cells called macrophages and T cells. Macrophages are infected first.



**FIGURE 33.7**

**The HIV infection cycle.** The cycle begins and ends with free HIV particles present in the bloodstream of its human host. These free viruses infect white blood cells called CD4<sup>+</sup> T cells.

**Entry into Macrophages.** After docking onto the CD4 receptor of a macrophage, HIV requires a second macrophage receptor, called CCR5, to pull itself across the cell membrane. After gp120 binds to CD4, it goes through a conformational change that allows it to bind to CCR5. The current model suggests that after the conformational change, the second receptor passes the gp120-CD4 complex through the cell membrane, triggering passage of the contents of the HIV virus into the cell by endocytosis, with the cell membrane folding inward to form a deep cavity around the virus.

**Replication.** Once inside the macrophage, the HIV particle sheds its protective coat. This leaves virus RNA floating in the cytoplasm, along with a virus enzyme that was also within the virus shell. This enzyme, called **reverse transcriptase**, synthesizes a double strand of DNA complementary to the virus RNA, often making mistakes and so introducing new mutations. This double-stranded DNA directs the host cell machinery to produce many copies of the virus. HIV does not rupture and kill the macrophage cells it infects. Instead, the new viruses are released from the cell by exocytosis. HIV synthesizes large numbers of viruses in this way, challenging the immune system over a period of years.

**Entry into T Cells.** During this time, HIV is constantly replicating and mutating. Eventually, by chance, HIV alters the gene for gp120 in a way that causes the gp120 protein to change its second-receptor allegiance. This new form of gp120 protein prefers to bind instead to a different second receptor, CXCR4, a receptor that occurs on the surface of T lymphocyte CD4<sup>+</sup> cells. Soon the body's T lymphocytes become infected with HIV. This has deadly consequences, as new viruses exit the cell by rupturing the cell membrane, effectively killing the infected T cell. Thus, the shift to the CXCR4 second receptor is followed swiftly by a steep drop in the number of T cells. This destruction of the body's T cells blocks the immune response and leads directly to the onset of AIDS, with cancers and opportunistic infections free to invade the defenseless body.

**HIV, the virus that causes AIDS, is an RNA virus that replicates inside human cells by first making a DNA copy of itself. It is only able to gain entrance to those cells possessing a particular cell surface marker recognized by a glycoprotein on its own surface.**

## The Future of HIV Treatment

New discoveries of how HIV works continue to fuel research on devising ways to counter HIV. For example, scientists are testing drugs and vaccines that act on HIV receptors, researching the possibility of blocking CCR5, and looking for defects in the structures of HIV receptors in individuals that are infected with HIV but have not developed AIDS. Figure 33.8 summarizes some of the recent developments and discoveries.

### Combination Drug Therapy

A variety of drugs inhibit HIV in the test tube. These include AZT and its analogs (which inhibit virus nucleic acid replication) and protease inhibitors (which inhibit the cleavage of the large polyproteins encoded by *gag*, *pol*, and *env* genes into functional capsid, enzyme, and envelope segments). When combinations of these drugs were administered to people with HIV in controlled studies, their condition improved. A combination of a protease inhibitor and two AZT analog drugs entirely eliminated the HIV virus from many of the patients' bloodstreams. Importantly, all of these patients began to receive the drug therapy within three months of contracting the virus, before their bodies had an opportunity to develop tolerance to any one of them. Widespread use of this **combination therapy** has cut the U.S. AIDS death rate by three-fourths since its introduction in the mid-1990s, from 49,000 AIDS deaths in 1995 to 36,000 in 1996, and just over 10,000 in 1999.

Unfortunately, this sort of combination therapy does not appear to actually succeed in eliminating HIV from the body. While the virus disappears from the bloodstream, traces of it can still be detected in lymph tissue of the patients. When combination therapy is discontinued, virus levels in the bloodstream once again rise. Because of demanding therapy schedules and many side effects, long-term combination therapy does not seem a promising approach.

### Using a Defective HIV Gene to Combat AIDS

Recently, five people in Australia who are HIV-positive but have not developed AIDS in 14 years were found to have all received a blood transfusion from the same HIV-positive person, who also has not developed AIDS. This led scientists to believe that the strain of virus transmitted to these people has some sort of genetic defect that prevents it from effectively disabling the human immune system. In subsequent research, a defect was found in one of the nine genes present in this strain of HIV. This gene is called *nef*, named for "negative factor," and the defective version of *nef* in the HIV strain that infected the six Australians seems to be missing some pieces. Viruses with the defective gene may have reduced reproductive capability, allowing the immune system to keep the virus in check.

This finding has exciting implications for developing a vaccine against AIDS. Before this, scientists have been unsuccessful in trying to produce a harmless strain of AIDS that can elicit an effective immune response. The Australian strain with the defective *nef* gene has the potential to be used in a vaccine that would arm the immune system against this and other strains of HIV.

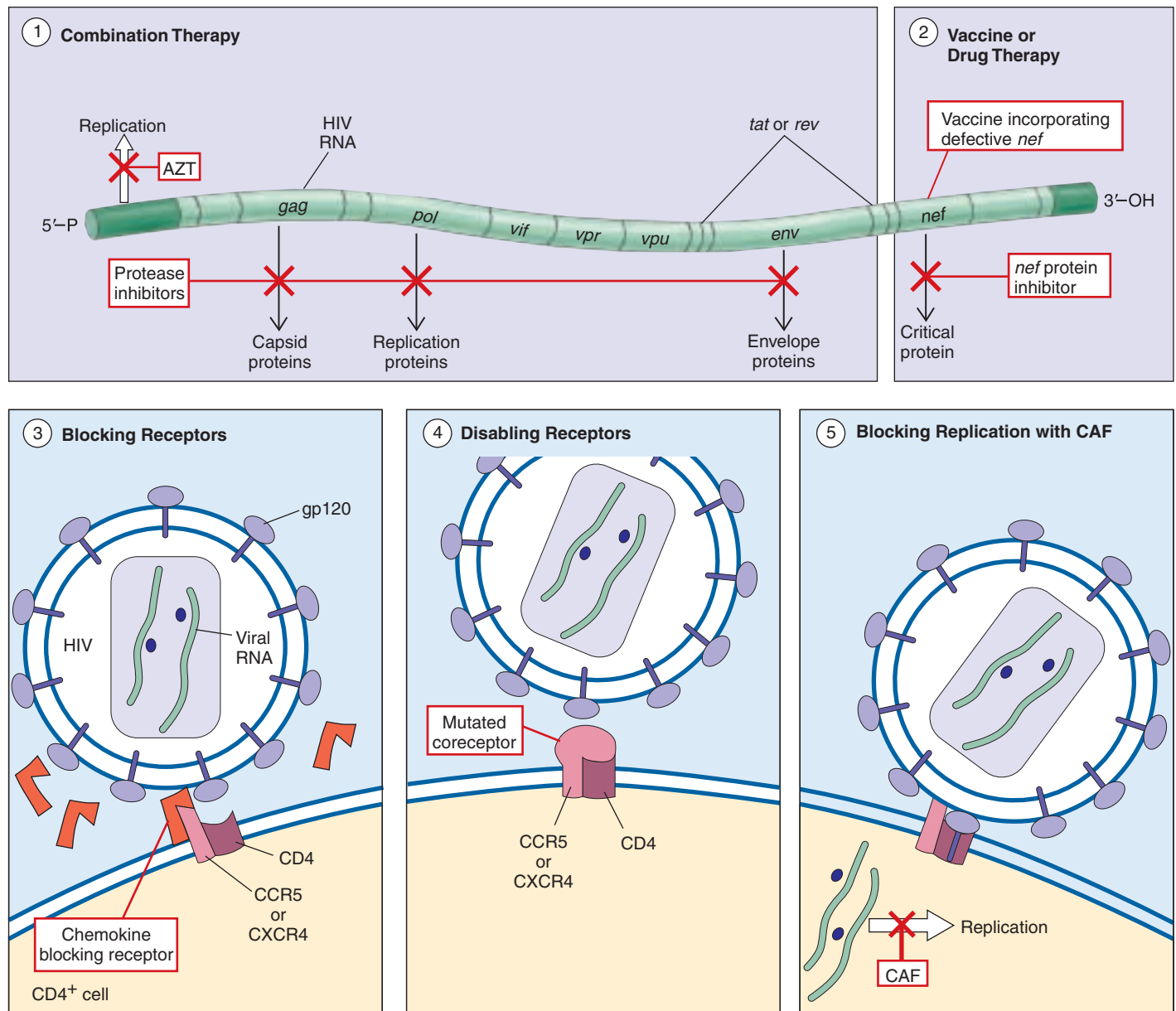
Another potential application of this discovery is its use in developing drugs that inhibit HIV proteins that speed virus replication. It seems that the protein produced from the *nef* gene is one of these critical HIV proteins, because viruses with defective forms of *nef* do not reproduce, as seen in the cases of the six Australians. Research is currently underway to develop a drug that targets the *nef* protein.

### Chemokines and CAF

In the laboratory, chemicals called **chemokines** appear to inhibit HIV infection by binding to and blocking the CCR5 and CXCR4 coreceptors. As you might expect, people long infected with the HIV virus who have not developed AIDS prove to have high levels of chemokines in their blood.

The search for HIV-inhibiting chemokines is intense. Not all results are promising. Researchers report that in their tests, the levels of chemokines were not different between patients in which the disease was not progressing and those in which it was rapidly progressing. More promising, levels of another factor called **CAF** (CD8<sup>+</sup> cell antiviral factor) *are* different between these two groups. Researchers have not yet succeeded in isolating CAF, which seems not to block receptors that HIV uses to gain entry to cells, but, instead, to prevent replication of the virus once it has infected the cells. Research continues on the use of chemokines in treatments for HIV infection, either increasing the amount of chemokines or disabling the CCR5 receptor. However, promising research on CAF suggests that it may be an even better target for treatment and prevention of AIDS.

One problem with using chemokines as drugs is that they are also involved in the inflammatory response of the immune system. The function of chemokines is to attract white blood cells to areas of infection. Chemokines work beautifully in small amounts and in local areas, but chemokines in mass numbers can cause an inflammatory response that is worse than the original infection. Injections of chemokines may hinder the immune system's ability to respond to local chemokines, or they may even trigger an out-of-control inflammatory response. Thus, scientists caution that injection of chemokines could make patients *more* susceptible to infections, and they continue to research other methods of using chemokines to treat AIDS.



**FIGURE 33.8**

Research is currently underway to develop new treatments for HIV. Among them are these five: (1) *Combination therapy* involves using two drugs, AZT to block replication of the virus and protease inhibitors to block the production of critical viral proteins. (2) Using a defective form of the viral gene *nef*, scientists may be able to construct an HIV *vaccine*. Also, *drug therapy* that inhibits *nef*'s protein product is being tested. (3) Other research focuses on the use of chemokine chemicals to *block receptors* (CXCR4 and CCR5), thereby disabling the mechanism HIV uses to enter CD4<sup>+</sup> T cells. (4) Producing mutations that will *disable receptors* may also be possible. (5) Lastly, CAF, an antiviral factor which acts inside the CD4<sup>+</sup> T cell, may be able to *block replication* of HIV.

### Disabling Receptors

A 32-base-pair deletion in the gene that codes for the CCR5 receptor appears to block HIV infection. Individuals at high risk of HIV infection who are homozygous for this mutation do not seem to develop AIDS. In one study of 1955 people, scientists found no individuals who were infected and homozygous for the mutated allele. The allele seems to be more common in Caucasian populations (10 to 11%) than in

African-American populations (2%), and absent in African and Asian populations. Treatment for AIDS involving disruption of CCR5 looks promising, as research indicates that people live perfectly well without CCR5. Attempts to block or disable CCR5 are being sought in numerous laboratories.

**A cure for AIDS is not yet in hand, but many new approaches look promising.**

## 33.4 Nonliving infectious agents are responsible for many human diseases.

### Disease Viruses

Humans have known and feared diseases caused by viruses for thousands of years. Among the diseases that viruses cause (table 33.1) are influenza, smallpox, infectious hepatitis, yellow fever, polio, rabies, and AIDS, as well as many other diseases not as well known. In addition, viruses have been implicated in some cancers and leukemias. For many autoimmune diseases, such as multiple sclerosis and rheumatoid arthritis, and for diabetes, specific viruses have been found associated with certain cases. In view of their effects, it is easy to see why the late Sir Peter Medawar, Nobel laureate in Physiology or Medicine, wrote, “A virus is a piece of bad news wrapped in protein.” Viruses not only cause many human diseases, but also cause major losses in agriculture, forestry, and in the productivity of natural ecosystems.

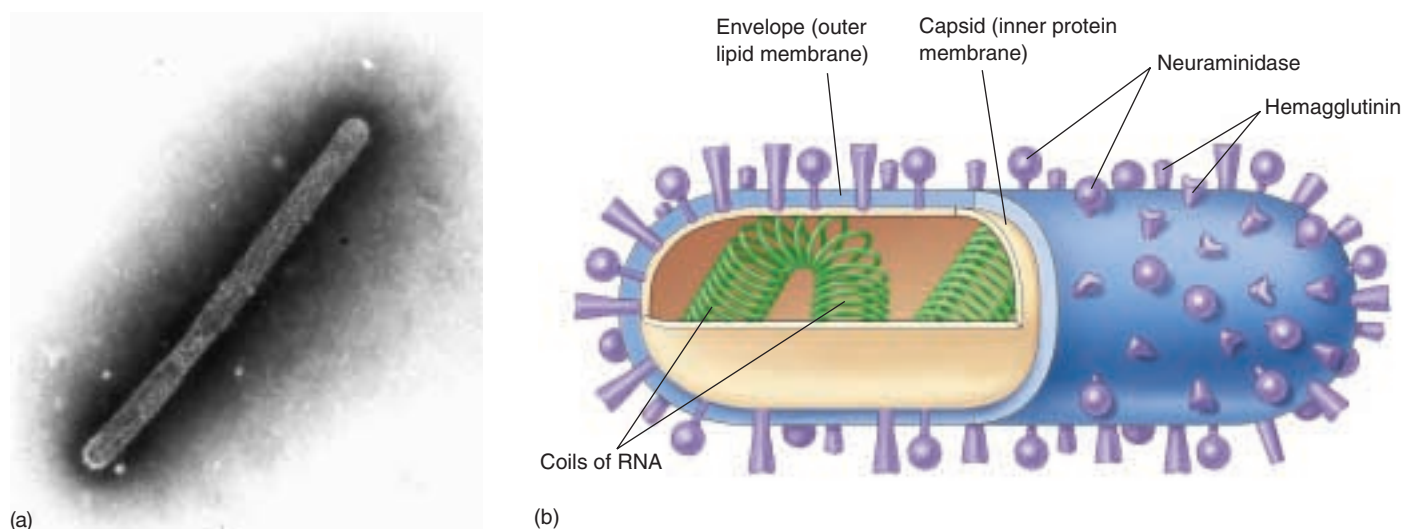
### Influenza

Perhaps the most lethal virus in human history has been the influenza virus. Some 22 million Americans and Europeans died of flu within 18 months in 1918 and 1919, an astonishing number.

**Types.** Flu viruses are animal retroviruses. An individual flu virus resembles a rod studded with spikes composed of two kinds of protein (figure 33.9). There are three general “types” of flu virus, distinguished by their capsid (inner membrane) protein, which is different for

each type: Type A flu virus causes most of the serious flu epidemics in humans, and also occurs in mammals and birds. Type B and Type C viruses, with narrower host ranges, are restricted to humans and rarely cause serious health problems.

**Subtypes.** Different strains of flu virus, called subtypes, differ in their protein spikes. One of these proteins, hemagglutinin (H) aids the virus in gaining access to the cell interior. The other, neuraminidase (N) helps the daughter virus break free of the host cell once virus replication has been completed. Parts of the H molecule contain “hot spots” that display an unusual tendency to change as a result of mutation of the virus RNA during imprecise replication. Point mutations cause changes in these spike proteins in 1 of 100,000 viruses during the course of each generation. These highly variable segments of the H molecule are targets against which the body’s antibodies are directed. The high variability of these targets improves the reproductive capacity of the virus and hinders our ability to make perfect vaccines. Because of accumulating changes in the H and N molecules, different flu vaccines are required to protect against different subtypes. Type A flu viruses are currently classified into 13 distinct H subtypes and 9 distinct N subtypes, each of which requires a different vaccine to protect against infection. Thus, the type A virus that caused the Hong Kong flu epidemic of 1968 has type 3 H molecules and type 2 N molecules, and is called A(H3N2).



**FIGURE 33.9**

**The influenza virus.** (a) TEM of the so-called “bird flu” influenza virus, A(H5N1), which first infected humans in Hong Kong in 1997. (b) Diagram of an influenza virus. The coiled RNA has been revealed by cutting through the outer lipid-rich envelope, with its two kinds of projecting spikes, and the inner protein capsid.

| Disease             | Pathogen                                | Reservoir                                                                    | Vector/Epidemiology                                                                                                                                                                                                                                                        |
|---------------------|-----------------------------------------|------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AIDS                | HIV                                     | STD                                                                          | Destroys immune defenses, resulting in death by infection or cancer. Over 33 million cases worldwide by 1998.                                                                                                                                                              |
| Chicken pox         | Human herpes-virus 3 (varicella-zoster) | Humans                                                                       | Spread through contact with infected individuals. No cure. Rarely fatal. Vaccine approved in U.S. in early 1995.                                                                                                                                                           |
| Ebola               | Filoviruses                             | Unknown                                                                      | Acute hemorrhagic fever; virus attacks connective tissue, leading to massive hemorrhaging and death. Peak mortality is 50–90% if the disease goes untreated. Outbreaks confined to local regions of central Africa.                                                        |
| Hepatitis B (viral) | Hepatitis B virus (HBV)                 | Humans                                                                       | Highly infectious through contact with infected body fluids. Approximately 1% of U.S. population infected. Vaccine available, no cure. Can be fatal.                                                                                                                       |
| Herpes              | Herpes simplex virus (HSV)              | Humans                                                                       | Fever blisters; spread primarily through contact with infected saliva. Very prevalent worldwide. No cure. Exhibits latency—the disease can be dormant for several years.                                                                                                   |
| Influenza           | Influenza viruses                       | Humans, ducks                                                                | Historically a major killer (22 million died in 18 months in 1918–19); wild Asian ducks, chickens, and pigs are major reservoirs. The ducks are not affected by the flu virus, which shuffles its antigen genes while multiplying within them, leading to new flu strains. |
| Measles             | Paramyxoviruses                         | Humans                                                                       | Extremely contagious through contact with infected individuals. Vaccine available. Usually contracted in childhood, when it is not serious; more dangerous to adults.                                                                                                      |
| Mononucleosis       | Epstein-Barr virus (EBV)                | Humans                                                                       | Spread through contact with infected saliva. May last several weeks; common in young adults. No cure. Rarely fatal.                                                                                                                                                        |
| Mumps               | Paramyxovirus                           | Humans                                                                       | Spread through contact with infected saliva. Vaccine available; rarely fatal. No cure.                                                                                                                                                                                     |
| Pneumonia           | Influenza virus                         | Humans                                                                       | Acute infection of the lungs, often fatal without treatment.                                                                                                                                                                                                               |
| Polio               | Poliovirus                              | Humans                                                                       | Acute viral infection of the CNS that can lead to paralysis and is often fatal. Prior to the development of Salk's vaccine in 1954, 60,000 people a year contracted the disease in the U.S. alone.                                                                         |
| Rabies              | Rhabdovirus                             | Wild and domestic Canidae (dogs, foxes, wolves, coyotes), bats, and raccoons | An acute viral encephalomyelitis transmitted by the bite of an infected animal. Fatal if untreated.                                                                                                                                                                        |
| Smallpox            | Variola virus                           | Formerly humans, now only exists in two research labs—may be eliminated      | Historically a major killer; the last recorded case of smallpox was in 1977. A worldwide vaccination campaign wiped out the disease completely.                                                                                                                            |
| Yellow fever        | Flavivirus                              | Humans, mosquitoes                                                           | Spread from individual to individual by mosquito bites; a notable cause of death during the construction of the Panama Canal. If untreated, this disease has a peak mortality rate of 60%.                                                                                 |

**Importance of Recombination.** The greatest problem in combating flu viruses arises not through mutation, but through recombination. Viral genes are readily reassorted by genetic recombination, sometimes putting together novel combinations of H and N spikes unrecognizable by human antibodies specific for the old configuration. Viral recombination of this kind seems to have been responsible for the three major flu pandemics

(that is, worldwide epidemics) that occurred in the last century, by producing drastic shifts in H N combinations. The “killer flu” of 1918, A(H1N1), killed 40 million people. The Asian flu of 1957, A(H2N2), killed over 100,000 Americans. The Hong Kong flu of 1968, A(H3N2), infected 50 million people in the United States alone, of which 70,000 died.

It is no accident that new strains of flu usually originate in the far east. The most common hosts of influenza virus are ducks, chickens, and pigs, which in Asia often live in close proximity to each other and to humans. Pigs are subject to infection by both bird and human strains of the virus, and individual animals are often simultaneously infected with multiple strains. This creates conditions favoring genetic recombination between strains, producing new combinations of H and N subtypes. The Hong Kong flu, for example, arose from recombination between A(H3N8) [from ducks] and A(H2N2) [from humans]. The new strain of influenza, in this case A(H3N2), then passed back to humans, creating an epidemic because the human population has never experienced that H N combination before.

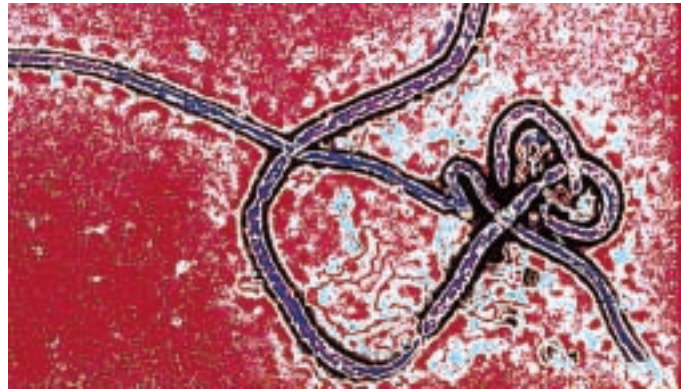
A potentially deadly new strain of flu virus emerged in Hong Kong in 1997, A(H5N1). Unlike all previous instances of new flu strains, A(H5N1) passed to humans directly from birds, in this case chickens. A(H5N1) was first identified in chickens in 1961, and in the spring of 1997 devastated flocks of chickens in Hong Kong. Fortunately, this strain of flu virus does not appear to spread easily from person to person, and the number of human infections by A(H5N1) remains small. Public health officials remain concerned that the genes of A(H5N1) could yet mix with those of a human strain to create a new strain that could spread widely in the human population, and to prevent this ordered the killing of all 1.2 million chickens in Hong Kong in 1997.

## Emerging Viruses

Sometimes viruses that originate in one organism pass to another, thus expanding their host range. Often, this expansion is deadly to the new host. HIV, for example, arose in chimpanzees and relatively recently passed to humans. Influenza is fundamentally a bird virus. Viruses that originate in one organism and then pass to another and cause disease are called **emerging viruses** and represent a considerable threat in an age when airplane travel potentially allows infected individuals to move about the world quickly, spreading an infection.

Among the most lethal of emerging viruses are a collection of filamentous viruses arising in central Africa that cause severe hemorrhagic fever. With lethality rates in excess of 50%, these so-called filoviruses are among the most lethal infectious diseases known. One, Ebola virus (figure 33.10), has exhibited lethality rates in excess of 90% in isolated outbreaks in central Africa. The outbreak of Ebola virus in the summer of 1995 in Zaire killed 245 people out of 316 infected—a mortality rate of 78%. The latest outbreak occurred in Gabon, West Africa, in February 1996. The natural host of Ebola is unknown.

Another type of emerging virus caused a sudden outbreak of a hemorrhagic-type infection in the southwestern United States in 1993. This highly fatal disease was soon attributed to the hantavirus, a single-stranded RNA virus



**FIGURE 33.10**

**The Ebola virus.** This virus, with a fatality rate that can exceed 90%, appears sporadically in West Africa. Health professionals are scrambling to identify the natural host of the virus, which is unknown, so they can devise strategies to combat transmission of the disease.

associated with rodents. The hantavirus is transmitted to humans through rodent fecal contamination in areas of human habitation. Although hantavirus has been known for some period of time, this particular outbreak was attributed to the presence of an unusually large rodent population in the area following a higher than normal amount of rainfall the previous winter.

## Viruses and Cancer

Through epidemiological studies and research, scientists have established a link between some viral infections and the subsequent development of cancer. Examples include the association between chronic hepatitis B infections and the development of liver cancer and the development of cervical carcinoma following infections with certain strains of papillomaviruses. It has been suggested that viruses contribute to about 15% of all human cancer cases worldwide. Viruses are capable of altering the growth properties of human cells they infect by triggering the expression of oncogenes (cancer-causing genes). Certain viruses can either activate host proto-oncogenes (see chapter 18) or bring in viral oncogenes that become incorporated into the host genome. Virus-induced cancer is not simply a matter of infection. The disease involves complex interactions with cellular genes and requires a series of events in order to develop.

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**Viruses are responsible for some of the most lethal diseases of humans. Some of the most serious examples are viruses that have transferred to humans from some other host. Influenza, a bird virus, has been responsible for the most devastating epidemics in human history. Newly emerging viruses such as Ebola have received considerable public attention.**

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## Prions and Viroids

For decades scientists have been fascinated by a peculiar group of fatal brain diseases. These diseases have the unusual property that it is years and often decades after infection before the disease is detected in infected individuals. The brains of infected individuals develop numerous small cavities as neurons die, producing a marked spongy appearance. Called **transmissible spongiform encephalopathies (TSEs)**, these diseases include scrapie in sheep, “mad cow” disease in cattle, and kuru and Creutzfeldt-Jakob disease in humans.

TSEs can be transmitted by injecting infected brain tissue into a recipient animal’s brain. TSEs can also spread via tissue transplants and, apparently, food. Kuru was common in the Fore people of Papua New Guinea, when they practiced ritual cannibalism, literally eating the brains of infected individuals. Mad cow disease spread widely among the cattle herds of England in the 1990s because cows were fed bone meal prepared from cattle carcasses to increase the protein content of their diet. Like the Fore, the British cattle were literally eating the tissue of cattle that had died of the disease.

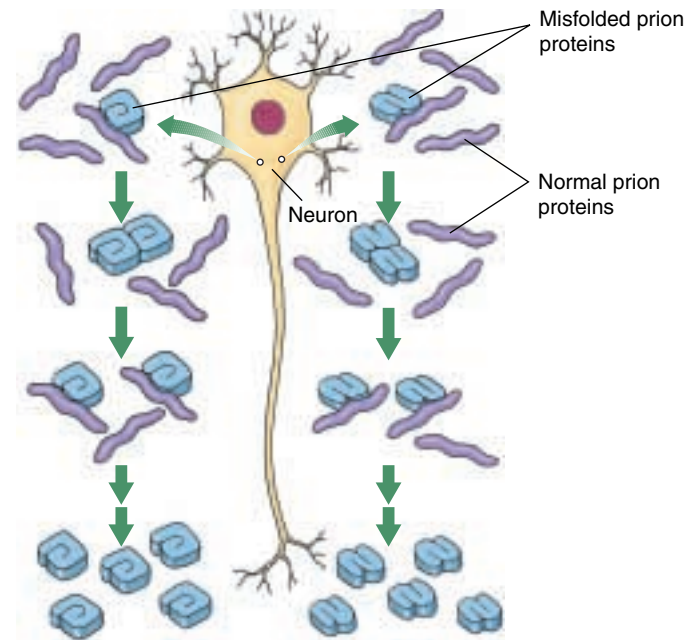
### A Heretical Suggestion

In the 1960s, British researchers T. Alper and J. Griffith noted that infectious TSE preparations remained infectious even after exposed to radiation that would destroy DNA or RNA. They suggested that the infectious agent was a protein. Perhaps, they speculated, the protein usually preferred one folding pattern, but could sometimes misfold, and then catalyze other proteins to do the same, the misfolding spreading like a chain reaction. This heretical suggestion was not accepted by the scientific community, as it violates a key tenant of molecular biology: only DNA or RNA act as hereditary material, transmitting information from one generation to the next.

### Prusiner’s Prions

In the early 1970s, physician Stanley Prusiner, moved by the death of a patient from Creutzfeldt-Jakob disease, began to study TSEs. Prusiner became fascinated with Alper and Griffith’s hypothesis. Try as he might, Prusiner could find no evidence of nucleic acids or viruses in the infectious TSE preparations, and concluded, as Alper and Griffith had, that the infectious agent was a *protein*, which in a 1982 paper he named a **prion**, for “proteinaceous infectious particle.”

Prusiner went on to isolate a distinctive prion protein, and for two decades continued to amass evidence that prions play a key role in triggering TSEs. The scientific community resisted Prusiner’s renegade conclusions, but eventually experiments done in Prusiner’s and other laboratories began to convince many. For example, when



**FIGURE 33.11**

**How prions arise.** Misfolded prions seem to cause normal prion protein to misfold simply by contacting them. When prions misfolded in different ways (*blue*) contact normal prion protein (*purple*), the normal prion protein misfolds in the same way.

Prusiner injected prions of a different abnormal conformation into several different hosts, these hosts developed prions with the same abnormal conformations as the parent prions (figure 33.11). In another important experiment, Charles Weissmann showed that mice genetically engineered to lack Prusiner’s prion protein are immune to TSE infection. However, if brain tissue with the prion protein is grafted into the mice, the grafted tissue—but not the rest of the brain—can then be infected with TSE. In 1997, Prusiner was awarded the Nobel Prize in Physiology or Medicine for his work on prions.

### Viroids

Viroids are tiny, naked molecules of RNA, only a few hundred nucleotides long, that are important infectious disease agents in plants. A recent viroid outbreak killed over ten million coconut palms in the Philippines. It is not clear how viroids cause disease. One clue is that viroid nucleotide sequences resemble the sequences of introns within ribosomal RNA genes. These sequences are capable of catalyzing excision from DNA—perhaps the viroids are catalyzing the destruction of chromosomal integrity.

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**Prions are infectious proteins that some scientists believe are responsible for serious brain diseases. In plants, naked RNA molecules called viroids can also transmit disease.**

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## Summary

## Questions

## Media Resources

## 33.1 Viruses are strands of nucleic acid encased within a protein coat.

- Viruses are fragments of DNA or RNA surrounded by protein that are able to replicate within cells by using the genetic machinery of those cells.
- The simplest viruses use the enzymes of the host cell for both protein synthesis and gene replication; the more complex ones contain up to 200 genes and are capable of synthesizing many structural proteins and enzymes.
- Viruses are basically either helical or isometric. Most isometric viruses are icosahedral in shape.

1. Why are viruses not considered to be living organisms?
2. How did early scientists come to the conclusion that the infectious agents associated with hoof-and-mouth disease in cattle were not bacteria?
3. What is the approximate size range of viruses and type of microscope is generally required to visualize viruses?



- Characteristics of Viruses

## 33.2 Bacterial viruses exhibit two sorts of reproductive cycles.

- Virulent bacteriophages infect bacterial cells by injecting their viral DNA or RNA into the cell, where it directs the production of new virus particles, ultimately lysing the cell.
- Temperate bacteriophages, upon entering a bacterial cell, insert their DNA into the cell genome, where they may remain integrated into the bacterial genome as a prophage for many generations.

4. What is a bacteriophage? How does a T4 phage infect a host cell?



- Life Cycle of Viruses

## 33.3 HIV is a complex animal virus.

- AIDS, a viral infection that destroys the immune system, is caused by HIV (human immunodeficiency virus). After docking on a specific protein called CD4, HIV enters the cell and replicates, destroying the cell.
- Considerable progress has been made in the treatment of AIDS, particularly with drugs such as protease inhibitors that block cleavage of HIV polyproteins into functional segments.

5. What specific type of human cell does the AIDS virus infect? How does it recognize this specific kind of cell?
6. How do many animal viruses penetrate the host cell? How does a plant virus infect its host? How does a bacterial virus infect its host?



- Bioethics Case Study: AIDS Vaccine
- On Science Articles: HIV's Waiting Game
- Drug Therapy for AIDS
- Curing AIDS Just Got Harder
- HIV Delivery Protein

## 33.4 Nonliving infectious agents are responsible for many human diseases.

- Viruses are responsible for many serious human diseases. Some of the most serious, like AIDS and Ebola, have only recently transferred to humans from some other animal host.
- Proteins called prions may transmit serious brain diseases from one individual to another.

7. Why is it so much more difficult to treat a viral infection than a bacterial one? Is this different from treating bacterial infections?
8. What is a prion? How does it integrate into living systems?



- Scientists on Science: Prions
- Book Review: *The Coming Plague* by Garrett
- On Science Articles: Smallpox: Tomorrow's Nightmare?
- Smallpox Questions
- Mad Cows and Prions
- Prions and Blood Supply
- Hepatitis C
- Increasing Mad Cow Diseases

# 34

## Bacteria

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### Concept Outline

#### 34.1 Bacteria are the smallest and most numerous organisms.

**The Prevalence of Bacteria.** The simplest of organisms, bacteria are thought to be the most ancient. They are the most abundant living organisms. Bacteria lack the high degree of internal compartmentalization characteristic of eukaryotes.

#### 34.2 Bacterial cell structure is more complex than commonly supposed.

**The Bacterial Surface.** Some bacteria have a secondary membranelike covering outside of their cell wall.

**The Cell Interior.** While bacteria lack extensive internal compartments, they may have complex internal membranes.

#### 34.3 Bacteria exhibit considerable diversity in both structure and metabolism.

**Bacterial Diversity.** There are at least 16 phyla of bacteria, although many more remain to be discovered.

**Bacterial Variation.** Mutation and recombination generate enormous variation within bacterial populations.

**Bacterial Metabolism.** Bacteria obtain carbon atoms and energy from a wide array of sources. Some can thrive in the absence of other organisms, while others must obtain their energy and carbon atoms from other organisms.

#### 34.4 Bacteria are responsible for many diseases but also make important contributions to ecosystems.

**Human Bacterial Diseases.** Many serious human diseases are caused by bacteria, some of them responsible for millions of deaths each year.

**Importance of Bacteria.** Bacteria have had a profound impact on the world's ecology, and play a major role in modern medicine and agriculture.



**FIGURE 34.1**

**A colony of bacteria.** With their enormous adaptability and metabolic versatility, bacteria are found in every habitat on earth, carrying out many of the vital processes of ecosystems, including photosynthesis, nitrogen fixation, and decomposition.

The simplest organisms living on earth today are bacteria, and biologists think they closely resemble the first organisms to evolve on earth. Too small to see with the unaided eye, bacteria are the most abundant of all organisms (figure 34.1) and are the only ones characterized by prokaryotic cellular organization. Life on earth could not exist without bacteria because bacteria make possible many of the essential functions of ecosystems, including the capture of nitrogen from the atmosphere, decomposition of organic matter, and, in many aquatic communities, photosynthesis. Indeed, bacterial photosynthesis is thought to have been the source for much of the oxygen in the earth's atmosphere. Bacterial research continues to provide extraordinary insights into genetics, ecology, and disease. An understanding of bacteria is thus essential.

## 34.1 Bacteria are the smallest and most numerous organisms.

### The Prevalence of Bacteria

Bacteria are the oldest, structurally simplest, and the most abundant forms of life on earth. They are also the only organisms with prokaryotic cellular organization. Represented in the oldest rocks from which fossils have been obtained, 3.5 to 3.8 billion years old, bacteria were abundant for over 2 billion years before eukaryotes appeared in the world (see figure 4.11). Early photosynthetic bacteria (cyanobacteria) altered the earth's atmosphere with the production of oxygen which lead to extreme bacterial and eukaryotic diversity. Bacteria play a vital role both in productivity and in cycling the substances essential to all other life-forms. Bacteria are the only organisms capable of fixing atmospheric nitrogen.

About 5000 different kinds of bacteria are currently recognized, but there are doubtless many thousands more awaiting proper identification (figure 34.2). Every place microbiologists look, new species are being discovered, in some cases altering the way we think about bacteria. In the 1970s and 80s a new type of bacterium was analyzed that eventually lead to the classification of a new prokaryotic cell type, the archeabacteria (or Archaea). Even when viewed with an electron microscope, the structural differences between different bacteria are minor compared to other groups of organisms. Because the structural differences are so slight, bacteria are classified based primarily upon their metabolic and genetic characteristics. Bacteria can be characterized properly only when they are grown on a defined medium because the characteristics of these organisms often change, depending on their growth conditions.

Bacteria are ubiquitous on Earth, and live everywhere eukaryotes do. Many of the other more extreme environments in which bacteria are found would be lethal to any other form of life. Bacteria live in hot springs that would cook other organisms, hypersaline environments that would dehydrate other cells, and in atmospheres rich in toxic gases like methane or hydrogen sulfide that would kill most other organisms. These harsh environments may be similar to the conditions present on the early Earth, when life first began. It is likely that bacteria evolved to dwell in these harsh conditions early on and have retained the ability to exploit these areas as the rest of the atmosphere has changed.

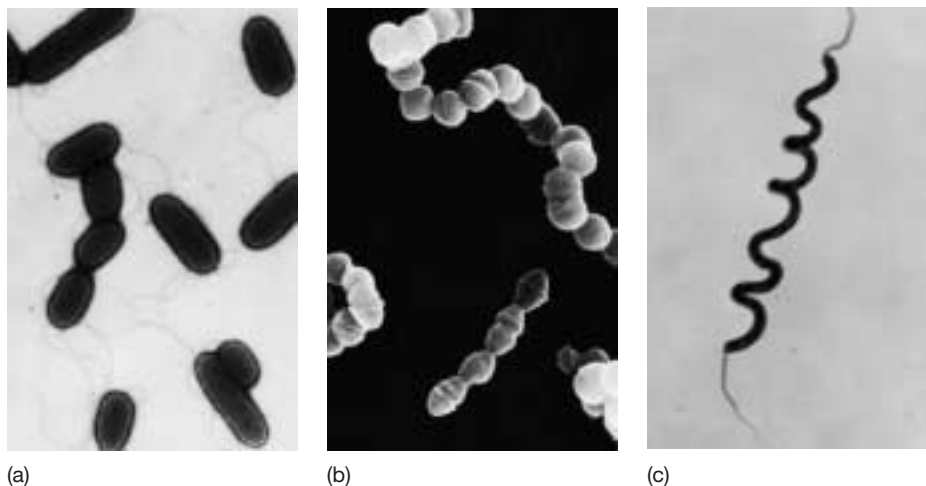


FIGURE 34.2

**The diversity of bacteria.** (a) *Pseudomonas aeruginosa*, a rod-shaped, flagellated bacterium (bacillus). *Pseudomonas* includes the bacteria that cause many of the most serious plant diseases. (b) *Streptococcus*. The spherical individual bacteria (cocci) adhere in chains in the members of this genus (34,000 $\times$ ). (c) *Spirillum volutans*, one of the spirilla. This large bacterium, which occurs in stagnant fresh water, has a tuft of flagella at each end (500 $\times$ ).

### Bacterial Form

Bacteria are mostly simple in form and exhibit one of three basic structures: **bacillus** (plural, **bacilli**) straight and rod-shaped, **coccus** (plural, **cocci**) spherical-shaped, and **spirillus** (plural, **spirilla**) long and helical-shaped, also called spirochetes. Spirilla bacteria generally do not form associations with other cells and swim singly through their environments. They have a complex structure within their cell membranes that allow them to spin their corkscrew-shaped bodies which propels them along. Some rod-shaped and spherical bacteria form colonies, adhering end-to-end after they have divided, forming chains (see figure 34.2). Some bacterial colonies change into stalked structures, grow long, branched filaments, or form erect structures that release **spores**, single-celled bodies that grow into new bacterial individuals. Some filamentous bacteria are capable of gliding motion, often combined with rotation around a longitudinal axis. Biologists have not yet determined the mechanism by which they move.

### Prokaryotes versus Eukaryotes

Prokaryotes—eubacteria and archaea—differ from eukaryotes in numerous important features. These differences represent some of the most fundamental distinctions that separate any groups of organisms.

1. **Multicellularity.** All prokaryotes are fundamentally single-celled. In some types, individual cells adhere to

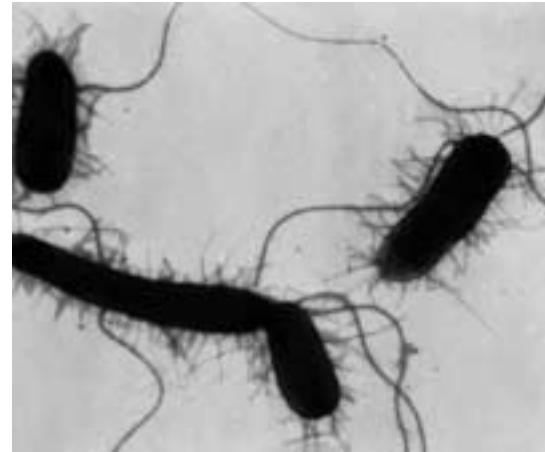
**FIGURE 34.3**  
**Approaches to multicellularity in bacteria.**

*Chondromyces crocatus*, one of the gliding bacteria. The rod-shaped individuals move together, forming the composite spore-bearing structures shown here. Millions of spores, which are basically individual bacteria, are released from these structures.



each other within a matrix and form filaments, however the cells retain their individuality. Cyanobacteria, in particular, are likely to form such associations but their cytoplasm is not directly interconnected, as often is the case in multicellular eukaryotes. The activities of a bacterial colony are less integrated and coordinated than those in multicellular eukaryotes. A primitive form of colonial organization occurs in gliding bacteria, which move together and form spore-bearing structures (figure 34.3). Such coordinated multicellular forms are rare among bacteria.

2. **Cell size.** As new species of bacteria are discovered, we are finding that the size of prokaryotic cells varies tremendously, by as much as five orders of magnitude. Most prokaryotic cells are only 1 micrometer or less in diameter. Most eukaryotic cells are well over 10 times that size.
3. **Chromosomes.** Eukaryotic cells have a membrane-bound nucleus containing chromosomes made up of both nucleic acids and proteins. Bacteria do not have membrane-bound nuclei, nor do they have chromosomes of the kind present in eukaryotes, in which DNA forms a structural complex with proteins. Instead, their naked circular DNA is localized in a zone of the cytoplasm called the **nucleoid**.
4. **Cell division and genetic recombination.** Cell division in eukaryotes takes place by mitosis and involves spindles made up of microtubules. Cell division in bacteria takes place mainly by binary fission (see chapter 11). True sexual reproduction occurs only in eukaryotes and involves syngamy and meiosis, with an alternation of diploid and haploid forms. Despite their lack of sexual reproduction, bacteria do have mechanisms that lead to the transfer of genetic material. These mecha-



**FIGURE 34.4**  
**Flagella in the common intestinal bacterium, *Escherichia coli*.**  
The long strands are flagella, while the shorter hairlike outgrowths are called pili.

nisms are far less regular than those of eukaryotes and do not involve the equal participation of the individuals between which the genetic material is transferred.

5. **Internal compartmentalization.** In eukaryotes, the enzymes for cellular respiration are packaged in mitochondria. In bacteria, the corresponding enzymes are not packaged separately but are bound to the cell membranes (see chapters 5 and 9). The cytoplasm of bacteria, unlike that of eukaryotes, contains no internal compartments or cytoskeleton and no organelles except ribosomes.
6. **Flagella.** Bacterial flagella are simple in structure, composed of a single fiber of the protein flagellin (figure 34.4; see also chapter 5). Eukaryotic flagella and cilia are complex and have a 9 + 2 structure of microtubules (see figure 5.27). Bacterial flagella also function differently, spinning like propellers, while eukaryotic flagella have a whiplike motion.
7. **Metabolic diversity.** Only one kind of photosynthesis occurs in eukaryotes, and it involves the release of oxygen. Photosynthetic bacteria have several different patterns of anaerobic and aerobic photosynthesis, involving the formation of end products such as sulfur, sulfate, and oxygen (see chapter 10). Prokaryotic cells can also be chemoautotrophic, using the energy stored in chemical bonds of inorganic molecules to synthesize carbohydrates; eukaryotes are not capable of this metabolic process.

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**Bacteria are the oldest and most abundant organisms on earth. Bacteria, or prokaryotes, differ from eukaryotes in a wide variety of characteristics, a degree of difference as great as any that separates any groups of organisms.**

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## 34.2 Bacterial cell structure is more complex than commonly supposed.

### The Bacterial Surface

The bacterial **cell wall** is an important structure because it maintains the shape of the cell and protects the cell from swelling and rupturing. The cell wall usually consists of **peptidoglycan**, a network of polysaccharide molecules connected by polypeptide cross-links. In some bacteria, the peptidoglycan forms a thick, complex network around the outer surface of the cell. This network is interlaced with peptide chains. In other bacteria a thin layer of peptidoglycan is found sandwiched between two plasma membranes. The outer membrane contains large molecules of lipopolysaccharide, lipids with polysaccharide chains attached. These two major types of bacteria can be identified using a staining process called a **Gram stain**. **Gram-positive** bacteria have the thicker peptidoglycan wall and stain a purple color (figure 34.5). The more common **gram-negative** bacteria contain less peptidoglycan and do not retain the purple-colored dye. Gram-negative bacteria stain red. The outer membrane layer makes gram-negative bacteria resistant to many antibiotics that interfere with cell wall synthesis in gram-positive bacteria. In some kinds of bacteria, an additional gelatinous layer, the capsule, surrounds the cell wall.

Many kinds of bacteria have slender, rigid, helical **flagella** (singular, **flagellum**) composed of the protein flagellin (figure 34.6). These flagella range from 3 to 12 micrometers in length and are very thin—only 10 to 20 nanometers thick. They are anchored in the cell wall and spin, pulling the bacteria through the water like a propeller.

**Pili** (singular, **pilus**) are other hairlike structures that occur on the cells of some bacteria (see figure 34.4). They are shorter than bacterial flagella, up to several micrometers long, and about 7.5 to 10 nanometers thick. Pili help the bacterial cells attach to appropriate substrates and exchange genetic information.

Some bacteria form thick-walled **endospores** around their chromosome and a small portion of the surrounding cytoplasm when they are exposed to nutrient-poor conditions. These endospores are highly resistant to environmental stress, especially heat, and can germinate to form new individuals after decades or even centuries.

**Bacteria are encased within a cell wall composed of one or more polysaccharide layers. They also may contain external structures such as flagella and pili.**

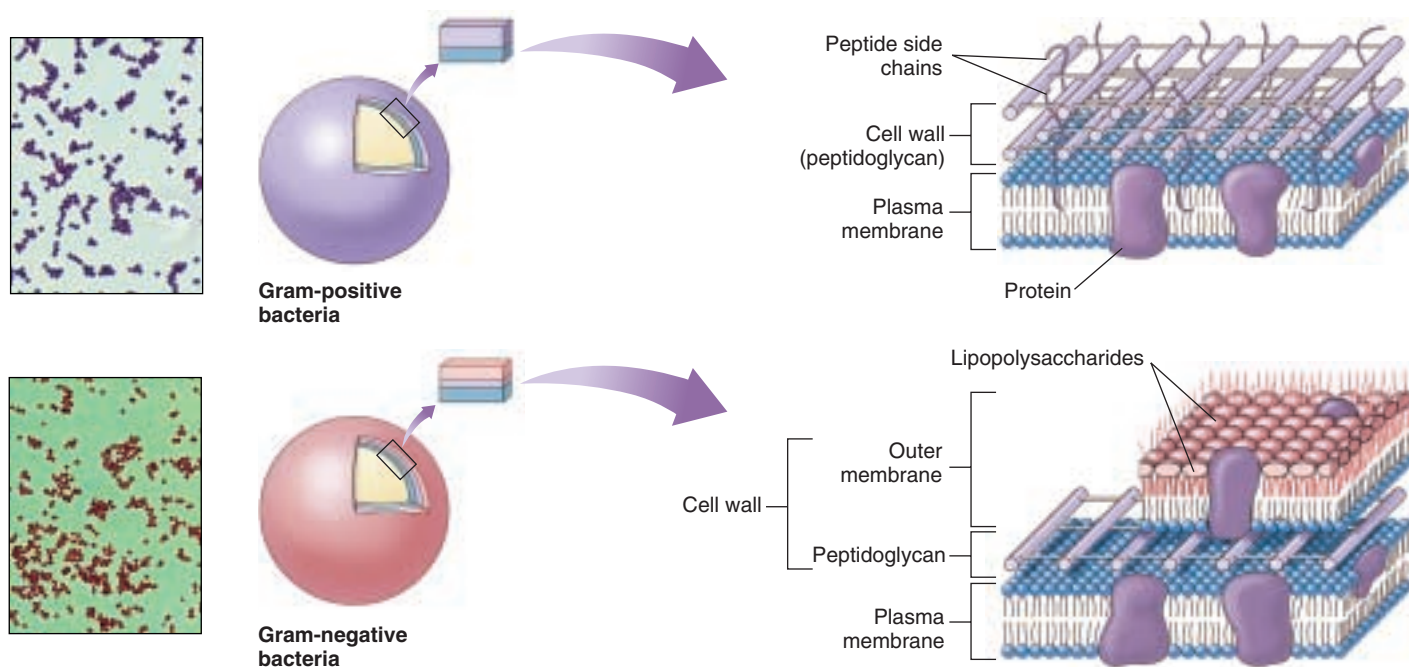
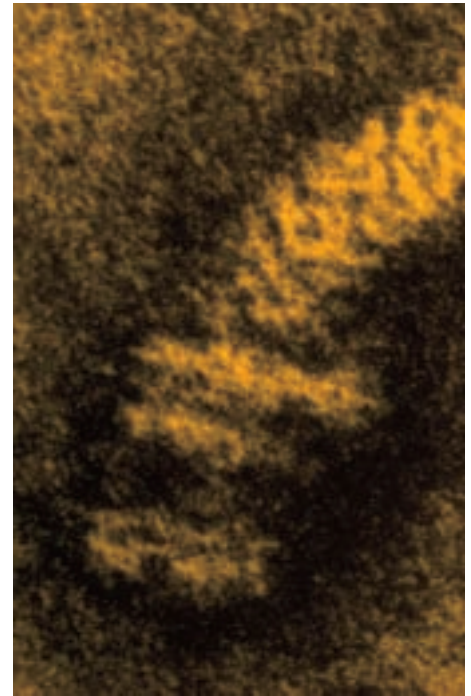
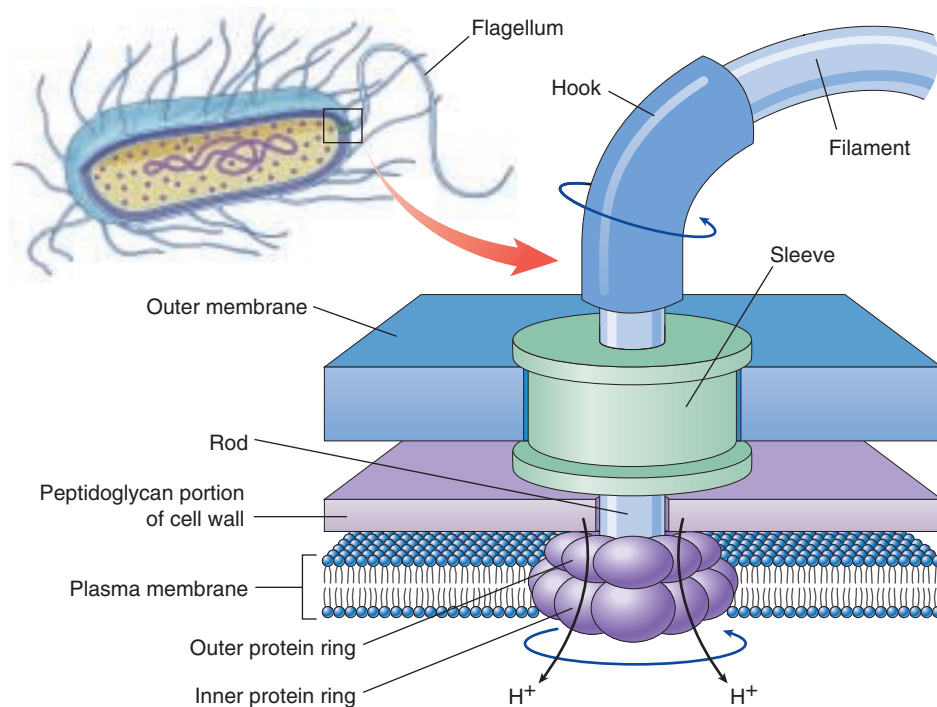


FIGURE 34.5

**The Gram stain.** The peptidoglycan layer encasing gram-positive bacteria traps crystal violet dye, so the bacteria appear purple in a Gram-stained smear (named after Hans Christian Gram, who developed the technique). Because gram-negative bacteria have much less peptidoglycan (located between the plasma membrane and an outer membrane), they do not retain the crystal violet dye and so exhibit the red background stain (usually a safranin dye).



**FIGURE 34.6**

**The flagellar motor of a gram-negative bacterium.** A protein filament, composed of the protein flagellin, is attached to a protein shaft that passes through a sleeve in the outer membrane and through a hole in the peptidoglycan layer to rings of protein anchored in the cell wall and plasma membrane, like rings of ballbearings. The shaft rotates when the inner protein ring attached to the shaft turns with respect to the outer ring fixed to the cell wall. The inner ring is an  $H^+$  ion channel, a proton pump that uses the passage of protons into the cell to power the movement of the inner ring past the outer one.

## The Cell Interior

The most fundamental characteristic of bacterial cells is their prokaryotic organization. Bacterial cells lack the extensive functional compartmentalization seen within eukaryotic cells.

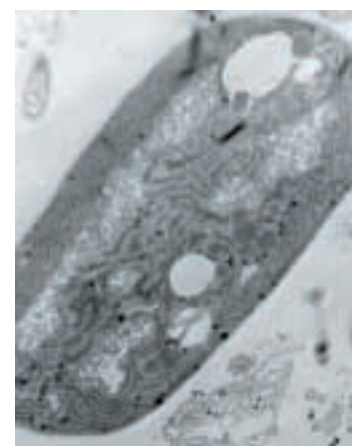
**Internal membranes.** Many bacteria possess invaginated regions of the plasma membrane that function in respiration or photosynthesis (figure 34.7).

**Nucleoid region.** Bacteria lack nuclei and do not possess the complex chromosomes characteristic of eukaryotes. Instead, their genes are encoded within a single double-stranded ring of DNA that is crammed into one region of the cell known as the **nucleoid region**. Many bacterial cells also possess small, independently replicating circles of DNA called plasmids. Plasmids contain only a few genes, usually not essential for the cell's survival. They are best thought of as an excised portion of the bacterial chromosome.

**Ribosomes.** Bacterial ribosomes are smaller than those of eukaryotes and differ in protein and RNA content. Antibiotics such as tetracycline and chloramphenicol can tell the difference—they bind to bacterial ribosomes and block protein synthesis, but do not bind to eukaryotic ribosomes.



(a)



(b)

**FIGURE 34.7**

**Bacterial cells often have complex internal membranes.** This aerobic bacterium (a) exhibits extensive respiratory membranes within its cytoplasm not unlike those seen in mitochondria. This cyanobacterium (b) has thylakoid-like membranes that provide a site for photosynthesis.

**The interior of a bacterial cell may possess internal membranes and a nucleoid region.**

## 34.3 Bacteria exhibit considerable diversity in both structure and metabolism.

### Bacterial Diversity

Bacteria are not easily classified according to their forms, and only recently has enough been learned about their biochemical and metabolic characteristics to develop a satisfactory overall classification comparable to that used for other organisms. Early systems for classifying bacteria relied on differential stains such as the Gram stain. Key bacterial characteristics used in classifying bacteria were:

1. Photosynthetic or nonphotosynthetic
2. Motile or nonmotile
3. Unicellular or multicellular
4. Formation of spores or dividing by transverse binary fission

With the development of genetic and molecular approaches, bacterial classifications can at last reflect true evolutionary relatedness. Molecular approaches include: (1) the analysis of the amino acid sequences of key proteins; (2) the analysis of nucleic acid base sequences by establishing the percent of guanine (G) and cytosine (C); (3) nucleic acid hybridization, which is essentially the mixing of single-stranded DNA from two species and determining the amount of base-pairing (closely related species will have more bases pairing); and (4) nucleic acid sequencing especially looking at ribosomal RNA. Lynn Margulis and Karlene Schwartz proposed a useful classification system that divides bacteria into 16 phyla, according to their most significant features. Table 34.1 outlines some of the major features of the phyla we describe.

#### Kinds of Bacteria

Although they lack the structural complexity of eukaryotes, bacteria have diverse internal chemistries, metabolisms and unique functions. Bacteria have adapted to many kinds of environments, including some you might consider harsh. They have successfully invaded very salty waters, very acidic or alkaline environments, and very hot or cold areas. They are found in hot springs where the temperatures exceed 78°C (172°F) and have been recovered living beneath 435 meters of ice in Antarctica!

Much of what we know of bacteria we have learned from studies in the laboratory. It is important to understand the limits this has placed on our knowledge: we have only been able to study those bacteria that can be cultured in laboratories. Field studies suggest that these represent but a small fraction of the kinds of bacteria that occur in soil, most of which cannot be cultured with existing tech-

niques. We clearly have only scraped the surface of bacterial diversity.

As we learned in chapter 32, bacteria split into two lines early in the history of life, so different in structure and metabolism that they are as different from each other as either is from eukaryotes. The differences are so fundamental that biologists assign the two groups of bacteria to separate domains. One domain, the Archaea, consists of the archaeobacteria (“ancient bacteria”—although they are actually not as ancient as the other bacterial domain). It was once thought that survivors of this group were confined to extreme environments that may resemble habitats on the early earth. However, the use of genetic screening has revealed that these “ancient” bacteria live in nonextreme environments as well. The other more ancient domain, the Bacteria, consists of the eubacteria (“true bacteria”). It includes nearly all of the named species of bacteria.

#### Comparing Archaeobacteria and Eubacteria

Archaeobacteria and eubacteria are similar in that they both have a prokaryotic cellular but they vary considerably at the biochemical and molecular level. There are four key areas in which they differ:

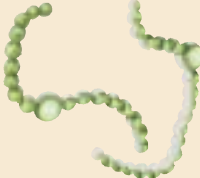
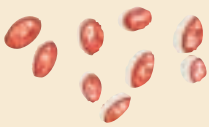
1. **Cell wall.** Both kinds of bacteria typically have cell walls covering the plasma membrane that strengthen the cell. The cell walls of eubacteria are constructed of carbohydrate-protein complexes called peptidoglycan, which link together to create a strong mesh that gives the eubacterial cell wall great strength. The cell walls of archaeobacteria lack peptidoglycan.
2. **Plasma membranes.** All bacteria have plasma membranes with a lipid-bilayer architecture (as described in chapter 6). The plasma membranes of eubacteria and archaeobacteria, however, are made of very different kinds of lipids.
3. **Gene translation machinery.** Eubacteria possess ribosomal proteins and an RNA polymerase that are distinctly different from those of eukaryotes. However, the ribosomal proteins and RNA of archaeobacteria are very similar to those of eukaryotes.
4. **Gene architecture.** The genes of eubacteria are not interrupted by introns, while at least some of the genes of archaeobacteria do possess introns.

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**While superficially similar, bacteria differ from one another in a wide variety of characteristics.**

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Table 34.1 Bacteria

| Major Group                  | Typical Examples                                            |                                                                                     | Key Characteristics                                                                                                                                                                                                                                                                                                                                                           |
|------------------------------|-------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ARCHAEBACTERIA</b>        |                                                             |                                                                                     |                                                                                                                                                                                                                                                                                                                                                                               |
| Archaeobacteria              | Methanogens, thermophiles, halophiles                       |    | Bacteria that are not members of the kingdom Eubacteria. Mostly anaerobic with unusual cell walls. Some produce methane. Others reduce sulfur.                                                                                                                                                                                                                                |
| <b>EUBACTERIA</b>            |                                                             |                                                                                     |                                                                                                                                                                                                                                                                                                                                                                               |
| Actinomycetes                | <i>Streptomyces</i> , <i>Actinomyces</i>                    |    | Gram-positive bacteria. Form branching filaments and produce spores; often mistaken for fungi. Produce many commonly used antibiotics, including streptomycin and tetracycline. One of the most common types of soil bacteria; also common in dental plaque.                                                                                                                  |
| Chemoautotrophs              | Sulfur bacteria, <i>Nitrobacter</i> , <i>Nitrosomonas</i>   |    | Bacteria able to obtain their energy from inorganic chemicals. Most extract chemical energy from reduced gases such as H <sub>2</sub> S (hydrogen sulfide), NH <sub>3</sub> (ammonia), and CH <sub>4</sub> (methane). Play a key role in the nitrogen cycle.                                                                                                                  |
| Cyanobacteria                | <i>Anabaena</i> , <i>Nostoc</i>                             |   | A form of photosynthetic bacteria common in both marine and freshwater environments. Deeply pigmented; often responsible for “blooms” in polluted waters.                                                                                                                                                                                                                     |
| Enterobacteria               | <i>Escherichia coli</i> , <i>Salmonella</i> , <i>Vibrio</i> |  | Gram-negative, rod-shaped bacteria. Do not form spores; usually aerobic heterotrophs; cause many important diseases, including bubonic plague and cholera.                                                                                                                                                                                                                    |
| Gliding and budding bacteria | Myxobacteria, <i>Chondromyces</i>                           |  | Gram-negative bacteria. Exhibit gliding motility by secreting slimy polysaccharides over which masses of cells glide; some groups form upright multicellular structures carrying spores called fruiting bodies.                                                                                                                                                               |
| Pseudomonads                 | <i>Pseudomonas</i>                                          |  | Gram-negative heterotrophic rods with polar flagella. Very common form of soil bacteria; also contain many important plant pathogens.                                                                                                                                                                                                                                         |
| Rickettsias and chlamydias   | <i>Rickettsia</i> , <i>Chlamydia</i>                        |  | Small, gram-negative intracellular parasites. <i>Rickettsia</i> life cycle involves both mammals and arthropods such as fleas and ticks; <i>Rickettsia</i> are responsible for many fatal human diseases, including typhus ( <i>Rickettsia prowazekii</i> ) and Rocky Mountain spotted fever. Chlamydial infections are one of the most common sexually transmitted diseases. |
| Spirochaetes                 | <i>Treponema</i>                                            |  | Long, coil-shaped cells. Common in aquatic environments; a parasitic form is responsible for the disease syphilis.                                                                                                                                                                                                                                                            |

## Bacterial Variation

Bacteria reproduce rapidly, allowing genetic variations to spread quickly through a population. Two processes create variation among bacteria: mutation and genetic recombination.

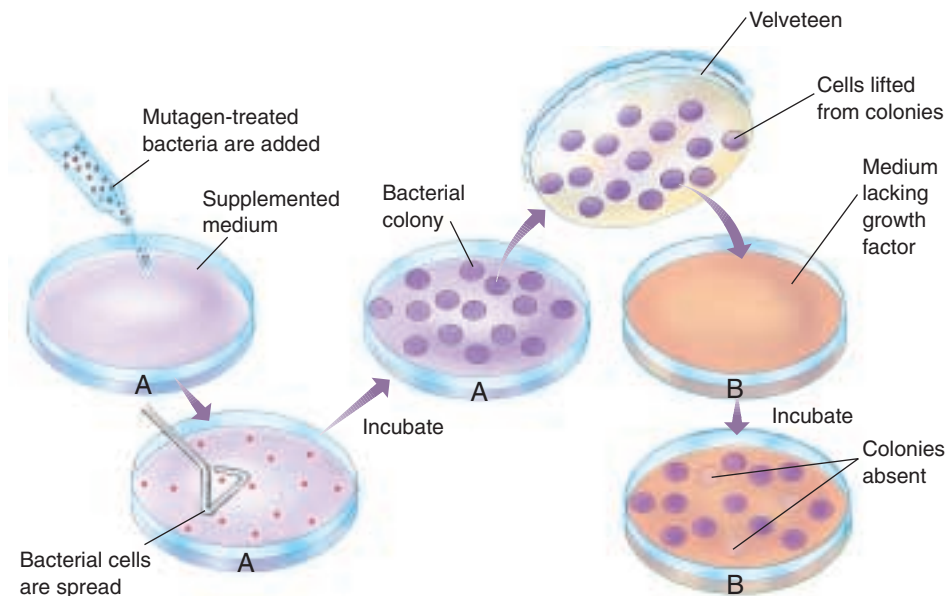
### Mutation

Mutations can arise spontaneously in bacteria as errors in DNA replication occur. Certain factors tend to increase the likelihood of errors occurring such as radiation, ultraviolet light, and various chemicals. In a typical bacterium such as *Escherichia coli* there are about 5000 genes. It is highly probable that one mutation will occur by chance in one out of every million copies of a gene. With 5000 genes in a bacterium, the laws of probability predict that 1 out of every 200 bacteria will have a mutation (figure 34.8). A spoonful of soil typically contains over a billion bacteria and therefore should contain something on the order of 5 million mutant individuals!

With adequate food and nutrients, a population of *E. coli* can double in under 20 minutes. Because bacteria multiply so rapidly, mutations can spread rapidly in a population and can change the characteristics of that population.

The ability of bacteria to change rapidly in response to new challenges often has adverse effects on humans. Recently a number of strains of *Staphylococcus aureus* associated with serious infections in hospitalized patients have appeared, some of them with alarming frequency. Unfortunately, these strains have acquired resistance to penicillin and a wide variety of other antibiotics, so that infections caused by them are very difficult to treat. *Staphylococcus* infections provide an excellent example of the way in which mutation and intensive selection can bring about rapid change in bacterial populations. Such changes have serious medical implications when, as in the case of *Staphylococcus*, strains of bacteria emerge that are resistant to a variety of antibiotics.

Recently, concern has arisen over the prevalence of antibacterial soaps in the marketplace. They are marketed as a means of protecting your family from harmful bacteria; however, it is likely that their routine use will favor bacteria that have mutations making them immune to the antibiotics contained in them. Ultimately, extensive use of antibacterial soaps could have an adverse effect on our ability to treat common bacterial infections.



**FIGURE 34.8**

**A mutant hunt in bacteria.** Mutations in bacteria can be detected by a technique called replica plating, which allows the genetic characteristics of the colonies to be investigated without destroying them. The bacterial colonies, growing on a semisolid agar medium, are transferred from A to B using a sterile velveteen disc pressed on the plate. Plate A has a medium that includes special growth factors, while B has a medium that lacks some of these growth factors. Bacteria that are not mutated can produce their own growth factors and do not require them to be added to the medium. The colonies absent in B were unable to grow on the deficient medium and were thus mutant colonies; they were already present but undetected in A.

### Genetic Recombination

Another source of genetic variation in populations of bacteria is recombination, discussed in detail in chapter 18. Bacterial recombination occurs by the transfer of genes from one cell to another by viruses, or through conjugation. The rapid transfer of newly produced, antibiotic-resistant genes by plasmids has been an important factor in the appearance of the resistant strains of *Staphylococcus aureus* discussed earlier. An even more important example in terms of human health involves the Enterobacteriaceae, the family of bacteria to which the common intestinal bacterium, *Escherichia coli*, belongs. In this family, there are many important pathogenic bacteria, including the organisms that cause dysentery, typhoid, and other major diseases. At times, some of the genetic material from these pathogenic species is exchanged with or transferred to *E. coli* by plasmids. Because of its abundance in the human digestive tract, *E. coli* poses a special threat if it acquires harmful traits.

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**Because of the short generation time of bacteria, mutation and recombination play an important role in generating genetic diversity.**

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## Bacterial Metabolism

Bacteria have evolved many mechanisms to acquire the energy and nutrients they need for growth and reproduction. Many are **autotrophs**, organisms that obtain their carbon from inorganic CO<sub>2</sub>. Autotrophs that obtain their energy from sunlight are called *photoautotrophs*, while those that harvest energy from inorganic chemicals are called *chemoautotrophs*. Other bacteria are **heterotrophs**, organisms that obtain at least some of their carbon from organic molecules like glucose. Heterotrophs that obtain their energy from sunlight are called *photoheterotrophs*, while those that harvest energy from organic molecules are called *chemoheterotrophs*.

**Photoautotrophs.** Many bacteria carry out photosynthesis, using the energy of sunlight to build organic molecules from carbon dioxide. The cyanobacteria use chlorophyll *a* as the key light-capturing pigment and use H<sub>2</sub>O as an electron donor, releasing oxygen gas as a by-product. Other bacteria use bacteriochlorophyll as their pigment and H<sub>2</sub>S as an electron donor, leaving elemental sulfur as the by-product.

**Chemoautotrophs.** Some bacteria obtain their energy by oxidizing inorganic substances. Nitrifiers, for example, oxidize ammonia or nitrite to obtain energy, producing the nitrate that is taken up by plants. This process is called nitrogen fixation and is essential in terrestrial ecosystems as plants can only absorb nitrogen in the form of nitrate. Other bacteria oxidize sulfur, hydrogen gas, and other inorganic molecules. On the dark ocean floor at depths of 2500 meters, entire ecosystems subsist on bacteria that oxidize hydrogen sulfide as it escapes from thermal vents.

**Photoheterotrophs.** The so-called purple nonsulfur bacteria use light as their source of energy but obtain carbon from organic molecules such as carbohydrates or alcohols that have been produced by other organisms.

**Chemoheterotrophs.** Most bacteria obtain both carbon atoms and energy from organic molecules. These include decomposers and most pathogens.

### How Heterotrophs Infect Host Organisms

In the 1980s, researchers studying the disease-causing species of *Yersinia*, a group of gram-negative bacteria, found that they produced and secreted large amounts of proteins. Most proteins secreted by gram-negative bacteria have special signal sequences that allow them to pass through the bacterium's double membrane. This key signal sequence was missing the proteins being secreted by *Yersinia*. These proteins lacked a signal-sequence that two known secretion mechanisms require for transport across the double membrane of gram-negative bacteria. The proteins must therefore have

been secreted by a third type of system, which researchers called the *type III system*.

As more bacteria species are studied, the genes coding for the type III system are turning up in other gram-negative animal pathogens, and even in more distantly related plant pathogens. The genes seem to be more closely related to one another than do the bacteria. Furthermore, the genes are similar to those that code for bacterial flagella.

The role of these proteins is still under investigation, but it seems that some of the proteins are used to transfer other virulence proteins into nearby eukaryotic cells. Given the similarity of the type III genes to the genes that code for flagella, some scientists hypothesize that the transfer proteins may form a flagellum-like structure that shoots virulence proteins into the host cells. Once in the eukaryotic cells, the virulence proteins may determine the host's response to the pathogens. In *Yersinia*, proteins secreted by the type III system are injected into macrophages; they disrupt signals that tell the macrophages to engulf bacteria. *Salmonella* and *Shigella* use their type III proteins to enter the cytoplasm of eukaryotic cells and thus are protected from the immune system of their host. The proteins secreted by *E. coli* alter the cytoskeleton of nearby intestinal eukaryotic cells, resulting in a bulge onto which the bacterial cells can tightly bind.

Currently, researchers are looking for a way to disarm the bacteria using knowledge of their internal machinery, possibly by causing the bacteria to release the virulence proteins before they are near eukaryotic cells. Others are studying the eukaryotic target proteins and the process by which they are affected.

### Bacteria as Plant Pathogens

Many costly diseases of plants are associated with particular heterotrophic bacteria. Almost every kind of plant is susceptible to one or more kinds of bacterial disease. The symptoms of these plant diseases vary, but they are commonly manifested as spots of various sizes on the stems, leaves, flowers, or fruits. Other common and destructive diseases of plants, including blights, soft rots, and wilts, also are associated with bacteria. Fire blight, which destroys pears, apple trees, and related plants, is a well-known example of bacterial disease. Most bacteria that cause plant diseases are members of the group of rod-shaped bacteria known as pseudomonads (see figure 34.2a).

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**While bacteria obtain carbon and energy in many ways, most are chemoheterotrophs. Some heterotrophs have evolved sophisticated ways to infect their hosts.**

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## 34.4 Bacteria are responsible for many diseases but also make important contributions to ecosystems.

### Human Bacterial Diseases

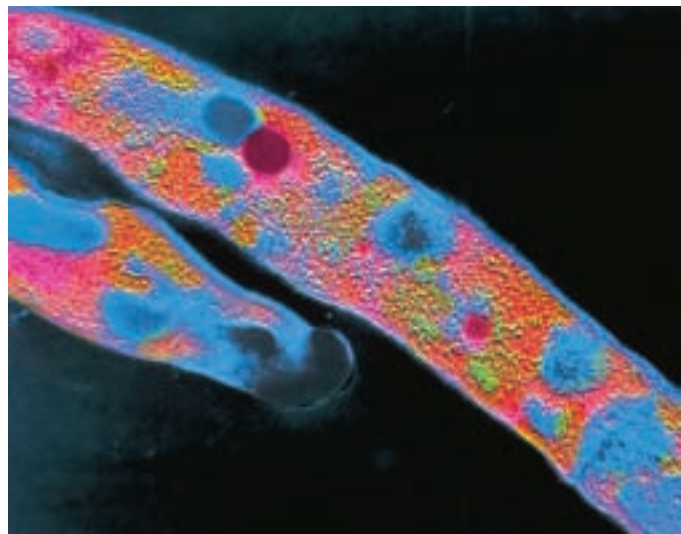
Bacteria cause many diseases in humans, including cholera, leprosy, tetanus, bacterial pneumonia, whooping cough, diphtheria and lyme disease (table 34.2). Members of the genus *Streptococcus* (see figure 34.2b) are associated with scarlet fever, rheumatic fever, pneumonia, and other infections. Tuberculosis (TB), another bacterial disease, is still a leading cause of death in humans. Some of these diseases like TB are mostly spread through the air in water vapor. Other bacterial diseases are dispersed in food or water, including typhoid fever, paratyphoid fever, and bacillary dysentery. Typhus is spread among rodents and humans by insect vectors.

#### Tuberculosis

Tuberculosis has been one of the great killer diseases for thousands of years. Currently, about one-third of all people worldwide are infected with *Mycobacterium tuberculosis*, the tuberculosis bacterium (figure 34.9). Eight million new cases crop up each year, with about 3 million people dying from the disease annually (the World Health Organization predicts 4 million deaths a year by 2005). In fact, in 1997, TB was the leading cause of death from a *single infectious agent* worldwide. Since the mid-1980s, the United States has been experiencing a dramatic resurgence of tuberculosis. TB afflicts the respiratory system and is easily transmitted from person to person through the air. The causes of this current resurgence of TB include social factors such as poverty, crowding, homelessness, and incarceration (these factors have always promoted the spread of TB). The increasing prevalence of HIV infections is also a significant contributing factor. People with AIDS are much more likely to develop TB than people with healthy immune systems.

In addition to the increased numbers of cases—more than 25,000 nationally as of March 1995—there have been alarming outbreaks of multidrug-resistant strains of tuberculosis—strains resistant to the best available anti-TB medications. Multidrug-resistant TB is particularly concerning because it requires much more time to treat, is more expensive to treat, and may prove to be fatal.

The basic principles of TB treatment and control are to make sure all patients complete a full course of medication so that all of the bacteria causing the infection are killed and drug-resistant strains do not develop. Great efforts are being made to ensure that high-risk individuals who are infected but not yet sick receive preventative therapy, which is 90% effective in reducing the likelihood of developing active TB.



**FIGURE 34.9**  
*Mycobacterium tuberculosis*. This color-enhanced image shows the rod-shaped bacterium responsible for tuberculosis in humans.

#### Dental Caries

One human disease we do not usually consider bacterial in origin arises in the film on our teeth. This film, or plaque, consists largely of bacterial cells surrounded by a polysaccharide matrix. Most of the bacteria in plaque are filaments of rod-shaped cells classified as various species of *Actinomyces*, which extend out perpendicular to the surface of the tooth. Many other bacterial species are also present in plaque. Tooth decay, or **dental caries**, is caused by the bacteria present in the plaque, which persists especially in places that are difficult to reach with a toothbrush. Diets that are high in sugars are especially harmful to teeth because lactic acid bacteria (especially *Streptococcus sanguis* and *S. mutans*) ferment the sugars to lactic acid, a substance that reduces the pH of the mouth, causing the local loss of calcium from the teeth. Frequent eating of sugary snacks or sucking on candy over a period of time keeps the pH level of the mouth low resulting in the steady degeneration of the tooth enamel. As the calcium is removed from the tooth, the remaining soft matrix of the tooth becomes vulnerable to attack by bacteria which begin to break down its proteins and tooth decay progresses rapidly. Fluoride makes the teeth more resistant to decay because it retards the loss of calcium. It was first realized that bacteria cause tooth decay when germ-free animals were raised. Their teeth do not decay even if they are fed sugary diets.

| Disease                    | Pathogen                                                       | Vector/Reservoir                          | Epidemiology                                                                                                                                                                                                                                                |
|----------------------------|----------------------------------------------------------------|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Anthrax                    | <i>Bacillus anthracis</i>                                      | Animals, including processed skins        | Bacterial infection that can be transmitted through contact or ingested. Rare except in sporadic outbreaks. May be fatal.                                                                                                                                   |
| Botulism                   | <i>Clostridium botulinum</i>                                   | Improperly prepared food                  | Contracted through ingestion or contact with wound. Produces acute toxic poison; can be fatal.                                                                                                                                                              |
| Chlamydia                  | <i>Chlamydia trachomatis</i>                                   | Humans, STD                               | Urogenital infections with possible spread to eyes and respiratory tract. Occurs worldwide; increasingly common over past 20 years.                                                                                                                         |
| Cholera                    | <i>Vibrio cholerae</i>                                         | Human feces, plankton                     | Causes severe diarrhea that can lead to death by dehydration; 50% peak mortality if the disease goes untreated. A major killer in times of crowding and poor sanitation; over 100,000 died in Rwanda in 1994 during a cholera outbreak.                     |
| Dental caries              | <i>Streptococcus</i>                                           | Humans                                    | A dense collection of this bacteria on the surface of teeth leads to secretion of acids that destroy minerals in tooth enamel—sugar alone will not cause caries.                                                                                            |
| Diphtheria                 | <i>Corynebacterium diphtheriae</i>                             | Humans                                    | Acute inflammation and lesions of mucous membranes. Spread through contact with infected individual. Vaccine available.                                                                                                                                     |
| Gonorrhea                  | <i>Neisseria gonorrhoeae</i>                                   | Humans only                               | STD, on the increase worldwide. Usually not fatal.                                                                                                                                                                                                          |
| Hansen's disease (leprosy) | <i>Mycobacterium leprae</i>                                    | Humans, feral armadillos                  | Chronic infection of the skin; worldwide incidence about 10–12 million, especially in Southeast Asia. Spread through contact with infected individuals.                                                                                                     |
| Lyme disease               | <i>Borrelia burgdorferi</i>                                    | Ticks, deer, small rodents                | Spread through bite of infected tick. Lesion followed by malaise, fever, fatigue, pain, stiff neck, and headache.                                                                                                                                           |
| Peptic ulcers              | <i>Helicobacter pylori</i>                                     | Humans                                    | Originally thought to be caused by stress or diet, most peptic ulcers now appear to be caused by this bacterium; good news for ulcer sufferers as it can be treated with antibiotics.                                                                       |
| Plague                     | <i>Yersinia pestis</i>                                         | Fleas of wild rodents: rats and squirrels | Killed ¼ of the population of Europe in the 14th century; endemic in wild rodent populations of the western U.S. today.                                                                                                                                     |
| Pneumonia                  | <i>Streptococcus</i> ,<br><i>Mycoplasma</i> , <i>Chlamydia</i> | Humans                                    | Acute infection of the lungs, often fatal without treatment                                                                                                                                                                                                 |
| Tuberculosis               | <i>Mycobacterium tuberculosis</i>                              | Humans                                    | An acute bacterial infection of the lungs, lymph, and meninges. Its incidence is on the rise, complicated by the development of new strains of the bacteria that are resistant to antibiotics.                                                              |
| Typhoid fever              | <i>Salmonella typhi</i>                                        | Humans                                    | A systemic bacterial disease of worldwide incidence. Less than 500 cases a year are reported in the U.S. The disease is spread through contaminated water or foods (such as improperly washed fruits and vegetables). Vaccines are available for travelers. |
| Typhus                     | <i>Rickettsia typhi</i>                                        | Lice, rat fleas, humans                   | Historically a major killer in times of crowding and poor sanitation; transmitted from human to human through the bite of infected lice and fleas. Typhus has a peak untreated mortality rate of 70%.                                                       |

## Sexually Transmitted Diseases

A number of bacteria cause sexually transmitted diseases (STDs). Three are particularly important (figure 34.10).

**Gonorrhea.** **Gonorrhea** is one of the most prevalent communicable diseases in North America. Caused by the bacterium *Neisseria gonorrhoeae*, gonorrhea can be transmitted through sexual intercourse or any other sexual contacts in which body fluids are exchanged, such as oral or anal intercourse. Gonorrhea can infect the throat, urethra, cervix, or rectum and can spread to the eyes and internal organs, causing conjunctivitis (a severe infection of the eyes) and arthritic meningitis (an infection of the joints). Left untreated in women, gonorrhea can cause pelvic inflammatory disease (PID), a condition in which the fallopian tubes become scarred and blocked. PID can eventually lead to sterility. The incidence of gonorrhea has been on the decline, but it remains a serious threat.

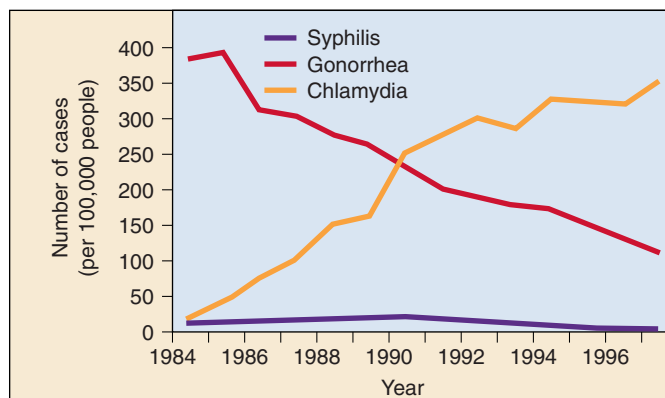
**Syphilis.** **Syphilis**, a very destructive STD, was once prevalent but is now less common due to the advent of blood-screening procedures and antibiotics. Syphilis is caused by a spirochete bacterium, *Treponema pallidum*, that is transmitted during sexual intercourse or through direct contact with an open syphilis sore. The bacterium can also be transmitted from a mother to her fetus, often causing damage to the heart, eyes, and nervous system of the baby.

Once inside the body, the disease progresses in four distinct stages. The first, or primary stage, is characterized by the appearance of a small, painless, often unnoticed sore called a chancre. The chancre resembles a blister and occurs at the location where the bacterium entered the body about three weeks following exposure. This stage of the disease is highly infectious, and an infected person may unwittingly transmit the disease to others.

The second stage of syphilis is marked by a rash, a sore throat, and sores in the mouth. The bacteria can be transmitted at this stage through kissing or contact with an open sore.

The third stage of syphilis is symptomless. This stage may last for several years, and at this point, the person is no longer infectious but the bacteria are still present in the body, attacking the internal organs. The final stage of syphilis is the most debilitating, however, as the damage done by the bacteria in the third stage becomes evident. Sufferers at this stage of syphilis experience heart disease, mental deficiency, and nerve damage, which may include a loss of motor functions or blindness.

**Chlamydia.** Sometimes called the “silent STD,” chlamydia is caused by an unusual bacterium, *Chlamydia trachomatis*, that has both bacterial and viral characteristics. Like a bacterium, it is susceptible to antibiotics, and, like a virus, it depends on its host to replicate its genetic material; it is an obligate internal parasite. The bacterium is transmitted through vaginal, anal, or oral intercourse with an infected person.



**FIGURE 34.10**  
**Trends in sexually transmitted diseases in the U.S.**

Source: CDC, Atlanta, GA.

**Chlamydia** is called the “silent STD” because women usually experience no symptoms until after the infection has become established. In part because of this symptomless nature, the incidence of chlamydia has skyrocketed, increasing by more than sevenfold nationally since 1984. The effects of an established chlamydia infection on the female body are extremely serious. Chlamydia can cause pelvic inflammatory disease (PID), which can lead to sterility.

It has recently been established that infection of the reproductive tract by chlamydia can cause heart disease. *Chlamydia* produce a peptide similar to one produced by cardiac muscle. As the body’s immune system tries to fight off the infection, it recognizes this peptide. The similarity between the bacterial and cardiac peptides confuses the immune system and T cells attack cardiac muscle fibers, inadvertently causing inflammation of the heart and other problems.

Within the last few years, two types of tests for chlamydia have been developed that look for the presence of the bacteria in the discharge from men and women. The treatment for chlamydia is antibiotics, usually tetracycline (penicillin is not effective against chlamydia). Any woman who experiences the symptoms associated with this STD should be tested for the presence of the chlamydia bacterium; otherwise, her fertility may be at risk.

This discussion of STDs may give the impression that sexual activity is fraught with danger, and in a way, it is. It is folly not to take precautions to avoid STDs. The best way to do this is to know one’s sexual partners well enough to discuss the possible presence of an STD. Condom use can also prevent transmission of most of the diseases. Responsibility for protection lies with each individual.

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**Bacterial diseases have a major impact worldwide. Sexually transmitted diseases (STDs) are becoming increasingly widespread among Americans as sexual activity increases.**

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## Importance of Bacteria

Bacteria were largely responsible for creating the properties of the atmosphere and the soil over billions of years. They are metabolically much more diverse than eukaryotes, which is why they are able to exist in such a wide range of habitats. The many autotrophic bacteria—either photosynthetic or chemoautotrophic—make major contributions to the carbon balance in terrestrial, freshwater, and marine habitats. Other heterotrophic bacteria play a key role in world ecology by breaking down organic compounds. One of the most important roles of bacteria in the global ecosystem relates to the fact that only a few genera of bacteria—and no other organisms—have the ability to fix atmospheric nitrogen and thus make it available for use by other organisms (see chapter 28).

Bacteria are very important in many industrial processes. Bacteria are used in the production of acetic acid and vinegar, various amino acids and enzymes, and especially in the fermentation of lactose into lactic acid, which coagulates milk proteins and is used in the production of almost all cheeses, yogurt, and similar products. In the production of bread and other foods, the addition of certain strains of bacteria can lead to the enrichment of the final product with respect to its mix of amino acids, a key factor in its nutritive value. Many products traditionally manufactured using yeasts, such as ethanol, can also

be made using bacteria. The comparative economics of these processes will determine which group of organisms is used in the future. Many of the most widely used antibiotics, including streptomycin, aureomycin, erythromycin, and chloromycetin, are derived from bacteria. Most antibiotics seem to be substances used by bacteria to compete with one another and fungi in nature, allowing one species to exclude others from a favored habitat. Bacteria can also play a part in removing environmental pollutants (figure 34.11)

### Bacteria and Genetic Engineering

Applying genetic engineering methods to produce improved strains of bacteria for commercial use, as discussed in chapter 19, holds enormous promise for the future. Bacteria are under intense investigation, for example, as non-polluting insect control agents. *Bacillus thuringiensis* attacks insects in nature, and improved, highly specific strains of *B. thuringiensis* have greatly increased its usefulness as a biological control agent. Bacteria have also been extraordinarily useful in our attempts to understand genetics and molecular biology.

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**Bacteria play a major role in modern medicine and agriculture, and have profound ecological impact.**

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**FIGURE 34.11**

**Using bacteria to clean up oil spills.** Bacteria can often be used to remove environmental pollutants, such as petroleum hydrocarbons and chlorinated compounds. In areas contaminated by the *Exxon Valdez* oil spill (rocks on the *left*), oil-degrading bacteria produced dramatic results (rocks on the *right*).



## Summary

## Questions

## Media Resources

**34.1 Bacteria are the smallest and most numerous organisms.**

- Bacteria are the oldest and simplest organisms, but they are metabolically much more diverse than all other life-forms combined.
- Bacteria differ from eukaryotes in many ways, the most important of which concern the degree of internal organization within the cell.

1. Structural differences among bacteria are not great. How are different species of bacteria recognized?
2. In what seven ways do prokaryotes differ substantially from eukaryotes?



- Enhancement Chapter: Extremophilic Bacteria, Introduction and Section 1

**34.2 Bacterial cell structure is more complex than commonly supposed.**

- Most bacteria have cell walls that consist of a network of polysaccharide molecules connected by polypeptide cross-links.
- A bacterial cell does not possess specialized compartments or a membrane-bounded nucleus, but it may exhibit a nucleoid region where the bacterial DNA is located.

3. What is the structure of the bacterial cell wall? How does the cell wall differ between gram-positive and gram-negative bacteria? In general, which type of bacteria is more resistant to the action of most antibiotics? Why?



- Characteristics of Bacteria



- Enhancement Chapter: Extremophilic Bacteria, Section 2

**34.3 Bacteria exhibit considerable diversity in both structure and metabolism.**

- The two bacterial kingdoms, Archaeobacteria and Eubacteria, are made up of prokaryotes, with about 5000 species named so far.
- The Archaeobacteria differ markedly from Eubacteria and from eukaryotes in their ribosomal sequences and in other respects.
- Mutation and genetic recombination are important sources of variability in bacteria.
- Many bacteria are autotrophic and make major contributions to the world carbon balance. Others are heterotrophic and play a key role in world ecology by breaking down organic compounds.
- Some heterotrophic bacteria cause major diseases in plants and animals.

4. How do the Archaeobacteria differ from the Eubacteria? What unique metabolism do they exhibit?
5. Why does mutation play such an important role in creating genetic diversity in bacteria?
6. How do heterotrophic bacteria that are successful pathogens overcome the many defenses the human body uses to ward off disease?



- Bacteria Diversity



- Scientists on Science: Marine Biotechnology
- Enhancement Chapter: Extremophilic Bacteria, Section 3

**34.4 Bacteria are responsible for many diseases but also make important contributions to ecosystems.**

- Human diseases caused by heterotrophic bacteria include many fatal diseases that have had major impacts on human history, including tuberculosis, cholera, plague, and typhus.
- Bacteria play vital roles in cycling nutrients within ecosystems. Certain bacteria are the only organisms able to fix atmospheric nitrogen into organic molecules, a process on which all life depends.

7. What are STDs? How are they transmitted? Which STDs are caused by viruses and which are caused by bacteria? Why is the cause of chlamydia unusual?



- Student Research: Improving Antibiotics
- On Science Article: Antibiotic Resistance

# 35

## Protists

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### Concept Outline

#### 35.1 Eukaryotes probably arose by endosymbiosis.

**Endosymbiosis.** Mitochondria and chloroplasts are thought to have arisen by endosymbiosis from aerobic bacteria.

#### 35.2 The kingdom Protista is by far the most diverse of any kingdom.

**The Challenge of Classifying the Protists.** There is no general agreement among taxonomists about how to classify the protists.

**General Biology of the Protists.** Protista contains members exhibiting a wide range of methods of locomotion, nutrition, and reproduction.

#### 35.3 Protists can be categorized into five groups.

**Five Groups of Protists.** The 15 major phyla of protists can be conveniently discussed in seven general groups that share certain characteristics.

**Heterotrophs with No Permanent Locomotor Apparatus.** Amoebas and other sarcodines have no permanent locomotor apparatus.

**Photosynthetic Protists.** The flagellates are photosynthesizers that propel themselves through the water with flagella. Diatoms are photosynthesizers with hard shells of silica. Algae are photosynthetic protists, some are multicellular.

**Heterotrophs with Flagella.** Flagellates propel themselves through the water. Single cells with many cilia, the ciliates possess highly complex and specialized organelles.

**Nonmotile Spore-Formers.** The sporozoans are nonmotile parasites that spread by forming spores.

**Heterotrophs with Restricted Mobility.** Heterotrophs with restricted mobility, molds have cell walls made of carbohydrate.

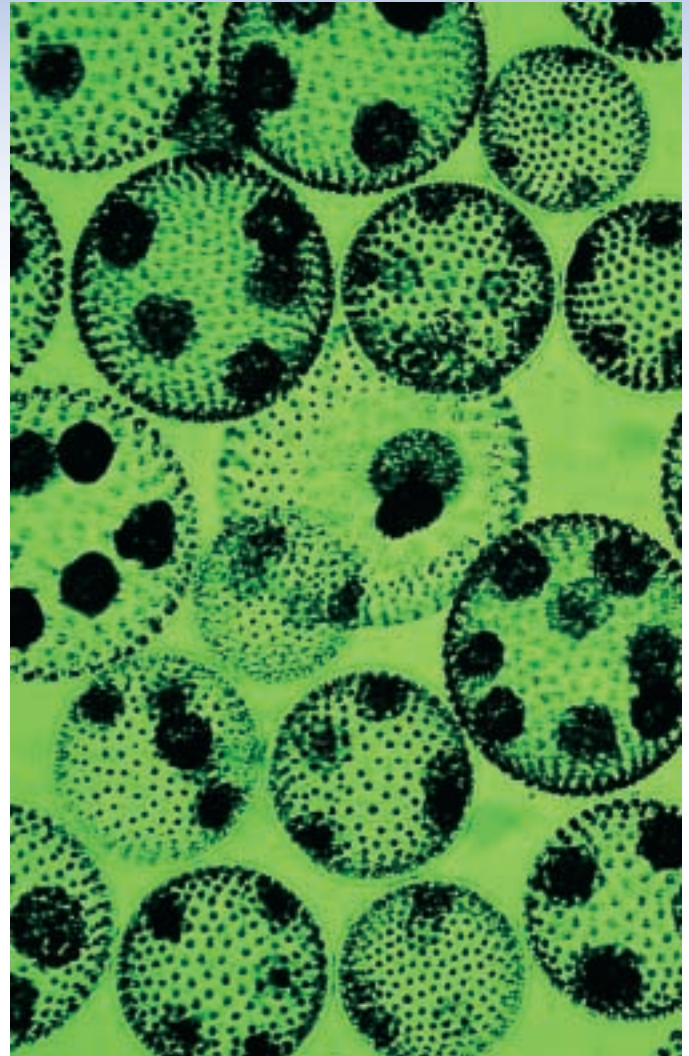


FIGURE 35.1

**Volvox, a colonial protist.** The protists are a large, diverse group of primarily single-celled organisms, a group from which the other three eukaryotic kingdoms each evolved.

For more than half of the long history of life on earth, all life was microscopic in size. The biggest organisms that existed for over 2 billion years were single-celled bacteria fewer than 6 micrometers thick. The first evidence of a different kind of organism is found in tiny fossils in rock 1.5 billion years old. These fossil cells are much larger than bacteria (some as big as 60 micrometers in diameter) and have internal membranes and what appear to be small, membrane-bounded structures. Many have elaborate shapes, and some exhibit spines or filaments. These new, larger fossil organisms mark one of the most important events in the evolution of life, the appearance of a new kind of organism, the eukaryote (figure 35.1). Flexible and adaptable, the eukaryotes rapidly evolved to produce all of the diverse large organisms that populate the earth today, including ourselves—indeed, all organisms other than bacteria are eukaryotes.

## 35.1 Eukaryotes probably arose by endosymbiosis.

### Endosymbiosis

What was the first eukaryote like? We cannot be sure, but a good model is *Pelomyxa palustris*, a single-celled, nonphotosynthetic organism that appears to represent an early stage in the evolution of eukaryotic cells (figure 35.2). The cells of *Pelomyxa* are much larger than bacterial cells and contain a complex system of internal membranes. Although they resemble some of the largest early fossil eukaryotes, these cells are unlike those of any other eukaryote: *Pelomyxa* lacks mitochondria and does not undergo mitosis. Its nuclei divide somewhat as do those of bacteria, by pinching apart into two daughter nuclei, around which new membranes form. Although *Pelomyxa* cells lack mitochondria, two kinds of bacteria living within them may play the same role that mitochondria do in all other eukaryotes. This primitive eukaryote is so distinctive that it is assigned a phylum all its own, Caryoblastea.

Biologists know very little of the origin of *Pelomyxa*, except that in many of its fundamental characteristics it resembles the archaeobacteria far more than the eubacteria. Because of this general resemblance, it is widely assumed that the first eukaryotic cells were nonphotosynthetic descendants of archaeobacteria.

What about the wide gap between *Pelomyxa* and all other eukaryotes? Where did mitochondria come from? Most biologists agree with the theory of **endosymbiosis**, which proposes that mitochondria originated as symbiotic, aerobic (oxygen-requiring) eubacteria (figure 35.3). Symbiosis (Greek, *syn*, “together with” + *bios*, “life”) means living together in close association. Recall from chapter 5 that mitochondria are sausage-shaped organelles 1 to 3 micrometers long, about the same size as most eubacteria. Mitochondria are bounded by two membranes. Aerobic eubacteria are thought to have become mitochondria when they were engulfed by ancestral eukaryotic cells, much like *Pelomyxa*, early in the history of eukaryotes.

The most similar eubacteria to mitochondria today are the nonsulfur purple bacteria, which are able to carry out oxidative metabolism (described in chapter 9). In mitochondria, the outer membrane is smooth and is thought to be derived from the endoplasmic reticulum of the host cell, which, like *Pelomyxa*, may have already



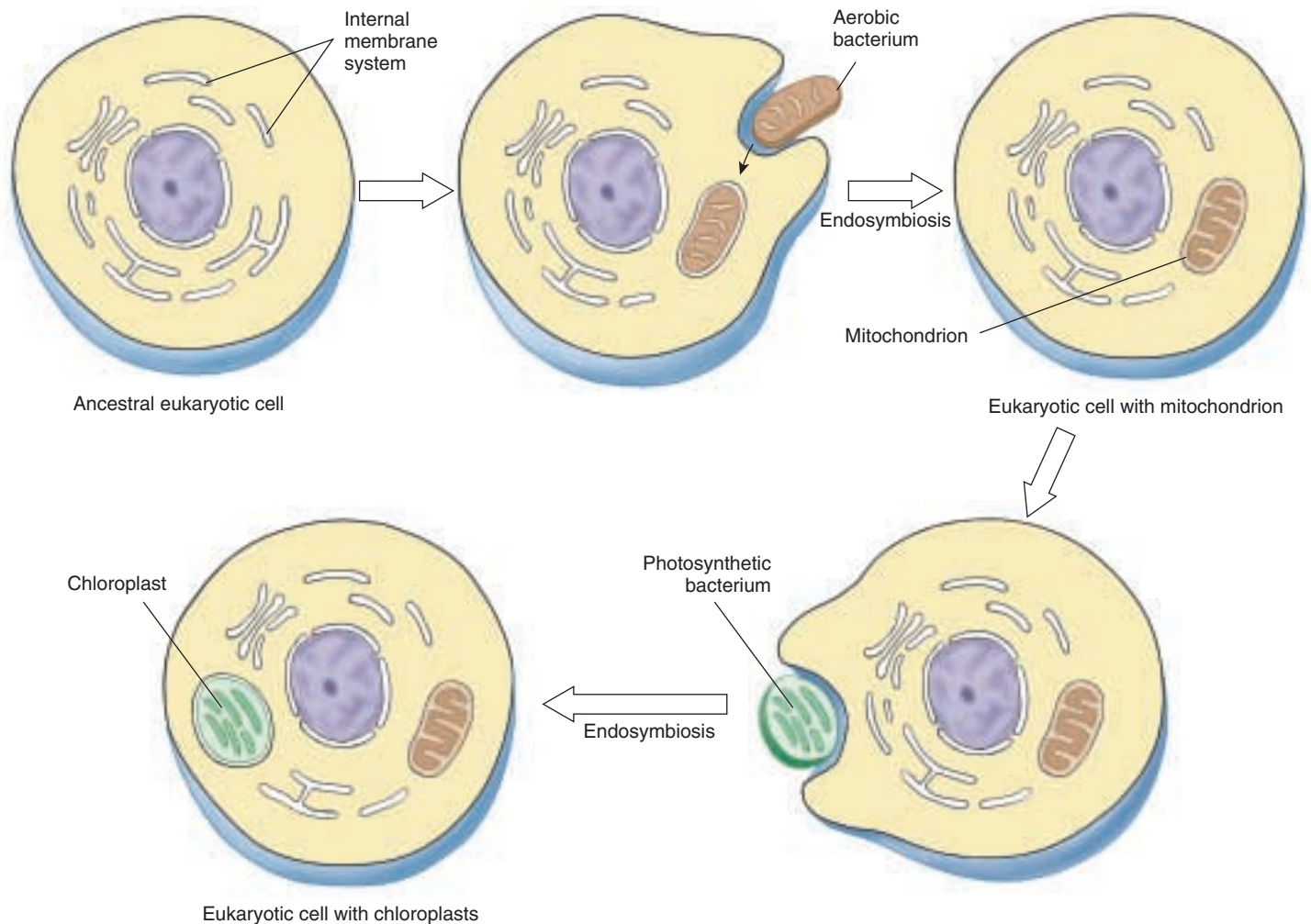
**FIGURE 35.2**  
*Pelomyxa palustris*. This unique, amoeba-like protist lacks mitochondria and does not undergo mitosis. *Pelomyxa* may represent a very early stage in the evolution of eukaryotic cells.

contained a complex system of internal membranes. The inner membrane of mitochondria is folded into numerous layers, resembling the folded membranes of nonsulfur purple bacteria; embedded within this membrane are the proteins that carry out oxidative metabolism. The engulfed bacteria became the interior portion of the mitochondria we see today. Host cells were unable to carry out the Krebs cycle or other metabolic reactions necessary for living in an atmosphere that contained increasing amounts of oxygen before they had acquired these bacteria.

During the billion and a half years in which mitochondria have existed as endosymbionts within eukaryotic cells, most of their genes have been transferred to the chromosomes of the host cells—but not all. Each mitochondrion still has its own genome, a circular, closed molecule of DNA similar to that found in eubacteria, on which is located genes encoding the essential proteins of oxidative metabolism. These genes are transcribed within the mitochondrion, using mitochondrial ribosomes that are smaller than those of eukaryotic cells, very much like bacterial ribosomes in size and structure. Mitochondria divide by simple fission, just as bacteria do, replicating and sorting their DNA much as bacteria do. However, nuclear genes direct the process, and mitochondria cannot be grown outside of the eukaryotic cell, in cell-free culture.

The theory of endosymbiosis has had a controversial history but has now been accepted by all but a few biologists. The evidence supporting the theory is so extensive that in this text we will treat it as established.

What of mitosis, the other typical eukaryotic process that *Pelomyxa* lacks? The mechanism of mitosis, now so common among eukaryotes, did not evolve all at once. Traces of very different, and possibly intermediate, mechanisms survive today in some of the eukaryotes. In fungi and some groups of protists, for example, the nuclear membrane does not dissolve and mitosis is confined to the nucleus. When mitosis is complete in these organisms, the nucleus divides into two daughter nuclei, and only then does the rest of the cell divide. This separate nuclear division phase of mitosis does not occur in most protists, or in plants or animals. We do not know if it represents an intermediate step on the evolutionary journey to the form of mitosis that is characteristic of most



**FIGURE 35.3**

**The theory of endosymbiosis.** Scientists propose that ancestral eukaryotic cells, which already had an internal system of membranes, engulfed aerobic eubacteria, which then became mitochondria in the eukaryotic cell. Chloroplasts may also have originated this way, with eukaryotic cells engulfing photosynthetic eubacteria.

eukaryotes today or if it is simply a different way of solving the same problem. There are no fossils in which we can see the interiors of dividing cells well enough to be able to trace the history of mitosis.

### Endosymbiosis Is Not Rare

Many eukaryotic cells contain other endosymbiotic bacteria in addition to mitochondria. Plants and algae contain chloroplasts, bacteria-like organelles that were apparently derived from symbiotic photosynthetic bacteria. Chloroplasts have a complex system of inner membranes and a circle of DNA. Centrioles, organelles associated with the assembly of microtubules, resemble in many respects spirochete bacteria, and they contain bacteria-like DNA involved in the production of their structural proteins.

While all mitochondria are thought to have arisen from a single symbiotic event, it is difficult to be sure with chloro-

plasts. Three biochemically distinct classes of chloroplasts exist, each resembling a different bacterial ancestor. Red algae possess pigments similar to those of cyanobacteria; plants and green algae more closely resemble the photosynthetic bacteria *Prochloron*; while brown algae and other photosynthetic protists resemble a third group of bacteria. This diversity of chloroplasts has led to the widely held belief that eukaryotic cells acquired chloroplasts by symbiosis at least three different times. Recent comparisons of chloroplast DNA sequences, however, suggest a single origin of chloroplasts, followed by very different postendosymbiotic histories. For example, in each of the three main lines, different genes became relocated to the nucleus, lost, or modified.

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**The theory of endosymbiosis proposes that mitochondria originated as symbiotic aerobic eubacteria.**

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## 35.2 The kingdom Protista is by far the most diverse of any kingdom.

### The Challenge of Classifying the Protists

Protists are the most diverse of the four kingdoms in the domain Eukaryota. The kingdom Protista contains many unicellular, colonial, and multicellular groups. Probably the most important statement we can make about the kingdom Protista is that it is an artificial group; as a matter of convenience, single-celled eukaryotic organisms have typically been grouped together into this kingdom. This lumps many very different and only distantly related forms together. The “single-kingdom” classification of the Protista is not representative of any evolutionary relationships. The phyla of protists are, with very few exceptions, only distantly related to one another.

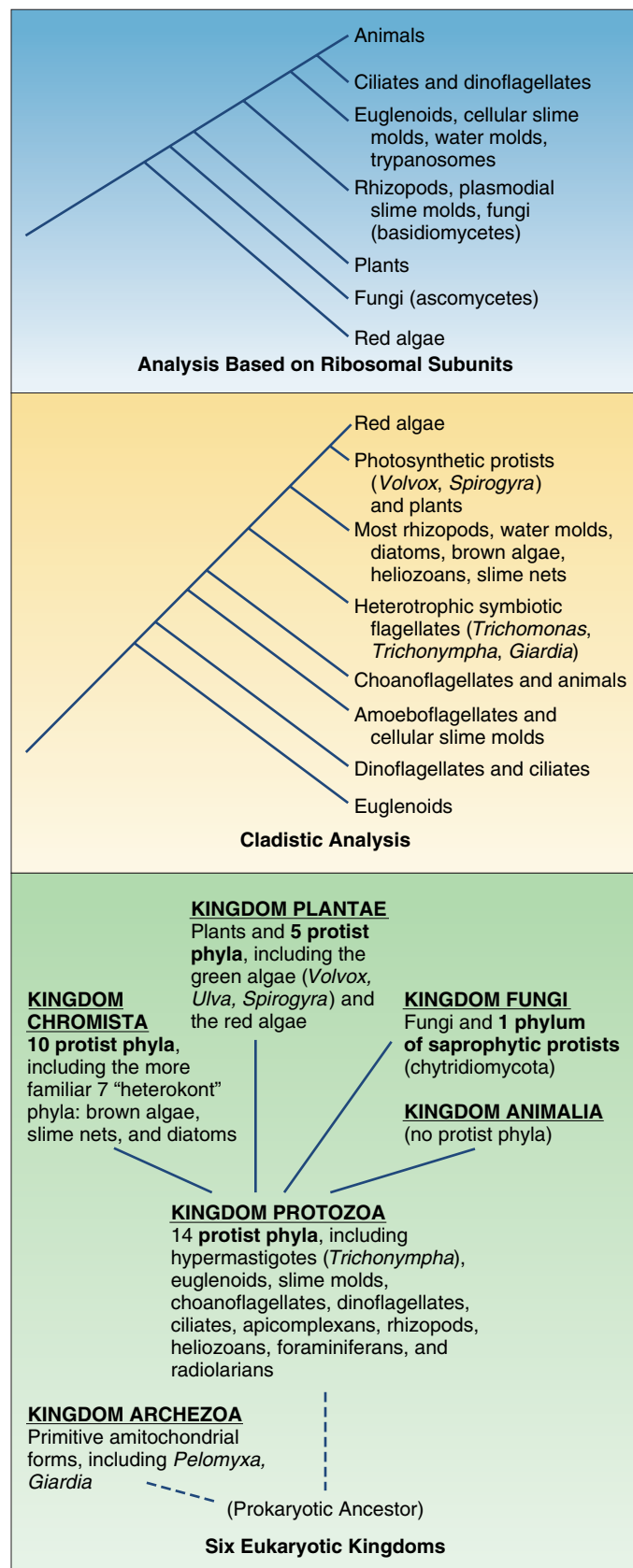
New applications of a wide variety of molecular methods are providing important insights into the relationships among the protists. Of all the groups of organisms biologists study, protists are probably in the greatest state of flux when it comes to classification. There is little consensus, even among experts, as to how the different kinds of protists should be classified. Are they a single, very diverse kingdom, or are they better considered as several different kingdoms, each of equal rank with animals, plants, and fungi?

Because the Protista are still predominantly considered part of one diverse, nonunified group, that is how we will treat them in this chapter, bearing in mind that biologists are rapidly gaining a better understanding of the evolutionary relationships among members of the kingdom Protista (figure 35.4). It seems likely that within a few years, the traditional kingdom Protista will be replaced by another more illuminating arrangement.

**The taxonomy of the protists is in a state of flux as new information shapes our understanding of this kingdom.**

FIGURE 35.4

**The challenge of protistan classification.** Three different suggestions for protistan classification are presented, each adapted from the work of an authority in the field. Their great differences attest to the wide divergence of opinion within the field itself. The classification on the top is based on molecular variation in ribosomal subunits. The classification in the middle presents a cladistic analysis of a broad range of characters (including ribosomal subunits). The classification on the bottom outlines a more revolutionary reevaluation of the protists. Comparison of the three schemes reveals that some groups are commonly recognized as related (like ciliates and dinoflagellates), while the classification of others (like *Giardia*) is clearly in a state of flux.



## General Biology of the Protists

Protists are united on the basis of a single negative characteristic: they are not fungi, plants, or animals. In all other respects they are highly variable with no uniting features. Many are unicellular (figure 35.5), but there are numerous colonial and multicellular groups. Most are microscopic, but some are as large as trees. They represent all symmetries, and exhibit all types of nutrition.

### The Cell Surface

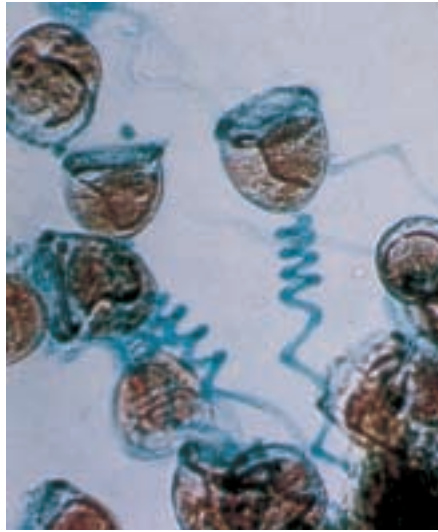
Protists possess a varied array of cell surfaces. Some protists, like amoebas, are surrounded only by their plasma membranes. All other protists have a plasma membrane but some, like algae and molds, are encased within strong cell walls. Still others, like diatoms and forams, secrete glassy shells of silica.

### Locomotor Organelles

Movement in protists is also accomplished by diverse mechanisms. Protists move chiefly by either flagellar rotation or pseudopodial movement. Many protists wave one or more flagella to propel themselves through the water, while others use banks of short, flagella-like structures called cilia to create water currents for their feeding or propulsion. Pseudopodia are the chief means of locomotion among amoeba, whose pseudopods are large, blunt extensions of the cell body called lobopodia. Other related protists extend thin, branching protrusions called filopodia. Still other protists extend long, thin pseudopodia called axopodia supported by axial rods of microtubules. Axopodia can be extended or retracted. Because the tips can adhere to adjacent surfaces, the cell can move by a rolling motion, shortening the axopodia in front and extending those in the rear.

### Cyst Formation

Many protists with delicate surfaces are successful in quite harsh habitats. How do they manage to survive so well? They survive inhospitable conditions by forming **cysts**. A cyst is a dormant form of a cell with a resistant outer covering in which cell metabolism is more or less completely shut down. Not all cysts are so sturdy. Vertebrate parasitic amoebae, for example, form cysts that are quite resistant to gastric acidity, but will not tolerate desiccation or high temperature.



**FIGURE 35.5**  
**A unicellular protist.** The protist kingdom is a catch-all kingdom for many different groups of unicellular organisms, such as this *Vorticella* (phylum Ciliophora), which is heterotrophic, feeds on bacteria, and has a retractable stalk.

### Nutrition

Protists employ every form of nutritional acquisition except chemoautotrophic, which has so far been observed only in bacteria. Some protists are photosynthetic autotrophs and are called **phototrophs**. Others are heterotrophs that obtain energy from organic molecules synthesized by other organisms. Among heterotrophic protists, those that ingest visible particles of food are called **phagotrophs**, or **holozoic feeders**. Those ingesting food in soluble form are called **osmotrophs**, or **saprophytic feeders**.

Phagotrophs ingest food particles into intracellular vesicles called **food vacuoles** or **phagosomes**. Lysosomes fuse with the food vacuoles, introducing enzymes that digest the food particles within. As the digested molecules are absorbed across the vacuolar membrane, the food vacuole becomes progressively smaller.

### Reproduction

Protists typically reproduce asexually, reproducing sexually only in times of stress. Asexual reproduction involves mitosis, but the process is often somewhat different from the mitosis that occurs in multicellular animals. The nuclear membrane, for example, often persists throughout mitosis, with the microtubular spindle forming within it. In some groups, asexual reproduction involves spore formation, in others fission. The most common type of fission is **binary**, in which a cell simply splits into nearly equal halves. When the progeny cell is considerably smaller than its parent, and then grows to adult size, the fission is called **budding**. In multiple fission, or **schizogony**, common among some protists, fission is preceded by several nuclear divisions, so that fission produces several individuals almost simultaneously.

Sexual reproduction also takes place in many forms among the protists. In ciliates and some flagellates, **gametic meiosis** occurs just before gamete formation, as it does in metazoans. In the sporozoans, **zygotic meiosis** occurs directly *after* fertilization, and all the individuals that are produced are haploid until the next zygote is formed. In algae, there is **intermediary meiosis**, producing an alternation of generations similar to that seen in plants, with significant portions of the life cycle spent as haploid as well as diploid.

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Protists exhibit a wide range of forms, locomotion, nutrition and reproduction.

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## 35.3 Protists can be categorized into five groups.

### Five Groups of Protists

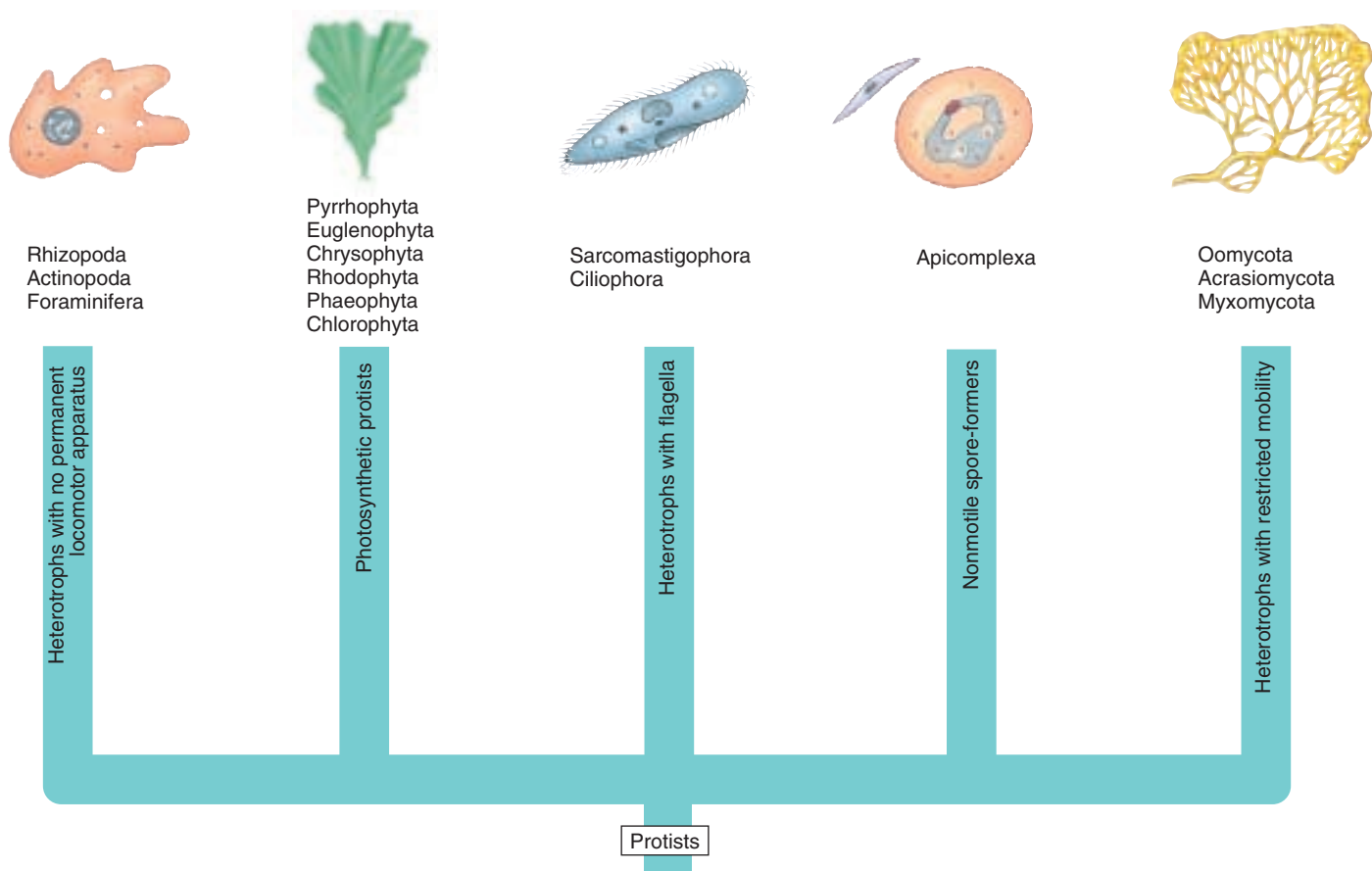
There are some 15 major phyla of protists. It is difficult to encompass their great diversity with any simple scheme. Traditionally, texts have grouped them artificially (as was done in the nineteenth century) into photosynthesizers (algae), heterotrophs (protozoa), and absorbers (funguslike protists).

In this text, we will group the protists into five general groups according to some of the major shared characteristics (figure 35.6). These are characteristics that taxonomists are using today in broad attempts to classify the kingdom Protista. These include (1) the presence or absence and type of cilia or flagella, (2) the presence and kinds of pigments, (3) the type of mitosis, (4) the kinds of cristae present in the mitochondria, (5) the molecular genetics of the ribosomal “S” subunit, (6) the kind of inclusions the protist may have, (7) overall body form (amoeboid, coccoid, and so forth), (8) whether the protist has any kind of shell or other body “armor,” and

(9) modes of nutrition and movement. These represent only some of the characters used to define phylogenetic relationships.

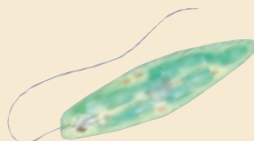
The five criteria we have chosen to define groups are not the only ones that might be chosen, and there is no broad agreement among biologists as to which set of criteria is preferable. As molecular analysis gives us a clearer picture of the phylogenetic relationships among the protists, more evolutionarily suitable groupings will without a doubt replace the one represented here. Table 35.1 summarizes some of the general characteristics and groupings of the 15 major phyla of protists. It is important to remember that while the phyla of protists discussed here are generally accepted taxa, the larger groupings of phyla presented are functional groupings.

**The 15 major protist phyla can be conveniently categorized into five groups according to major shared characteristics.**



**FIGURE 35.6**  
**Five general groups of protists.** This text presents the 15 major phyla of protists in five groups that share major characteristics.

Table 35.1 Kinds of Protists

| Group                                              | Phylum            | Typical Examples               |                                                                                      | Key Characteristics                                                                                                                                                  |
|----------------------------------------------------|-------------------|--------------------------------|--------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| HETEROTROPHS WITH NO PERMANENT LOCOMOTOR APPARATUS |                   |                                |                                                                                      |                                                                                                                                                                      |
| Amoebas                                            | Rhizopoda         | <i>Amoeba</i>                  |    | Move by pseudopodia                                                                                                                                                  |
| Radiolarians                                       | Actinopoda        | Radiolarians                   |                                                                                      | Glassy skeletons; needlelike pseudopods                                                                                                                              |
| Forams                                             | Foraminifera      | Forams                         |                                                                                      | Rigid shells; move by protoplasmic streaming                                                                                                                         |
| PHOTOSYNTHETIC PROTISTS                            |                   |                                |                                                                                      |                                                                                                                                                                      |
| Dinoflagellates                                    | Pyrrhophyta       | Red tides                      |    | Photosynthetic; unicellular; two flagella; contain chlorophylls <i>a</i> and <i>b</i>                                                                                |
| Euglenoids                                         | Euglenophyta      | <i>Euglena</i>                 |                                                                                      | Some photosynthetic; others heterotrophic; unicellular; contain chlorophylls <i>a</i> and <i>b</i> or none                                                           |
| Diatoms                                            | Chrysophyta       | <i>Diatoma</i>                 |    | Unicellular; manufacture the carbohydrate chrysolaminarin; unique double shells of silica; contain chlorophylls <i>a</i> and <i>c</i>                                |
| Golden algae                                       | Chrysophyta       | Golden algae                   |                                                                                      | Unicellular, but often colonial; manufacture the carbohydrate chrysolaminarin; contain chlorophylls <i>a</i> and <i>c</i>                                            |
| Red                                                | Rhodophyta        | Coralline algae                |   | Most multicellular; contain chlorophyll <i>a</i> and a red pigment                                                                                                   |
| Brown                                              | Phaeophyta        | Kelp                           |                                                                                      | Multicellular; contain chlorophylls <i>a</i> and <i>c</i>                                                                                                            |
| Green                                              | Chlorophyta       | <i>Chlamydomonas</i>           |                                                                                      | Unicellular or multicellular; contain chlorophylls <i>a</i> and <i>b</i>                                                                                             |
| HETEROTROPHS WITH FLAGELLA                         |                   |                                |                                                                                      |                                                                                                                                                                      |
| Zoomastigotes                                      | Sarcomastigophora | Trypanosomes                   |  | Heterotrophic; unicellular                                                                                                                                           |
| Ciliates                                           | Ciliophora        | <i>Paramecium</i>              |                                                                                      | Heterotrophic unicellular protists with cells of fixed shape possessing two nuclei and many cilia; many cells also contain highly complex and specialized organelles |
| NONMOTILE SPORE-FORMERS                            |                   |                                |                                                                                      |                                                                                                                                                                      |
| Sporozoans                                         | Apicomplexa       | <i>Plasmodium</i>              |  | Nonmotile; unicellular; the apical end of the spores contains a complex mass of organelles                                                                           |
| HETEROTROPHS WITH RESTRICTED MOBILITY              |                   |                                |                                                                                      |                                                                                                                                                                      |
| Water molds                                        | Oomycota          | Water molds, rusts, and mildew |  | Terrestrial and freshwater                                                                                                                                           |
| Cellular slime molds                               | Acrasiomycota     | <i>Dictyostelium</i>           |                                                                                      | Colonial aggregations of individual cells; most closely related to amoebas                                                                                           |
| Plasmodial slime molds                             | Myxomycota        | <i>Fuligo</i>                  |                                                                                      | Stream along as a multinucleate mass of cytoplasm                                                                                                                    |

## Heterotrophs with No Permanent Locomotor Apparatus

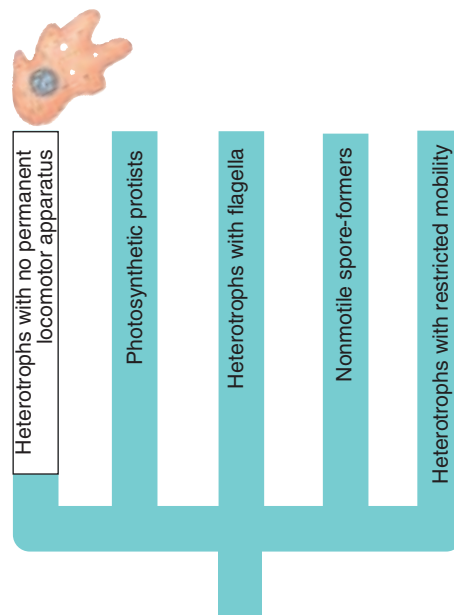
The largest of the five general groups of protists are primarily unicellular organisms with amoeboid forms. There are three principle phyla: the forams and the radiolarians have carbonate shells and the rhizopods lack shells.

### Rhizopoda: The Amoebas

Hundreds of species of amoebas are found throughout the world in both fresh and salt waters. They are also abundant in soil. Many kinds of amoebas are parasites of animals. Reproduction in amoebas occurs by fission, or the direct division into two cells of equal volume. Amoebas of the phylum Rhizopoda lack cell walls, flagella, meiosis, and any form of sexuality. They do undergo mitosis, with a spindle apparatus that resembles that of other eukaryotes.

Amoebas move from place to place by means of their **pseudopods**, from the Greek words for “false” and “foot” (figure 35.7). Pseudopods are flowing projections of cytoplasm that extend and pull the amoeba forward or engulf food particles, a process called cytoplasmic streaming. An amoeba puts a pseudopod forward and then flows into it. Microfilaments of actin and myosin similar to those found in muscles are associated with these movements. The pseudopodia can form at any point on the cell body so that it can move in any direction.

Some kinds of amoebas form resistant cysts. In parasitic species such as *Entamoeba histolytica*, which causes amoebic dysentery, cysts enable the amoebas to resist digestion by their animal hosts. Mitotic division takes place within the cysts, which ultimately rupture and release four, eight, or even more amoebas within the digestive tracts of their host animals. The primary infection takes place in the intestine, but it often moves into the liver and other parts of the body. The cysts are dispersed in the feces and may be transmitted from person to person in infected food or water, or by flies. It is estimated that up to 10 million people in the United States have infections of parasitic amoebas, and some 2 million show symptoms of the disease, ranging from abdominal discomfort with



**FIGURE 35.7**  
*Amoeba proteus*. This relatively large amoeba is commonly used in teaching and for research in cell biology. The projections are pseudopods; an amoeba moves by flowing into them. The nucleus of the amoeba is plainly visible.

slight diarrhea to much more serious conditions. In some tropical areas, more than half of the population may be infected. The spread of amoebic dysentery can be limited by proper sanitation and hygiene.

### Actinopoda: The Radiolarians

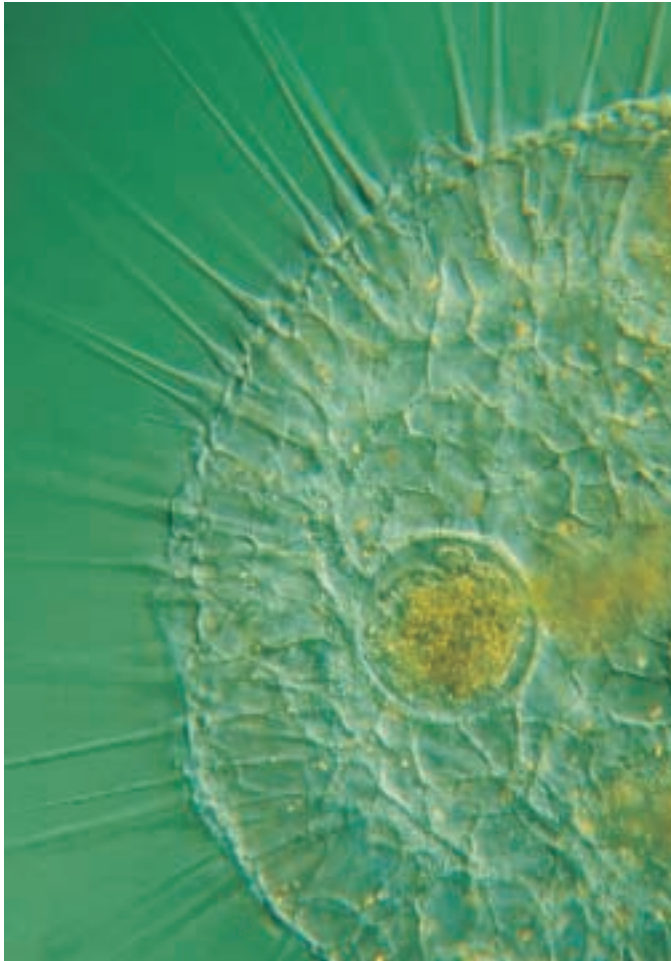
The pseudopodia of amoeboid cells give them truly amorphous bodies. One group, however, have more distinct structures. Members of the phylum Actinopoda, often called radiolarians, secrete glassy exoskeletons made of silica. These skeletons give the unicellular organisms a distinct shape, exhibiting either bilateral or radial symmetry. The shells of different species form many elaborate and beautiful shapes and its pseudopodia extrude outward along spiky projections of the skeleton (figure 35.8). Microtubules support these cytoplasmic projections.

### Foraminifera: Forams

Members of the phylum Foraminifera are heterotrophic marine protists. They range in diameter from about 20 micrometers to several centimeters. Characteristic of the group are pore-studded shells (called **tests**) composed of organic materials usually reinforced with grains of inorganic matter. These grains may be calcium carbonate, sand, or even plates from the shells of echinoderms or spicules (minute needles of calcium carbonate) from sponge skeletons. Depending on the building materials they use, foraminifera—often informally called “forams”—may have shells of very different appearance. Some of them are brilliantly colored red, salmon, or yellow-brown.

Most foraminifera live in sand or are attached to other organisms, but two families consist of free-floating planktonic organisms. Their tests may be single-chambered but more often are multichambered, and they sometimes have a spiral shape resembling that of a tiny snail. Thin cytoplasmic projections called **podia** emerge through openings in the tests (figure 35.9). Podia are used for swimming, gathering materials for the tests, and feeding. Forams eat a wide variety of small organisms.

The life cycles of foraminifera are extremely complex, involving an alternation between haploid and diploid generations (sporic meiosis). Forams have contributed massive



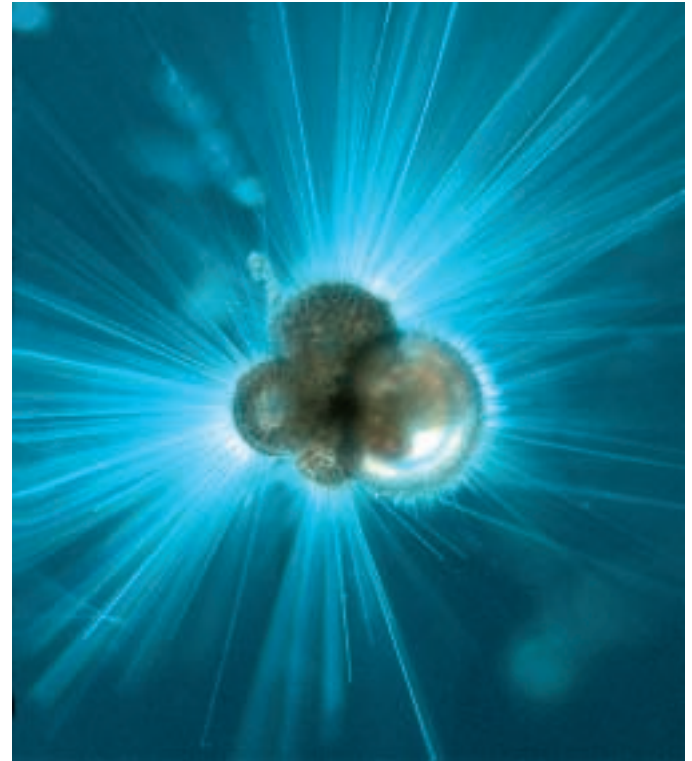
**FIGURE 35.8**  
*Actinosphaerium*, a protist of the phylum Actinopoda (300×). This amoeba-like radiolarian has striking needlelike pseudopods.

accumulations of their tests to the fossil record for more than 200 million years. Because of the excellent preservation of their tests and the often striking differences among them, forams are very important as geological markers. The pattern of occurrence of different forams is often used as a guide in searching for oil-bearing strata. Limestones all over the world, including the famous white cliffs of Dover in southern England, are often rich in forams (figure 35.10).

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Amoebas, radiolarians, and forams are unicellular, heterotrophic protists that lack cell walls, flagella, meiosis, and sexuality. Amoebas move from place to place by means of extensions called pseudopodia. The pore-studded tests, or shells, of the forams have openings through which podia extend that are used for locomotion.

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**FIGURE 35.9**  
A representative of the Foraminifera (90×). A living foram with podia, thin cytoplasmic projections that extend through pores in the calcareous test, or shell, of the organism.



**FIGURE 35.10**  
**White cliffs of Dover.** The limestone that forms these cliffs is composed almost entirely of fossil shells of protists, including coccolithophores (a type of algae) and foraminifera.

## Photosynthetic Protists

### Pyrrophyta: The Dinoflagellates

The dinoflagellates consist of about 2100 known species of primarily unicellular, photosynthetic organisms, most of which have two flagella. A majority of the dinoflagellates are marine, and they are often abundant in the plankton, but some occur in fresh water. Some planktonic dinoflagellates are luminous and contribute to the twinkling or flashing effects that we sometimes see in the sea at night, especially in the tropics.

The flagella, protective coats, and biochemistry of dinoflagellates are distinctive, and they do not appear to be directly related to any other phylum. Plates made of a cellulose-like material encase the cells. Grooves form at the junctures of these plates and the flagella are usually located within these grooves, one encircling the body like a belt, and the other perpendicular to it. By beating in their respective grooves, these flagella cause the dinoflagellate to rotate like a top as it moves. The dinoflagellates that are clad in stiff cellulose plates, often encrusted with silica, may have a very unusual appearance (figure 35.11). Most have chlorophylls *a* and *c*, in addition to carotenoids, so that in the biochemistry of their chloroplasts, they resemble the diatoms and the brown algae, possibly acquiring such chloroplasts by forming endosymbiotic relationships with members of those groups.

Some dinoflagellates occur as symbionts in many other groups of organisms, including jellyfish, sea anemones, mollusks, and corals. When dinoflagellates grow as symbionts within other cells, they lack their characteristic cellulose plates and flagella, appearing as spherical, golden-brown globules in their host cells. In such a state they are called **zooxanthellae**. Photosynthetic zooxanthellae provide their hosts with nutrients. It is the photosynthesis conducted by zooxanthellae that



**FIGURE 35.11**  
**Some dinoflagellates: *Noctiluca*, *Ptychodiscus*, *Ceratium*, and *Gonyaulax*.**  
*Noctiluca*, which lacks the heavy cellulose armor characteristic of most dinoflagellates, is one of the bioluminescent organisms that causes the waves to sparkle in warm seas. In the other three genera, the shorter, encircling flagellum is seen in its groove, with the longer one projecting away from the body of the dinoflagellate. (Not drawn to scale.)

makes coral reefs one of the most productive ecosystems on earth. Corals primarily live in warm tropical seas that are typically extremely low in nutrients; without the aid of their photosynthetic endosymbionts, they would not be able to form large reefs in the nutrient-poor environment. Most of the carbon that the zooxanthellae fix is translocated to the host corals.

The poisonous and destructive “red tides” that occur frequently in coastal areas are often associated with great population explosions, or “blooms,” of dinoflagellates. The pigments in the individual, microscopic cells of the dinoflagellates are responsible for the color of the water. Red tides have a profound, detrimental effect on the fishing industry in the United States. Some 20 species of dinoflagellates are known to produce powerful toxins that inhibit the diaphragm and cause respiratory failure in many vertebrates. When the toxic dinoflagellates are abundant, fishes, birds, and marine mammals may die in large numbers.

More recently, a particularly dangerous toxic dinoflagellate called *Pfiesteria piscicida* is reported to be a carnivorous, ambush predator. During blooms, it stuns fish with its toxin and then feeds on the prey’s body fluids.

Dinoflagellates reproduce primarily by asexual cell division. But sexual reproduction has been reported to occur under starvation conditions. They have a unique form of mitosis in which the permanently condensed chromosomes divide longitudinally within the confines of a permanent nuclear envelope. After the numerous chromosomes duplicate, the nucleus divides into two daughter nuclei. Also the dinoflagellate chromosome is unique among eukaryotes in that the DNA is not complexed with histone proteins. In all other eukaryotes, the chromosomal DNA is complexed with histones to form nucleosomes, which represents the first order of DNA packaging in the nucleus. How dinoflagellates are able to maintain distinct chromosomes without histones and nucleosomes remains a mystery.

## Euglenophyta: The Euglenoids

Most of the approximately 1000 known species of euglenoids live in fresh water. The members of this phylum clearly illustrate the impossibility of distinguishing “plants” from “animals” among the protists. About a third of the approximately 40 genera of euglenoids have chloroplasts and are fully autotrophic; the others lack chloroplasts, ingest their food, and are heterotrophic. These organisms are not significantly different from some groups of zoomastigotes (see next section), and many biologists believe that the two phyla should be merged into one.

Some euglenoids with chloroplasts may become heterotrophic if the organisms are kept in the dark; the chloroplasts become small and nonfunctional. If they are put back in the light, they may become green within a few hours. Normally photosynthetic euglenoids may sometimes feed on dissolved or particulate food.

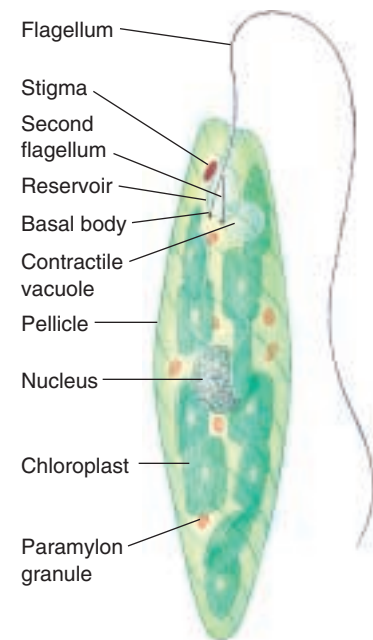
Individual euglenoids range from 10 to 500 micrometers long and are highly variable in form. Interlocking proteinaceous strips arranged in a helical pattern form a flexible structure called the **pellicle**, which lies within the cell membrane of the euglenoids. Because its pellicle is flexible, a euglenoid is able to change its shape. Reproduction in this phylum occurs by mitotic cell division. The nuclear envelope remains intact throughout the process of mitosis. No sexual reproduction is known to occur in this group.

In *Euglena* (figure 35.12), the genus for which the phylum is named, two flagella are attached at the base of a flask-shaped opening called the **reservoir**, which is located at the anterior end of the cell. One of the flagella is long and has a row of very fine, short, hairlike projections along one side. A second, shorter flagellum is located within the reservoir but does not emerge from it. Contractile vacuoles collect excess water from all parts of the organism and empty it into the reservoir, which apparently helps regulate the osmotic pressure within the organism. The **stigma**, an organ that also occurs in the green algae (phylum Chlorophyta), is light-sensitive and aids these photosynthetic organisms to move toward light.

Cells of *Euglena* contain numerous small chloroplasts. These chloroplasts, like those of the green algae and plants, contain chlorophylls *a* and *b*, together with carotenoids. Although the chloroplasts of euglenoids differ somewhat in structure from those of green algae, they probably had a common origin. It seems likely that euglenoid chloroplasts ultimately evolved from a symbiotic relationship through ingestion of green algae.



(a)



(b)

**FIGURE 35.12**

**Euglenoids.** (a) Micrograph of individuals of the genus *Euglena* (Euglenophyta). (b) Diagram of *Euglena*. Paramylon granules are areas where food reserves are stored.

## Chrysophyta: The Diatoms and Golden Algae

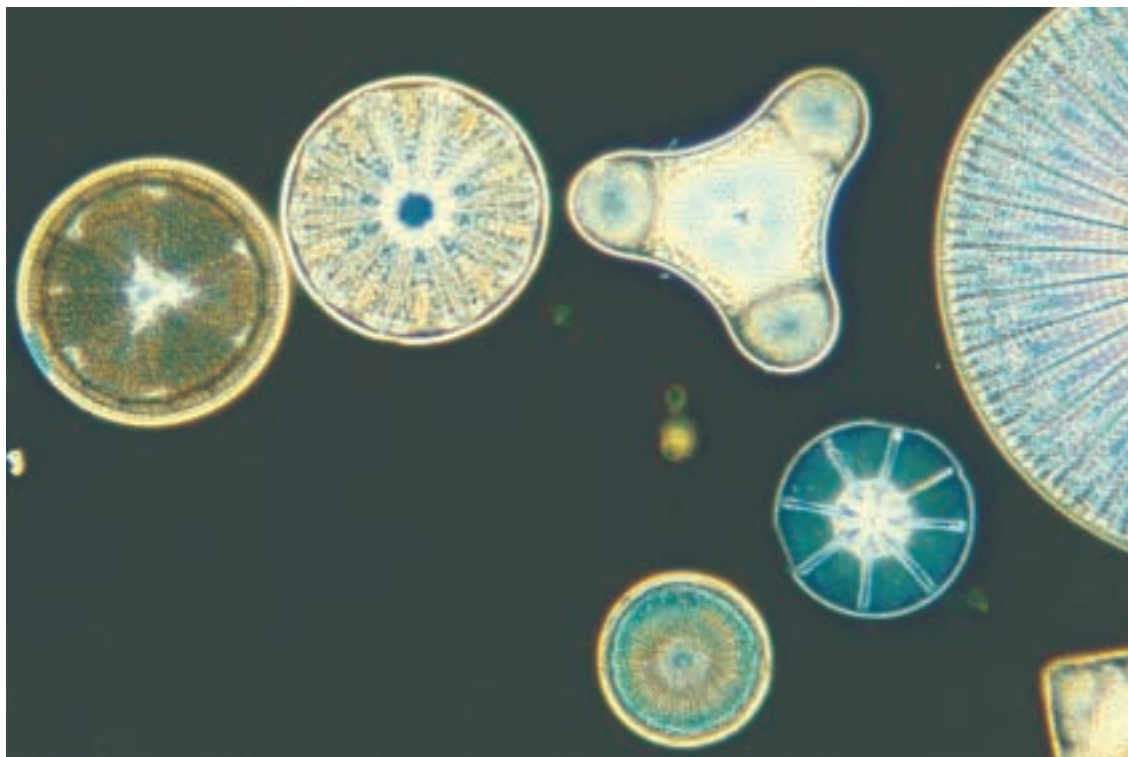
**The Diatoms.** Diatoms, members of the phylum Chrysophyta, are photosynthetic, unicellular organisms with unique double shells made of opaline silica, which are often strikingly and characteristically marked. The shells of diatoms are like small boxes with lids, one half of the shell fitting inside the other. Their chloroplasts, with chlorophylls *a* and *c*, as well as carotenoids, resemble those of the brown algae and dinoflagellates. In other respects, however, there are few similarities between these groups, and they probably do not share an immediate common ancestor. Another member of the phylum Chrysophyta is the golden algae. Diatoms and golden algae are grouped together because they both produce a unique carbohydrate called chrysolaminarin.

There are more than 11,500 living species of diatoms, with many more known in the fossil record. The shells of fossil diatoms often form very thick deposits, which are sometimes mined commercially. The resulting “diatomaceous earth” is used as an abrasive or to add the sparkling quality to the paint used on roads, among other purposes. Living diatoms are often abundant both in the sea and in fresh water, where they are important food producers. Diatoms occur in the plankton and are attached to submerged objects in relatively shallow water. Many species are able to move by means of a secretion that is produced from a fine groove along each shell. The diatoms exude and perhaps also retract this secretion as they move.

There are two major groups of diatoms, one with radial symmetry (like a wheel) and the other with bilateral (two-sided) symmetry (figure 35.13). Diatom shells are rigid, and the organisms reproduce asexually by separating the two halves of the shell, each half then regenerating another half shell within it. Because of this mode of reproduction, there is a tendency for the shells, and consequently the individual diatoms, to get smaller and smaller with each asexual reproduction. When the resulting individuals have diminished to about 30% of their original size, one may slip out of its shell, grow to full size, and regenerate a full-sized pair of new shells.

Individual diatoms are diploid. Meiosis occurs more frequently under conditions of starvation. Some marine diatoms produce numerous sperm and others a single egg. If fusion occurs, the resulting zygote regenerates a full-sized individual. In some freshwater diatoms, the gametes are amoeboid and similar in appearance.

**The Golden Algae.** Also included within the Chrysophyta are the golden algae, named for the yellow and brown carotenoid and xanthophyll accessory pigments in their chloroplasts, which give them a golden color. Unicellular but often colonial, these freshwater protists typically have two flagella, both attached near the same end of the cell. When ponds and lakes dry out in summer, golden algae form resistant cysts. Viable cells emerge from these cysts when wetter conditions recur in the fall.



**FIGURE 35.13**  
**Diatoms (Chrysophyta).** Several different centric (radially symmetrical) diatoms.

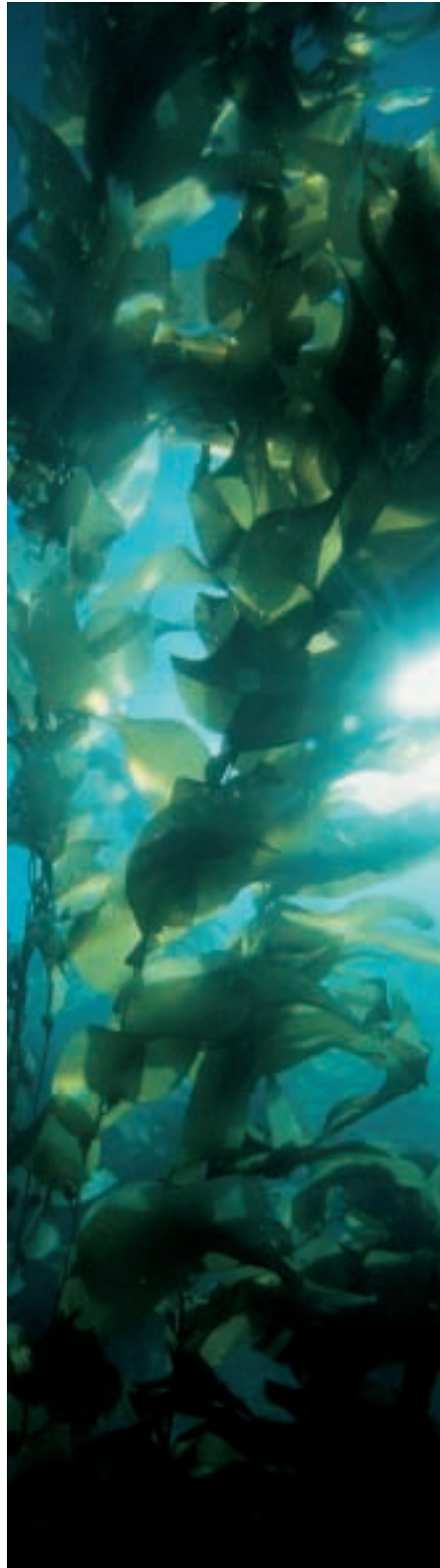
## Rhodophyta: The Red Algae

Along with green algae and brown algae, red algae are the seaweeds we see cast up along shores and on beaches. Their characteristic colors result from phycoerythrin, a type of phycobilin pigment. Phycobilins are responsible for the colors of the cyanobacteria. Chlorophyll *a* also occurs with the phycobilins in red algae, just as it does in cyanobacteria. These similarities with cyanobacteria make it likely that the rhodophyta evolved when their heterotrophic eukaryotic ancestor developed an endosymbiotic relationship with a cyanobacteria which eventually gave rise to their chloroplasts.

The great majority of the estimated 4000 species of red algae occur in the sea, and almost all are multicellular.

Red algae have complex bodies made up of interwoven filaments of cells. In the cell walls of many red algae are sulfated polysaccharides such as agar and carrageenan, which make these algae important economically. Agar is used to make gel capsules, as material for dental impressions, and as a base for cosmetics. It is also the basis of the laboratory media on which bacteria, fungi, and other organisms are often grown. In addition, agar is used to prevent baked goods from drying out, for rapid-setting jellies, and as a temporary preservative for meat and fish in warm regions. Carrageenan is used mainly to stabilize emulsions such as paints, cosmetics, and dairy products such as ice cream. In addition to these uses, red algae such as *Porphyra*, called “nori,” are eaten and, in Japan, are even cultivated as a human food crop.

The life cycles of red algae are complex but usually involve an alternation of generations (sporic meiosis). None of the red algae have flagella or cilia at any stage in their life cycle, and they may have descended directly from ancestors that never had them, especially as the red algae also lack centrioles. Together with the fungi, which also lack flagella and centrioles, the red algae may be one of the most ancient groups of eukaryotes.



**FIGURE 35.14**  
**Brown algae (Phaeophyta).** The massive “groves” of giant kelp that occur in relatively shallow water along the coasts of the world provide food and shelter for many different kinds of organisms.

## Phaeophyta: The Brown Algae

The phaeophyta, or brown algae, consist of about 1500 species of multicellular protists, almost exclusively marine. They are the most conspicuous seaweeds in many northern regions, dominating rocky shores almost everywhere in temperate North America. In habitats where large brown algae known as **kelps** (order Laminariales) occur abundantly in so-called kelp forests (figure 35.14), they are responsible for most of the food production through photosynthesis. Many kelps are conspicuously differentiated into flattened blades, stalks, and grasping basal portions that anchor them to the rocks.

Among the larger brown algae are genera such as *Macrocystis*, in which some individuals may reach 100 meters in length. The flattened blades of this kelp float out on the surface of the water, while the base is anchored tens of meters below the surface. Another ecologically important member of this phylum is sargasso weed, *Sargassum*, which forms huge floating masses that dominate the vast Sargasso Sea, an area of the Atlantic Ocean northeast of the Caribbean. The stalks of the larger brown algae often exhibit a complex internal differentiation of conducting tissues analogous to that of plants.

The life cycle of the brown algae is marked by an alternation of generations between a sporophyte and a gametophyte. The large individuals we recognize, such as the kelps, are sporophytes. The gametophytes are often much smaller, filamentous individuals, perhaps a few centimeters across. Sporangia, which produce haploid, swimming spores after meiosis, are formed on the sporophytes. These spores divide by mitosis, giving rise to individual gametophytes. There are two kinds of gametophytes in the kelps; one produces sperm, and the other produces eggs. If sperm and eggs fuse, the resulting zygotes grow into the mature kelp sporophytes, provided that they reach a favorable site.

## Chlorophyta: The Green Algae

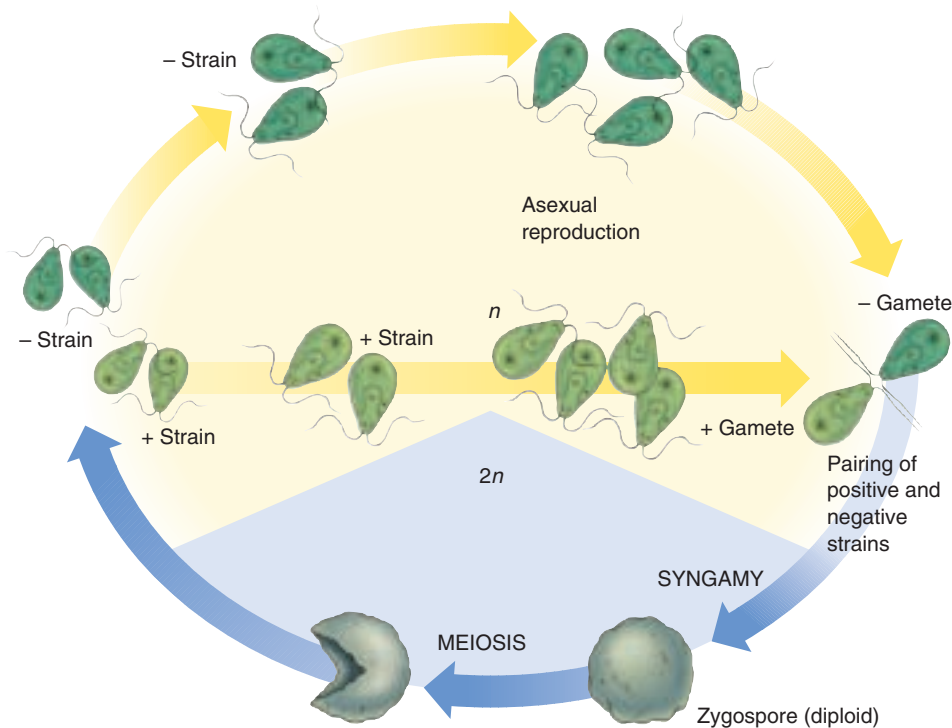
Green algae are an extremely varied group of more than 7000 species. The chlorophytes have an extensive fossil record dating back 900 million years. They are mostly aquatic, but some are semiterrestrial in moist places, such as on tree trunks or in soil. Many are microscopic and unicellular, but some, such as sea lettuce, *Ulva* (see figure 35.16), are tens of centimeters across and easily visible on rocks and pilings around the coasts.

Green algae are of special interest, both because of their unusual diversity and because the ancestors of the plant kingdom were clearly multicellular green algae. Many features of modern green algae closely resemble plants, especially their chloroplasts which are biochemically similar to those of the plants. They contain chlorophylls *a* and *b*, as well as carotenoids. Green algae include a very wide array of both unicellular and multicellular organisms.

Among the unicellular green algae, *Chlamydomonas* (figure 35.15) is a well-known genus. Individuals are microscopic (usually less than 25 micrometers long), green, rounded, and have two flagella at the anterior end.

They move rapidly in water by beating their flagella in opposite directions. Each individual has an eyespot, which contains about 100,000 molecules of rhodopsin, the same pigment employed in vertebrate eyes. Light received by this eyespot is used by the alga to help direct its swimming. Most individuals of *Chlamydomonas* are haploid. *Chlamydomonas* reproduces asexually (by cell division) as well as sexually. In sexual reproduction, two haploid individuals fuse to form a four-flagellated zygote. The zygote ultimately enters a resting phase, called the **zygospore**, in which the flagella disappear and a tough protective coat is formed. Meiosis occurs at the end of this resting period and results in the production of four haploid cells.

*Chlamydomonas* probably represents a primitive state for green algae and several lines of evolutionary specialization have been derived from organisms like it. The first is the evolution of nonmotile, unicellular green algae. *Chlamydomonas* is capable of retracting its flagella and settling down as an immobile unicellular organism if the ponds in which it lives dry out. Some common algae of soil and bark, such as *Chlorella*, are essentially like *Chlamydomonas* in this



**FIGURE 35.15**

**Life cycle of *Chlamydomonas* (Chlorophyta).** Individual cells of this microscopic, biflagellated alga, which are haploid, divide asexually, producing identical copies of themselves. At times, such haploid cells act as gametes—fusing, as shown in the lower right-hand side of the diagram, to produce a zygote. The zygote develops a thick, resistant wall, becoming a zygospore; this is the only diploid cell in the entire life cycle. Within this diploid zygospore, meiosis takes place, ultimately resulting in the release of four haploid individuals. Because of the segregation during meiosis, two of these individuals are called the (+) strain, the other two the (–) strain. Only + and – individuals are capable of mating with each other when syngamy does take place, although both may divide asexually to reproduce themselves.

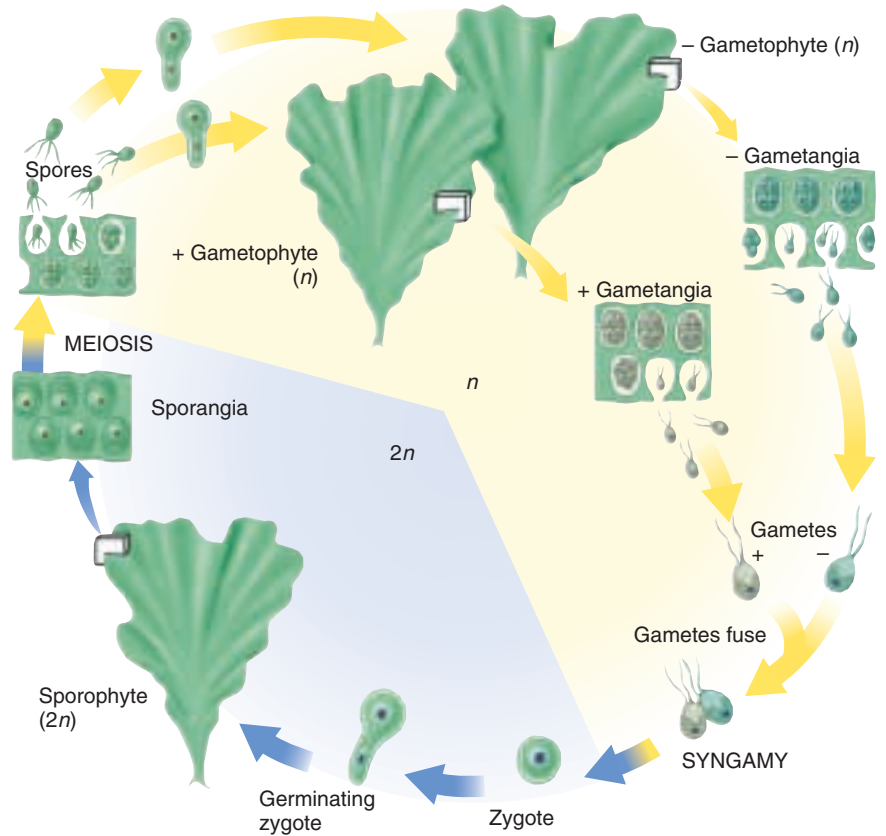
trait, but do not have the ability to form flagella. *Chlorella* is widespread in both fresh and salt water as well as soil and is only known to reproduce asexually. Recently, *Chlorella* has been widely investigated as a possible food source for humans and other animals, and pilot farms have been established in Israel, the United States, Germany, and Japan.

Another major line of specialization from cells like *Chlamydomonas* concerns the formation of motile, colonial organisms. In these genera of green algae, the *Chlamydomonas*-like cells retain some of their individuality. The most elaborate of these organisms is *Volvox* (see figure 35.1), a hollow sphere made up of a single layer of 500 to 60,000 individual cells, each cell with two flagella. Only a small number of the cells are reproductive. The colony has definite anterior and posterior ends, and the flagella of all of the cells beat in such a way as to rotate the colony in a clockwise direction as it moves forward through the water. The reproductive cells of *Volvox* are located mainly at the posterior end of the colony. Some may divide asexually, bulge inward, and give rise to new colonies that initially remain within the parent colony. Others produce gametes. In some species of



**FIGURE 35.16**

**Life cycle of *Ulva*.** In this green alga, the gametophyte and the sporophyte are identical in appearance and consist of flattened sheets two cells thick. In the haploid ( $n$ ) gametophyte, gametangia give rise to haploid gametes, which fuse to form a diploid ( $2n$ ) zygote. The zygote germinates to form the diploid sporophyte. Sporangia within the sporophyte give rise to haploid spores by meiosis. The haploid spores develop into haploid gametophytes.



*Volvox*, there is a true division of labor among the different types of cells, which are specialized in relation to their ultimate function throughout the development of the organism.

In addition to these two lines of specialization from *Chlamydomonas*-like cells, there are many other kinds of green algae of less certain derivation. Many filamentous genera, such as *Spirogyra*, with its ribbon-like chloroplasts, differ substantially from the remainder of the green algae in their modes of cell division and reproduction. Some of these genera have even been placed in separate phyla. The study of the green algae, involving modern methods of electron microscopy and biochemistry, is beginning to reveal unexpected new relationships within this phylum.

*Ulva*, or sea lettuce (figure 35.16), is a genus of marine green algae that is extremely widespread. The glistening individuals of this genus, often more than 10 centimeters across, consist of undulating sheets only two cells thick. Sea lettuce attaches by protuberances of the basal cells to rocks or other substrates. The reproductive cycle of *Ulva* involves an alternation of generations (sporic meiosis; figure 35.16) as is typical among green algae. Unlike most organisms that undergo sporic meiosis, however,

the **gametophytes** (haploid phase) and **sporophytes** (diploid phase) resemble one another closely.

The stoneworts, a group of about 250 living species of green algae, many of them in the genera *Chara* and *Nitella*, have complex structures. Whorls of short branches arise regularly at their nodes, and the gametangia (structures that give rise to gametes) are complex and multicellular. Stoneworts are often abundant in fresh to brackish water and are common as fossils.

**Dinoflagellates** are primarily unicellular, photosynthetic, and flagellated. Euglenoids (phylum Euglenophyta) consist of about 40 genera, about a third of which have chloroplasts similar biochemically to those of green algae and plants. Diatoms and golden algae are unicellular, photosynthetic organisms that produce a unique carbohydrate. Diatoms have double shells made of opaline silica. Nonmotile, unicellular algae and multicellular, flagellated colonies have been derived from green algae like *Chlamydomonas*—a biflagellated, unicellular organism. The life cycle of brown algae is marked by an alternation of generations between the diploid phase, or sporophyte, and the haploid phase, or gametophyte.

## Heterotrophs with Flagella

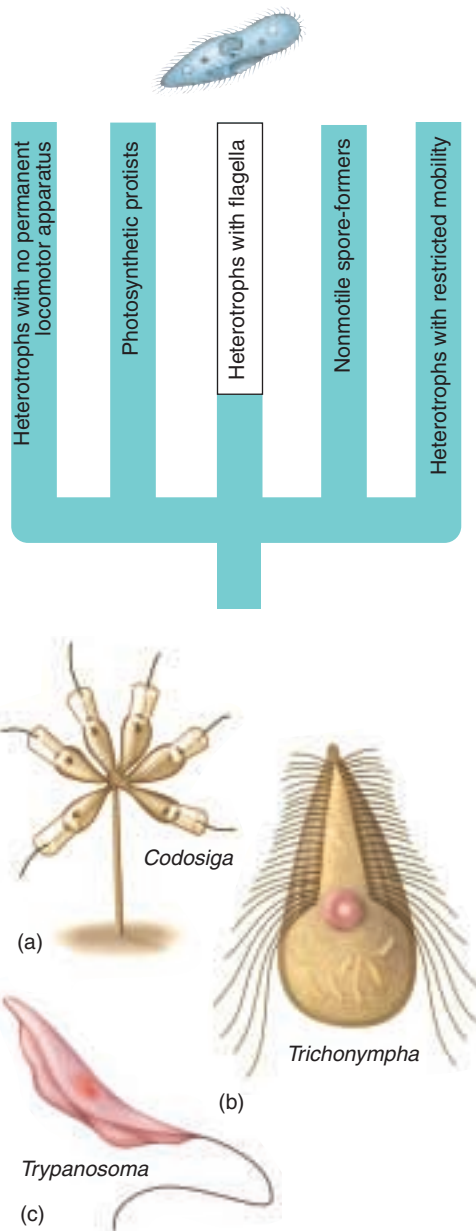
The phylum Sarcomastigophora contains a diverse group of protists combined into one phylum because they all possess a single kind of nucleus and use flagella or pseudopodia (or both) for locomotion. We will focus on the class Zoomastigophora.

### Zoomastigophora: The Zoomastigotes

The class Zoomastigophora is composed of unicellular, heterotrophic organisms that are highly variable in form (figure 35.17). Each has at least one flagellum, with some species having thousands. They include both free-living and parasitic organisms. Many zoomastigotes apparently reproduce only asexually, but sexual reproduction occurs in some species. The members of one order, the kinetoplastids, include the genera *Trypanosoma* (figure 35.17c) and *Crithidia*, pathogens of humans and domestic animals. The euglenoids could be viewed as a specialized group of zoomastigotes, some of which acquired chloroplasts during the course of evolution.

Trypanosomes cause many serious human diseases, the most familiar of which is trypanosomiasis also known as African sleeping sickness (figure 35.18). Trypanosomes cause many other diseases including East Coast fever, leishmaniasis, and Chagas' disease, all of great importance in tropical areas where they afflict millions of people each year. Leishmaniasis, which is transmitted by sand flies, afflicts about 4 million people a year. The effects of these diseases range from extreme fatigue and lethargy in sleeping sickness to skin sores and deep eroding lesions that can almost obliterate the face in leishmaniasis. The trypanosomes that cause these diseases are spread by biting insects, including tsetse flies and assassin bugs.

A serious effort is now under way to produce a vaccine for trypanosome-caused diseases. These diseases make it



**FIGURE 35.17**  
**Three genera of zoomastigotes (Zoomastigophora), a highly diverse group.** (a) *Codosiga*, a colonial choanoflagellate that remains attached to its substrate; other colonial choanoflagellates swim around as a colony, resembling the green alga *Volvox* in this respect. (b) *Trichonympha*, one of the zoomastigotes that inhabits the guts of termites and wood-feeding cockroaches and digests cellulose there. *Trichonympha* has rows of flagella in its anterior regions. (c) *Trypanosoma*, which causes sleeping sickness, an important tropical disease. It has a single, anterior flagellum.

impossible to raise domestic cattle for meat or milk in a large portion of Africa. Control is especially difficult because of the unique attributes of these organisms. For example, tsetse fly-transmitted trypanosomes have evolved an elaborate genetic mechanism for repeatedly changing the antigenic nature of their protective glycoprotein coat, thus dodging the antibodies their hosts produce against them (see chapter 57). Only a single one out of some 1000 to 2000 variable antigen genes is expressed at a time. Rearrangements of these genes during the asexual cycle of the organism allow for the expression of a seemingly endless variety of different antigen genes that maintain infectivity by the trypanosomes.

When the trypanosomes are ingested by a tsetse fly, they embark on a complicated cycle of development and multiplication, first in the fly's gut and later in its salivary glands. It is their position in the salivary glands that allows them to move into their vertebrate host. Recombination has been observed between different strains of trypanosomes introduced into a single fly, thus suggesting that mating, syngamy, and meiosis occur, even though they have not been observed directly. Although most trypanosome reproduction is asexual, this sexual cycle, reported for the first time in 1986, affords still further possibilities for recombination in these organisms.

In the guts of the flies that spread them, trypanosomes are noninfective. When they are ready to transfer to the skin or bloodstream of their host, trypanosomes migrate to the salivary glands and acquire the thick coat of glycoprotein antigens that protect them from the host's antibodies. When they are taken up by a fly, the trypanosomes again shed their coats. The production of vaccines against such a system is complex, but tests are underway. Releasing sterilized flies to impede the reproduction of populations is another technique used to try to control the fly population. Traps made of dark cloth and scented like cows, but

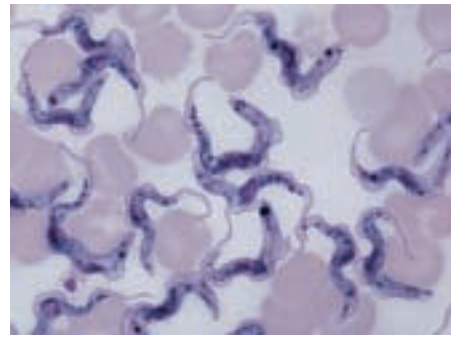
poisoned with insecticides, have likewise proved effective. Research is proceeding rapidly because the presence of tsetse flies with their associated trypanosomes blocks the use of some 11 million square kilometers of potential grazing land in Africa.

Some zoomastigotes occur in the guts of termites and other wood-eating insects. They possess enzymes that allow them to digest the wood and thus make the components of the wood available to their hosts. The relationship is similar to that between certain bacteria and protozoa that function in the rumens of cattle and related mammals (see chapter 51).

Another order of zoomastigotes, the choanoflagellates, is most likely the group from which the sponges (phylum Porifera) and probably all animals arose. Choanoflagellates have a single emergent flagellum surrounded by a funnel-shaped, contractile collar composed of closely placed filaments, a unique structure that is exactly matched in the sponges. These protists feed on bacteria strained out of the water by the collar.

**Hiker's Diarrhea.** *Giardia lamblia* is a flagellate protist (belonging to a small order called diplomonads) found throughout the world, including all parts of the United States and Canada (figure 35.19). It occurs in water, including the clear water of mountain streams and the water supplies of some cities. It infects at least 40 species of wild and domesticated animals in addition to humans. In 1984 in Pittsburgh, 175,000 people had to boil their drinking water for several days following the appearance of *Giardia* in the city's water system. Although most individuals exhibit no symptoms if they drink water infested with *Giardia*, many suffer nausea, cramps, bloating, vomiting, and diarrhea. Only 35 years ago, *Giardia* was thought to be harmless; today, it is estimated that at least 16 million residents of the United States are infected by it.

*Giardia* lives in the upper small intestine of its host. It occurs there in a motile form that cannot survive outside the host's body. It is spread in the feces of infected individuals in the form of dormant, football-shaped cysts—sometimes at levels as high as 300 million individuals per gram of feces. These cysts can survive at least two months in cool water, such as that of mountain streams. They are relatively resistant to the usual water-treatment agents such as chlorine and iodine but are killed at temperatures greater than about 65°C. Apparently, pollution by humans seems to be the main way *Giardia* is released into stream water. There are at least three species of *Giardia* and many distinct strains; how many of them attack humans and under what circumstances are not known with certainty.



(a)

20  $\mu$ m



(b)

**FIGURE 35.18**

***Trypanosoma* is the zoomastigote that causes sleeping sickness.** (a) *Trypanosoma* among red blood cells. The nuclei (dark-staining bodies), anterior flagella, and undulating, changeable shape of the trypanosomes are visible in this photograph (500 $\times$ ). (b) The tsetse



**FIGURE 35.19**

***Giardia lamblia*.** *Giardia* are flagellated unicellular parasites that infect the human intestine. *Giardia* are very primitive, having only a rudimentary cytoskeleton and lacking mitochondria and chloroplasts. Sequencing of ribosomal RNA suggests that *Giardia* and *Pelomyxa*, the eukaryotes most closely related to prokaryotes, should be grouped together. The name Archezoa (Greek *arkhaios*, "ancient") has been suggested for the group, stressing its early divergence from bacteria as long as 2 billion years ago.

In the wilderness, good sanitation is important in preventing the spread of *Giardia*. Dogs, which readily contract and spread the disease, should not be taken into pristine wilderness areas. Drinking water should be filtered—the filter must be capable of eliminating particles as small as 1 micrometer in diameter—or boiled for at least one minute. Water from natural streams or lakes should never be consumed directly, regardless of how clean it looks. In other regions, good sanitation methods are important to prevent not only *Giardia* infection but also other diseases.

## Ciliophora: The Ciliates

As the name indicates, most members of the Ciliophora feature large numbers of cilia. These heterotrophic, unicellular protists range in size from 10 to 3000 micrometers long. About 8000 species have been named. Despite their unicellularity, ciliates are extremely complex organisms, inspiring some biologists to consider them organisms without cell boundaries rather than single cells.

Their most characteristic feature, cilia, are usually arranged either in longitudinal rows or in spirals around the body of the organism (figure 35.20). Cilia are anchored to microtubules beneath the cell membrane, and they beat in a coordinated fashion. In some groups, the cilia have specialized locomotory and feeding functions, becoming fused into sheets, spikes, and rods which may then function as mouths, paddles, teeth, or feet. The ciliates have a tough but flexible outer covering called the pellicle that enables the organism to squeeze through or move around many kinds of obstacles.

All ciliates that have been studied have two very different types of nuclei within their cells, small **micronuclei** and larger **macronuclei** (figure 35.21). The micronuclei, which contain apparently normal diploid chromosomes, divide by meiosis and are able to undergo genetic recombination. Macronuclei are derived from certain micronuclei in a complex series of steps. Within the macronuclei are multiple copies of the genome, and the DNA is divided into small pieces—smaller than individual chromosomes. In one group of ciliates, these are equivalent to single genes. Macronuclei divide by elongating and constricting and play an essential role in routine cellular functions, such as the production of mRNA to direct protein synthesis for growth and regeneration.

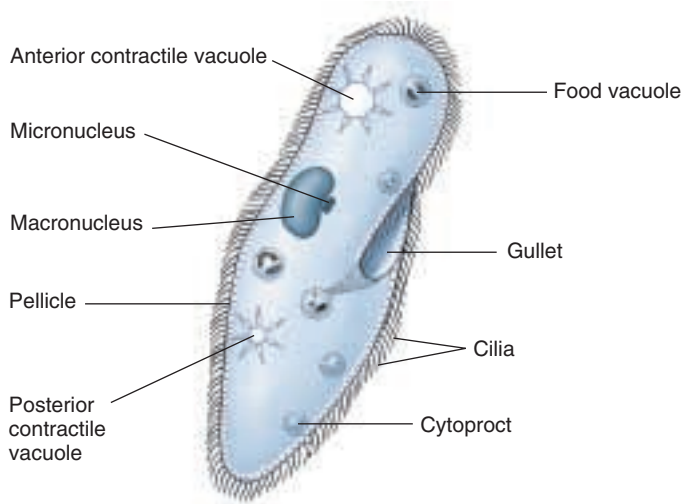
Ciliates form vacuoles for ingesting food and regulating their water balance. Food first enters the gullet, which in the well-known ciliate *Paramecium* is lined with cilia fused into a membrane (figure 35.21). From the gullet, the food passes into food vacuoles, where enzymes and hydrochloric acid aid in its digestion. After the digested material has been completely absorbed, the vacuole empties its waste contents through a special pore in the pellicle known as the **cytoproct**. The cytoproct is essentially an exocytotic vesicle that appears periodically when solid particles are ready to be expelled. The contractile vacuoles, which function in the regulation of water balance, periodically expand and contract as they empty their contents to the outside of the organism.

Ciliates usually reproduce by transverse fission of the parent cell across its short axis, thus forming two identical individuals (figure 35.22*a*). In this process of cell division, the mitosis of the micronuclei proceeds normally, and the macronuclei divide as just described.

In *Paramecium*, the cells divide asexually for about 700 generations and then die if sexual reproduction has not occurred. Like most ciliates, *Paramecium* has a sexual process called **conjugation**, in which two individual cells remain



**FIGURE 35.20**  
**A ciliate (Ciliophora).**  
*Stentor*, a funnel-shaped ciliate, showing spirally arranged cilia (120 $\times$ ).



**FIGURE 35.21**  
*Paramecium*. The main features of this familiar ciliate are shown.

attached to each other for up to several hours (figure 35.22*b,c*). Only cells of two different genetically determined mating types, *odd* and *even*, are able to conjugate. Meiosis in the micronuclei of each individual produces several haploid micronuclei, and the two partners exchange a pair of these micronuclei through a cytoplasmic bridge that appears between the two partners.

In each conjugating individual, the new micronucleus fuses with one of the micronuclei already present in that individual, resulting in the production of a new diploid micronucleus in each individual. After conjugation, the macronucleus in each cell disintegrates, while the new diploid micronucleus undergoes mitosis, thus giving rise to two new identical diploid micronuclei within each individual. One of these micronuclei becomes the precursor of the future micronuclei of that cell, while the other micronucleus undergoes multiple rounds of DNA replication, becoming the new macronucleus. This kind of complete segregation of the genetic material is a unique feature of the



(a)



(b)

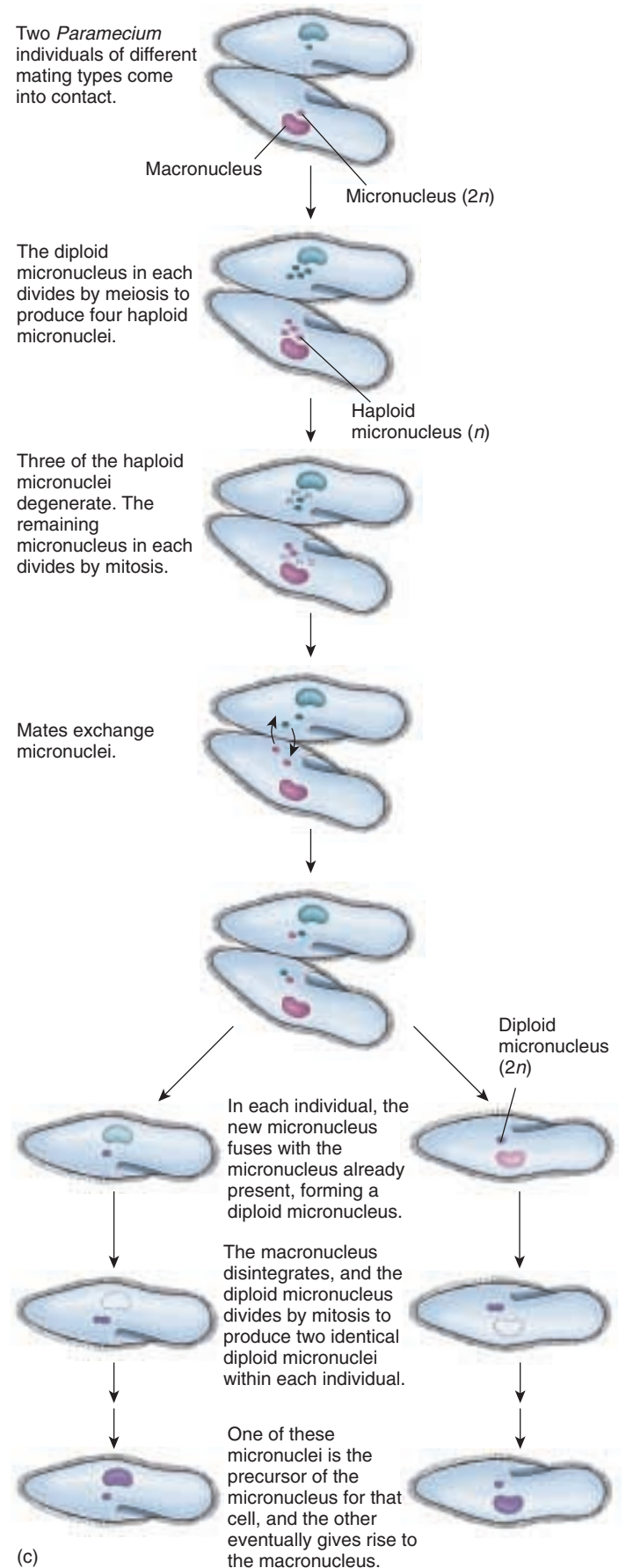
### FIGURE 35.22

**Life cycle of *Paramecium*.** (a) When *Paramecium* reproduces asexually, a mature individual divides, and two complete individuals result. (b,c) In sexual reproduction, two mature cells fuse in a process called conjugation (100 $\times$ ).

ciliates and makes them ideal organisms for the study of certain aspects of genetics.

Progeny from a sexual division in *Paramecium* must go through about 50 asexual divisions before they are able to conjugate. When they do so, their biological clocks are restarted, and they can conjugate again. After about 600 asexual divisions, however, *Paramecium* loses the protein molecules around the gullet that enable it to recognize an appropriate mating partner. As a result, the individuals are unable to mate, and death follows about 100 generations later. The exact mechanisms producing these unusual events are unknown, but they involve the accumulation of a protein, which is now being studied.

The zoomastigotes are a highly diverse group of flagellated unicellular heterotrophs, containing among their members the ancestors of animals as well as the very primitive *Giardia*. Ciliates possess characteristic cilia, and have two types of nuclei. The macronuclei contain multiple copies of certain genes, while the micronuclei contain multigene chromosomes.



## Nonmotile Spore-Formers

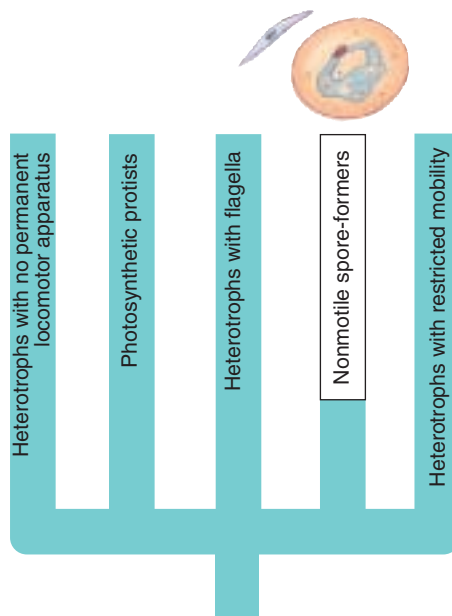
### Apicomplexa: The Sporozoans

All sporozoans are nonmotile, spore-forming parasites of animals. Their spores are small, infective bodies that are transmitted from host to host. These organisms are distinguished by a unique arrangement of fibrils, microtubules, vacuoles, and other cell organelles at one end of the cell. There are 3900 described species of this phylum; best known among them is the malarial parasite, *Plasmodium*.

Sporozoans have complex life cycles that involve both asexual and sexual phases. Sexual reproduction involves an alternation of haploid and diploid generations. Both haploid and diploid individuals can also divide rapidly by mitosis, thus producing a large number of small infective individuals. Sexual reproduction involves the fertilization of a large female gamete by a small, flagellated male gamete. The zygote that results soon becomes an **oocyst**. Within the oocyst, meiotic divisions produce infective haploid spores called **sporozoites**.

An alternation between different hosts often occurs in the life cycles of sporozoans. Sporozoans of the genus *Plasmodium* are spread from person to person by mosquitoes of the genus *Anopheles* (figure 35.23); at least 65 different species of this genus are involved. When an *Anopheles* mosquito penetrates human skin to obtain blood, it injects saliva mixed with an anticoagulant. If the mosquito is infected with *Plasmodium*, it will also inject the elongated sporozoites into the bloodstream of its victim. The parasite makes its way through the bloodstream to the liver, where it rapidly divides asexually. After this division phase, **merozoites**, the next stage of the life cycle, form, either reinvading other liver cells or entering the host's bloodstream. In the bloodstream, they invade the red blood cells, dividing rapidly within them and causing them to become enlarged and ultimately to rupture. This event releases toxic substances throughout the body of the host, bringing about the well-known cycle of fever and chills that is characteristic of malaria. The cycle repeats itself regularly every 48 hours, 72 hours, or longer.

*Plasmodium* enters a sexual phase when some merozoites develop into **gametocytes**, cells capable of producing gametes. There are two types of gametocytes: male and female. Gametocytes are incapable of producing gametes within their human hosts and do so only when they are extracted from an infected human by a mosquito. Within the gut of the mosquito, the male and female gametocytes form sperm and eggs, respectively. Zygotes de-



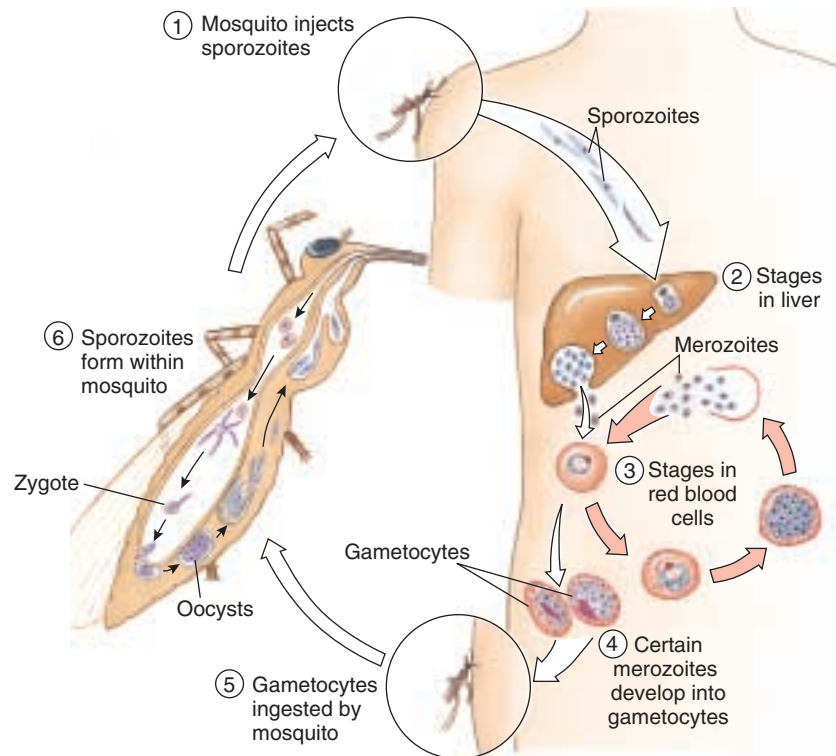
velop within the mosquito's intestinal walls and ultimately differentiate into oocysts. Within the oocysts, repeated mitotic divisions take place, producing large numbers of sporozoites. These sporozoites migrate to the salivary glands of the mosquito, and from there they are injected by the mosquito into the bloodstream of a human, thus starting the life cycle of the parasite again.

**Malaria.** Malaria, caused by infections by the sporozoan *Plasmodium*, is one of the most serious diseases in the world. According to the World Health Organization, about 500 million people are affected by it at any one time, and approximately 2 million of them, mostly children, die each year. Malaria kills most children

under five years old who contract it. In areas where malaria is prevalent, most survivors more than five or six years old do not become seriously ill again from malaria infections. The symptoms, familiar throughout the tropics, include severe chills, fever, and sweating, an enlarged and tender spleen, confusion, and great thirst. Ultimately, a victim of malaria may die of anemia, kidney failure, or brain damage. The disease may be brought under control by the person's immune system or by drugs. As discussed in chapter 21, some individuals are genetically resistant to malaria. Other persons develop immunity to it.

Efforts to eradicate malaria have focused on (1) the elimination of the mosquito vectors; (2) the development of drugs to poison the parasites once they have entered the human body; and (3) the development of vaccines. The widescale applications of DDT from the 1940s to the 1960s led to the elimination of the mosquito vectors in the United States, Italy, Greece, and certain areas of Latin America. For a time, the worldwide elimination of malaria appeared possible, but this hope was soon crushed by the development of DDT-resistant strains of malaria-carrying mosquitoes in many regions; no fewer than 64 resistant strains were identified in a 1980 survey. Even though the worldwide use of DDT, long banned in the United States, nearly doubled from its 1974 level to more than 30,000 metric tons in 1984, its effectiveness in controlling mosquitoes is dropping. Further, there are serious environmental concerns about the use of this long-lasting chemical anywhere in the world. In addition to the problems with resistant strains of mosquitoes, strains of *Plasmodium* have appeared that are resistant to the drugs that have historically been used to kill them.

As a result of these problems, the number of new cases of malaria per year roughly doubled from the mid-1970s to



**FIGURE 35.23**

**The life cycle of *Plasmodium*, the sporozoan that causes malaria.** *Plasmodium* has a complex life cycle that alternates between mosquitoes and mammals.

the mid-1980s, largely because of the spread of resistant strains of the mosquito and the parasite. In many tropical regions, malaria is blocking permanent settlement. Scientists have therefore redoubled their efforts to produce an effective vaccine. Antibodies to the parasites have been isolated and produced by genetic engineering techniques, and they are starting to produce promising results.

**Vaccines against Malaria.** The three different stages of the *Plasmodium* life cycle each produce different antigens, and they are sensitive to different antibodies. The gene encoding the sporozoite antigen was cloned in 1984, but it is not certain how effective a vaccine against sporozoites might be. When a mosquito inserts its proboscis into a human blood vessel, it injects about a thousand sporozoites. They travel to the liver within a few minutes, where they are no longer exposed to antibodies circulating in the blood. If even one sporozoite reaches the liver, it will multiply rapidly there and cause malaria. The number of malaria parasites increases roughly eightfold every 24 hours after they enter the host's body. A compound vaccination against sporozoites, merozoites, and gameto-

cytes would probably be the most effective preventive measure, but such a compound vaccine has proven difficult to develop.

However, research completed in 1997 brings a glimmer of hope. An experimental vaccine containing one of the surface proteins of the disease-causing parasite, *P. falciparum*, seems to induce the immune system to produce defenses that are able to destroy the parasite in future infections. In tests, six out of seven vaccinated people did not get malaria after being bitten by mosquitoes that carried *P. falciparum*. Although research is still underway, many are hopeful that this new vaccine may be able to fight malaria, especially in Africa, where it takes a devastating toll.

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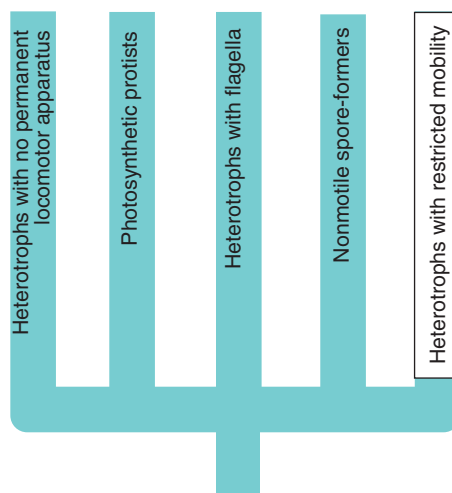
**The best known of the sporozoans is the malarial parasite *Plasmodium*. Like other sporozoans, *Plasmodium* has a complex life cycle involving sexual and asexual phases and alternation between different hosts, in this case mosquitoes and humans. Malaria kills about 2 million people each year.**

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## Heterotrophs with Restricted Mobility

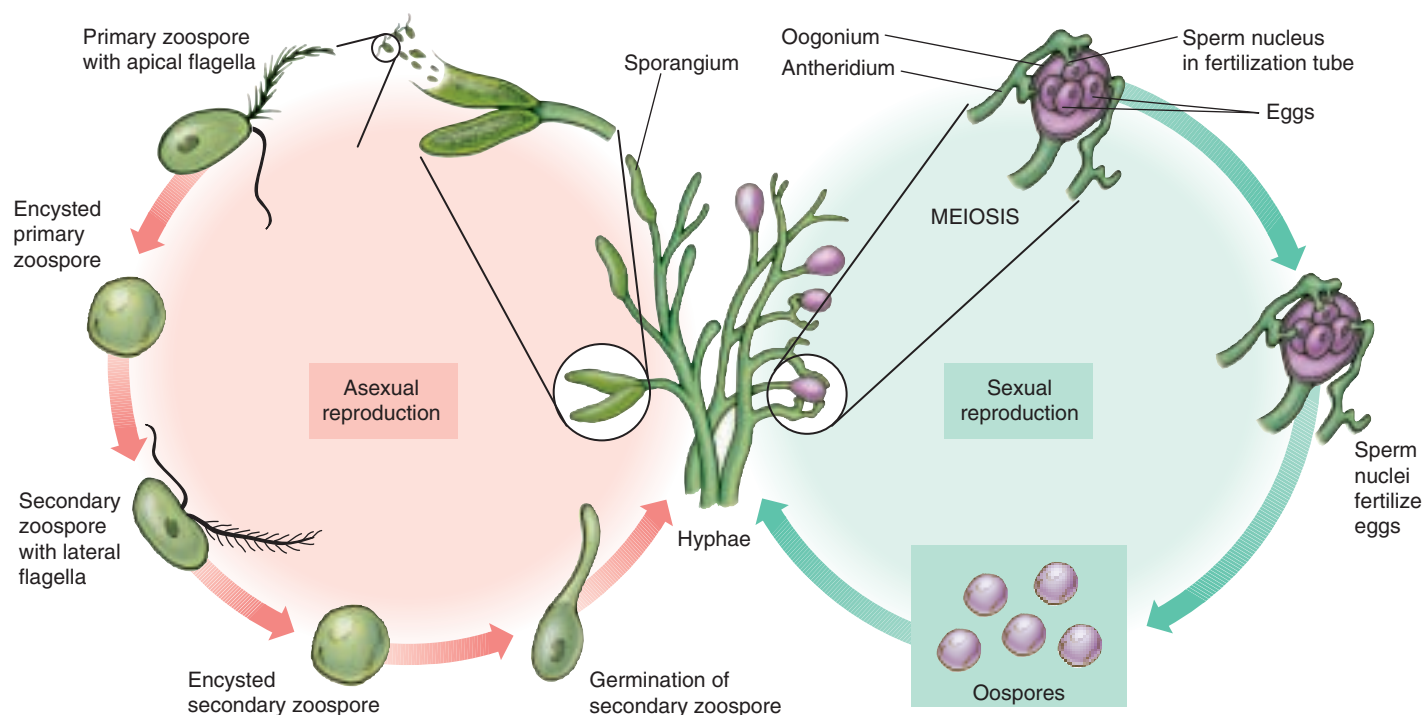
### Oomycota

The oomycetes comprise about 580 species, among them the water molds, white rusts, and downy mildews. All of the members of this group are either parasites or **saprobies** (organisms that live by feeding on dead organic matter). The cell walls of the oomycetes are composed of cellulose or polymers that resemble cellulose. They differ remarkably from the chitin cell walls of fungi, with which the oomycetes have at times been grouped. Oomycete life cycles are characterized by gametic meiosis and a diploid phase; this also differs from fungi. Mitosis in the oomycetes resembles that in most other organisms, while mitosis in fungi has a number of unusual features, as you will see in chapter 36. Filamentous structures of fungi and, by convention, those of oomycetes, are called **hyphae**. Most oomycetes live in fresh or salt water or in soil, but some are plant parasites that depend on the wind to spread their spores. A few aquatic oomycetes are animal parasites.



Oomycetes are distinguished from other protists by the structure of their motile spores, or **zoospores**, which bear two unequal flagella, one of which is directed forward, the other backward. Such zoospores are produced asexually in a sporangium. Sexual reproduction in the group involves **gametangia** (singular, **gametangium**)—gamete-producing structures—of two different kinds. The female gametangium is called an **oogonium**, and the male gametangium is called an **antheridium**. The antheridia contain numerous male nuclei, which are the functional male gametes; the oogonia contain from one to eight eggs, which are the female gametes. When the contents of an antheridium flow into an oogonium, it leads to the individual fusion

of male nuclei with eggs. This is followed by the thickening of the cell wall around the resulting zygote or zygotes. This produces a special kind of thick-walled cell called an **oospore**, the structure that gives the phylum its name. Details from the life cycle of one of the oomycetes, *Saprolegnia*, are shown in figure 35.24.



**FIGURE 35.24**

**Life cycle of *Saprolegnia*, an oomycete.** Asexual reproduction by means of flagellated zoospores is shown at left, sexual reproduction at right. Hyphae with diploid nuclei are produced by germination of both zoospores and oospores.

Aquatic oomycetes, or water molds, are common and easily cultured. Some water molds cause fish diseases, producing a kind of white fuzz on aquarium fishes. Among their terrestrial relatives are oomycetes of great importance as plant pathogens, including *Plasmopara viticola*, which causes downy mildew of grapes, and *Phytophthora infestans*, which causes the late blight of potatoes. This oomycete was responsible for the Irish potato famine of 1845 and 1847, during which about 400,000 people starved to death or died of diseases complicated by starvation. Millions of Irish people emigrated to the United States and elsewhere as a result of this disaster.

### Acrasiomycota: The Cellular Slime Molds

There are about 70 species of cellular slime molds. This phylum has extraordinarily interesting features and was once thought to be related to fungi, “mold” being a general term for funguslike organisms. In fact, the cellular slime molds are probably more closely related to amoebas (phylum Rhizopoda) than to any other group, but they have many special features that mark them as distinct. Cellular slime molds are common in fresh water, damp soil, and on rotting vegetation, especially fallen logs. They have become one of the most important groups of organisms for studies of differentiation because of their relatively simple developmental systems and the ease of analyzing them (figure 35.25).

The individual organisms of this group behave as separate amoebas, moving through the soil or other substrate and ingesting bacteria and other smaller organisms. At a certain phase of their life cycle, the individual organisms aggregate and form a moving mass, the “slug,” that eventually transforms itself into a spore-containing mass, the **sorocarp**. In the sorocarp the amoebas become encysted as spores. Some of the amoebas fuse sexually to form **macro-cysts**, which have diploid nuclei; meiosis occurs in them after a short period (zygotic meiosis). The sporocarp develops a stalked structure with a chamber at the top which releases the spores. Other amoebas are released directly, eventually aggregating again to form a new slug.

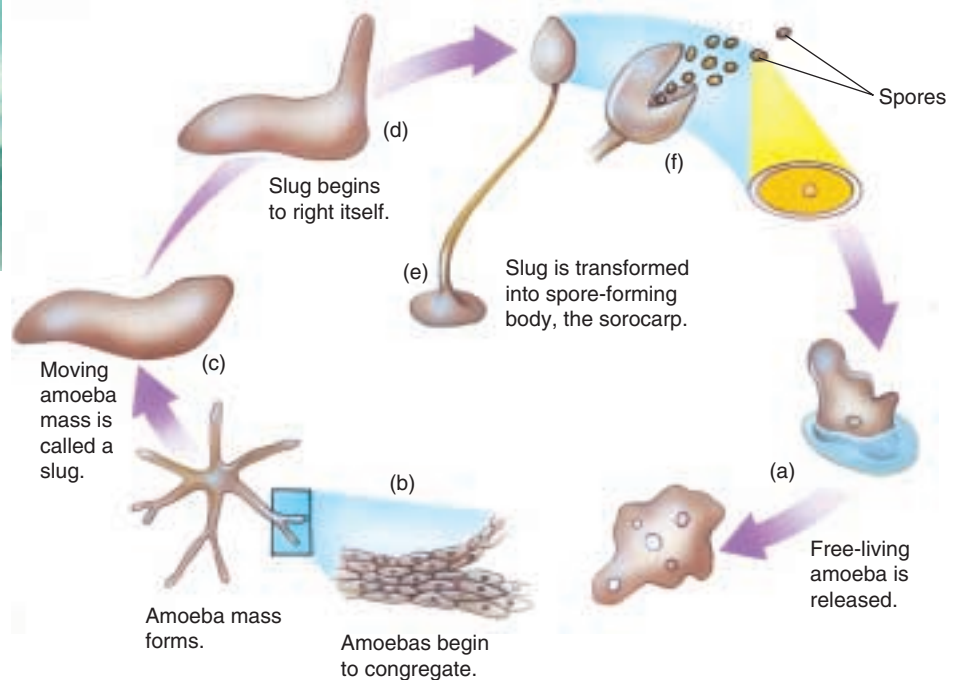
The development of *Dictyostelium discoideum*, a cellular slime mold, has been studied extensively because of the implication its unusual life cycle has for understanding the developmental process in general. When the individual amoebas of this species exhaust the supply of bacteria in a given area and are near starvation, they aggregate and form a compound, motile mass. The aggregation of the individual amoebas is induced by pulses of cyclic adenosine monophosphate (cAMP), which the cells begin to secrete when they are starving. The cells form an aggregate organism that moves to a new area where food is more plentiful. In the new area, the colony differentiates into a multicellular sorocarp within which spores differentiate. Each of these spores, if it falls into a suitably moist habitat, releases a new amoeba, which begins to feed, and the cycle is started again.



**FIGURE 35.25**

#### Development in *Dictyostelium discoideum*, a cellular slime mold.

(a) First, a spore germinates, forming amoebas. These amoebas feed and reproduce until the food runs out. (b) The amoebas aggregate and move toward a fixed center. (c) Next, they form a multicellular “slug” 2 to 3 mm long that migrates toward light. (d) The slug stops moving and begins to differentiate into a spore-forming body, called a sorocarp (e). (f) Within heads of the sorocarps, amoebas become encysted as spores.





**FIGURE 35.26**

**A plasmodial protist.** This multinucleate plasmodium moves about in search of the bacteria and other organic particles that it ingests.

### Myxomycota: The Plasmodial Slime Molds

Plasmodial slime molds are a group of about 500 species. These bizarre organisms stream along as a **plasmodium**, a nonwalled, multinucleate mass of cytoplasm, that resembles a moving mass of slime (figure 35.26). This is called the feeding phase, and the plasmodia may be orange, yellow, or another color. Plasmodia show a back-and-forth streaming of cytoplasm that is very conspicuous, especially under a microscope. They are able to pass through the mesh in cloth or simply flow around or through other obstacles. As they move, they engulf and digest bacteria, yeasts, and other small particles of organic matter. Plasmodia contain many nuclei (multinucleate), but these are not separated by cell membranes. The nuclei undergo mitosis synchronously, with the nuclear envelope breaking down, but only at late anaphase or telophase. Centrioles are lacking in cellular slime molds. Although they have similar common names, there is no evidence that the plasmodial slime molds are closely related to the cellular slime molds; they differ in most features of their structure and life cycles (figure 35.27).

When either food or moisture is in short supply, the plasmodium migrates relatively rapidly to a new area. Here it stops moving and either forms a mass in which spores differentiate or divides into a large number of small mounds, each of which produces a single, mature **sporangium**, the

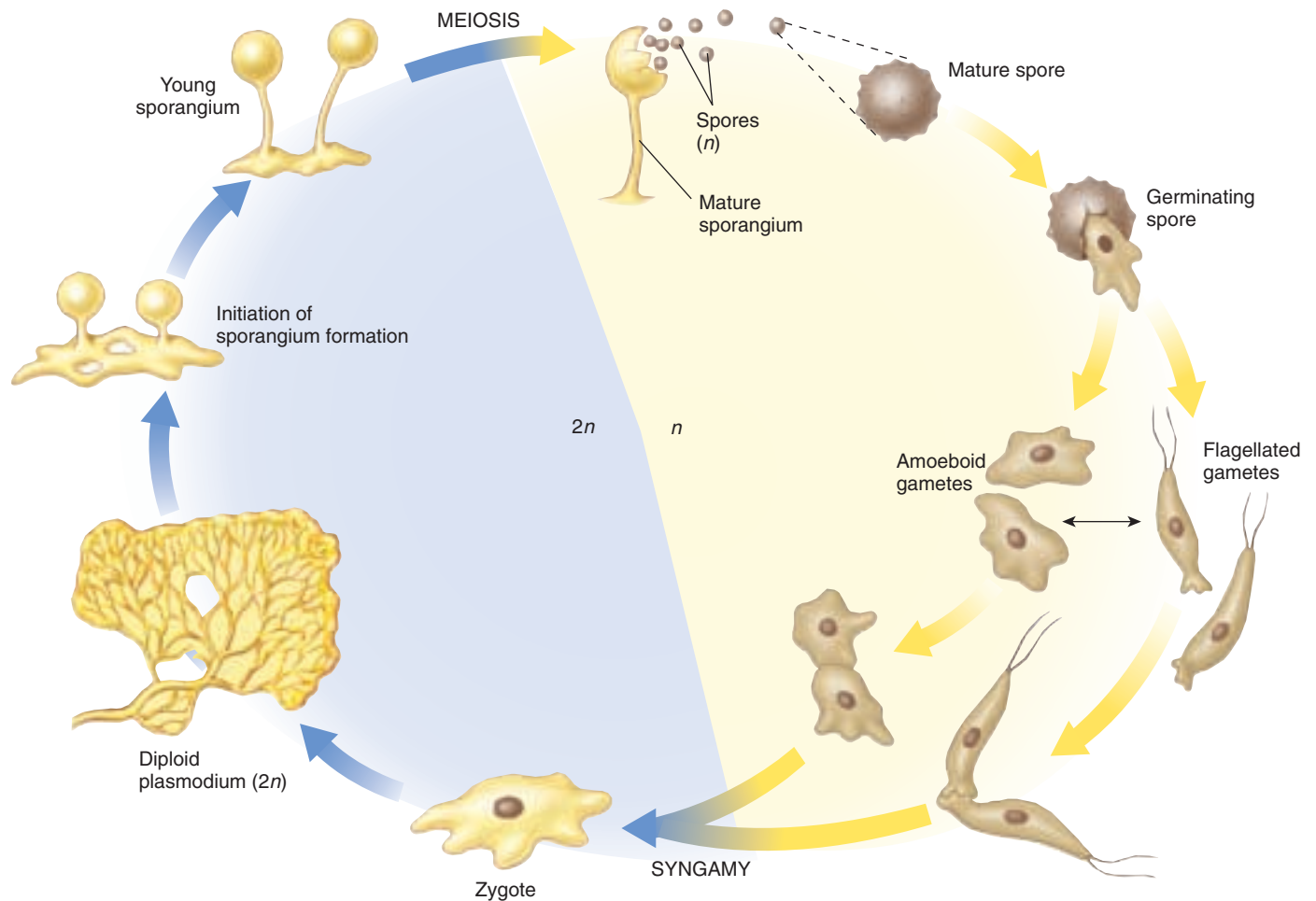
structure in which spores are produced. These sporangia are often extremely complex in form and beautiful (figure 35.28). The spores can be either diploid or haploid. In most species of plasmodial slime molds with a diploid plasmodium, meiosis occurs in the spores within 24 hours of their formation. Three of the four nuclei in each spore disintegrate, leaving each spore with a single haploid nucleus.

The spores are highly resistant to unfavorable environmental influences and may last for years if kept dry. When conditions are favorable, they split open and release their protoplast, the contents of the individual spore. The protoplast may be amoeboid or bear two flagella. These two stages appear to be interchangeable, and conversions in either direction occur readily. Later, after the fusion of haploid protoplasts (gametes), a usually diploid plasmodium may be reconstituted by repeated mitotic divisions.

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**Molds are heterotrophic protists, many of which are capable of amoeba-like streaming. The feeding phase of plasmodial slime molds consists of a multinucleate mass of protoplasm; a plasmodium can flow through a cloth mesh and around obstacles. If the plasmodium begins to dry out or is starving, it forms often elaborate sporangia. Meiosis occurs in the spores once they have formed within the sporangium.**

---



**FIGURE 35.27**

**Life cycle of a plasmodial slime mold.** When food or moisture is scarce, a diploid plasmodium stops moving and forms sporangia. Haploid spores form by meiosis. The spores wait until conditions are favorable to germinate. Spores can give rise to flagellated or amoeboid gametes; the two forms convert from one to the other readily. Fusion of the gametes forms the diploid zygote, which gives rise to the mobile, feeding plasmodium by mitosis.



**FIGURE 35.28**

**Sporangia of three genera of plasmodial slime molds (phylum Myxomycota).** (a) *Arcyria*. (b) *Fuligo*. (c) Developing sporangia of *Tubifera*.



## Summary

## Questions

## Media Resources

## 35.1 Eukaryotes probably arose by endosymbiosis.

- The theory of endosymbiosis, accepted by almost all biologists, proposes that mitochondria and chloroplasts were once aerobic eubacteria that were engulfed by ancestral eukaryotes.
- There is some suggestion that centrioles may also have an endosymbiotic origin.

1. What kind of bacteria most likely gave rise to the chloroplasts in the eukaryotic cells of plants and some algae?

## 35.2 The kingdom Protista is by far the most diverse of any kingdom.

- The kingdom Protista consists of predominantly unicellular phyla, together with three phyla that include large numbers of multicellular organisms.
- The catch-all kingdom Protista includes all eukaryotic organisms except animals, plants, and fungi.

2. Why is the kingdom Protista said to be an artificial group? How is this different from the other kingdoms?



- Characteristics of Protists

## 35.3 Protists can be categorized into five groups.

- Dinoflagellates (phylum Dinoflagellata) are a major phylum of primarily unicellular organisms that have unique chromosomes and a very unusual form of mitosis. They are the only eukaryotes known to lack histones and nucleosomes.
- Euglenoids (phylum Euglenophyta) have chloroplasts that share the biochemical features of those found in green algae and plants.
- Diatoms (phylum Chrysophyta) are unicellular, photosynthetic protists with opaline silica shells. They include the golden algae.
- Brown algae (phylum Phaeophyta) are multicellular, marine protists, some reaching 100 meters in length. The kelps contribute greatly to the productivity of the sea, especially along the coasts in relatively shallow areas.
- The zoomastigotes (phylum Sarcomastigophora) are a group of heterotrophic, mostly unicellular protists that includes the organism responsible for sleeping sickness.
- There are about 8000 named species of ciliates (phylum Ciliophora); these protists have a very complex morphology with numerous cilia.
- The malarial parasite, *Plasmodium*, is a member of the phylum Apicomplexa. Carried by mosquitoes, it multiplies rapidly in the liver of humans and other primates and brings about the cyclical fevers characteristic of malaria by releasing toxins into the bloodstream of its host.

3. Why is mitosis in dinoflagellates unique? What are zooxanthellae?

4. What determines whether a collection of individuals is truly multicellular?

5. What unique characteristic differentiates the members of Ciliophora from other protists? What is the function of two vacuoles exhibited by most members of Ciliophora?

6. Why has it been so difficult to produce a vaccine for trypanosome-caused diseases?

7. What differentiates the oomycetes from the kingdom Fungi, in which they were previously placed? What is the feeding strategy of this phylum? Why are these organisms generally considered harmful?



- Protozoa
- Photosynthetic Protists
- Fungus-like Protists

# 36

## *Fungi*

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### Concept Outline

#### 36.1 Fungi are unlike any other kind of organism.

**A Fungus Is Not a Plant.** Unlike any plant, all fungi are filamentous heterotrophs with cell walls made of chitin.

**The Body of a Fungus.** Cytoplasm flows from one cell to another within the filamentous body of a fungus.

**How Fungi Reproduce.** Fungi reproduce sexually when filaments of different fungi encounter one another and fuse.

**How Fungi Obtain Nutrients.** Fungi secrete digestive enzymes and then absorb the products of the digestion.

**Ecology of Fungi.** Fungi are among the most important decomposers in terrestrial ecosystems.

#### 36.2 Fungi are classified by their reproductive structures.

**The Three Phyla of Fungi.** There are three phyla of fungi, distinguished by their reproductive structures.

**Phylum Zygomycota.** In zygomycetes, the fusion of hyphae leads directly to the formation of a zygote.

**Phylum Ascomycota.** In ascomycetes, hyphal fusion leads to stable dikaryons that grow into massive webs of hyphae that form zygotes within a characteristic saclike structure, the ascus. Yeasts are unicellular fungi, mostly ascomycetes, that play many important commercial and medical roles.

**Phylum Basidiomycota.** In basidiomycetes, dikaryons also form, but zygotes are produced within reproductive structures called basidia.

**The Imperfect Fungi.** Fungi that have not been observed to reproduce sexually cannot be classified into one of the three phyla.

#### 36.3 Fungi form two key mutualistic symbiotic associations.

**Lichens.** A lichen is a mutualistic symbiotic association between a fungus and a photosynthetic alga or cyanobacterium.

**Mycorrhizae.** Mycorrhizae are mutualistic symbiotic associations between fungi and the roots of plants.



**FIGURE 36.1**

**Spores exploding from the surface of a puffball fungus.** The fungi constitute a unique kingdom of heterotrophic organisms. Along with bacteria, they are important decomposers and disease-causing organisms.

Of all the bewildering variety of organisms that live on earth, perhaps the most unusual, the most peculiarly different from ourselves, are the fungi (figure 36.1). Mushrooms and toadstools are fungi, multicellular creatures that grow so rapidly in size that they seem to appear overnight on our lawns. At first glance, a mushroom looks like a funny kind of plant growing up out of the soil. However, when you look more closely, fungi turn out to have nothing in common with plants except that they are multicellular and grow in the ground. As you will see, the more you examine fungi, the more unusual they are.

## 36.1 Fungi are unlike any other kind of organism.

### A Fungus Is Not a Plant

The fungi are a distinct kingdom of organisms, comprising about 77,000 named species (figure 36.2). **Mycologists**, scientists who study fungi, believe there may be many more species in existence, as many as 1.2 million. Although fungi have traditionally been included in the plant kingdom, they lack chlorophyll and resemble plants only in their general appearance and lack of mobility. Significant differences between fungi and plants include the following:

1. **Fungi are heterotrophs.** Perhaps most obviously, a mushroom is not green. Virtually all plants are photosynthesizers, while no fungi have chlorophyll or carry out photosynthesis. Instead, fungi obtain their food by secreting digestive enzymes onto the substrate, and then absorbing the organic molecules that are released by the enzymes.
2. **Fungi have filamentous bodies.** Fungi are basically filamentous in their growth form (that is, their bodies consist of long slender filaments called hyphae), even though these hyphae may be packed together to form complex structures like the mushroom. Plants, in contrast, are made of several types of cells organized into tissues and organs.
3. **Fungi have unusual reproductive modes.** Some plants have motile sperm with flagella. No fungi do.

Most fungi reproduce sexually with nuclear exchange rather than gametes.

4. **Fungi have cell walls made of chitin.** The cell walls of fungi are built of polysaccharides (chains of sugars) and chitin, the same tough material a crab shell is made of. The cell walls of plants are made of cellulose, also a strong building material.
5. **Fungi have nuclear mitosis.** Mitosis in fungi is different from that in plants or most other eukaryotes in one key respect: the nuclear envelope does not break down and re-form. Instead, mitosis takes place *within* the nucleus. A spindle apparatus forms there, dragging chromosomes to opposite poles of the *nucleus* (not the cell, as in most other eukaryotes).

You could build a much longer list, but already the take-home lesson is clear: fungi are not like plants at all! Their many unique features are strong evidence that fungi are not closely related to any other group of organisms. DNA studies confirm significant differences from other eukaryotes.

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**Fungi absorb their food after digesting it with secreted enzymes. This mode of nutrition, combined with a filamentous growth form, nuclear mitosis, and other traits, makes the members of this kingdom highly distinctive.**

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(a)



(c)



(b)

**FIGURE 36.2**

**Representatives of the three phyla of fungi.** (a) A cup fungus, *Cookeina tricholoma*, an ascomycete, from the rain forest of Costa Rica. (b) *Amanita muscaria*, the fly agaric, a toxic basidiomycete. In the cup fungi, the spore-producing structures line the cup; in basidiomycetes that form mushrooms, like *Amanita*, they line the gills beneath the cap of the mushroom. All visible structures of fleshy fungi, such as the ones shown here, arise from an extensive network of filamentous hyphae that penetrates and is interwoven with the substrate on which they grow. (c) *Pilobolus*, a zygomycete that grows on animal feces. Stalks about 10 millimeters long contain dark spore-bearing sacs.

## The Body of a Fungus

Fungi exist mainly in the form of slender filaments, barely visible to the naked eye, which are called **hyphae** (singular, **hypha**). These hyphae are typically made up of long chains of cells joined end-to-end divided by cross-walls called **septa** (singular, **septum**). The septa rarely form a complete barrier, except when they separate the reproductive cells. Cytoplasm characteristically flows or streams freely throughout the hyphae, passing right through major pores in the septa (figure 36.3). Because of this streaming, proteins synthesized throughout the hyphae may be carried to their actively growing tips. As a result, fungal hyphae may grow very rapidly when food and water are abundant and the temperature is optimum.

A mass of connected hyphae is called a **mycelium** (plural, **mycelia**). This word and the term *mycologist* are both derived from the Greek word for fungus, *myketos*. The mycelium of a fungus (figure 36.4) constitutes a system that may, in the aggregate, be many meters long. This mycelium grows through and penetrates its substrate, resulting in a unique relationship between the fungus and its environment. All parts of such a fungus are metabolically active, continually interacting with the soil, wood, or other material in which the mycelium is growing.

In two of the three phyla of fungi, reproductive structures formed of interwoven hyphae, such as mushrooms, puffballs, and morels, are produced at certain stages of the life cycle. These structures expand rapidly because of rapid elongation of the hyphae. For this reason, mushrooms can appear suddenly on your lawn.

The cell walls of fungi are formed of polysaccharides and chitin, not cellulose like those of plants and many groups of protists. Chitin is the same material that makes up the major portion of the hard shells, or exoskeletons, of arthropods, a group of animals that includes insects and crustaceans (see chapter 46). The commonality of chitin is one of the traits that has led scientists to believe that fungi and animals share a common ancestor.

Mitosis in fungi differs from that in most other organisms. Because of the linked nature of the cells, the cell itself is not the relevant unit of reproduction; instead, the nucleus is. The nuclear envelope does not break down and re-form; instead, the spindle apparatus is formed *within* it. Centrioles are lacking in all fungi; instead, fungi regulate the formation of microtubules during mitosis with small, relatively amorphous structures called **spindle plaques**. This unique combination of features strongly suggests that fungi originated from some unknown group of single-celled eukaryotes with these characteristics.

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Fungi exist primarily in the form of filamentous hyphae, typically with incomplete division into individual cells by septa. These and other unique features indicate that fungi are not closely related to any other group of organisms.

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**FIGURE 36.3**  
**A septum (45,000 $\times$ ).** This transmission electron micrograph of a section through a hypha of the basidiomycete *Inonotus tomentosus* shows a pore through which the cytoplasm streams.



**FIGURE 36.4**  
**Fungal mycelium.** This mycelium, composed of hyphae, is growing through leaves on the forest floor in Maryland.

## How Fungi Reproduce

Fungi are capable of both sexual and asexual reproduction. When a fungus reproduces sexually it forms a diploid zygote, as do animals and plants. Unlike animals and plants, all fungal nuclei except for the zygote are haploid, and there are many haploid nuclei in the common cytoplasm of a fungal mycelium. When fungi reproduce sexually, hyphae of two genetically different mating types come together and fuse. In two of the three phyla of fungi, the genetically different nuclei that are associated in a common cytoplasm after fusion do not combine immediately. Instead, the two types of nuclei coexist for most of the life of the fungus. A fungal hypha containing nuclei derived from two genetically distinct individuals is called a **heterokaryotic** hypha. If all of the nuclei are genetically similar to one another, the hypha is said to be **homokaryotic**. If there are two distinct nuclei within each compartment of the hyphae, they are **dikaryotic**. If each compartment has only a single nucleus, it is **monokaryotic**. Dikaryotic hyphae have some of the genetic properties of diploids, because both genomes are transcribed. These distinctions are important in understanding the life cycles of the individual groups.

Cytoplasm in fungal hyphae normally flows through perforated septa or moves freely in their absence. Reproductive structures are an important exception to this general pattern. When reproductive structures form, they are cut off by complete septa that lack perforations or have perforations that soon become blocked. Three kinds of reproductive structures occur in fungi: (1) **sporangia**, which are involved in the formation of spores; (2) **gametangia**, structures within which gametes form; and (3) **conidiophores**, structures that produce **conidia**, multinucleate asexual spores.

Spores are a common means of reproduction among fungi. They may form as a result of either asexual or sexual processes and are always nonmotile, being dispersed by wind. When spores land in a suitable place, they germinate, giving rise to a new fungal hypha. Because the spores are very small, they can remain suspended in the air for long periods of time. Because of this, fungal spores may be blown great distances from their place of origin, a factor in the extremely wide distributions of many kinds of fungi. Unfortunately, many of the fungi that cause diseases in plants and animals are spread rapidly and widely by such means. The spores of other fungi are routinely dispersed by insects and other small animals.

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Fungi reproduce sexually after two hyphae of opposite mating type fuse. Asexual reproduction by spores is a second common means of reproduction.

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**FIGURE 36.5**

**A carnivorous fungus.** The oyster mushroom, *Pleurotus ostreatus*, not only decomposes wood but also immobilizes nematodes, which the fungus uses as a source of nitrogen.

## How Fungi Obtain Nutrients

All fungi obtain their food by secreting digestive enzymes into their surroundings and then absorbing back into the fungus the organic molecules produced by this **external digestion**. The significance of the fungal body plan reflects this approach, the extensive network of hyphae providing an enormous surface area for absorption. Many fungi are able to break down the cellulose in wood, cleaving the linkages between glucose subunits and then absorbing the glucose molecules as food. That is why fungi so often grow on dead trees.

It might surprise you to know that some fungi are predatory (figure 36.5). For example, the mycelium of the edible oyster fungus, *Pleurotus ostreatus*, excretes a substance that anesthetizes tiny roundworms known as nematodes (see chapter 44) that feed on the fungus. When the worms become sluggish and inactive, the fungal hyphae envelop and penetrate their bodies and absorb their nutritious contents. The fungus usually grows within living trees or on old stumps, obtaining the bulk of its glucose through the enzymatic digestion of cellulose from the wood, so that the nematodes it consumes apparently serve mainly as a source of nitrogen—a substance almost always in short supply in biological systems. Other fungi are even more active predators than *Pleurotus*, snaring, trapping, or firing projectiles into nematodes, rotifers, and other small animals on which they prey.

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Fungi secrete digestive enzymes onto organic matter and then absorb the products of the digestion.

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**FIGURE 36.6**

**World's largest organism?**

*Armillaria*, a pathogenic fungus shown here afflicting three discrete regions of coniferous forest in Montana, grows out from a central focus as a single circular clone. The large patch at the bottom of the picture is almost 8 hectares in diameter. The largest clone measured so far has been 15 hectares in diameter—pretty impressive for a single individual!



## Ecology of Fungi

Fungi, together with bacteria, are the principal decomposers in the biosphere. They break down organic materials and return the substances locked in those molecules to circulation in the ecosystem. Fungi are virtually the only organisms capable of breaking down lignin, one of the major constituents of wood. By breaking down such substances, fungi release critical building blocks, such as carbon, nitrogen, and phosphorus, from the bodies of dead organisms and make them available to other organisms.

In breaking down organic matter, some fungi attack living plants and animals as a source of organic molecules, while others attack dead ones. Fungi often act as disease-causing organisms for both plants (figure 36.6) and animals, and they are responsible for billions of dollars in agricultural losses every year. Not only are fungi the most harmful pests of living plants, but they also attack food products once they have been harvested and stored. In addition, fungi often secrete substances into the foods that they are attacking that make these foods unpalatable, carcinogenic, or poisonous.

The same aggressive metabolism that makes fungi ecologically important has been put to commercial use in many ways. The manufacture of both bread and beer depends on the biochemical activities of **yeasts**, single-celled fungi that produce abundant quantities of ethanol and carbon dioxide. Cheese and wine achieve their delicate flavors because of the metabolic processes of certain fungi, and others make possible the manufacture of soy sauce and other fermented foods. Vast industries depend on the biochemical manufacture of organic substances such as citric

acid by fungi in culture, and yeasts are now used on a large scale to produce protein for the enrichment of animal food. Many antibiotics, including the first one that was used on a wide scale, penicillin, are derived from fungi.

Some fungi are used to convert one complex organic molecule into another, cleaning up toxic substances in the environment. For example, at least three species of fungi have been isolated that combine selenium, accumulated at the San Luis National Wildlife Refuge in California's San Joaquin Valley, with harmless volatile chemicals—thus removing excess selenium from the soil.

Two kinds of mutualistic associations between fungi and autotrophic organisms are ecologically important. **Lichens** are mutualistic symbiotic associations between fungi and either green algae or cyanobacteria. They are prominent nearly everywhere in the world, especially in unusually harsh habitats such as bare rock. **Mycorrhizae**, specialized mutualistic symbiotic associations between the roots of plants and fungi, are characteristic of about 90% of all plants. In each of them, the photosynthetic organisms fix atmospheric carbon dioxide and thus make organic material available to the fungi. The metabolic activities of the fungi, in turn, enhance the overall ability of the symbiotic association to exist in a particular habitat. In the case of mycorrhizae, the fungal partner expedites the plant's absorption of essential nutrients such as phosphorus. Both of these associations will be discussed further in this chapter.

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**Fungi are key decomposers and symbionts within almost all terrestrial ecosystems and play many other important ecological and commercial roles.**

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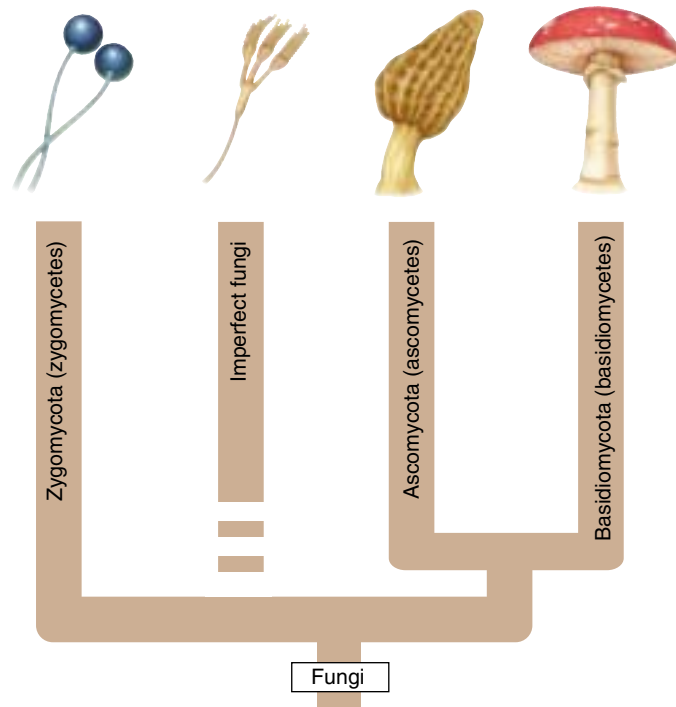
## 36.2 Fungi are classified by their reproductive structures.

### The Three Phyla of Fungi

There are three phyla but actually four groups of fungi: phylum Zygomycota, the **zygomycetes**; phylum Ascomycota, the **ascomycetes**; phylum Basidiomycota, the **basidiomycetes**, and the imperfect fungi (figure 36.7 and table 36.1). Several other groups that historically have been associated with fungi, such as the slime molds and water molds (phylum Oomycota; see chapter 35), now are considered to be protists, not fungi. Oomycetes are sharply distinct from fungi in their (1) motile spores; (2) cellulose-rich cell walls; (3) pattern of mitosis; and (4) diploid hyphae.

The three phyla of fungi are distinguished primarily by their sexual reproductive structures. In the zygomycetes, the fusion of hyphae leads directly to the formation of a zygote, which divides by meiosis when it germinates. In the other two phyla, an extensive growth of dikaryotic hyphae may lead to the formation of structures of interwoven hyphae within which are formed the distinctive kind of reproductive cell characteristic of that particular group. Nuclear fusion, followed by meiosis, occurs within these cells. The imperfect fungi are either asexual or the sexual reproductive structures have not been identified.

**Sexual reproductive structures distinguish the three phyla of fungi.**



**FIGURE 36.7**

**The four major groups of fungi.** The imperfect fungi are not a true phylum, but rather a collection of fungi in which sexual structures have not been identified.

**Table 36.1 Fungi**

| Phylum          | Typical Examples                        |                                                                                     | Key Characteristics                                                                                                                                                                                              | Approximate Number of Living Species |
|-----------------|-----------------------------------------|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| Ascomycota      | Yeasts, truffles, morels                |  | Develop by sexual means; ascospores are formed inside a sac called an ascus; asexual reproduction is also common                                                                                                 | 32,000                               |
| Imperfect fungi | <i>Aspergillus</i> , <i>Penicillium</i> |  | Sexual reproduction has not been observed; most are thought to be ascomycetes that have lost the ability to reproduce sexually                                                                                   | 17,000                               |
| Basidiomycota   | Mushrooms, toadstools, rusts            |  | Develop by sexual means; basidiospores are borne on club-shaped structures called basidia; the terminal hyphal cell that produces spores is called a basidium; asexual reproduction occurs occasionally          | 22,000                               |
| Zygomycota      | <i>Rhizopus</i> (black bread mold)      |  | Develop sexually and asexually; multinucleate hyphae lack septa, except for reproductive structures; fusion of hyphae leads directly to formation of a zygote, in which meiosis occurs just before it germinates | 1050                                 |

## Phylum Zygomycota

The zygomycetes (phylum Zygomycota) lack septa in their hyphae except when they form sporangia or gametangia. Zygomycetes are by far the smallest of the three phyla of fungi, with only about 1050 named species. Included among them are some of the more common bread molds (figure 36.8), as well as a variety of other microscopic fungi found on decaying organic material. The group is named after a characteristic feature of the life cycle of its members, the production of temporarily dormant structures called **zygosporangia**.

In the life cycle of the zygomycetes (figure 36.8b), sexual reproduction occurs by the fusion of gametangia, which contain numerous nuclei. The gametangia are cut off from the hyphae by complete septa. These gametangia may be formed on hyphae of different mating types or on a single hypha. If both + and – mating strains are present in a colony, they may grow together and their nuclei may fuse. Once the haploid nuclei have fused, forming diploid zygote nuclei, the area where the fusion has taken place develops into an often massive and elaborate zygosporangium. A zygosporangium may contain one or more diploid nuclei and acquires a thick

coat. The zygosporangium helps the species survive conditions not favorable for growth. Meiosis occurs during the germination of the zygosporangium. Normal, haploid hyphae grow from the haploid cells that result from this process. Except for the zygote nuclei, all nuclei of the zygomycetes are haploid.

Asexual reproduction occurs much more frequently than sexual reproduction in the zygomycetes. During asexual reproduction, hyphae grow over the surface of the bread or other material on which the fungus feeds and produce clumps of erect stalks, called **sporangophores**. The tips of the sporangiophores form **sporangia**, which are separated by septa. Thin-walled haploid spores are produced within the sporangia. Their spores are thus shed above the substrate, in a position where they may be picked up by the wind and dispersed to a new food source.

lloid spores are produced within the sporangia. Their spores are thus shed above the substrate, in a position where they may be picked up by the wind and dispersed to a new food source.

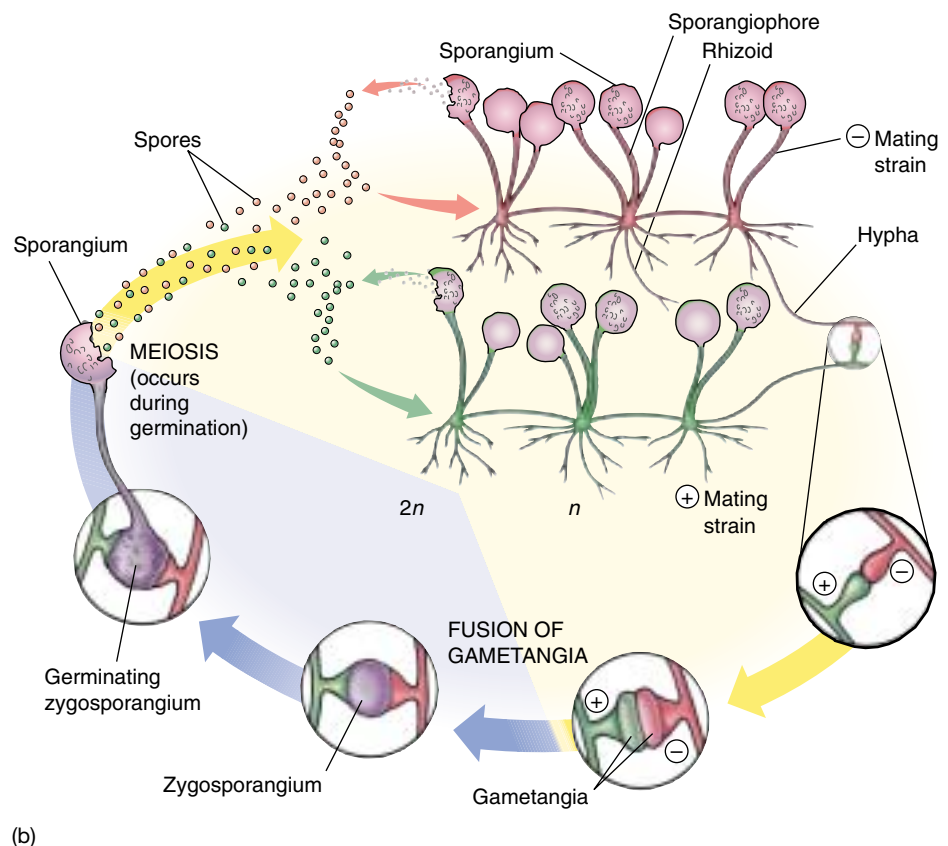
**Zygomycetes form characteristic resting structures, called zygosporangia, which contain one or more zygotic nuclei. The hyphae of zygomycetes are multinucleate, with septa only where gametangia or sporangia are separated.**



(a)

**FIGURE 36.8**

*Rhizopus*, a zygomycete that grows on moist bread and other similar substrates. (a) The dark, spherical, spore-producing sporangia are on hyphae about a centimeter tall. The rootlike hyphae anchor the sporangia. (b) Life cycle of *Rhizopus*, a zygomycete. This phylum is named for its characteristic zygosporangia.



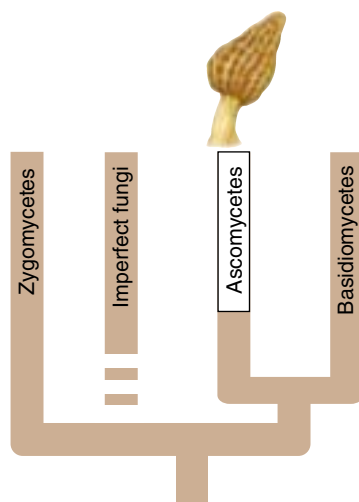
(b)

## Phylum Ascomycota

The second phylum of fungi, the ascomycetes (phylum Ascomycota), is a very large group of about 32,000 named species, with more being discovered each year. Among the ascomycetes are such familiar and economically important fungi as yeasts, common molds, morels (figure 36.9a,b), and truffles. Also included in this phylum are many serious plant pathogens, including the chestnut blight, *Cryphonectria parasitica*, and Dutch elm disease, *Ophiostoma ulmi*.

The ascomycetes are named for their characteristic reproductive structure, the microscopic, saclike **ascus** (plural, **asci**). The zygotic nucleus, which is the only diploid nucleus of the ascomycete life cycle (figure 36.9c), is formed within the ascus. The asci are differentiated within a structure made up of densely interwoven hyphae, corresponding to the visible portions of a morel or cup fungus, called the **ascocarp**.

Asexual reproduction is very common in the ascomycetes. It takes place by means of **conidia** (singular,



**conidium**), spores cut off by septa at the ends of modified hyphae called **conidiophores**. Conidia allow for the rapid colonization of a new food source. Many conidia are multinucleate. The hyphae of ascomycetes are divided by septa, but the septa are perforated and the cytoplasm flows along the length of each hypha. The septa that cut off the asci and conidia are initially perforated, but later become blocked.

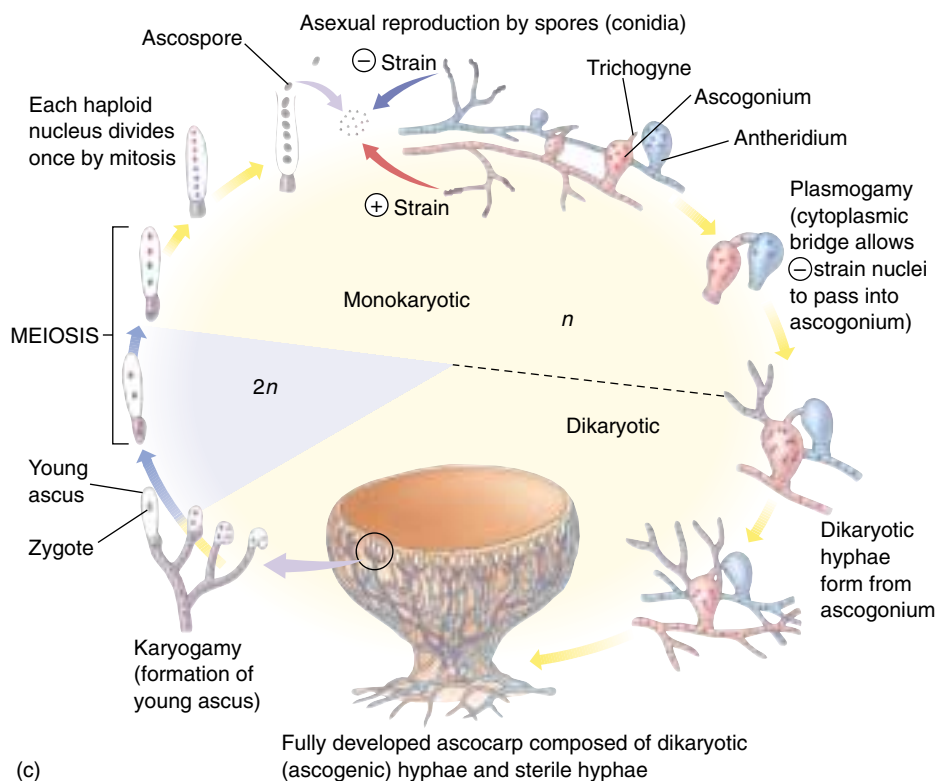
The cells of ascomycete hyphae may contain from several to many nuclei. The hyphae may be either homokaryotic or heterokaryotic. Female gametangia, called **ascogonia**, each have a beaklike outgrowth called a **trichogyne**. When the **antheridium**, or male gametangium, forms, it fuses with the trichogyne of an adjacent ascogonium. Initially, both kinds of gametangia contain a number of nuclei. Nuclei from the antheridium then migrate through the trichogyne into the ascogonium and pair with nuclei of the opposite mating type. Dikaryotic hyphae then arise from the area of the fusion. Throughout such hyphae,



(a)



(b)



(c)

**FIGURE 36.9**

**An ascomycetes.** (a) This morel, *Morchella esculenta*, is a delicious edible ascomycete that appears in early spring. (b) A cup fungus. (c) Life cycle of an ascomycete. The zygote forms within the ascus.

nuclei that represent the two different original mating types occur. These hyphae are thus both dikaryotic and heterokaryotic.

Asci are formed at the tips of the dikaryotic hyphae and are separated by the formation of septa. There are two haploid nuclei within each ascus, one of each mating type represented in the dikaryotic hypha. Fusion of these two nuclei occurs within each ascus, forming a zygote. Each zygote divides immediately by meiosis, forming four haploid daughter nuclei. These usually divide again by mitosis, producing eight haploid nuclei that become walled **ascospores**. In many ascomycetes, the ascus becomes highly turgid at maturity and ultimately bursts, often at a pre-formed area. When this occurs, the ascospores may be thrown as far as 30 centimeters, an amazing distance considering that most ascospores are only about 10 micrometers long. This would be equivalent to throwing a baseball (diameter 7.5 centimeters) 1.25 kilometers—about 10 times the length of a home run!

## Yeasts

Yeasts, which are unicellular, are one of the most interesting and economically important groups of microscopic fungi, usually ascomycetes. Most of their reproduction is asexual and takes place by cell fission or budding, when a smaller cell forms from a larger one (figure 36.10).

Sometimes two yeast cells will fuse, forming one cell containing two nuclei. This cell may then function as an ascus, with syngamy followed immediately by meiosis. The resulting ascospores function directly as new yeast cells.

Because they are single-celled, yeasts were once considered primitive fungi. However, it appears that they are actually reduced in structure and were originally derived from multicellular ancestors. The word *yeast* actually signifies only that these fungi are single-celled. Some yeasts have been derived from each of the three phyla of fungi, although ascomycetes are best represented. Even yeasts that were derived from ascomycetes are not necessarily directly related to one another, but instead seem to have been derived from different groups of ascomycetes.

**Putting Yeasts to Work.** The ability of yeasts to ferment carbohydrates, breaking down glucose to produce ethanol and carbon dioxide, is fundamental in the production of bread, beer, and wine. Many different strains of yeast have been domesticated and selected for these processes. Wild yeasts—ones that occur naturally in the areas where wine is made—were important in wine making historically, but domesticated yeasts are normally used now. The most important yeast in all these processes is *Saccharomyces cerevisiae*. This yeast has been used by humans throughout recorded history. Other yeasts are important pathogens and cause diseases such as thrush and



**FIGURE 36.10**

Scanning electron micrograph of a yeast, showing the characteristic cell division method of budding (19,000 $\times$ ). The cells tend to hang together in chains, a feature that calls to mind the derivation of single-celled yeasts from multicellular ancestors.

cryptococcosis; one of them, *Candida*, causes common oral or vaginal infections.

Over the past few decades, yeasts have become increasingly important in genetic research. They were the first eukaryotes to be manipulated extensively by the techniques of genetic engineering, and they still play the leading role as models for research in eukaryotic cells. In 1983, investigators synthesized a functional artificial chromosome in *Saccharomyces cerevisiae* by assembling the appropriate DNA molecule chemically; this has not yet been possible in any other eukaryote. In 1996, the genome sequence of *S. cerevisiae*, the first eukaryote to be sequenced entirely, was completed. With their rapid generation time and a rapidly increasing pool of genetic and biochemical information, the yeasts in general and *S. cerevisiae* in particular are becoming the eukaryotic cells of choice for many types of experiments in molecular and cellular biology. Yeasts have become, in this respect, comparable to *Escherichia coli* among the bacteria, and they are continuing to provide significant insights into the functioning of eukaryotic systems.

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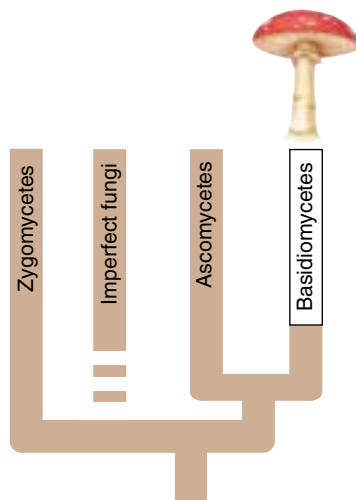
**Ascomycetes form their zygotes within a characteristic saclike structure, the ascus. Meiosis follows, resulting in the production of ascospores. Yeasts are unicellular fungi, mainly ascomycetes, that have evolved from hypha-forming ancestors; not all yeasts are directly related to one another. Long useful for baking, brewing, and wine making, yeasts are now becoming very important in genetic research.**

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## Phylum Basidiomycota

The third phylum of fungi, the basidiomycetes (phylum Basidiomycota), has about 22,000 named species. These are among the most familiar fungi. Among the basidiomycetes are not only the mushrooms, toadstools, puffballs, jelly fungi, and shelf fungi, but also many important plant pathogens including rusts and smuts (figure 36.11). Many mushrooms are used as food, but others are deadly poisonous.

Basidiomycetes are named for their characteristic sexual reproductive structure, the **basidium** (plural, **basidia**). A basidium is club-shaped. Karyogamy occurs within the basidium, giving rise to the zygote, the only diploid cell of the life cycle (figure 36.11*b*). As in all fungi, meiosis occurs immediately after the formation of the zygote. In the basidiomycetes, the four haploid products of meiosis are incorporated into **basidiospores**. In most members of this phylum, the basidiospores are borne at the end of the basidia on slender projections called **sterigmata** (singular, **sterigma**). Thus the structure of a basidium differs from that of an ascus, although functionally the two are identical. Recall that the ascospores of the ascomycetes are borne internally in asci.



The life cycle of a basidiomycete continues with the production of homokaryotic hyphae after spore germination. These hyphae lack septa at first. Eventually, septa form between the nuclei of the monokaryotic hyphae. A basidiomycete mycelium made up of monokaryotic hyphae is called a **primary mycelium**. Different mating types of monokaryotic hyphae may fuse, forming a dikaryotic or **secondary mycelium**. Such a mycelium is heterokaryotic, with two nuclei, representing the two different mating types, between each pair of septa. The maintenance of two genomes in the heterokaryon allows for more genetic plasticity than in a diploid cell with one nucleus. One genome may compensate for mutations in the other. The **basidiocarps**, or mushrooms, are formed entirely of secondary (dikaryotic) mycelium. Gills on the undersurface of the cap of a mushroom form vast numbers of minute spores. It has been estimated that a mushroom with a cap that is 7.5 centimeters across produces as many as 40 million spores per hour!

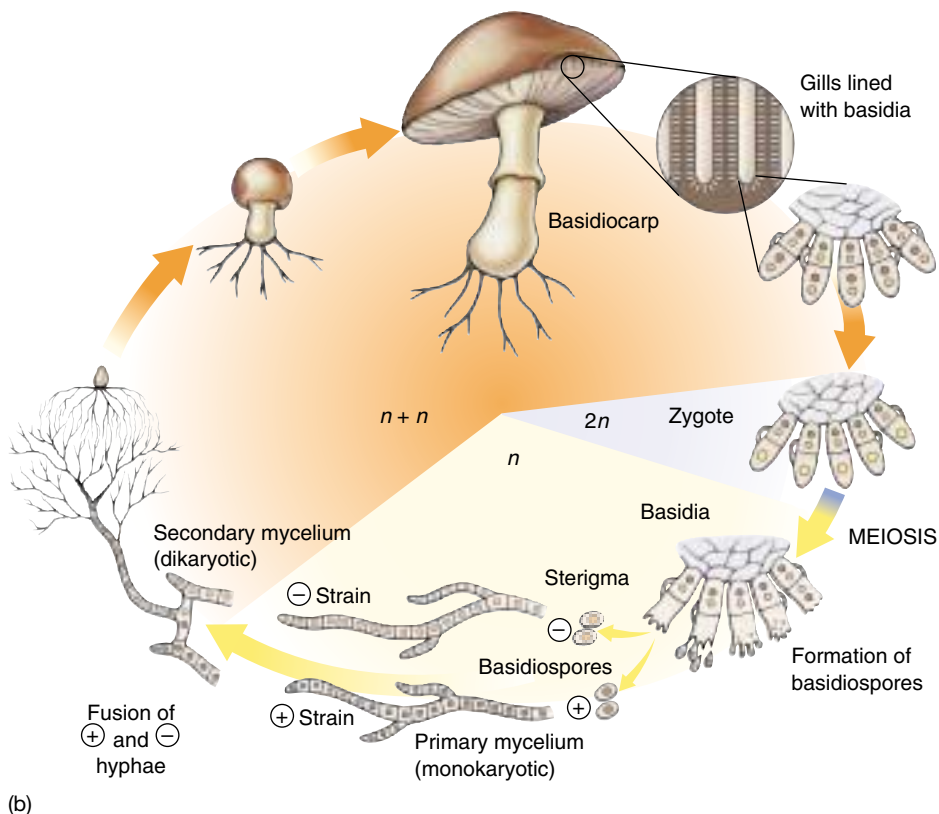
**Most basidiomycete hyphae are dikaryotic. Ultimately, the hyphae fuse to form basidiocarps, with basidia lining the gills on the underside. Meiosis immediately follows syngamy in these basidia.**



(a)

**FIGURE 36.11**

**Basidiomycetes.** (a) Death cap mushroom, *Amanita phalloides*. When eaten, these mushrooms are usually fatal. (b) Life cycle of a basidiomycete. The basidium is the reproductive structure where syngamy occurs.

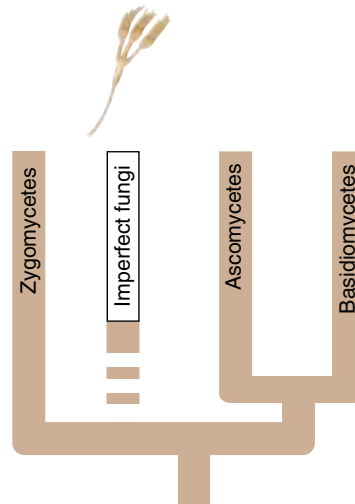


(b)

## The Imperfect Fungi

Most of the so-called imperfect fungi, a group also called deuteromycetes, are those in which the sexual reproductive stages have not been observed. Most of these appear to be related to ascomycetes although some have clear affinities to the other phyla. The group of fungi from which a particular nonsexual strain has been derived usually can be determined by the features of its hyphae and asexual reproduction. It cannot, however, be classified by the standards of that group because the classification systems are based on the features related to sexual reproduction. One consequence of this system is that as sexual reproduction is discovered in an imperfect fungus, it may have two names assigned to different stages of its life cycle.

There are some 17,000 described species of imperfect fungi (figure 36.12). Even though sexual reproduction is absent among imperfect fungi, a certain amount of genetic recombination occurs. This becomes possible when hyphae of different genetic types fuse, as sometimes happens spontaneously. Within the heterokaryotic hyphae that arise from such fusion, a special kind of genetic recombination called **parasexuality** may occur. In parasexuality, genetically distinct nuclei within a common hypha exchange portions of chromosomes. Recombination of this sort also occurs in other groups of fungi and seems to be responsible for some of the new pathogenic strains of wheat rust.



## Economic Importance

Among the economically important genera of the imperfect fungi are *Penicillium* and *Aspergillus*. Some species of *Penicillium* are sources of the well-known antibiotic penicillin, and other species of the genus give the characteristic flavors and aromas to cheeses such as Roquefort and Camembert. Species of *Aspergillus* are used to ferment soy sauce and soy paste, processes in which certain bacteria and yeasts also play important roles. Citric acid is produced commercially with members of this genus under highly acidic conditions. Some species of both *Penicillium* and *Aspergillus* form ascocarps, but the genera are still classified primarily as imperfect fungi because the ascocarps are found rarely in only a few species. Most of the fungi that cause skin diseases in humans, including athlete's foot and ringworm, are also imperfect fungi.

A number of imperfect fungi occur widely on food. *Fusarium* species growing on spoiled food produce highly toxic substances such as trichothecenes. Aflatoxins, among the most carcinogenic compounds known, are produced by some *Aspergillus flavus* strains growing on corn, peanuts, etc. Most developed countries have legal limits on the concentration of aflatoxin permitted in different foods.

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**Imperfect fungi are fungi in which no sexual reproduction has been observed. For this reason, they cannot be classified by the standards applied to the three phyla of fungi. The great majority of the imperfect fungi are clearly ascomycetes.**

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(a)



(b)



(c)

**FIGURE 36.12**

**The imperfect fungi.** (a) *Verticillium albo-atrum* (1350 $\times$ ), an important pathogen of alfalfa, has whorled conidiophores. The single-celled conidia of this member of the imperfect fungi are borne at the ends of the conidiophores. (b) In *Tolypocladium inflatum*, the conidia arise along the branches. This fungus is one of the sources of cyclosporin, a drug that suppresses immune reactions and thus assists in making human organ grafts possible; the drug was put on the market in 1979. (c) This scanning electron micrograph of *Aspergillus* shows conidia, the spheres at the end of the hyphae.

### 36.3 Fungi form two key mutualistic symbiotic associations.

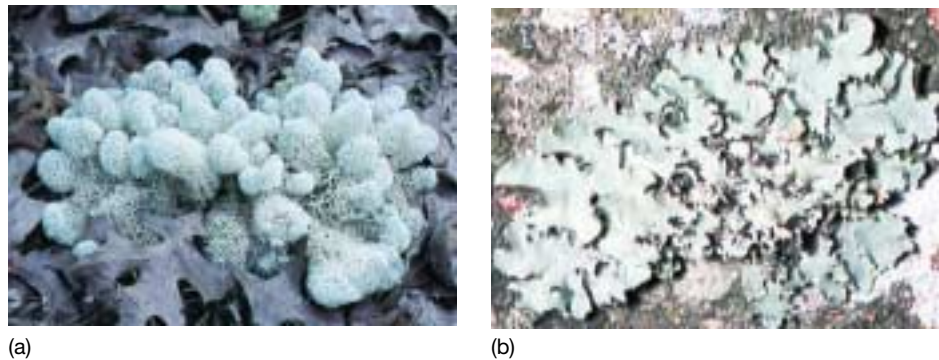
#### Lichens

Lichens (figure 36.13) are symbiotic associations between a fungus and a photosynthetic partner. They provide an outstanding example of mutualism, the kind of symbiotic association that benefits both partners. Ascomycetes (including some imperfect fungi) are the fungal partners in all but about 20 of the approximately 15,000 species of lichens estimated to exist; the exceptions, mostly tropical, are basidiomycetes. Most of the visible body of a lichen consists of its fungus, but within the tissues of that fungus are found cyanobacteria, green algae, or sometimes both (figure 36.14). Specialized fungal hyphae penetrate or envelop the photosynthetic cells within them and transfer nutrients directly to the fungal partner. Biochemical “signals” sent out by the fungus apparently direct its cyanobacterial or green algal component to produce metabolic substances that it does not produce when growing independently of the fungus. The photosynthetic member of the association is normally held between thick layers of interwoven fungal hyphae and is not directly exposed to the light, but enough light penetrates the translucent layers of fungal hyphae to make photosynthesis possible. The fungi in lichens are unable to grow normally without their photosynthetic partners and the fungi protect their partners from strong light and desiccation.

The durable construction of the fungus, combined with the photosynthetic properties of its partner, has enabled lichens to invade the harshest habitats at the tops of mountains, in the farthest northern and southern latitudes, and on dry, bare rock faces in the desert. In harsh, exposed areas, lichens are often the first colonists, breaking down the rocks and setting the stage for the invasion of other organisms.

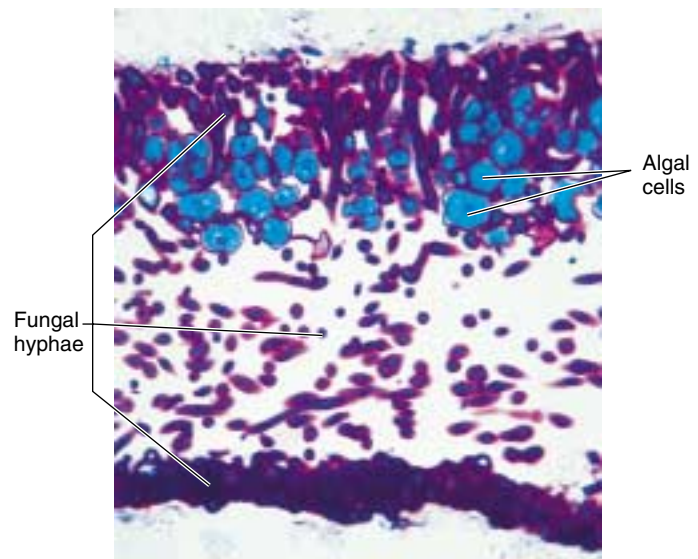
Lichens are often strikingly colored because of the presence of pigments that probably play a role in protecting the photosynthetic partner from the destructive action of the sun’s rays. These same pigments may be extracted from the lichens and used as natural dyes. The traditional method of manufacturing Scotland’s famous Harris tweed used fungal dyes.

Lichens are extremely sensitive to pollutants in the atmosphere, and thus they can be used as bioindicators of air quality. Their sensitivity results from their ability to absorb substances dissolved in rain and dew. Lichens are generally absent in and around cities because of automo-



**FIGURE 36.13**

**Lichens are found in a variety of habitats.** (a) A fruticose lichen, *Cladonia evansii*, growing on the ground in Florida. (b) A foliose (“leafy”) lichen, *Parmotrema gardneri*, growing on the bark of a tree in a mountain forest in Panama.



**FIGURE 36.14**

**Stained section of a lichen (250×).** This section shows fungal hyphae (purple) more densely packed into a protective layer on the top and, especially, the bottom layer of the lichen. The blue cells near the upper surface of the lichen are those of a green alga. These cells supply carbohydrate to the fungus.

bile traffic and industrial activity, even though suitable substrates exist.

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**Lichens are symbiotic associations between a fungus—an ascomycete in all but a very few instances—and a photosynthetic partner, which may be a green alga or a cyanobacterium or both.**

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## Mycorrhizae

The roots of about 90% of all kinds of vascular plants normally are involved in mutualistic symbiotic relationships with certain kinds of fungi. It has been estimated that these fungi probably amount to 15% of the total weight of the world's plant roots. Associations of this kind are termed **mycorrhizae** (from the Greek words for “fungus” and “roots”). The fungi in mycorrhizae associations function as extensions of the root system. The fungal hyphae dramatically increase the amount of soil contact and total surface area for absorption. When mycorrhizae are present, they aid in the direct transfer of phosphorus, zinc, copper, and other nutrients from the soil into the roots. The plant, on the other hand, supplies organic carbon to the fungus, so the system is an example of mutualism.

There are two principal types of mycorrhizae (figure 36.15): **endomycorrhizae**, in which the fungal hyphae penetrate the outer cells of the plant root, forming coils, swellings, and minute branches, and also extend out into the surrounding soil; and **ectomycorrhizae**, in which the hyphae surround but do not penetrate the cell walls of the roots. In both kinds of mycorrhizae, the mycelium extends far out into the soil.

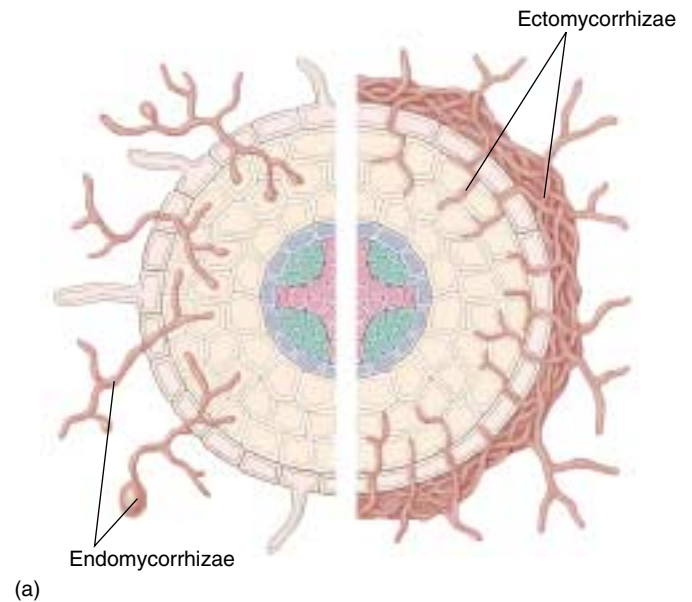
### Endomycorrhizae

Endomycorrhizae are by far the more common of these two types. The fungal component in them is a zygomycete. Only about 100 species of zygomycetes are known to be involved in such relationships throughout the world. These few species of zygomycetes are associated with more than 200,000 species of plants. Endomycorrhizal fungi are being studied intensively because they are potentially capable of increasing crop yields with lower phosphate and energy inputs.

The earliest fossil plants often show endomycorrhizal roots. Such associations may have played an important role in allowing plants to colonize land. The soils available at such times would have been sterile and completely lacking in organic matter. Plants that form mycorrhizal associations are particularly successful in infertile soils; considering the fossil evidence, the suggestion that mycorrhizal associations found in the earliest plants helped them succeed on such soils seems reasonable. In addition, the most primitive vascular plants surviving today continue to depend strongly on mycorrhizae.

### Ectomycorrhizae

Ectomycorrhizae (figure 36.15*b*) involve far fewer kinds of plants than do endomycorrhizae, perhaps a few thousand. They are characteristic of certain groups of trees and shrubs, particularly those of temperate regions, including pines, firs, oaks, beeches, and willows. The fungal components in most ectomycorrhizae are basidiomycetes, but



**FIGURE 36.15**

**Endomycorrhizae and ectomycorrhizae.** (a) In endomycorrhizae, fungal hyphae penetrate and branch out in the root cells of plants. In ectomycorrhizae, fungal hyphae do not penetrate root cells but grow around and extend between the cells. (b) Ectomycorrhizae on roots of pines. From left to right are yellow-brown mycorrhizae formed by *Pisolithus*, white mycorrhizae formed by *Rhizopogon*, and pine roots not associated with a fungus.

some are ascomycetes. Several different kinds of ectomycorrhizal fungi may form mycorrhizal associations with one plant. Different combinations have different effects on the physiological characteristics of the plant and its ability to survive under different environmental conditions. At least 5000 species of fungi are involved in ectomycorrhizal relationships, and many of them are restricted to a single species of plant.

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**Mycorrhizae are symbiotic associations between plants and fungi.**

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## Summary

## Questions

## Media Resources

## 36.1 Fungi are unlike any other kind of organism.

- The fungi are a distinct kingdom of eukaryotic organisms characterized by a filamentous growth form, lack of chlorophyll and motile cells, chitin-rich cell walls, and external digestion of food by the secretion of enzymes.
- Fungal filaments, called hyphae, collectively make up the fungus body, which is called the mycelium.
- In many fungi, the two kinds of nuclei that will eventually undergo syngamy occur together in hyphae for a long period before they fuse. Meiosis occurs immediately after the formation of the zygote in all fungi; the zygote, therefore, is the only diploid nucleus of the entire life cycle in these organisms.

1. What is a hypha? What is the advantage to having incomplete septa?
2. What is the composition of the fungal cell wall? Why is this composition an advantage to the fungi?
3. Which fungal nuclei are diploid? Which are haploid? To what do the following terms refer: *heterokaryotic*, *homokaryotic*, *dikaryotic*, and *monokaryotic*?



- Characteristics of Fungi

## 36.2 Fungi are classified by their reproductive structures.

- There are three phyla of fungi: Zygomycota, the zygomycetes; Ascomycota, the ascomycetes; and Basidiomycota, the basidiomycetes.
- Zygomycetes form septa only when gametangia or sporangia are cut off at the ends of their hyphae; otherwise, their hyphae are multinucleate. Most hyphae of ascomycetes and basidiomycetes have perforated septa through which the cytoplasm, but not necessarily the nuclei, flows freely.
- Cells within the heterokaryotic hyphae of ascomycetes are multinucleate; those within the heterokaryotic hyphae of the basidiomycetes are dikaryotic. Zygotes in ascomycetes form within sac-like structures known as asci, and those in basidiomycetes form within structures known as basidia.
- Asexual reproduction in zygomycetes takes place by means of spores from multinucleate sporangia; in ascomycetes, it takes place by means of conidia. Asexual reproduction in basidiomycetes is rare.

4. What are the three reproductive structures that occur in fungi? How do they differ?
5. Fungi are nonmotile. How are they dispersed to new areas?
6. What are the ascomycete asexual spores called? Do the nonreproductive hyphae of this division have septa?
7. To what phyla do the yeasts belong? How do they differ from other fungi? Is it more likely that this characteristic is primitive or degenerate?
8. What are the imperfect fungi? Which phylum seems to be best represented in this group? By what means can individuals in this group be classified?



- Diversity of Fungi



- Student Research: Mushroom Spore Germination

## 36.3 Fungi form two key mutualistic symbiotic associations.

- Lichens are mutualistic symbiotic systems involving fungi (almost always ascomycetes), which derive their nutrients from green algae, cyanobacteria, or both.
- Mycorrhizae are mutualistic symbiotic associations between fungi and plants. Endomycorrhizae, more common, involve zygomycetes, while ectomycorrhizal fungi are mainly basidiomycetes.

9. What are lichens? Which fungal phylum is best represented in the lichens?
10. What are mycorrhizae? How do endomycorrhizae and ectomycorrhizae differ?

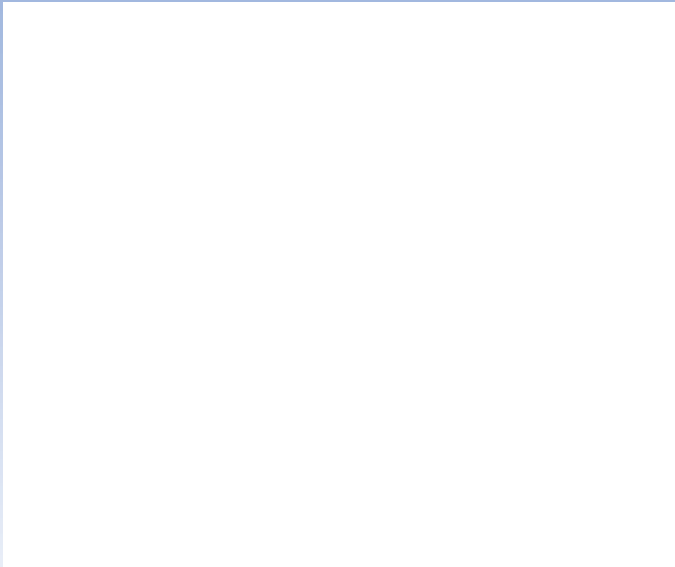
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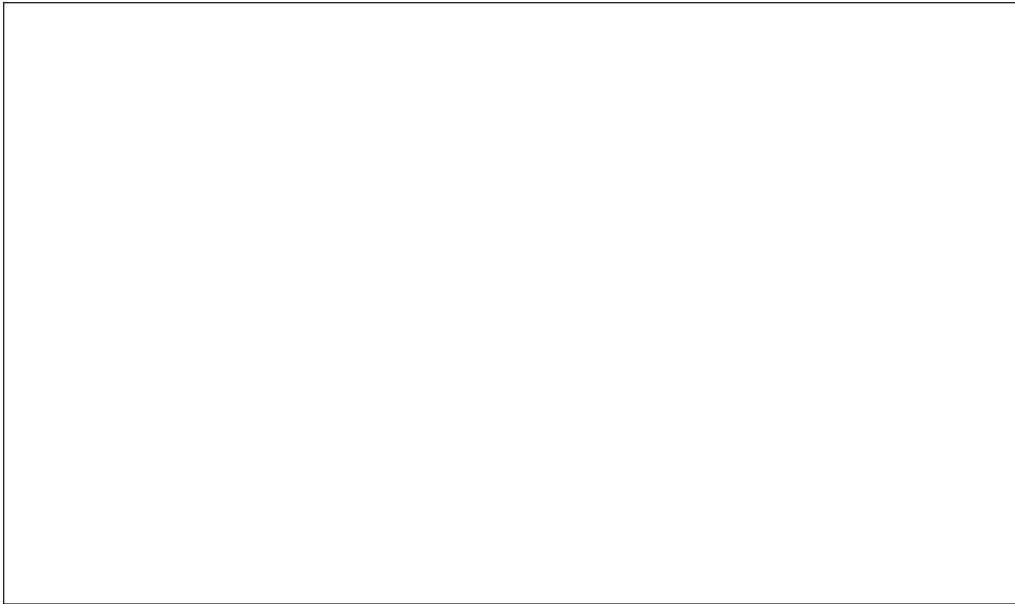
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# 37

## *Evolutionary History of Plants*

### Concept Outline

#### 37.1 Plants have multicellular haploid and diploid stages in their life cycles.

**The Evolutionary Origins of Plants.** Plants evolved from freshwater green algae and eventually developed cuticles, stomata, conducting systems, and reproductive strategies that adapt them well for life on land.

**Plant Life Cycles.** Plants have haplodiplontic life cycles. Diploid sporophytes produce haploid spores which develop into haploid gametophytes that produce haploid gametes.

#### 37.2 Nonvascular plants are relatively unspecialized, but successful in many terrestrial environments.

**Mosses, Liverworts, and Hornworts.** The most conspicuous part of a nonvascular plant is the green photosynthetic gametophyte, which supports the smaller sporophyte nutritionally.

#### 37.3 Seedless vascular plants have well-developed conducting tissues in their sporophytes.

**Features of Vascular Plants.** In vascular plants, specialized tissue called xylem conducts water and dissolved minerals within the plant, and tissue called phloem conducts sucrose and plant growth regulators within the plant.

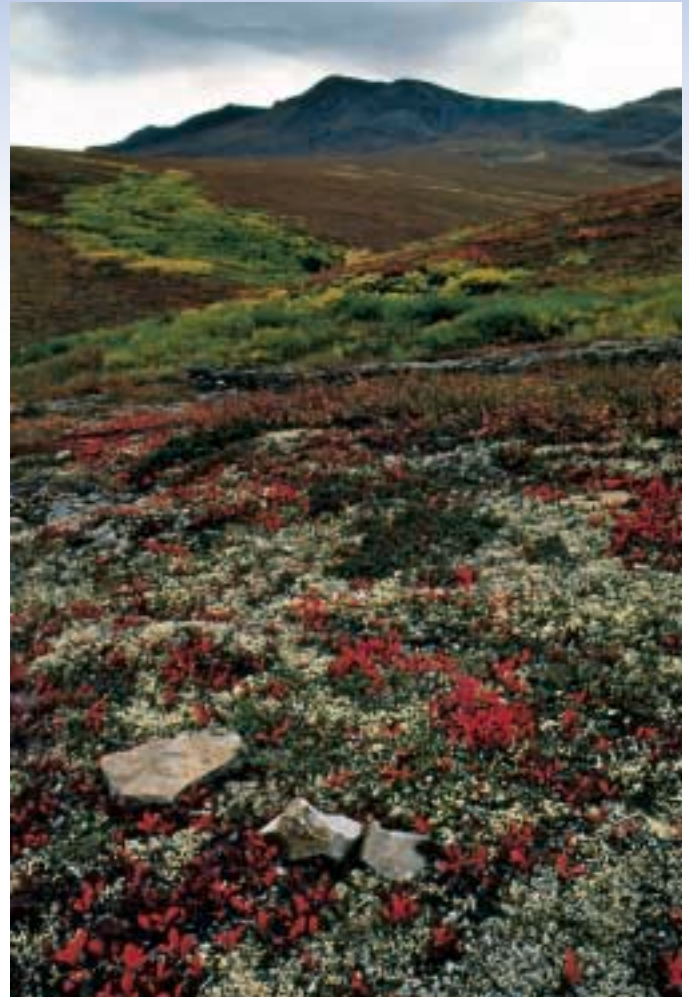
**Seedless Vascular Plants.** Seedless vascular plants have a much more conspicuous sporophyte than nonvascular plants do, and many have well-developed conducting systems in stem, roots, and leaves.

#### 37.4 Seeds protect and aid in the dispersal of plant embryos.

**Seed Plants.** In seed plants, the sporophyte is dominant. Male and female gametophytes develop within the sporophyte and depend on it for food. Seeds allow embryos to germinate when conditions are favorable.

**Gymnosperms.** In gymnosperms, the female gametophyte (ovule) is not completely enclosed by sporophyte tissue at the time of pollination by male gametophytes (pollen).

**Angiosperms.** In angiosperms, the ovule is completely enclosed by sporophyte tissue at the time of pollination. Angiosperms, by far the most successful plant group, produce flowers.



**FIGURE 37.1**

**An arctic tundra.** This is one of the harshest environments on earth, yet a diversity of plants have made this home. These ecosystems are fragile and particularly susceptible to global change.

**P**lant evolution is the story of the conquest of land by green algal ancestors. For about 500 million years, algae were confined to a watery domain, limited by the need for water to reproduce, provide structural support, prevent water loss, and provide some protection from the sun's ultraviolet irradiation. Numerous evolutionary solutions to these challenges have resulted in over 300,000 species of plants dominating all terrestrial communities today, from forests to alpine tundra, from agricultural fields to deserts (figure 37.1). Most plants are photosynthetic, converting light energy into chemical-bond energy and providing oxygen for all aerobic organisms. We rely on plants for food, clothing, wood for shelter and fuel, chemicals, and many medicines. This chapter explores the evolutionary history and strategies that have allowed plants to inhabit most terrestrial environments over millions of years.

## 37.1 Plants have multicellular haploid and diploid stages in their life cycles.

### The Evolutionary Origins of Plants

Biologists have long suspected that plants are the evolutionary descendants of green algae. Now we are sure. The evolutionary history of plants was laid bare at the 1999 International Botanical Congress held in St. Louis, Missouri, by a team of 200 biologists from 12 countries that had been working together for five years with U.S. federal funding. Their project, Deep Green (more formally known as The Green Plant Phylogeny Research Coordination Group), coordinated the efforts of laboratories using molecular, morphological, and anatomical traits to create a new “Tree of Life.” Deep Green confirmed the long-standing claim that green algae were ancestral to plants. More surprising was the finding that just a single species of green algae gave rise to the entire terrestrial plant lineage from moss through the flowering plants (angiosperms). Exactly what this ancestral alga was is still a mystery, but close relatives are believed to exist in freshwater lakes today. DNA sequence data is consistent with the claim that a single “Eve” gave rise to the entire kingdom Plantae 450 million years ago. At each subsequent step in evolution, the evidence suggests that only a single family of plants made the transition. The fungi appear to have branched later than the plants and are more closely related to us.

There are 12 plant phyla, all of which afford some protection to their embryos. All plants also have a haploid and a diploid stage that is multicellular. The trend over time has been toward increasing embryo protection and a smaller haploid stage in the life cycle. The plants are divided into two groups based on the presence or absence of vascular tissues which facilitate the transport of water and nutrients in plants. Three phyla (mosses, liverworts, and hornworts) lack vascular tissue and are referred to as the **nonvascular plants**. Members of 9 of the 12 plant phyla are collectively called **vascular plants**, and include, among others, the ferns, conifers, and flowering plants. Vascular plants have water-conducting xylem and food-conducting phloem strands of tissues in their stems, roots, and leaves. Vascular plants can be further grouped based on how much protection embryos have. The seedless vascular plants (ferns) provide less protection than the seeds of the gymnosperms (conifers) and angiosperms (flowering plants) (figure 37.2). About 150 million years ago the angiosperms arose with further innovations—flowers to attract pollinators and fruit surrounding the seed to protect the embryo and aid in seed dispersal. Many of these lineages have persisted. If you could travel back 65 million years to the dinosaur era, you would encounter oak, walnut, and sycamore trees!

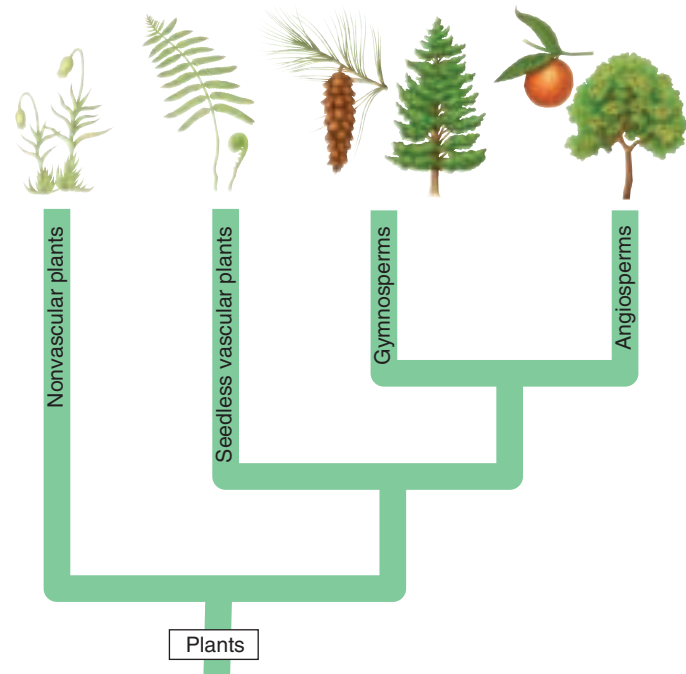


FIGURE 37.2

**The four major groups of plants.** In this chapter we will discuss four major groups of plants. The green algae, discussed in chapter 35, are the protists from which plants are thought to have evolved.

### Adaptations to Land

Plants and fungi are the only major groups of organisms that are primarily terrestrial. Most plants are protected from **desiccation**—the tendency of organisms to lose water to the air—by a waxy **cuticle** that is secreted onto their exposed surfaces. The cuticle is relatively impermeable and provides an effective barrier to water loss.

This solution creates another problem by limiting gas exchange essential for respiration and photosynthesis. Water and gas diffusion into and out of a plant occurs through tiny mouth-shaped openings called **stomata** (singular, **stoma**).

The evolution of leaves resulted in increased photosynthetic surface area. The shift to a dominant diploid generation, accompanied by the structural support of vascular tissue, allowed plants to take advantage of the vertical dimension of the terrestrial environment, resulting in trees.

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Plants evolved from freshwater green algae and eventually developed cuticles, stomata, conducting systems, and reproductive strategies that adapt them well for life on land.

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## Plant Life Cycles

All plants undergo mitosis after both gamete fusion *and* meiosis. The result is a multicellular haploid and a multicellular diploid individual, unlike us where gamete fusion directly follows meiosis. We have a **diploontic** life cycle (only the diploid stage is multicellular), but the plant life cycle is **haplodiplontic** (with multicellular haploid and diploid stages). The basic haplodiplontic cycle is summarized in figure 37.3. Brown, red, and green algae are also haplodiplontic (see chapter 35). While we produce gametes via meiosis, plants actually produce gametes by mitosis in a multicellular, haploid individual. The diploid generation, or **sporophyte**, alternates with the haploid generation, or **gametophyte**. Sporophyte means “spore plant,” and gametophyte means “gamete plant.” These terms indicate the kinds of reproductive cells the respective generations produce.

The diploid sporophyte undergoes meiosis to produce haploid spores (*not* gametes). Meiosis takes place in structures called **sporangia**, where diploid **spore mother cells** (**sporocytes**) undergo meiosis, each producing four haploid **spores**. Spores divide by mitosis, producing a multicellular, haploid gametophyte. Spores are the first cells of the gametophyte generation.

In turn, the haploid gametophyte produces haploid gametes by mitosis. When the gametes fuse, the zygote they form is diploid and is the first cell of the next sporophyte generation. The zygote grows into a diploid sporophyte that produces sporangia in which meiosis ultimately occurs.

While all plants are haplodiplontic, the haploid generation consumes a much larger chunk of the life cycle in mosses than in gymnosperms and angiosperms. In mosses, liverworts, and ferns, the gametophyte is photosynthetic and free-living; in other plants it is either nutritionally dependent on the sporophyte, or **saprobic** (deriving its energy directly from nonliving organic matter). When you look at moss, what you see is largely gametophyte tissue; their sporophytes are usually smaller, brownish or yellowish structures attached to or enclosed within tissues of the gametophyte. In most vascular plants the gametophytes are much smaller than the sporophytes. In seed plants, the gametophytes are nutritionally dependent on the sporophytes and are enclosed within their tissues. When you look at a gymnosperm or angiosperm, what you see, with rare exceptions, is a sporophyte.

The difference between dominant gametophytes and sporophytes is key to understanding why there are no moss trees. What we identify as a moss plant is a gameto-

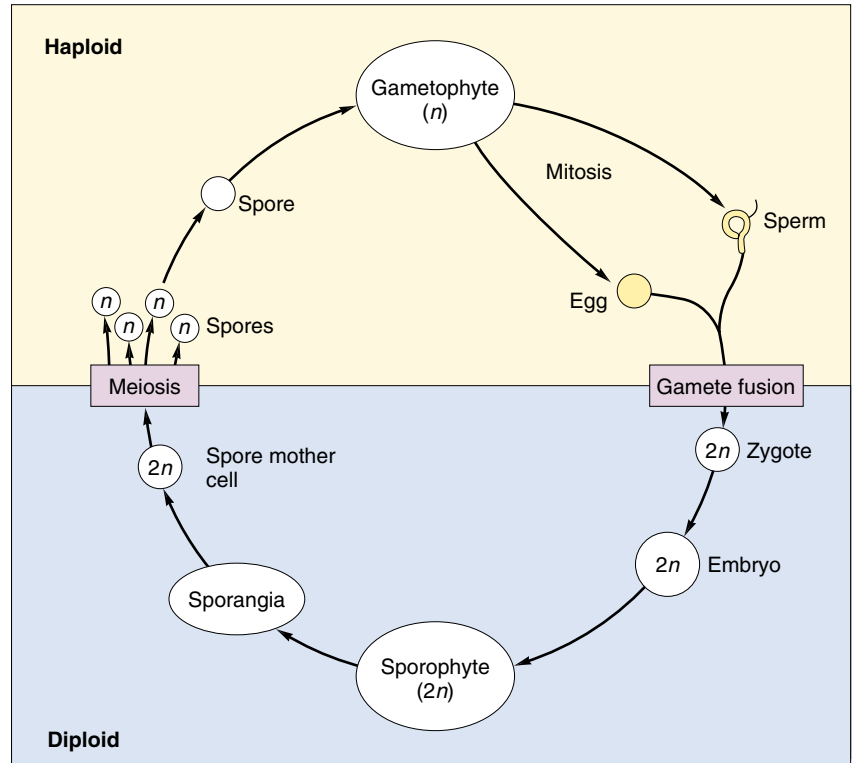


FIGURE 37.3

**A generalized plant life cycle.** Anything yellow is haploid and anything blue is diploid. Note that both haploid and diploid individuals can be multicellular. Also, spores are produced by meiosis while gametes are produced by mitosis.

phyte and it produces gametes at its tip. The egg is stationary and sperm lands near the egg in a droplet of water. If the moss were the height of a sequoia, not only would it need vascular tissue for conduction and support, the sperm would have to swim up the tree! In contrast, the fern gametophyte develops on the forest floor where gametes can meet. Fern trees abound in Australia and the haploid spores fall to the ground and develop into gametophytes.

Having completed our overview of plant life cycles, we will consider the major plant groups. As we do, we will see a progressive reduction of the gametophyte from group to group, a loss of multicellular **gametangia** (structures in which gametes are produced), and increasing specialization for life on the land, culminating with the remarkable structural adaptations of the flowering plants, the dominant plants today. Similar trends must have characterized the progression to seed plants over the hundreds of millions of years since a freshwater alga made the move onto land.

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**Plants have haplodiplontic life cycles. Diploid sporophytes produce haploid spores which develop into haploid gametophytes that produce haploid gametes.**

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## 37.2 Nonvascular plants are relatively unspecialized, but successful in many terrestrial environments.

### Mosses, Liverworts, and Hornworts

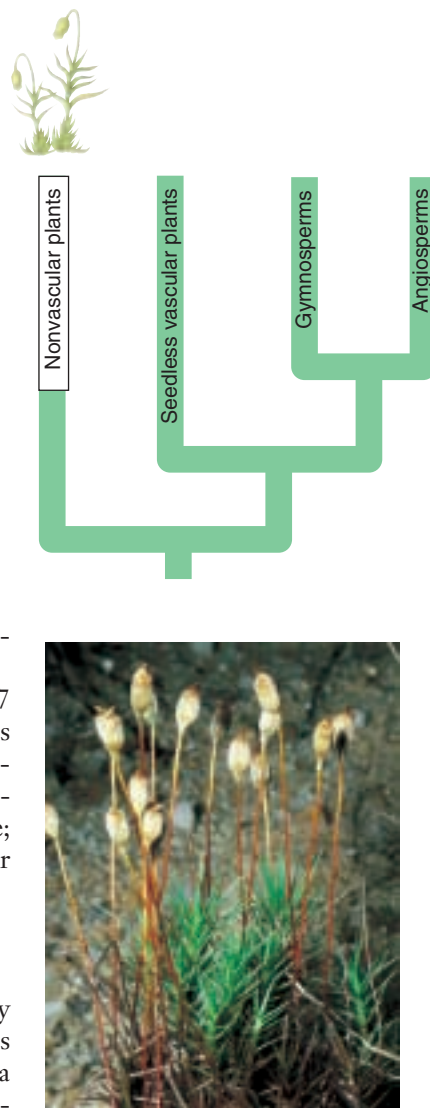
There are about 24,700 **bryophytes**—mosses, liverworts, and hornworts—that are simply but highly adapted to a diversity of terrestrial environments (even deserts!). Scientists now agree that bryophytes consist of three quite distinct phyla of relatively unspecialized plants. Their gametophytes are photosynthetic. Sporophytes are attached to the gametophytes and depend on them nutritionally to varying degrees. Bryophytes, like ferns and certain other vascular plants, require water (for example, rainwater) to reproduce sexually. It is not surprising that they are especially common in moist places, both in the tropics and temperate regions.

Most bryophytes are small; few exceed 7 centimeters in height. The gametophytes are more conspicuous than the sporophytes. Some of the sporophytes are completely enclosed within gametophyte tissue; others are not and usually turn brownish or straw-colored at maturity.

#### Mosses (Bryophyta)

The gametophytes of mosses typically consist of small leaflike structures arranged spirally or alternately around a stemlike axis (figure 37.4); the axis is anchored to its substrate by means of **rhizoids**. Each rhizoid consists of several cells that absorb water, but nothing like the volume of water absorbed by a vascular plant root. Moss “leaves” have little in common with true leaves, except for the superficial appearance of the green, flattened blade and slightly thickened midrib that runs lengthwise down the middle. They are only one cell thick (except at the midrib), lack vascular strands and stomata, and all the cells are haploid.

Water may rise up a strand of specialized cells in the center of a moss gametophyte axis, but most water used by the plant travels up the outside of the plant. Some mosses



**FIGURE 37.4**  
A hair-cup moss, *Polytrichum* (phylum Bryophyta). The “leaves” belong to the gametophyte. Each of the yellowish-brown stalks, with the capsule, or sporangium, at its summit, is a sporophyte.

also have specialized food-conducting cells surrounding those that conduct water.

Multicellular gametangia are formed at the tips of the leafy gametophytes (figure 37.5). Female gametangia (**archegonia**) may develop either on the same gametophyte as the male gametangia (**antheridia**) or on separate plants. A single egg is produced in the swollen lower part of an archegonium while numerous sperm are produced in an antheridium. When sperm are released from an antheridium, they swim with the aid of flagella through a film of dew or rainwater to the archegonia. One sperm (which is haploid) unites with an egg (also haploid), forming a diploid zygote. The zygote divides by mitosis and develops into the sporophyte, a slender, basal stalk with a swollen capsule, the **sporangium**, at its tip. As the sporophyte develops, its base becomes embedded in gametophyte tissue, its nutritional source. The sporangium is often cylindrical or club-shaped. Spore mother cells within the sporangium undergo meiosis, each becoming four haploid spores. At maturity, the top of the sporangium pops off, and the spores are released. A spore that lands in a suitable damp location may germinate and grow into a threadlike structure that branches to form rhizoids and “buds” that grow upright. Each bud develops into a new gametophyte plant consisting of a leafy axis.

In the Arctic and the Antarctic, mosses are the most abundant plants, boasting not only the largest number of individuals in these harsh regions, but also the largest number of species. Many mosses are able to withstand prolonged periods of drought, although they are not common in deserts. Most are remarkably sensitive to air pollution and are rarely found in abundance in

or near cities or other areas with high levels of air pollution. Some mosses, such as the peat mosses (*Sphagnum*), can absorb up to 25 times their weight in water and are valuable commercially as a soil conditioner, or as a fuel when dry.

## Liverworts (Hepaticophyta)

The old English word *wyrt* means “plant” or “herb.” Some common liverworts have flattened gametophytes with lobes resembling those of liver—hence the combination “liverwort.” Although the lobed liverworts are the best-known representatives of this phylum, they constitute only about 20% of the species (figure 37.6). The other 80% are leafy and superficially resemble mosses. Liverworts are less complex than mosses. Gametophytes are prostrate instead of erect, and the rhizoids are one-celled.

Some liverworts have air chambers containing upright, branching rows of photosynthetic cells, each chamber having a pore at the top to facilitate gas exchange. Unlike stomata, the pores are fixed open and cannot close.

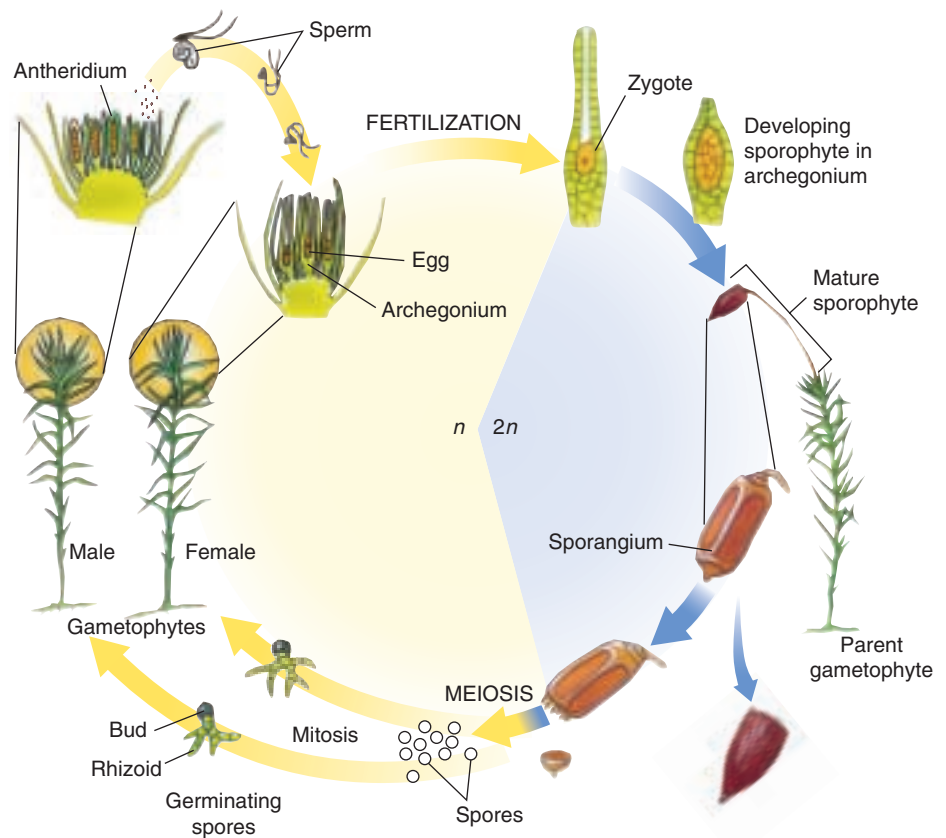
Sexual reproduction in liverworts is similar to that in mosses. Lobed liverworts form gametangia in umbrella-like structures. Asexual reproduction occurs when lens-shaped pieces of tissue that are released from the gametophyte grow to form new gametophytes.

## Hornworts (Anthocerotophyta)

The origins of hornworts are a puzzle. They are most likely among the earliest land plants, yet the earliest fossil spores date from the Cretaceous period, 65 to 145 million years ago, when angiosperms were emerging.

The small hornwort sporophytes resemble tiny green broom handles rising from filmy gametophytes usually less than 2 centimeters in diameter (figure 37.7). The sporophyte base is embedded in gametophyte tissue, from which it derives some of its nutrition. However, the sporophyte has stomata, is photosynthetic, and provides much of the energy needed for growth and reproduction. Hornwort cells usually have a single chloroplast.

**The three major phyla of nonvascular plants are all relatively unspecialized, but well suited for diverse terrestrial environments.**



**FIGURE 37.5**

**Life cycle of a typical moss.** The majority of the life cycle of a moss is in the haploid state. The leafy gametophyte is photosynthetic, while the smaller sporophyte is not, and is nutritionally dependent on the gametophyte. Water is required to carry sperm to the egg.



**FIGURE 37.6**

**A common liverwort, *Marchantia* (phylum Marchantiophyta).** The sporophytes are borne within the tissues of the umbrella-shaped structures that arise from the surface of the flat, green, creeping gametophyte.



**FIGURE 37.7**

**Hornworts (phylum Anthocerotophyta).** Hornwort sporophytes are seen in this photo. Unlike the sporophytes of other bryophytes, these are photosynthetic, but they also depend on the gametophyte for nutrition.

## 37.3 Seedless vascular plants have well-developed conducting tissues in their sporophytes.

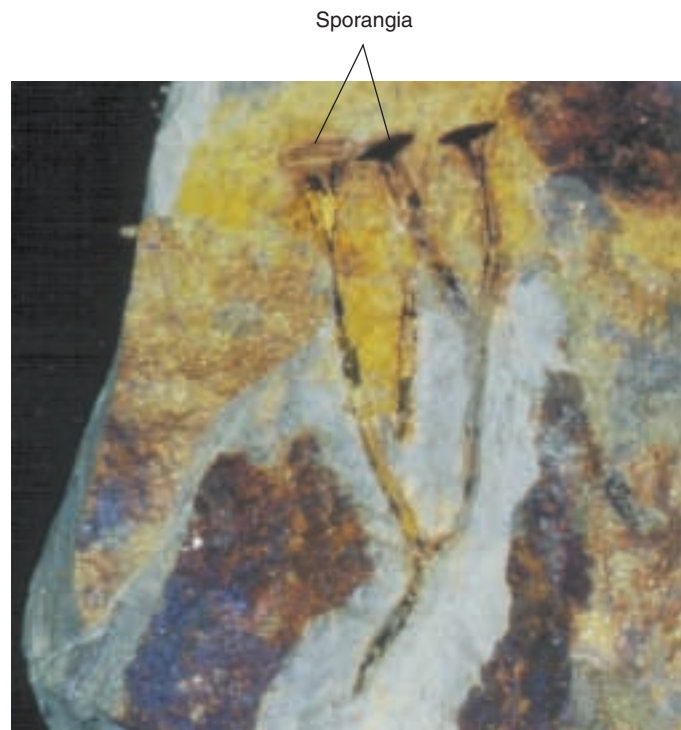
### Features of Vascular Plants

The first vascular plants for which we have a relatively complete record belonged to the phylum Rhyniophyta; they flourished some 410 million years ago but are now extinct. We are not certain what the very earliest of these vascular plants looked like, but fossils of *Cooksonia* provide some insight into their characteristics (figure 37.8). *Cooksonia*, the first known vascular land plant, appeared in the late Silurian period about 420 million years ago. It was successful partly because it encountered little competition as it spread out over vast tracts of land. The plants were only a few centimeters tall and had no roots or leaves. They consisted of little more than a branching axis, the branches forking evenly and expanding slightly toward the tips. They were **homosporous** (producing only one type of spore). Sporangia formed at branch tips. Other ancient vascular plants that followed evolved more complex arrangements of sporangia. Leaves began to appear as protuberances from stems.

*Cooksonia* and the other early plants that followed it became successful colonizers of the land through the development of efficient water- and food-conducting systems known as **vascular tissues**. The word *vascular* comes from the Latin *vasculum*, meaning a “vessel or duct.” These tissues consist of strands of specialized cylindrical or elongated cells that form a network throughout a plant, extending from near the tips of the roots, through the stems, and into true leaves. One type of vascular tissue, the *xylem*, conducts water and dissolved minerals upward from the roots; another type of tissue, *phloem*, conducts sucrose and hormonal signals throughout the plant. It is important to note that vascular tissue developed in the sporophyte, but (with few exceptions) not the gametophyte. (See the discussion of vascular tissue structure in chapter 38.) The presence of a cuticle and stomata are also characteristic of vascular plants.

The nine phyla of vascular plants (table 37.1) dominate terrestrial habitats everywhere, except for the highest mountains and the tundra. The haplodiplontic life cycle persists, but the gametophyte has been reduced during evolution of vascular plants. A similar reduction in multicellular gametangia has occurred.

Accompanying this reduction in size and complexity of the gametophytes has been the appearance of the seed. Seeds are highly resistant structures well suited to protect a plant embryo from drought and to some extent from predators. In addition, most seeds contain a supply of food for the young plant. Seeds occur only in **heterosporous** plants (plants that produce two types of spores). Heterospory is believed to have arisen multiple



**FIGURE 37.8**

*Cooksonia*, the first known vascular land plant. Its upright, branched stems, which were no more than a few centimeters tall, terminated in sporangia, as seen here. It probably lived in moist environments such as mudflats, had a resistant cuticle, and produced spores typical of vascular plants. This fossil represents a plant that lived some 410 million years ago. *Cooksonia* belongs to phylum Rhyniophyta, consisting entirely of extinct plants.

times in the plants. Fruits in the flowering plants add a layer of protection to seeds and attract animals that assist in seed dispersal, expanding the potential range of the species. Flowers, which evolved among the angiosperms, attract pollinators. Flowers allow plants to overcome limitations of their rooted, immobile nature and secure the benefits of wide outcrossing in promoting genetic diversity.

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**Most vascular plants have well-developed conducting tissues, specialized stems, leaves, roots, cuticles, and stomata. Many have seeds which protect embryos until conditions are suitable for further development.**

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